



NURSING PRACTICE I

Community Health Nursing

Community Health Nursing is a synthesis of nursing knowledge and practice, as well as the science and practice of public health, implemented through the systematic use of the nursing process and other processes to promote health and prevent illness in population groups. (Outside the hospital)

- (1) Promotion of health (increase level of health through healthy lifestyle) and prevention of disease/illness prevention (through health teachings)
 - **Health promotion**
 - Aim IBW (Ideal Body Weight)
 - <10 - growth chart; >10 BMI
 - (N) 18-25 (WHO); 18-23 (APOG)
 - Build nutrition practices
 - Choose food wisely
 - Diet
 - Exercise
 - Rest
 - Stop Smoking
 - **Illness prevention**
 - Specific protection
- (2) Comprehensive, general, continual and not episodic
- (3) Recognize the population as a whole
- (4) The nurse and the client have greater control in making decisions related to health care, and collaborate as equals
- (5) Recognizes the impact of different factors on health and has greater awareness of the client's life and situation

Goal of CHN

- Optimum level of functioning (highest level of wellness)

Subspecialty:

- **School Health Nursing** (Health assessment once a year/safety of the students)
- **Occupational Health Nursing** (Company nurse/health and safety of employee)
 - 50 or more employees - requires company nurse
- **Mental Health Nursing** (mental hygiene)
 - Addresses mental health issues in the community
 - **Mental Health Law (RA 11036)**
 - First ever mental health act
 - 4 facets of mental health burden
 - **Defined burden** - affects individual



- **Undefined burden** - affects other individual
- **Hidden burden** - “stigma” (mark of shame) and violation of human rights
- **Future burden** - next generation

Clients of Community Health Nurses

Individual - considered as the “entry point” (primary care)

Family - collection of people who are integrated, interacting and interdependent (basic unit of service/care)

Group/Aggregated/Population - a group of people who share common characteristics, developmental stage or common exposure to particular environment factors and consequently common health problems, issues and concerns

Community - a group of people sharing common geographic boundaries and/or common values and interests within a specific social system

Goal of PHC

- Self-reliance in health

Goal of COPAR

- **Community Health Development** (social transformation)

Health

1. Viewed as a continuum, is considered as the goal of public health in general and community health nursing
2. A state of complete physical, mental and social well-being and not merely the absence of disease (WHO)
3. An integrated method of functioning which is oriented toward maximizing the potential which the individual is capable (Dunn & Pender)
4. Quality of life involving social, emotional, mental, spiritual and biological fitness on the part of the individual which results from adaptations to the environment (Dubos)
5. Health and illness in relation to the environment (Florence Nightingale)
6. A state characterized by soundness and wholeness of human structures and bodily and mental functions (Dorothea Orem)

Determinants of Health - factors that make an individual healthy or not

- Genetics
- Income
- Education
- Behavior
- Health care delivery system



Primary Health Care

- Essential health care/basic health care based on practical, scientifically sound and socially acceptable methods and technology made universally accessible to individuals and families in the community through their full participation and at a cost that the community can afford to maintain at every stage of their development in the spirit of self-reliance and self-determination (WHO 1978, UNICEF)
Alma Ata Conference (Kazakhstan)
- **LOI (Letter Of Instruction) 949 - Legal Basic of PHC (Marcos - Oct. 1979)**
 - **Vision:** “Health in the hands of the people”
 - **Mission:** “Increase opportunities that people will manage their own health care”
 - **2 Core Concept of PHC:**
 - **Partnership** (Active)
 - **Empowerment** (Transfer **K**nowledge, **S**kills and **A**ttitude)
 - **4 Pillars of PHC:**
 - Active community participation (most important)
 - Intra & Intersectional linkages
 - **Intrasectoral** - within the HCDS (primary, secondary & tertiary)
 - **Intersectoral** - outside the HCDS (NGO, government organization)
 - Use of appropriate technology (local indigenous resources - herbal)
 - Support mechanism made available (support from the local government)
 - **Characteristics of PHC:**
 - Community-based
 - Accessible (within 3-5 km, within 30 mins)
 - Sustainable
 - Affordable (low cost)
 - Self-reliance
 - Available

Essential Elements of PHC

- **EDUCATION** about prevailing health problems, including methods of prevention and control (competency-based teaching and learning)
 - Principles
 - Learning is intellectual & emotional process
 - Learning is an individual matter (different learning style)
 - Learning is facilitated when experience have meaning to the learner
 - **Level of Prevention** (health teachings can be rendered in all levels)
 - (1) **Primary level** - health education, immunization (patient is healthy) - health promotion & illness prevention



- (2) **Secondary level** - can detect this early stage so that prompt treatment can be initiated to avoid progression of the disease (high-risk individuals) - diagnosis, screening and treatment (prevent complications)
- (3) **Tertiary level** - applicable to limit the disability and restore the functional capabilities of the patient (post-treatment patients) - rehabilitation (return to health state), palliative care (supportive quality of life - pain management)
- Promotion of adequate food supply and **PROPER NUTRITION**
 - **Malnutrition**
 - **Macronutrients:** CHO, CHON
 - **Micronutrients:** vitamins and minerals
 - **Marasmus:** low carbs, protein, skin & bones (wrinkled)
 - **Kwashiorkor:** normal carbs, low protein, thin extremities, fluid shifting, moon face, ascites, thin sparse hair
 - **Mid Upper Arm Circumference (MUAC)** - diagnostic
 - **Red** - severe malnutrition <115 mm
 - **Yellow** - moderate malnutrition 115 - 125mm
 - **Green** - normal malnutrition >125mm
 - **Ready to Use Therapeutic Food (RUTF)**
 - Target: Macronutrients
 - **Food Fortification Program (Sangkap Pinoy)** - added nutrition
 - April & October (twice a year)
 - **Target:** Micronutrients
 - **Vitamin A (Retinol) deficiency**
 - Bitot's spots
 - Corneal clouding
 - Xerophthalmia
 - Night blindness

Age	Normal	Vitamin A Deficiency
6-12 months	100,000 IU	100,000 IU
12 - 60 months	200,000 IU	200,000 IU
Schedule	Today, 6 months after	Today, Tomorrow & 2 weeks after

- **Iron** - Iron Deficiency Anemia
 - Best time: empty stomach (acidic environment)
 - (+) Orange juice/calamansi
- **Iodine** - goiter and mental retardation (once/year)
- **IMMUNIZATION** (Expanded Program on Immunization) against major infectious diseases



- **DOH: National Immunization Program (NIP)**
- **PD 996 (1976):** all children below 8 years old will receive free vaccine
- **RA 10152 (2011):** mandatory immunization for children below 5 years old
 - **Pentavalent** (5 in 1 vaccine) - Diphtheria, Pertussis, Tuberculosis, Hepatitis B, HiB (Haemophilus influenza type B - to prevent meningitis)
 - **Pneumococcal Conjugate Vaccine (PCV)** - to prevent meningitis
 - **Inactivated Polio Vaccine (IPV)**
 - **Rota Vaccine** - to prevent gastroenteritis
 - **Measles Mumps Rubella (MMR)**
- **5 Elements EPI**
 - **Surveillance** - search for cases as a form of evaluation (last part)
 - **Information Education Communication (IEC)** - people should be aware that there is free vaccine
 - National Immunization Day: Wednesday
 - BHS - monthly
 - Geographically Isolated and Disadvantaged Area (GIDA) - quarterly
 - **Cold chain and logistic management** - all vaccines are sensitive to heat
 - Most sensitive to heat (Freezer: -15C to -25C): **Varicella, OPV, MMR**
 - BCG, Hepa B, Pentavalent, PCV, Rota Vaccine, IPV, TT/TD (Body: +2C & +8C)
 - **Vaccine Vial Monitor (VVM)** - checks the vaccines are damaged by heat
 - **Note:** if the square is lighter than the circle - okay to give
 - **Assessment & Evaluation**
 - **RA 10152 - Mandatory Infants and Children Health Immunization Act of 2011**

Vaccine	Dose	Schedule	Route	Dosage	Nursing Intervention
BCG	1	At birth	Intradermal	0.05	● (-) AIDS, leukemia, chemotherapy ● (+) Hepatitis B ● Clean site: water only ● Koch's phenomenon ● Deep abscess
Hepa B			Intramuscular	0.5	● Recombinant RNA
Pentavalent	3	4 weeks 6 weeks 10 weeks	Intramuscular	0.5	● Side effects: fever (paracetamol RTC every 6 hours for 24



					hours), local tenderness (cold compress) ● Adverse effects: seizure/convulsion due to pertussis) within 3 days (-) pentavalent 2, 3
OPV			Oral	2 drops	● Live attenuated ● On the tongue ● Vomits within 30 mins (Give) ● Vomits after 30 mins (do not give) ● NPO for 30 mins
PCV			Intramuscular	0.5	-
Rota V	2	6 weeks 10 weeks	Oral	1.5	
IPV		6 weeks 14 weeks	Intramuscular	0.5	● Killed ● Vastus lateralis (IM)
MMR		9 months and 12 months	Subcutaneous	0.5	● Live attenuated ● Fever - paracetamol ● Rashes - keep it dry

- **Target Setting** - primary element
 - **Eligible population:**
 - **Infant:** total population x 2.7%
 - **Pregnant:** total population x 3.5%
 - **Annual Vaccine Required:** EP (eligible population) x Dose x WF (wastage factor) 1/(1-wastage %)
 - 5% - pentavalent, PCV, Rota V
 - 15% - hepa B, IPV, TT/TD, Flu vaccine
 - 25% - MMR, MR, Japanese encephalitis, OPV
 - 60% - BCG (most sensitive in all vaccines) 4-6 hours
 - **Note:** change percentage to decimal to get the wastage factor
- **SANITATION Environment**
 - **PD 856:** Sanitation Code of the Philippines
 - **PD 825:** Garbage Disposal Act
 - **Food sanitation**



- **4 Rights of Food Safety**
 - **Right Source:** fresh
 - **Right Preparation:** wash
 - **Right Cooking:** 70C
 - **Right Storage:** Cool: 10C; Warm: 60C
 - Room temperature: maximum of 2 hours
- **Food Establishment**
 - **Owner:** sanitary permit
 - **Food handlers:** health certificate
 - **Class Rating:**
 - **Class A:** Excellent
 - **Class B:** Very Satisfactory
 - **Class C:** Satisfactory
- **Water Facility**

Level 1	Level 2	Level 3
● Point source ● Protected well (deep well) ● Developed spring ● >250 m ● Rural	● Communal faucet - stand posts ● 1:4-5 household	● Individual household connection ● Waterworks system ● Cleanest

- **Toilet Facility**

Level 1	Level 2	Level 3
● Non-water carriage ● Pit latrines ● Small amount water toilet facility ● Pour flush toilet	● Water-sealed ● Flushed style ● Septic tank	● Level 2 + sewage system ● Treatment plant (waste water treatment) ● Cleanest

- Prevention and control of **LOCALLY ENDEMIC DISEASE**
 - **Epidemiology** - study of occurrence and distribution of diseases (backbone of preventing disease)
 - **Based on Occurrence of Disease**
 - **Sporadic disease** - occurs only occasionally
 - **Endemic disease** - constantly present in the population, country or community



- **Epidemic disease/outbreak**- acquire disease in a relatively short period greater than normal number of cases in an area within a short period of time
 - **Point source epidemic** - simultaneous exposure (single event)
 - **Propagated epidemic** - person to person (flu, COVID)
 - **Cyclical/Seasonal epidemic** - rainy: leptospirosis, dengue; summer: conjunctivitis, heat stroke
 - **Secular epidemic** (variation) - change of disease overtime
- **Pandemic disease** - epidemic disease that occurs worldwide
- **Field Health Service Information System (FHSIS)** - recording and the reporting system of DOH
 - **Objective**
 - **Data summary** (1st cause of death in the Philippines: cardiac problems, 2nd: cancer, 3rd: disease of the blood vessel - hypertension, aneurysm)
 - **Program monitoring & evaluation** - follow-up for vaccination
 - **Standardized facility database**
 - **Accurate and timely data**
 - **Minimize recording and reporting burden**
 - **4 Components of FHSIS**
 - **Family Treatment Record** - fundamental block of FHSIS (sort per family)
 - Individual health record
 - **Target Client List** - 2nd building block (per DOH program) - use for follow-up and monitoring, making of reports (total cases)
 - **Reporting Form** - tally report, transmitted from one facility to another
 - **Reporting unit** - BHS (midwife) & RHU/HC (PHN) deliver report to PHO (**monthly, quarterly** (3 months), **annually**)
 - **M1 infectious disease report** - weekly
 - **Output Report** - consolidate reporting form produced by PHO, delivers report to regional DOH, then DOH Manila
 - **Record** - raw data, **Report** - processed data
- **Community Health Information Tracking System (CHITS)** - electronic-FHSIS
- **Vital Statistics** - study of vital events (birth date, marriage)

Title	Significance	Formula
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Crude Birth Rate	Measures natural increase in population	Total birth/Total population x 1000
Crude Death Rate	Measures natural decrease in population	Total death/Total population x 1000
Maternal Mortality Rate	Index of prenatal care, death due to pregnancy or labor	Total death due to pregnancy/Live birth x 1000
Neonatal Mortality Rate	Death of <28 days of age, Index of postpartum care	Total death below 28 days/Live birth x 1000
Infant Mortality Rate	Death of >28 days to < 1 year of age, Index health status	Total death above 28 days and below 1 year of age/Live birth x 1000
Fetal Death Rate	Pregnancy wastage	Total death intrautero/Live birth x 1000
Incidence Rate	Measures acute disease in the community (<6 months)	Total number of new cases/Total population x 1000
Prevalence Rate	Measures chronic disease in the community (>6 months)	Total number of old and new cases/Total persons examined x 100
Sex Ratio	Measures the balance of male over female	Male/Female x 100
Swaroop's Index	Measures proportionate mortality indicator, longevity of life, lifestyle, Best index of health status of the whole community	Total death of >50/Total Death x 100

- Appropriate treatment of common diseases (**COMMUNICABLE & NON-COMMUNICABLE DISEASE**) and **EMERGENCY SERVICES**
 - **Non-communicable disease** - lifestyle related to disease
 - **Disasters:** disruption in society
 - **Natural:** act of God
 - **Man-made:** human-related activities
 - **Onset**
 - **Acute:** within days and weeks
 - **Chronic/creeping:** months or years
 - **RA 10121: NDRRMC**
 - National government budget: 2%
 - LGU budget: 5%



- **Principles**
 - It's the responsibility of all
 - Capability building (drills)
 - **First priority:** people; **second priority:** resources
- **MATERNAL AND CHILD HEALTH**
 - Reduced MMR
 - **Strategies**
 - **Prenatal visit:** 8 (WHO); 4 (DOH) **sentrong sigla** (quality assurance program)
 - **Basic Emergency Obstetrical Care (BEmOc)** - concept: all pregnant women are high-risk pregnancy
 - Address common problems of pregnancy
 - Capability building training
 - Oxytocin
 - Antibiotic
 - Anticonvulsant: Magnesium sulfate (Pre-eclampsia)
 - Steroids - preterm labor, promote lung maturity (bethamethasone)
 - Assisted delivery (forceps delivery, vacuum-assisted delivery)
 - Manual removal of placenta/retained products of conception (prevent bleeding and DIC)
 - **Comprehensive Emergency Obstetrical Care (CEmOC)**
 - (+) Blood transfusion & CS
 - **Activities**
 - **Mother & Child (MC) booklet** - all information about pregnancy, immunization
 - **Micronutrient supplements**
 - **Vitamin A (retinol)** - teratogenic in first trimester, 4th month - 10,000 IU twice a week, postpartum - 20,000 IU within 6 weeks
 - **Iron** (Ferrous sulfate + Folic acid) - daily, taken with empty stomach
 - **Iodine** - to prevent goiter (mother), mental retardation (baby); once/pregnancy
 - **Tetanus Toxoid/Tetanus Diphtheria** - 2 TT in pregnancy for active immunity

TT/TD	Schedule	Mother (Active)	Baby (Passive)
1	Anytime/ 5th-6th months of pregnancy	None (0%)	None



2	1 month after TT 1	3 years (80%)	6 months to 1 year (Need own TT injection)
3	6 months after TT2	5 years (95%)	
4	1 year after TT3	10 years (99%)	
5	1 year after TT4	Lifetime (99%)	

- **Essential Intrapartal Newborn Care (EINC)**

- During labor
- (-) NPO & IVF, shave of pubic hair (causes abrasion), lithotomy
- (+) ambulation, companion of choice, any position of comfort, restrictive episiotomy (less blood loss, fast recovery)
- **Delivery of placenta** (Active Management of Third Stage of Labor)
 - IM Oxytocin
 - Controlled cord traction (signs of placental separation-gushing of blood, lengthening of cord, fundus rises)
 - Uterine massage

- **Essential Newborn Care (Unang Yakap)**

Time-Bound	Non-Time bound	Unnecessary Intervention
Dry infant immediately	● Hepa B ● Vitamin K ● Erythromycin ointment	● Suctioning ● slap/stimulate heel ● foot printing ● milking of the cord ● immediate clamping ● wash immediately ● cover stump ● antiseptic (alcohol/betadine)
Early skin-to-skin contact	Anthropometric measurement (weight, height, head and abdominal circumference)	
Delayed cord clamping (1-3 minutes) - once the pulsation has stopped	Physical assessment	
Breastfeeding (exclusive for the first 6 months)	Wash after 6 hours	
	APGAR Scoring (1 minute and 5 minutes after birth)	

- **Newborn Screening** - to detect genetic disorder

- Prevent complications
 - Mental retardation
 - Premature death



- **RA 9288**
- After 24 hours through heel prick (lateral)
- **2 classification of NBS**
 - **Basic standard** (6 genetic disorders)
 - (1) **Congenital hypothyroidism**
 - (2) **Congenital adrenal hyperplasia**
 - (3) **Galactosemia**
 - (4) **G6PD deficiency**
 - (5) **Phenylketonuria (PKU)**
 - (6) **Maple-syrup urine**
 - **Expanded** (+22 genetic disorders, total of 28)
 - (1) Cystic fibrosis
 - (2) Organic acid disorder
 - (3) Amino acid disorder
 - (4) Fatty acid disorder

Strategies

- (1) Reorientation and reorganization of the national health care system with the establishment of functional support mechanism in support of the mandate of devolution under the Local Government Code 1991
- (2) Effective preparation and enabling process for health action at all levels
- (3) Mobilization of the people to know their communities and identifying their basic health needs with the end in view of providing appropriate solutions (including legal measures) leading to self-reliance and self-determination
- (4) Development and utilization of appropriate technology focusing on local indigenous resources available in and acceptable to the community
- (5) Organization of communities arising from their unexpressed needs which they have decided to address and that this is continually evolving in pursuit of their own development
- (6) Increase opportunities for community participation in local level planning, management, monitoring and evaluation within the context of regional and national objectives
- (7) Development of intra-sectoral linkages with other government and private agencies so that programs of the health sector is closely linked with those of other socio-economic sectors at the national, intermediate and community levels
- (8) Emphasizing partnership so that the health workers and the community leaders/members view each other as partners rather than merely providers and receiver of health care respectively

Health Care Delivery System

- the totality of societal services and activities designed to protect or restore the health of individuals, families, groups and communities.



(1) **public sector** which is largely financed through tax-based budgeting system at both national and local levels and where health care generally given free at the point of service

(2) the **private sector** which is largely market-oriented and where health care is paid through user fees at the point of service.

Levels of Health Care

- (1) **Primary level of care** (Accessible) - health care provided by center physicians, PHNs, RH midwives, BHWs, traditional healers and others at the barangay health stations (BHS) and RHU. The primary health facility is usually the first contact between the community members and the other levels of the health facility. Provides basic health care services.
- (2) **Secondary level of care** - usually given in health facilities either privately owned or government operated such as infirmaries, municipal and district hospitals and out-patient departments or provincial hospitals. It is capable of performing minor surgeries and performing some simple laboratory examinations.
- (3) **Tertiary level of care** - rendered by specialists including medical centers as well as regional and provincial hospitals and specialized hospitals. Complicated cases and intensive care requires tertiary care.

2 way referral system - Intrasectoral linkage

Facilities

- **BHS** - satellite station/extension; manned by PHM, BHW
- **HC/RHU** - main center/intermediate primary center; complete health care team
 - **Frontline:** PHM
 - **Supervisor:** PHN
 - **Manager:** RHP (overall activities of the center)
 - **Sanitary inspector:** environment sanitation
 - **Medtech:** laboratory
 - **Dentist:** Dental Health Program

Levels of Primary Health Care Workers

- (1) **Village or BHWs** - trained community health workers or health auxiliary volunteer or a traditional birth attendant or healer
- (2) **Intermediate level health workers** - general medical practitioners or their assistants. PHNs, Rural sanitary inspectors and midwives may compose these groups

Health Worker to Patient Ratio (DOH)

Category	Ratio	Legal Basis
PHN	1:10,000	RA 7160, RA 7305



Midwife	1:5,000	RA 1082, RA 7305
RHU/Physician	1:20,000	RA 1082, RA 7160

Department of Health Programs

- The national agency mandated to lead the health sector towards assuring quality health care for all Filipinos
- **Vision:** “Filipinos are among the **healthiest** people in Southeast Asia by 2022 and Asia by 2040”
- **Mission:** “To lead the country in the development of a **productive, resilient, equitable and people-centered** health systems”
- **E.O 102:** 3 functions
 - **Leadership in health** (policies)
 - **Enabler & Capacity builder** (new strategy & training)
 - **Administrative** (manage tertiary services/emergency/specialty hospital)

Goal of DOH

- National objective for health (roadmap for all stakeholders) evaluate every 5 years

Local Government Code of 1991 (RA 7160)

- Decentralization & Devolution (transfer of power)
- **DOH:** formulate programs
- **LGU:** implement programs
 - **Provincial government:** governor (chairman) - governs secondary facilities; vice chairman: (PHO) provincial health officer (doctor)
 - **Municipal/city government:** mayor (chairman) - governs primary facilities; vice chairman: MHO/CHO

Health Sector Reform Agenda (HSRA)

- The overriding goal of DOH
- (1) Provide fiscal autonomy to government hospitals
- (2) Secure funding for priority public health programs
- (3) Promote the development of local health systems and ensure its effective performance
- (4) Strengthen the capacities of health regulatory agencies
- (5) Expand the coverage of the National Health Insurance Programs

Millennium Development Goals (MDGs)

- (1) Eradicate extreme poverty and hunger
- (2) Achieve universal primary education
- (3) Promote gender equality and empower women
- (4) Reduce child mortality
- (5) Improve maternal health



- (6) Combat HIV/AIDS, malaria and other diseases
- (7) Ensure environmental sustainability
- (8) Develop global partnership for development

Sustainable Development Goals (SDGs) 2030

- (1) No poverty
- (2) Zero hunger
- (3) Good health and well-being
- (4) Quality education
- (5) Gender equality
- (6) Clean water and sanitation
- (7) Affordable and clean energy
- (8) Decent work and economic growth
- (9) Industry, innovation and infrastructure
- (10) Reduced inequalities
- (11) Sustainable cities and communities
- (12) Responsible consumption and production
- (13) Climate action
- (14) Life below water
- (15) Life on land
- (16) Peace, justice and strong institutions
- (17) Partnership for the goals

FOURmula One for Health (Arroyo)

Goals:

- (1) Better health outcomes
- (2) More responsive health systems
- (3) Equitable health care financing

Elements:

- (1) Health financing
- (2) Health regulation
- (3) Health service delivery
- (4) Good governance

Universal Health Care Act (RA 11223)

- Theme: Pangkalahatang kalusugan
- All Filipinos are automatically enrolled to PhilHealth
- **Strategic Thrusts:**
 - Financial risk protection through expansion in NHIP enrollment and benefit delivery
 - Improved access to quality hospitals and health care facilities



- Attainment of the health-related MDGs
 - **Strategic Instruments**
 - Health financing
 - Service delivery
 - Policy, standards and regulation
 - Governance for health
 - Human resource for health
 - Health information
 - **Public Health Programs**
 - Reproductive and Maternal Health
 - Expanded Garantisadong Pambata (child health)
 - Control of communicable disease
 - Control of non-communicable or lifestyle disease
 - Environmental health

F1PLUS (FOURmula One Plus) (Duterte)

- Theme: boosting universal health care

Pillars:

- (1) **Health financing** (better investment in health)
 - NHIP (National Health Insurance Program) - PhilHealth (Case rate method - per diagnosis)
- (2) **Health regulation** - affordable, quality health care services
- (3) **Health services delivery** - Accessible and available health services
- (4) **Good governance** - (stakeholders)
 - Transparent (Statement of assets, liability and network)
 - Accountable
 - Efficient
- (5) **Performance accountability** - outcome based approach

Family Nursing Care Plan (APIE)

Nursing process: the main framework or guide in nursing practice and the means by which nurses work with client-partners to enhance wellness or address the health needs and problems of clients.

Phases of Nursing Process

- (1) **Assessment:** process of collecting, organizing and analyzing data/information about the client
 - Interview, observation
 - **First level assessment:** determine “what” are the problems
 - 4 categories of health problem:
 - Wellness state (healthy)
 - Health deficit (disability & disease)



- Health threat (increase risk for illness & injury)
 - Foreseeable crisis/Stress points (anticipated problems that are developmental or social in nature - **ADJUST**)
 - **Second level of assessment:** determine “why” (reason behind the problem) - **INABILITY** to recognize the presence of the problem
- (2) **Planning:** a logical step-by-step process in designing a plan of action to accomplish specific goal and objectives (Specific, Measurable, Attainable, Reliable, Time bound)
- Goal setting
 - Construct plan of action
 - Choose nursing intervention
 - Prioritization (Maglaya - 4 Criteria) - highest score is 5
 - The highest score, the highest priority; the lowest score, the lowest priority
 - **Nature of the problem** (Wt: 1) - type of nursing problem
 - Wellness state: 3
 - Health deficit: 3
 - Health threat: 2
 - Foreseeable crisis - 1
 - **Modifiability of the problem** (Wt: 2) - the solution of the problem (most important)
 - Easily modifiable: 2
 - Partially modifiable: 1
 - Not modifiable: 0
 - **Preventive potential** (Wt: 1) - future re-occurrence of the problem
 - Highly preventable: 3
 - Moderately preventable: 2
 - Low preventive potential: 1
 - **Salience** (Wt: 1) - perception of the client about the problem
 - Immediate: 2
 - Not immediate: 1
 - Not a problem: 0
 - **Formula:** score/highest score x wt
- (3) **Implementation:** consists of doing or carrying out the interventions specified in the care plan in partnership with client-partners and/or other members of the team
- **Independent** - scope of nursing practice
 - **Dependent** - doctor’s order
 - **Interdependent** - collaborative care
- (4) **Evaluation:** a planned, ongoing, purposeful activity in which the nurse and the client-partner determine the client’s progress toward achievement of goals and outcomes
- **Outcome: Quality, Assurance, Elements**
 - Elements: SOP



- **Structural elements:** physical settings (Manpower, Money, Material/equipment)
- **Outcome elements:** changes resulting from nursing intervention
- **Process elements:** steps of nursing process (APIE)

Family Data Analysis: is done by comparing findings with accepted standards for individual family members and for the family unit'

System of Organizing Family Data

- **Family Structure and Characteristics**
 - Data on household membership
 - Demographic characteristics
 - Family members living outside the household
 - Family mobility
 - Family dynamics
 - Data on family structure can be visualized clearly through graphic tools such as genogram, ecomap and family tree
- **Socioeconomic characteristics:**
 - Data on social integration
 - Educational experiences and literacy
 - Work history
 - Financial resources and leisure time interests
 - Cultural influences
 - Spirituality or religious affiliation
- **Family environment:**
 - Refers to the physical environment inside the family's home/residence and its neighborhood
- **Family Health and Health Behavior**
 - Family's activities of daily living
 - Self-care
 - Risk behaviors
 - Health history
 - Current health status
 - Health care resources

Family Nursing Diagnosis

- As an individual family member
- As a family unit
- As the family in relation to its environment/community
- **International (NANDA-I, 2011)**
 - Serve as a common framework of expressing human responses to actual and potential health problems



- **Family Coping Index**

- This tool is based on premise that nursing action may help a family in providing for a health need or resolving a health problem by promoting the family's coping capacity

Nine Areas of Assessment of the Family Coping Index

- I. **Physical Independence** - family members' mobility and ability to perform activities of daily living
- II. **Therapeutic Competence** - ability to comply with prescribed or recommended procedures and treatment to be done at home
- III. **Knowledge of Health Condition** - understanding of the health condition or essentials of care according to the developmental stages of family members
- IV. **Application of Principles of Personal and General Hygiene** - practice of general health promotion and recommended preventive measures
- V. **Health Care Attitudes** - family's perception of health care in general
- VI. **Emotional Competence** - degree of emotional maturity of family members according to their developmental stages
- VII. **Family Living Patterns** - interpersonal relationships among family members, management of the family finances, and the type of discipline in the home
- VIII. **Physical Environment** - includes home, school, work, and community environment that influence the health of family members
- IX. **Use of Community Facilities** - ability of the family to seek and utilize, as needed both environment-run and private health

Community Organizing Participatory Action Research (COPAR)

Community Organization - plan solution

Participatory Action Research - active participation

Action Research - promotes development, the people are the researcher themselves

Role of the nurse: Facilitator, assist, teach

Role of the people: leader, assess, plan, implementation, evaluation

Emphases of Community Organizing in PHC

- (1) People from the community working together to solve their own problems
- (2) Internal organizational consolidation as a pre-requisite to external expansion
- (3) Social movement first before technical change
- (4) Health reforms occurring within the context of broader social transformation

Community organizing as a process

- Development of the community's collective capacities
- Continuous process of educating the community
- Process of education and mobilizing members of the community



Community Development entails a process of assessment of the current situation, the identification of needs, deciding on appropriate courses of actions or response, mobilization of resources to address these needs, and monitoring and evaluation by the people

Community organizing is a value-based process

- Human rights
- Social justice
- Social responsibility

Community organizing is people-centered

- The people are the means and ends of development and community empowerment is the process and the outcome

Community organizing is participative

- Participation of the community in the entire process is considered as the prime mover and determinant rather than beneficiaries and recipients of development efforts including health care

Community organizing is democratic

- It is a process that allows the majority of people to recognize and critically analyze their difficulties and articulate their aspiration

Community organizing is developmental

- Should be directed towards changing current undesirable conditions

Community organizing is process-oriented

- It is dynamic and goals of empowerment and development are achieved through a process of change

Phases of COPAR

- **Pre-Entry**
 - Simplest phase
 - Initial steps involved in establishing partnership with the people and identifying the needs, problems, issues affecting the community
 - **Preliminary Social Investigation** - collection of information in a systematic manner to generate community profile (different barangays)
 - **Site selection**
 - **Site Selection Criteria: GIDA**
 - Geographically inaccessible disadvantaged area
 - A community with a population of 100 or more families
 - Marginalized or in need
 - Peace and order situation is generally safe
 - People are receptive and willing to participate
 - **Baseline** - courtesy call (leaders of the community), networking, information dissemination
- **Entry/Social Preparation**
 - Establish the foundation/ground for the organization



- **Deepening Social Investigation** - identify the pressing needs
 - **Immersion/integration**
 - Imbibe the life of people in the community
 - Identifying potential leaders, formation of core groups, organization and implementation and evaluation and phase out
 - **Criteria for selecting potential leaders:**
 - Active listener
 - A resident of the community in good standing
 - Has moral ascendancy
 - Influential and respected by the people (charismatic)
 - Currently not holding a leadership post in other organizations like the barangay
 - Willing to work for the community
 - Desire for change
 - **Core Group Formation (5-8)** - the foundation of the organization also referred to as executive committee or council of elders in indigenous communities
 - **Self-Awareness Leadership Training** - DOH, PHC, leadership training
- **Community Study/Assessment/Diagnosis/Situational Analysis**
 - Profiling of the community (BioData)
 - Research team - AD HOC/temporary committee
 - **Community diagnosis**
 - **Comprehensive** - general information
 - **Problem-oriented** - specific needs
 - **Population pyramid** - age and sex distribution
 - **Expansive population pyramid** - increased birth rate, increased death rate
 - **Stationary population pyramid** - steady growth of population
 - **Constrictive population pyramid** - low birth rate, aging population (long life expectancy)
- **Organizational Phase**
 - Plan solution for community problems
 - Build CHO (Community Health Organization)
 - Election of officers
 - Training of officers
 - Team building of officers
- **Action Phase**
 - **Mobilization** - exercise of people's power
 - Program
 - Implementation



- **Monitoring**
- **Evaluation**
- Education & Training - BHW
- **Sustenance and Strengthening Phase**
 - Continue education and training of officers/BHW(secondary)
 - Develop medium and long term goals
- **Turnover**
 - Self-reliance
 - **Types of Health Problems:**
 - **Health Status** - morbidity, mortality
 - **Health Resources** - manpower, money and material
 - **Health-related** - socio-economic, environmental, cultural, political

Public Health Nursing

- The science and art of preventing disease, prolonging life and promoting health and efficiency through organized community effort (Winslow)
- (1) **Assessment** - the regular collection and analysis of health data
- (2) **Policy development** - involves advocacy and political action to develop policies in various levels of decision making
- (3) **Assurance** - making sure that health services are effective, available and accessible to the people

Roles and Function of PHN

(1) Clinician/Health Care Provider

- Ensure that health services are provided, not just to individuals and families but also to groups and population

(2) Educator

- Communicating information to help patients make an informed choice regarding their health

(3) Advocate

- Acting on behalf of the client group

(4) Manager

- Oversee client care
- Supervise ancillary staff
- Do case management
- Manage case loads, run clinics or conduct community health needs, assessment projects

(5) Case management

- Assess clients' needs
- Plans for and coordinates services
- Monitors and evaluates progress to ensure that clients multiple service needs are met



(6) Collaborator

- Work with many people, including clients, nurses, physicians, social workers, community leaders etc. as member of the health team

(7) Leader

- Directs, influences and to effect change

(8) Researcher/Epidemiologist

- Utilizes research findings to provide evidence to convince decision-makers to enact policies for health

Clinic and Home Visits

Clinic Visit: patients visit to avail services offered by the facility primarily for consultation on matters that allied them physically.

Standard Procedures Performed during Clinic Visits

I. Registration/Admission

- A. Greet the client upon entry and establish rapport
- B. Prepare the family record of new patients or retrieve records of old clients
- C. Elicit and record the client's chief complaint and clinical history
- D. Perform physical examination on the client and record accordingly

II. Waiting time

- A. Give priority numbers to clients
- B. Implement the first come first served policy except for emergency/urgent cases

III. Triageing

- A. Manage program-based cases
- B. Refer all non-program-based cases to the physician
- C. Provide first-aid treatment to emergency cases and refer when necessary to the next level of care

IV. Clinical Evaluation

- A. Validate clinical history and physical examination
- B. The nurse arrives at evidence-based diagnosis and provides rational treatment based on DOH programs
- C. Inform the client on the nature of illness, the appropriate treatment and prevention and control measures

V. Laboratory and other diagnostic examinations

- A. Identify a designated referral laboratory when needed

VI. Referral system

- A. Refer the patient if he need further management following the two-way referral system (BHS to RHU, RHU to RHU, RHU to Hospital)
- B. Accompany the patient when an emergency referral is needed



VII. Prescription/Dispensing

- A. Give proper instructions on drug intake

VIII. Health Education

- A. Conduct one-on-one counselling with the patient
- B. Reinforce health education and counselling messages
- C. Give appointments for the next visit

Home Visit: family-nurse contact which allows the health worker to assess the home and family situation in order to provide the necessary nursing care and health related activities.

Purpose of Home Visit

- (1) To give nursing care to the sick, to a postpartum mother and her newborn with the view to teach a responsible family member to give the subsequent care
- (2) To assess the living condition of the patient and his family and their health practices in order to provide appropriate health teaching
- (3) To give health teachings regarding the prevention and control of diseases
- (4) To establish close relationship between the health agencies and the public for the promotion of health
- (5) To make use of the inter-referral system and to promote the utilization of community services

Steps in Conducting Home Visits

- (1) Greet the patient and introduce yourself
- (2) State the purpose of the visit
- (3) Observe the patient and determine the health needs
- (4) Put the bag in a convenient place then proceed to perform the bag technique
- (5) Perform the nursing care needed and give health teachings
- (6) Record all important data, observation and care rendered
- (7) Make appointment for a return visit

Public Health Nurse Bag Technique

Bag Technique is a tool by which the nurse, during her visit will enable her to perform a nursing procedure with ease and deftness, to save time and effort, with the end view of rendering effective nursing care to clients

Principles of Bag Technique

- (1) Performing the bag technique will minimize, if not prevent the spread of any infection
- (2) It saves time and effort in the performance of nursing procedures
- (3) The bag technique should show effectiveness of total care given to an individual or family



- (4) The bag technique can be performed in a variety of ways depending on the agency's policy, the home situation, or as long as principles of avoiding transfer of infection is always observed

Contents of the Public Health Bag

Paper lining	Extra paper for making waste bag
Plastic/linen lining	Apron
Hand towel	Soap in a soap dish
Thermometers (oral and rectal)	2 pairs of scissors (bandage and surgical)
2 pairs of forceps (curved and straight)	Disposable syringes with needles (g. 23 & 25)
Hypodermic needles (g. 19, 22, 23, 25)	Sterile dressing
Cotton balls (dry and with alcohol)	Cord clamp
Micropore plaster	Tape measure
1 pair of sterile gloves	Baby's scale
Alcohol lamp	Test tube holders
Solutions of betadine, zephiran solution, spirit of ammonia, acetic acid, 70% alcohol, hydrogen peroxide, ophthalmic ointment, benedict's solution	Sphygmomanometer and stethoscope

Important Points to Consider in the Use of the Bag

- (1) The bag should contain all the necessary articles, supplies and equipment that will be used to answer emergency needs
- (2) The bag and its contents should be cleaned very often, the supplies replaced and ready for use anytime
- (3) The bag and its contents should be well-protected from contact with any article in the patient's home. Consider the bag and its contents clean and sterile, while the articles that belong to the patients as dirty and contaminated
- (4) The arrangement of the contents of the bag should be the one most convenient to the user, to facilitate efficiency and avoid confusion

Steps in Performing the Bag Technique Actions	Rationale
1. Upon arrival at the patient's home, place the bag on the table lined with a clean paper. The clean side must be out and the	To protect the bag from getting contaminated



folded part, touching the table	
2. Ask for a basin of water or a glass of drinking water if tap water is not available	To be used for handwashing
3. Open the bag and take out the towel and soap	To prepare for handwashing
4. Wash hands using soap and water, wipe to dry	To prevent infection from the care provider to the client
5. Take out the apron from the bag and put it on with the right side out	To protect the nurse's uniform
6. Put out all the necessary articles needed for the specific care	To have them readily accessible
7. Close the bag and put it in one corner of the working area	To prevent contamination
8. Proceed in performing the necessary nursing care and treatment	To give comfort and security and hasten recovery
9. After giving the treatment, clean all things that were used and perform handwashing	To protect the caregiver and prevent infection
10. Open the bag and return all things that were used in their proper places after cleaning them	
11. Remove the apron, folding it away from the person, the soiled side in and the clean side out. Place it in the bag	
12. Fold the lining, place it inside the bag and close the bag	
13. Take the record and have a talk with the mother. Write down all the necessary data that were gathered, observations, nursing care and treatment rendered. Give instructions for care of patients in the absence of the nurse	For reference in the next visit
14. Make appointment for the next visit (either home or clinic) taking note of the date and time	For follow-up care



Epidemiology

- The study of the distribution and determinants of health-related states or events in specified populations, and the application of this study to the prevention and control of health problems

Distribution - refers to the analysis by time, places and classes of people affected

Determinants - include all the biological, chemical, physical, social, cultural, economic, genetic and behavioral factors that influence health

Practical Applications of Epidemiology

- (1) Assessment of the health status of the community or community diagnosis
- (2) Elucidation of the natural history of disease
- (3) Determination of disease causation
- (4) Prevention and control of disease
- (5) Monitoring and evaluation of health interventions
- (6) Provision of evidence for policy formulation

Methods of Epidemiology

- epidemiological study requires systematic collection of health data, its analysis, description of health needs and health problems, identification of factors associated with the problems and stating the hypothesis

Observational studies - people are observed in their natural state

- **Descriptive studies** - observing distribution of disease or health-related characteristics in human populations and identifying the characteristics with which the disease in question seems to be associated
- **Analytical Studies**
 - **Case control study** - retrospective studies, proceeds backwards from the effect to cause
 - **Cohort study** - proceeds forward from cause to effect
 - Prospective cohort studies
 - Retrospective cohort studies
 - A combination of retrospective and prospective cohort studies

Experimental studies - referred as clinical trials, are studies in which treatment or procedure is intentionally introduced by the research and result is observed

- Randomized control trials
- Non-randomized Trials
 - Uncontrolled trials
 - Natural experiments
 - Before & after comparison studies



Data Presentation

- **Line listing** - organizes information being collected during surveillance or an outbreak investigation
- **Table** - used to demonstrate patterns, exceptions, differences and other relationships
- **Graph** - displays numerical data in visual form
- **Histogram** - displays frequency distribution of a continuous variable
- **Population pyramid** - presents the count or percentage of a population by age and sex
- **Scatter diagram** - displays relationship between two continuous variables, with the x and y axes each representing a variable
- **Bar chart** - displays categories of a variable in a comparative manner in which the frequency of the event is represented on opposite axes
- **Pie chart** - displays the data into slices or segments that display the proportion of each component
- **Maps** - used to display the geographic location of epidemiological event

Sources of Health Data

- Census, vital registration system, disease notification, disease registers, surveillance system, hospital data, health insurance, school health program, downloadable data sets, surveys

Disease Registry - a compilation of information about a particular disease

Disease Surveillance Systems in the Philippines

- (1) Notifiable Disease Reporting System (NDRS)
- (2) Field Health Service Information System (FHSIS)
- (3) National Epidemiology Sentinel Surveillance System (NESSS)
- (4) Expanded Program on Immunization Surveillance System (IPE Surveillance)
- (5) HIV/AIDS Registry

Communicable Disease

Infection - the growth of microorganisms in body tissue where they are not usually found

Disease - a detectable alteration in normal tissue function

- **Communicable disease**
 - Leading causes of illnesses in the country today, often afflict the most vulnerable, the young and the elderly
 - The infectious agent can be transmitted to an individual by direct or indirect contact or as an airborne infection
 - About 60% of deaths caused by communicable diseases can be attributed to:
 - HIV/AIDS
 - Malaria



- Tuberculosis
- Measles
- Diarrheal disease
- Acute respiratory infection

Goal of WHO

- (1) Prevention of disease
- (2) Prevention of disability and death from infection
- (3) Prevention through immunization

Classification of Infectious Diseases

Based on Manifestation in the Body

- **Local** - limited, confined to one specific area, localized sign
- **Systemic** - the microorganisms spread and damage different parts of the body, generalized sign

Based on Behavior within the Host

- **Infectious disease** - any disease caused by invasion and multiplication of microorganism
 - **Endogenous** - comes from the patient
 - **Exogenous** - comes from the environment
- **Contagious disease** - disease that easily spreads from one person to another

Based on Severity or Duration of Disease

- **Acute disease** - develops rapidly but lasts only a short time (days - weeks)
 - **Measles, Mumps, Rubella, Varicella**
 - **Hepatitis A** - infectious liver disease (with vaccine) endemic hepatitis
 - **Hepatitis E** (perinatal - 3rd trimester) - fulminant (severe/lethal)
- **Chronic disease** - develops slowly, milder but longer lasting clinical manifestations (months, years, lifelong)
 - *Mycobacterium leprae*, *Mycobacterium tuberculosis*
 - Herpes Simplex 1 & 2 (life-long)
 - **Hepatitis B/Serum Hepatitis** - with vaccine, fulminant (severe/lethal), possible life-long (blood-borne, sex),
 - **Hepatitis C/Inoculation** (uncommon in PH, drugs) - re-infection
 - **Hepatitis D** (need Hepatitis B - co-infection-Hep B+Hep D/superinfection-after infection of Hep B)
 - Interferon IM (B,C,D)

Based on State of Host Resistance

- **Primary infection** - acute infection that causes the initial illness
- **Secondary infection** - one caused by an opportunistic pathogen after primary infection has weakened the body's defenses

Healthcare Associated Infection



- **Nosocomial (Hospital/Facility-based)** - develop during a client's stay in a facility or manifest after discharge (48 hours after admission/discharge)
- **Iatrogenic (procedure-based)** - the direct result of diagnostic or therapeutic or invasive procedures

The Epidemiological Triad

- disease result from the interaction between the agent and the susceptible host in an environment that supports transmission of the agent from a source to that host
 - **Host** - influences an individual exposure, susceptibility or response to causative agent. Host can be the patient, carrier, suspect or contact
 - **Environment** - the sum total of all external conditions that affects the development of an organism and serves as the opportunity of the hosts exposure to harmful microorganisms.
 - Physical
 - Biological
 - Socio-economic
 - **Infectious Agent** - are the microorganisms or pathogen that cause infection
 - **Factors influencing the Cause of Disease**
 - **Pathogenicity** - ability of an organism to cause a disease, create damage to the host (immune system)
 - **Infectivity** - ability of an organism to infect (multiple within the host)
 - **Virulence** - ability of an organism to produce disease (severity)
 - **Infective dose** - number of organism and amount of toxin released by the organism to induce disease
 - **Influenza** - high infectivity, moderate pathogenicity, low to moderate virulence
 - **Rabies** - low infectivity, very high pathogenicity, extremely virulent
 - **HIV** - low to moderate infectivity, high pathogenicity, high virulence over time

Stages of Disease

Incubation period - time interval between the initial infection and the first appearance of any signs and symptoms

- **Start** - upon exposure
- **End** - first signs and symptoms

Prodromal period - early, mild symptoms of disease

- **Start** - first signs and symptoms
- **End** - general signs and symptoms



Period of illness - overt signs and symptoms of disease

- All signs and symptoms appear
- Peak/acme
- Pathognomonic sign - specific

Period of convalescence - signs and symptoms are disappearing (recovery)

Period of Defervescence - onset of complications leading to death

Chain of Infection

- conditions that must be met in order for the infectious diseases to spread

Pathogen or Causative Agent

- Biologic agent (organism) capable of causing disease
 - **Bacteria** - most common (DPT)
 - **Virus** - most microscopic (MMRV)
 - **Fungi** - least common (buni, alipunga, had had, anan)
 - **Parasites** - protozoans, helminths, flukes (filariasis and malaria)
- Eliminate organism by:
 - Sterilizing surgical instruments and anything that touches sterile spaces of the body
 - Using good food safety methods
 - Providing safe drinking water
 - Vaccinating people so they do not become reservoirs of illness
 - Treating people who are ill

Reservoir

- Any persons, animal, arthropod, plant, soil, or substance in which a causative agent normally lives and multiplies, on which it depends primarily for survival, and where it reproduces in such numbers that it can be transmitted to a susceptible host
- Eliminate reservoir by:
 - Treating people who are ill
 - Vaccinating people
 - Handling and disposing body fluids responsibly
 - Monitoring soil and contaminated water in sensitive areas of the hospital and washing hands carefully after contact with others

Portal of Exit

- The way the causative agent gets out of the reservoir
- Reduce risk from portal of exit by:
 - Covering coughs and sneezes with a tissue
 - Handling body fluids with gloves, then doing hand hygiene
 - Keeping draining wounds covered with a dressing
 - Not working when you have exudative (wet) lesions or weeping dermatitis

Mode of Transmission



- Any mechanism by which a pathogen is spread from a source or reservoir to a person, unwashed hands, things which are not cleaned between patients, droplets, or, for a few diseases, air
- Eliminate the mode of transmission by:
 - Hand hygiene
 - Wearing gloves to minimize contamination of hands and discarding them after each patient
 - cleaning , disinfection, or sterilization of equipment used by more than one patient
 - Cleaning of the environment, especially high-touch surfaces
- **Contact Transmission**
 - **Direct contact** - person to person transmission of an agent by physical contact between its source and susceptible host
 - **Horizontal Transmission** - immediate transfer from person to person
 - **Vertical Transmission** - refers to pathogen that is passed from mother to child either from utero (placenta) or during child birth
 - **Indirect contact** - reservoir to susceptible host by means of non-living object
 - **Vehicle Transmission** - through fomites
 - **Vector-borne Transmission** - occurs when a living organism carries a disease pathogen and passes it to another living organism
- **Droplet Transmission** - can occur when a person comes within 1 meter of an infected person
 - Source and the host are within 1 m (3ft)
 - >5 microns
 - PIMP - Pertussis, Influenza, Meningitis, Pneumonia
- **Airborne Transmission** - occurs when infectious agents are carried in the air via aerosol
 - Occurs regardless of distance
 - <5 microns
 - MTV - Measles, Tuberculosis and Varicella
- **Enteric Transmission** - involves infectious agents that are spread via from feces of an infected individual to the mouth of another individual
- **Blood-borne transmission** - occur from the introduction of pathogen via blood transfusion

Portal of Entry

- Hole in the skin that allows the infectious agent to get into the body (mouth, nose, eyes, rashes, cuts, needlestick injuries, surgical wounds and IV sites)
- Protect portals of entry by:
 - Dressing on surgical wounds
 - IV site dressings and care
 - Elimination of tubes as soon as possible
 - Masks, goggles, and face shields
 - Keeping unwashed hands and objects away from mouth
 - Actions and devices to prevent needlesticks
 - Food and water safety



Susceptible Host

- A person or animal lacking effective resistance to a particular infectious agent
- Minimize risk to susceptible hosts by:
 - Vaccinating people
 - Preventing new exposure to infection in people who are already ill, receiving immunocompromising treatment or infected with HIV
 - Maintaining good nutrition
 - Maintaining good skin condition
 - Covering skin breaks
 - Encouraging rest and balance in our lives
- **Determinants:**
 - **Poor primary defenses** - impaired skin integrity, poor skin moisture
 - **Poor secondary defenses** - low levels of antibodies
 - **Poor tertiary defenses** - immature immune system, deteriorating immune system, immunosuppression

How to Stop the Infection (Break the Chain)

- **Cleaning** - visibly without dirt (soap & water)
- **Disinfection** - kill all microbes except SPORES
- **Sterilization** - kill all microbes (wipe out)

Disinfecting Agents

- **Antiseptic** - usually used for skin to prevent and reduce growth of harmful microorganisms
 - Alcohol - minimum 60% according to CDC
 - Iodine based solution (betadine)
 - Chlorhexidine
 - Hydrogen peroxide
 - Silver sulfadiazine
- **Disinfectant** - chemicals that are used to destroy or inactivate harmful microorganisms on non-living surface
 - Example: bleach, quaternary ammonium compounds, phenols, formaldehyde
 - **Concurrent disinfection** - while the patient is still a source of infection
 - **Terminal disinfection** - once the patient is no longer the source of infection or has been removed from the environment
 - **Prophylactic disinfection** - preventive disinfection of environments to avoid the spread of occurrence of infection

Methods of Sterilization

Autoclave	Radiation	Gas	Dry Heat
• Use of moist	Ionizing radiation (X-ray,	• Ethylene	• use of dry

heat higher than 121°C - 134°C • Steam under pressure • 15 to 30 minutes • Used for: surgical instruments, dressings, culture media, glassware	gamma rays) • Used for: disposable medical equipment (syringes, catheters) • Pharmaceuticals • Tissue grafts No-ionizing radiation (UV light) • Surface sterilization (biosafety cabinets, operating rooms)	oxide (EtO) Gas • Effective against: all microorganisms and spores • Used for: heat-sensitive medical equipment (endoscope, plastics)	heat at 150-170°C • Hot air oven • 1-2 hours • Used for: glassware, metal instruments, oils and powders
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Isolation - people who are infected and are known risk to others, separates sick people with a disease

Quarantine - for people who have been exposed to a disease but not yet have shown symptoms, separates and restricts the movement of people who were exposed to a contagious disease to see if they become sick

Cohorting - refers to the grouping of patients with a same infection within isolated-area and it is sort of strict isolation

Infection Control

Standard Precaution - used for all patient care regardless of their diagnosis or possible infection status

- Elements:

- **Hand hygiene**

- Most effective and most practical method of controlling the spread of microorganisms
- Amount of liquid soap: 4 ml
- Focus/important element: friction
- Time handwashing: 15-20 seconds (if not visibly soiled)
- Hand rubbing: 30 seconds friction (3-5 ml of alcohol)
- World Handwashing Day: October 15

- **Wearing of PPE**

- **Gloves**

- Disposable gloves - latex, nitrile, vinyl (hospital)
- Chemical resistant gloves - labs, hazardous chemicals
- Cut resistant gloves

- **Gown**

- Aprons/gowns - to prevent contamination of uniform

- **Mask**

- Droplet - surgical mask



- Airborne - N95 respiratory, N99, N100 - high risk settings
- **Goggles/Face shield**
 - Indirectly vented goggles - with hole
 - Non-vented goggles - no hole
- **Donning - GOWN, MASK, GOGGLES, GLOVES** (GowMaGogGlov)
- **Doffing - GLOVES, GOGGLES, GOWN, MASK** (GlovGogGowMa)
- **Safe injection practices**
 - NEVER
 - Recap needle
 - Bend/break needle
 - Reuse/share contaminated needles
 - Throw used needles in a puncture resistant container

Color of container/bag	Type of Waste
Black	Non-infectious dry waste
Green	Non-infectious wet waste
Yellow	Infectious and pathological waste
Yellow with black band	Chemical waste including those with heavy metals
Orange	Radioactive waste
Red	Sharpe's and pressurized containers

- **Safe handling of contaminated materials**
- **Respiratory hygiene/cough etiquette**

Transmission-based Precaution - second tier of basic infection control and are to be used in addition to standard precautions for patients who may be infected or colonized with certain infectious agents for which additional precautions are needed to prevent infection transmission

- **Contact precaution:** gloves, gown, single room occupancy, >3 feet away same CA/infection
 - Enteric precaution - same as contact with handwashing
 - Enteric + special handling - hepatitis A, shigella, rotavirus
 - Avoid contact with wound drainage, fecal incontinence or other discharges post risk for contamination
 - Vancomycin-resistant enterococci (VRE), methicillin resistant staphylococcus (MRSA), major wound infection such as scabies, cutaneous Diphtheria, herpes Simplex virus
- **Droplet precaution:** surgical mask, single room occupancy, spatial separation - keeping individuals physically apart, same CA/infection - cohort >3ft with curtain and divider
 - Prevent transmission through close respiratory or mucous membrane contact
 - Influenza, pertussis, mumps, meningitis, rubella, respiratory Diphtheria



- **Airborne precaution:** N95 mask, negative pressured room, no visitor/door closed, number of air changes 6-12 air changes per hour
 - Restrict susceptible personnel from entering the room, limit number of people needed to enter the room
 - Tuberculosis, measles, varicella, SARS, smallpox
- **Protective environment/reversed isolation:** private room with HEPA filter, positive pressure >12 air changes per hour, protective environment designed for allogenic patients, fungal spore counts in the air and reduce the risk of invasive environmental fungal infections, potted plants, flower (avoid)
 - For compromised immune systems, such as those undergoing chemotherapy, organ transplant

Immunity

Active Immunity - results when exposure to a disease organism triggers the immune system to produce antibodies to that disease

- **Natural active immunity** - acquired from exposure to the disease organism through infection with the actual disease
- **Artificial active immunity** - acquired through introduction of a killed or weakened form of the disease organism through vaccination

Passive Immunity - provided when a person is given antibodies to a disease rather than producing them through his/her own immune system

- **Natural passive immunity** - passed from mother to child either through placenta during pregnancy or through breastmilk
- **Artificial passive immunity** - receiving antibodies from an external source such as immune globulin injection

Immunization (Active artificial immunity)

- (1) **Live attenuated vaccine** - use live, but weakened pathogens (not safe for pregnant women/immunocompromised)
- (2) **Inactivated vaccine** - use pathogens that have been killed (safe for immunocompromised), can cause allergic reaction
- (3) **Subunit, recombinant or conjugate vaccine** - contain specific parts of the pathogen such as proteins or sugars that trigger immune response
- (4) **Toxoid vaccine** - used inactivated bacterial toxins that no longer cause disease but still induce an immune response
- (5) **messenger RNA (mRNA) vaccines** - use a small piece of genetic material to instruct cells to produce a protein similar to one found on the surface of pathogen

VACCINE	DISEASE/S	COMPONENTS
BACILLUS CALMETTE GUERIN	TUBERCULOSIS	live attenuated strain of <i>Mycobacterium bovis</i>
HEPATITIS B VACCINE (MONOVALENT)	HEPATITIS B	RNA RECOMBINANT, The vaccine contains purified Hepatitis B surface antigen (HBsAg), which is a protein found on the surface of the actual Hepatitis B virus.
PENTAVALENT (DPT – HEPB – HAEMOPHILUS INFLUENZA	DIPHTHERIA PERTUSIS TETANUS HEPATITIS B PNEUMONIA MENINGITIS	D – WEAKENED TOXINS P – KILLED BACTERIA T – WEAKENED TOXINS, LIQUID CLEAR
ORAL POLIO VACCINE (OPV)	POLIOMYELITIS	LIVE ATTENUATED VIRUS
MEASLES MUMPS RUBELLA VARICELLA (MMR) QUAD VACCINE	MEASLES MUMPS RUBELLA CHICKEN POX	LIVE ATTENUATED VIRUS DRIED FREEZE WITH SPECIAL DILUENT
INACTIVATED POLIO VACCINE	POLIOMYELITIS	killed-virus vaccine that protects against poliomyelitis (polio), a disease caused by the poliovirus.
PNEUMOCOCCAL CONJUGATE VACCINE	PNEUMONIA MENINGITIS	conjugate vaccine designed to protect against pneumococcal disease caused by the bacterium <i>Streptococcus pneumoniae</i> .

VACCINES	ROUTE	AMOUNT	STORAGE	SCHEDULE	DOSE	IF UP
BCG	ID	0.05ML	BODY	At Birth	1	WALA
HEPA B	IM	0.5ML	BODY	At Birth	1	WALA
PENTAVALENT	IM	0.5ML	BODY	6 weeks	3	6, 10, 14
OPV	PO	2-3 gtt/s	FREEZER	6 weeks	3	6, 10, 14
MMR	SUBQ	0.5ML	FREEZER	6 months	2	1 YEAR
IPV	IM	0.5ML	BODY	14 weeks	2	9 MONTH 10, 14
PCV	IM	0.5ML	BODY	6 weeks	3	6, 10, 14

Fully Immunize Child (FIC) >>> a child who has received all immunization that should be given before reaching the 1st year of life.

Completely Immunize Child (CIC) >>> a child who has completed all immunization scheduled at the age of 12 – 23 months

Child Protected at Birth (CPAB)
>>> 2 doses of DT during pregnancy or 3 doses of DT given anytime prior to pregnancy

DOH Recommended Diphtheria & Tetanus Immunization for Women			
DOSE	SCHEDULE	% OF PROTECTION	DURATION OF PROTECTION
1	ASAP		
2	4 WEEKS after 1 st dose within the same pregnancy	80%	3 years
3	6 months after DT2	95%	5 years
4	1 year after DT3	99%	10 years
5	1 year after DT4	99%	Lifetime

Most sensitive to heat (-15 to -25C) - OPV and MMR

Least sensitive to heat (+2 to +8C) - pentavalent, Hep B, BCG, TT

RA 11332 - Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act

- **Category 1** - report within 24 hours
 - Meningococcal disease
 - Hand-foot-and-mouth disease
 - Acute flaccid paralysis
 - Rabies
 - Severe Acute Respiratory Syndrome (SARS)
 - COVID 19
 - Human avian influenza
 - Adverse event following immunization
 - Measles
 - Paralytic shellfish poisoning
 - Neonatal tetanus
 - Anthrax
- **Category II** - weekly (Friday)
 - Bacterial meningitis
 - Acute encephalitis syndrome
 - Dengue
 - Cholera
 - Acute hemorrhagic fever syndrome
 - Influenza-like illness
 - Leptospirosis



- Diphtheria
- Pertussis
- Acute bloody diarrhea
- Malaria
- Non-neonatal tetanus
- Typhoid and paratyphoid fever
- Acute viral hepatitis

Common Infectious Diseases

Endophagic - feed indoor

Exophagic - feed outdoor

Endophily - rest inside dwellings

Exophily - rest outdoors

Circulatory System

DISEASE	AGENT	MOSQUITO	AREA	BEHAVIOR
MALARIA	PLASMODIUM	ANOPHELES FLAVIROSTRIS (Pale Golden Patch)	Rural Area; Slow Flowing, Clear, partially shaded stream	Nigh Flying
DENGUE / CHIKUNGUNYA ZIKA	DENGUE VIRUS / CHIKUNGUNYA VIRUS	AEDES AEGYPTI "Tiger Mosquito" (Silver Markings, White Bands) AEDES ALBOPICTUS (RURAL)	Urban and Rural; Clear, stagnant water	Day Flying
FILARIASIS	WUCHERERIA / BRUGIA	AEDES POECILUS (Broad and White Scales)	Rural Area: Swampy and Forested Area (Abaca, Banana, Gabi- raising areas)	Night Flying 10pm to 12am <u>Mansonia Bonneae</u> (1am – 2am)
JAPANESE ENCEPHALITIS	JE VIRUS	CULEX TRITAENIORYNCHUS (very small dark brown to black scales)	Rural Area: <u>Ricefields</u>	Night Flying (6pm to 7pm)

Dengue Fever

- Fever, headache, anorexia, malaise + muscle/joint pain + rashes + bleeding
- An RNA virus of the family Flaviviridae; genus Flavivirus
- DENV-1, DENV-2, DENV-3, DENV-4
- Fifth type announced in 2013
- Incubation period: 5 - 7 days
- Mode of Transmission: vector-borne
- Aedes aegypti



- Day biting (2 hours after sunrise and 2 hours before sundown)
- Breeds on stagnant water
- Limited and low-flying movement
- fine white dots at the base of the wings with white bands on the legs
- **Disease Process**
 - (1) Initial infection and viral replication
 - (2) Immune response and cytokine storm
 - (3) Increased vascular permeability
 - (4) Platelet destruction and bone marrow suppression
- **Clinical Manifestations**
 - **Febrile** (Day 1 to 3)
 - FHAMS + muscle/joint pain (+) tourniquet test
 - Biphase fever (40C/104F)
 - Severe headache; retro-orbital eye pain; muscle, joint, and bone pain; macular or maculopapular rash
 - **Toxic** (Day 4 to 7)
 - All bleeding symptoms
 - Hemorrhagic symptoms such as hematemesis, bloody stool, or menorrhagia, especially if they have been in prolonged shock
 - **Convalescent** (Day 8 to recovery)
 - Hemodynamic status stabilizes and diuresis ensues
- **Dengue Case Classification and Level of Severity**
 - Dengue without warning signs - **suspect**: (+) TT, **probable**: dengue duo, **confirmatory** test: PCR
 - **Tourniquet Test/Rumpel-Leeds Test/Capillary Fragility Test** - only applicable in febrile state - 20 or more petechiae per 1 square inch
 - **Dengue Duo/Rapid Diagnostic Test** (blood sample)
 - NS 1 Antigen
 - Antibodies (IgG - past infection; IgM - new infection)
 - Dengue with warning signs
 - Severe dengue
- **Dengue Management**
 - Fever - TSB (25 to 35C), paracetamol 4-6 hours (-) aspirin, NSAIDS (risk for bleeding), corticosteroids (suppress immune system, hyperglycemia)
 - Headache - bed rest, paracetamol
 - Anorexia - SFF, no dark-colored food, low fat, low fiber, non-irritating, non-carbonated
 - Malaise - bed rest
 - Bleeding episodes
 - Nose bleeding - upright or seated position with trunk slightly leaning forward
 - Gum bleeding - suck ice chips, cold compress on cheeks



- GI bleeding - NPO, lie in bed

Treatment, Report (Weekly), Early Detection, Casefinding (TREC)

Anti-Mosquito Repellants

- DEET: Diethyltoluamide
- Picaridin
- Oil of Lemon Eucalyptus)

Sanitation of Environment and Home – 4’S

- Search and Destroy
- Seek Early Consultation
- Self-Protection Measures
- Support Spraying and Fogging

Mosquito Nets – Treat the net twice a year

Indoor Spraying and Fogging

- only during outbreaks or when there is a sudden rise in adult mosquito populations.
- early morning or late afternoon when mosquitoes are active.
- 2 to 3 times a week until outbreak subsides

Neem Tree, Eucalyptus and Fish and Frog Propagation (Gambusia Affinis (Mosquito Fish), Tilapia)

Chikungunya/Viral Arthritis

- Fever, headache, anorexia, malaise + muscle/joint pain + rashes
- **Causative Agent:** Chikungunya virus
- **Vector:** Aedes Aegypti – Urban Area; Aedes Albopictus – Rural Areas;
- **Incubation Period:** 3 to 7 days (range 1 – 12 days)
- **Clinical Manifestation**
 - **F** – Flu-like symptoms (FHAMS)
 - **A** – Arthralgia (severe joint pain, especially in small joints)
 - **R** – Rash (maculopapular rash, often starting on the face)
 - **M** – Muscle pain (myalgia)
 - **S** – Swelling (joint swelling)
- **Diagnosis:** Serological test – ELISA ; Presence of IgM and IgG Anti-chikungunya antibodies
Virological methods – RT PCR (for genotyping)
- **Treatment “FECCS”**
 - **Fever and Pain Management:** Paracetamol (acetaminophen) Avoid NSAIDs (like ibuprofen, aspirin)
 - **Encourage oral fluids** (ORS, water, juice)
 - **Complete rest** during the acute phase (Fatigue may persist for weeks)
 - **Chikungunya arthritis** can persist for weeks or months; Paracetamol or low-dose NSAIDs (if dengue has been ruled out)
 - Severe pain or Neurologic symptoms (rare) Refer to hospital



- No Specific Antiviral drug Treatment
- No Commercially available Chikungunya Vaccine
- One chikungunya vaccine (IXCHIQ) available in the US only.

Go to the RHU – TREC Treatment, Report (Weekly), Early Detection, Casefinding

Anti-Mosquito Repellants

- DEET: Diethyltoluamide
- Picaridin
- Oil of Lemon Eucalyptus)

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Neem Tree, Eucalyptus and Fish and Frog Propagation (Gambusia Affinis (Mosquito Fish), Tilapia)

Zika

- Fever, headache, anorexia, malaise + muscle/joint pain + rashes + conjunctivitis
- **Causative Agent:** Zika virus
- **Incubation Period:** 3 to 14 Days
- **Mode of Transmission:** Congenital or Intrauterine (in Utero); Perinatal, Breastfeeding, Sex, Transfusion (Platelet) and Mosquito Bite
- **Vector:** Ae. aegypti and Ae. Albopictus
- **Diagnostic Test:** Polymerase chain reaction test; Zika virus nucleic acid amplification testing (NAAT)
- **Clinical Manifestation**
 - Mostly Asymptomatic;
 - Onset: Fever, Arthralgia, Nonpurulent Conjunctivitis, and Maculopapular rash.
 - Other symptoms: Edema, Headache, Lymph Adenopathy, Myalgia, Retro- orbital Pain, and Vomiting.
 - **F** – Flu-like Symptoms FHAMS
 - **R** - Rash (maculopapular)
 - **E** - Extreme Joint Pain (joint pain, particularly in the hands and feet)
 - **C** - Conjunctivitis (Red or sore eyes) In



- Pregnancy: Brain Defect – Microcephaly Miscarriage, Stillbirth and other birth defect Guillain – Barre Syndrome (Rare)
- Symptomatic Treatment (No Specific Cure) For Mild Cases (most common)
 - Paracetamol – for fever and body aches
 - Rest – adequate sleep and low activity
 - Hydration – drink plenty of fluids
 - Avoid NSAIDs (ibuprofen, aspirin)
 - Special Care for Pregnant Women → Congenital Zika Syndrome (e.g., microcephaly).
- **DOH Guidelines:**
 - Immediate referral to OB-GYN if a pregnant woman shows symptoms (rash, fever, red eyes)
 - Serial ultrasounds to monitor fetal development
 - Counseling and education about risks and options
 - Possible referral to tertiary hospitals for high-risk pregnancy care

Treatment, Report (Immediate), Early Detection, Casefinding

Anti-Mosquito Repellants

- DEET: Diethyltoluamide
- Picaridin
- Oil of Lemon Eucalyptus)

Sanitation of Environment and Home – 4’S

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Neem Tree, Eucalyptus and Fish and Frog Propagation (Gambusia Affinis (Mosquito Fish), Tilapia)

Japanese Encephalitis

- Causative Agent: Japanese encephalitis (JE) virus
- Mode Of Transmission: Mosquito Bite
- Vector: Culex species mosquitoes: Culex Tritaeniorhynchus
- Incubation Period: 5 to 15 days
- Diagnosis: IgG and IgM Antibodies, ELISA, Cerebrospinal Fluid (CSF)
- Clinical Manifestation
 - Mild Symptoms: Fever, Headache, and Vomiting



- <1% Develop encephalitis: Disorientation, weakness, and coma. Seizures are common, especially among children.
- **F** – Fever
- **I** – Irritability (altered mental status)
- **E** – Encephalopathy (brain dysfunction)
- **R** – Rigidity (Parkinsonism-like symptoms)
- **C** – Convulsions (seizures)
- **E** – Encephalitis (inflammation of the brain)
 - Classical Sign of a parkinsonian syndrome with mask - like facies, tremor, cogwheel rigidity, and choreoathetoid movements. Acute flaccid paralysis, with clinical and pathological features like those of poliomyelitis
- Treatment - Hospital admission, Patients showing: Altered mental status, Seizures, High fever and neurologic signs
 - Monitoring of vital signs, oxygen level, and neurologic function
 - Fever control with paracetamol
 - Anticonvulsants (e.g., diazepam, phenytoin) for seizures
 - Intravenous fluids and nutrition
 - Respiratory support (oxygen, intubation if needed)
 - Brain swelling: Elevate the head of the bed
 - Reduce stimulation
 - Monitor for signs of herniation or deterioration

Table 5-15 Administration information for the inactivated Vero cell culture–derived Japanese encephalitis vaccine, IXIARO

AGE	DOSE	ROUTE	SCHEDULE	BOOSTER ¹
2 months–2 years	0.25 mL	IM	0, 28 days	≥1 year after primary series
3–17 years	0.5 mL	IM	0, 28 days	≥1 year after primary series
18–65 years	0.5 mL	IM	0, 7–28 days	≥1 year after primary series
>65 years	0.5 mL	IM	0, 28 days	≥1 year after primary series

Go to the RHU – TREC Treatment, Report (Weekly), Early Detection, Casefinding

Anti-Mosquito Repellants

- DEET: Diethyltoluamide
- Picaridin
- Oil of Lemon Eucalyptus

Sanitation of Environment and Home – 4’S

- Search and Destroy



- Seek Early Consultation
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Neem Tree, Eucalyptus and Fish and Frog Propagation (Gambusia Affinis (Mosquito Fish), Tilapia)

Malaria

- Causative Agent: Plasmodium
- Mode Of Transmission: Mosquito Bite
- Vector: Female Anopheles Flavirostris Mosquito
 - Breed in clear, flowing and shaded streams
 - Brown colored and bigger than other species
 - Night Biting 10pm – 4am; Don't bite person in Motion
- INCUBATION PERIOD: 5 to 15 days
- Diagnosis: Malarial Blood Smear

CAUSATIVE AGENT	DESCRIPTION	INCUBATION PERIOD	PAROXYMS
P. Falciparum	Most common and deadliest	8 to 15 days	
P. Vivax	Second most common	12 to 20 days	
<u>P. Ovale</u>	Tertian malaria	11 to 16 days	
P. Malariae –	Quartan malaria; rare	18 to 40 days	

Paroxysm of Malaria

COLD STAGE	• Sudden Inappropriate feeling of coldness and Apprehension. Mild Shivering turns to violent teeth chattering and shaking of the entire body (15 to 60mins)
HOT STAGE	• Headache, Palpitations, Tachypnea, Epigastric Discomfort, Thirst, Nausea and Vomiting, Temp: 41°C, Patient becomes confused or Delirious, Skin flushed (2 to 6hours)
SWEAT STAGE	• Defervescence and Diaphoresis or Profuse Sweating (2 to 4 hours)

Treatment of Malaria Artemisinin – Based Combination Therapy

CONDITION OF PATIENT	CAUSATIVE AGENT	TREATMENT REGIMENT
UNCOMPLICATED MALARIA	P.Falcifarum	AL + PQ (1 Day) Artemether – Lumefantrine + PrimaQuine
	P. Vivax P.Ovale	AL + PQ (14 days) Artemether – Lumefantrine + PrimaQuine
SEVERE MALARIA	P.Falcifarum	Artesunate IV for 24 hours then A-L+ PQ anti-gametocyte (1 day)
	P.Vivax	Artesunate IV for 24 hours then A-L+ PQ anti-relapse (14 days) Or QN + antibiotic + PQ anti relapse (14 days)
	Pre – Referral	Children -Artesunate suppository Adults - Artesunate im/iv

- Nursing Management

- **Cold Stage** (15-60 mins)
 - Warm blankets / environment - Prevent hypothermia, provide comfort
 - Paracetamol (acetaminophen) - Preempt fever, reduce chills-related pain
 - Hydration (oral or IV fluids) - Prevent dehydration from upcoming fever
 - Start antimalarial treatment - Begin ACT
- **Hot Stage** (2-6 hours)
 - Tepid Sponge Bath, Cooling Measures and Hydration
 - Paracetamol (acetaminophen)
 - Antimalarial drugs (continued)
- **Sweating Stage** (2-4 hours)
 - Rehydration Replace fluid lost through sweating
 - Rest and monitoring; Patient may feel very weak and exhausted
 - Nutritional support Boost recovery from systemic stress

Go to the RHU – TREC Treatment, Report (Weekly), Early Detection, Casefinding

Anti-Mosquito Repellants

- DEET: Diethyltoluamide
- Picaridin



- Oil of Lemon Eucalyptus

Sanitation of Environment and Home – 4’S

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Neem Tree (Azadirachta indica) leaves, oil, smoke, Eucalyptus (plant and repellent)

Frog - Eat mosquito larvae and adult mosquitoes, Natural predators in wetland ecosystems

Fish Propagation - Larvivorous fish, Common in biological mosquito control programs, Gambusia

Affinis (Mosquito Fish), Tilapia, Gourami, Guppies, 2 to 4 fish per sq.m or 200-400 fish per hectare

Malaria Prophylaxis

- Atovaquone-proguanil
- Doxycycline
- Mefloquine

Treatment of Malaria		
Artemisinin – Based Combination Therapy		
CONDITION OF PATIENT	CAUSATIVE AGENT	TREATMENT REGIMENT
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	P. Malariae	AL + PQ (14 days) Artemether – Lumefantrine + PrimaQuine
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	P.Vivax	Artesunate IV for 24 hours then A-L+ PQ anti-relapse (14 days) Or QN + antibiotic + PQ anti relapse (14 days)
	Pre – Referral	Children -Artesunate suppository Adults - Artesunate im/iv

Ebola

- 5 Causative Agents (strains)
 - Bundibugyo
 - Zaire
 - Sudan
 - Reston (present in Philippines, affects pigs)
 - Tai Forest
- Incubation Period: 2-21 days



- Mode of Transmission:
 - Human to human exposure to body secretions (except sweat)
 - Vector animals

Filariasis

- The disease often progresses to become chronic, debilitating and disfiguring disease since its symptoms are unnoticed or unfamiliar to health workers
- High rates in region V (Bicol), VIII (Samar, Leyte), XI (Davao)
- *Wuchereria bancrofti* and *Bugia malayi*
- Transmitted to the bite of infected female mosquitoes (*Aedes*, *Anopheles*, *Mansonia*)
- The larvae are carried in the blood stream and lodged in lymphatic vessels and lymph glands where they mature in adult form
- Two biological type:
 - Nocturnal - microfilaria circulate in peripheral blood at night (10 pm - 2am)
 - Diurnal - microfilaria circulate in greater concentration at daytime
- Clinical Manifestations:
 - Acute Stage:
 - Filarial fever and lymphatic inflammation that occurs frequently as 10 times per year and usually abates spontaneously after 7 days
 - Lymphadenitis (inflammation of lymph nodes)
 - Lymphangitis (inflammation of the lymph vessels)
 - Chronic Stage (10-15 years from the onset of the first attack):
 - Hydrocele (swelling of the scrotum)
 - Lymphedema (temporary swelling of the upper and lower extremities)
 - Elephantiasis (enlargement and thickening of the skin of the lower and upper extremities)
- Laboratory Diagnosis
 - Blood smear - presence of microfilaria
 - Immunochromatographic test (ICT)
 - Eosinophil count
- Management
 - Specific Therapy
 - Diethylcarbamazine (DEC) 6mg/KBW in divided doses for 12 consecutive days
 - Ivermectin (Mectizan)
 - Supportive Therapy
 - Paracetamol
 - Antihistamine for allergic reaction due to DEC
 - Vitamin B complex
 - Elevation of infected limb, elastic stocking



- DEC should be taken immediately after meals
- It may cause loss of vision, night blindness, or tunnel vision with prolonged use
- Ivermectin is best taken as single dose with a full glass of water in an empty stomach
- Cannot be used in patient with asthma
- Preventive measure
 - Health teachings
 - Environmental sanitation

Respiratory System

Tuberculosis

- The world's deadliest disease and remains as a major public health problem
- Badly nourished, neglected and fatigued individuals are more prone
- Susceptibility is highest in children under 3 years old
- AKA: Koch's disease: galloping consumption/phthis
- Causative agent: Mycobacterium tuberculae, M. africanum, M. bovis
- Incubation period: 2 - 10 weeks
- Mode of transmission: droplet or airborne (absence of sunlight)
- Signs and symptoms: cough (>2 weeks), anorexia, chest pain
- Diagnostic test
 - Gene Xpert - confirmatory
 - Sputum smear microscopy - alternative
 - BHW collects, medtech analyzes
 - Contraindication: hemoptysis, massive hemoptysis (REFER)
 - PPD Test/Mantoux/Tuberculin
 - Test for exposure (48-72 days)
 - General population (>10mm)
 - Immunocompromised (>5mm)
 - X-ray - most common
- Control measures
 - vaccine BCG
 - Exposure to direct sunlight
 - Reusable utensils
- Management:
 - WOF side effects: RIPE
 - Rifampicin - red to orange secretion (avoid use of contact lens)
 - Isoniazid - peripheral neuritis (vitamin B6/pyridoxine)
 - Pyrazinamide - hyperuremia (increase OFI)



- Ethambutol - optic neuritis/temporary color blindness
- Streptomycin - ototoxicity (hearing loss)
- Check the liver function
- Diet: High caloric diet
- Avoid physiotherapy
- TB DOTS (direct observation treatment short course therapy) before meals
 - New: has been treated for less than 1 month
 - Retreatment: treated for more than 1 month
 - Relapse
 - Loss for follow-up
 - Treatment after failure

The National Tuberculosis Control Program

- **Vision:** a country where TB is no longer a public health problem
- **Mission:** ensure that TB DOTS services are available to the communities
- **Goal:** to reduce the prevalence and mortality from TB by half by the year 2015
- **Targets:**
 - To cure at least 85% of the sputum smear positive TB patient discovered
 - Detect at least 70% of the estimated new sputum smear positive TB cases

Old Classification

Category	Type of TB Patient
I	● New smear-positive PTB ● New smear-negative PTB with extensive parenchymal lesions on CXR as assessed by the TBDC ● EPTB ● Severe concomitant HIV disease
II	● Treatment failure ● Relapse ● Return after default ● Other
III	● New smear-negative PTB with minimal parenchymal lesions on CXR assessed by the TBDC
IV	● Chronic (still smear-positive after supervised re-treatment)

Severe Acute Respiratory Syndrome (SARS)

- Causative agent: novel coronavirus



- Incubation period: 2-10 days (2002, 2012) 2-14 days (2019)
- Vector:
 - Civet cat (2002)
 - Bats/camels (2012) - MERS Cov
 - Bats/Pangolin (2019)
- Origin:
 - Guandong, China (2002)
 - Middle East (2012)
 - Wuhan (2019)
- Mode of transmission: droplet (2002, 2012) airborne (2019)
- Home quarantine: 5-7 days
- Facility quarantine: 14 days
- Philippines (COVID 19)
 - 1st death: January 30, 2020
 - 1st confirmed: February 1, 2020
 - Community outbreak: March 7, 2020
- Signs and symptoms
 - Pathognomonic sign: ageusia (loss of taste)
 - Anosmia (loss of smell)
 - Dysgeusia (dysfunctional taste)
 - Other: cough, dyspnea, exacerbation of co-morbidity
- Diagnostic test
 - RT-PCR
- Control measure
 - Vaccine
 - Face mask/shield
 - Isolation and quarantine
- Management
 - Symptomatic approach

Influenza

- Acute highly communicable disease characterized by abrupt onset with fever which lasts 1 day to 6 days, chilly sensation or chills, aches, or pains in the back and limbs with prostrations.
- Etiological agent: Influenza virus A, B, C
- Source of infection: discharges from the mouth and nose of infected persons
- MOD: direct contact through droplet infection
- Incubation period: short, 24-72 hours

Avian Influenza (Bird Flu)

- H5N1 infection



- Clinical manifestations:
 - Fever, sore throat, cough, severe respiratory distress

Swine Influenza (H1N1)

- Affects pig

Pertussis

- Whooping cough (successive coughing then inhales)
- Causative agent: Bordetella pertussis
- Incubation period: 7-10 days
- Mode of transmission: droplet
- signs and symptoms
 - Catarrhal stage - watery/swampy stage (most communicable)
 - Coryza
 - Sneezing
 - Lacrimation
 - Paroxysmal stage
 - Whooping cough
 - Ear and nose bleeding
 - Abdominal hernia
 - Asthma
 - Convalescent stage - no longer communicable
 - WOF: residual paroxysm
- Diagnostic test
 - Bordet gengou agar
- Control measure
 - DPT/pentavalent
- Management
 - Rest
 - Position: Prone (during attack)
 - Abdominal binder/hug a teddy bear
 - Fruit juice > milk
 - Emergency suction
 - Avoid stressing the child: trigger residual paroxysm
 - DOC: erythromycin & azithromycin

Diphtheria

- Description: acute febrile infection of the tonsils, throat, nose, larynx or wound marked by patches of grayish membrane from which the Diphtheria bacillus is readily cultured



- CA: *Corynebacterium diphtheria* (Klebs-Loeffler bacillus)
- Sources of infection:
 - Discharges and secretion from mucus surface of nose and nasopharynx and from skin and other lesions
 - Reservoir: man
- MOT:
 - Contact with a patient or carrier or with articles soiled with discharges of infected persons
 - Milk (vehicle)
- IP: 2-5 days
- Period of communicability:
 - 2 weeks to more than 4 weeks
 - Variable until virulent bacilli has disappeared from secretions and lesions
- Types:
 - Nasal - with foul-smelling serosanguinous secretions from the nose
 - Tonsillar - lesions are confined to the tonsils only but tend to spread over the pillars, into the soft palate and uvula
 - Nasopharyngeal - cervical lymph nodes are swollen, neck tissues are edematous
 - Laryngeal
 - Most commonly found in children ages 2-5 years old
 - It is considered as most severe and more fatal type due to anatomical reason
 - There is moderate hoarseness; voice is diminished until it is finally absent
 - Wound/Cutaneous - affects to mucous membrane and any break in the skin
- Clinical Manifestations:
 - Bull neck formation
 - Exudates forming the membrane are grayish in appearance (pseudomembrane)
 - Fatigue/malaise
 - Slight sore throat
 - breathing difficulty
 - Husky voice
 - swelling of the palate
 - Low grade fever
- Prevention:
 - active immunization of all infants and children with 3 doses of DPT
 - Pasteurization of milk
 - education of patients
 - reporting of case to the health officer of proper medical care
- Diagnostic tests:
 - Swab from nose and throat
 - Schick test
 - Involves giving an injection of 0.1 ml of dilute diphtheria toxin intradermally



- Area is checked in 3-4 days and the reaction is documented
- A positive test is indicated by inflammation or induration at the point of injection.
This indicates that the client lacks antibodies to Diphtheria
- Virulence test
- Moloney test - detects a high degree of sensitivity to diphtheria toxoid is given intradermally
- Treatment:
 - Penicillin
 - Anti-toxin
 - Erythromycin
- Nursing care:
 - Follow prescribed dosage and correct technique in administering anti-toxin
 - Provide comfort
 - Absolute bed rest for at least 2 weeks
 - Soft-food diet; SFF
 - Ice collar applied to neck
 - Visiting bag should be set up outside the room of the patient or should be far from the bedside of the patient
 - WOF shock which can develop suddenly as a result of systemic vascular collapse Airway Obstruction or anaphylaxis
 - If neuritis develops, tell the patient it's usually transient. Be aware that peripheral neuritis may not develop until 2-3 months after the onset of the illness
 - Explain how to properly dispose of nasopharyngeal secretion and teach proper infection precautions

Anthrax

- other terms: Wool-sorter's disease/Ragpicker's disease
- Description: an acute bacterial disease usually affecting the skin but which may be very rarely involve the oropharynx, lower respiratory tract, mediastinum or intestinal tract
- CA: Bacillus anthracis
- MOT:
 - Cutaneous infection is by contact with:
 - Tissues of animals dying of the disease
 - contaminated hair, wool, or products made from them such as drums or brushes
 - Soil associated with infected animals or contaminated bone meal used in gardening
- IP:
 - Inhalation anthrax: 1-7 days usually within 48 hours
 - Cutaneous anthrax: 1-7 days rarely up to 7 weeks



- Ingestion: 1-7 days
- Clinical manifestations:
 - Cutaneous anthrax
 - Most common (over 90% cases)
 - Infection is through skin
 - Over the a few days a sore which begins as a pimple, grows, ulcerates and forms a black scab around which are purplish vesicles
 - Systemic symptoms may include rigor headache and vomiting
 - The sore usually diagnostic: 20% cases are fatal
 - Inhalational anthrax
 - Spores are inhaled with subsequent invasion of mediastinal lymph nodes
 - Abrupt onset of flu-like illness, rigors, dyspnea and cyanosis followed by shock and usually death over the next 2-6 days
 - Most fatal
 - Intestinal anthrax
 - occurs following ingestion of meat from infected animals and is manifested as violent gastroenteritis with fever, vomiting, bloody stools and then septicemia
 - Poor prognosis
- Diagnostic Tests
 - Polymerase Chain Reaction (PCR) - definitive test for B. anthracis
 - Swabs from cutaneous lesions
 - Blood cultures
 - Lymph node or spleen aspirates
 - CSF shows characteristic bacilli on staining with polychrome methylene blue
 - Chest radiology may show fluid surrounding the lungs or widening of the mediastinum
- Treatment:
 - Antibiotics (Penicillin, Ciprofloxacin - DOC, doxycycline)
 - Treatment of cutaneous anthrax is oral antibiotic for 7-10 days
 - length of treatment for GI anthrax is 60 days, but safety has not been evaluated beyond 14 days
- Complications:
 - Cutaneous anthrax - septicemia
 - Inhalational anthrax
 - hemorrhagic meningitis
 - Pleural effusion
 - Mediastinitis
 - shock
 - ARDS
 - GI anthrax
 - Hemorrhage



- shock
- Nursing consideration:
 - obtain culture specimens before starting antibiotic therapy
 - Supportive measures are geared toward the type of anthrax exposure
 - teach the patient and family that anyone who has been exposed to anthrax must see a doctor immediately
 - Instruct the patient to take antibiotics as prescribed and until completed
 - Instruct the patient with cutaneous anthrax not to scratch at lesions
 - Alcohol-based hand sanitizers do not kill anthrax spores; wash hands with soap and water
- Prevention:
 - Pretreatment of animal product and good occupational health cover are the mainstays of control
 - Animals believed to have died of anthrax should be disposed of under supervision
 - Mass vaccination of animals may reduce disease spread
 - Non-cellular vaccines for human use are available for individuals at risk from occupational exposure
 - Workers handling potentially infectious raw materials should be aware of the risks

Bird Flu

- Other terms: avian influenza
- Description: infectious disease of birds ranging from mild to severe form of illness
- Source of infection: viruses that normally infect only birds and less commonly pigs
- IP: 3-5 days
- Clinical manifestations:
 - Fever
 - Body weakness/muscle pain
 - Cough
 - sore throat
 - May have difficulty breathing in severe cases
 - Sore eyes
- Susceptibility, resistance & occurrence: all birds are susceptible to infection but domestic poultry flocks are especially vulnerable to infection that can rapidly reach epidemic proportion
- Control measures:
 - rapid destruction, proper disposal of carcasses and quarantining and rigorous disinfection of farms
 - Restrictions on the movement of live poultry
- Nursing care:
 - Isolation precaution



- Infected control
- Early recognition of cases of highly pathogenic avian influenza during outbreak among poultry

Pneumonia

- Description: an acute infectious disease of the lungs usually caused by the pneumococcus resulting in the consolidation of one or more lobes of either one or both lungs
- CA:
 - Streptococcus pneumoniae
 - Staphylococcus aureus
 - Haemophilus influenzae
 - Pneumococcus of Friedlander
- IP: 2-3 days
- MOT:
 - droplet
 - indirect contact (fomites)
- Clinical manifestations:
 - Rhinitis
 - chest indrawing
 - Rusty sputum
 - Productive cough
 - High fever
 - Vomiting
 - Convulsions
 - flushed face
 - Dilated pupils
 - Pain over affected lung
 - Highly colored urine with reduced chlorides and increased urates
- Complications:
 - Emphysema
 - endocarditis
 - pneumococcal meningitis
 - Otitis media
 - jaundice
- Diagnostic test:
 - Chest x-ray
 - sputum analysis
 - Blood /serologic exam
 - Dull percussion noted on affected side



- Management:
 - Bed rest
 - Adequate salt, fluid, calorie and vitamin intake
 - TSB
 - Frequent turning from side to side
- Prevention:
 - Prevent common colds, influenza, and other URTI
 - Immunization with pneumonia vaccine
 - Eliminate contributory factors such as exposure to cold, pollution and physical conditions of fatigue and alcoholism

Nervous System

Meningitis

- Inflammation of the meninges (covers the brain and spinal cord)
- AKA: cerebrospinal fever
- **Causative agent:** Neisseria meningitidis
- **Incubation period:** 2 - 10 days
- **Mode of transmission:** droplet
- **Signs and symptoms:**
 - Pathognomonic sign: nuchal rigidity (stiff neck)
 - Photosensitivity
 - Increased ICP
 - Decreased LOC (irritability, lethargic)
 - (+) Kernig's - Knee
 - (+) Brudzinkis - Back of the head
- **Diagnostic Test**
 - **Lumbar tap/lumbar puncture**
 - **Cloudy/turbid:** bacterial
 - **Clear:** viral
 - **NR:** Perform MRI first, Increased ICP (diuretics)
- **Control Measures**
 - Prophylaxis: Rifampicin
 - PPE: face mask
- **Nursing Management**
 - **Frequent neurological check:** (VS, GCS, PERRLA)
 - **WOF danger signs (REPORT)**
 - Remittent fever (continuously rising of temperature)



- Onset of seizure
- Altered respiration (usually bradypneic)
- **Position:** semi-fowler's
- **Environment:** non-stimulating
- **Avoid:** valsalva maneuver (bearing down, lift heavy objects)
- **Diet:** high fiber
- **DOC:** Penicillin G, Cephalosporin or Chloramphenical
- **DOC for inflammation:** corticosteroids

Meningococemia

- Caused by Neisseria meningitides, a gram-negative diplococcus
- Transmitted through airborne or close contact
- Incubation: 1-3 days
- Natural reservoir is human nasopharynx

Clinical manifestations: sudden onset of high-grade fever, rash, and rapid deterioration of clinical condition within 24 hours

Signs and symptoms

- (1) Meningococemia
 - (a) Spiking fever
 - (b) Chills
 - (c) Arthralgia
 - (d) Sudden appearance of hemorrhagic rash
- (2) Fulminant meningococemia (Waterhouse Friderichsen)
 - (a) Septic shock
 - (b) Hypotension
 - (c) Tachycardia
 - (d) Enlarging petechial rash
 - (e) Adrenal insufficiency

Laboratory

- Blood culture
- Gram stain of peripheral smear, CSF, and skin lesions
- CBC

Treatment

- Antimicrobial
 - Benzyl penicillin 250-400,000 u/kg/day
 - Chloramphenicol 100 mg/kg/day
- Symptomatic and supportive
 - Fever
 - Seizures



- Hydration
- Respiratory function

Chemoprophylaxis

- (1) Rifampicin 300-600 mg q 12 hours x 4 doses
- (2) Ofloxacin 400 mg single dose
- (3) Ceftriaxone 125-250 mg IM single dose

Nursing Intervention

- Provide strict isolation
- Wearing of PPE
- Health teaching
- Contact tracing
- Prophylaxis
- Meningococcal vaccine for high risk patient

Rabies

- Myeloencephalitis- inducing viral infection (brain & muscle)
- Acute viral encephalomyelitis
- **AKA:** Hydrophobia, Lyssa, La rage
- **Causative agent:** rhabdovirus
- **Incubation period:** 2 - 8 weeks
- **Risk for developing rabies:** face bite 60%, upper extremities 15-40%, lower extremities 10%, 100% fatal
- **Mode of Transmission:** droplet (saliva)
- **Types:**
 - **Canine** - pet (dogs)
 - **Sylvatic** - non-pet animals (bats)
- **Signs and symptoms**

First Stage	
Dog: Dumb	Human: Prodromal
- Withdrawn - Hiding (photophobia)	- Tingling at site - photophobia
Second Stage	
Dog: Furious	Human: Excitement
- Fights & bites - Hypersalivation	- Laryngospasm/dysphagia - Hydrophobia



- Hydrophobia	- Aerophobia (avoid scent & noise) - Maniacal behavior
Third Stage	
Dog & Human: Paralytic	
- Cardiopulmonary arrest	

- **Diagnostic Test**
 - 10 day observation of the dog (CHN)
 - Brain biopsy (negri bodies)
 - Fluorescent rabies antibody test
- **Control Measures**
 - Dog vaccination (Department of Agriculture)
 - Pet owner responsibility (provide anti-rabies to pet every year)
 - Dog bite victim management
- **Management (first aid)**
 - Wash: detergent soap & running water for 10 - 15 minutes
 - Bayabas - anti-septic
 - Consult ABC (Animal Bite Center)
 - ERlg: Equine Rabies Immunoglobulin (horse)
 - HRlg: Human Rabies Immunoglobulin (human)
 - PVRV - Purified Vero Rabies Vaccine
 - **Category I** - reddened, no visible wound
 - **Category II** - bite body part (extremities)
 - **Category III** - bite neck and head, any body part: oozing
- **Management (+ rabies)**
 - Monitor CP status
 - Position: away from CR/sink
 - Cover: IV/IV lines
 - Restraint (maniacal)
 - Emotional support (palliative)

Tetanus

- Spasm inducing bacterial infection (involuntary)
- AKA: lockjaw
- **Causative agent:** Clostridium tetani (anaerobic)
- **Incubation period:** 3 days to 1 month
- **Mode of Transmission:** entry to intact skin
- **Signs and symptoms:**



- **Earliest:** lockjaw
- **Pathognomonic sign:** Rissus Sardonius (smiles like a day)
- Other:
 - **larygospasm** - DOB
 - **Opisthotonus** - exaggerated arching of the back muscle (cremation)
 - **Spastic paralysis**
- **Diagnostic tests**
 - **Wound culture** - confirmatory
 - **Spatula test** - alternative
 - **Negative:** (+) gag reflex
 - **Positive:** bites the material (tongue depressor) (+) spasm)
- **Control Measures**
 - **Vaccination**
 - DPT/pentavalent - 6,10, 14 weeks
 - TT - at least 2 shots for protection (complete 5 for lifetime)
 - Given (-) wound
 - Tetanus anti-toxin - (+) wound
 - **Wound care**
 - Hydrogen peroxide
 - Betadine - thin dressing
- **Nursing Management**
 - **Environment** - non-stimulating (quiet, dimlit and well-ventilated)
 - **Safety measures**
 - **(+) Seizure:** stay
 - **(+) Opisthotonus:** side-lying/add supporting pillows
 - **Diet** - high calorie diet (hypermetabolism)
 - **Drug of Choice:**
 - **Metronidazole:** anti-microbial, anti-seizure
 - NR: avoid alcohol beverages - inhibits drug absorption
 - **Tetracycline:** alternative
 - NR: avoid calcium-rich foods/beverages

Poliomyelitis

- Viral infection causing flaccid paralysis
- **AKA:** Heine-Medin disease; infantile paralysis
- **Causative agent:** Legio debilitans
- **Incubation period:** 7 - 21 days
- **Mode of transmission:** fecal-oral
- **Signs and symptoms**



- **Minor**
 - Subclinical: asymptomatic
 - Abortive: lasts 72 hours, fever & sore throat
- **Major**
 - Non-paralytic: Pathognomonic sign - spine stiffness, tripod position, Hoyne sign (head lag), paresis (weakness), paresthesia (tingling)
 - Paralytic - same with non-paralytic, (-) tendon reflex, (+) kernig's & brudzinki
 - **Bulbar** - brain stem
 - **CN IX:** (-) swallow reflex
 - **CN X:** (-) cardiopulmonary reflex
 - **Spinal** - spine
 - Paralysis of limbs, intercostal muscles
 - **Bulbospinal** - both
- **Diagnostic Test**
 - **Stool exam** (fecal-oral)
 - **Lumbar tap** (spinal/bulbospinal)
 - **Electromyelogram** - to check for muscle activity
 - **Muscle grading** (palpate muscle left and right, polio = muscle atrophy)
- **Control Measure**
 - **Vaccination**
 - OPV: sabin, oral
 - IPV: salk, IM
 - **Enteric precaution:** proper disposal of feces
- **Management**
 - Avoid massage
 - Warm compress (vasodilation)
 - Bed: Firm, non-soggy
 - **DOC:** bethanicol chloride (prevent respiratory depression), pleconaril (inhibits viral entry to nervous system)

Leptospirosis (Weil's Disease)

- A zoonotic systemic infection caused by leptospira, that penetrate intact and abraded skin through exposure to water, wet soil contaminated with urine of infected animals

2 types

- Anicteric type (without jaundice)
 - Manifested by fever, conjunctival injection
 - Signs of meningeal irritation
- Icteric type (weil syndrome)
 - Hepatic and renal manifestation



- Jaundice, hepatomegaly
- Oliguria, anuria which progress to renal failure
- Shock, coma, CHF
- Convalescent period

Diagnosis

- Clinical history and manifestation
- Culture
- Blood during the first week
- CSF: from the 5th and 12th day
- Urine: after the 1st week of convalescent period
- LAAT (leptospira agglutination test)
- Other laboratory tests: BUN, CREA, liver enzymes

Treatment

- Specific - penicillin 50,000 units/kg/day & tetracycline 20-40 mg/kg/day
- Non-specific - supportive and symptomatic
- Administration of fluids
- Peritoneal dialysis for renal failure
- Educate public regarding the MOT, avoid swimming or wadding in potentially contaminated waters and use proper protective equipment

Nursing Responsibilities

- Dispose and isolate urine of patient
- Environmental sanitation
- Give support and asymptomatic therapy
- Administration of fluids and electrolytes
- Assist in peritoneal dialysis for renal failure patients

Sexually Transmitted Diseases

HIV/AIDS

- Retrovirus (HIV1 & HIV2)
- Attacks and kills CD4+ lymphocytes (T-helper)
- Capable of replicating in the lymphocytes undetected by the immune system
- Immunity declines and opportunistic microbes set in
- No known cure
- HIV/AIDS reverses development and poses serious threats to future generations

Mode of Transmission

- Sexual intercourse
- Exposure to contaminated blood, semen, breastmilk, and other bodily fluids



- Blood transfusion
- IV drug use
- Transplacental
- Needlestick injuries

High Risk Group

- Homosexual or bisexual
- Intravenous drug users
- BT recipient before 1985
- Sexual contact with HIV+
- Babies of mothers who are HIV+

Signs and symptoms

- Acute viral illness (1 month after initial exposure) - fever, malaise, lymphadenopathy
- Clinical latency - 8 years w/o sex; towards end - bacterial and skin infections and constitutional sex - AIDS related complex; CD4 counts 400-200
- AIDS - 2 years; CD4 T lymphocytes <200 w/ (+) ELISA or western blood and opportunistic infections

HIV Classification

- Category 1 - CD4+ 500 or more
- Category 2 - CD4+ 200 - 499
- Category 3 - CD4+ less than 200

HIV Test

- ELISA
- Western blot test
- Rapid HIV test

CD4 T cells count

- HIV+ - CD4+ count below 500
- Full blown AIDS - CD4+ less than 200

Exhibits one or more of the ff:

- Extreme fatigue
- Intermittent fever
- Night sweats
- Chills
- Lymphadenopathy
- Enlarged spleen
- Anorexia
- Weight loss
- Severe diarrhea
- Apathy and depression
- PTB
- Kaposi sarcoma



- Pneumocystitis carinii
- AIDS dementia

Treatment

- Anti-retroviral therapy - ziduvirine
 - Prolong life
 - Reduce risk of opportunistic infection
 - Prolong incubation period

Prevention

- Abstinence
- Be faithful
- Condoms
- Don't use drugs

Syphilis

- Treponema pallidum, spirochete
- Beautiful fast moving but delicate spiral thread
- Incubation period: 10-90 days

Primary (3-6 weeks after contact) - non-tender lymphadenopathy and chancre, most infectious; resolves 4-6 weeks

- Chancre - painless ulcer with heaped up firm edges appears at the site where the treponema enters. Related to pattern of sexual behavior
- BUBO - swelling of the regional lymph node

Secondary - systemic; generalized macular papular rash including palms and soles and painless wartlike lesions in vulva or scrotum (condylomata lata) and lymphadenopathy

Tertiary (6-40 years) - neurosyphilis/permanent damage (insanity), gumma (necrotic granulomatous lesions), aortic aneurysm

Diagnosis

- Dark-field examination of lesions - 1st and 2nd stage
- Non-specific VDRL and RPR
- FTA-ABS

Management

- Primary and secondary - Pen G
- Tertiary - IV Pen G

Trichomoniasis

- Other terms: trich
- CA: Trichomonas vaginalis
- MOT:



- Direct sexual contact
- indirect contact (towels, wash clothes, douching equipment)
- IP: 5-21 days
- Clinical manifestations:
 - Female: white or greenish - yellow odorous discharge; vaginal itching and soreness, painful urination
 - Male: slight itching of penis, painful urination, clear discharge from penis
- Diagnosis:
 - Microscopic slide of discharge
 - Culture of urethral tissue, urine or semen
 - Physical examination
 - The OSOM Trichomonas rapid test identifies within 10-45 minutes, but it is less sensitive and specific than culture
- Treatment:
 - Metronidazole (Flagyl) - DOC
 - Tinidazole (Tindamax)
- Complication:
 - Cervical cancer
- Nursing considerations:
 - follow standard precautions
 - Assist in obtaining appropriate specimen for culture and testing
 - tell the patient to avoid ingesting alcohol while taking metronidazole as the combination may cause severe nausea and vomiting, abdominal pain, headaches and flushing

Chlamydia

- CA: Chlamydia trachomatis
- MOT:
 - Vaginal/rectal intercourse
 - Oral/genital contact
- IP: 7-14 days
- Clinical manifestations:
 - Cervical erosion
 - Mucopurulent discharges
 - Dyspareunia
 - Pain and tenderness of the abdomen
 - Chills
 - Fever
 - dysuria
 - Urinary frequency



- painful scrotal swelling
- Diarrhea
- tenesmus
- Diagnostic test:
 - culture of the site of infection will reveal C. Trachomatis
 - Nucleic acid probe will be positive for C. Trachomatis
- Treatment:
 - Tetracycline
 - Erythromycin
 - Azithromycin
- Complications:
 - Sterility
 - Prematurity
 - stillbirths
 - Infant pneumonia
 - Eye infections (infants)
- Nursing management:
 - observe standard precaution
 - HIV testing for both partners
 - Assess newborn for signs of chlamydial infection
 - Urge patient to inform sexual contacts of his/her infection so they can receive appropriate treatment
 - Stress the importance of completing the course of antibiotics even after symptoms subside
 - Teach the patient to follow meticulous personal hygiene measures
 - Instruct patient to avoid touching any discharge and to wash and dry hands thoroughly before touching the eyes to prevent eye contamination

Gonorrhea

- Other terms: clap/Flores blancas/gleet/drip
- Description: a sexually transmitted disease involving the mucosal lining of the genito-urinary tract, the rectum and pharynx
- CA: Neisseria gonorrhoeae
- IP: 2-5 days
- MOT:
 - Direct contact through sexual intercourse
 - Direct contact with contaminated secretions of the mother during vaginal delivery
 - Indirect contact (fomites)
- Clinical manifestations:



- Female:
 - 80% are asymptomatic
 - Burning sensation and frequent urination
 - Yellowish purulent vaginal discharge
 - Redness and swelling of the genitals
- Male:
 - Dysuria with purulent discharge
 - Rectal infection
 - Inflammation of the urethra
 - Prostatitis
 - Pelvic pain
- Complications:
 - Sterility
 - Pelvic infection
 - Epididymitis
 - Arthritis
 - endocarditis
 - conjunctivitis
 - Meningitis
- Diagnostic tests:
 - Gram staining
 - Culture or cervical & urethral smear
- Treatment:
 - Ceftriaxone (IM)
 - Azithromycin or doxycycline (PO)
- Nursing considerations:
 - standard precautions
 - sexual abstinence until he/she recovers
 - For gonococcal arthritis (apply moist heat to relieve pain)
- Prevention:
 - Sex education
 - Case finding
 - Report cases of gonorrhea

Candidiasis

- Other terms: candidiasis/moniliasis
- Description: superficial fungal infection that usually infects the skin, nails, mucous membrane, vagina, esophagus and GI tract
- CA: Candida albicans



- Sources of infection: Candida is part of the normal flora in GI tract, mouth, vagina and skin. They cause infection when some changes in the body such as increased blood glucose or immunocompromised occurs
- Clinical manifestations:
 - Skin - scaly, erythematous, papular rash, sometimes covered with exudates, appearing below the breast between the fingers and the axillae, groin and umbilicus
 - Nails - red, swollen, darkened nail bed, occasionally purulent discharge and the separation of a pruritic nail from the nail bed
 - Oropharyngeal mucosa (thrush) - cream-colored or bluish white curd-like patches of exudates on the tongue, mouth, or pharynx that reveal bloody engorgement when scraped
 - Esophageal mucosa - dysphagia, retrosternal pain, regurgitation, occasionally scales in the mouth and throat
 - Vaginal mucosa - white or yellow discharge, with pruritus and local excoriation, white or gray raised patches on vaginal walls, with local inflammation, dyspareunia
 - Lungs - hemoptysis, cough, fever
 - Kidney - fever, flank pain, dysuria, hematuria, pyuria, cloudy urine
 - Brain - headache, nuchal rigidity, seizures, focal neurologic deficits
 - Endocardium -systolic or diastolic murmur, fever, chest pain, embolic phenomena
 - Eye - endophthalmitis, blurred vision, orbital or periorbital pain, scotoma, exudates
- Diagnostic Procedure:
 - blood culture
 - Culture of vaginal scraping
 - Echocardiography if there is cardiac involvement
 - Fundoscopy for patients with endophthalmitis
- Management:
 - Antifungal: Nystatin, clotrimazole, miconazole
- Diagnostic tests:
 - ELISA (presumptive test)
 - Western blot (confirmatory)
 - Particle agglutination
 - Immunofluorescent test
- Treatment:
 - Reverse transcriptase inhibitors (Zidovudine)
 - Protease inhibitors (Ritonavir)
- Nursing management:
 - Avoid accidental wounds from infectious materials used in HIV patients
 - Avoid contact of open skin lesions
 - Gloves should be worn when handling blood specimens
 - Handwashing
 - Blood and other specimens should be labeled prominently



- Instruments with lenses should be sterilized after use on AIDS patient
- Needles should not be bent after use, placed it under puncture-resistant
- Patients with active AIDS should be isolated
- Care of thermometer - wash with warm soapy water, soak in 70% alcohol for 10 minutes, dry and store

Integumentary System

Chicken Pox

- other term: Varicella zoster
- Description: acute infectious disease of sudden onset with slight fever, mild constitutional symptoms and eruptions which are maculopapular for a few hours, vesicular for 3-4 days and leaves granular scabs
- Etiologic agent: Human (alpha) herpes 3 (Varicella-zoster virus)
- Sources of infection:
 - secretions of respiratory tract of infected persons
 - Lesions (little consequences)
 - Scabs are not infective
- MOT:
 - Direct contact
 - contact with contaminated linen and fomites
 - Airborne
- Incubation Period:
 - 2-3 weeks
- Period of communicability:
 - Cases are infectious up to 2 days before the onset of the rash until 5 days after the first crop of vesicles
- Diagnostic Test:
 - Isolation of the virus from the vesicular fluid within the 3 to 4 days of the rash
 - Serum antibodies is present in 7 days after onset
- Congenital varicella results in:
 - Hypoplastic, deformities and scarring of limb
 - Retarded growth
 - CNS and ophthalmic manifestation
- Nursing considerations:
 - strict isolation
 - exclusion from school for 1 week after eruption first appears and avoid contact with susceptible



- Concurrent disinfection if throat and nose discharge
- Tell the patient not to scratch the lesions
- Teach the child and family to apply topical antipruritic medication correctly
- Susceptibility, resistance and occurrence
 - Universal among those not previously attacked
 - Severe in adults
 - An attack covers long immunity
 - second attacks are rare
- Prevention:
 - Vaccine: varicella-zoster immune globulin (VZIG) it should be given within 10 days of exposure

Measles

- Other terms: Rubeola/Morbili/7-day measles
- Description: it is an acute contagious and exanthematous disease that usually affects children who are susceptible to URTI
- Etiological agent: filterable virus of measles (paramyxoviridae)
- Source of infection: secretions of nose and throat infected persons
- MOT:
 - Droplet spread/direct contact with infected person
 - Indirect contact (articles with secretions)
 - Airborne
- IP: 1-2 weeks
- Period of communicability: starts just before the prodrome and lasts until 4 days after the rash appear
- Clinical manifestations: Koplik's spots (pathognomonic sign)
 - Pre-eruptive stage
 - Fever
 - catarrhal symptoms (cough, conjunctivitis, coryza)
 - Photophobia
 - stimson's line (red line on the lower conjunctiva)
 - Eruptive stage
 - maculo-papular rash
 - High grade fever
 - Anorexia and irritability
 - throat is red and extremely sore
 - Convalescence stage
 - rashes fade away
 - Fever subsides



- Desquamation begins
- Symptoms subside and appetite restored
- Diagnostic Tests
 - Throat and nose swab
 - Urinalysis
 - Blood eams (single-raised IgM or rise on IgG)
- Treatment:
 - Antiviral drug (isoprenosine)
 - Antibiotics
 - oxygen inhalation
 - IV fluids
- Complications:
 - Bronchopneumonia
 - otitis media
 - Pneumonia
 - nephritis
 - Encephalitis
- Nursing management:
 - Isolation
 - Maintain standard and airborne precautions
 - Place the patient on a negative pressure room
 - Tepid sponge bath
 - Skin care
 - Oral and nasal hygiene
 - Eye care (photosensitivity)
 - Ear care
 - Daily elimination (mild laxative)
 - During febrile stage, limit diet to fruit juice, milk and water
 - Give medication as ordered by the physician (penicillin)
- Prevention:
 - Immunization
 - Anti-measles at the age of 9 months as a single dose
 - MMR vaccine (15 months); second dose (11-12 years old)
 - Measle vaccine should not be given to pregnant women or to persons with active tuberculosis, leukemia, lymphoma or depressed immune system

German Measles

- Other terms: rubella/3-day measles
- Description:



- it is mild viral illness caused by rubella virus
- It causes mild feverish illness associated with rashes and aches in joints
- It has teratogenic effect on the fetus
- Etiologic agent: rubella virus
- MOT:
 - Droplet transmission
 - Transplacental transmission in congenital rubella
- IP: 2-3 weeks
- Clinical manifestations:
 - Prodromal period
 - Low grade fever
 - Headache
 - Malaise
 - mild coryza
 - Conjunctivitis
 - Post-auricular, sub-occipital and posterior cervical lymphadenopathy which occurs on the 3rd to 5th day after onset
 - Eruptive period
 - Forchheimer's spot (pinkish rash on the soft palate)
 - Eruption appears after the onset of adenopathy
 - children usually present less or no constitutional symptoms
 - The rash may last for one to five days and leaves no pigmentation nor desquamation
 - Testicular pain in young adults
 - Transients polyarthralgia and polyarthritis may occur in adults and occasionally in children
 - Congenital rubella
 - Classic congenital rubella syndrome
 - Intrauterine growth retardation
 - Infant has low birth weight
 - Thrombocytopenic purpura known as "blueberry muffin" skin
 - Intrauterine infection
 - May result in spontaneous abortion
 - Birth may result in spontaneous abortion or multiple birth anomalies
 - Cleft palate, talipes and eruption of teeth
 - cardiac defects (patent ductus arteriosus, atrial septal defect)
 - Eye defects (glaucoma, retinopathy, microphthalmia)
 - Neurologic (microcephaly, mental retardation, psychomotor retardation, vasomotor instability)
- Diagnostic Test:



- Clinical observation
- cell cultures of the throat, blood, urine and CSF confirm the presence of the virus
- Convalescent serum that shows a fourfold rise antibody titer supports the diagnosis
- Treatment:
 - Acetaminophen for fever and joint pain
 - Isolation
- Complications:
 - Encephalitis
 - Neuritis
 - Arthritis
 - Arthralgias
 - Rubella syndrome
 - Microcephaly
 - mental retardation
 - Cataract
 - Deaf-mutism
 - Heart disease
- Nursing considerations:
 - Provide comfort
 - Make sure female patients understand how important it is to avoid exposure to this disease when pregnant
 - Report confirmed cases of rubella to local public health officials
 - Warn the patient about possible mild fever, slight rash, transient arthralgia and arthritis
 - If lymphadenopathy persists after the initial 24 hours, suggest a cold compress to promote vasoconstriction and prevent antigenic cyst formation
 - Patient's room must be darkened to avoid photophobia
 - Patient's eyes should be irrigated with warm saline to relieve irritation
 - Good ventilation is necessary
- Prevention:
 - administration of live attenuated vaccine (MMR)
 - Pregnant women should avoid exposure to patients infected with rubella virus
 - Administration of immune serum globulin one week after exposure to rubella

Pediculosis

- Description: any human infestation of lice anywhere on the body
- Types:
 - Pediculosis capitis - lice feed on the scalp and rarely on the skin under the eyebrows, eyelashes and beard
 - Pediculosis corporis - lice live next to the skin in clothing seams



- Pediculosis pubis - lice found primary in pubic hairs but may extend to the eyebrows, eyelashes and axillary or body hair
- MOT:
 - Head to head contact
 - Fomites
 - sexual activity
- IP: 3-7 days
- Clinical manifestations:
 - Pruritus
 - Tickling sensation of something moving in the hair may be noticed
 - Head lice and their nits are most commonly found behind the ears and on the hairs of the neck and occiput
 - Body lice are found on clothing seams
 - Pubic lice will be found attached to the base of the pubic hair and the infestation generally results in severe itching
- Diagnostic test:
 - Wood's light examination (fluorescence of the adult lice)
 - Microscopic examination (presence of nits on the hair shaft)
- Treatment:
 - Permethrin (Eliminate)/Pyrethrin (Rid Mousse)
 - Initial treatment of choice
 - topical insecticide
 - For pediculosis capitis & pubis
 - Fine tooth comb dipped in vinegar
 - Washing hair with ordinary shampoo
 - Oral anthelmintics (Ivermectin, levamisole, albendazole) are effective against lice infestation
 - Prevention of head reinfestation
 - Clothes and bed linens must be washed in hot water, ironed or dry cleaned
 - Storing clothes or linens for more than 30 days or placing them in dry heat of 140F (60C) kills lice
- Complications:
 - Excoriation
 - secondary bacterial infections
 - If left untreated, pediculosis may result in dry, hyperpigmented, thickly encrusted, scaly skin with residual scarring
- Nursing considerations:
 - Contact precautions should be maintained until treatment is complete to prevent spreading the infection



- Have the patient's fingernails cut short to prevent skin breaks and secondary bacterial infections caused by scratching
- Be alert for possible adverse reactions to treatment with an antiparasitic, including sensitivity reactions and in some cases, CNS toxicity
- To prevent self-infestation, avoid direct contact with the patient's hair, clothing and bedsheets
- After each treatment, inspect the patient for remaining lice or eggs
- Use gloves, gown and protective head covering when administering delousing treatment
- Teach the patient and family how to inspect and identify lice, eggs, and related lesions
- Instruct the patient and family about the use of the creams, lotions, powders and shampoos that eliminate lice
- Instruct the patient in the proper application of lindane, which can be absorbed by the skin and cause CNS complications

Leprosy

- AKA: Hansen's disease/hansenosis
- Causative agent: *Mycobacterium leprae*
- Incubation period: 6 months - 8 years or more
- Chronic infectious and communicable disease
- **Mode of transmission:** droplet, contact (prolonged exposure)
- **Signs and symptoms:**
 - **Early:** anesthesia, skin discoloration, anhidrosis
 - **Late:** madarosis (eyebrow disappearance), lagophthalmos (opened eyes), lion-like appearance, wound that doesn't heal, contractures, gynecomastia
 - **Terminal sign:** amputation, blindness, luscious phenomenon
- Diagnostic Test
 - Slit skin smear - confirmatory
 - LRT (Lepromin Reaction Test) - test communicability of disease
 - (+) Rashes - not communicable
 - (-) Rashes - communicable
 - 8-10 day
- Control measure
 - Vaccine BCG
 - Prophylaxis: Rifampicin, Ofloxacin, Minocycline
- Management
 - Hygiene
 - Artificial tears (lagophthalmus)
 - Closed shoes
 - Avoid sharps and rough objects



- Wooden utensils
- DOC
 - Rifampicin
 - Dapsone
 - Clofazimine

Herpes Zoster

- Other terms: shingles
- Description: it is an acute unilateral and segmented inflammation of the dorsal root ganglia caused by reactivation of the herpes varicella-zoster virus which also causes chicken pox, usually occurs in adults
- CA: varicella virus
- IP: unknown, but believed to be 13-17 days
- Period of communicability: communicable a day before the appearance of the first rash until 5-6 days after the last crust
- MOT:
 - Airborne
 - Droplet
 - Direct contact
- Clinical manifestations:
 - Begins with fever and malaise
 - Severe deep pain, pruritus, and paresthesia and hyperesthesia, usually on the trunk and occasionally on the arms and legs
 - Small, red, nodular skin lesions (unilateral) erupt on the painful areas up to 2 weeks after the first symptoms
 - Vesicles filled with fluid or pus
 - cranial nerve involvement
- Complications:
 - Generalized CNS infection
 - Acute transverse and ascending myelitis
 - Intractable neurologic pain
- Diagnostic test:
 - Differentiation of herpes zoster from herpes simplex through fluorescent light
 - Tissue culture technique
 - Smear of vesicle fluid
 - microscopy
- Management:
 - Antiviral therapy - acyclovir
 - Analgesics to control pain



- Anti-inflammatory
- Nursing interventions:
 - Airborne and contact precautions
 - If vesicles rupture, apply cold compress as ordered
 - To minimize neuralgic pain, administer analgesics as ordered and evaluate their effects
 - Instruct the patient to avoid scratching the lesions
 - Keep the patient comfortable and maintain meticulous hygiene
 - Encourage sufficient bed rest and give supportive care
- Prevention:
 - Vaccination against varicella
 - Avoid exposure to patients with varicella infection

Gastrointestinal System

Cholera

- *Vibrio cholerae* (gram -)
- Cholera toxin induces active secretion of NaCl
- Active immunization
- Incubation period: few hours to 5 days
- MOT: oral - fecal route
- Signs and symptoms: rice watery stool with flecks of mucus, severe dehydration
- Diagnosis: stool culture
- Management: IV fluids, tetracycline, doxycycline, erythromycin, quinolones, furazolidone, and sulfonamides (children)

Typhoid Fever

- Description: it is a systemic infection characterized by continued fever, anorexia, involvement of lymphoid tissue especially ulceration of Peyer's patches
- CA: *Salmonella typhi* or Typhoid bacillus
- Sources of infection: feces and urine of infected persons
- MOT:
 - Fecal-oral transmission
 - Contaminated urine
 - Direct/indirect contact with infected person
 - Ingestion of contaminated food, water, milk
- IP: 1-3 weeks (average: 2 weeks)
- Period of communicability: as long as typhoid bacilli appears in excreta



- Clinical manifestations:
 - Onset:
 - Headache
 - Nausea and vomiting
 - ladder-like fever
 - Rose spots on the abdomen
 - Typhoid state
 - coma vigil
 - Subsultus tendinum
 - carpholagia
 - Delirium
- Complications:
 - Hemorrhage/perforation
 - Peritonitis
 - Bronchitis and pneumonia
 - Typhoid spine
 - Septicemia
 - Reiter's syndrome - joint pain, eye irritation
- Diagnostic tests:
 - Typhidot - confirmatory
 - ELISA
 - Widal
 - Rectal swab
 - Bone marrow aspiration
- Treatment:
 - Chloramphenicol - DOC
 - Ampicillin
 - Co-trimoxazole
 - Ciprofloxacin
 - Cefixime/Azithromycin
 - Ceftriaxone (recommended for complicated cases)
- Nursing Management:
 - Isolation
 - Maintain standard precautions unless the patient is incontinent or in diapers or if an outbreak develops in an institution
 - Give nourishment fluids in small quantities at frequent intervals
 - Monitor VS
 - Prevent further injury
 - WOF: intestinal bleeding/bowel perforation including sudden pain in the lower right side of the abdomen and abdominal rigidity



- Provide good skin and mouth care
- Turn the patient frequently and perform mild passive exercise as indicated
- Apply mild heat to the abdomen to relieve cramps
- Prevention:
 - Sanitary and proper disposal of excreta
 - Proper supervision of food handlers
 - Enteric isolation
 - Prevention of safe drinking water supply
 - Detection and supervision of typhoid carriers

Bacillary Dysentery

- Other terms: shigellosis/bloody flux
- Description: a acute bacterial infection of the intestine characterized by diarrhea, fever, tenesmus and in severe cases bloody and mucoid stools
- CA:
 - *Shigella sonnei* (most common in Western Europe)
 - *Shigella flexneri*
 - *Shigella boydii*
 - *Shigella dysenteriae*
- IP: 12-96 hours but may be up to 1 week
- Period of communicability: the patient can transmit the microorganisms during the acute infection until the feces are negative of the organism
- MOT:
 - Ingestion of contaminated food
 - drinking of contaminated water/milk
 - Fecal-oral transmission
- Clinical manifestations:
 - Fever
 - Tenesmus
 - Nausea and vomiting
 - Colicky or cramping abdominal pain associated with anorexia and body weakness
 - Bloody mucoid stools
 - rapid dehydration
- Diagnostic test:
 - Microscopic examination
 - Direct immunofluorescence
 - Sigmoidoscopy or proctoscopy
 - Stool culture
- Treatment:



- Antibiotics
 - Ampicillin
 - Ceftriaxone
 - Trimethoprim-sulfamethoxazole
 - Ciprofloxacin
- IV therapy
- Low residue diet
- CI: anti-diarrheal drugs (delays fecal excretion that can lead to prolonged fever)
- Prevention:
 - Sanitary disposal of human feces
 - Adequate personal hygiene
 - sanitary supervision of processing, preparation and serving food
 - Fly control and protection against fly contamination
 - Isolation (acute stage)
 - Protection and purification of public water supply
 - Routine cooking kills shigella

Paragonimiasis

- CA:
 - Lung fluke
 - Paragonimus westermani
 - Paragonimus siamensis
- MOT:
 - Ingestion of raw/uncooked crabs/crayfish
 - Contamination of food
 - Using meat/juice of infected animals
- Reservoir:
 - cats
 - Dogs
 - rats
 - pigs
- Clinical manifestations:
 - Cough of long duration
 - Recurrent blood-streaked sputum
 - Chest/back pain
 - PTB
- Diagnostic test:
 - Sputum microscopy
 - Immunology



- Cerebral paragonimiasis
- Treatment:
 - Praziquantel (Billtrizide)
- Prevention:
 - treatment of infected person
 - Anti-mollusk campaign
 - Education of the population
 - Avoid eating infected foods

Mumps

- other terms: infectious parotitis/epidemic parotitis
- Description: an acute viral disease manifested by swelling of one or both parotid glands, with occasional involvement of the glandular structures, particularly the testes in males
- CA: paramyxoviridae
- Source of infection: secretion of the mouth and nose
- MOT:
 - Direct contact
 - Indirect contact with the articles freshly soiled with secretion from the nasopharynx
- Period of communicability:
 - Cases are infectious for up to a week (normally 2 days) before parotid swelling until 9 days after
 - 48 hours period immediately preceding onset of swelling is considered the time of highest communicability
- Clinical manifestations:
 - Sudden headache
 - Earache
 - Loss of appetite
 - Fever
 - swelling of parotid gland (between the earlobe and angle of the mandible)
- Complications:
 - Orchitis
 - Oophoritis
 - Mastitis
 - Nuchal rigidity
 - Deafness
 - Meningoencephalitis
 - Pancreatitis
 - Myocarditis
 - Nephritis



- Diagnostic Tests:
 - Serum amylase determination
 - Complement fixation test
 - hemo-agglutination inhibition test - used to determine immune status
 - Neutralization test - determines immunity to mumps
 - Viral culture
- Treatment:
 - Analgesics for pain
 - antipyretics for fever
 - IV fluid replacement
 - Hold and cold application
- Nursing management:
 - Medical Aseptive Protective Care
 - single occupancy room
 - Oral and personal hygiene (warm salt water gargle)
 - General Management of the Disease
 - Bed rest
 - Diversional activities
 - Eye care
 - provide extra fluids
 - Diet:
 - No restriction of food
 - Soft bland and semi-solid
 - Acid foods (-)
- Prevention:
 - Active immunization (MMR)
 - reporting of cases to health authorities
 - Isolation of patient

Botulism

- Description:
 - Rare but severe form of poisoning caused by gram-positive, anaerobic bacteria
 - An illness of descending paralysis and autonomic dysfunction due to neurotoxin
- CA: Clostridium botulinum
 - Foodborne botulism
 - Wound botulism
 - Infant botulism
- Source of Infection:
 - Untreated water



- undercooked and improperly preserved canned foods, especially those with a low acid content
- home-canned vegetables
- cured pork and ham
- smoked or raw fish
- honey and corn syrup
- MOT:
 - Ingestion (or injection) of preformed toxin
 - Spores may resist 100 degree celcius for many hours
 - Inhalation of toxin can also cause the disease
 - Introduction of spores into wound
- IP: 12-72 hours but extremes of 2 hours to 10 days are reported
- Clinical manifestations:
 - double or blurred vision
 - Droopy eyelids
 - dry mouth
 - Difficulty swallowing or talking
 - difficulty breathing
 - Flaccid paralysis (descending)
 - Decreased or absent DTR
 - Initial vomiting or diarrhea followed by constipation
- Diagnostic test:
 - toxicity screen
 - stool culture
 - suspected food may also be cultured
 - Electromyography
 - Mouse-onoculation test
- Complications:
 - Aspiration
 - weakness and nervous system problems can be permanent
 - Death
- Treatment:
 - Botulinus antitoxin (IV,IM)
 - infant-inducing vomiting or giving an enema
 - IV fluids
 - Nasogastric tube
 - Endotracheal intubation (respiratory distress)
- Nursing consideration:
 - obtain careful history of food eaten in the past several days
 - Monitor respiratory and cardiac function carefully



- Perform frequent neurologic checks
- purge the GI tract as ordered
- Check for patient's allergies, perform skin test first (administration of botulinus toxin)
- Educate the patient and family about importance of proper hand hygiene
- Teach the patient and family to cook food thoroughly before ingesting
- Instruct the patient who eats home canned foods to boil the food for 10 minutes before eating to ensure that it is safe to consume
- Teach patient and family to see their doctor promptly for infected wounds and avoid injectable street drugs
- Suggested on-call action
 - Ensure that the case is admitted to hospital
 - Obtain food history as a matter of urgency
 - Obtain suspect foods
 - Identify others at risk
 - Inform appropriate local and national authorities

Amoebiasis

- Description: protozoal infection that initially involves the colon but may spread into the liver and lungs by lymphatic dissemination
- CA: *Entamoeba histolytica*
 - Cyst - considered to be the infective stage and the resistance to environmental conditions and can survive for few days outside the body
 - Trophozoites/vegetative form - facultative parasites that invade the tissue
- Source of infection: contaminated food and water, flies
- MOT: fecal-oral, oral-anal
- IP: 2-4 weeks (severe infection: 3 days)
- Period of communicability: communicable for the entire duration of the illness or until cysts are present in the stool
- Clinical Manifestations:
 - Acute amoebic dysentery
 - Slight attack of diarrhea altered with PD of constipation
 - Watery foul-smelling stools containing blood streaked mucus
 - gaseous distention of the lower abdomen
 - nausea, flatulence
 - Tenderness in the right iliac region
 - Chronic amoebic dysentery
 - diarrhea for several days, succeeded by constipation
 - Anorexia, weight loss, weakness, fatigue
 - Watery, bloody mucoid stool



- Flatulence and irregular bowel movement
- abdomen loses its elasticity
- severe cases - scattered ulceration is seen through sigmoidoscopy
- Diagnostic test:
 - Stool exam
 - blood exam (leucytosis)
 - Sigmoidoscopy
- Management:
 - Metronidazole (Flagyl) 800 mg TID x 5 days
 - Tetracycline, ampicillin, streptomycin, chloramphenicol
- Nursing interventions:
 - Observe isolation and enteric precautions
 - Proper collection of stool specimen
 - No oil prep for 48 hours
 - Large portion of stools containing blood mucus
 - Label specimen properly
 - Send specimen immediately to the laboratory
 - Provide skin care and hygiene
 - Provide optimum comfort dysenteric patient should never allowed to feel cold
 - Diet fluid should be forced
 - cereals and strained meat broth without fats
 - Bland diet without cellulose or bulk producing foods
 - chicken and fish may be added when convalescence is established
- Prevention:
 - health education and fly control
 - Sanitary disposal of feces
 - Safe drinking water
 - proper food preparation and food handling
 - Detection and treatment of carriers

Schistomiasis

- Caused by blood flukes Schistomas
- Has 3 species *S. Haematobium*, *S. Mansoni* and *S. Japonicum*
- *S. Japonicum* is endemic in the Philippines (Leyte, Samar, Sorsogon, Mindoro, Bohol)
- Intermediate host: *Oncomelania Quadrasi*
- Diagnosis
 - Schistosoma eggs in stool
 - Rectal biopsy
 - Kato Katz



- ultrasound of HBT
- Clinical manifestations
 - Severe jaundice
 - Edema
 - Ascites
 - Hepatosplenomegaly
 - Signs and symptoms of portal hypertension
- Management
 - Praziquantrel 60mg/kg once dosing
 - Supportive and symptomatic
- Methods of control
 - Educate the public regarding the mode of transmission and methods of protection
 - Proper disposal of feces and urine
 - Prevent exposure to contaminated water. To minimize penetration after accidental water exposure, towel dry and apply 70% alcohol
- The organism is pathogenic only in man

NURSING PRACTICE II

Obstetric Nursing

Obstetric - the care of women during childbirth

Obstetric nursing - also known as OB or perinatal nursing, focuses on providing care to women throughout their reproductive cycle, including pregnancy, childbirth, and postpartum periods.

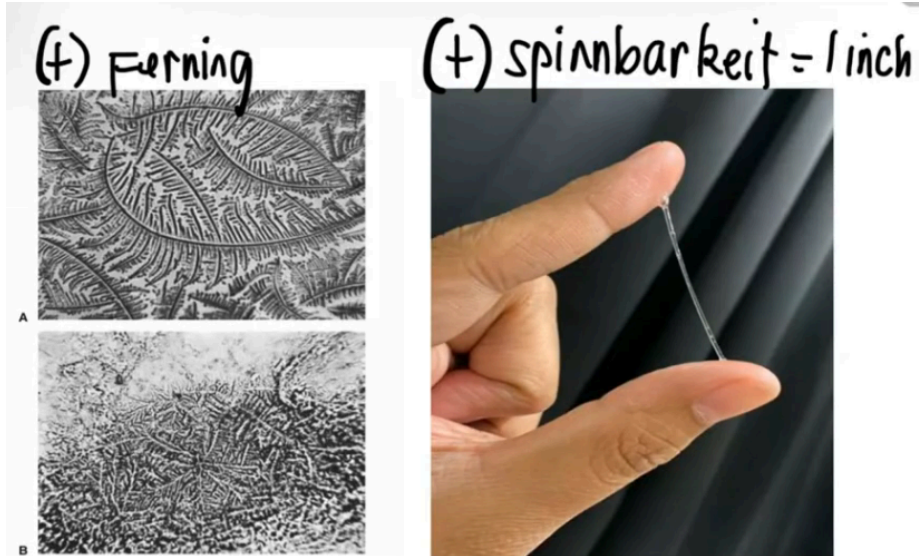
Goal of Maternal and Child Health Nursing

- Promotion and maintenance of optimal family health to ensure cycles of optimal childbearing and childrearing.

Ovaries

- Almond-shaped female sex glands for production of estrogen and progesterone
- **Estrogen** - development of secondary sexual characteristics
 - Growth spurt
 - Fat distribution in the hips and legs
 - Thelarche - breast development
 - Adrenarche - pubic hair
 - Menarche - first mens
 - Adrenarche - axillary hair
 - Vaginal secretions
 - Inhibits FSH, stimulates LH

- Growth of long bones (12 y/o)
- Early closure of epiphysis of long bones (16-18 y/o)
- Sexual desires
- Vaginal lubrication
- Fertile cervical mucus (wet, alkaline, slippery, thin, clear, watery, abundant, stretchy, (+)spinnbarkeit, (+) ferning)



- Billing's Method - Abstain from coitus if cervical mucus is stretchy, wet and thin until 4 days after the last episode of wet mucus
- **Progesterone** - prepares the endometrium for implantation (hormone of the mother/pregnant)
 - Increases BBT (.3-.6C/.5-1F) during and after ovulation
 - BBT method: no sex if BBT is increased for 3 days
 - Inhibits LH
 - Mammary gland development - lactation
 - Mood swings - after ovulation
 - Muscle relaxants - reduced peristalsis
 - Infertile cervical mucus (thick, sticky, dry, dense, opaque, scanty, (-) ferning, (-) spinnbarkeit)

Menstruation

Menstrual cycle - episodic uterine bleeding in response to a cycle hormonal changes

- Normal menstrual cycle - 24-38 days
- Menstrual period - 2-7th day
- Blood loss - 5-80ml
- Color: dark red



- Iron loss: 11 mg
- **What to report**
 - (1) Heaving bleeding (>80 ml, >7 days, changing napkins in every <2 hours)
 - (2) Getting a fever
 - (3) A change from regular to irregular cycles
 - (4) Severe pain during menses
 - (5) No mens for 3 months or more
 - (6) During menses (decreased estrogen and progesterone) -> hypothalamus -> GnRH -> APG -> ovaries -> production of estrogen and progesterone -> maturation of ovum
- Day 12/13 (increased estrogen, mature ovum, decreased progesterone) -> hypothalamus -> GnRH -> APG -> LH -> ovary (increased progesterone) -> ovulation (day 14)
- **Mittelschmerz** - pain during ovulation (mid-cycle pain)
- 28-day cycle ovulation occurs on day 14; 32-day cycle ovulation occurs on day 18
- **Ovum survives** - 24-48 hours
- **Sperm survives** - 48-72 hours
- **Irregular cycle** (responsible FSH)
 - 6-8 consecutive cycles
 - Shortest cycle - 18 days
 - Longest cycle - 11 days
 - Results: fertile days

Shortest = 25 days Longest = 30 days
- 18 - 11

Day 7 → Day 19

Hypothalamus	GnRH
Anterior Pituitary Gland	FSH - Follicle Stimulating Hormone - Maturation of ovum - Stimulates ovary to produce estrogen LH - Luteinizing Hormone (Primordial follicle -> Graafian follicle) - Ovulation (release of mature ovum) - Stimulates ovary to produce progesterone
Ovaries	Estrogen, Progesterone



Uterus	
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Menstrual Hormones

- **Estrogen:** Produced by the ovaries, estrogen is crucial for developing and thickening the uterine lining (endometrium) and plays a role in follicle development.
- **Progesterone:** Also produced by the ovaries (specifically after ovulation), progesterone maintains the uterine lining and prepares the uterus for a potential pregnancy.
- **Follicle-Stimulating Hormone (FSH):** Released by the pituitary gland, FSH stimulates the growth of follicles (which contain eggs) in the ovaries.
- **Luteinizing Hormone (LH):** Another hormone released by the pituitary gland, LH triggers ovulation (the release of an egg from the ovary).
- **Gonadotropin-Releasing Hormone (GnRH):** Produced by the hypothalamus (a part of the brain), GnRH stimulates the pituitary gland to release FSH and LH, initiating the menstrual cycle.

Phases of Menstruation

- **Follicular Phase (1-14):** Follicles in the ovaries start to develop, and the uterine lining begins to thicken. This phase begins with the end of menstruation and ends with ovulation. (FSH)
 - **Menstrual** - bleeding (day 1-6) decreased estrogen and progesterone
 - **Proliferative** - thickening (day 7-13) increased estrogen and decreased progesterone
- **Ovulation:** An egg is released from an ovary and travels down the fallopian tube, making it available for fertilization. Ovulation typically occurs around day 14 of a 28-day cycle.
 - If fertilization has occurred -> after 6-8 days -> implantation (fundus) -> hCG production -> maintains corpus luteum for 1st trimester -> production of estrogen and progesterone -> to maintain endometrial lining -> amenorrhea -> pregnancy
- **Luteal Phase (15-28):** After ovulation, the body prepares for a potential pregnancy. If fertilization doesn't occur, the uterine lining sheds, and the cycle restarts. This phase is characterized by rising progesterone levels. (LH)
 - **Secretory** - twisted and spongy (day 15-26) increased progesterone, decreased estrogen
 - **Ischemic** - vasoconstriction (day 27-28) corpus luteum -> whitish corpus albicans -> no estrogen and progesterone production -> menstruation
- **Menstruation (Menstrual Phase):** This is when the uterine lining (endometrium) sheds, resulting in menstrual bleeding. This phase typically lasts 3-7 days

Primordial Follicle	Follicular	Low E Low P	Menstrual	Bleeding
Graafian Follicle		High E Low P	Proliferative	Thickening



Corpus Luteum	Luteral	High P Low E	Secretory	Twisted/Spongy/C orkscrew
Corpus Albicans		Low P Low E	Ischemic	Constricted

Related Terminologies

- **Menarche** - first menstruation
- **Dysmenorrhea** - excessive menstrual pain
 - **Primary** - idiopathic
 - **Secondary** - due to pelvic pathology (most common endometriosis)
- **Amenorrhea** - absence of menstruation
 - **Primary** - menses have never occurred
 - **Secondary** - previously menstrual bleeding has occurred
 - **Regular cycles** in 3 missed periods
 - **Irregular cycles** in 6 missed periods
- **Metrorrhagia** - bleeding in between menses/bleeding at irregular intervals
- **Menorrhagia** - heavy menstrual bleeding
- **Menopause** - cessation of menses (average age 51)

Pregnancy

Fertilization: the beginning of pregnancy

- Also referred to conception and impregnation/fecundation
- 22 pairs of autosomes and 1 pair of sex chromosomes = 46 diploids chromosomes
 - X, female, acid resistant, slow
 - Y, male, dies in acid, fast
- The union of an ovum and spermatozoa (zygote) that occurs in the outer third of a fallopian tube (ampulla)
- Dizygotic twins/fraternal twins - multiple ovulation, assisted reproductive technology, use of clomid (fertility drug)
 - Same sex/different sex
 - 2 BOW, 2 placenta
- Monozygotic twins/identical twins/same sex
 - Dichorionic diamnionic (1-3)
 - Monochorionic diamnionic (4-7)
 - Monochorionic monoamnionic (8-13)
 - Conjoined twins (>13 days)



Implantation/Nidation (6-8 days after fertilization)

- A zygote (union of ovum and sperm) migrates over the next 3 to 4 days toward the body of the uterus, aided by currents initiated by the muscular contractions of the fallopian tube
- Contact between the growing structure and the uterus and the uterine endometrium occurs approximately 8 to days after fertilization
- Signs of implantation
 - Vaginal spotting
 - Slight pain
- Site: fundus (uppermost portion)

Embryonic and Fetal Structures

Decidua

- Endometrium continues to grow and thickens and vascularity, instead of sloughing off as in a usual menstrual cycle

Chorionic villi

- Once implantation is complete, the trophoblastic layer of cells of the blastocysts begins to mature rapidly as early as 11th or 12th day

Amnion - innermost, contains the umbilical cord and BOW, facing the fetus, thin

Chorion - develops from the trophoblast, outermost and thick, contains the placenta, facing the uterus

- **Cytotrophoblast (Langhan's layer)**
 - **Function:** to protect fetus against toxoplasmosis and syphilis on the first 16-18 weeks of pregnancy
- **Syncytiotrophoblast**
 - **Function:** to produce hCG, for nutrition, production of hormones

Placental Hormones

Human Chorionic Gonadotropin (hCG)

- The first placental hormone produced found in the maternal blood as early as the first missed menstrual period
- The purpose is to act as fail-safe measure to ensure the corpus luteum of the ovary continues to produce progesterone and estrogen
- Appears in the maternal serum as early as 24-48 hours after implantation
- 60,000-140,000mIU/ml
- **Function**
 - Prolongs corpus luteum for 1st 6-8 weeks
 - Basis for pregnancy
- **False positive result**
 - Taking contraceptive pills
 - Women who have proteinuria



Estrogen

- Estriol
- Produced as the second product of the syncytial cells of the placenta
- Contributes to the woman's mammary gland development in preparation for lactation and stimulates uterine growth to accommodate developing fetus
- Expansion of blood
- Vascularization = increase in blood supply
 - Chadwick's sign - bluish discoloration of vagina/cervix/vulva/perineum
 - Nasal congestion (hearing difficulty early in pregnancy), softening of gums (gingival hypertrophy)
 - Palmar erythema (pruritus)
- **Secretions**
 - Leucorrhea (whitish vaginal discharge)
 - Excessive salivation (ptyalism)
 - Operculum (cervical mucus plug) - bacteriostatic effect
- **Enlargement of Uterus**
 - Hypertrophy to myometrium
 - Non-pregnant: 50-60g, inverted avocado
 - Pregnant: 1000-1100g, ovoid

Progesterone

- Hormone of women/mothers
- Maintains endometrial lining of the uterus during pregnancy
- Increases progressively during the remainder of pregnancy
- Reduce contractility of uterus, preventing premature labor
- **Function**
 - Muscle relaxant
 - Decreases peristalsis
 - Maintains endometrial lining
 - Fluid retention
 - Increase BBT
 - Prepares mammary gland
 - 4th month: colostrum production begins

Trimester	Estrogen	Progesterone
1st	High	High
2nd	High	High
3rd	High	Low



Human Placental Lactogen (Human Chorionic Somatomamotropin)

- With growth-promoting and lactogenic properties
- Produced by the placenta as early as 6th week of pregnancy, at peak after 20 weeks
- Promotes mammary gland
- Regulates maternal glucose, protein and fat levels
- Function and effect

(1) Insulin antagonistic hormone

- (a) Effect: the diabetogenic effect of pregnancy
- (b) Note: all placental hormones increase insulin resistance

(2) Lactogenic properties

Signs of Pregnancy

Presumptive (subjective)	Probable (objective)	Positive
- Breast changes (starting 6 weeks) - Nausea and vomiting/morning sickness (increased HCG, E, Pregnancy hormones) - Amenorrhea (can be documented) - Frequent urination (enlarged uterus) - Fatigue (increased metabolic requirement) - Quickening (Fluttering sensation) 5 months & above - Linea nigra (dark line) - Striae gravidarum (stretch marks) - Chloasma/Melasma (mask of pregnancy) - Uterine enlargement Cannot be documented	Starting 6 weeks - Chadwick's sign (bluish discoloration) - Goodell's sign (softening of the cervix) - Hegar's sign (softening of lower uterine segment) Starting 4-5 months - Ballottement (fetal rebound 4-5 months) - Braxton hicks contraction (painless, irregular false labor) - Elevated BBT (increased P) - HCG test - Abdominal enlargement	- Evidence on ultrasound of fetal outline - Abdominal = full bladder - Transvaginal = empty bladder with consent - Fetal heart audible - Fetal movement felt by examiner

Ambivalence (mixed emotions) - first feeling upon learning of pregnancy

Couvade syndrome (sympathetic pregnancy) - signs of pregnancy felt by the husband



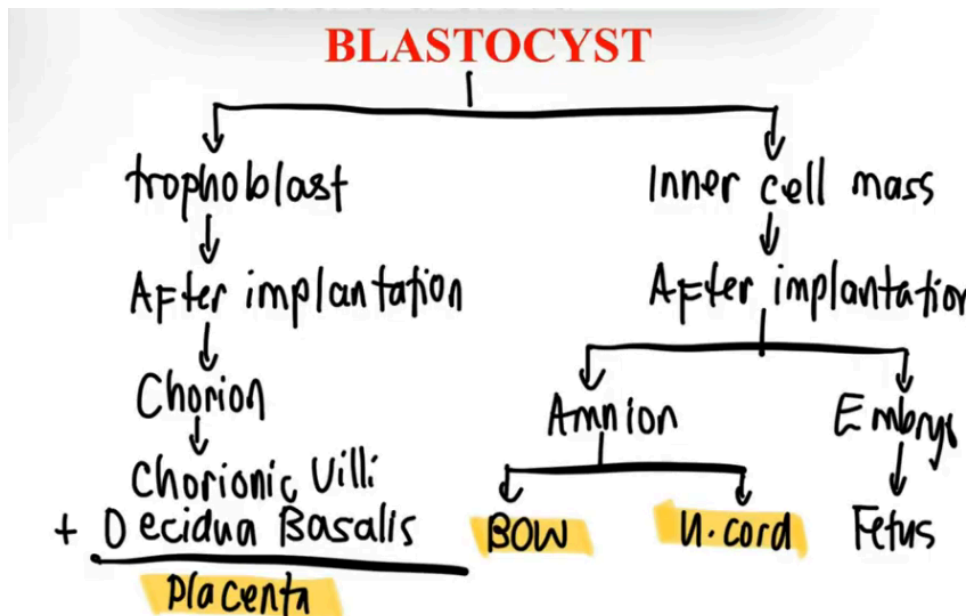
Physiologic Changes of Pregnancy

Location of Change	1st Trimester	2nd Trimester	3rd Trimester
Cardiovascular system	- Blood volume increasing - Pseudoanemia - Clotting factors increasing	Blood pressure slightly decreased	Blood pressure returns to pre-pregnancy levels
Ovary	Corpus luteum active	Corpus luteum fading	
Uterus	Increased growth	Placenta forming estrogen and progesterone	
Cervix	Softening progressive		“Ripe”
Vagina	White discharge present		Increasing
Musculoskeletal system		- Progressive cartilage softening - Lordosis increasing	
Pigmentation		Progressively increasing	
Kidney	- Maternal GFR increasing - Aldosterone increased, retaining sodium and fluid	glycosuria	
Gastrointestinal system		Slowed peristalsis	
Thyroid	Increased		

	metabolic rate		
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Fetal Development and Development

1. Zygote - fertilization - 2 weeks
2. Cleavage - 2 cell
3. Blastomeres (after 3-4 days) 4 cell, 8 cell
4. Morula (after 3-4 days) 16-50 cells
5. Blastocysts - floating in the uterus



6. Implantation
7. Embryo (2-8 weeks) - period of organogenesis (most critical stage)
8. Fetus (8 weeks to term)

End of 4th Gestational Week (1 month)

- Length: 0.75-1 cm
- Weight: 400 mg
- The spinal cord is formed and fused at the midpoint
- Lateral wings that will form the body are folded forward to fuse at the midline
- The head folds forward and becomes prominent, presenting about one-third of the entire structure
- The back is bent so that the head almost touches the tip of the tail
- The rudimentary heart appears as a prominent bulge on the anterior surface
- Arms and legs are budlike structures
- Rudimentary eyes, ears and nose are discernible

End of 8th Gestational Week (2 months)



- Length: 2.5 cm (1 in)
- Weight: 20 g
- Organogenesis is complete
- The heart with a septum and valves is beating rhythmically
- Facial features are definitely discernible
- Arms and legs have developed
- External genitalia are forming, but sex is not yet distinguishable by simple observation
- The primitive tail is regressing
- The abdomen bulges forward because the fetal intestine is growing so rapidly
- An ultrasound shows a gestational sac, diagnostic of pregnancy

End of the 12th Gestational Week (First Trimester) (3 months)

- Length: 7-8 cm
- Weight: 45 g
- Nail beds are forming on fingers and toes
- Spontaneous movements are possible, although they are usually too faint to be felt by the mother
- Some reflexes, such as the Babinski reflex are present
- Bone ossification centers begin to form
- Tooth buds are present
- Sex is distinguishable by outward appearance
- Urine secretion begins but may not yet be evident in amniotic fluid
- The heartbeat is audible through Doppler technology

End of 16th Gestational Week (4 months)

- Length: 10-17 cm
- Weight: 55-120 g
- Fetal health sounds are audible by an ordinary stethoscope
- Lanugo is well formed
- Liver and pancreas are functioning
- Fetus actively swallows amniotic fluid, demonstrating an intact but uncoordinated swallowing reflex, urine is present in amniotic fluid
- Sex can be determined by ultrasound

End of 20th Gestational Week (5 months)

- Length: 25 cm
- Weight: 223 g
- Spontaneous fetal movements can be sensed by the mother
- Antibody production is possible
- The hair forms on the head, extending to include eyebrows
- Meconium is present in the upper intestine
- Brown fat, a special fat that will aid in temperature regulation at birth, begins to be formed behind the kidneys, sternum, and posterior neck
- Vernix caseosa begins to form and cover the skin



- Passive antibody transfer from mother to fetus begins
- Definite sleeping and activity patterns are distinguishable (the fetus has developed biorhythms that will guide sleep/wake patterns throughout life)

End of 24th Gestational Week (Second Trimester) (6th months)

- Length: 28-36 cm
- Weight: 550 g
- Meconium is present as far as the rectum
- Active production of lung surfactant begins
- Eyebrows and eyelashes become well-defined
- Eyelids, previously fused since the 12th week, now open
- Pupils are capable of reacting to light
- When fetuses reach 24 weeks or 601 g, they have achieved a practical low-end age of viability (earliest age at which fetuses could survive if born at that time), if they are cared for after birth in a modern intensive care facility
- Hearing can be demonstrated by response to sudden sound

End of 28th Gestational Week (7 months)

- Length: 35-38 cm
- Weight: 1200 g
- Lung alveoli begin to mature, and surfactant can be demonstrated in amniotic fluid
- Testes begin to descend into the scrotal sac from the lower abdominal activity
- The blood vessels of the retina are formed but thin and extremely susceptible to damage from high oxygen concentrations (an important consideration when caring for preterm infants who need oxygen)

End of 32nd Gestational Week (8 months)

- Length: 38-43 cm
- Weight: 1600 g
- Subcutaneous fat begins to be deposited (the former stringy, "little old man", appearance is lost)
- Fetus responds by movement to sounds outside the mother's body
- Active moro reflex is present
- Iron stores, which provide iron for the time during which the neonate will ingest only milk after birth, are beginning to be developed
- Fingernails grow to reach the end of fingertips

End of 36th Gestational Week (9 months)

- Length: 42-48 cm
- Weight: 1800-2700 g (5-6 lbs)
- Body stores of glycogen, iron, carbohydrate, and calcium are deposited
- Additional amounts of subcutaneous fat are deposited
- Sole of the foot has only one or two crisscross creases, compared with the full crisscross pattern that will be evident at term
- Amount of lanugo begins to diminish



- Most babies turn into a vertex (head down) presentation during this month

End of 40th Gestational Week (Third Trimester) (10 months)

- Length: 48-52 cm (crown to rump, 35-37 cm)
- Weight: 3000 g (7-7.5 lbs)
- Fetus kicks actively, hard enough to cause the mother considerable discomfort
- Fetal hemoglobin begins its conversion to adult hemoglobin. The conversion is so rapid that, at birth, about 20% of hemoglobin will be adult in character
- Vernix caseosa is fully formed
- Fingernails extend over the fingertips
- Creases on the soles of the feet cover at least two thirds of the surface

Primary Germ Layers

- **Ectoderm** - epidermis, sweat glands, sebaceous glands, hair follicles, sense organs, mouth, oral glands, tooth enamel, pituitary gland, mammary gland, CNS
- **Mesoderm** - walls of digestive tract, dermis, kidneys, ureter, reproductive organs, musculoskeletal system, cardiovascular system, spleen, pleura, blood, bone marrow
- **Endoderm** - respiratory system, lungs, pharynx, tongue, tonsils, thyroid, parathyroid, thymus, liver, pancreas, urethra, urinary bladder

Meconium Stool

- 1st day - dark green, thick, tarry
- 2nd-3rd day - green & loose
- Absence of meconium stool 1st 24-48 hours
 - Imperforate anus - no anal opening
 - Hirschsprung's diseases - no ganglion cells, no peristalsis
 - Cystic fibrosis - no pancreatic enzyme -> meconium ileus (hardened)
- Meconium staining - mostly post-term (36 weeks) -> placental functions begin to diminish -> placental insufficiency -> chronic hypoxia -> meconium is released -> meconium staining -> fetal distress (except in breech presentation)

Amniotic Fluid

- Major sources of amniotic fluid after 20 weeks: fetal kidneys
- Normal volume: 500-1000 ml/800-1200 ml
- pH: 7-7.25
- **Normal color:** clear
- **If BOW ruptures**
 - Check for FHR



- Ask the time
- **Functions**
 - Protection from trauma (major)
 - Maintains temperature
 - Prevent cord compression
 - In cord compression: reposition
 - Sim's
 - Trendelenburg
 - Knee-chest
 - Hands and knees
 - If cord is exposed: cover with wet gauze
 - Cord drying - atrophy of umbilical vessels
 - Helps in delivery
 - For musculoskeletal development
- **Oligohydramnios** <500 cc after 20 weeks, risk for pulmonary hypoplasia, fetal injury
 - Indication: fetal kidney malfunctions, renal agenesis
- **Polyhydramnios** >1500 cc, risk for preterm delivery
 - Indication: fetal GIT obstruction (esophageal atresia, duodenal atresia), twin pregnancy, diabetes (fetal polyuria)
- **Abnormal colors of amniotic fluid**
 - **Green:** meconium staining
 - **Golden yellow:** due to destruction of fetal RBCs, related to Rh incompatibility

Umbilical Cord/Funis

- Color: grayish-white, no nerve supply
- Average length: 50-55 cm long
- Less than 35 cm = short: at risk for abruptio placenta
- More than 70 cm = long: at risk for cord prolapse
- Arteries = 2 u oxygenated, waste products
- Vein = 1 oxygenated blood
- Wharton's jelly to protect the vessels from rupture

Fetal Circulation

Main shunts

- **Ductus venosus** - connects umbilical vein to inferior vena cava
- **Foramen ovale** - right atrium to left atrium
- **Ductus arteriosus** - connects pulmonary artery to ovarian artery



Oxygenated blood from placenta -> umbilical vein -> ductus venosus -> inferior vena cava -> right atrium -> foramen ovale -> left atrium -> left ventricle -> ascending aorta -> different organs of the fetus

Unxygenated blood & waste -> right atrium -> right ventricle -> pulmonary artery -> ductus arteriosus -> descending aorta -> 2 umbilical arteries -> placenta -> uterus -> maternal circulation

Placenta

- Attaches to the uterine wall and allows metabolic exchange between the fetus and the mother
- Expands on the first 5 months (placental migration)
 - No diagnosis of placenta previa earlier than 5 months
- Round, 22 cm in diameter, 2-2.5 cm thick
- 500-600g at term
- Blood supply at 10 weeks: 50-60 ml per min
- Blood supply at Term: 500-600 ml per min
- Cotyledons: 30
- **Functions**
 - Exchange of O₂ and CO₂
 - Transport nutrients
 - Waste excretion
 - Production of hormones
 - Serves as protective barrier to some organisms
- **Infections that can cross the placenta**
 - Toxoplasmosis
 - Other infections
 - Rubella
 - Cytomegalovirus
 - Herpes

Antepartal Period

- Days: 257-280
- Weeks: 37-42
- Calendar months: 9
- Lunar months: 10
- Age of reproductivity: 15-44 y/o
- High risk of pregnancy: <18 >35 y/o
- Ideal age: 20-30 y/o

Prenatal Visits (per DOH protocol) - at least 8 visits



- First Trimester: 1
- Second Trimester: 2
- Third Trimester: 5
- Old/Default
 - 1st 7 months: monthly
 - 8th to 9th months: twice a month
 - At term: weekly
 - Post term: twice a week

Prenatal Diagnostic Tests

1. **Chorionic Villi Sampling:** obtaining a sample of chorionic villi and analyzed for abnormalities
 - a. Routes:
 - i. **Transabdominal:** using spinal needle
 - ii. **Transcervical:** using 20 mm catheter
 - b. Done ideally at 10-13 weeks AOG earlier than 10 weeks = fetal limb reduction
 - c. Ultrasound is performed prior to the procedure
 - d. Position:
 - i. **Transabdominal:** supine
 - ii. **Transcervical:** lithotomy
 - e. Most common complication: post procedure bleeding
 - f. Additional risks:
 - i. ROM
 - ii. Leakage of AF
 - iii. Intrauterine infection
 - iv. RH immunization
 - v. Fetal loss
 - g. Check FHR (twice) in 30 mins after the procedure
2. **Amniocentesis:** transabdominal needle aspiration of amniotic fluid for analysis
 - a. 2nd trimester is preferred for genetic screening
 - b. 15-20 AOG for AFP screening
 - c. 35-36 weeks and above for lung maturity studies
 - d. Early amniocentesis between 11-14 weeks results in:
 - i. Increase pregnancy loss
 - ii. Leakage of AF
 - iii. Talipes equinovarus
 - e. Ultrasound is done prior to amniocentesis
 - f. Position:
 - i. Left lateral tilt
 - ii. Semi-fowlers



- g. Risk:
 - i. Abortion
 - ii. Trauma to fetus and placenta
 - iii. Bleeding
 - iv. Preterm labor
 - v. Maternal infections
 - vi. RH sensitization
 - h. Check FHR after the procedure
 - i. No lifting anything heavy for 2 days after the procedure
 - j. Report fever, severe cramps, or vaginal bleeding
3. **Ultrasound:** uses high frequency soundwaves
- a. Diagnose pregnancy at 6 weeks
 - b. Visualize fetus, placenta, umbilical cord, BOW
 - c. Gestational age
 - d. Fetal growth and abnormalities
 - e. Number of fetuses
 - f. To confirm cardiac activity
 - g. To rule out ectopic pregnancy, Hmole, abortion, fetal demise
 - h. To determine presentation
 - i. Types:
 - i. Transabdominal
 - ii. Transvaginal
4. **Alphafetoprotein test (AFP):** AFP is a substance produced by fetal liver that is present in the amniotic fluid and maternal serum starting 15 weeks and above
- a. Increased AFP = NTD
 - b. Decreased AFP = Down syndrome
 - c. False high result:
 - i. AFP gone after 20 weeks
 - ii. Bloody tap
5. **Non-Stress Test** (28 weeks and above): measures the response of FHR to fetal movements
- a. Position: reclining
 - b. Goal: 2 FM in 20 minutes
 - c. Interpretations:
 - i. **Reactive:** baseline FHR lasting 15 secs in each fetal movement
 - ii. **Non-reactive**
6. **Contraction Stress Test:** to measure the response of FHR to contractions
- a. Position: reclining
 - b. Other terms: OCT, BSST
 - c. Goal: 3 contractions in 10 mins
 - d. Interpretations:



- i. **Negative:** no late decelerations
 - ii. **Positive:** if 50% or more of the contractions cause a late deceleration
- e. Indications:
 - i. IUGR
 - ii. Diabetes during pregnancy
 - iii. Post-term
 - iv. Non-reactive NST
 - v. Abnormal BPP
- f. Contraindications:
 - i. Previous classical CS
 - ii. History of abruptio placenta, PROM, preterm labor
 - iii. Current placenta previa
 - iv. Multiple gestation
- 7. **Biophysical Profile (BPP):** Ultrasound 20-30 minutes with a combination of 5 parameters to check the fetal well-being
 - a. **Fetal breathing movement:** at least 1 episode of 30 secs FBM
 - b. **Gross fetal movement:** 3 FMS
 - c. **Fetal tone:** at least 1 flexion and 1 extension
 - d. **Amniotic fluid volume:** 1 or more pockets
 - e. **NST:** reactive

Partograph

- Tool to help in management of labor
- What to monitor:
 - Progress of labor (dilatation, contractions)
 - Maternal well-being (VS, urine)
 - Fetal well-being (FHR, color of amniotic fluid)
- Conditions not needing partograph
 - Antepartum hemorrhage
 - Severe pre-eclampsia and eclampsia
 - Fetal distress
 - Previous CS
 - Multiple pregnancy
 - Malpresentation
 - Very premature baby
 - Obvious obstructed labor
- Plotting the progress labor:
 - Plot only the cervical dilatation using the symbol x
 - Start when the woman is in active labor

- Start plotting on the alert line
- Perform IE every 4 hours
- Plotting that passes the alert line, more so if it reaches action line indicates abnormal progress of labor
- If plotting passes the alert line:
 - Reassess woman, monitor intensively
 - Empty the bladder
 - Ensure adequate hydration and omit food
 - Encourage upright position and walking

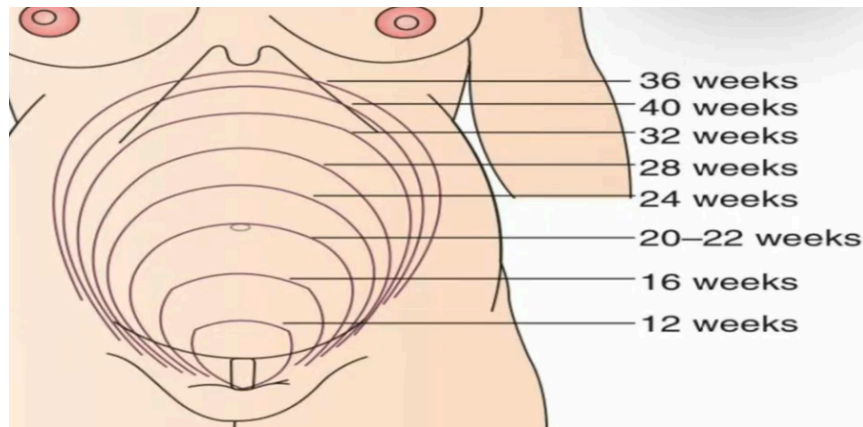
OB-related Calculations

1. Naegele's Rule (Expected Date of Confinement/Birth)

- LMP (first day of last menstrual period)
 - If LMP is unknown, use Date of Quickening
 - Primi: date of quickening + 4 months = EDB
 - Multi: date of quickening + 5 months = EDB
- January to March: +9 +7
- April to December: -3 +7 +1

2. Bartholomew's Rule (Estimating the AOG)

- Starts on 3rd month
- Rule of 4
- Void
- Position = dorsal recumbent (to relax the abdomen)



3. Fundic height in centimeters

- 20-36 cm = 20-36 weeks

Affecting Factors

1. Obesity
2. Diabetes



3. Volume of amniotic fluid
4. Multiple pregnancy

Non-Pregnant Weight	Weight Gain During Pregnancy
Average (BMI 18.5-24.9)	25-35 lbs
Underweight (BMI < 18.5)	28-40 lbs
Overweight (BMI 25-29.9)	15-25 lbs
Obese (BMI > 29.9)	<15 lbs

Distribution of weight gain for singleton pregnancy

- First Trimester: 1 lb/month
- Second Trimester: 1 lb/week
- Third Trimester: 1 lb/week

The average maternal weight gain is distributed as follows:

- 11 lbs - fetus, placenta, amniotic fluid
- 2 lbs - uterus
- 4 lbs - increased blood volume
- 3 lbs - breast tissues
- 5-10 lbs - maternal stores

	Twins	Triplets	Quadruplets
Weight gain 1st 5 months (per week)	1 lbs	1 ½ lbs	2 lbs
Weight gain 6th-10th month (per week)	2 lbs	2 ½ lbs	3 lbs
Total weight gain	40-50 lbs	50-60 lbs	65-80 lbs
Average AOG	36 weeks	32 weeks	30 weeks

Nutrients Needed During Pregnancy

Calories per day

- Non-pregnant: 2200/day
- Pregnant
 - 1st: 2200/day
 - 2nd: +300/day = 2500/day
 - 3rd: +300/day = 2500/day
- During lactation: +500 = 2700/day



Protein	60g/day or 1g/kg/day
Carbohydrates	175g/day
Iron	Non-pregnant: 15 mg/day Pregnant: 30 mg/day Twin pregnancy: 60-100 mg/day Vitamin C: 80-100 mg/day to absorb iron Caffeinated beverages should be avoided
Fats	Omega 6 and Omega 3, a daily quota of 90g/day
Folate/B9 Non-pregnant: 50 ug/day	400 mcg/day with no history of NTD 4 mg/day with history of NTD
Zinc	11-12 mg/day
Sodium	1.5-2.3g/day/3g/day (moderate)
Calcium Non-pregnant: 800 mg/day	Pregnant: 1000-12000 mg/day Greater than 25000mg/day: kidney stones
Phosphorus	1200 mg/day
Vitamin A	2500 IU/day
Vitamin D	400-800 IU/day with sun exposure
Fluids	3L/day
Vitamin B12	4 ug/day
Iodine Non-pregnant: 150 mcg/day	Pregnant: 175 mcg/day Lactating: 200 mcg/day

Intrapartal Period

Theories of Labor

- (1) Uterine stretch theory
- (2) Prostaglandin theory
- (3) Progesterone deprivation theory
- (4) Aging placenta starting on 36 weeks
- (5) Oxytocin theory

Preliminary Signs of Labor

- Lightening - descent of fetal presenting part into the pelvis



- Increase in level of activity - related to increase in epinephrine release initiated by a decrease in progesterone
- Slight loss of weight - as progesterone falls, body fluid is more easily excreted from the body
- Braxton hicks contractions - causes true discomfort
- Ripening of cervix - becomes “buttersoft”, internal announcement that labor is close at hand

Duration of Labor

- Primipara: 14-20 hours
- Multipara: 8-14 hours

Factors in Prolonged Labor

- Malpresentation
- Inadequate pelvis
- weak contractions
- Delayed dilation
- Number of fetuses
- Macrosomia

Rate of Dilatation

- Primi: 1cm/hr
- Multi: 1.5 to 2cm/hr

NOTE: Do IE every 4 hours

False Contractions	True Contractions
• Begin and remain irregular • Felt first abdominally and remain confined to the abdomen and groin • Often disappears with ambulation or sleep • Do not increase in duration, frequency or intensity • Do not achieve cervical dilatation	• Begin irregularly but become regular and predictable (3 contractions in 10 minutes) • Felt first in lower back and sweep around to the abdomen in a wave • Continue no matter what the woman's level of activity • Increase in duration, frequency and intensity • Achieve cervical dilatation

Signs of True Labor

- Uterine contractions - surest sign of labor
- Bloody show - mucus plug that filled the cervical canal during pregnancy is expelled with blood
- Rupture of membranes - sudden gush or a scanty slow seeping of clear fluid from the vagina

Tocolytic Agents - use to stop/halt labor for short time (up to 48 hours) if you begin labor too early

- Contraindications:

- Dilatation is more than 3 cm
- Cardiac disease
- Fetal distress
- Abruptio placenta



- PROM
- Chorioamnionitis
- **Types:**
 - Indomethacin
 - Nifedipine (Procardia)
 - Yutopar (Ritodrine)
 - Magnesium sulfate
 - Terbutaline (Breathing)

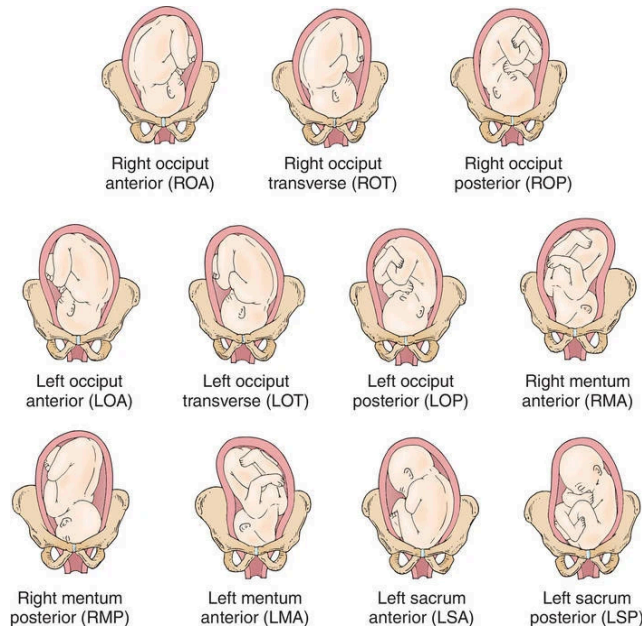
Components of Labor

Passage - refers to the route a fetus must travel from the uterus to the cervix and vagina to the external perineum

Passenger - the fetus

- **Molding** - change in the shape of fetal skull produced by the force of uterine contractions pressing the vertex of the head against the not-yet-dilated cervix
- **Attitude** - describes the degree of flexion a fetus assumes during labor or the relation of the fetal parts to each other
- **Engagement** - refers to the settling of the presenting part of a fetus far enough into the pelvis to be at the level of the ischial spines, a midpoint of the pelvis
- **Fetal Station** - refers to the relationship of the presenting part of the fetus to the level of ischial spines
- **Fetal lie** - the relationship between the long (cephalocaudal) axis of the fetal body and the long (cephalocaudal) axis of a woman's body
- **Fetal Presentation** - denotes the body part that will first contact the cervix or be born first
 - **Cephalic Presentation** - the fetal head is the body part that will first contact the cervix
 - **Breech Presentation** - either the buttocks or the feet are the first body parts that will contact the cervix
 - **Complete breech presentation** - fetus has thighs tightly flexed on the abdomen, both the buttocks and the tightly flexed feet present to the cervix
 - **Frank breech presentation** - the hips are flexed but the knees are extended to the rest on the chest
 - **Footling** - neither the thighs nor lower legs are flexed
 - **Shoulder Presentation** - presenting part is usually the shoulders, iliac crest, a hand or elbow
- **Fetal Position** - relationship of the presenting part to a specific quadrant of the woman's pelvis
 - **Vertex Presentation (Occiput)**
 - Left occipitoanterior (LOA)
 - Left occipitoposterior (LOP)
 - Left occipitotransverse (LOT)

- Right occipitoanterior (ROA)
- Right occipitoposterior (ROP)
- Right occipitotransverse (ROT)
- **Breech Presentation (Sacrum)**
 - Left sacroanterior (LSaA)
 - Left sacroposterior (LSaP)
 - Left sacrotransverse (LSaT)
 - Right sacroanterior (RSaA)
 - Right sacroposterior (RSaP)
 - Right sacrotransverse (RSaT)
- **Face Presentation (mentum)**
 - Left mentoanterior (LMA)
 - Left mentoposterior (LMP)
 - Left mentotransverse (LMT)
 - Right mentoanterior (RMA)
 - Right mentoposterior (RMP)
 - Right mentotransverse (RMT)
- **Shoulder Presentation (acromion process)**
 - Left scapuloanterior (LAA)
 - Left scapuloposterior (LAP)
 - Right scapuloanterior (RAA)
 - Right scapuloposterior (RAP)



Management for Fetal Malposition: to help rotate the head from posterior to anterior

- Hands and knees position



- All four position
- Squatting
- Knee-chest

Leopold's Maneuver - series of four abdominal palpations performed by HP to assess fetal position, presentation, and engagement within the uterus during pregnancy

- **Fundal grip** - fetal lie
- **Umbilical grip** - fetal position
- **Pawlik's grip** - fetal presentation
- **Pelvic grip** - fetal attitude

Power - force supplied by the fundus of the uterus, implemented by uterine contractions, a natural process that causes cervical dilatation and then expulsion of the fetus from the uterus

- **Uterine contractions** - begin at a pacemaker point located in the uterine myometrium near one of the uterotubal junctions
- **Cervical changes**
 - **Effacement** - shortening and thinning of the cervical canal
 - **Dilatation** - refers to the enlargement or widening of the cervical canal from an opening a few millimeters wide to one large enough to permit passage of the fetus

Psyche - a woman's psychological outlook refers to the psychological state or feelings that a woman brings into labor

Stages of Labor

First Stage of Labor

- **Latent Phase/Preparatory** - begins at the onset of regularly perceived uterine contractions and ends when the rapid cervical dilatation begins
- **Active Phase** - cervical dilatation occurs more rapidly, increasing from 4-7 cm
- **Transition Phase** - contractions reach their peak of intensity, occurring every 2 to 3 minutes with the duration of 60-90 seconds and causing maximum cervical dilatation of 8-10 cm

	Latent (6 hours for Primi; 4.5 hours for Multi)	Active (3 hours for Primi; 2 hours for Multi)	Transition
Dilatation	0-3 cm	4-7 cm	8-10 cm
Frequency	Every 5-10 minutes	Every 3-5 minutes	Every 2-3 minutes
Duration	20-40 secs	40-60 secs	60-90 secs
Intensity	Mild	Moderate	Strong



Behavior	Excited	Fear of losing control	Resistance to touch
Considerations	Give instructions	Give analgesia	Reassurance
Monitor Fetal Heart Rate Every 15-30 minutes			

Unnecessary Interventions during Labor (WHO protocol)

- fundal pushing
- Perineal shaving
- IV infusion
- NPO
- Strict supine position
- Artificial ROM
- Enemas
- Episiotomy

Second Stage of Labor - period from full dilatation and cervical effacement to birth of the infant (FETAL EXPULSION STAGE - 1 HOUR)

Temperature of delivery room	25-28C
Transfer to delivery room	Primi: 10 cm Multi: 7-8 cm
Wear	Double sterile gloves
Most common birthing position	lithotomy
Surest sign that the baby is about to be delivered	Bulging of the perineum

When to push: during contractions, fully dilated cervix

When not to push: not fully dilated cervix, intervals, delivery of the head

Mechanisms (Cardinal Movements) of Labor

- **Descent** - downward movement of the biparietal diameter of the fetal head to within the pelvic inlet
- **Flexion** - the head bends forward onto the chest making the smallest anteroposterior diameter present to the birth canal
- **Internal rotation** - head flexes as it touches the pelvic floor and the occiput rotates to bring the head into the best relationship to the outlet of pelvis
- **Extension** - the back of the neck stops beneath the pubic arch and acts as a pivot for the rest of the head
- **External rotation** - the head rotates back to the diagonal or transverse position of the early part of the labor
- **Expulsion** - end of pelvic division of labor



Third Stage of Labor/Placental Stage (3-10 minutes) - begins with the birth of the infant and ends with the delivery of placenta

- **Signs of placental separation**
 - Lengthening of the umbilical cord
 - Sudden gush of vaginal blood
 - Change in the shape of uterus
 - Firm contraction of the uterus
 - Appearance of the placenta at the vaginal opening
- **Placental presentation**
 - **Schultz** - appearing shiny and glistening from the fetal membrane
 - **Duncan** - raw, red and irregular, with ridges or cotyledons that separate blood collection spaces showing (maternal surface)
- **Active Management (WHO-DOH)** - decreases the risk of postpartum hemorrhage
 - Administration of uterotonic - 10 units oxytocin IM, administered to the mother within 1 minute of baby's birth
 - Controlled cord traction with counter traction - wait a strong contraction (2-5 minutes), push the fundus upward with one hand while applying continuous, steady, non-directional traction on the cord with the other hand
 - uterine massage - after placental delivery

NOTE: do not wait for signs of placental separation

Fourth Stage of Labor (1-2 hours)) - critical stage/recovery stage

- VS checking for the first 2 hours every 15 minutes

Temperature	36.2-38C for the first 24 hours due to dehydration
BP	Should remain consistent with baseline BP during pregnancy
RR	16-22 cpm
PR	May be bradycardia (50-70 bpm) for one week

Report:

- Hypo - tachy- tachy = shock
- Increased temperature after 24 hours = infection

Normal characteristics of fundus after delivery: Firm, contracting, midline, below the umbilicus

Fundus is soft, relaxed and boggy	Uterine Atony
Deviated to the side	Full bladder
Palpable above the umbilicus	Over distention of the uterus
Firm but there is steady trickle of blood	Laceration

Vagina, fourchette, skin of perineum	1st degree
Muscles of perineum	2nd degree
Anal sphincter	3rd degree
Rectal mucosa	4th degree
Location of the fundus after delivery	
After delivery of the placenta	Midway between the umbilicus and the symphysis pubis
After 1 to 2 hours	At the level of umbilicus
After 24 hours	One fingerbreadth below the umbilicus
10th day	Non-palpable

Fetal Distress

Causes (hypoxia)

- Prolonged cord compression
- Antepartal and Intrapartal hemorrhage
- Pre-eclampsia and eclampsia
- Induction or augmentation of labor
- Heart conditions complicating pregnancy and labor
- Hypotensive effects of anesthesia

Signs

- Hyperactive fetus
- Fetal tachycardia and bradycardia
- Severe variable and persistent late deceleration
- Meconium staining, except for breech presentation

Complications

- Mental retardation
- Seizures
- Cerebral palsy
- IUFD

Types	Cause	Image	Management
Early (Normal)	Head Compression	Mirror	Document the finding
Late (Abnormal)	Uteroplacental	Uniform	Left side



	Insufficiency		
Variable (Abnormal)	Cord Compression	Unpredictable	Modified trendelenburg

Postpartal Period

Involution - the process whereby the reproductive organs return to their non-pregnant state (6 weeks)

4th Stage of Labor	1000-1100g
After 2 weeks	500g
After 3-4 weeks	300g
After 5-6 weeks	50-60g

Subinvolution - incomplete return of the uterus to non-pregnant state after 5-6 weeks

- Most common cause: uterine infection

Lochia - uterine flow, consisting of blood, fragments of decidua, WBCs, mucus and some bacteria

- **Lochia rubra** - 1-3 days after birth, red in color
- **Lochia serosa** - 3-10 days pink or 7-9 days brownish
- **Lochia alba** - 10-21 days (may last until 6 weeks), amount of flow decreases and becomes colorless or white
 - Odor: fleshy, musky, non-offensive, non-foul, sour
 - Amount: moderate, light, scanty
 - Should not exceed 4-8 pads per day
- **Report**
 - fishy odor
 - More than 8 pads
 - Clots larger than 1cm

Notes:

- A new endometrium is formed 3 weeks after delivery
- The placental site is healed by 6 weeks
- Postpartum hemorrhage >500 cc blood loss (NSD); >1000 cc (CS)
 - Early (1st 24 hours): uterine atony
 - Late (After 24 hours): retained secundines
- Sex can be resumed when lochia flow has stopped and episiorrhaphy is healed
- Cervix after delivery - transversed slit
- Vagina returns to pre-pregnant state by 6 weeks
- For non-lactating mother, ovulation resumes after 6 weeks
- For lactating mother, ovulation resumes after 6 months because prolactin inhibits ovulation
- Lactation amenorrhea method is only effective on the first 6 months



- All contraception with estrogen are not safe for lactating mothers
- Progesterone only pill is safe for lactating mothers
- Women who have rubella titer of less than 1:10 are usually given rubella vaccine in the postpartum period and instruct the mother that pregnancy is not allowed for 3 months
- All Rh negative women who meet specific criteria should receive RhoGAM
- Routine antibiotic prophylaxis is not recommended for women with uncomplicated vaginal birth and for those women with episiotomy and 1st and 2nd degree lacerations
- Hunger will start to increase 1-2 hours after birth
- First bowel movement normally occurs by 2-3 days postpartum
- Immediate weight loss after delivery 12-15 lbs
- Weight loss in the 1st week 5 lbs
- Loss in the next 6 weeks 10 lbs
- Most women return to their pre-pregnant weight by 6 months
- The best schedule for breastfeeding per demand
- 3E's of breastfeeding
 - Early (ASAP)
 - Exclusive (1st 6 months)
 - Extended (up to 2 years or beyond)

Phases of the Puerperium

- **Taking-in** - a time of reflection, 2-3 day period, mother is passive
- **Taking-hold** - initiate action, expressed interest in caring for the child
- **Letting-go** - mother redefines her new role, requires some grief work and readjustment of relationships

Taking-In (first 3 days)	Dependent phase (passive)	Personal care
Taking-Hold (4th-6th day)	Independent phase (active)	Newborn care
Letting Go (7th day and above)	Interdependent phase (collaborative)	Parenting and family planning

1st Trimester Complications

Abortion

- Medical term for any interruption of a pregnancy before a fetus is viable (20 weeks)
 - **Spontaneous** - miscarriage between 16-24 weeks
 - **Threatened** - begins as vaginal bleeding, initially only scant and usually bright red, no cervical dilatation
 - **Imminent/Inevitable** - (+) uterine contractions and cervical dilation, no FHT



- **Complete** - entire products of conception are expelled without assistance
- **Incomplete** - part of the conceptus is expelled, but the membrane or placenta is retained in the uterus
- **Missed** - fetus dies in utero but is not expelled
- **Septic** - due to infection
- **Habitual** - recurrent, with dilation
- **Induced**
 - **Therapeutic**
 - **Intentional/elective**

Methods of Abortion

- Medical termination
- Avoids an anesthesia
- More effective at earlier gestation (49 days or earlier)

Surgical Termination

- **Suction evacuation (menstrual extraction)**
 - Performed 5-7 weeks after LMP
 - A rigid or flexible suction is used
- **Dilatation and Curettage**
 - If the AOG is less than 13 weeks
 - Needs anesthetic block and cervical dilators
- **Dilatation and Vacuum Extraction**
 - Those between 12 and 16 weeks
 - Dilatation of the cervix is begun
 - Negative pressure of suction pump

Ectopic/Tubal Pregnancy

- implantation occurs outside the uterine cavity
- most common site - ampulla of the fallopian tube

Type	Duration
Ampullar	6-12 weeks
Isthmic	6-8 weeks
Fimbriated	6-12 weeks
Interstitial	12-14 weeks

- **Risk factors**
 - Previous PID (pelvic inflammatory disease)
 - Previous ectopic pregnancy, previous BTL, previous tubal or pelvic surgeries



- Maternal smoking at the time of conception
- **Signs and symptoms**
 - Classic: lower and unilateral abdominal pain, amenorrhea, abdominal vaginal bleeding on 6-8th week (decreased progesterone levels)
 - Low hCG, low estrogen, low progesterone
- **After tubal rupture**
 - Deep, sharp, stabbing, unilateral, acute, lower quadrant abdominal pain
 - **Cullen's sign** - bluish navel/hematoperitoneum indicative of internal bleeding
 - **Kehr's sign** - referred pain (shoulders) due to stimulation to phrenic nerve

Management:

- **Unruptured ectopic pregnancy:** oral administration of methotrexate (anti-folic acid) and followed by Leucovorin (to decrease toxic effects of methotrexate)
- **Prerequisites:**
 - Ectopic sac is smaller than 3.5 cm in diameter
 - Serum hCG levels <5000mIU/ml
 - Normal kidney and liver functions
 - No evidence of thrombocytopenia
- Hysterosalpingography or ultrasound is performed after therapy - to check tubal patency

Second Trimester Complications

Hydatidiform Mole/Gestational Trophoblastic Disease

- Abnormal proliferation and then degeneration of the trophoblastic villi developing into avascular transparent grapelike vesicles
- **Signs and symptoms**
 - Big uterus
 - Snowstorm pattern in the ultrasound
 - No FHT
 - Excessive hCG (1-2 million IU)
 - Hyperemesis gravidarum
 - Hypertension before 20 weeks
 - Ovarian enlargement
 - Dark brown vaginal discharge
 - Passage of clear fluid-filled vesicles

Types of H-Mole

- **Partial**
 - Localized areas of chorionic villi
 - 69, (XXY, XXX, XYY) triploid karyotype



- 1 set of chromosomes of maternal in origin and 2 sets of paternal in origin (Dispermy)
- **Complete**
 - Generalized areas of the chorionic villi
 - A sperm fertilizes an ovum with no genetic material
 - Androgenesis
 - 46 diploids karyotype
 - History of clomid therapy
 - 20% risk for developing cancer

Diagnostic Test: transvaginal ultrasound and serum hCG

Management

- Remove the moles - suction evacuation
- Oxytocin after evacuation
- follow up visit - check hCG levels monthly for 6-12 months
- Regular chest X-ray and regular pelvic examination
- DOC: methotrexate
 - If metastasis occurs, dactinomycin is added to the regimen
- Other considerations:
 - No pregnancy for at least 1 year
 - Recommend to use contraception
 - Abdominal hysterectomy (optional)

Premature Cervical Dilatation/Incompetent Cervix

- refers to a cervix that dilates prematurely and therefore cannot hold a fetus until term
- Usually painless, occurring before 20 weeks
- Due to
 - Congenital incompetence: short cervix (<2cm)
 - Acquired incompetence: trauma, laceration, infection
 - First symptom - bloody show before 20 weeks
- **Best diagnostic tool to detect:** transvaginal ultrasound
- **Management:** Cervical Cerclage (12-14 weeks) under regional anesthesia
 - **McDonald** - simple procedure (temporary)
 - **Shirodkar** - complicated operation
 - Success rate 85-90% either technique
 - **NOTE:** following the Shirodkar operation, the sutures may be left in place and CS delivery performed
 - **After Cerclage:**
 - Position: modified/slight trendelenburg for few days to reduce pressure
 - Observe ROM, contraction, VS, infections
 - No sex, heavy lifting



- No prolonged sitting and standing

Third Trimester Complications

Placenta Previa

- A condition of pregnancy in which the placenta is implanted abnormally in the uterus
- Cannot be diagnosed on the first 5 months of pregnancy due to placental migration
 - **Low-lying placenta** - implantation in the lower rather than in the upper portion of the uterus
 - **Marginal placenta** - the placenta edge approaches that of the cervical os
 - **Partial placenta** - implantation occludes a portion of the cervical os
 - **Total placenta** - implantation totally obstructs the cervical os

Management

- Sufficient migration of placenta away from the cervix
- prevent and control bleeding
- Bed rest
- Bethamethasone may be administered
- Double set up

Abruptio Placenta

- **Premature separation of placenta**
 - **Partial Abruptio:** The placenta detaches partially from the uterine wall.
 - **Complete Abruptio:** The placenta detaches completely from the uterine wall.
 - **Revealed Abruptio:** Bleeding is visible, meaning blood escapes through the cervix.
 - **Concealed Abruptio:** Bleeding is hidden behind the placenta, and blood does not escape through the cervix.

	Grade 1	Grade 2	Grade 3
Placental Separation	Less than 1/6	half of the placenta	More than half
Blood Loss	<500 ml	1000-1500 ml	>1500 ml
Fetal Distress	NONE	POSSIBLE	YES
Board like Abdomen	NONE	YES	YES
Painful Contractions	NONE	YES	YES

Complications of Abruptio Placenta

- Prematurity



- Hemorrhage
- Couvelaire uterus/uteroplacental insufficiency
- infection

Placenta Previa		Abruptio Placenta
abnormal	Implantation	normal
1. Complete 2. Partial 3. Marginal 4. Low lying	Types	1. Covert 2. Overt
Sudden external bleeding (around 30 weeks)	Bleeding	Internal/external
Bright red	Color	Dark red or bright red
Soft and non-tender	Uterus	Board like and tender
Painless	Pain	Painful
Multiparity	Most Common Cause	Hypertension
Smoking Previous CS/abortion Age >35 y/o multiple pregnancy	Other Causes	Smoking Short cord Cocaine use Age >35 y/o Accidents Multiple Pregnancy

Other Pregnancy Complications

Pregnancy-Induced Hypertension/Toxemia

- Condition which vasospasm occurs during pregnancy in both small and large arteries, occurs after 20 weeks

Hypertension Type	Symptoms
Gestational hypertension	Blood pressure 140/90 or systolic pressure elevated 30 mm Hg or diastolic pressure elevated 15 mm Hg above prepregnancy level; no proteinuria or edema; blood pressure returns to normal after birth
Mild pre-eclampsia	Blood pressure 140/90 or systolic pressure elevated 30 mm Hg or



	diastolic pressure elevated 15 mm Hg above prepregnancy level; proteinuria of 1–2 on a random sample; weight gain over 2 lb per wk in second trimester and 1 lb per wk in third trimester; mild edema in upper extremities or face
Severe pre-eclampsia	Blood pressure of 160/110; proteinuria 3–4 on a random sample and 5 g on a 24-hour sample; oliguria (500 mL or less in 24 hours or altered renal function tests; elevated serum creatinine more than 1.2 mg/dL); cerebral or visual disturbances (headache, blurred vision); pulmonary or cardiac involvement; extensive peripheral edema; hepatic dysfunction; thrombocytopenia; epigastric pain
Eclampsia	Seizure or coma accompanied by signs and symptoms of pre-eclampsia

Major Body Parts affected:

- Brain (CNS irritation and hyperreflexia)
- Uterus (Uteroplacental insufficiency)
- Liver and Pancreas (ischemia)
- Kidneys (Oliguria and proteinuria)

	Mild Pre-eclampsia	Severe Pre-eclampsia
Weight gain (first sign)	2 lbs per week	3 lbs per week
Urine output	>500 cc in 24 hours	<500 cc in 24 hours
Headache	Occasional	persistent
Visual disturbances	none	present
Epigastric pain (late sign)	none	Aura for convulsion

Complications:

- **HELLP Syndrome**
 - Hemolysis
 - Elevated
 - Liver Enzymes
 - Low
 - Platelet Count

Management

- **Criteria for Home management**
 - BP <150/100
 - Proteinuria <1g/24 hours
 - Normal platelet count
 - No fetal growth restriction



- **Hospital care: bed rest**
 - Position: left lateral recumbent (increase venous return)
 - Diet: increased protein (70-80 g/day), moderate sodium (3g/day)
 - Room: non-stimulating (dim & quiet)
- **Anti-hypertensive therapy**
 - Hydralazine (Apresoline), Aldomet (Methyldopa), Normodyne (Labetadol), Nifedipine (Procardia)
 - Given to severe pre-eclampsia
 - Therapeutic goal: to decrease BP slightly lower than 140/90 mmHg
- **Anticonvulsant: magnesium sulfate**
 - CNS depressant, muscle relaxant
 - If RR <12/<14 do not give
 - IV piggyback/IM gluteal muscles, Z-track (do not massage)
 - Obtain serum magnesium levels every 6-8 hours
 - Therapeutic level: 4-7 mEq/L or 5.8 mEq/L
 - Ankle Clonus - monitor every hour
 - Patellar reflex - monitor every hour (N) +2
 - Magnesium Toxicity Parameters
 - BP: hypotension
 - Urine output: <30 cc/hr
 - RR: <12
 - Patellar reflex: absent
 - Somnolence: sleepiness (magnesium has sedative effects)
 - Antidote: calcium Gluconate
 - 9-12 mEq/L - patellar reflex disappear
 - 15-20 mEq/L - RR depression
 - >20 mEq/L - cardiac arrest
- **Seizure precaution**
 - During seizure: safety
 - After seizure: airway
 - Side rails should be up
 - Position on her side
 - MgSO4 IV STAT
 - Airway should be maintained
 - Observed for onset of labor
 - Monitor FHR
- Monitor for signs of abruptio placenta
- Check every 15 minutes for vaginal bleeding

Intrapartal Management

- Real cure: birth



- Labor may be induced by IV oxytocin when there is evidence of fetal maturity and cervical readiness (buttersoft)
- In severe cases, CS
- oxygen administration via facemask

Postpartum Management

- Monitor the patient for 48 hours after delivery
- Continue MgSO₄ for about 24 hours postpartum

Gestational Diabetes Mellitus

- A form of hyperglycemia that develops during pregnancy and poses risks to both mother and fetus
- **Maternal risks**
 - Polyhydramnios
 - Pre-eclampsia
 - Shoulder dystocia
 - Maternal Infections
- **Fetal risks**
 - Congenital anomalies
 - LGA/Macrosomia
 - Increased risk for birth trauma
 - Increased incidence of RDS
 - Newborn: at risk for hypoglycemia

Screening for GDM

- Between 24-28 weeks
- 50g oral glucose tolerance test
- If serum glucose is >140 mg/dL, scheduled for a 100g 3 hour OGTT

Test Type	Pregnant Glucose Level (mg/dl)
Fasting	95
1 hour	180
2 hours	155
2 hours	140

- GDM first therapy is diet: 1800-2400 cal/day
- If diet is inadequate - insulin
- Oral hypoglycemics are NEVER used during pregnancy

Perinatal Insulin Requirements

- **First Trimester:** decreased
- **Second Trimester:** increased



- **Third trimester:** increased
- **Labor and delivery:** decreased
- **Immediate postpartum:** rapid return to pre-pregnant levels
- **Breastfeeding:** decreased

Premature Rupture of Membranes (PROM)

- Rupture of fetal membranes with loss of amniotic fluid during pregnancy before 37 weeks
- **Cause:**
 - Unknown
 - Chorioamnionitis (infection of the membranes)
- **Complications:**
 - **Infection**
 - Gold standard is 24 hours
 - If more than 24 hours, there will be sepsis
 - **Cord prolapse**
 - Extension of the cord out of the uterine cavity into vagina
 - This condition could interfere with fetal circulation
- **Management:**
 - Bed rest
 - Corticosteroids such as bethamethasone to hasten fetal lung maturity
 - Do not reinsert the cord
 - Moisten gauze with NSS and cover the cord
 - Provide oxygenation
 - Get fetal heart tone
 - Outright delivery may be necessary if there is presence of maternal infection, fetal distress and labor

Fetal Alcohol Syndrome

- Fetus cannot remove the breakdown of alcohol from the body leading to vitamin B deficiency resulting to neurologic damage
- Craniofacial deformity:
 - Flat and upturned nose
 - Thin upper lip
 - Small eyes
 - Microcephaly
 - Big ears
 - Cognitively challenged



Vulvovaginal Infections

- **Candida**
 - Vulvar itching and reddening
 - Thick white, cheese-like vaginal discharge
 - **Management:** Nystatin, Myconazole, Fluconazole
- **Trichomonas**
 - Thin, irritating, frothy, gray green discharge
 - Strong rotten odor
 - Itching
 - **Management:** metronidazole (Flagyl), douching with vaginal solution
- **Gardnerella**
 - Edema and reddening of the vulva
 - Milky gray discharge
 - Fish-like odor
 - **Management:** metronidazole and clindamycin
- **Chlamydia trachomatis**
 - Watery graywhite vaginal discharge
 - Vulvar itching
 - **Management:** tetracycline/doxycycline
 - During pregnancy: erythromycin
- **Neisseria gonorrhea**
 - Possibly symptomless
 - May have profuse yellow-green vaginal discharge
 - **Management:** oral ampicillin, ceftriaxone, and doxycycline
- **Treponema pallidum**
 - Painless ulcer on vulvar or vagina
 - **Management:** benzathine penicillin
- **Streptococcus**
 - Vaginitis, vulvar itching
 - Edema and reddening of vulva
 - **Management:** amoxicillin
- **Herpes virus**
 - Painful pinpoint vesicles on an erythematous base with watery vaginal discharge
 - Voiding may be irritating and painful
 - **Management:** acyclovir helps heal lesions, bathing with dilute sodium bicarbonate solution, oral analgesics



Essential Intrapartum and Newborn Care

- a set of evidence-based practices focused on improving newborn care and reducing infant mortality.
- **Core Principles:**
 - EINC emphasizes immediate and thorough drying of the newborn, followed by skin-to-skin contact with the mother, delayed cord clamping (1-3 minutes after birth), and the initiation of breastfeeding without separating the baby from the mother.
- **Evidence-Based:**
 - These practices are supported by research and recommended by organizations like the Department of Health (DOH) and the World Health Organization (WHO).
- **Impact:**
 - EINC aims to reduce neonatal mortality rates by ensuring consistent and high-quality care for newborns during the critical first moments after birth.
- **Implementation:**
 - The program is designed to be implemented by skilled birth attendants in all settings, including both government and private facilities.
- **"Unang Yakap":**
 - EINC is also known as "Unang Yakap" in the Philippines, where it is a key component of the Department of Health's efforts to improve maternal and newborn health.
- **Four time-bound interventions:**
 1. **Immediate drying**
 - a. Using a clean, dry cloth, thoroughly dry the baby, wiping the face, eyes, head, front, back, arms and legs
 2. **Skin-to-skin contact**
 - a. If a baby is crying and breathing normally, avoid any manipulation such as routine suctioning that may cause trauma or introduce infection. Place the newborn prone on the mother's abdomen or chest skin-to-skin
 - b. Cover the newborn's back with a blanket and head with a bonnet. Place identification band on ankle
 3. **Proper cord clamping and cutting**
 - a. Clamp and cut the cord after cord pulsations have stopped (1-3 minutes)
 - b. Put ties tightly around the cord at 2cms and 5cms from the newborn's abdomen
 - c. Cut between the ties with sterile instrument
 - d. Observe for oozing blood
 - e. Do not milk the cord towards the newborn
 - f. After cord clamping, ensure oxytocin 10IU IM given to the mother
 4. **Non-separation of baby from mother & breastfeeding initiation**



- a. Observe the newborn. Only when the newborn shows feeding cues (opening mouth, tonguing, licking, rooting), make verbal suggestions to the mother to encourage her newborn to move toward the breast
- b. Counsel on positioning and attachment
- c. When the baby is ready, advise the mother to:
 - i. Make sure the newborn's neck is neither flexed nor twisted
 - ii. Make sure the newborn is facing the breast, with the newborn's nose opposite her nipple and chin touching the breast
 - iii. Hold the newborn's body close to her body
 - iv. Support the newborn's body, not just the neck and shoulder
 - v. Wait until her newborn's mouth is opened wide
 - vi. Move her newborn into her breast, aiming the infant's lower lip well below the nipple
 - vii. Look for signs of good attachment and suckling:
 1. Mouth wide open
 2. Lower lip turned outward
 3. Baby's chin touching the breast
 4. Suckling is slow, deep and some pauses
 5. If the attachment or suckling is not good, try again and reassess

Notes:

- Health workers should not touch the newborn unless there is a medical indication
- Do not give sugar water, formula or other prelacteals
- Do not give bottles or pacifiers
- Do not throw away colostrum

Teratogenicity of Drug

Drug	Drug Use	Teratogenic Effect
Isotreteneoin (Accutane) Etritinate (Tegison)	Acne Psoriasis	Craniofacial, cardiac and CNS anomalies
Wine, whiskey	Social use	Fetal alcohol syndrome
Acetylsalicylic acid (Aspirin) NSAIDs	Minor pain relief	Prolonged pregnancy, maternal bleeding, ductus arteriosus
Methotrexate Cyclophosphamide	Chemotherapy	Multiple anomalies
Danazol	Endometriosis	Masculinization of female fetus



Phenytoin (Dilantin) Valproic acid Carbamazepine Lamotrigine	Seizures	Fetal hydantoin syndrome, NTDs, possibly fetal anomalies
Warfarin	Anticoagulant	Fetal bleeding or anomalies
Tofranil	Elevate mood	Cardiovascular anomalies
Lithium	Schizophrenia	Hydramnios
Methimazole	Hypothyroidism	Hypothyroidism in fetus
Ribavirin Sulfonamides Tetracycline	Respiratory infection Infection Infection	Multiple anomalies Hyperbilirubinemia Teeth and bone deformities
Enapril Captopril	Reduce hypertension	Oligohydramnios
Caffeine	Coffee, softdrinks, chocolates	Low birth weight
Tolbutamide (Orinase)	Type II diabetes	Profound hypoglycemia in the fetus
Nicotine	Cigarette smoke	Growth restrictions
Thalidomide	Antiemetic	Amelia, phocomelia
Steroids	Anti-inflammatory	Cleft lip, cleft palate, abortion
Streptomycin/Quinine	Antibiotic	Damage to 8th cranial nerve leading to poor hearing and deafness

Pediatric Nursing

- Focuses on the care of infants, children, and adolescents, encompassing health promotion, illness management, and health restoration
- Cephalocaudal
- **Terms:**
 - **Growth** - physical change and quantitative increase in size of the whole body or any of its parts
 - **Development** - changes that mark an increase in function, complexity and progression of skill



- **Maturation** - involves intrinsic processes of development that are genetically or organically programmed
- **Concepts:**
 - **Asynchronous growth**
 - Whole body does not grow at once
 - Different regions and systems develop at different rates and times
 - **The pace of growth and development is uneven**
 - Growth is greater/very rapid in two periods: infancy and adolescence
 - **All body systems do not develop at the same rate**
 - Neurologic tissues grow during the first year of life while genital tissue grows until puberty
 - **Development proceeds from gross to refined skills**
 - This principle parallels the preceding one, once children are able to control distal body parts such as fingers, they are able to perform fine motor skills
- **Principles of Growth and Development:**
 - Growth and development are continuous processes from conception until death
 - At all times a child is growing new cells and learning new skills
 - Growth and development proceed in an orderly sequence
 - Growth in height occurs in only one sequence from smaller to larger
 - Development proceeds from gross to refined skills
 - There is an optimum time for initiation of experiences or learning
 - Children cannot learn new tasks until their nervous system is mature enough to allow that particular learning
 - All body systems do not develop at the same rate
 - Certain body tissues mature more rapidly than others
 - Development is cephalocaudal
 - Development proceeds from the head to tail
 - Development proceeds from proximal to distal body parts
 - Development starts from the midline of the body and progresses towards the extremities

Basic Division of Childhood	
Neonate	First 28 days
Infant	1 month to 1 year
Toddler	1 to 3 years
Preschooler	4 to 6 years
School-aged child	7 to 12 years



Adolescent	13 to 18 years
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Stage	Fears	Play	Hazards
Infant	Fear of stranger	Solitary play	Aspiration, falls
Toddler	Separation anxiety	Parallel play	Accidental ingestions (poisoning), aspiration
Preschool	Punishment, mutilation and castration	Associative/imitation	Safety
School-age	Death, pain, and displacement from school	Indoor competitive	Safety
Adolescence	Displacement from peers	Outdoor competitive	Motor vehicles, homicide, drowning, suicide

Psychosocial Theories

Psychosexual Development

- Sigmund Freud based his theory of childhood development on the belief that sexual energy, termed “libido” was the driving force of human behavior

Phase	Age	Focus
Oral	Infant	Major site of tension and gratification is the mouth, lips, and tongue; includes biting and sucking activities Id is present at birth Ego develops gradually from rudimentary structure present at birth
Anal	Toddler	Anus and surrounding area are major source of interest Voluntary sphincter control (toilet training) is acquired
Phallic	Preschool	Genital is the focus of interest, stimulation, and excitement



		Penis is organ of interest for both sexes Masturbation is common Penny envy (wish to possess penis) is seen in girls Oedipal complex - for boys involves desire for mother Electra complex - for girls, involves desire for father
Latency	School-age	Resolution of oedipal/electra complex Sexual drive channeled into socially accepted activities such as school work and sports Formation of superego Final stage of psychosexual development
Genital	Adolescent	Begins with puberty and the biologic capacity for orgasm; involves the capacity for true intimacy

Developmental Theories

Psychosocial Stages of Development

- Erik Erikson views psychosocial growth occurs in sequential phases and each stage is dependent on completion of the previous stage or life task

Stage	Virtue	Task
Trust vs. Mistrust (Infant)	Hope	Viewing the world as safe and reliable; relationships as nurturing, stable and dependable
Autonomy vs. Shame and Doubt (Toddler)	Will	Achieving a sense of control and free will
Initiative vs. Guilt (Preschool)	Purpose	Beginning development of a conscience; learning to manage conflict and anxiety
Industry vs. Inferiority (School Age)	Competence	Emerging confidence in own abilities; taking pleasure in accomplishments
Identity vs. Role Confusion (Adolescence)	Fidelity	Formulating a sense of self and belonging
Intimacy vs. Isolation (Young Adult)	Love	Forming adult, loving relationships and meaningful attachments to others



Generativity vs. Stagnation (Middle Adult)	Care	Being creative and productive; establishing the next generation
Ego Integrity vs. Despair (Maturity)	Wisdom	Accepting responsibility for one's self and life

Cognitive Stages of Development

- Human intelligence progresses through series of stages based on age, with the child at each successive stage demonstrating a higher level of functioning than of previous stages
- **Jean Piaget's Four Stages of Cognitive Development**
 - (1) **Sensorimotor (birth to 2 years old):** the child develops a sense of self separate from the environment and the concept of object permanence; that is, tangible objects do not cease to exist just because they are out of sight. He/she begins to form mental images
 - (2) **Preoperational (2-6 years old):** the child develops the ability to express self with language, understands the meaning of symbolic gestures, and begins to classify objects
 - (3) **Concrete operational (6-12 years):** the child begins to apply logic to thinking, understands spatially and reversibility and is increasingly social and able to apply rules; however thinking is still concrete
 - (4) **Formal operational (12-15 years old):** the child learns to think and reason in abstract terms, further develops logical thinking and reasoning, and achieves cognitive maturity

Moral Development

- Lawrence Kohlberg recognized moral reasoning helps determine whether children can be depended on to carry out self-care activities

Age (Year)	Stage	Description
Preconventional (Level I)		
2-3	1	Punishment/Obedience orientation: child does right because a parent tells him or her to and to avoid punishment
4-7	2	Individualism: instrumental purpose and exchange. Carries out actions to satisfy own needs rather than society's
Conventional (Level II)		
7-10	3	Orientation to interpersonal relations of mutuality: Child follows rules because of a need to be a "good" person in



		own eyes of others
10-12	4	Maintenance of social order, fixed rules and authority: child finds following rules satisfying
Postconventional (Level III)		
Older than 12	5	Social contract, utilitarian law-making perspective: follows standards of society for good of all people
	6	Universal ethical principle orientation: follows internalized standards of conduct

Growth and Development

Infant

Month	Motor Development	Fine Motor Development	Socialization and Language	Play
0-1	Largely reflex	Keeps hands fisted; able to follow object to midline		Enjoys watching face of primary caregiver, listening to soothing sounds
2	Holds head up when prone	Has social smile	Makes cooing sounds; differentiates cry	Enjoys bright-colored mobiles
3	holds head and chest up when prone	Follows object past midline	Laughs out loud	spends time looking at hands or uses them as toy during the month (hand regard)
4	Grasp, stepping, tonic neck reflexes are fading			needs space to turn
5	Turns front to back, no longer has head lag when pulled			Handles rattles well



	upright; bears partial weight on feet when held upright			
6	Turns both ways; moro reflex fading	Uses palmar grasp	May say vowel sounds (oh-oh)	Enjoys bathtub toys, rubber ring for teething
7	Reaches out in anticipation of being picked up; first tooth (central incisor) erupts; sits unsteadily (still needs support)	Transfers objects hand to hand	shows beginning fear of strangers	Likes objects that are good size for transferring
8	Sits securely without support		Has peak fear of strangers	Enjoys manipulation, rattles and toys of different textures
9	Creeps or crawls (abdomen off floor)		Says first word	Needs space for creeping
10	Pulls self to standing	Uses pincer grasp to pick up small objects		Plays games like patty-cake and peek-a-boo
11	“Cruises” (walks with support)			“Cruises”
12	Stands alone; some infants take first step	holds cup and spoon well; helps to dress	says two words	Likes toys that fit inside each other; nursery rhymes; will like pull toys as soon as walking

- **Fear:** Stranger and anxiety
- **Play:** solitary
- **Toys:** mobile, rattle, musical toys, crib (sensory toys)
- Rapid growth and development
- Birth until 1 year
- **Freud’s psychoanalytic theory: oral phase**
 - Meet the oral needs of the infant: provide safe and washable toys such as pacifier



- Feed on demand: feed according to the child's biologic need for food
- When oral feeding is contraindicated but sucking is not, give a pacifier to suck
- **Piaget's cognitive development**
 - 1-4 months: primary circular reaction (body is center of attention)
 - 4-8 months: secondary circular reaction (from body to environment)
 - 8-12 months: coordination of secondary reaction
 - 12-18 months: tertiary circular reaction (trial and error)
- **Kohlberg's theory of moral development**
 - The infancy period is pre-religious state. Infants learn that when they do certain actions, parents give them affection and approval; for other actions, parents scold and label the behavior "bad"

Toddler

- **Fear:** separation anxiety
- **Play:** parallel
- **Toys:** push and pull toys
- **Age 1-3 years old**
 - Characterized by alternating rapid and slow rate of growth and development
 - Pot-belly
 - Bow-legged
 - Ask questions constantly, up to 400 a day
 - A consistent bedtime ritual helps prepare the toddler for sleep
 - Security object at bedtime may assist in sleep
 - Children generally have 20 of their deciduous teeth by 3 years of age
- **Signs of readiness for toilet training**
 - Child is able to stay dry for 2 hours
 - Child is waking up dry from a nap
 - Child is able to sit, squat, and walk
 - Child is able to remove clothing
 - Child recognizes urge to defecate or urinate
 - Child expresses willingness to please parent
 - Child is able to sit on toilet for 5-10 minutes without fussing or getting off
- **Negativism** - toddlers often say "no" when asked to do something or may not obey request from people other than their parents because they do not view their authority as being at the same level as their parents' authority
 - **Nursing intervention**
 - Offer choices to them
 - Avoid using close-ended questions, use open-ended questions instead
- **Freud's psychoanalytic theory: anal phase**



- Focus on the anal region as they begin toilet training
- **Piaget's cognitive development**
 - **Preoperational thought: 2-7 years**
 - Thought becomes more symbolic; can arrive at answers mentally instead of through physical attempt. Comprehends simple abstractions but thinking is basically concrete and literal. Child is egocentric (unable to see the viewpoint of another)
- **Kohlberg's moral development**
 - **Pre-conventional (2-3 years old): stage 1**
 - Morals are thought to be motivated by punishment and rewards

Preschooler

- **Fear:** mutilation and castration
- **Play:** Associative and imitation/make-believe
- **Toys:** a simple puzzle, dolls, coloring book
- **Age 3-5 years**
 - The preschool grows 2 ½ to 3 inches per year
 - By 5 years old, the child tends to focus on social aspects of eating, table conversations, manners and willingness to try new foods
 - **Oedipus complex:** refers to strong emotional attachment a preschool boy demonstrates toward his mother
 - **Electra complex:** attachment of a preschool girl to her father
 - **Centering**
 - Children tend to look at an object and see only one of its characteristics
 - They see that a banana is yellow but not notice it is also long
 - **Magical thinking:** perceive animals and even inanimate objects capable of thought or feeling
 - **Egocentrism:** perceiving one's thoughts and needs are better or more important than those of other
- **Kohlberg's moral development**
 - **Pre-conventional (4-7) stage 2**
 - Individualism carries out actions to satisfy own needs rather than society's

School Age

- **Fear:** displacement from school
- **Play:** indoor competitive
- **Toys:** computer games and table games
- **Age 6-12 years old**



- Characterized by having a slow period of growth and development patterns
- **Freud's psychosexual stage: latent phase** - a time which children's libido appears to be diverted into concrete thinking
- **Piaget's cognitive development: concrete operational (7-12 years old)**
 - Include systematic reasoning. Classifications involve sorting objects according to attributes such as color. Child is aware of reversibility, an opposite operation or continuation of reasoning back to a starting point
- **Kohlberg's moral development: Level II: conventional (7-10 years old)**
 - Orientation to interpersonal relations of mutuality. Child follows rules because of the need to be a "good" person in own eyes and others
- Reasoning during school age tends to be inductive, proceeding from specific to general: school-age children tend to reason that a toy they are holding is broken, the toy is made of plastic so all plastic toys break easily

Adolescence

- **Fear:** displacement from peers
- **Play:** outdoor competitive (athletic and sports)
- **Toys:** basketball, etc.
- **Age 13-18 years old**
 - Accelerated growth and maturation
 - Influenced by hormonal changes characterized by growth spurt which begins early in girls, about 1-2 years ahead than boys
 - Sebaceous and sweat glands become active and fully functional
- **Freud's psychosexual stage: genital phase** - establishment of new sexual aims and finding of new love objects
- **Piaget's cognitive development: formal operational: 12 years**
 - Can solve hypothetical problems with scientific reasoning
 - Understands thought casualty and can deal with past, present and future
 - Adult or mature thought, good activity this period "talk time" to sort through attitudes
- **Kohlberg's moral development: post-conventional (older than 12 years old)**
 - Social contract: utilitarian law-making perspective, follows standard of society for the good of all people
 - Universal ethical principle: orientations, follow internalized standards of conduct
- Adolescents can be responsible for self-care because they view this as a standard of adult behavior
- Many adult do not reach this level of moral development



Newborn

APGAR Screening Test

- Done twice at 1 and 5 minutes

Criteria	0	1	2
Appearance	Blue	Acrocyanosis	Pink
Pulse	Absent	<100	>100
Grimace	No response	Grimace	Vigorous cry
Activity	Limp/flaccid	Minimal	Full flexion
Respiration	Absent	Weak cry	Vigorous cry

Interpretation:

- **1-3:** POOR - needs immediate resuscitation (CPR)
- **4-7:** FAIR - needs further observation and stimulation, needs suctioning (rubber)
- **8-10:** GOOD - healthy

Estimation of Age of Gestation Criteria

	Pre-Mature	Term	Full-Term
Skin	Very thin, gelatinous, visible blood vessels	Smooth, thick, less visible blood vessels	Leathery, cracked, wrinkled
Lanugo	Abundant	Thinning	Bald
Plantar creases	Anterior transverse	2/3 with creases	Entire sole with creases
Breast	Stripped areola	Raised areola	Full areola
Ear	Flat and folded	Thin and soft	Thick and firm
Genital (M)	Undescended testis	Intermediate	Fully descended
Genital (F)	Prominent labia and clitoris	Labia minora and clitoris partially covered by labia majora	Completely covered minora and clitoris



Profile

- Normal birth weight: 2.5-4 kg
 - Doubles at 6 months
 - Triples at 12 months
 - Quadruplets at 2 ½ years
- Second born usually weights more than first born, continues to increase with each succeeding child in a family
- During the first few days after birth, a newborn loses 5% to 10% of birth weight due to newborn is no longer under the influence of salt and fluid retaining maternal hormones
- Low birth weight: <2500 g
- Large for gestational age: >4000 g
- Birth length: 46-54 cm
- Head circumference: 33-35 cm
- Chest circumference: 31-33 cm

Neurological Assessment

Reflexes

- **Extrusion**
 - Food placed on infant's tongue is thrust forward and out of the mouth
- **Tonic neck**
 - As head is turned to one side, arm and leg on that side extends and opposite extremities in flexion
 - Response usually disappears within 3-4 months
- **Palmar grasp**
 - Elicited by placing finger in newborn's palm
 - Palmar response lessens within 3-4 months
- **Moro**
 - Place the newborn on a flat surface and strike the surface or make a loud abrupt noise to startle the newborn
 - The newborn systematically abducts and extends the arms
 - A persistent response lasting more than 6 months may indicate occurrence of brain damage during pregnancy
- **Sucking and rooting**
 - Touch the newborn's lip, cheek or corner of the mouth with a nipple
 - Newborn turns head toward the nipple, opens the mouth, takes hold of the nipple and sucks
 - Usually disappears after 3-4 months but may persist for up to 1 year
- **Stepping or walking**



- Hold the newborn in a vertical position, allowing one foot to touch a table surface
- The newborn stimulates walking, alternately flexing and extending the feet
- Present for 3-4 months
- **Babinski Sign: Plantar Reflex**
 - Beginning at the heel of the foot, gently stroke upward along the lateral aspect of the sole, and then move the finger along the ball of the foot
 - The newborn's toes hyperextend while the big toe dorsiflexes
 - Disappears after the newborn is 1 year old
 - Absence of this reflex indicates the need for neurological examination

Vital Signs

- **Temperature:** Axillary - 97.9-98F
- **Apical rate:** 120-160 bpm
- **Respirations:** 30-60 cpm
- **Blood pressure:** 73/55 mmHg

Newborns can conserve heat by constricting blood vessels and moving away from the skin. Brown fat, a special tissue found in mature newborns, apparently helps to conserve or produce body heat by increasing metabolism

Physical Assessment

- **Head**
 - **Anterior fontanel**
 - Soft, flat, diamond shaped 3-4 cm wide by 2-3 long
 - Closes between 12-18 months
 - **Posterior fontanel**
 - Triangular shaped 0.5-1 cm
 - Closes at 2-3 months
 - **Caput succedaneum**
 - Swelling of scalp caused by prolonged labor crosses over the suture line
 - Gradually disappears at about third day of life
 - **Cephalhematoma**
 - Collection blood caused by increased pressure of birth
 - Caused by rupture of periosteal capillary
 - Absorbed within 3-6 weeks
 - **Craniotabes**
 - Localized swelling of the cranial bones caused by pressure of the fetal skull against the mother's pelvic bone in uterus
 - Condition corrects itself without treatment



- **Eyes**
 - Infant eyes assume their permanent color between 3-12 months of age
 - Lacrimal ducts do not fully mature until about 3 months of age
 - Strabismus is normal until 6 months
 - Subconjunctival hemorrhage - a red spot on sclera on inner aspect of eye due to pressure at birth (absorbed 2-3 weeks)
- **Ears**
 - The pinna normally align from inner to outer canthus of the eye
 - The low set ears indicate chromosomal disease
 - Trisomy 21 (down syndrome)
 - Kidney anomaly
 - Test newborn hearing by ringing a bell held 6 inches from each ear
- **Nose**
 - Nasal flaring is the enlargement of the opening of the nostrils during breathing
 - Nasal flaring indicates respiratory distress
 - Test for choanal atresia (blockage at the rear of the nose) by closing the newborn's mouth and compressing one nares at a time with fingers
 - Note any discomfort or distress while breathing this way
- **Neck**
 - Short chubby with creases skin folds
 - Rigidity of the neck may indicate congenital torticollis/meningitis
 - The trachea may be prominent on the front of the neck, and the thymus gland may be enlarged because of the rapid growth of glandular tissue
 - The thymus gland will triple in size by 3 years of age; it remains at the size until the child is about 10 years old, and then shrinks
- **Chest**
 - It is approximately 2 inches smaller than head circumference
 - Retractions or drawing in of the chest during inspiration should not be observed, it could indicate respiratory distress
- **Abdomen**
 - The abdomen of the child should look slightly protuberant, a scaphoid or sunken appearance could indicate missing abdominal contents
 - Bowel sounds should be present 1 hour after birth
 - Umbilical cord
 - Stump should appear as a white, gelatinous structure with blue and red streaks of the umbilical vein and arteries
 - 2 arteries and 1 vein
 - Single artery could signify congenital heart or kidney anomaly
 - Umbilical cord should break free by day 6 to 10



- If umbilical hernia is present, taping or putting buttons or coins on the cord do not help defects to close
- **Anogenital Area**
 - Inspect the anus of a newborn to be certain it is present, patent and not covered by a membrane (imperforate anus)
 - If a newborn does not do so in the first 24 hours, suspect imperforate anus or meconium ileus
- **Male genitalia**
 - Both testes should be present in the scrotum
 - If one or both testicles are not present (cryptorchidism) caused by agenesis (absence of an organ)
 - Ectopic testes (the testes cannot enter the scrotum because the opening to the scrotal sac is closed) or undescended testes
 - Newborns with agenesis of the testes are usually referred for investigation of kidney anomalies because the testes arise from the same germ tissue as the kidneys
 - Elicit a cremastetic reflex - a test for the integrity of spinal nerves T8-T10. The response may be absent in newborns who are younger than 10 days
 - Urethral opening should be on the tip of the glans, not on the dorsal surface (epispadias) or on the ventral surface (hypospadias)
- **Female genitalia**
 - The vulva in female newborns may be swollen because of the effect of maternal hormones
 - Pseudomenstruation: female newborns have a mucus vaginal secretion which is sometimes blood tinged which is normal
- **Back**
 - Inspect the base of the newborn's spine carefully to be sure there is no pinpoint opening, dimpling, or sinus tract in the skin which would suggest a dermal sinus or spinal bifida occulta
- **Extremity**
 - Unusually short arms may signify achondroplastic dwarfism
 - Inspect the palm for a simian crease (down syndrome)
 - Assess webbing (syndactyly)
 - Extra toes or fingers (polydactyly)
 - Both hips can flexed and abducted to such an extent (180 degrees) that the knees touch or nearly touch the surface of the bed if the hip joint seems to lock short of this distance (160-170 degrees), hip subluxation (a shallow and poorly formed acetabulum) is suggested



Congenital Anomaly Appraisal

Procedure	Abnormalities Considered
Inquire for hydramnios or oligohydramnios	Presence of hydramnios suggests congenital gastrointestinal obstruction Oligohydramnios suggests genitourinary obstruction or extreme prematurity
Appearance of abdomen	Distended abdomen suggests ascites, possible bowel obstruction or tumor Empty abdomen suggests diaphragmatic hernia
Passage of nasogastric tube through nares into stomach	Failure to pass nasogastric tube through nares on either side establishes choanal atresia Failure to pass into the stomach confirms presence of esophageal atresia
Counting of umbilical arteries	The presence of one artery suggests possible congenital urinary or cardiac anomalies, chromosomal trisomy

Breast feeding facts

- Almost all drugs pass into breastmilk. A breastfeeding mother must be certain not to take any medication without contacting her primary care provider to be certain it is compatible with breastfeeding
- If a baby will be formula fed, be certain the parents understand the potential danger of warming bottles in a microwave oven (the inner core of milk may grow very hot)
- Caution parents not to prop bottles, because it increases the risk for aspiration and otitis media. It also deprives infants of the pleasure of being held for feedings
- To avoid baby bottle syndrome (cavities of the lower teeth), infants should not be put to bed with a bottle

Neurologic Disorders

Increased Intracranial Pressure

- **Normal:** 5-15 mmHg; >20 mmHg (life-threatening)
- **Cushing's triad**
 - Hypertension
 - Bradycardia
 - Bradypnea
- **Widened pulse pressure:** difference between systolic and diastolic
 - **Normal:** 30-40 mmHg



- **Signs and symptoms**

- Bulging fontanel: aggravated by crying
 - **Anterior:** diamond, closes 12-18 months
 - **Posterior:** triangle, closes 2-3 months
- High pitch/shrill/irritable cry: early sign
- Increased head circumference: tape measure every shift (8 hours)
- Headache: initial sign
- Projectile vomiting: compression of medulla (chemoreceptor trigger zone: vomiting center) -> cerebral edema
- Diplopia (double vision): increased ICP = increased IOP (proximity of brain and eyes) -> optic nerve damage -> blurring of vision -> blindness
- Pupillary changes (**Pupil Equally Round and Reactive to Light and Accommodation** (normal) - consensual)
 - **Anisocoria:** uneven pupils -> brain damage
 - **Dilated pupils** -> shock
 - **Constricted** -> narcotic overdose
- Sunset eyes: black portion of the eyes is pushed downwards
- Anorexia, nausea, weight loss
 - Weight: important indicator of health
- Seizures
 - Increased neuronal firing -> erratic transmission of electrical impulses
 - Vomiting -> decreased fluid levels -> dehydration -> increased temperature -> convulsions

- **Management:**

- **Position:** semi-fowler (decreased ICP = gravity)
 - Where sodium goes, water follows
 - Increased fluid = Increased pressure
- Avoid coughing and sneezing
- Limit fluid intake: 1200-1500 ml/day
- **Pharmacotherapy**
 - Diuretics (potassium wasting -> monitor for hypokalemia)
 - Lasix (loop diuretic) -> loop of Henle -> localized
 - Mannitol (osmotic diuretic) -> osmosis (pulling pressure) -> generalized
 - Decadron (dexamethasone) -> anti-inflammatory -> prevent cerebral edema
 - Anti-convulsants: prevent seizure episodes
 - Antacids: neutralizes acidity
 - Stress -> GI compensates and increases HCl
 - Counteract the effects of decadron
 - Types:
 - Magnesium hydroxide: diarrhea



- Aluminum hydroxide: constipation
- Anticoagulants: prevents clumping of blood
 - Increased ICP -> damage the brain blood vessel walls -> bleeding -> clotting of blood -> formation of thrombus -> dislodges (emboli) -> CVA -> multiple organ failure
 - Types:
 - Heparin: IV/SQ - pTT (partial thromboplastin time)
 - Warfarin: oral - pt (prothrombin time)
- **REMEMBER:** opiates and sedatives (depressants) are contraindicated in increased ICP (due to bradypnea and bradycardia = cardiac and respiratory depression)

Seizure Disorder/Epilepsy

- Erratic transmission of electrical impulses -> muscle spasms
- **Types:**
 - **Grand Mal** (generalized)
 - **Tonic** (mild) - **clonic** (severe) **phase:** “during” -> dyspnea, salivation, urination
 - **Position:** flat/supine
 - Protect the head
 - **Post ictal phase** (exhaustion phase): “after”
 - **Position:** side-lying/recovery - to prevent aspiration
 - **Petite Mal** (absent seizure)
 - Blank facial expression
 - Automatisms: repeated purposeless behavior
 - Lip smacking
 - **Jacksonian:** tonic-clonic of a group of muscle then progress to grand mal seizure
 - **Psychomotor:** mental clouding and intoxication
 - Common in older people
 - **Febrile** (under 5): 38.5 - 40C
 - Common in children
 - Paracetamol RTC
 - 42C -> brain damage
 - Under 5 year olds - immature hypothalamus (thermoregulation)
 - **Status epilepticus:** most dangerous
 - Lasts for 30 minutes (can be continuous or intermittent) -> brain damage could occur
- **Management:** not all seizure can be managed with medication
 - **Pharmacotherapy:**
 - Hydantoin: phenytoin



- WOF: gingival hyperplasia (soft bristled toothbrush/meticulous oral care), pinkish red urine - normal (inform significant other)
- Benzodiazepines “pam/lam”
 - Minimizes seizure episode
- Iminostilbenes: carbamazepine (refractory seizures - recurrent seizures)
- Valproic acids/Valproates: last resort for pedia (immature liver)
 - Side effect: hepatotoxicity
 - NEVER GIVEN in pregnancy (especially first trimester) - causes NTDs (neural tube defects)
- **Surgical:** last resort
 - Neurectomy: removal/surgical resection of cranial nerve involved in the seizure

Bacterial Meningitis

- Infection of the meninges (supports and nourishes the brain)
- **Diagnostics:** lumbar puncture (L3, L4, L5)
 - Position: C-shaped, fetal, genupectoral/knee chest
 - Normal CSF: clear
 - With meningitis: cloudy (elevated WBC & protein - by product/waste of bacteria)
- **Causative agent:** Neisseria meningitides
- **Signs and symptoms:**
 - Kernig's: pathognomonic sign
 - Knee flex -> pain in the hamstring, back, neck
 - Brudzinkski: pathognomonic sign
 - Nape flex -> pain in the neck, back
 - Nuchal rigidity (stiff neck)
 - Seizures
 - Opisthotonus: arching of the back
 - **Position:** side-lying
- **Management:**
 - Antibiotic therapy: finish the duration of the drug (to prevent drug resistance)

Head Trauma

- **Types:**
 - **Concussion:** jarring of the brain -> sudden forceful contact in a rigid skull
 - Transient loss of consciousness
 - **Contusion** (bruising): structural alteration -> extravasation of blood
- **Management:**
 - **Safety:** bike helmets, seat belts, safe driving



- Proper usage of infant car seat
 - < 3 years old: rear facing (to protect the spine)
 - > 3 years old: front/forward facing booster seat
- **Assess for cerebral functioning**
 - PERRLA
 - GCS -> LOC (most important prognostic indicator)
- **Assess for cervical neck injury**
 - (+): do not move the client (immobilize) - to prevent further damage
 - (-): elevated HOB - to decrease ICP (gravity)

Cardiovascular System

- **Layers of the heart:**
 - Endocardium: innermost
 - Myocardium: muscle -> contract -> cardiac output (tissue perfusion/oxygenation)
 - Pericardium: outermost
 - Visceral: inner
 - Parietal: outer
 - Pericardial space: in between visceral and parietal, with pericardial fluid
- **4 chambers of the heart**
 - Right atrium
 - Right ventricle
 - Left atrium
 - Left ventricle
- **4 heart valves**
 - Tricuspid valve
 - Pulmonary valve
 - Bicuspid/Mitral valve
 - Aortic valve
- **Heart circulation**
 - Superior/Inferior vena cava (deoxygenated) -> right atrium -> tricuspid valve -> right ventricle -> pulmonary valve -> pulmonary artery -> lungs (pulmonary circulation) -> pulmonary vein (oxygenated) -> left atrium -> mitral/bicuspid valve -> left ventricle -> aortic valve -> aorta -> body (systemic circulation)
- **Fetal heart circulation**
 - Placenta (oxygenated blood) -> umbilical vein -> ductus venosus -> inferior vena cava -> right atrium
 - -> foramen ovale -> left atrium -> left ventricle -> ascending aorta -> upper body
 - -> right ventricle -> pulmonary artery -> ductus arteriosus -> descending aorta -> lower body



- Umbilical arteries (deoxygenated blood) -> placenta -> oxygen exchange with the mother

Congenital Cardiovascular Defects

- **Types of congenital defects:**
 - **Acyanotic:** absence of cyanosis (oxygenated > deoxygenated), 1 problem only
 - **Cyanotic:** presence of cyanosis (deoxygenated > oxygenated), 2 or more problem

Acyanotic Congenital Defect

Patent Ductus Arteriosus

- Point of intersection of aorta and pulmonary artery
- Ductus arteriosus did not close after birth
- **Signs and symptoms:**
 - Machine-like murmur (pathognomonic sign)
 - Signs and symptoms of heart failure
 - Poor feeding
 - Poor weight gain
 - **6 months:** birth weight is doubled
 - **12 months:** birth weight is tripled
 - Irritability: cerebral hypoxia
- **Management:**
 - DOC: Indomethacin (prostaglandin inhibitor) - facilitates closure of the PDA
 - Secondary: Ibuprofen

Septal Defects

- **Types:**
 - **Atrial septal defect (ASD):** a hole between atrium
 - **Ventricular septal defect (VSD):** a hole between ventricles
- **Signs and Symptoms:**
 - Dyspnea on exertion
 - Fatigue -> poor feeding
 - Failure to thrive (delayed milestones)
 - Signs and symptoms of heart failure
- **Management:**
 - Surgery by suture (by pass) - through large blood vessel



- Dacron patch: for large hole
 - **Tissue:** decreased rejection rate (favorable)
 - **Plastic:** increased rejection rate (if with increased pressure of the heart, more durable)

Coarctation of Aorta

- Narrowing of aorta (descending aorta)
- Increased pressure, decreased cardiac output
- VS (upper - increased; lower - decreased) - due to proximity
- **Signs and symptoms:**
 - Upper extremities: elevated BP, bounding pulse
 - Lower extremities: decreased BP, weak/absent pulse
 - Rib notching = rib deformity
- **Management:**
 - Surgery
 - **Balloon angioplasty with coronary stenting** - prevent the blood vessel to collapse
 - Repair of artery (aorta) using a balloon and installs a coronary stent (serve as a support)

Cyanotic Congenital Defect

Tetralogy of Fallot (TOF)

- **Problems:**
 - **Pulmonary stenosis:** primary problem
 - **Right ventricular hypertrophy:** coping mechanism
 - **Overriding of aorta** (anatomical defect): allows mixing of blood
 - **Ventricular septal defects:** keeps the patient alive, relieves the pressure in the right ventricle
- **Diagnostic test:** 2D echo: boot-shaped heart
- **Signs and symptoms:**
 - Cyanosis: ineffective tissue perfusion (blue babies)
 - Squatting: allow for compensation
 - Decrease venous return -> decrease cardiac workload -> cardiac rest
 - Conserve oxygen in the upper body (vital organs)
 - Tet spells: group of signs and symptoms -> decreased oxygenation (pathognomonic sign)
 - Irritability, blackouts, convulsions, pallor (hypoxemia)



- Cardiomegaly
- Clubbing of fingernails -> chronic hypoxia
- Pansystolic murmur
- **Management:**
 - Allow the child to squat
 - **Surgery**
 - **Palliative:** relieves sign and symptoms -> GOAL: oxygenated blood > unoxygenated blood
 - **Blalock Taussig Shunt:** anastomosis of pulmonary artery and aorta using subclavian artery
 - **Curative:** resolve the problems, cures the condition
 - **Intra-cardiac surgery/Brock's procedure:** combination of two surgery
 - **Balloon angioplasty:** pulmonary stenosis, right ventricular hypertrophy
 - **Dacron patch:** VSD, overriding of aorta

Transposition of the Great Arteries (TOGA/TGA)

- **Mechanism:**
 - Right ventricle -> aorta
 - Left ventricle -> pulmonary artery
- **PDA:** keeps the patient alive (1-2 weeks survival)
- **Signs and symptoms:**
 - Severe respiratory depression -> cyanosis
 - Fatigue
 - Failure to thrive
 - No murmur
- **Management:**
 - Prostaglandin: keeps the PDA open/maintains PDA
 - **Surgery**
 - **Arterial switch/Jatene surgery** (during the first week of life - most crucial due to increased survival)

Infectious Cardiovascular Disease

Rheumatic Heart Fever

- **Cause:** GABHS (group A beta hemolytic streptococcus) -> sore throat/acute glomerulonephritis
- **Jones criteria:** 2 major + history or 1 major + 2 minor + history



- **Major:**
 - Carditis (inflammation of heart)
 - Polyarthritis (multiple joint inflammation)
 - Chorea/St. Vitus Dance (worm-like movements)
 - Subcutaneous nodules (knuckles, elbows, knees)
 - Erythema marginatum (redness in the truncal area)
- **Minor:**
 - Low grade fever
 - Arthralgia (painful joints without swelling)
 - Elevated of ASO (anti-streptolysin) titer - GABHS specific
 - Elevated of inflammatory markers
 - ESR: erythrocyte sedimentation rate
 - CRP: C-reactive protein
- **Management:**
 - **DOC:** penicillin (narrowed-spectrum bacteria) - finish the duration: 5 to 10 days to prevent drug resistance
 - **If with allergy:** clindamycin or erythromycin
 - **If with exacerbation/remission:** treatment is continued up to 10 years (continuous follow-up/monitoring, medication: monthly)
 - Salicylates: for pain and swelling (4 A's of Aspirin)
 - Anti-platelet -monitor for bleeding
 - Anti-pyretic
 - Analgesics (RHF)
 - Anti-inflammatory (RHF)
 - Corticosteroids (anti-inflammatory): to relieve carditis

Heart Failure

- Insufficient cardiac output
- **Types:**
 - **Right sided HF:** systemic manifestation
 - Jugular vein distention
 - Hepatomegaly (enlargement of liver) -> portal hypertension (increased BP in portal vein)
 - Esophageal varices
 - Ascites (fluid deposition in the abdomen)
 - Hemorrhoids
 - Peripheral/dependent/pitting edema
 - Weight gain (fluid retention)
 - Body weakness (actual weakness)



- Anorexia, nausea
- Tachycardia (compensation) -> early sign
- **Left sided HF:** pulmonary manifestation
 - Crackles/rales
 - Dyspnea on exertion
 - Orthopnea (difficulty of breathing when lying down)
 - Pulmonary edema
 - Cough
 - Tachycardia (compensation) -> early sign
- **Diagnostic tests:**
 - CXR: cardiomegaly
 - 2D echo: hypokinetic heart (slow movement of the heart) -> late sign
 - Pulse oximetry: decreased oxygen saturation (Normal: 95-100%)
 - Pulmonary Capillary Wedge Pressure (Normal: 4-12 mmHg) -> LSHF
 - Central Venous Pressure (Normal: 8-12 mmHg) -> RSHF
- **Management:**
 - Fowler's position: HOB elevated -> maximize lung expansion -> increased oxygenation
 - Administer high O2: venturi mask- accurate/precise
 - Inotropic drugs -> strengthen heart contraction -> maximize cardiac output
 - Lanoxin (digoxin) - WOF: signs of toxicity
 - Nausea
 - Anorexia
 - Visual disturbances/vomiting
 - Diarrhea
 - Abdominal cramps
 - Antidote: Digibind/Digoxin immune fab
 - Urine output and intake monitoring
 - Record weight daily
 - Administer diuretics

Kawasaki Disease

- **Comprises of 2 conditions:**
 - Mucocutaneous lymph nodes syndrome
 - Altered immune response
 - Multisystemic vasculitis
 - Inflammation of blood vessels
- **Signs and symptoms:**
 - High spiking fever (hyperpyrexia)
 - Strawberry red tongue (pathognomonic sign)



- Photophobia (extreme sensitivity to light) -> hyperemia of conjunctiva -> blood pooling (dark colored glasses/large brim hats/sun visors)
- Polymorphous rash (different type of rashes in a point of time)
- Palmar desquamation
- **Diagnostics:** elevated ESR
- **Management:**
 - Immunoglobulin as ordered (enhanced immune response)
 - Aspirin as ordered
 - Clear liquid diet: to monitor bleeding in the stool
 - CPR (complication: CAD)

Gastrointestinal Disorders

Cleft Lip/Palate

Cleft Lip	Cleft Palate
Male	Female Speech problem
Hereditary Maternal smoking Difficulty of feeding Risk for aspiration Risk for infection (URTI)	
Responsibilities	
Large nipples - easier to suck, criss-cross cut	Cup/medicine dropper/asepto syringe - precise fluid flow, rubber tipped
Surgery	
Cheiloplasty - Rule of 10 - 10 weeks - 10 lbs - 10K WBC - 10 g/dl hemoglobin - Post-op: unaffected side, if bilateral - HOBE, monitor for frequent swallowing (bleeding)	Palatorrhapy/Palatoplasty - Rule: 18-24 months - Not too early -> re-open - Not too late -> speech problem - Post-op: prone, monitor for frequent swallowing (bleeding)



Gastroesophageal Reflux Disease (GERD)

- Also known as chalasia
- Incompetent LES (lower esophageal sphincter/cardiac sphincter)
- **Signs and symptoms:** esophageal lining damage
 - Forceful vomiting
 - Heartburn/pyrosis (chest pain)
 - Bitter taste in the mouth (due to HCl) -> oral care
 - Dysphagia: difficulty swallowing
 - Odynophagia: painful swallowing
 - Hoarseness: laryngeal affectation
- **Diagnostic test:**
 - Barium swallow (should not stay for > 2 days: obstruction) -> excretion (increased OFI and laxatives as ordered)
- **Management:**
 - Diet modification: low fat (fats are hard to digest), high fiber (enhances digestion and increases peristalsis)
 - Small frequent feeding
 - Avoid: spicy, tobacco, alcohol, caffeine (gastric irritants)
 - Pharmacological
 - Antacids: neutralize acidity
 - H2 blockers: "tidine" blocks production of HCl
 - PPI: "prazole" blocks HCl production + protective coating
 - HOBE: 6-8 inches during sleeping (2 pillows)

Pyloric Stenosis

- Narrowing of pyloric sphincter (controls gastric emptying) -> abdominal distention (olive-shaped mass) -> projectile vomiting without bile
- **Signs and symptoms:**
 - Vomiting: blood tinged with gastric contents, without bile
 - No anorexia with good appetite but with vomiting
 - Weight loss
 - Upper abdominal distention (olive-shaped mass)
 - Visible peristaltic movement (left to right)
 - Malnutrition and dehydration
 - Metabolic alkalosis
- **Management:**
 - Monitor feeding patterns
 - Assess the vomitus (without bile)



- Increase OFI
- Prevent aspiration: feeding slowly
- Burp frequently to prevent gas retention
- High fowler's position to prevent regurgitation
- Surgery
 - **Pyloromyotomy**: incision that splits the obstruction

Celiac Disease

- Also known as celiac sprue or gluten sensitive enteropathy
- Irreversible but manageable
- **Cause**: malabsorption of gluten
- **Signs and symptoms**:
 - Acute diarrhea
 - Steatorrhea (foul fatty feces)
 - Anorexia
 - Vomiting
 - Severe abdominal distention (gas retention)
 - Body wasting (cachexia)
 - Retarded growth
 - Failure to thrive (delayed milestones)
- **Diagnostic tests**:
 - Bowel biopsy: flat mucosal surface with hyperplastic villous atrophy
 - Elevated IgG and IgA levels
- **Management**:
 - Avoids: BROW (barley, rye, oat, wheat)
 - Health teaching: read the nutrient value before feeding the child
 - Allow: meat, eggs, milk products (cream, cheese, milk), all fruits and vegetables, rice, corn
 - Birthday party: bring own cake (gluten-free)
 - Gluten-free diet for life

Hirschprung's Disease

- AKA congenital anganglionic megacolon
- Cause: absence of ganglions (cells/nerves responsible for peristalsis)
- Manifestations: megacolon
 - Pencil/pellet like stool or ribbon-like stool (pathognomonic sign)
 - Problem: failure to pass meconium for the first 24-48 hours
- Diagnostic tests:



- Barium enema: visualizes megacolon, and its location
- Rectal biopsy: confirmatory, absence of ganglionic cells
- Management
 - **Surgery:** Swenson Pull Through (end to end anastomosis with temporary colostomy)
 - 1-3 months -> 6 months (maximum)
 - **Colostomy care**
 - **Right**
 - Liquid
 - Without odor
 - No irrigation
 - Continue appliance of the bag
 - **Transverse**
 - Mushy
 - Slight odor
 - Irrigate if solid
 - **Left**
 - Solid
 - With odor
 - With irrigation
 - Continue appliance of the bag
 - **Assess the characteristics of the stoma**
 - **Color:** pinkish; bluish -> lack of circulation (notify)
 - **Moist:** (+); dry -> dehydration (notify)
 - **Elevated:** slightly; concave/depressed -> atrophy (notify)
 - Avoid foods that can obstruct the stoma (gas forming)
 - Foods to allow: spinach, parsley, yogurt, broccoli
 - Swimming allowed
 - Stool softeners as ordered

Intussusception

- Telescoping of the colon
- Sausage-shaped mass
- **Signs and symptoms:**
 - Colicky pain (gas formed pain)
 - Sausage-shaped mass
 - Bile-stained fecal emesis
 - Currant jelly like stool (blood mucoid stool)
- **Diagnostic tests:**
 - Barium enema: determines the location



- Guaiac's test/occult blood test
 - Dark colored foods: false (+)
 - Vit. C rich foods: false (-)
 - Agent: hydrogen peroxide: blue ring = (+) bleeding
- **Management:**
 - Auscultate bowel sounds
 - Assess for abdominal distention (gas retention)
 - NGT insertion: Feeding, Irrigation, Decompression, Medication
 - **Surgery:** Swenson Pull Through

Respiratory Disorders

Croup

- Narrowing of airway due to viral inflammation
- **Signs and symptoms:**
 - Barking seal like cough (pathognomonic)
 - Inspiratory stridor (passage of air in a narrowed airway)
 - Dyspnea -> cyanosis
 - No fever/low grade fever
- **WOF:** drooling, nasal flaring, use of accessory muscles while breathing -> Epiglottitis/ARDS
- **Management:**
 - Increase humidity in the room (cool mist): enhances gas exchanges
 - Inhale cool night air/warm bathroom air (bronchodilation)
 - Tracheostomy set at the bedside
 - Pharmacologic
 - Antivirals "vir"
 - Antibiotics for prophylaxis
 - Bronchodilators: to dilate bronchus
 - Side effect: tachycardia

Cystic Fibrosis

- Blockage of the exocrine glands (mucus)
- Organs affected: pancreas, lungs, intestines, sweat glands
- **Signs and symptoms:**
 - Pancreatitis
 - Dyspnea: ineffective airway clearance
 - Meconium ileus



- Increased salty sweat
- **Diagnostic tests:**
 - Sweat chloride test
 - Increased chloride levels in the sweat
- **Management:**
 - CPT (chest physiotherapy)
 - Process of removing mucus in the lungs
 - Percussion, vibration, postural drainage
 - Done on empty stomach/before meals/2 hours after meals
 - Oral care
 - Pharmacologic
 - Pancreatic enzymes
 - Given with meals/never double the dosage

Asthma

- Reversible
- Cause: allergens (substances that can cause allergic and hypersensitivity reactions)
- If exposed to allergens -> hyperresponsiveness -> bronchospasms
- **Common allergens:**
 - Pollens
 - Molds
 - Dust
 - Seaweeds
 - Pet danders
 - Egg white
 - Sea food
- **Exacerbation:** air pollutants, cold heat, weather changes, strong odors, extreme emotions
- **Signs and symptoms:** wheezes (expiratory)
 - During asthmatic attack -> sudden loss of wheezes -> complete obstruction of airway (status asthmaticus)
- **Management:**
 - Bronchodilators
 - Corticosteroids - side effect: immunosuppression/weight gain
 - Oxygen therapy
 - Avoid allergens



NURSING PRACTICE III

Emergency Nursing

Emergency - any situation that is sudden/unexpected that needs immediate action (threat to life)

- Rendered without delay
- Small scale
- Priority: SAFETY

Triage

- French word “trier” means to sort based on severity (seriousness or extent of illness) and immediacy (urgency - treatment needed)/prioritize

	Routine Triage	Field/Reverse Triage
Principle	Sickest (1st)	Utilitarianism
Example	3 level, 5 level (ESI)	Mass casualty incidents, NATO triage system, START, military triage

3-Level Triage - based on urgency

Level	Definition	Examples
I - Emergent	Life-threatening - ABC problem	Cardiac arrest, severe trauma, acute stroke, respiratory failure, anaphylaxis
II - Urgent	Serious, but not immediately life-threatening	Severe pain, high fever, acute infections
III - Non - urgent	Episodic illnesses, elective procedures	Chronic conditions, minor injuries, routine check-ups

5-Level Triage - based on acuity and sources (ESI - Emergency Severity Index)

Level	Color	Definition	See Patient	Examples
I - Resuscitation	Red	Require immediate life-saving intervention (ABC)	STAT	Cardio-respiratory arrest, severe trauma, unconscious, shock, respiratory failure



II - Emergent	Orange	High-risk situation, ALOC, severe pain or distress, VS danger	Within 15 minutes	Severe pain, signs of stroke, chest pain, suicidality, profuse, bleeding, febrile infant (immature thermoregulation)
III - Urgent	Yellow	Multiple resources, normal vitals	Within 30 minutes	Abdominal pain, active vomiting, mild asthma/allergy, diarrhea, fever, palpitation
IV - Less Urgent	Green	1 resource only	Within 60 minutes	Back pain, headache, rashes, painful urination, ankle injury/minor trauma
V - Non - Urgent	Blue	No resource	Within 120 minutes	Sore throat, rabies vaccine, simple wound, sore eyes, procedures for chronic patient, prescription refill

ESI Algorithm

- Require immediate life-saving treatment (ABC) - I - RED
- In High-risk situation or ALOC or severe pain or distress - II - ORANGE
- Count resources if more than 1 and with normal VS - III - YELLOW
- If one resource - IV - GREEN
- If none - V - BLUE

Primary Survey

- to stabilize patient
- (1) **Airway** - goal: patent airway
 - (a) **Reposition the head**
 - (i) Usual: head tilt chin lift maneuver
 - (ii) Spinal cord injury: jaw thrust maneuver
- (2) **Oropharyngeal Airway (OPA)** - location: back of the tongue
 - (a) **Indication:** unconscious, breathing spontaneously
 - (b) **CI:** conscious, semi - conscious (triggers gag reflex)
 - (c) **Insert:** upside down into the mouth then rotate 180 degrees upon insertion
- (3) **Nasopharyngeal Airway (NPA)** - location: nares
 - (a) **Indication:** same with OPA, tolerated to semi- conscious



- (b) **CI**: facial trauma, basilar skull fracture (enters the brain)
- (4) **Endotracheal Intubation** - location: trachea via vocal cords
 - (a) Indication: absent ventilation, persistent hypoxia
- 1. **Assessment (Artificial Airway)**
 - **Proper ventilation**: bilateral breath sounds
 - **WOF**: dislodgement, pneumothorax
- 2. **Breathing** - goal: to provide adequate ventilation
 - **Trauma** - priority:
 - **SCI** - immobilize spine (C3 - C5 - phrenic nerve - diaphragm - respiratory failure)
 - **Chest injuries** - 3 - sided occlusive dressing (to prevent tension pneumothorax)
- 3. **Circulation** - cardiovascular system
 - Priority: control bleeding
 - **Pressure** (to promote coagulation)
 - **Immobilize** (prevent further bleeding)
 - **Elevate** (to decrease arterial blood flow)
- 4. **Disability** - LOC
 - AVPU (Alert, Verbal, Pain, Unresponsive)
 - GCS (Glasgow Coma Scale/Score)
 - Eye Opening 4
 - Verbal Response 5
 - Motor Response 6
- 5. **Exposure** (DCAP BTLS) - undress patient, cut off clothing
 - **Discoloration/Deformity**
 - **Contusion**
 - **Abrasion**
 - **Penetration/Puncture** - NI: do not remove the object
 - **Burn**
 - **Tenderness**
 - **Laceration** - Protrusion - NI: saline dressing
 - **Swelling/Edema**
 - Once exposed: warm the patient (to prevent hypothermia)

Secondary Survey

- Full set of VS, family presence
- Get monitoring devices, give comfort (analgesics)
- History taking, head to toe assessment (mechanism of injury)
- Input diagnostics, laboratory studies



- Wound management: prevent tetanus (vaccine)

BLS, ACLS

Clinical death: no pulse, no breathing (cardiopulmonary arrest)

- Respiratory arrest - no breathing with pulse
- Cardiac arrest - no pulse with breathing
- Time frame: 4-6 minutes STAT post-arrest: brain damage (permanent)

Basic Life Support

- Chest compressions
- Rescue breaths
- Defibrillation

Advanced Cardiac Life Support

- BLS + intubation
- Drugs (epinephrine)
- Equipment

Basic Life Support

1. **Danger** - scene, safety
 - a. Survey the scene
 - b. Priority: safety
2. **Response**
 - a. First patient assessment: LOC then shout “are you okay?”
 - i. Look for signs of life (movement, breathing)
3. **Activate emergency response system**
 - a. Action: call 911 and get AED
 - b. NEVER: leave patient
4. **CAB - Compression, Airway, Breathing**
 - **Assess:** pulse and breathing - <10 seconds - perfusion
 - i. Adult, children: carotid (1 side)
 - ii. Infant: brachial, femoral
 - **Chest compressions** - push hard and fast (allow complete recoil of the chest)

Adults	Children (1-8)	Infants
Both hands, heel at sternum (non-dominant over dominant)	Heel of one hand	2 fingers (index & middle/ 2 thumb encircling the chest)
Lower half of sternum, avoid xiphoid process		
Depth: 2 inches		Depth: 1.5 inches



- **Airway**
 - **Breathing** - if available: Bag valve mask (BVM)
 - Watch for: chest rise
 - Avoid: excessive ventilation - decreases cardiac output, gastric inflation
 - **Ratio:** 1 rescuer - 30:2
 - Compression rate: 100 - 120 bpm
 - Ventilation rate: 10 - 12 cpm/ 1 breath every 6 seconds
5. **Defibrillation** - automated external defibrillators (AED)
- a. Monophasic - 360J
 - b. Biphasic - 120-200J
- Proper AED placement**
- **Clothing:** remove
 - **Chest:** dry
 - **Placement of pads:** to sandwich the heart
 - **Right:** upper chest above right nipple
 - **Left:** lower chest below left nipple
 - **Hairy chest:** clippers
 - **Pacemaker/implantable devices:** 1 inch away from the pacemaker (to avoid malfunction)
 - **Avoid:** placing above the pacemaker
 - **Arrest:**
 - **Monitored:** defibrillation before CPR
 - **Unmonitored:** CPR before defibrillation
 - **Rhythms:**
 - **Shockable:** ventricular fibrillation, ventricular tachycardia (all clear)
 - **Non-shockable:** asystole, pulseless electrical activity (CPR)
6. **Return of Spontaneous Circulation (ROSC)**
- a. **Position:** recovery - side-lying (prevent aspiration)

Disaster Nursing

Disaster - any event that is sudden/unexpected that interrupts the functioning of entire community

- Capacity of community: exceeded
- Requires: external assistance
- Leads to: multiple injuries/deaths, property damage

Types:

- **Natural** - caused by natural processes
- **Man-made** - caused by human actions

Law: **RA 10121 - Philippine DRRM Act of 2010**

- Decrease disaster risk



- Increase resilience of people (NDRRMC) - Department of National Defense (DND)

Disaster Phases

Mitigation	Preparedness	Response	Recovery
Pre - Disaster		Post - Disaster	
To reduce likelihood or impact of having the disaster from occurring	To enhance the ability to respond to disaster	To provide emergency assistance	To return/rebuild to normalcy
Prevention	Trainings and drills	Short-term needs of community	Long-term needs
DOST (Department of Science and Technology)	DILG (Department of Interior and Local Government)	DSWD (Department of Social Welfare and Development)	NEDA (National Economic Development Authority)

NATO Triage Classification

- Triage: Reverse, Field - prioritize likelihood of survival, resources available
- Principle: utilitarianism “greatest good for the greatest number”

Priority	Tag	Name	Definition
1	Red	Immediate	Life-threatening, but survivable with minimal or minor interventions
2	Yellow	Delayed	Serious injuries but can wait hours for treatment
3	Green	Minimal	Minor injuries, “walking wounded”, can wait hours to days for treatment
4	Black	Expectant	Fatal injuries, unlikely to survive even with advanced care

Start Triage Algorithm

Simple Triage and Rapid Treatment (START) - Primary Survey

- (1) **Walk** (green), if not check respiration
- (2) **Respiration** - reposition airway, if breathing (red), if not (black)
 - Respiratory rate - >30 (red), < 30 check pulse
- (3) **Pulse** - > 2 seconds capillary refill (delayed) - red (control bleeding), < 2 seconds check LOC



(4) **LOC** - obeys command (yellow), if not (red)

Jump START - Primary triage in children < 8 years old

Secondary Assessment of Victim Endpoint (SAVE)

- **Purpose:** secondary survey
- **Disaster:** catastrophic in an extended period of time

Burns

- Injury of skin layers and underlying tissues caused by:
 - **Thermal:** hot objects
 - **Chemical:** strong acids and alkali
 - **Electrical:** electrical source
 - Priority: spinal cord injury, arrhythmia
 - **Radiation:** sunlight, ionizing radiation

Skin layers

- **Epidermis** - outer/superficial
- **Dermis** - with nerves and blood vessels
- **Subcutaneous** - fat layer
- **Muscle/Fascia/Bones**

Degree	Name (thickness)	Structure	Keywords	Pain
1st	Superficial	Epidermis	Reddish (due to inflammation), intact nerves	Painful
2nd	Partial	Dermis	Moist (leakage of capillaries), blisters (accumulation of fluids) - weeping	Painful
3rd	Full	Subcutaneous	Pearly white/yellow, damaged nerves	Painless
4th	Deep Full	Muscles, bones	Blackish, charred leathery 3rd and 4th degree burns complications (1) Eschar - crusty, rigid dead tissue (a) circumferential (i) Compartment syndrome (extremities) (ii) Airway obstruction	Painless



			(neck & chest), mgt: escharotomy and fasciotomy to relieve pressure (2) Acute tubular necrosis - intrarenal acute kidney injury (a) Muscle breakdown - myoglobin - obstruct tubules - cell death	
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First Aid for Burns

- **Priority:** stop the burning process
 - **Active flames**
 - Do: stop, drop and roll
 - Throw: wet blanket
 - Remove: clothings, jewelries
 - **No flame**
 - Immerse: cool water
 - DO NOT apply: ice (vasoconstriction - decreased perfusion - tissue damage)
- Chemical burns: flushing/irrigation with water for more than 15 minutes

Priority during Fire

- Rescue patient
- Activate fire alarm
- Confine fire by closing the door
- Extinguish the fire

Using Fire Extinguisher

- Pull the pin
- Aim base of the fire
- Squeeze the handle
- Sweep side to side

Phases of Burns

1. **Emergent Phase (resuscitative)** - priority: ABC
 - a. Occur 48 hours
 - b. Fluid shifts IVC (intravascular) to ISC (interstitial) (capillary leak) - sign: edema
 - i. Fluid volume: hypovolemia (burn shock)
 - ii. RBC: hemoconcentration (increased hematocrit - RBC over plasma)



- iii. UO: oliguria (low urine output)
 - c. Electrolyte imbalance
 - i. Potassium: hyperkalemia (cell damage)
 - ii. Sodium: hyponatremia (capillary leakage)
 - d. Acid-base balance: metabolic acidosis (due to increase of lactic acid)
2. **Acute Phase (Intermediate)** - priority: wound healing (reduce infection)
- a. Occur: after 48 hours
 - b. Fluid shift ISC to IVC
 - i. Fluid volume: hypervolemia
 - ii. RBC: hemodilution (low hematocrit)
 - iii. UO: diuresis
 - c. Electrolyte imbalance
 - i. Potassium: hypokalemia
 - ii. Sodium: hyponatremia
 - d. Acid-base balance: metabolic acidosis (excretion of bicarbonate due to lack of sodium)
3. **Recovery Phase (Rehabilitation)**
- a. **Goals:** physical and psychosocial adjustment
 - b. **Occur:** after 5 days
 - c. **Electrolyte imbalance**
 - i. **Calcium:** hypocalcemia (depletes due to scab formation/granulation)
 - ii. **Potassium:** hypokalemia
 - d. **Wound care priority:** prevent permanent scarring (body image disturbance)

Burn Management

1. **Airway, Respiratory Function**
- a. **Establish patent airway**
 - i. Face/neck burns: intubation (before laryngeal edema)
 - b. **Position:** high fowlers (to promote lung expansion)
 - c. **Inhalation injury:** carbon monoxide poisoning
 - i. **Sign:** cherry red lips
 - ii. **Treatment:** 100% O₂ (non-rebreather mask), hyperbaric oxygen therapy
 - d. Singed hair, black/sooty sputum, hoarseness, stridor (severe airway problem - **REPORT**)
2. **Fluid Replacement** - priority: prevent hypovolemic shock
- a. Best assessment: urine output >30 ml/hr
 - i. **Adults:** 0.5-1 ml/kg/hr
 - ii. **Children:** 1 - 1.5 ml/kg/hr
 - b. **Types of fluid**
 - i. **Resuscitation** - crystalloid: PLR (isotonic, closest to plasma, (+) lactate - treats acidosis)



1. **Avoid:** PNSS (contains NaCl leads to hyperchloremic metabolic acidosis)
 2. **WOF:** pulmonary edema - respiratory distress/SOB, crackles, pink and frothy sputum (pathognomonic sign)
 - ii. **After 24 hours** - colloids: albumin, plasma expander (oncotic/osmotic pressure - pull)
- c. **Rules of 9s (TBSA)** - not counted: superficial
- i. **Head, face, neck:** 9
 - ii. **Arms:** 18
 - iii. **Anterior trunk:** 18
 - iv. **Posterior trunk:** 18
 - v. **Legs:** 36
 - vi. **Genital:** 1
- d. **Parkland Formula:** (24 hours fluid) $4\text{ml} \times \text{TBSA} \times \text{kg}$ (lbs/2.2 lbs)
- i. **1st 8 hours:** $1/2$
 - ii. **2nd 16 hours:** $1/2$

RECAP

A 45-year-old male (weight: 110 lbs) was involved in a warehouse explosion. On arrival at the emergency department, you assess the following:

- Entire anterior torso (full thickness) $9 + 9 = 18$
- ~~Entire left arm (superficial)~~
- Anterior surface of both legs (partial) $9 + 9 = 18$

$\frac{110\text{ lbs}}{2.2\text{ lbs}} = 50\text{ kg}$

1. % TBSA? $18 + 18 = 36$

2. Total Amount of Fluid? $= 4 \times 36 \times 50 = 7200\text{ ml}$

3. Fluid Amount for the 1st 8 hrs? $\frac{7200\text{ ml}}{2} = 3600\text{ ml}$

4. Fluid Rate for the 1st 8 hrs? $\frac{3600\text{ ml}}{8\text{ hrs}} = 450\text{ ml/hr}$

3. Pain Management

- a. Contraindicated route: IM (no absorption in muscles)
 - b. DOC: morphine (opioids)
 - i. Route: IV
 - ii. WOF: respiratory depression (<12 cpm)
 - iii. Antidote: naloxone (Narcan) - intranasal
4. **Nutrition** - avoid 48 hours: oral (should be NPO) - paralysis ileus (due to absence of peristalsis - decreased GI blood flow) (use TPN)
- a. Diet: after 48 hours
 - i. Calorie, carbohydrate: increased (due to hypermetabolism)



- ii. Protein: increased (for tissue repair)
- b. Vitamins
 - i. A (retinol) - for skin integrity
 - ii. B - for metabolism
 - iii. C - boost immune system to prevent infection (skin infection)
- c. Ulcer: Curling's - PPI (Omeprazole, pantoprazole), Histamine II blockers (ranitidine) - decrease HCl

5. Thermoregulation

- a. Emergent Phase: hypothermia (skin damage)
- b. Acute, recovery: low grade fever (inflammation)
- c. Hypothermia Prevention: warm
 - i. Remove: wet clothing
 - ii. Cover: warm blanket
 - iii. Room temperature: increased
 - iv. Bedside: heat lamp
 - v. Fluids: warm

6. Preventing Infection: damaged first line defense (skin)

- a. Precaution: reverse isolation
- b. Most effective: handwashing
- c. Vaccine: tetanus
- d. Wound irrigation: NSS (isotonic)
- e. Most common bacteria:
 - i. Gram-positive: Staphylococcus aureus
 - ii. Gram-negative: Pseudomonas aeruginosa
- f. Debridement: to remove dead tissues (necrotic tissues/gangrene)
 - i. Wet to dry dressing
 - ii. 1st action: give analgesics
- g. Full thickness burn - skin grafting
 - i. Isograft - identical twin
 - ii. Allograft - other humans
 - iii. Xenograft - animals (common: pig) (temporary)
 - iv. Autograft - self (#1 donor site - anterolateral aspect of the thigh)
 - 1. Easy access

Shock

Perfusion: decreased (lack of oxygen - damaged tissues) - multi-organ failure - DEATH

Shock	Involved	Cause	BP	HR	RR
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Hypovolemic (most common)	Blood	Decreased blood volume	low (Hypo)	high (Tachy)	high (Tachy)
Cardiogenic	Heart	Decreased cardiac output	low (Hypo)	high (Tachy)	high (Tachy)
Distributive	Blood vessels	Massive vasodilation	-	-	-
A. Septic (most fatal)		Infection (cytokines)	low (Hypo)	high (Tachy)	high (Tachy)
B. Anaphylactic		Allergic (Hypersensitivity I) - excessive release of histamine causing vasodilation	low (Hypo)	high (Tachy)	high (Tachy)
C. Neurogenic		No SNS response	low (Hypo)	low (Brady)	low (Brady)

Stages of Shock

1. **Compensatory** - decreased perfusion causing low oxygen - homeostasis
 - a. SNS (Increased epi & norepi)
 - i. Increased HR & RR - respiratory alkalosis
 - ii. Shunting of blood flow to vital organs: brain and heart
 - iii. Vasoconstriction - cool skin, decreased peristalsis leading to normalize blood pressure
 - b. ADH (Pituitary Gland), RAAS
 - i. Increased sodium (increased fluid volume - oliguria), decreased potassium
2. **Progressive**
 - a. Decompensation - severe decreased perfusion - severe decreased oxygen
 - i. Organ dysfunction (increased lactic acid - metabolic acidosis)
 1. Brain: ALOC
 2. Heart: MI
 3. Kidneys: oliguria
 - ii. BP: low
3. **Irreversible** (point of no return) - prolonged hypoxia - multi-organ failure - DEATH
 - a. Asystole (absent rhythm)
 - b. Apnea (absence of breathing)
 - c. Anuria (absence of urine)
 - d. Comatose



- e. Metabolic acidosis (accumulation of lactic acid)

Shock Management

1. **Respiratory Support:** intubation, 100% oxygen, NRB, Mechanical ventilation
2. **Fluid Replacement**
 - a. IV Access: 2 large bore (<18)
 - i. 26 - purple
 - ii. 24 - yellow
 - iii. 22 - blue
 - iv. 20 - pink
 - v. 18 - green
 - vi. 16 - gray
 - vii. 14 - orange
 - b. Best Fluids: crystalloids
 - i. 1st: PLR
 - ii. 2nd: PNSS
3. **Position:** modified trendelenburg (head of the bed flat, elevate legs) - to increase venous return to vital organs, (-) cardiogenic shock
4. **Acidosis:** Sodium bicarbonate (NaHCO_3)
5. **Treat Underlying Cause**
 - a. Hypovolemic: control bleeding
 - b. Cardiogenic: (+) inotropes (dobutamine, dopamine)
 - c. Anaphylactic: IM Epinephrine
 - d. Septic: antibiotic
 - e. Neurogenic: SCI - immobilize
6. **Vasopressor**
 - a. DOC: Norepinephrine
 - b. Target MAP: >65 mmHg

Airway Obstruction

Basic concepts

- **Common cause:**
 - Allergic reaction (Anaphylaxis)
 - Infection (Epiglottitis)
 - Angioedema (ACE inhibitor “-pril”)
- **Disease process:**
 - Airway obstruction -> absent ventilation -> impaired gas exchange
 - -> hypoxemia -> hypoxia



- -> hypercapnia -> acidosis
- Cardiopulmonary arrest -> brain damage -> death

Partial Airway Obstruction	Complete Airway Obstruction
Able to cough	Unable to cough, breathe and speak
Wheezing	Clutching of the neck (levine's), cyanosis, stridor
Encourage patient to cough forcefully	Heimlich maneuver (Abdominal thrust) 1. Stand behind the back of the victim 2. Wrap around the waist 3. Form a fist above the umbilicus, below xiphoid process 4. Wrap the fist 5. Quick upward thrust

Conscious:

- Adult/children: abdominal thrust
- Infant (<1 year old)
 - Back blows
 - Chest thrust
- Pregnant, Obese: chest thrust

Unconscious:

- Method: continue CPR, finger sweep (only if visible)
- Contraindication: invisible foreign object

Heat-Induced Illness

Heat exhaustion, Heat Stroke

- Patient population: older adults (decreases ability to sweat resulting to heat loss)
- Cause: hot environment, physical activity, inadequate fluid intake
- Heat illness: loss of fluid & electrolytes due to hot environment, physical activity & inadequate fluid intake
- **Heat exhaustion:** prolonged exposure to heat (hours to days) -> loss of F&E
- **Heat stroke:** untreated heat exhaustion -> hypothalamus failure -> absence of sweat (anhidrosis) -> hyperthermia (>40C) -> protein denaturation
 - -> brain encephalopathy (damage of brain) -> severe CNS signs and symptoms
 - -> systemic inflammatory response -> excessive release of cytokines -> massive vasodilation -> shock (distributive) -> multiple organ failure

	Heat Stroke	Heat Exhaustion
Body temperature	>40C	<40C



Skin	Anhidrosis -> dry, hot	Diaphoresis
CNS	Severe: confusion, coma, seizure	Mild: restlessness, anxiety, headache
Other symptoms	Hypo-Tachy, Tachy	Goosebumps (SNS response), orthostatic hypotension/orthostasis, muscle cramps (heat cramps) -> fatigue
Severity	Emergency	Less severe

Heat-Induced Illness Management

Prevention

- AVOID hot environment, 10 AM - 2PM, caffeine & alcohol (diuresis -> F&E loss)
- Stay hydrated
- Replace electrolytes: sport drinks
- Clothing: loose, light colored
- Room: air-conditioned, shaded

Heat Exhaustion Management

1. #1 Priority: rapid temperature reduction -> goal: 39C
 - a. Environment Interventions: cool
 - i. Passive: environment
 - ii. Active: equipment/actions

Heat Stroke Management

1. Cooling
 - a. Remove clothing
 - b. Most effective: cool water immersion/bath -> most rapid
 - c. Prevent hypothermia: STOP 38C
 - d. Monitor: core temperature (rectal)
2. Airway breathing support - intubation, 100% oxygen, NRB, mechanical ventilation
3. Circulation: fluids, electrolytes, continuous ECg (monitor arrhythmia)

Submersion Injuries

- Submersion injury - hypoxia -> submersion to water
- Non - fatal drowning - survival: >24 hours from a submersion that cause respiratory arrest
- Patient population: children -> best intervention: supervision

Freshwater Aspiration	Saltwater Aspiration
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Hypotonic (usually contaminated) -> enter the pulmonary circulation -> damage surfactant -> alveolar collapse -> atelectasis -> impaired gas exchange

- -> hypoxemia -> hypoxia
- -> hypercapnia -> acidosis
 - Cardiopulmonary arrest -> brain damage -> death

Hypertonic -> osmosis (low to high) -> pulmonary edema -> impaired gas exchange

- -> hypoxemia -> hypoxia
- -> hypercapnia -> acidosis
 - Cardiopulmonary arrest -> brain damage -> death

Submersion Injuries Management

- **Main goal:** adequate oxygenation
- Provide immediate CPR -> increased survival
- Priority assessment: lung sounds
- Hypothermia: warm patient
- **Teachings:**
 - Swimming pool gates: locked
 - Watercraft: life jacket
 - Learn: water survival skills
 - AVOID alcohol, drugs

Snake Bites

Envenomation - injection of venomous substance through bite

- **Most common site:** upper extremity (improper handling of snakes)
- **Signs:**
 - Skin: puncture/fang marks
 - Taste: metallic
- **Local effect:**
 - Edema (swelling)
 - Hemorrhagic bullae (fluid-filled blisters that can rupture causing necrosis)
 - Ecchymosis (bruising)
- **Systemic effect:**
 - Cardiotoxic -> causes arrhythmia
 - Hemolytic -> anemia related to bleeding
 - Myotoxic (rhabdomyolysis)
 - Neurotoxic -> ALOC, paralysis -> respiratory failure



Snake Bite Management

- Priority: move to safe area (to avoid further bites)
- Activity: rest
- **Position:**
 - Ideal: lie down
 - Part: immobilize
 - Level: below the heart (decreased venous return)
- Remove: constrictive items (edema)
- Wound: cleanse to prevent infection
- Assess: circumference -> every 15-10 minutes
- Administer: Antivenin/antivenom -> Ig (antibody)
 - Philippines: purified cobra antivenom (PCAV)
 - USA: Crotalidae polyvalent immune Fab antivenom (CroFab)
 - WOF: anaphylaxis -> premedication: antihistamine (diphenhydramine - Benadryl)
- **Avoid**
 - Suction (at risk for infection)
 - Corticosteroids (deactivate antivenom)
 - Heparin (prevent bleeding)
 - Ice (tissue damage)
 - Tourniquet (risk for amputation)
 - Incision (risk for infection)

Ingested Poison

- Poisoning by mouth
- Patient population: children

Management

1. Call: Poison Control Center
2. Maintain: ABC
3. **Interventions:**
 - a. assess: identify the poison -> DOC: antidote
 - b. GI -> done within 1-2 hours (to prevent the poison to travel to circulatory system)
 - i. Dilute: 1-2 glasses of water
 - ii. Decreased absorption: activated charcoal -> CI: ALOC (risk for aspiration)
 1. GI
 - a. Gastric: lavage
 - i. Tube: large
 - ii. Fluid: warm NSS
 - iii. CI: corrosive (damages the esophagus)



- iv. WOF: aspiration
- b. Never: induce vomiting (aspiration, damages esophagus)
- 4. Unresponsive treatment: dialysis, exchange blood transfusion

Common Antidotes

Drug	Antidote	Memory Trick
Acetaminophen (Paracetamol)	Acetyl cysteine	ACE
Opioids (Narcotics)	Naloxone (Narcan)	Nar
Beta Blockers	Glucagon	BB - Gun
Benzodiazepines	Flumazenil	AZE
Digoxin	Digibind	Dig
Iron -> chelation	Deferoxamine	Fe
Heparin	Protamine sulfate	H+
Warfarin	Vitamin K	War -> Kills
Potassium (K+)	Calcium Gluconate Beta II Agonist Insulin Glucose Kayexalate	Full -> K -> cells
Anticholinergics	Physostigmine	Anti -> stigma
Cholinergic	Atropine sulfate	A - INE
Lead	Calcium EDTA	EAD

Cardiology

Cardiovascular System - closed system

- **Responsible for tissue perfusion**
 - **Blood** (5-6L) - carries oxygen
 - Hypovolemic shock/Hemorrhagic shock
 - **Heart** - pumps blood
 - Cardiogenic shock



- Causes: coronary causes, non-coronary, (obstructive shock (-) modified trendelenburg - REFER)
- **Blood vessels** - distribute blood to different parts of body
 - **Arteries** - carries oxygenated blood except pulmonary artery
 - **Vein** - carries deoxygenated blood except pulmonary vein
 - Maintains vascular tone (vasodilation)
 - Distributive/circulatory shock - massive vasodilation
 - Infection (septic shock)
 - Allergy (anaphylactic shock)
 - SCI (neurogenic shock) - no SNS

Blood Vessels

- **Arteries** - resistance vessels (Aorta - biggest)
 - Pulses - assess arterial function
 - Pulse grading: (0 - absent, +1 - weak, +2 - normal, +3 - increased, +4 - bounding)
 - Layers
 - Tunica intima, Tunica media (muscle), Tunica externa
- **Veins** - capacitance vessels (Vena Cava - biggest)
- **Capillaries** - exchange vessels

Vascular Disorders

Angiography - most definitive diagnostic test, invasive

- Left cardiac catheterization (artery) - arterography
- Right cardiac catheterization (vein) - venography
- NPO to prevent aspiration
- Local anesthesia - patient must be awake during the procedure
- Check for allergy

Ultrasound - non-invasive diagnostic test

- **2D Echo** - ascending AA, cardiac
- **Abdominal ultrasound**
- **Doppler ultrasound/duplex ultrasound** - peripheral vascular diseases

Aneurysm

Congenital aneurysm - cerebral arteries (most common)

- Saccular
- Fusiform
- Ruptured (most dangerous)



Acquired aneurysm - aorta (common)

- **Aortic aneurysm** - localized sac or dilation formed at a weak point in the wall of an artery
 - **Atherosclerosis** (cause) - plaque formation & hardened blood vessels, big arteries
 - Risk factors: age, male, family history, sedentary lifestyle, smoking, alcohol, obesity, stress, high cholesterol diet, hypertension, DM
 - Tunica intima distends
 - Forms at descending and ascending aorta
 - i. **Ascending AA** - asymptomatic
 - Severe chest pain
 - ii. **Descending AA**
 - **Abdominal AA** - pulsating mass in the abdomen, with thrill and bruit
 - Severe lower back pain
 - **Thoracic AA** - asymptomatic
 - Severe epigastric pain
 - iii. **Dissecting AA**
 - **Type A:** in the ascending aorta
 - Type I: ascending & part of descending
 - Type II ascending only
 - **Type B:** in the descending aorta
 - Type III: descending

Management

- Type A - surgery
- Type B - control BP
- Monitor BP/VS
- Provide rest (decrease cardiac workload)
- Avoid stress
- Support cardiac function
- Surgery
- Cardiac rehabilitation
- Manage risk factors
- Live a healthy lifestyle

Peripheral Vascular Diseases (PVD)

- a. **Arterial insufficiency/Peripheral arterial disease/Peripheral arterial occlusive disease** (downward blood flow)
 - i. **Arteriosclerosis obliterans** - lower extremities
 - 1. Irreversible, small arteries
 - 2. Management



- Position the legs (dependent - reversed trendelenburg)
 - Avoid trauma
 - Skin care
 - Wound care
 - SET (supervised exercise therapy)
 - Drug therapy
 - Cilostazol (improve capillary flow)
 - Antiplatelet drugs like aspirin, clopidogrel (prevent thrombus formation)
 - Pain reliever (NSAIDS)
 - Surgery (revascularization)
 - Amputation
 - Rehabilitation therapy
 - Manage risk factors
- ii. **Raynaud's disease** - upper extremities (hands & fingers) - **BLUE** - WHITE - **RED** DISEASE
- Arteriolar vasospasm -> pooling of venous blood (blue) -> severe vasospasm (white) -> reversible -> vasodilate -> increased blood flow (red)
 - Exposure to cold temperature (risk factors)
 - Young adult female (common)
 - **Signs/symptoms:** pain, numbness and tingling sensation (paresthesia)
 - **Management**
 - Avoid exposure to cold temperature
 - Wear gloves/boots
 - Increase heat in the environment
 - Avoid smoking, stress, trauma
 - Skin and wound care
 - Calcium-channel blockers - vasodilators (as ordered) - diltiazem (monitor BP)
 - **Raynaud's Phenomenon** - manifesting signs and symptoms of Raynaud's disease in autoimmune disease
 - Rheumatoid arthritis
 - SLE
 - Scleroderma - hardening of the skin
 - i. **CREST syndrome** (calcinosis, raynaud's phenomenon, esophageal dysmotility, syndactyl, telangiectasia)
- b. **Venous insufficiency** - lower extremities (upward blood flow)
- i. **Varicose veins** - abnormally dilated, tortuous superficial veins



- Irreversible
 - Etiology: presence of incompetent valves
 - **Risk factors:** prolonged sitting, standing, crossing of legs/pregnancy/obesity
 - Venous congestion, phlebitis
 - **Management**
 - **Prevention:**
 - i. Avoid risk factors
 - ii. Position the legs (elevated)
 - iii. Use of compression stockings
 - **Interventions:**
 - i. Thermal ablation
 - ii. Laser therapy
 - iii. Microphlebotomy
 - iv. Sclerotherapy
 - v. Vein stripping & ligation
- ii. **Deep vein thrombosis (DVT)/Venous thromboembolism (VTE)**
- **Risk factor:** immobilization, DM, oral contraceptive pills
 - -> stasis/stagnation -> hypercoagulable -> endothelial injury (Virchow's Triad - thrombus formation)
 - **Types**
 - **Thrombophlebitis** - thrombus with inflammation
 - i. Signs and symptoms: redness, swelling
 - **Phlebothrombosis** - thrombus without inflammation
 - i. Asymptomatic, sudden onset of leg pain (assess Homan's sign) if **Homan's sign is (+), do not repeat the test (it can cause embolus)**
 - 1. Dyspnea, tachypnea, chest pain (elevate HOB (fowler's), REFER)
 - **Management**
 - **Prevention:** mobilization
 - i. Early ambulation
 - ii. Passive ROM (for paralyzed patients)
 - iii. Massage
 - iv. TED (thromboembolic deterrent) hose
 - v. Elevate the legs
 - **Intervention:** immobilization
 - i. Complete bed rest without bathroom privilege
 - ii. Heparin IV (prevent thrombus)



- iii. Pain reliever (NSAIDS)
- iv. Thrombolytic therapy (dissolve thrombus) - Urokinase, Streptokinase, Alteplase
 - 1. WOF: bleeding
- v. TED hose (depends on doctor's order)
 - 1. Can be taken off during night, just elevate legs when sleeping
- vi. Warfarin (anticoagulant - prevention)
- vii. Mechanical thrombolectomy

Arterial Insufficiency	Venous Insufficiency
- Leg pain (ischemia) - painful - Claudication (pain relieved by rest) - Thinning of hair - Pale - Cool to touch - Pulses may be abnormal - No edema - Arterial ulcer (gangrene) - Small but deep, well-circumscribed edges, no granulation tissue (clean) - Dependent position	- Leg pain (congestion) - Dark - Cyanotic - Pigmented - Thickening of the skin - Pulses are always normal but difficult to palpate - With edema - Venous ulcer (gangrene) - Dark skin, large ulcer, irregular border, several granulation tissues (dirty) - Elevate legs

- c. **Buerger's disease/thromboangiitis obliterans** - inflammation & thrombus formation involving the small and medium sized arteries & veins
 - i. Irreversible
 - ii. Only in smokers
 - iii. Upper & lower (common)
 - iv. Young adult men, stress, smoking, family history
 - v. Both arterial and venous insufficiency
 - vi. **Management**
 - Avoid smoking, stress, trauma
 - Skin & wound care
 - Pain reliever (NSAIDS)
 - Vasodilators are RARELY given
 - Amputation
 - Rehabilitation therapy

Heart

- Muscular tissue (cardiac muscle - involuntary)
 - **Contractility**
 - **Conductivity** (pacemaker)
 - SA Node
 - **Rhythmicity**
 - **Automacity**
 - **Excitability**
- **Layers of the heart**
 - **Endocardium** - inner
 - **Myocardium** - middle
 - **Epicardium** - outer
 - **Visceral pericardium** - membrane that covers the heart
 - **Parietal pericardium** - pericardial sac
- **Cardiac Circulation**
 - Superior (upper) and inferior (lower) vena cava -> right atrium -> tricuspid valve -> right ventricle -> pulmonary artery -> lungs (PULMONARY CIRCULATION - 25/9 mmHg)
-> pulmonary vein -> left atrium -> mitral/bicuspid valve -> left ventricle -> aortic valve
-> aorta -> SYSTEMIC CIRCULATION (110/70 mmHg)
- **Heart Sounds**
 - **S1** (lub) **AV**
 - **S2** (dub) **SL**
 - **S3** - rapid ventricular filling
 - Enlarged ventricles (CHF)
 - Ventricular gallop (lub dub dub)
 - **S4** - atrial systole/contraction
 - Atrial gallop (lub lub dub)
 - **Murmur** - defected valves
 - Increased turbulence in the flow of blood inside the heart
- **Assessment (Valves)**
 - **Aortic** - 2nd intercostal space, right parasternal
 - **Pulmonic** - 2nd intercostal space, left parasternal
 - **Tricuspid** - 4th intercostal space, left parasternal
 - **Mitral** - 5th intercostal space, left midclavicular line (PMI)

Valvular Diseases

1. Etiology:

- a. Congenital



- b. Acquired
 - i. Rheumatic Heart Disease
 - ii. Endocarditis
- 2. **Type**
 - a. **Valvular insufficiency** - inability of the valves to close completely, regurgitation (murmur)
 - b. **Valvular stenosis** - inability of the valves to open completely (murmur)
 - c. **Mitral valve prolapse** - ballooning/bulging of the mitral valve onto the left atrium (systolic/mitral click)
 - i. **Signs and symptoms:** tachycardia, fatigue, palpitation, chest pain
 - ii. **Cause**
 - 1. Mitral regurgitation
 - 2. Dysrhythmia
- 3. **Definitive test:** 2D Echo
- 4. **Management**
 - a. Valve dilation
 - b. Valvuloplasty
 - c. Valvular replacement
 - i. Mechanical
 - ii. fibrous
 - d. Support cardiac function
 - e. Provide rest
 - f. Avoid stress
 - g. Manage heart failure
 - h. Prevent and manage shock (cardiogenic)

Pericarditis

- Inflammation of the pericardium
- Idiopathic, infection, chest trauma, MI (MI induced pericarditis), SLE, malignancy (sign of metastasis)
- **Prominent symptom:**
 - Chest pain (subjective cue) - worsens with deep inspiration, lying down or turning and relieved by sitting or leaning forward - orthopneic position/tripod
 - Friction rub (objective cue) - auscultate at the 4th ICS left parasternal border, creaky/leathery/scratching sound heard best at the end of expiration and when patient is sitting or leaning forward
- **Definitive diagnostic test:** 2D Echo
- **Complication**
 - **Pericardial effusion** - accumulation of fluids in pericardial sac



- **Pericardiocentesis** - assist the physician, low to semi-fowler's position, ultrasound/x-ray/ecg guided (to prevent puncture of organs)
- **Cardiac tamponade** - life - threatening, the heart cannot pump anymore (Beck's triad)
 - Hypotension
 - Increased venous pressure
 - Distant heart sound

Endocarditis

- Inflammation of the endocardium
- Bacterial endocarditis - group A, streptococcus
- **Risk factors:**
 - Existing cardiac disease
 - Immunocompromised
 - Invasive procedure or surgery
- **Systemic inflammation:** fever (intermittent), chills, night sweats, fatigue, malaise, chest pain, anemia
 - Cardiogenic & septic shock (severe)
 - **Vegetation** - made up of small thrombi (can destroy the valves)
 - Splinter hemorrhages - bleeding under the nails
 - **Roth's spots** - white centered retinal hemorrhages
 - **Osler's nodes** - painful nodules on finger pads and toes
 - **Janeway lesions** - painless macules on palms and soles
- **Definitive test:** culture & sensitivity, CBC, elevated ESR (systemic inflammation), 2D Echo (vegetation)
- **Nursing Management**
 - Manage fever
 - Assess changes in murmur (during endorsement)
 - Monitor VS
 - Provide rest
 - Avoid stress
 - Monitor for signs and symptoms of heart failure
 - Monitor for signs and symptoms of shock
- **Medical Management**
 - Prevention - prophylaxis (penicillin/erythromycin)
 - Antipyretic
 - Antibacterial drug
 - Support cardiac function
 - Prevent and manage heart failure
 - Prevent and manage shock



Cardiomyopathy

- Disease of the heart muscle associated with the cardiac function
 - **Dilated CDM** - most common. Significant dilation without hypertrophy and systolic dysfunction
 - (-) CCB
 - Associated with diffused necrosis
 - Risk factors: viral infection, alcoholism, pregnancy
 - **Hypertrophic CDM** - autosomal dominant disorder that leads to increase heart muscle size and mass especially the septum
 - Family history (autosomal dominant - 50%)
 - (-) Digoxin, cardiotonic
 - **Restrictive CDM** - least common, caused by rigid ventricular wall
 - Family history
 - **Arrhythmogenic right ventricular CDM** - infiltration of fibrous and adipose tissue starting from the right ventricle until entire heart is affected
 - Family history
- **Definitive test:** 2D Echocardiography
- **Cure:** Heart transplant
- **Management**
 - Heart transplant
 - Ventricular assistive device
 - Support cardiac function
 - Provide rest
 - Avoid stress
 - Manage heart failure
 - Prevent shock

Hypertension

Category	Systolic	Diastolic
Normal	<120	<80
Elevated	120-129	<80
Stage 1	130-139	80-89
Stage 2	>140	>90
Hypertensive Crisis	>180	>120

- “Silent killer”



- **Primary** - idiopathic, disease
 - Family history
- **Secondary** - a sign of disease
 - DM
 - Renal disease
 - Pheochromocytoma (tumor in adrenal gland - catecholamines)
- Hypertension is also a risk factor - modifiable
- **Clinical Manifestation:** asymptomatic, headache, dizziness, blurred vision, epistaxis
- **Management**
 - To control BP - maintenance
 - Diet: low sodium, low fat, low cholesterol, low sugar
 - Exercise: 15-30 minutes every other day (minimum)
 - Avoid stress
 - Avoid smoking
 - Avoid alcohol
 - Restrict caffeine
 - Relaxation technique - to decrease cardiac workload
 - Deep breathing (improves circulation)
 - Guided imagery
 - Music therapy
 - Art therapy
 - Pet therapy
 - Massage
 - Aromatherapy
 - Drug therapy - anti-hypertensive drug
 - Primary adverse effects: hypotension
 - Secondary adverse effects: side effects
 - Agonist - stimulate
 - Antagonist - blockers
 - Increased SNS = increases BP
 - (1) Alpha 1 antagonist (BV) - vasodilator (-zosin)
 - (a) Avoid warm shower
 - (b) Avoid prolonged standing (orthostatic hypotension)
 - (2) Alpha 2 agonist (CNS) - decrease the flow of norepinephrine from the CNS (Clonidine, Methyldopa)
 - (a) Drowsiness
 - (3) Beta antagonist/blockers (-olol) - decrease HR, contractility
 - (a) Non-selective BB - blocks both beta 1 & 2 (lungs)
 - (i) CI: asthma, COPD
 - (ii) Propranolol, pindolol



(b) Selective beta 1 blocker - blocks beta 1

(i) Metoprolol, atenolol

- **RAAS**

(4) ACE inhibitors (-pril) - vasodilators

(a) Catopril, enalapril

(b) Side effects: angioedema, cough, elevated potassium

(5) Angiotensin II receptor blockers (-sartan)

(a) S/E: GI toxicity, elevated potassium

(6) Diuretics (thiazide)

(a) Hydrochlorothiazide

(7) Vasodilators- smooth muscles of blood vessels

(a) Direct-acting - directly relax the smooth muscles of the BV

(i) Hydralazine (apresoline)

(ii) Nitrates (sensitive to light and heat)

(b) Indirect-acting - decrease calcium in the smooth muscles

(i) Calcium - channel blockers

1) Short-acting (-ipine) - nifedipine, amlodipine, felodipine

2) Long-acting - diltiazem, verapamil

Coronary Artery Diseases/Ischemic Heart Disease

- **Coronary arteries** - blood supply of the heart

- **Left CA** - supply the anterior and lateral wall of the heart

- **Right CA** - supply the posterior and inferior wall of heart

- Most commonly affected of MI: anterior wall

- Atherosclerosis + thrombus -> ischemia (chest pain due to ischemia - angina)

- Common in elderly (atypical: confusion, epigastric pain, SOB)

Types of Angina

(1) **Stable/exertional angina** - increased cardiac workload

(2) **Unstable angina** - pre - infarction angina, due to decrease oxygen supply (severe atherosclerosis), unpredictable (irreversible)

(3) **Prinzmetal/variant type** - decreased oxygen supply, reversible due to coronary vasospasm but unpredictable

(4) **Intractable angina** - severe MI pain, excruciating pain, no objective sign except the presence of Levine's sign

(5) **Silent ischemia** - asymptomatic (traitor disease)



- **PQRST Assessment**

- **Position/location** - chest/substernal
- **Precipitating factors/provocation**
- **Quality** - constant, description (stabbing pain, crushing, heaviness)
- **Radiation** - left arm, neck, shoulder, jaw
- **Severity** - pain scale (subjective)
- **Timing/duration of pain**

Angina Pectoris	Myocardial Infarction
- Imbalance between oxygen supply and cardiac workload - Reversible - < 15 minutes - Relieving factors: rest, nitroglycerine	- Ischemia and necrosis of cardiac cells - Irreversible - > 30 minutes - No relieving factors

Angina Pectoris

- **Nursing Diagnosis**

- Ineffective myocardial tissue perfusion (priority)
- Acute pain
- Anxiety
- Ineffective health maintenance
- Non - compliance

- **Priority intervention:**

- **Stable angina:** rest, semi-fowler (independent)
- **Unstable angina:** nitroglycerine (dependent), oxygen (independent)
- **Prinzmetal angina:** nitroglycerine (dependent), oxygen (independent)

- **Definitive test:**

- Coronary angiography (invasive)
- Blood test
 - Elevated homocysteine level (risk for CAD)
 - Elevated C-reactive protein
- ECG - T wave inversion

- **Drug Therapy**

- (1) Nitroglycerine (acute - fast onset) - sublingual 3 times every 5 minutes
 - (a) Coronary vasodilator
 - (b) Peripheral vasodilator
- (2) Isosorbide nitrate (slow onset) - long-term treatment
- (3) Beta blockers - decrease heart rate
- (4) Calcium - channel blocker - peripheral vasodilator



- (5) Ranolazine - anti-anginal drugs (slow down electrical activity of the heart)
- **Procedure**
 - **Percutaneous transluminal coronary angioplasty/intervention (PTCA)/(PCI)**
 - Live a healthy lifestyle
 - Avoid straining

Myocardial Infarction

- **Nursing Diagnosis**
 - Acute pain (Morphine)
 - Ineffective myocardial tissue perfusion (Oxygen)
 - Anxiety
 - Risk for dysrhythmias (PVC - premature ventricular contraction/complex (6 or more per min)) administer anti-dysrhythmic drug as ordered
 - Risk for cardiogenic shock
 - Risk for decreased cardiac output
- **Definitive test:**
 - Angiography (left)
 - ECG (during the pain) - ST segment
 - Zone of infarction: ST segment elevation (early sign of acute MI)
 - Zone of injury: ST segment depression (myocardial injury)
 - Zone of ischemia: T wave inversion
 - Late/Old/Scar sign - pathologic Q wave
 - Cardiac markers
 - CK MB isoenzyme (most specific enzyme) 3 - 4 hours
 - Total CK/CPK
 - LDH
 - Troponin I protein (most important, reliable and sensitivity blood test) (lasts for 2 weeks) - 3 - 4 hours
 - Myoglobin (injury of muscle) - the first to elevate within 2 hours
- **2 types**
 - STEMI
 - NSTEMI - atypical sign (elderly)
- **Management**
 - Pain control (morphine)
 - Oxygen
 - Thrombolytic drugs
 - Anti - thrombotic drugs (blood thinners)
 - Anti - platelet
 - Anti - coagulant (heparin, warfarin)



- MONA + TASS (Thrombolytics, anticoagulant, stool softeners, sedatives)
- **Surgery**
 - PCI
 - CABG (Coronary Artery Bypass Graft) - general anesthesia/open heart surgery
 - Saphenous vein (most used)
 - Veins from peripheral vessels

Heart Failure

- Inability of the heart to pump effectively
 - Cardiac decompensation
- **Causes**
 - Cardiac
 - Non-cardiac: COPD (RHF)
- **Types**
 - **Left sided HF** - left ventricle fails, right ventricle will follow
 - Pulmonary edema
 - Progressive cough
 - Paroxysmal nocturnal dyspnea
 - Orthopnea
 - Dyspnea
 - Crackles/Rales
 - **Right sided HF** - right ventricle fails
 - Systemic edema
 - Ascites
 - Hepatomegaly
 - Splenomegaly
 - Jugular vein distention

NYHA classification of heart failure

Class I	No limitation of physical activity. Ordinary physical activity does not cause undue breathlessness, fatigue, or palpitations.
Class II	Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in undue breathlessness, fatigue, or palpitations.
Class III	Marked limitation of physical activity. Comfortable at rest, but less than ordinary physical activity results in undue breathlessness, fatigue, or palpitations.
Class IV	Unable to carry on any physical activity without discomfort. Symptoms at rest can be present. If any physical activity is undertaken, discomfort is increased.

- **Diagnostic test**
 - 2D Echo
 - Chest - X-ray
 - Blood test (BNP - brain natriuretic peptide (N) <100 pcg/ml)
- **Management: Airway/Breathing**
 - Position: High fowler's
 - Oxygen
 - IV line
 - Furosemide 40 mg IV STAT (DOC: pulmonary edema)
 - Catheterize patient
 - Monitor I&O
 - Monitor BP/VS
- **Management: Fluid Volume Excess**
 - Restrict fluid
 - Restrict sodium
 - Monitor I&O
 - Monitor VS
 - Weight daily (same time, same clothes, same weighing scale)
 - Diuretics (furosemide/spironolactone)
 - Monitor serum potassium level
- **Management: decreased cardiac output related to inability of the heart to pump effectively**
 - Goal: to increase cardiac output
 - Decrease cardiac workload
 - Provide rest
 - Avoid stress
 - Support cardiac function



- ACE inhibitors (captopril)
- Beta blockers (metoprolol)
- Angiotensin II receptor blockers (losartan)
- Calcium - channel blockers (diltiazem, verapamil)
- Increase cardiac contractility
 - Cardiotonic drugs
 - Phosphodiesterase inhibitors (milrinone, amrinone - IV through infusion pump) (+) inotropic effect (increased calcium in cardiac cells)
 - Peripheral vasodilation -monitor BP
 - Sympathomimetic (adrenergic agonist) - increased HR (+) chronotropic effect, (+) inotropic (dobutamine, dopamine)
 - Cardiac glycosides (digoxin, digitalis)
 - (+) Inotropic, (-) chronotropic
 - Nursing considerations:
 - Monitor HR prior to administration
 - Monitor serum potassium level (hypokalemia causes toxicity)
 - Monitor ECG
 - Maintain therapeutic level: 0.5 - 2 ng/ml
 - Do not combine with CCB
 - Do not combine with BB
 - Do not combine with amiodarone (anti-arrhythmic drug)
 - Monitor for signs of symptoms of toxicity
 - Bradycardia
 - GI toxicity
 - Visual disturbance
 - ECG changes
 - Withhold then refer
 - Antidote: digoxin immune Fab/digibind

Oncology

- A branch of medicine focused on the prevention, diagnosis, treatment and study of cancer
- **Oncology nursing** are as diverse and complex as those of any nursing specialty
- Nursing management of the patient with oncologic disorders includes care of patient throughout the cancer trajectory from prevention through end-of-life care



Cancer

- A disease process that begins when an abnormal cell is transformed by the genetic mutation of cellular DNA
- Uncontrolled cellular division
- **Cell Cycle:**
 - **Interphase:** longest phase
 - **G1:** cellular growth
 - **S:** DNA synthesis
 - **G2:** Preparation for mitosis
 - **Mitosis:** cell division (one-way phase, irreversible) - 2 separate daughter cells
 - **Karyokinesis:** nuclear division
 - **Cytokinesis:** cytoplasmic division (fluid portion of the cell)
 - **G0:** resting phase (cells that do not regenerate - example: neurons, sensory cells)
- **Carcinogenesis:** production of cancer, theory of cancer causation
 - **Initiation (transformed cell):** initial cell mutation occurs, spontaneous mutation, exposure to carcinogens, irreversible
 - **Promotion (Neoplastic lesion):** cancer cells stimulated to divide even further, reversible
 - **Progression (malignant tumor):** cancer cells compete with one another to survive, aggressive, malignant, poor prognosis
- **Oma:** tumor
 - **Carcinoma:** epithelial tissues (lining organs; glands); 80-90% cases
 - **Sarcoma:** connective tissues; osteosarcoma (bones); chondrosarcoma (cartilage); leiomyosarcoma (smooth muscle)
 - **Leukemia:** cancer of the blood
 - **Lymphoma:** cancer of lymphatic tissues, nodes
 - **Myeloma:** plasma cells (matured b cells); produced antibodies/immunoglobulin
- **Differentiation:** cells are transformed into different and more specialized cell types as they proliferate from a single stem cell
 - (1) **Hyperplasia:** irreversible increase number of cells in an organ or tissue, pre-malignant (BPH)
 - (2) **Metaplasia:** substitution of one cell with another, one cell type is converted/replace/transformed into another cell
 - (3) **Dysplasia:** deranged/bizarre cellular growth, complete alteration of cells in terms of size, shape, color, amount, etc., pre-malignant
 - (4) **Hypertrophy:** increase in cell size
 - (5) **Atrophy:** decrease in size
 - (a) Decrease oxygen, nerve, blood, nutrient supply
 - (b) Aging
 - (c) Prolonged immobilization



(d) decreased use

(6) **Neoplasia/Anaplasia**: automatic malignant

Characteristics	Benign	Malignant
Cell characteristics	Well-differentiated cells that resemble normal cells of the tissue from which the tumor originated	Cells are undifferentiated and often bear little resemblance to the normal cells of the tissue from which they arose
Mode of growth	Tumor grows by expansion and does not infiltrate the surrounding tissues; usually encapsulated	Grows at the periphery and sends out processes that infiltrate and destroy the surrounding tissues
Rate of growth	Rate of growth is usually slow	Rate of growth is variable and depends on level of differentiation; the more anaplastic the tumor, the faster its growth
Metastasis	Does not spread by metastasis	Gains access to blood and lymphatic channels and metastasizes to other areas of the body
General effects	Usually a localized phenomenon that does not cause generalized effects unless its location interferes with vital functions	Often causes generalized effects such as anemia, weakness, and weight loss
Tissue destruction	Does not usually cause tissue damage unless its location interferes with blood flow	Often causes extensive tissue damage as the tumor outgrows its blood supply or encroaches on blood flow to the area; may also produce substances that cause cell damage
Ability to cause death	Does not usually cause death unless its location interferes with vital functions	Usually causes death unless growth can be controlled

Diagnosis of Cancer

- A cancer diagnosis is based on assessment for physiological and functional changes and results of diagnostic evaluation



- (1) Determine the presence of tumor and its extent
- (2) Identify possible spread (metastasis) of the disease or invasion of other body tissue
- (3) Evaluate the function involved and uninvolved body systems and organs
- (4) Obtain tissue and cells for analysis, including evaluation of tumor stage and grading

Tumor Staging and Grading

- **Staging:** determines the size of the tumor and the existence of metastasis, identify the extent of involvement of cancer
 - **TNM system**
 - **T:** refers to the extent of primary tumor
 - Tx: primary tumor cannot be assessed
 - T0: no evidence of primary tumor
 - Tis: carcinoma in situ
 - T1, T2, T3, T4: increasing size and/or local extent of primary tumor
 - **N:** refers to lymph node involvement
 - Nx: regional lymph nodes cannot be assessed
 - N0: No regional lymph node metastasis
 - N1, NP2, NP3: increasing involvement of regional lymph nodes
 - **M:** refers to extent of metastasis
 - Mx: distant metastasis cannot be assessed
 - M0: no distant metastasis
 - M1: distant metastasis
 - Most common metastatic sites:
 - Lungs, liver, lymph nodes, brain and bones
- **Grading:** refers to the classification of the tumor cells (levels of differentiation)
 - **Grade 0:** grade cannot be assessed
 - **Grade I (low):** cells look relatively normal and well-differentiated (benign)
 - **Grade II (intermediate):** cells are moderately differentiated
 - **Grade III (high):** cells are poorly differentiated
 - **Grade IV (high):** cells are undifferentiated and are fast-growing (malignant)
 -

Cancer General Clinical Manifestations

- (1) **Pain:** a complex sensation that reflects both damage to the body and the body's response to the damage
 - (a) Acute pain: starts suddenly
 - (b) Chronic pain: lasts more than 6 months
- (2) **Bleeding:** a decrease in the circulating platelet count



- (3) **Infection:** streptococcus and staphylococcus species
- (4) **Anorexia-Cachexia Syndrome:** unexplained weight loss, wasted appearance

Warning Signs of Cancer

1. **Change in bowel or bladder movement/habit**
 - a. Alteration of diarrhea and constipation - colorectal cancer
 - i. Blood in stool (hematochezia)
 - ii. Dark starry stool (Melena)
 - iii. Change in the shape of stool (Pencil shaped stool)
 - b. Urinary frequency, urgency, hesitancy, dysuria (burning sensation), hematuria, obstruction - bladder cancer (cigarette smoking - strongest risk factor)
2. **A sore/wound that does not heal**
 - a. Oral cancer (risk factor: cigarette smoking, 30 y/o above)
 - b. Melanoma (most severe skin cancer)
 - i. ABCDE Assessment tool
 1. Asymmetry
 2. Irregular borders
 3. Color
 4. Diameter (N: <5 mm)
 5. Evolving
3. **Unusual bleeding or discharge**
 - a. Post menopausal bleed
 - b. Post coital bleed
 - c. Bleeding in between periods
 - d. Abnormal nipple discharge
 - e. Prolactinoma: benign pituitary gland tumor
 - f. Breast Cancer
 - g. Abnormal vaginal discharge (check color, smell)
 - h. Abnormal penile discharge
4. **Thickening of breast/lump or elsewhere (breast, testicles, lymph nodes)**
5. **Indigestion (dyspepsia)/difficulty swallowing (dysphagia) - head and neck cancer**
6. **Obvious change in wart/mole**
 - a. Elderly: scalp, face, behind the ear, nape
 - b. Adults: upper back
7. **Nagging/hoarseness**
 - a. Dry, persistent cough
 - b. Lung, laryngeal, head and neck cancer
8. **Unexplained anemia**
 - a. Colon cancer



9. Sudden weight loss (cachexia)

Cancer Prevention

- (1) **Primary Prevention:** health promotion and risk reduction strategies
 - (a) Assisting patients to avoid known carcinogens
 - (b) Adopting dietary and various lifestyle changes
 - (c) Immunization
 - (d) Limit consumption of alcohol beverages
- (2) **Secondary Prevention:** involves screening and early detection activities
 - (a) Mammography: to detect cyst and tumors that are non-palpable during physical exam
 - (b) Pap smear: for cervical cancer and HPV detection
 - (c) Prostate exam: for BPH
 - (d) Breast self-exam: do it every month, no BSE week before menstruation, only a week after menstruation
- (3) **Tertiary Prevention**
 - (a) Preventing recurrence
 - (b) Assessment for development of second malignancies

Management of Cancer

Surgery

1. **Diagnostic:** biopsy
 - **Fine needle aspiration:** solid tumors (breast, lungs, liver, thyroid, brain)
 - **Punch:** deeper tissue sample (skin, vulva)
 - **Shave:** superficial tissue sample
 - **Incisional:** very large tumors; portion of tumor removed
 - **Excisional:** entire tumor removed
 - **Bone marrow aspiration:** leukemia, lymphoma
 - Posterior iliac crest: most preferred site
 - Sternum/breast bone (avoid for pedia <12 years old)
 - Tibia: for pedia
2. **Primary treatment:**
 - Chemosurgery
 - Electrosurgery
 - Cryosurgery
 - Laser surgery



3. **Prophylactic surgery:** involves removing non-vital tissues or organs that are likely to develop cancer (risk reduction of symptoms)
4. **Palliative surgery:** when cure is not possible, the goals of treatment are to make the patient as comfortable as possible and promote a satisfying and productive life for as long as possible (improving the quality of life)
5. **Reconstructive surgery:** carried out in an attempt to improve function or obtain a more desirable cosmetic effect

Radiation Therapy

- **Radiosensitive cancer cells:** responsive to radiation
 - Normal cells: highly actively dividing cells
 - **Hair cells:** alopecia (improve self-esteem: offer wigs, scarves, hats); temporary hair loss (improve self-care: brush, shampoo, condition hair)
 - **GI tract cells:** stomatitis, oral mucositis, esophagitis (avoid alcoholic mouthwashes, avoid hot, rough, spicy, acidic foods: bland diet), radiation induced enteritis (diarrhea: low fiber diet)
 - **Blood cells:** Anemia (risk for fatigue, injury), leukopenia (risk for infection), thrombocytopenia (risk for bleeding) -> immunosuppression
 - **Skin cells:** radiation induced dermatitis (avoid wearing powders, lotions, cosmetics, etc., avoid extreme temperature -> promote dryness -> pruritus)
 - **Reproductive tract cells:** sterility, decreased libido
- **Radioresistant cancer cells:** difficult to kill; malignant
 - Normal cells: connective tissues, nerve cells, sensory cells, brain cells
- **External Radiation (Teletherapy):** more common than internal; patient is not radioactive; happens in sessions
- **Internal Radiation (Brachytherapy):**
 - **Sealed:** “implant”; localized; CBR w/o BP to avoid dislodgement; pick up the implant using long handled forceps and place it in a lead lined container -> refer; if not located: notify the safety radiation team
 - **Unsealed:** oral, IV; systemic; patient himself is radioactive, even body fluids; private CR; inform the patient to flush the toilet twice; 48 hours after radiation administration, patient is no longer radioactive
 - **DTS Principles**
 - **Distance:** 6 feet away; private room, away from the nurses station
 - **Time:** 30 mins/shift; cluster activities
 - **Shield:** Lead shield; dosimeter badge (indicator for radiation exposure)
 - Avoid elderly, pedia, pregnancy, immunocompromised visitors
- **Side effects:**
 - Alopecia

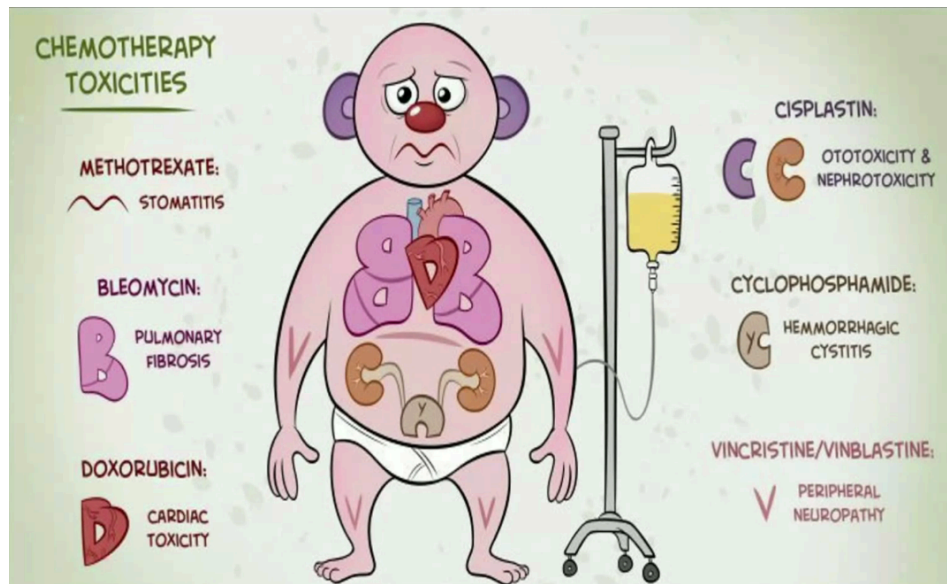


- Stomatitis
- Immunosuppression
- Dermatitis
- Sterility
- Anorexia
- Cachexia
- Fatigue (common of S/E)
- Nausea and vomiting

Chemotherapy

- Administration of antineoplastic agents to manage cancer
 - **Antineoplastic agents** are used in attempt to destroy tumor cells by interfering with cellular functions and reproduction (photosensitive)
- **Cell cycle specific (S Phase) - DNA Synthesis**
 - **Topoisomase I inhibitor:** Irinotecan, Topotecan
 - **Topoisomerase II inhibitor:** Etoposide, Teniposide
 - **Anti-metabolites:** Cytarabine, 5-fluorouracil, hydroxyurea, methotrexate
- **Cell cycle specific (M Phase) - Mitosis**
 - **Mitotic spindle inhibitors**
 - plant alkaloids: Vinblastine, Vincristine, Vinorelbine
 - Taxane: Paclitaxel, Docetaxel
- **Cell cycle non-specific** (all cell cycle - most dangerous)
 - **Alkylating agent**
 - Busulfan, Carboplatin, Cisplatin, Cyclophosphamide, Nitrogen mustard, Thiopeta
 - Nitrosureas
 - Carmustine, Lomustine
 - **Anti-tumor antibiotics**
 - Bleomycin, dactinomycin, daunorubicin, doxorubicin, epirubicin, idarubicin
 - **Hormonal agents**
 - Tamoxifen, Fulvestrant, Goserelin
- **Side effects:**
 - Alopecia (2nd)
 - Immunosuppression
 - Dermatitis
 - Sterility
 - Pulmonary toxicity
 - Nausea and vomiting (most common S/E)
 - Give ondansetron IV, metoclopramide IV push (30 minutes before chemotherapy)
 - Stomatitis

- Anemia, leukemia, thrombocytopenia (pancytopenia)
- Nephrotoxicity
- Neurotoxicity “chemo fog”
- Cardiac toxicity
- Infertility
- Peripheral neuropathy
- Cognitive impairment
- Fatigue
- Anorexia
- Cachexia
- **Before chemotherapy**, patients are stable and healthy



- **Extravasation** - infiltration of chemo agents; STOP THE INFUSION
 - **Irritants** - localized tissue damage
 - **Vesicants** - permanent tissue damage -> tissue necrosis -> amputation

Nursing Care for Patients with Cancer

Risk for infection

- Place patient in a private room
- Inform patient to avoid contact with people who have known or recent infection
- Instruct staff in careful hand hygiene before and after entering the room
- Avoid rectal or vaginal procedures, IMs, insertion of urinary catheter
- Encourage patient to ambulate in room unless contraindicated
- Assess IV sites for evidence of infection

Risk for Impaired Skin Integrity



- avoid the use of soaps, cosmetics, perfumes, lotions, powders in erythematous area
- Avoid rubbing or scratching skin
- Avoid applying hot water bottles, heating pads
- Avoid exposing the area to sunlight or cold weather
- Avoid tight clothing; use cotton clothing

Impaired Oral Mucous Membrane Integrity

- assess oral cavity daily using the same assessment criteria or rating scale
- Instruct patient to report oral burning, pain, areas of redness, open lesions on oropharyngeal mucosa and lips, pain associated with swallowing or decreased tolerance to temperature extremes of food
- Encourage and assist as needed in oral hygiene
- Advise patient to avoid irritants such as commercial mouthwashes, alcoholic beverages and tobacco
- Avoid irritating foods (acidic, hot, rough and spicy)
- Use water-based moisturizers to protect lips

Impaired Tissue Integrity: Alopecia

- Discuss potential hair loss and regrowth with patient and family; advise that hair loss may occur on body parts other than the head
- Explore potential impact of hair loss on self-image, interpersonal relationships and sexuality
- Explain that hair growth usually begins again once therapy is completed
- Prevent or minimize hair loss through the following:
 - Use mild shampoo and conditioner, gently pat dry, and avoid excessive shampoo
 - Avoid excessive combing or brushing; use wide toothed comb
- Suggest ways to assist in coping with hair loss:
 - Purchase wig or hairpiece before hair loss
 - Begin to wear wig before hair loss

Impaired Nutritional Status (Anorexia, Cachexia, Malabsorption)

- Assess and address factors that interfere with oral intake or are associated with increased risk of decreased nutritional status
- Initiate appropriate referrals for interdisciplinary collaboration to manage factors that interfere with oral intake
- suggest smaller, more frequent meals
- Educate patient to avoid unpleasant sights, odors and sounds in the environment during mealtime
- If patient desires, serve alcoholic beverage at mealtime with foods
- Consider cold foods if desired

Impaired Nutritional Status (Nausea and Vomiting)

- Assess the patient's previous experiences and expectations of nausea and vomiting, including causes and interventions used
- Adjust diet before and after drug administration according to patient preference and tolerance
- Prevent unpleasant sights, odors and sounds in the environment



- Administer prescribed antiemetics, sedatives and corticosteroids before chemotherapy and afterward as needed
- Ensure adequate fluid hydration before, during, and after drug administration; assess intake and output
- Encourage frequent oral hygiene

Fatigue

- Encourage balance of rest and exercise; avoiding extended periods of inactivity. At minimum, promote patient's normal sleep habits
- Encourage protein, fat, calorie intake at least equal to that recommended for the general public
- Encourage participation in planned exercise programs involving aerobic, resistance and flexibility training based on individual limitations and safety measures
- Encourage the use of relaxation technique and guided imagery

Chronic Pain

- Use pain scale to assess pain and discomfort characteristics: location, quality, frequency, duration at baseline and on the ongoing basis
- Assure patient that you know the pain is real and will assist them in reducing it
- Address myths or misconceptions and lack of knowledge about the use of opioid analgesics
- Collaborate with patient, primary provider, and other health care team members when changes in pain management are necessary
- Explore nonpharmacologic and complementary strategies to relieve pain and discomfort: distraction, imagery, relaxation, cutaneous stimulation, acupuncture

Grief

- Encourage verbalization of fears, concerns and questions regarding the disease, treatment and future implications
- Explore previous successful coping strategies
- Encourage ventilation of negative feelings, including projected anger and hostility within acceptable limits
- Involve spiritual advisor as desired by the patient and family
- Allow for progression through the grieving process at the individual pace of the patient and family

Disturbed Body Image

- Identify threats to patient's self-esteem, validate concerns with patient
- Encourage continued participation in activities and decision making
- Encourage patient to verbalize concern
- Assist patient in self-care when fatigue, lethargy, nausea, vomiting and other symptoms prevent independence
- Assist patient in selecting and using cosmetics, scarves, hair pieces, hats, and clothing that increase their sense of attractiveness



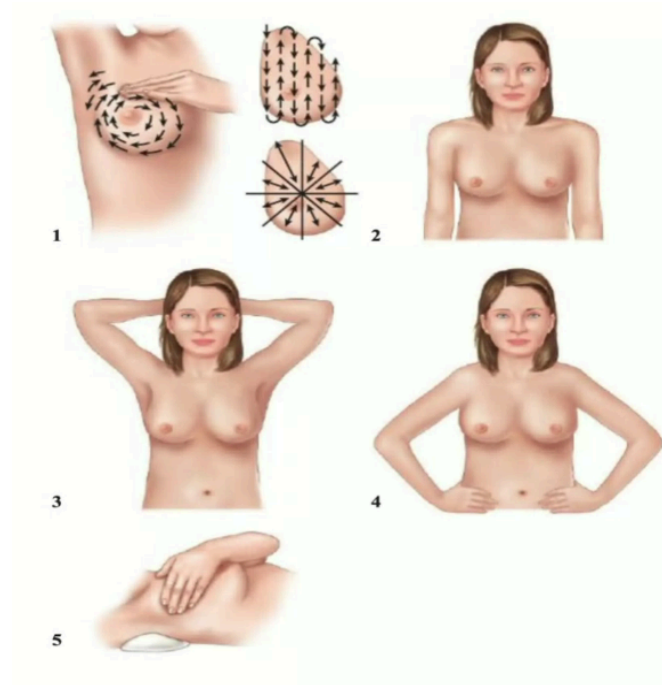
Common Cancers Worldwide

Breast Cancer

- Most common cancer worldwide
- Terminal duct lobular unit (site of milk release - mature cells)
- Upper outer quadrant
- **Risk factors:**
 - **Predisposing:**
 - Gender: 99% Females; 1% Males (BRCA 2 Gene, Klinefelter syndrome: XXY chromosome)
 - Age (55-65 y/o)
 - Family history
 - Race: Caucasians
 - **Precipitating:**
 - Previous cancer of the breast, uterus or ovaries
 - Obesity -> increased subcutaneous fats -> increased estrogen hormones
 - Nulliparity
 - Early menarche (<12 y/o)
 - Late menopause (>55 y/o)
 - Pills
 - High-dose radiation exposure to chest
 - Smoking
 - Alcohol drinking
- **Diagnostic Test:**
 - **Mammography** - initial screening test (X-ray of the breast)
 - 40-44 years old: option to start screening
 - 45-54 years old: required annually
 - >55 years old: biennially; every two years; every other year
 - High risk: on the age 30 years old annually
 - **Tumor Marker test** - CA 125; screening procedure only, unreliable; ovarian cancer
 - **Genetic testing** - BRCA 1 and BRCA 2 Gene (ovarian cancer, prostate cancer), HER 2 Gene -> HER 2 proteins
 - **Imaging test**
 - **CT Scan:** radio waves; fastest, least expensive
 - **MRI:** magnetic waves; provides detailed pictures than CT Scan; remove metallic objects (jewelries, implants), hearing aids; CI: pacemaker; if claustrophobic: sedate; braces are not magnetic (except head MRI)
 - **PET Scan:** radioisotopes; identifies metabolic abnormalities; most expensive



- **Ultrasound:** sound waves; drink 6-8 glasses of water without voiding (pelvic/abdominal only)
- **Biopsy** (confirmatory)
- **Signs and symptoms:** Upper outer quadrant
 - Non tender, firm, fixed, irregular nodules
 - Peau D'orange (Orange peel skin) - large pores near the areola
 - Abnormal nipple discharge, inversion, retraction, ulceration
 - Skin dimpling
 - Pain (severe)
- **Interventions:**
 - Radiation therapy
 - Chemotherapy
 - Progesterone therapy
 - Surgery
 - **Mastectomy**
 - **Modified radical:** removal of breast tissues, nipple areola complex and axillary lymph nodes; pectoralis major + minor intact
 - **Radical:** remove everything
 - **Total:** prophylactic purpose; benign - removal of breast tissues + nipple areola complex
 - **Lumpectomy:** everything is intact, tumor only removed
- **Post-operative complications:**
 - **Lymphedema:** semi fowlers with affected arm elevated above the level of the heart; avoid BP taking, venipuncture
 - **Hematoma formation:** Jackson-Pratt drainage (negative pressure suction)
 - **Infection:** Culture first, antibiotics after (7-14 days)
- **Post-operative interventions:**
 - Monitor VS
 - Encourage deep breathing
 - Assess for infection, color changes, bleeding in surgical site
 - Maintain F&E balance, diuretics and low salt diet for severe lymphedema
- **Breast Self Exam**
 - Pre-menopausal: 5-7 days after menstrual period
 - Post-menopausal: same day each month
 - Irregular: same day each month



- Stand in front of the mirror
- Circular motion outer to inner, up down motion or spiral motion
- If lying down, place a folded towel underneath the shoulders for even distribution of tissues across the chest wall

Lung Cancer

- Bronchogenic carcinoma (2nd common cancer)
- **Risk factors:**
 - >60 y/o; M>F
 - Family history
 - Cigarette smoking
 - Secondhand smoke
 - Inhalational chemical carcinogens
 - Environmental and occupational pollutants (silica dust, asbestos, radon gas)
- **Diagnostic Test:**
 - **Tumor Marker** - CEA (carcinoembryonic antigen) initial screening; colon cancer
 - **Imaging Test**
 - Low Dose CT Scan - 50-80 y/o annually (smokers); 20 smoking pack year history
 - 1 pack year = 1 pack (20 cigarettes per day for a year)
 - 1 pack /day for straight 20 years
 - 2 packs/day for straight 10 years



- **Bronchoscopic biopsy** (confirmatory test)
 - Pre-endoscopy: suppresses gag reflex (administer lidocaine spray)
 - Post-endoscopy: assess return of gag reflex
- **Clinical manifestations:**
 - Nagging cough (dry persistent cough) -> hemoptysis
 - Dyspnea -> fatigue -> anorexia -> cachexia
 - Nausea and vomiting, dysphagia, hoarseness, chest pain
 - Diminished or absent breath sounds, respiratory changes
- **Interventions:**
 - **Surgery:**
 - Thoracotomy: general surgical procedure; surgical incision into the thoracic cavity
 - Lobectomy: removal of a lung lobe; remaining lobes will re expand
 - Position: unaffected side to promote lung expansion
 - Pneumonectomy: removal of entire lung
 - Position: affected side to promote maximum lung expansion
 - Wedge resection: incision of a small part of of tissue (biopsy)
 - **Radiation therapy**
 - **Chemotherapy**
 - **Incentive Spirometry**
 - To prevent atelectasis
 - To strengthen intercostal muscles
 - 10-15x/waking hours
 - Inhale & suck
 - **Nursing management:**
 - Monitor VS, breath sounds and oxygen saturation
 - Fowlers position and administer oxygen PRN
 - High calorie, high protein, high vitamin diet
 - Administer bronchodilators, corticosteroids, analgesics as prescribed
 - Provide activity and rest periods as tolerated

Common Blood Cancers

Hodgkin's Lymphoma

- Rare; high cure rate
- **Risk Factors:**
 - 15-35 years old & >60 years old
 - Male



- Family history
- History of viral infections
- Long term use of immunosuppressants
- **Diagnostic test:**
 - Reed Sternberg Cells
 - PE: palpable lymph nodes
 - PET Scan
 - Bone marrow aspiration biopsy
- **Signs and symptoms:**
 - Painless, firm and large lymphatic nodules (cervical lymph node)
 - B symptoms:
 - Fever
 - Night sweats
 - Weight loss
- **Management:**
 - Chemotherapy

Non-Hodgkin's Lymphoma: absence of Reed Sternberg cells

Multiple Myeloma

- **Risk factors:**
 - >65 y/o, male
- **Diagnostic test:**
 - Bence Jones Proteins (urine) - WOF: infection, injures
 - Renal function test
 - Imaging test
 - X-ray
 - Bone marrow aspiration biopsy
- **Signs and symptoms: CRAB**
 - Hypercalcemia (encourage increased OFI)
 - Renal dysfunction
 - Anemia
 - Bone destruction (osteoporosis, multiple fractures)
- **Management: SAFETY is the PRIORIT**
 - Chemotherapy



Common Reproductive Cancers

Prostate Cancer

- **Prostate gland:** seminal fluid production; below the neck of the bladder
- **Risk factors:** lifestyle related
 - Unclear cause
 - Advancing age (>50 y/o; average 67 y/o); men
 - Heavy metal exposure
 - Red meat diet
 - Smoking
 - History of STI
 - Race: African American descent
- **Diagnostic test:**
 - Tumor marker: PSA (Prostate Specific Antigen)
 - Digital rectal exam
 - Imaging tests: rectal ultrasound
 - Biopsy
- **Clinical manifestation:**
 - Asymptomatic in early stage
 - Changes in bladder habits (painless hematuria, nocturia, urinary obstruction)
 - Pelvic/abdominal/back pain
- **Management:**
 - **Surgery**
 - **Prostatectomy:** removal of prostate gland
 - **Transurethral resection of the prostate (TURP):** the procedure involves insertion of a scope into the urethra to excise prostatic tissue
 - **Suprapubic prostatectomy:** removal of prostate gland by an abdominal incision with a bladder incision
 - **Retropubic prostatectomy:** removal of prostate gland by a low abdominal incision without opening the bladder
 - **Perineal prostatectomy:** the prostate gland is removed through an incision made between the scrotum and anus (risk for sterility, bladder incontinence)
 - **Continuous bladder irrigation**
 - Decrease bleeding and to keep the bladder free from clots
 - 1 lumen is for inflating the balloon (30ml)
 - 1 lumen is for instillation (inflow) - PNSS



- 1 lumen is for outflow
 - Maintain traction on the catheter, if applied. Instruct patient to keep the leg straight
 - Run the solution at a rate as prescribed to keep the urine pink (clear)
 - Avoid heavy lifting, stressful exercise, driving, the valsalva maneuver and sexual intercourse for 2 to 6 weeks to prevent strain and call the if bleeding occurs
-
- Radiation therapy
 - Chemotherapy
 - Androgen suppression therapy
 - Orchiectomy - removal of testicles

Testicular Cancer

- Testicles: sperm production
- One of the best prognosis: 95% chance recovery
- Thermoregulation: 3 degrees less than the body temperature for the sperm to survive
- Risk factors:
 - Young adult males 15-35/17-45 years old
 - Family history
 - Personal history
 - History of cryptorchidism (undescended testicles) - late pregnancy
 - Orchiopexy - surgical manipulation of testicles
 - Inguinal hernia
 - Klinefelter syndrome
- **Diagnostic tests:**
 - Tumor marker test: HCG, AFP, LDH
 - Testicular self exam (after warm shower, same day each month)
 - Gently roll and palpate using the thumb and forefinger
 - Hold the penis upward
 - Imaging test: ultrasound
 - Biopsy
- **Signs and symptoms:**
 - Painless testicular swelling
 - Sudden feelings of heaviness scrotum
 - Fluid collection scrotum
 - Pelvic, abdominal, scrotal pain
 - Low back pain (bone metastasis)



- Gynecomastia
- Management:
 - Radiation therapy or chemotherapy as needed
 - Testosterone Replacement therapy (oral, IM, topical)
 - Unilateral orchiectomy (risk for sterility); radical orchiectomy; retroperitoneal lymph node dissection
 - Post-operative interventions:
 - Monitor for signs of bleeding, infection, I&O, pain management
 - Instruct patient to avoid lifting and strenuous activity
 - Discuss reproduction, sexuality, and fertility information and options with the client

Cervical Cancer

- **Risk factors:**
 - Average: 52 y/o; 35-45 y/o
 - Family history
 - Human papillomavirus (HPV) infection
 - HPV types 6, 11: STI
 - HPV types 16,18: Cervical cancer
 - Cigarette smoking
 - Risky sexual behavior (early coitus <17 y/o)
 - Unprotected sex, multiple sexual partners
 - Multiparity
 - Poor utilization of pap smear, HPV vaccine
- **Diagnostic Test:**
 - Pap Smear: is not biopsy
 - Identify cervical lesions, inflammations
 - 25-65 y/o: with HPV test every 5 years
 - HPV test only: every 5 years
 - Pap smear only: every 3 years
 - Avoid sexual intercourse before pap smear (false positive)
 - Avoid douching, contraceptive creams for 2-3 days before pap smear (false negative)
 - Colposcopy: “acetowhite lesions”
- **Clinical manifestations:**
 - Unusual bleedings or discharge
 - Painless vaginal postmenstrual and postcoital bleeding
 - Foul-smelling or serosanguinous vaginal discharge
 - Pelvic, abdominal lower back, leg or groin pain, dyspareunia



- Leg edema
- Hydronephrosis
- Anorexia and weight loss
- Leakage of urine and feces from vagina (vesicovaginal fistula/rectovaginal fistula)
- Changes of bladder habits
 - Dysuria and hematuria, urinary obstruction
- **Management:**
 - **Surgery**
 - **Cryosurgery:** freezing the cancer tissues
 - **Conization:** cone-shaped cancerous area of cervix is removed; allows the woman to retain reproductive capacity
 - **Hysterectomy:** removal of whole uterus; discuss reproductive, sexuality, fertility issues
 - **Pelvic exenteration:** the removal of all pelvic contents, for recurrent cancer if no evidence of tumor outside the pelvis and no lymph node involvement exist
 - **Types of pelvic exenteration**
 - **Anterior:** removal of uterus, ovaries, fallopian tubes, vagina, bladder, urethra and pelvic lymph nodes
 - **Posterior:** removal of uterus, ovaries, fallopian tubes, descending colon, rectum and anal canal
 - **Total:** combination of anterior and posterior
 - **Radiation and Chemotherapy**
 - **Laser therapy:** boundaries of the lesions are visible during colposcopic examination
- **Prevention:**
 - Pap smear
 - HPV vaccine
 - 9-14 y/o - 2 doses; day 0 - 6-12 months
 - 15-45 y/o - 3 doses; day 0, 1-2 months, 6 months
 - Gardasil: HPV Type 6, 11, 16, 18
 - Cervarix: HPV Type 16, 18

Ovarian Cancer

- Clinical manifestations:
 - abdominal discomfort or swelling
 - GI disturbance
 - Dysfunctional vaginal bleeding
 - Abdominal mass
 - Elevated tumor marker (CA 125)
- Interventions:



- Radiation, chemotherapy
- Total abdominal hysterectomy and bilateral salpingo-oophorectomy (TAHBSO)

Endometrial (Uterine) Cancer

- Risk factors:
 - use of estrogen replacement therapy
 - Nulliparity
 - polycystic ovary disease
 - Increased age
 - late menopause
 - Family history of uterine cancer
 - Obesity
 - Hypertension
 - diabetes mellitus
- Clinical manifestations:
 - abnormal bleeding especially post menopausal women
 - Vaginal discharge
 - Low back, pelvic, or abdominal pain
 - enlarged uterus
- Interventions:
 - Radiation, chemotherapy, progesterone therapy
 - TAHBSO

Orthopedics

Musculoskeletal Nursing - focuses on the care of patients with conditions affecting the body's bones, muscles, joints and related soft tissues

Orthopedics Nursing - a specialized field focused on providing care to patients with musculoskeletal conditions and injuries

Structures & Functions

Muscles

- **Group of muscles:** Skeletal (striated), Cardiac (striated) & Smooth
- **Bundle/parallel groups of muscles cells** - Fascicle
- Fibrous tissue encases these Perinysium & Epinysium (Fascia)
- **Main responsibilities**



- Heat production (Shivering mechanism - adults/brown fats - newborn)
- Movement
- Posture
- **Skeletal Muscle Contractions**
 - Muscle cell/fiber
 - Sarcomere - contractile units
 - Effector (motor) nerve cells at motor end-plate
 - Action potential -> acetylcholine -> nicotinic muscarinic receptors on neuromuscular junction -> sodium influx -> release calcium from sarcoplasmic reticulum
 - Increased calcium
 - Sliding filament theory - myosin and actin slide across one another (contraction/depolarization)
 - H-zone - myosin
 - I-band - actin
 - A-band
 - Calcium decreased in sarcomere
 - Muscle relaxation
 - Without calcium, myosin and actin will cease to interact
- **Types of muscle contraction during movement**
 - **Isometric contraction** - length of muscle remains constant but force generated is increased
 - pushing against an immovable wall
 - **Isotonic contraction** - shortening of muscle without increase in tension within the muscle
 - flexion of forearm
 - **Concentric** - shortening + tension
 - **Eccentric** - lengthening + tension
- **Energy utilization**
 - Every contraction and relaxation > energy utilization
 - Main muscle energy source > adenosine triphosphate (ATP) > generated through cellular oxidative metabolism
 - Sedentary activity (low activity level) > oxidation of glucose to H₂O and CO₂ = ATP
 - Strenuous activity > insufficient oxygen > glucose (muscle glycogen) is metabolized to lactic acid > depletion of glycogen & accumulation of lactic acid > muscle fatigue

Bones

- Total: **206 bones**, 4 main types
 - Rods or shafts with rounded edges - **long bones**



- **Diaphysis** - shaft of the aforementioned bone and composition (Ewing's sarcoma)
- **Epiphysis** - ends of the aforementioned bone and composition (Osteosarcoma)
- **Epiphyseal cartilage** - cartilage separating epiphysis from diapiphysis, responsible for longitudinal growth in childhood
- **Articular cartilage** - cartilage covering ends of long bones at the joints (articulation)
- Bone marrow aspiration for infants
- Irregularly shaped bones in ankles and hands - **short bones**
- Bones where extensive protection - **flat bones**
 - Hematopoiesis (RBC bone production) - bone marrow - posterior iliac crest (flat bones)
- Bones such as vertebrae and jaw - **irregular bone**
- **Osteoblast** - bone cells for bone formation secreting bone matrix
- **Bone matrix** - made up of collagen, structural framework
- **Ossification** - calcium & phosphorus
- **Osteocytes** - bone signaling
- **Osteoclast** - bone resorption
- **Periosteum** - houses the nerves, blood vessels and lymphatics
- **Endosteum** - vascular membrane covering marrow cavity
- **Red Bone Marrow** - RBC production
- **Yellow Bone Marrow** - fats/adipose
- **Bone remodeling** - removing old bones

Bone Maintenance

- **Calcium:** 1000-1200 mg (adults)
 - Dairy (milk, cheese, yogurt)
 - Small fishes
- **Vitamin D (calcitriol)** - absorption of calcium from intestine & mineralization of osteoid tissue
 - 600 IU - young adults, 800-1500 IU - 50 y/o and up
- **Major hormones for bone mineralization and matrix formation**
 - **Parathyroid** - moves of calcium from bones to blood (parathyroid gland) - hypocalcemia
 - Demineral of bones and formation bone cyst
 - **Calcitonin** - moves deposition from blood to bones (thyroid glands) - hypercalcemia
- **Other factors influencing bone maintenance:**
 - **Thyroid hormones & steroids (glucocorticoids)**
 - Stimulates osteoclastic activity (bone weakening) - risk for fractures
 - **Aging**
 - Accelerates resorption: bone weakening
 - **Estrogen**
 - Osteoblastic activity (bone strengthening)



- **Growth hormone & Testosterone** - increases musculoskeletal health and growth

Articular System

- **Joints/Articulation** - junction of two or more bones
 - **Synarthrosis** - immovable joints/fibrous joints
 - **Amphiarthrosis** - allowed limited motion/catilogenous joints
 - **Diarthrosis** - freely movable joints/synovial joints
 - Types
 - **Ball-and-socket** - full freedom of movement (shoulder & hips)
 - **Hinge** - bends only in one direction (elbows - ulna)
 - **Saddle** - movement in two planes at right angle (base of thumb)
 - **Pivot** - turn doorknobs (elbow - radius)
 - **Gliding** - gliding against each other (wrist)
- **Joint Capsule** - fibrous sheath surrounding articulating bones
- **Ligaments** - bone to bone
- **Tendons** - bone to muscles
- **Synovium** - secretes lubricating and shock-absorbing synovial fluid
- **Bursae** - sac filled with synovial fluid

Diagnostic Tests

1. **Bone X-ray Studies** - determines bone density, textures, erosions and changes in bone relationship
 2. **Computed tomography** - 3D
 - a. more detailed corss-sectional image of body
 - b. indication: tumors, injury to soft tissues, ligaments, or tendons and severe trauma to chest, abdomen, pelvis, head, spinal cord
 - c. Contrast agent: iodine/seafoods
 - i. Assess for allergies
 - ii. Monitor serum creatinine (kidney function)
 - iii. NPO 4-6 hours
 3. **Magnetic Resonance Imaging**
 - a. uses magnetic fields and radio waves
 - b. better contrast resolution
 - c. Indication: torn muscles, cartilage, herniated discs, hip and pelvic
- **Interventions**
 - Preprocedure
 - Ci: pacemakers, implanted defibrillators, hip prosthesis, vascular clips
 - Remove metal objects



- Assess claustrophobia, administer medication (sedation - fentanyl)
- Explain MRI is noisy and may take 30-90 minutes (stay still)
- If with contrast agent: assess allergies and renal functionA
- Administer analgesics for pain
- HCp wears lead apron
- Post-procedure
 - May return to ADLs unless sedated and fully alert
 - Increased OFI
- 4. Arthrocentesis (Joint aspiration)**
 - a. Aspiration of synovial fluid, blood, or pus via needle inserted into joint cavity
 - b. relieves pain due to effusion
 - i. Synovial fluid - clear, pale straw color (rice straw), scanty in volume
 - c. Anti-inflammatories like corticosteroids may be instilled as necessary
- **Interventions:**
 - Before:
 - Informed consent
 - Remove hair as necessary
 - Antispasmodic may be given - decrease pain
 - After:
 - Bleeding: plastic compression + ice compression (vasoconstriction) + rest for 24-48 hours
 - Pain: analgesics
 - If with DM: monitor blood glucose
 - Notify PHCP if fever or joint swelling occur
- 5. Arthroscopy**
 - a. Endoscopic examination of joints
 - b. Repairs tears and remove loose bodies
 - c. OR Procedure
 - i. Insert large-bore needles
 - ii. Distend saline
 - iii. Insert arthroscope
 - iv. Close with adhesive strips or sutures + sterile dressing
- **Interventions:**
 - **Before:**
 - Informed consent
 - NPO 4-6 hours
 - **After:**
 - Monitor neurovascular status
 - Pain medications post-procedure
 - Elastic compression bandage for 2-4 days



- Walking: once sensation returns
- Swelling: Elevate extremities above heart level for 24 hours + ice compress for 12-24 hours
- Notify physician: fever or increased joint pain, drainage/edema for 3 days

6. Arthrography

- a. Inject of radiopaque contrast agent or air into joint cavity + joint performs range of motion + series of x-rays
 - i. Positive tear if agents leaks out of joint cavity

- Interventions:

- Tingling or discomfort (normal)
- Clicking or crackling (24-28 hours)
- Strenuous activity (avoid until approved)
- Elastic bandage + rest for 12 hours

7. Bone Densitometry

- a. Painless procedures which measure amount of minerals in bones
 - i. Dual-energy X-ray absorptiometry (DEXA Scan)
 1. Measures bone mass of spine, wrist, hip, total body
 2. Minimal radiation exposure
 3. Diagnose metabolic bone disease and monitor progress of treatment
 - ii. Quantitative ultrasound
 1. evaluates strength, density, and elasticity of bones using ultrasounds rather than radiation

8. Bone Scan

- a. Identify, evaluate and stage bone cancer before and after treatment
- b. Detect fractures or loosening of total joint replacement that x-rays don't show
- c. How:
 - i. Injection of IV radioisotopes
 - ii. Scan is performed 2 to 3 hours after
 - iii. Areas of abnormal bone formation - increased uptake and appear brighter - osteosarcoma, metastatic bone diseases

- Intervention:

- Before:

- informed consent
- NPO (-)
- Remove metals
- Assess allergies/pregnancy/breastfeeding
- Fluid intake & voiding: increased OFI & and to excrete radioisotope and for better visualization
- Position: Supine

- After:



- Radionuclide excretion: not harmful to others, increase OFI for 48 hours

9. **Bone or muscle biopsy** - detects malignancy

- **Intervention:**

- Informed consent
- Mild to moderate discomfort is normal
- Monitor for bleeding, swelling, hematoma, or severe pain
- Elevate site for 24 hours post procedure to reduce edema
- Ice pack post procedure to prevent hematoma and decrease discomfort
- Malignant hyperthermia - exposure to anesthesia (muscle rigidity, tachypnea, tachycardia, hyperthermia >40C)
 - Antidote: dantrolene (muscle relaxant)
 - Autosomal dominant

10. **Electromyography**

- Electrical potential of muscles and nerves
- Evaluate muscle weakness, pain and disability
- Differentiates muscle and nerve problems
- How:
 - Needles are inserted into muscle and recordings of muscular electrical activity traced on recording paper through oscilloscope

- **Interventions:**

- **Before:**

- Informed consent
- Avoid 24 hours (sedatives, stimulants)
- CI: infection, tumor

- **After:**

- Analgesia & warm compress

Musculoskeletal Trauma

Strain, Sprains & Contusions

1. **Strain (muscle pull)** - overuse /overstretch/stress of muscle or tendon

a. **Classifications:**

i. **First degree: mild + no loss of ROM**

- Slight overstretch or minor tear
- Mild pain, tenderness and swelling
- Little to no loss of strength or function

ii. **Second degree: moderate or partial tearing**

- Noticeable pain, swelling and bruising



2. Limited ROM and strength
3. possible instability of the affected area

iii. Third degree: severe with rupturing & complete tearing

1. immediate pain: sharp and intense, may lessen due to nerve damage
2. significant swelling, bruising, and muscle deformity
3. Complete loss of function and strength

b. Diagnosis:

- i. MRI or ultrasound
- ii. X-ray

2. Sprain

- a. Injury to ligaments and tendons surrounding joints due to twisting or hyperextension

b. Classifications:

- i. **Grade I** - stretching or slight tearing = mild localized hematoma
 1. Mild pain, edema, local tenderness
- ii. **Grade II** - partial tearing
 1. Increased pain with motion, edema, tenderness, joint instability, ecchymosis, partial loss of normal joint function
- iii. **Grade III** - complete tear or rupture; may also cause avulsion fracture
 1. severe pain, edema, tenderness, ecchymosis, abnormal joint motion

3. Contusion

- a. Soft tissue injury due to blunt force
- b. Small blood vessels rupture and bleed into soft tissue - ecchymosis or bruising
- c. more severe - black and blue appearance or hematoma
- d. Self-limiting: heals over time with conservative treatment

Interventions:

- Protection
- Rest
- Ice
- Compression
- Elevation
- NSAIDS

Dislocation

- Complete displacement of bone from its joints
- results in loss of joint function, pain, deformity and swelling
- Often requires manual reduction and immobilization

Subluxation

- Partial or incomplete dislocation
- Bone ends still maintain some contact
- Less severe than full dislocation, but may recur if ligaments are weak



Ligaments, Tendons, Menisci

Rotator Cuff Tears

- Rotator cuff attaches humerus scapula bone - allows arm lifting and rotation
- Rip in tendon connecting rotator cuff muscles to humeral head
- **Risk factors:**
 - Age
 - fracture
 - Overuse
 - acute or chronic stress
- **Assessment:**
 - Tenderness to palpation
 - Decreased ROM and strength
 - Aching pain worsening with use
 - Difficulty sleeping on affected side
- **Interventions:**
 - Conservative
 - Rest
 - NSAIDS & corticosteroids
 - Physical therapy
 - Surgical (Post-operative)
 - Immobilize - joint is immobilized with a sling or brace for several weeks to protect the repair
 - Rehabilitation - begins gradually with passive movements, progressing to active exercise to restore strength and function

Lateral & Medial Epicondylitis

- Tears and inflammation of tendons on epicondyles of the humerus
- **Lateral Epicondylitis (Tennis elbow)**
 - pain on the outside of the elbow
 - Involves proximal wrist extensor muscles
- **Medial Epicondylitis (Golfer's elbow)**
 - pain on the inside of the elbow
 - Involves proximal wrist flexor muscles
- **Interventions:**
 - Rest, support, ice compress and NSAIDS
 - Corticosteroids for severe pain
 - Physical therapy and surgery as necessary

Lateral & Medial Collateral Ligament Injury

- occurs when the knee experiences excessive force or twist often during sports or trauma
 - **Medially** - stretch and tear to medial collateral ligament (MCL)



- **Laterally** - stretch and tear to lateral collateral ligament (LCL)
- **Symptoms:** pain, tenderness, joint instability, and inability to walk without assistance
- **Intervention:**
 - P.R.I.C.E. + analgesics
 - Brace and/or use of crutches
 - Exercise to strengthen supporting muscles
 - Joint fluid aspiration (if swelling is significant)
- **Severe MCL**
 - May require surgical repair if completely torn or associated with other ligament injuries
 - followed by prolonged immobilization and structured rehabilitation
- **Severe LCL**
 - Often requires surgery due to poor blood supply and potential involvement of other structures
 - Post-op recovery includes immobilization, physical therapy and gradual return to activity

Anterior & Posterior Cruciate Ligament Injury

- Occurs when the knee experiences sudden twisting, hyperextension, or direct trauma such as in sports injuries or accidents
- ACL injury often results from sudden stop or changes in direction; PCL injury usually from a direct blow to the front of the knee
- **Interventions:**
 - **Early General**
 - P.R.I.C.E. + joint stabilization with a brace
 - Joint aspiration if swelling is present + elastic compression
 - Use of crutches to reduce weight bearing
 - **Surgical reconstruction**
 - tendon repair with grafting done arthroscopically to restore ligament function
 - **Post-surgery**
 - Oral analgesics and cryotherapy to control the pain and swelling
 - physical therapy focusing on progressive ROM and gradual weight bearing

Meniscus Injuries

- Two crescent shaped (semilunar) cartilages in the knee (right & left side of proximal tibia) acting as shock absorbers
- **Occurs when:**
 - Twisting or sudden pivoting motion of the knee
 - Direct trauma or impact to the knee
 - Repetitive squatting or heavy lifting
 - Degenerative changes in aging
 - Leads to swelling, pain and edematous knees (swollen knee)
- **Lateral tear (bucket-handle tear)**



- Large, displaced tear where a portion of the meniscus flips into the joint like a handle of a bucket
- Causes locking, clicking, or catching sensation in the knee
- Often requires arthroscopic surgery to repair or removal
- **Intervention:**
 - initial conservative
 - Immobilization + cryotherapy + crutches + anti-inflammatory + analgesia + activity modification
 - Surgical
 - Arthroscopic meniscectomy
 - Non-weight bearing for 4-6 weeks post-surgery to full weight bearing in a locked (extension) knee brace

Rupture of Achilles Tendon

- Attaches soleus and gastrocnemius muscles to calcaneus (heel bone)
- occurs when there is a sudden, forceful plantar flexion or overstretching of the tendon, often during sports or jumping
- common in:
 - men more than women
 - Age group: typically 30-50 years old
- **Diagnosis:**
 - Calf muscle test (thompson test): squeezing the calf causes no plantar flexion if tendon is ruptured
- **Interventions:**
 - **Nonoperative**
 - Casting + non-weight bearing for 2-8 weeks
 - Use of boot and functional brace with heel lift
 - Progressive physical therapy to restore strength and mobility
 - **Operative**
 - Indicated for complete ruptures, young active patients or failed conservative treatment
 - Postoperative casting followed by structural joint rehabilitation and physical therapy

Metabolic Bone Disorders

1. **Osteomalacia** - deficient activated vitamin D - inadequate bone mineralization = softening and weakening of skeleton (rickets in pedia)
 - a. **Risk factors:**
 - i. Insufficient vitamin D in diet/malnutrition
 - ii. Gastrointestinal disorders



- iii. Liver and kidney diseases
 - 1. Vitamin D -> hydroxylation -> liver -> calcidiol -> kidney -> calcitriol
- iv. Sunlight
 - 1. Skin (7 α -dehydrocholesterol) -> UVB (thermal energy) -> pre vitamin D3 -> Vitamin D3 (Cholecalciferol)
- v. Acidosis
 - 1. Calcium acts as buffers against acidosis
- vi. Hyperthyroidism
- vii. Prolonged use of anticonvulsants (phenytoin, phenobarbital)
 - 1. Increases vitamin D breakdown

b. Diagnosis:

- i. X-ray
- ii. Bone density tests
- iii. Laboratory tests: decreased calcium & phosphorus, increased PTH, ALP
- iv. Bone biopsy

c. Assessment

- i. Bowing of legs (waddling gait)
- ii. Kyphosis
- iii. Pathological fractures
- iv. Pain and tenderness to touch

d. Interventions

- i. Treat underlying cause
 - 1. Supplement vitamin D & calcium
 - 2. Sunlight
 - 3. Treat underlying diseases

2. Paget Disease (Osteitis Deformans)

- a. Idiopathic localized rapid bone turnover = skull, tibia, femur, vertebrae
- b. initially primary proliferation of osteoclast or osteoblast occur = bone resorption
- c. Compensatory increase in osteoblast
- d. Diseased bone is structurally weak and highly vascularized
- e. Mosaic pattern - Hallmark sign

f. Risk factors:

- i. >50 years old/Men
- ii. Family history

g. Assessment:

- i. Thickened skull or cranium
- ii. Impaired hearing (due to compression of cranial nerves)
- iii. Bowing for femur and tibia
- iv. Spine bending forward and rigid
- v. Thorax become immobile during respiration



- vi. Shortened trunk
- vii. Tenderness and warmth over bones

h. Diagnosis:

i. Laboratory results:

- 1. Normal: calcium
- 2. Elevated: alkaline phosphatase (increased during osteoblastic activity) for calcium deposition in bones

ii. X-ray

i. Interventions:

i. Pain - NSAIDS

ii. Gait

- 1. walking aids, shoe lifts, physical therapy
- 2. Control weight to reduce stress on weakened bones

iii. Diet - adequate calcium and vitamin D

iv. Pharmacology

- 1. Bisphosphonates (anti-resorptive agent)
- 2. IV Pamidracin (Mithracin) (cytotoxic antibiotics - inhibits PTH activity)

3. **Osteoporosis** - bone demineralization = loss calcium of and phosphorus salts = fragile bones = risk for fractures

a. **Primary** - occurs in premenopausal women (>50) and older men (decreased estrogen & testosterone)

b. **Secondary** - results from diseases or prolonged medications

i. Risk factors:

- 1. Alcohol intake
- 2. Corticosteroids
- 3. Smoking and alcohol
- 4. Family history
- 5. Inactive or sedentary lifestyle and inadequate calcium and vitamin D intake
- 6. Low body mass index & small frame
- 7. Malabsorption disorders
- 8. Men >60 y/o/ Women postmenopausal or >50 y/o
- 9. Asian and Caucasian

ii. Diagnosis:

1. DEXA Scan

a. T-score: postmenopausal and older men (primary)

- i. >-1.0 (N)
- ii. -1.0 to -2.5 (osteopenia)
- iii. <-2.5 (osteoporosis)

b. Z-score: premenopausal (secondary)



- i. <-2.0 (osteoporosis)

iii. Assessment

1. Asymptomatic until fractures begin
2. Pain: lower back pain after bending, sitting, stooping
3. Dowager's hump - kyphosis of dorsal spine
4. Problems with balance
5. Decreased in height

iv. Interventions:

1. Lifestyle
 - a. increased vitamin C: 1000-1200 mg/day
 - b. increased vitamin D: 400-1000 IU/day
 - c. Sunlight exposure: 15-30 minutes before 10am - 4pm
 - d. Proper body mechanics and weight bearing exercise daily
 - e. Avoid:
2. Safety
3. Pharmacological
 - a. **Bisphosphonates** - binds to hydroxyapatite receptor on bones -> increased uptake of osteoclasts
 - i. Side effects: nausea, diarrhea, constipation
 - ii. Adverse effects: esophageal or gastric ulcer
 - iii. Interventions
 1. Take early in the morning (at least 30 minutes before meals)
 2. Only take it with water
 3. Remain upright or avoid lying down for at least 30 minutes
 4. Take vitamin D and calcium supplements 4 hours after intake
 - b. **Estrogen Agonist/Antagonist (Selective Estrogen Receptor Modulator)**
 - i. Generic name: Raloxifene
 - ii. Indication: postmenopausal
 - iii. Side effects: hot flashes and leg cramps
 - iv. Adverse effects: venous thromboembolism (DVT, MI, CVA, Pulmonary embolism)
 - v. Interventions:
 1. PO OD in tandem with Calcium & Vitamin D
 - c. **Receptor Activator of Nuclear Factor Kappa-B Ligand (RANKL) Inhibitors** PTH -> binds to osteoblasts -> release RANKL -> binds to RANKL receptors on osteoclasts ->



osteoclastic growth and proliferation -> resulting in increased bone resorption

- i. Generic name: Denosumab
- ii. Indication: postmenopausal women
- iii. Side effects: skin rashes
- iv. Adverse effects: hypocalcemia
- v. Intervention:

1. Once every 6 months SQ

d. PTH Analogue -> stimulates osteoclastic activity -> compensatory bone formation -> increased in bone strength (low dose and short amount of time)

- i. Generic name: Teriparatide
- ii. Indication: postmenopausal and older men
- iii. Interventions:

1. stored in refrigerator
2. Administer SQ daily for 2 years

Generic Name	Indication	Nursing Consideration
Alendronate Risedronate	- Postmenopausal women - Men and women taking corticosteroids	- Drug holiday (1-2 year break for 4-5 years of treatment)
Ibandronate	- Postmenopausal women - intolerant of PO biphosphonates or non-adherent to therapy	- PO every month - IV every 3 months
Zoledronic Acid (strongest)	- postmenopausal women - Men and women taking corticosteroids for at least 12 months	- Assess for kidney function (serum creatinine) - Osteoporosis (IV every year) - Osteoporosis risk (IV every 2 years)

Fracture

- A break in continuity of bone due to stress greater than it can absorb; caused by:
 - Trauma
 - twisting
 - Bone disease or degeneration
- **Assessment:**



- Pain or tenderness
- Decrease or loss of muscular strength or function
- obvious deformity and shortening
- Crepitation
- localized edema and ecchymosis
- **Diagnosis:**
 - X-ray
 - MRI or arthroscopy
- **Types by exposure**
 - **Closed or simple** - skin over fracture remains intact
 - **Compound or open** - bone is exposed to air through break in skin
- **Types by extent of break**
 - **Complete** - bone is separated completely, breaks into two parts
 - **Incomplete** - fracture line does not extend through full transverse width of bone
- **Types by pattern**
 - **Comminuted** - bone is shattered or crushed, creates numerous fragments
 - **Compression** - fractured bone is compressed by other bones
 - **Depressed** - bone fragments are driven inward, commonly seen in skull fractures
 - **Greenstick** - one side of the bone is broken and the other is bent, occurs mostly in children
 - **Impacted** - part of the fracture bone is driven into another bone
 - **Oblique** - fracture lines run at an angle across the axis of the bone
 - **Pathological/Spontaneous** - results from weakening of bone structure by pathological processes
 - **Spiral** - break partially encircles the bone caused by twisting bone
 - **Stress** - occurs in normal or abnormal bone exposed to repeated stress, common in athletes or military recruits
 - **Transverse** - bone is fractured straight across (horizontal line across the bone)
- **Emergency management (nonsurgical)**
 - Immobilize immediately + apply splinting to prevent further injury
 - Open fracture
 - Cover wound with sterile dressing and splint the limb
 - Do not attempt to manually reduce the fracture outside of the hospital setting
 - In the ED, remove or cut the clothes around the injury and assess and treat
- **Surgical management**
 - Fracture reduction - restoration of bone fragments to anatomic realignment and positioning with immobilization
- **Interventions:**



- Consent
- Analgesia
- Open fractures are usually left unclosed and treated with VAC (Vacuum-Assisted Closure)

Fracture Interventions

Reduction

- **Closed reduction**
 - Bringing bone fragments into anatomic alignment through manipulation and manual traction
 - Consent
 - Anesthesia
 - Anatomical alignment
 - Immobilizing device
 - X-rays
- **Open reduction**
 - Fracture fragments are aligned surgically and stabilized using fixation devices
 - Internal fixation devices
 - Provide immediate bone stabilization
 - Common methods: pins, plates, screws, wires, nails, or rods
 - Removal
 - May be temporary or permanent
 - Removal is considered if:
 - There is infection, loosening or irritation
 - Growth interference in children
 - Pain or functional impairment
 - Applications
 - Used for complex, unstable fractures
 - Often applied in fractures of the hip, femur, humerus or forearm
 - Promotes early mobilization and faster recovery
 - **External fixation devices**
 - Metal pins or screws are inserted into bone and connected to an external frame
 - Allow more freedom of movement than traction
 - Useful in cases of massive tissue trauma, open fractures, bone lengthening, or infection
 - Applications



- Severe comminuted or open fractures
 - Fractures with extensive soft tissue damage
 - Situations where internal fixation is contraindicated: infection, poor skin integrity
- **Post op care:**
- Edema - elevation and ice compress
 - Isometric and muscle setting exercises

Traction

- Can be short-term or long term
- The exertion of a pulling force applied in two directions to reduce and immobilize fractures
- **Goals:**
 - Realign fracture fragments
 - Correct or prevent deformity
 - Decrease muscle spasm and pain
- **Principles of effective traction**
 - Whenever traction is applied, countertraction should also be used = provides a force acting in the opposite direction (usually the patient's body weight or bed elevation)
 - Traction must be continuous
 - Skeletal traction is never interrupted
 - Skin traction may be removed for skin care but must be reapplied promptly
 - Weights should not be removed unless prescribed
 - Good body alignment must be maintained at all times
 - Ropes must be unobstructed (move smoothly and not fray)
 - Weights must hang freely, never rest on the bed or floor
 - Knots in the rope or foot plate must not touch the puky or foot of the bed (prevents interference with traction pull)
- **Types of traction:**
 - **Skeletal traction**
 - Provides continuous traction for fractures of the femur, tibia and cervical spine
 - Applied mechanically to the bone using pins, wires or tongs
 - Typical weight: 25-40 lbs (11-18 kg) depending on the fracture and patient size
 - How:
 - Maintain surgical asepsis
 - Perform surgical scrub using chlorhexidine solution
 - Infiltrate local anesthetic at insertion site and periosteum
 - Make a small skin incision -> drill sterile pin or wire through bone
 - After insertion, pin, or wire is attached to traction bow or caliper
 - Cover ends of pin or wire with protective caps



- Attach weights to pin or wire bow via rope and pulley system
- Initial weights must overcome shortening spasms then gradually reduced to prevent fracture dislocation and promote healing
- Skeletal traction care
 - Pins in areas with soft tissue are at greatest risk for infection
 - Goal: avoid infection -> prevent osteomyelitis
 - Interventions:
 - Inspect pin sites every 8 hours
 - Inflammatory signs (expected): redness, warmth, serosanguineous drainage -> usually subside after 72 hours
 - Infective signs (abnormal): purulent drainage, pin loosening, tenting of skin, malodor, fever
 - Prophylactic IV antibiotics may be given 24-48 hours post-insertion
 - First 48 hours: cover site with sterile absorbent non-stick dressing and secure with rolled gauze or ace bandage
 - After 48-72 hours: perform pin site care daily or weekly
 - Clean using chlorhexidine (2 mg/ml) solution; if contraindicated, use sterile saline solution
- **Skin traction**
 - Relief of muscle spasm or immobilization before surgery
 - Applied using elastic bandages, adhesive, velcro, tape, straps, foam boot or sling
 - Type of skin traction
 - Cervical skin traction: alleviate chronic neck pain and upper extremity symptoms
 - Use a head halter and chin pad to attach traction
 - Apply powder to protect ears from friction rub
 - Position with head of bed elevated to 40 degrees
 - Attach weights to a pulley system over head of bed
 - Buck's ((extension) traction: alleviate muscle spasms and immobilize lower limb by maintaining a straight pull
 - Apply boot appliance and attach weight to a pulley, hanging freely over edge of bed
 - Ensure weight is 2-3.5 (4.5-8 lbs) only
 - Elevate foot of bed to provide countertraction
 - Russell's skin (sling) traction: uses sling to raise lower leg and maintain the knee in a flexed position while exerting pull on both knee and hip
 - Pelvic skin traction: relieve low back, hip, or leg pain; reduce muscle spasm



- Method: traction belt snugly over pelvis and iliac crest, attached to weights (approximately 4.5-9 kg/10-20 lbs)
- Balanced suspension traction
 - Provides countertraction using client's body or an additional system
 - Allows greater mobility and alignment compared to straight/running traction
- Skin traction care
 - Skin breakdown
 - Number of nurses: 2 nurses when removing and reapplying appliances
 - Removing appliance: at least once every 8 hours for skin inspection and care
 - Repositioning: do not turn side to side -> use draw sheet to lift, not drag
 - Bed type: firm mattress or overhead frame with pressure-relieving devices
 - Never damage (Buck's traction focus)
 - Assess dorsiflexion and plantar flexion
 - Dorsiflexion -> integrity of peroneal nerve -> deficit = footdrop and inversion
 - Plantar flexion -> integrity of tibia nerve -> deficit = foot eversion
 - Assess for alterations in sensation under appliance = burning or tingling -> notify physician immediately (possible nerve compression)
 - Circulatory impairment
 - Compartment syndrome - assess 5P's: pain, pallor, pulselessness, paresthesia, paralysis
 - Deep vein thrombosis (DVT): look for unilateral calf tenderness, warmth, redness and swelling

Cast

- Rigid external immobilizing device molded to contours of body
- Permits mobilization while restricting movement of affected part
- Utilizes plaster, fiberglass or air casts
- **Types of casts by location:**
 - **Arm casts**
 - Below elbow to palmar crease



- Secured around base of thumb
- Thumb spica or gauntlet cast (includes thumb)
- **Long arm casts**
 - Above forearm to proximal palmar crease
 - Elbow is immobilized at right angle
- **Leg casts**
 - **Short leg cast**
 - Below knee to base of toes
 - Foot is fixed at right angle in neutral position
 - **Long leg cast**
 - Junction of upper and middle third of thigh to base of toes
 - Knees may be slightly flexed
 - **Walking cast or boot**
 - Short or long leg cast reinforced for strength
- **Body or spica casts**
 - **Body cast**
 - Encircles the trunk
 - **Shoulder spica cast**
 - Body jacket enclosing trunk, shoulder, and elbow
 - **Hip spica cast**
 - Encloses trunk and lower extremities
 - Double hip spica (includes both legs)
- **Cast care**
 - **Plaster of paris**
 - **Drying**
 - Handle with palms not fingertips
 - Use hair dryer in cool setting to dry
 - Turn extremity every 2-4 hours
 - **Skin breakdown**
 - Petal or apply moleskin to edges
 - **Circulatory impairment**
 - Bluish and cold color
 - **Infection and pressure spot**
 - Monitor for increased temperature, hot spots, foul odor, pain
 - Window if necessary
 - **Fiberglass**
 - Gore-Tex - waterproof lining permitting patient to get wet
 - If wet:
 - Dry by blotting or patting with towel
 - Or sweeping motion with blow dryer on cool setting for 20 minutes



Cast Complications

Compartment Syndrome

- Medical, painful emergency condition mostly in lower leg
- Pressure increases within one or more compartments, usually in extremities = decreased blood flow -> tissue ischemia and neurologic impairment
- Common among young adults; may take up to 48 hours for symptoms to be present
- Neurologic damage may be irreversible if untreated within 4-12 hours
- **Assessment (6P's):**
 - Pain (severe, unrelieved by opioids, disproportionate)
 - Pallor
 - Poikilothermia (coolness)
 - Paralysis (loss of movement)
 - Paresthesia (tingling/numbness)
 - Pulselessness (late sign)
- **Intervention:**
 - Univalve or bivalve the cast/dressing
 - Fasciotomy (anterolateral or posteromedial removal of fascia)
 - Wound left unclosed for 3-5 days
 - Cover with moist, sterile saline dressing or aseptic sheet
 - Negative-pressure wound therapy (VAC)
 - Splint in functional position + elevate + passive ROM
 - Once resolved, debride + close with skin graft

Pressure Ulcer

- Pressure on soft tissues causing anoxia and pressure ulcers
- Typically reports very painful "hot spot" and tightness under cast
- May feel warmth in affected area
- Drainage may stain cast and emit unpleasant odor
- Extensive loss of tissue with skin breakdown and necrosis
- Intervention:
 - Windowing - ensure prevention of window edema

Disuse Syndrome

- Muscle atrophy and loss of strength = deterioration of body system
- Intervention:
 - Isometric muscle contraction exercises/muscle setting exercises



- **How:**
 - Arm
 - Contract and relax arm muscles against resistance
 - Quadriceps & gluteal setting exercise
 - Position: supine (lying flat)
 - Push knee back onto mattress
 - How? Contract anterior thigh muscles
 - Or contract muscle of buttocks
 - Hold for 5 to 10 seconds
 - Relax and repeat 10 times each hour when awake

Fracture Complications

Fat Embolism Syndrome

- Movement of fat globules/foreign particles into pulmonary circulation
- Occurs within 24 - 72 hours of injury; mostly in males (10-40 years old) with closed fractures
- **Assessment (classic triad):**
 - Hypoxemia
 - Neurologic changes
 - Petechial rash (upper body, conjunctiva, oral mucosa)
 - Sudden chest pain
 - Dyspnea/ difficulty breathing
 - Restlessness and confusion
 - Coughing with bilateral crackles
- **Diagnosis**
 - Clinical suspicion (triad) + imaging (CXR, CT)
- **Intervention (supportive)**
 - Immobilization of fracture
 - Oxygenation (humidified O₂)
 - IV fluids to maintain perfusion
 - IV anticoagulant (heparin)
 - Intubation + mechanical ventilation if ABG indicates severe hypoxemia

Infection & Osteomyelitis

- Introduction of microorganisms into bone -> inflammation and localized bone infection
- **Assessment**
 - Tenderness and fever



- Edema and pain
- Leukocytosis and elevated ESR
- **Intervention**
 - Aggressive, long term IV antibiotics via central venous access
 - Surgery for resistant infection with sequestrectomy and/or bone grafts
 - Sequestrum = fragment of dead bone that has separated
 - Debridement with sterile irrigating solution in OR
 - Hyperbaric oxygen therapy (HBOT) for unrelenting infection

Avascular Necrosis (Osteonecrosis)

- Loss of blood supply -> bony collapse and joint destruction
- Common in: femoral head, scaphoid, talus, humeral head
- **Assessment:**
 - Pain with movement -> progresses to pain at rest
- **Diagnosis:**
 - X-ray, CT scan, bone scan
- **Interventions:**
 - Joint preservation approach
 - Nonoperative: activity modification (reduced weight bearing)
 - Surgical:
 - Drilling avascular segment (core decompression) or bone marrow graft
 - Remove necrotic fragment + joint reconstruction

Complex Regional Pain Syndrome

- Disproportionate regional pain in limb following fracture, soft tissue injury, or surgery
- Chronic; may extend to adjacent areas
- Common in women > 40 years
- **Types:**
 - **Type I** - reflex sympathetic dystrophy (no confirmed nerve injury)
 - **Type II** - causalgia (confirmed nerve injury)
- **Assessment:**
 - Severe burning pain
 - Vasomotor skin changes (mottled, shiny skin)
 - Hyperesthesia (increased sensitivity)
 - Local edema
 - Stiffness
- **Diagnosis:**
 - Physical history and exam + ruling out other causes



- **Interventions:**

- Pain relief: NSAIDs, opioids, topical anesthetics, corticosteroids
- Neuropathic pain: anticonvulsants, antidepressants
- Sympathetic nerve blocks, neural stimulation
- Novel treatment: IVIG infusion, ketamine, hyperbaric oxygen therapy
- Psychiatric nursing support + rehabilitation

Hip Fracture

- **Risk factors**

- Age > 65 years old
- Women (osteoporosis risk)
- Higher risk for falls

- **Types:**

- **Intracapsular** - fracture of femoral head and neck
 - High risk of vascular damage on neck of femur = ischemia -> avascular necrosis
- **Extracapsular** - fracture outside joint capsule = base of femoral neck to distal lesser trochanter

- **Assessment:**

- Pain over outer thigh or groin
- Discomfort with hip flexion and rotation
- Shortening, adduction, external rotation

- **Diagnosis:** X-ray

- **Intervention:**

- **Intracapsular fracture** - broken femoral head within joint capsule
 - Preoperatively: reduce fracture and decrease spasms with skin traction
 - Surgical: hip replacement (hemi/total arthroplasty) or ORIF
 - Postoperatively: prevent dislocation
 - Avoid hip flexion >90 degrees
 - Avoid crossing legs or pointing toes inward
 - Ensure correct positioning
- **Extracapsular fracture** - fracture outside joint capsule
 - Preoperatively: reduce pain and spasms with skin traction or balanced suspension
 - Surgical: ORIF with nail plate or intramedullary fixation
 - Postoperatively:
 - Institute safety precautions
 - Maintain leg and hip alignment
 - Prevent internal or external rotation
 - Turn only to unaffected side if needed
 - Elevate HOB 30 to 45 degrees



- Educate on walker use to avoid weight bearing
- Keep operative leg extended, straight, elevated -> prevents hip flexion >90 degrees
- Avoid low chairs, crossing legs, bending over
- Use anti-embolism stockings or sequential compression devices
- Encourage flexing and extending of feet

Total Hip Arthroplasty

- Replacement of severely damaged hip with an artificial joint
- **Indications:**
 - Femoral fracture
 - Osteoarthritis
 - Rheumatoid arthritis
 - Avascular necrosis of hips
- **Interventions (postoperative):**
 - Maintain femoral head in acetabular cup for approximately 4 months
 - Avoid hip full flexion, adduction and internal rotation
 - Position:
 - Supine
 - Abduction splint = wedge pillow or two pillows between legs
 - Continuous boot = prevents leg rotation and prevents pressure ulcer on heels
 - Never flex hip >90 degrees
 - Movement: turning or bedpan
 - Keep operative hip in abduction
 - Don't turn on operative side
 - No bending forward
 - High seat (orthopedic) chairs
 - Hips higher (not lower) than knees
 - No leg crossing
 - Allowed to sit for <1 hour only
 - Activity resumption
 - Stair climbing and driving resumes within 3-6 months
 - Sexual activity resumes at about 3 months

Total Knee Arthroplasty

- **Interventions (postoperative):**
 - **General**



- Compression bandage and cold packs
- Assess neurovascular status every 2 hours
- **Infection**
 - Monitor drainage and infection
 - Wound suction drain for 24-48 hours
 - Prophylactic antibiotic for 24 hours
- **Mobility**
 - Ambulate by first postoperative day
 - Continuous passive motion started immediately if prescribed or active flexion of foot if awake = give analgesics
 - Avoid leg dangling, knee gatch position (flexed knee) and pillows behind knees
 - Protect with knee immobilizer and elevate legs when sitting
 - Administer tranexamic acid to reduce bleeding
- **Rehabilitation**
 - Acute rehab: 1-2 weeks
 - Total recovery: approximately 6 weeks
 - Prosthesis may last 15-20 years

Intervertebral Disc Herniation

- Nucleus of disc protrudes into annulus -> nerve compression
- Cervical disc herniation: most common at C5-C6 and C6-C7
 - **Assessment**
 - Pain and paresthesia
 - Neck, top of shoulders, scapula
 - Arm, hand, pectoral muscles
 - Myelopathy - bilateral arm and leg weakness
 - Lhermitte's sign - electric-like shock in extremities or spine with neck flexion
 - **Diagnosis:** cervical MRI
 - **Intervention**
 - Conservative management
 - Bed rest
 - Ice and heat compress -> hot, moist compress (10-20 mins) to back of neck, several times/day
 - Maintain head alignment
 - Pharmacological
 - Muscle relaxants
 - NSAIDs and corticosteroids
 - Head alignment



- Cervical collar: limits neck movement and holds head neutral or slightly flexed
 - Wear intermittently or 24 hours/day
 - Inspect skin for irritation
 - After pain decreases -> cervical pain exercise to strengthen muscles
- Other interventions
 - Minimize long sitting/standing (<30 mins)
 - Avoid neck twisting, flexion, extension, rotation
 - Avoid sleeping in prone; maintain neutral position (neck, spine, hips) with pillows to prevent flexion
 - Place wrinkle-free sheet under collar
- **Lumbar disc herniation:** most common at L4-L5 or L5-S1 interspace
 - **Assessment**
 - Pain and spasms
 - Relieved by bed rest
 - Aggravated by movement, lifting, straining, coughing
 - Low back pain
 - Radiating into one hip and leg (sciatica)
 - Straight-leg raising test = pain radiates when leg is raised
 - **Interventions**
 - Conservative management
 - Sleeping: firm mattress
 - Pelvic traction as prescribed
 - Avoid lifting above elbow level
 - Corset or brace for support
 - Ice and heat compress
 - Progressive ambulation once inflammation/edema/pain subside
 - Analgesics, sedatives, NSAIDs, corticosteroids
 - Proper posture in sitting, standing, walking, working
 - Weight control and exercise program
- **Disc surgery:** indicated when conservative treatments fail
 - Discectomy
 - Discectomy with fusion
 - Laminectomy
 - Laminotomy
- **Post-operative interventions**
 - **Cervical**
 - Assess/manage dressings and drains
 - Monitor respiratory difficulty -> hoarseness and inability to cough properly
 - Encourage coughing, deep breathing, early ambulation



- Throat sprays or lozenges for sore throat; avoid anesthetic lozenges (mask swallowing difficulty)
- Soft diet if dysphagia
- Watch return of radicular pain
- **Lumbar**
 - Assess/manage dressings and drains
 - Monitor lower extremities for neurovascular complications
 - Monitor for urinary retention, paralytic ileus, constipation
 - High fiber diet, fluids, stool softener as prescribed
 - Opioids and sedatives for pain/anxiety
 - Back brace or corset + cotton underwear to prevent skin irritation
 - Immediately lie supine post-op + pillow under neck and slightly flex knees
 - Avoid spinal flexion or twisting
 - Minimize sitting + avoid extreme hip flexion when lying on side
 - Avoid heavy work for 2-3 months

Amputation

- Surgical removal of a limb or part of a limb
- **Post-operative interventions**
 - Evaluate for phantom limb sensation and pain
 - First 24 hours: elevate foot of bed (to reduce edema, not the stump -> prevents contracture)
 - After 24 to 48 hours: place in prone position (to prevent hip flexion contracture, especially in above-knee amputation)
- **Types of amputation**
 - **Below-knee amputation (BKA)**
 - Prevent knee flexion contracture
 - Encourage extension exercises and prone positioning
 - **Above-knee amputation (AKA)**
 - Prevent hip flexion contracture
 - Prone position several times a day
- **Emergency amputation (traumatic)**
 - Obtain emergency medical assistance (call 911) and stay with the victim
 - Apply direct pressure with gauze or cloth + elevate injured part
 - If finger(s) amputated:
 - Place in waterproof sealed plastic bag
 - Then place the bag in ice water (not directly on ice -> prevents tissue damage)
 - Transport to emergency department with victim



Assistive Devices

- Crutches

- Principles

- Client

- Distance between axilla and axillary pad on crutch should be 2-3 finger widths in axilla
- Elbow should be slightly flexed = 20-30 degrees
- Should look up and outward when ambulating
- Place crutches 6-10 inches diagonally in front of foot
- Stop ambulating if with numbness or tingling in hands or arms

- Nurse

- Stand on affected side when ambulating with client

- Crutch measurement

- If standing: set crutch 5 cm (2 in) below axilla
- If lying down: measure from axillary fold of axilla to sole + 2.5 cm (1 in)
- If using patient's height: subtract 40 cm (16 in), adjust hand grip to allow 20-30 degrees elbow flexion

- Crutch gaits

- 4-point gait: partial weight bearing both feet; provides maximal support; constant shift of weight
- 2-point gait: partial weight bearing both feet; provides less support; faster than 4-point gait
- 3-point gait: non-weight bearing; requires good balance and arm strength; faster gait
- Swing-to-gait: for partial weight bearing limitations with bilateral lower extremities prostheses
- Swing-through gait: requires adequate muscle power and balance in arms and legs

- Crutch movement

- Sitting and standing

- Place unaffected leg against front of the chair
- Move crutches at unaffected side
- Grasp arm of chair with hand of affected side
- Flex knee of unaffected leg to lower self into chair while placing affected leg straight out in front
- Going up stairs "Up with the GOOD" (move unaffected leg first)
- Going down stairs "Down with the BAD" (move affected leg first)

- Canes

- Principles



- Stand at affected side of the client; use gait or transfer belt as necessary
 - Handle at greater trochanter level
 - Elbow flexed at 20-30 degrees
 - Hold cane 6-10 inches to side of foot
 - Hold cane in hand of unaffected side
 - Move cane together with affected leg
 - Inspect rubber tips regularly
- **Walkers**
- Principles
 - Stand adjacent on affected side
 - Pull all four points flat on floor
 - Move walker first
 - Then affected/weaker foot
 - Then unaffected/stronger foot

Musculoskeletal Disorders

Upper Extremity Disorder

1. Bursitis and tendonitis

- a. **Bursitis:** inflammation of bursae
- b. **Tendonitis:** inflammation of muscle tendon sheaths with repetitive stretching
- c. **Intervention**
 - i. Self-limiting + pain relief
 - 1. Conservation treatment = rest + intermittent ice and heat to joint + NSAIDs
 - 2. Corticosteroid injection
 - 3. Arthroscopic surgery if shoulder pain and weakness persist

2. Loose bodies (joint mice)

- a. Occurs in joint spaces as result of articular cartilage wear and bone erosion
- b. Fragments interfere with joint movement
- c. **Intervention**
 - i. Arthroscopic surgery if fragments cause pain or mobility issues

3. Impingement syndrome

- a. General term for impaired movement of rotator cuff of shoulder
- b. Occurs from repetitive overhead movement of arm or acute trauma
- c. Results in rotator cuff tendons or subacromial bursa grinding against coracoacromial arch
- d. **Assessment**
 - i. Early



1. Edema
2. Muscle spasm
3. Limited movement
4. Tender and painful shoulder

- ii. Eventual disuse atrophy

e. Intervention

- i. Rest, NSAIDs, corticosteroid injection
- ii. Heat/ice
- iii. Physical therapy
- iv. Surgical decompression if persistent

4. Carpal Tunnel Syndrome

- a. Entrapment neuropathy when median nerve is compressed by thickened flexor tendon sheath, skeletal encroachment, edema, or soft tissue mass

b. Risk factors

- i. Women, 30-60 years old
- ii. Comorbidities: RA, diabetes, acromegaly, hyperthyroidism, trauma

c. Interventions

- i. Non-surgical
 1. Acupuncture
 2. Wrist splints (prevent hyperextension)
 3. Oral or intra-articular corticosteroids/NSAIDs
- ii. Surgical - nerve release surgery
 1. Recovery will take several weeks or months
 2. Local anesthesia
 3. Incision into affected wrist
 4. Cutting carpal ligament to widen carpal tunnel
 5. Splint and limit hand movement during healing

5. Ganglion

- a. Collection of neurological gelatinous material near tendon sheath and joints
- b. Common in women <50 years old

c. Assessment

- i. Round, firm, cystic swelling on dorsum of wrist
- ii. Local tenderness and aching pain

d. Interventions

- i. Aspiration
- ii. Surgical excision
- iii. Corticosteroid injection
- iv. Compression dressing and splint

6. Dupuytren Disease

- a. Slowly progressive contracture (abnormal shortening) of palmar fascia



- b. Starts in one hand but may affect both
- c. Pattern of inheritance/risk factors
 - i. Men >50 years old
 - ii. Scandinavian or celtic heritage
 - iii. Cigarette smoking and alcoholism
 - iv. Comorbidities: arthritis, diabetes, gout
- d. **Assessment**
 - i. Starts as nodule -> flexion of 4th, 5th and even 3rd finger
 - ii. Morning numbness
 - iii. Dull and aching discomfort
 - iv. Stiffness in affected fingers
- e. **Interventions**
 - i. Non-surgical
 - 1. Finger stretching exercise
 - 2. Intranodular corticosteroid injections
 - ii. Surgical
 - 1. Palmar and digital fasciotomy

Lower Extremity Disorder

1. Corn

- a. Area of hyperkeratosis by internal pressure
- b. 5th toe most commonly involved
- c. Interventions
 - i. Podiatrist: soak and scrape horny layer
 - ii. Protective shield/pad + separation by lamb's wool or gauze
 - iii. Surgical modification of underlying offending osseous structure

2. Hallux valgus (bunion)

- a. Great toe deviates laterally due to marked prominence of medial aspect of first metatarsophangeal joint
- b. Osseous enlargement (exostosis) of medial site -> bursitis
- c. Intervention
 - i. Shoe that conforms to foot shape
 - ii. Corticosteroid injection (for acute inflammation)
 - iii. Surgical excision of exostosis + toe realignment

3. Hammer toe

- a. Flexion deformity of interphangeal joint
- b. Toes pulled upward while metatarsophangeal joints forced down
- c. Corns = top, callus = underneath
- d. Intervention



- i. Open-toes footwear, pads on protruding joints, orthotics
- ii. Manipulation exercises
- iii. Osteotomy

4. Ingrown toenail (Onychocryptosis)

- a. Free edge of nail plate penetrates surrounding skin
- b. May cause secondary infection/granulation tissue
- c. Caused by improper trimming, trauma, pressure, infection
- d. Intervention
 - i. Proper nail trimming
 - ii. Active treatment: warm soaks, antibiotic ointments, surgical excision of nail edge or matrixectomy

5. Morton neuroma (Plantar digital neuroma/neurofibroma)

- a. Swelling of third branch of medial plantar nerve
- b. Throbbing, burning pain in foot -> relieved by rest and massage
- c. Intervention
 - i. Innersoles and metatarsal pads
 - ii. corticosteroids/anesthetic injections
 - iii. Surgical excision of neuroma

6. Plantar fasciitis

- a. Inflammation of foot-supporting fascia
- b. Acute heel pain with first steps in morning (anterior medial aspect) -> improves with stretching of foot and achilles tendon
- c. Intervention
 - i. Stretching
 - ii. Supportive, cushioned shoes
 - iii. Orthotic devices
 - iv. Corticosteroid injections

7. Pes cavus (clawfoot)

- a. Abnormally high arch + fixed equinus deformity of forefoot
- b. Produces metatarsal and dorsum calluses
- c. Causes: charcot-marie-tooth disease, diabetes, progressive neurologic disorders
- d. Intervention
 - i. Exercises
 - ii. Orthotic devices
 - iii. Arthrodesis (fusion) to reshape and stabilize foot

8. Pes planus (flatfoot)

- a. Diminished longitudinal arch
- b. Causes: congenital abnormality, excessive weight, muscle fatigue, improper footwear, arthritis
- c. Symptoms: burning sensation, fatigue, clumsy gait, edema, pain



- d. Intervention
 - i. Exercises
 - ii. Foot orthoses

Arthritis Disorders

1. Rheumatoid Arthritis (RA)

- a. Autoimmune destruction of cartilage tissue and synovial membrane within joints
- b. exacerbation occur during periods of physical or emotional stress
- c. **Risk factors:**
 - i. female thrice than males
 - ii. 3rd and 6th decade of life
 - iii. genetics and family history
- d. **Mechanism:** autoimmune reaction in synovial tissue
 - i. collagen breakdown -> edema/proliferation of synovial membrane -> pannus formation (Hallmark) -> destroys cartilages/bone erosion -> decreased joint ROM
 - ii. Small joints -> synovial membrane inflammation -> compression of nerves -> neuropathy (risk for carpal tunnel syndrome) 1. Tinel - percuss inner wrist, 2. Phalen - wrist flexion, 3. reversed phalen - wrist extension, 4. durkan - wrist compression (+) Carpal Tunnel Syndrome (median nerve)- within 60 mins tingling & numbness (paresthesia)
- e. **Assessment:**
 - i. Acute bilateral & symmetrical starting from small joints
 - ii. Initial: Symmetrical joints pain & morning stiffness 1 hour (>30 minutes)
 - iii. Progressive:
 - 1. Spongy or boggy joints on palpitations
 - 2. Contractures & soft tissue deformities
 - iv. Deformities:
 - 1. Ulnar deviation & swan neck (fingers)
 - 2. Rheumatoid nodules - nontender & movable in subcutaneous tissue
- f. **Diagnosis:**
 - i. Joint involvement: small joints
 - ii. Serology:
 - 1. Rheumatoid factor
 - 2. Anti-citrullinated peptide antibody (anti-CCP) - increased even 2-3 years before inflammation
 - 3. ESR & CRP
 - iii. Duration of symptoms >6 weeks
 - iv. X-ray, ultrasound, MRI



- g. **Interventions:** Early Rheumatoid Arthritis
- i. Pharmacological
 - 1. Nonbiologic vs Biologic Disease-Modifying Anti-rheumatic Drugs (DMARD) - suppresses immune system (cytokine signaling) - takes effect within 6 weeks
 - a. Initially nonbiologic first
 - i. Non-biological DMARD - methotrexate
 - 1. adverse effect: immunosuppression, hepatotoxic, nephrotoxic
 - a. Assess for TB and Hep B
 - b. CI: pregnant
 - c. Assess liver function (AST, ALT)
 - d. Assess kidney function (BUN, creatinine)
 - e. WOF: bone marrow suppression
 - f. CBC monthly (first 3 months)
 - g. CBC every 2-3 months (after 3 months)
 - h. Toxicity: antidote: Leucovorin
 - i. GI problems
 - ii. Oral throats
 - ii. Biologic DMARDS - first targeted therapy for RA
 - 1. faster to take effect however it is more expensive + shorter years of usage
 - iii. Janus Kinase (JAK) inhibitor
 - 1. Monotherapy or with DMARDS
 - 2. WOF: signs of infection
 - iv. NSAIDS (Cox-2 blockers) - prostaglandin -> epithelial cells (gut) -> gastric mucus -> protection from acids/enzymes
 - 1. Cox - 1 - GI irritation/ulceration
 - 2. Cox - 2 - cardiovascular (HF)
 - v. Systematic corticosteroids “bridging medication”
 - 1. While waiting for DMARDS to take effect
2. **Osteoarthritis (OA)**
- a. Non-inflammatory degenerative disorder of joints = progressive deterioration of articular cartilage in proximal (lower legs) and axial (vertebrae) joints
 - b. affects weight-bearing joints and with greatest stress
 - c. Begins 3rd decade; peaks between 5th to 6th decades
 - d. **Risk factors:**
 - i. Women



- ii. Hispanic or African American
- iii. Age >65 y/o
- iv. Obesity
- v. Laborious occupations and sport activities
- vi. History of previous injuries and genetics
- e. **Mechanism:** genetic and hormonal factors/mechanical injury/previous joint damage
 - i. Chondrocyte response -> cytokines release -> proteolytic enzymes, collagenases and metalloprotease release - articular cartilage breakdown -> results in subchondral bone damage -> compensatory bone formation (osteophytes/bone spurs) -> subchondral bone thickening -> joint space narrowing -> decreased joint ROM
- f. **Assessment:**
 - i. Pain aggravated by movement or exercise & relieved by rest + morning stiffness <30 mins
 - ii. Enlarged joint + <ROM in weight bearing joints:
 - 1. Bouchard's node: BP: proximal interphalangeal joints
 - 2. Heberden's node: HD: distal interphalangeal joints
 - iii. Crepitation on palpation over affected knee
 - iv. Mild joint effusion but no systematic manifestations
- g. **Diagnosis:**
 - i. X-ray
 - 1. subchondral bone thickens
 - 2. joint space narrows
 - 3. osteophytes formation
- h. **Intervention:**
 - i. Nonpharmacological
 - 1. Position joint in functional position - extended with light flexion, minimally externally rotated
 - 2. Avoid large pillows under head or knees (to prevent flexion contractures)
 - ii. Pharmacological
 - 1. Viscosupplementation (hyaluronates) - to nourish articular cartilage
 - iii. Surgical
 - 1. Osteotomy - bone resection
 - 2. Arthroplasty - Total joint replacement
- 3. **Gout** - systemic disease where urate crystals deposits in joints and other body tissues
 - a. **Types:**
 - i. **Primary** - starvation/increased purine diet
 - ii. **Secondary** - uric acid under excretion
 - b. **Risk factors:**
 - i. Men 3-4x more than women

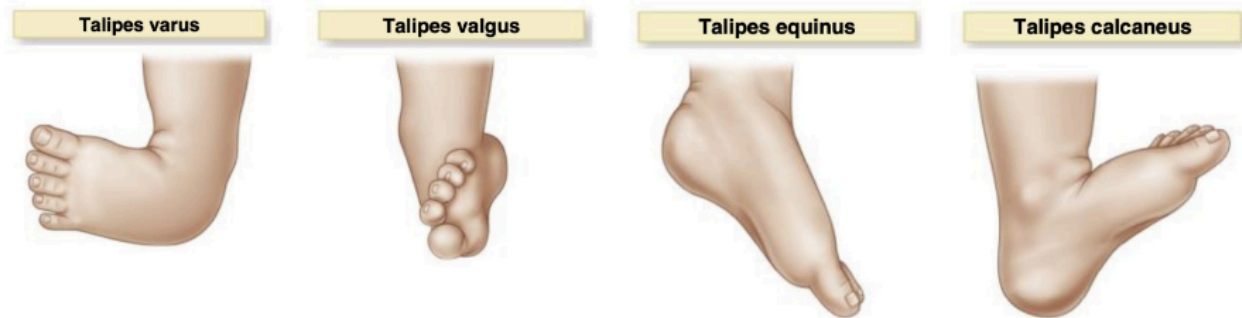


- ii. Age (increased)
- iii. BMI (increased)
- iv. Diet: red meats, organ meats, legumes
- c. **Mechanism:** hyperuricemia -> uric acid deposition -> stimulates phagocytosis by macrophages -> interleukin release -> inflammation -> gout
- d. **Assessment:**
 - i. Hyperuricemia >6.8 mg/dl
 - ii. Gouty arthritis - pain, occurs at night, awakening the patient lasting 3-10 days
 - iii. Tophi - crystalline deposition in articular tissues, cartilage
 - iv. Uric acid urinary calculi & gouty nephropathy
- e. **Stages:**
 - i. Asymptomatic hyperuricemia
 - ii. Acute gouty arthritis
 - iii. Intercritical gout (interval between gout attacks)
 - iv. Chronic tophaceous gout (decreased joint ROM)
- f. **Diagnosis:**
 - i. Polarized light microscopy of synovial fluid (needle-like crystals)
- g. **Intervention:**
 - i. **Acute attacks:**
 1. Colchicine - anti-mitosis - inhibits B-tubulin polymerization into microtubules (prevents activation of neutrophils)
 2. NSAIDS (indomethacin) or corticosteroids
 - ii. **Lowering uric acid: (intercritical phase)**
 1. Xanthines oxidase inhibitors (Allopurinol & Febuxostat) - blocks enzyme xanthine oxidase to prevent conversion of hypoxanthine to uric acid
 2. Uricosuric agents (Probenecid) - blocks urate transporter 1 protein to prevent proximal tubule reabsorption which results in uric acid secretions
 3. Lifestyle changes (low purine diet & increased OFI)

Pediatric Disorders

Congenital Clubfoot (Talipes Equinovarus)

- **Types**
 - Equinus (plantar flexion)
 - Varus (heel inward)
 - Adductus (forefoot curved in)
 - Cavus (high arch)



- **Risk factors**
 - Family history
 - Oligohydramnios
 - Neuromuscular disorders
- **Assessment**
 - Affected foot smaller and shorter
 - Heel in varus, sole facing inward and downward
 - Rigid, cannot be manipulated into neutral
- **Interventions**
 - **Ponseti method**
 - Weekly gentle manipulation and serial casting (5-7 casts)
 - Percutaneous achilles tenotomy if equinus persists
 - Abduction brace for 23 hours/day x 3 months, then naps/night until 4-5 years

Legg-Calve-Perthes Disease

- Idiopathic avascular necrosis of femoral head in children
- **Risk factors**
 - Boys (4-8 years old)
 - Low birth weight
 - Family history
 - Coagulopathy
- **Mechanism:** temporary interruption of blood supply to femoral head -> necrosis -> bone resorption -> remodeling
- **Stages**
 - Stage I Avascular (Month-1 year): necrosis, femoral head flattened
 - Stage II Fragmentation (up to 2 years): revascularization, bone resorption -> weak femoral head
 - Stage III Reparative (1-3 years) new bone formation
 - Stage IV Regenerative (several years): remodeling until healing complete



- **Assessment**
 - Limping (painless or painful)
 - Groin/thigh/knee pain
 - Limited hip abduction and internal rotation
 - Shortened leg
- **Diagnosis**
 - X-ray: flattened femoral head, fragmentation
 - MRI: early changes
- **Intervention**
 - Conservative (initial)
 - Bed rest, NSAIDs
 - Traction, bracing to keep femoral head in acetabulum
 - Surgical (if >8 years old or severe)
 - Osteotomy, femoral/pelvic reconstruction

Fibrodysplasia Ossificans Progressiva

- Rare genetic disorder -> soft tissues turn into bone
- **Affected gene:** ACVR1 (ALK2) mutation
- **Pattern of inheritance:** autosomal dominant
- **Assessment**
 - Congenital malformations of great toes
 - Progressive heterotopic ossification (muscle, tendons, ligaments)
 - Episodes triggered by trauma or infection
 - Progressive immobility -> respiratory compromise
- **Intervention**
 - Avoid intramuscular injections, trauma, biopsies
 - Corticosteroids during flare ups
 - NSAIDs, leukotriene inhibitors, bisphosphonates (experimental)
 - Supportive: physical therapy (gentle), respiratory care
 - Surgery contraindicated (leads to more bone formation)

Marfan's Syndrome

- Connective tissue disorder affecting skeleton, eyes, cardiovascular system
- **Affected gene:** FBN1 (fibrillin-1) mutation
- **Pattern of inheritance:** autosomal dominant
- **Assessment**
 - Tall stature, long arms/legs/fingers (arachnodactyly)
 - Hypermobile joints



- Lens dislocation (ectopia lentis)
- Pectus deformities (excavatum/carinatum)
- Kyphoscoliosis
- Cardiovascular: aortic root dilation, aneurysm, dissection, mitral valve prolapse
- **Intervention**
 - **Cardiac:** beta blockers or ARBS to reduce aortic dilation
 - **Orthopedic:** braces, surgery for scoliosis/pectus deformity
 - **Ophthalmic:** corrective lens, lens surgery
 - **Lifestyle:** avoid high-intensity contact sports
 - **Surgical:** aortic root replacement if aneurysm develops

Respiratology/Pulmonology

Respiratology/Pulmonology - a medical specialty focused on respiratory system

Respiratology nursing - focuses on the specialized care of patients with respiratory illnesses, encompassing both acute and chronic conditions

Respiratory System

Anatomy

- comprised of the upper and lower airway structures, responsible for ventilation
- **Upper respiratory system** - filters, moistens and warms air during inspiration
 - **Nose** - serves as a passageway for air to pass to and from the lungs. It filters impurities and humidifies and warms the air as it is inhaled
 - **Paranasal sinuses** - prominent function of the sinuses is to serve as resonating chamber in speech
 - **Pharynx** - throat, a tube-like structure that connects the nasal and oral cavities to the larynx
 - **Larynx** - voice organ, a cartilaginous epithelium lined structure that connects the pharynx and the trachea
- **Lower respiratory system** - enables the exchange of gases to regulate serum PaO₂, PaCO₂ and pH
 - **Trachea** (windpipe) - serves as the passage between larynx and bronchi
 - **Lungs** - paired elastic structures enclosed in the thoracic cage, which is an airtight chamber with distensible walls
 - **Pleural** - serous membrane that lined the lungs and wall of the thorax
 - **Bronchi and Bronchioles**
 - **Alveoli**
 - Basic gas exchange unit of the respiratory system (simple diffusion)



- Alveolar stretch receptors respond to inspiration by sending signals to inhibit inspiratory neurons in the brain stem to prevent lung over distention
- During expiration stretch receptors stop sending signals to inspiratory neurons and inspiration is ready to start again
- Oxygen and carbon dioxide are exchanged across the alveolar capillary membrane by process of diffusion
- Neural control of respiration is located in the medulla. The respiratory center in the medulla is stimulated by the concentration of carbon dioxide in the blood
- Chemoreceptors, a secondary feedback system, located in the carotid arteries and aortic arch respond to hypoxemia. These chemoreceptors stimulate the medulla

Gas Exchange

- The delivery of oxygen from the lungs to the bloodstream and the elimination of carbon dioxide from the blood stream to the lungs. Occurs in the alveoli through passive diffusion
- **Types:**
 - **Normal ventilation** - for healthy lungs
 - **High ventilation-low perfusion ratio** - dead space, pulmonary emboli, pulmonary infarction, cardiogenic shock
 - **Low ventilation-high perfusion ratio** - client with pneumonia, atelectasis, mucus plug, tumor, COPD
 - **Low ventilation-low perfusion ratio** - silent, pneumothorax, ARDS

Arterial Blood Gas Analysis

- **Three parameters:**
 - **pH level:** 7.35-7.45
 - **PCO₂:** 35-45
 - **HCO₃:** 22-26
- **Steps:**
 - **Step 1:** know the normal values
 - **Step 2:** look at the pH, it indicates presence of acidemia or alkalemia or normal pH
 - **Step 3:** look at the PCO₂ - respiratory indicator (Respiratory Opposite)
 - **Step 4:** look at the HCO₃ - metabolic indicator (Metabolic Equal)
 - **Step 5:** determine the compensation
 - if pH normal: fully compensated



- if pH abnormal, PCO₂ or HCO₃ abnormal: uncompensated
- if pH abnormal, both PCO₂ and HCO₃ abnormal: partially compensated

Metabolic Acidosis - an acid-base imbalance resulting from excessive absorption or retention of acid or excessive excretion of bicarbonate

- Heart problem (ischemia) - production of lactic acid
- Aspirin, metformin - lactic acid accumulation
- Diabetes Mellitus Type 1: Diabetic Ketoacidosis
- Gastrointestinal - diarrhea, ileostomy, enema (excrete too much bicarbonate)
- Renal problem/failure/damage - may not be able to remove acid from the body

Metabolic Alkalosis - occurs when your blood becomes overly alkaline

- Vomiting/NGT suctioning = removes stomach acid which is rich in hydrogen ion causing increased pH level or bicarbonate in the body
- Diuretics - increased pH level
- Alkaline loading - water, nuts and seeds, apple, avocado and banana

Respiratory Acidosis - when the lungs can't remove enough carbon dioxide from the body, making the blood too acidic (hypoventilation)

- Anesthesia
- COPD - retention of carbon dioxide
- Increased ICP - can affect the brain region which is responsible for respiration (medulla & pons)
- Drug overdose - morphine (opioids) -> respiratory depression

Respiratory Alkalosis - hyperventilation

- Anxiety, fever, chest pain
- Living in a high altitude
- Pregnant mom - increased progesterone (stimulate respiratory center - increased ventilation, decreased carbon dioxide in the blood)
- Analeptic overdose - stimulates CNS -> seizure (hyperventilation)

Restrictive Disorders

- The problem is with the parenchyma of the lungs, interstitial tissue also the inflammatory situation like fibrosis means lot of collagen deposited to lung parenchyma results in lungs stiffness and restrict expansion of lungs

Pneumothorax

- Accumulation of atmospheric air in the pleural space, which results in a rise in the intrathoracic pressure and reduced vital capacity
- **Types:**
 - Spontaneous pneumothorax: entry of air in pleural space without obvious cause



- Open pneumothorax: air enters pleural space through an opening in the chest wall (stab/gunshot wound)
- Tension pneumothorax: air enters pleural space with at inspiration and can't escape leading to over distention of thoracic cavity resulting to shifting of mediastinum content to unaffected side (flail chest - paradoxical breathing pattern)
- **Predisposing factors:**
 - Chest trauma
 - Inflammation lung conditions
 - tumor
- **Signs and symptoms:**
 - Sudden sharp chest pain
 - Unexplained dyspnea or SOB
 - Cyanosis
 - diminished or decreased breath sound of affected lung
 - Cool moist-skin - initial sign of shock
 - Mild restlessness/apprehension, anxiety
 - Resonance to hyperresonance
 - Decreased tactile fremitus
- **Diagnosis:**
 - ABG
 - CXR - confirms pneumothorax/collapse of lung
- **Nursing management:**
 - Assist in endotracheal intubation
 - Assist in thoracentesis
 - Administer meds: morphine sulfate due to pain, antimicrobial agents due to bacteria
 - Assist in test tube thoracotomy attached to water sealed drainage system

Pleural Conditions

Pleural Effusion

- Is the collection of fluid in the pleural space
- **Causes:**
 - Tumor
 - Heart failure
 - Pulmonary embolus
 - Chest injury
 - Kidney failure
- **Types:**



- Hemothorax - blood in pleural space
- Pyothorax - pus in plural space
- Hydrothorax - water in plural space
- **Signs and Symptoms**
 - Chest pain (pleuritic pain)
 - SOB
 - Dry, unproductive cough
 - tachycardia
 - Elevated temperature (pyothorax)
 - decreased breath sounds
- **Diagnosis:** CXR and CT Scan
- **Management:**
 - Remain with the client, stay calm
 - High fowlers
 - O2 therapy
 - Thoracentesis
 - CXR
 - Antibiotics

Hemothorax

- Blood in pleural space as a result of penetrating or blunt chest trauma
- Accompanies in high percentage of chest injuries
- Can result in hidden blood loss
- Patient may be asymptomatic, dyspneic, apprehensive or in shock
- **Management:**
 - Assist with thoracentesis to aspirate blood from pleural space
 - Assist with chest tube insertion and set up drainage system for complete and continuous removal of blood or air
 - Auscultate lungs and monitor for relief of dyspnea
 - Monitor amount of blood loss in drainage
 - Replace volume with IV fluids or blood products

Pleurisy (Pleuritis)

- Inflammation of both layers of the pleurae (parietal and visceral)
- May develop in conjunction with pneumonia or an upper respiratory tract, TB, or collagen disease
- When the inflamed pleural membranes rub together during respiration (intensified on inspiration), the result is severe, sharp, knife-like pain
- **Clinical manifestations:**



- Pleuritic pain during deep breath, coughing or sneezing
- Pain is limited in distribution rather than diffuse
- Pleural friction rub can be heard with stethoscope
- **Diagnostic tests:**
 - CXR
 - Sputum analysis
 - Thoracentesis
 - Pleural biopsy
- **Management:**
 - Treatment of underlying condition causing pleurisy
 - Topical applications of heat or cold
 - Indomethacin for pain relief
 - Intercostal nerve block if pain is severe
- **Nursing intervention:**
 - Instruct the patient in heat/cold application for pain relief
 - Instruct the patient to turn onto the affected side to splint the chest wall and reduce the stretching of the pleurae
 - Teach the patient to use hands or pillow to splint the ribcage while coughing

Empyema Thoracis

- Accumulation of purulent fluid in the pleural space
- Occur as complication of bacterial pneumonia, lung abscess or chest trauma
- Patient is acutely ill and has signs and symptoms similar to acute respiratory infection
- **Diagnostic test:** chest CT
- **Main objective:** to drain the fluid in the pleural cavity
- **Thoracentesis** is done if the fluid is not too thick
- **Tube thoracostomy** is done to patients with loculated or complicated pleural effusions
- **Open chest drainage** via thoracotomy is done to remove thickened pleura, pus and debris
- **Nursing intervention:** provide specific to the method of drainage of pleural fluid

Upper Respiratory System Disorders

Rhinitis

- A group of disorders characterized by inflammation and irritation of the mucous membranes of the nose
- **Allergic rhinitis**



- Further classified as seasonal rhinitis (occurs during pollen seasons) or perennial rhinitis (occurs throughout the year)
- Commonly associated with exposure to airborne particles such as dust, dander or plant pollens in people who are allergic to these substances
- **Clinical manifestations:**
 - Rhinorrhea (excessive drainage or runny nose)
 - Nasal congestion
 - Sneezing
 - pruritus of the nose, roof of the mouth, throat, eyes and ears
- **Management:**
 - Antihistamines
 - corticosteroids nasal sprays
 - Desensitizing immunizations
- **Nursing Intervention:**
 - Instruct the patient with allergic rhinitis to avoid or reduce exposure to allergens and irritants
 - instruct the patient in correct administration of nasal medications
 - To achieve maximal relief, the patient is instructed to blow the nose before applying any medication into the nasal cavity
- **Viral rhinitis (Common cold)**
 - Most frequent viral infection in the general population caused by coronavirus
 - Highly contagious because the virus is shed for 2 days before the symptoms appear and during the first part of symptomatic phase
 - **Clinical manifestations:**
 - Low grade fever
 - Nasal congestion
 - Rhinorrhea and nasal discharge
 - Halitosis, sneezing
 - Tearing watery eyes
 - Scratchy or sore throat
 - general malaise, chills
 - Headache and muscle aches
 - **Management:**
 - symptomatic therapy
 - adequate fluid intake and rest
 - Prevention of chilling
 - warm salt-water gargles to soothe the sore throat
 - NSAIDS to relieve aches and pains
 - antihistamines are used to relieve sneezing, rhinorrhea and nasal congestion
 - Inhalation of steam or heated, humidified air



Acute Pharyngitis

- a sudden painful inflammation of the pharynx, the back portion of the throat that includes the posterior third of the tongue, soft palate, and tonsils
- Commonly referred as sore throat
- **Clinical manifestations:**
 - fiery-red pharyngeal membrane and tonsils
 - Swollen lymphoid follicles
 - Enlarged and tender cervical lymph nodes
 - Fever
 - Malaise
 - sore throat
- **Pharmacologic therapy**
 - Penicillin (DOC)
 - Cefasporin
 - Macrolides
 - Gargles with benzocaine may relieve symptoms
- **Nursing interventions**
 - Liquid or soft diet is provided during the acute stage
 - Cool beverage , warm liquids and flavored frozen dessert such as popsicles are often soothing
 - Warm saline gargles or throat irrigations
 - Increase oral fluid intake
 - Ice collar can relieve severe sore throats
 - CBR during febrile stage
 - Instruct the patient about preventive measures

Chronic Pharyngitis

- chronic pharyngitis is a persistent inflammation of the pharynx. It is common in adults who work in dusty surroundings, use their voice to excess, suffer from chronic cough, or habitually use alcohol and tobacco
- **Three types:**
 - **Hypertrophic** - characterized by general thickening and congestion of the pharyngeal mucous membrane
 - **Atrophic** - late stage of the first type (the membrane is thin, whitish, glistening and at times wrinkled)
 - **Chronic granular** - “clergyman’s sore throat”, characterized by numerous swollen lymph follicles on the pharyngeal wall
- **Clinical manifestations:**



- constant sense of irritation or fullness in the throat
- Mucus that collects in the throat
- Difficulty swallowing
- **Pharmacological management:**
 - Nasal sprays or medications containing ephedrine sulfate or phenylephrine hydrochloride
 - Antihistamine decongestant medications
 - Acetaminophen
- **Nursing management:**
 - instruct patient to avoid contact with others until the fever subsides to prevent the spread of infection
 - Avoidance of alcohol, tobacco, secondhand smoke, and exposure to cold or to environment or occupational pollutants

Tonsillitis and Adenoiditis

- The tonsils are composed of lymphatic tissue and are situated on each side of the oropharynx
- The adenoids or pharyngeal tonsils consist of lymphatic tissue near the center of the posterior wall of nasopharynx
- **Clinical manifestations:**
 - Sore throat, fever, snoring and difficulty swallowing
 - Enlarged adenoids may cause mouth-breathing, earache, draining ears, frequent head colds, bronchitis, foul-smelling breath, voice impairment and noisy respiration
- **Management:**
 - Penicillin (first line) or cephalosporins
 - Tonsillectomy or adenoidectomy is indicated if the patient has had repeated episodes of tonsillitis despite antibiotic therapy
- **Nursing Intervention (post-op):**
 - In the immediate post-operative period, the most comfortable position is prone, with head turned to the side to allow drainage from the mouth and pharynx
 - apply ice collar to the neck
 - assess for post op bleeding much as frequent swallowing
 - Ice chips may be given to the patient
 - Alkaline mouthwashes and warm saline solutions are useful in coping with the thick mucus and halitosis that may be present after surgery
 - Milk and milk products (ice cream and yogurt) may be restricted
 - provide soft, adequate diet
 - instruct the patient to avoid vigorous tooth brushing or gargling
 - Encourage the use of cool-mist vaporizer or humidifier in the home
 - Instruct patient to avoid smoking and heavy lifting or exertion for 10 days



Peritonsillar Abscess (Quinsy)

- Most common major suppurative complication of sore throat/tonsillitis. This collection of purulent exudate between the tonsillar capsule and the surrounding tissues including the soft palate, may develop after an acute tonsillar infection that progress to local cellulitis and abscess
- **Clinical manifestations:**
 - Severe sore throat, fever, trismus (inability to open the mouth), and drooling
 - Severe pain, raspy voice
 - Odynophagia (a severe sensation of burning, squeezing pain while swallowing)
 - Dysphagia (difficulty swallowing)
 - Otagia(pain in the air), tender and enlarged cervical lymph nodes
 - Airway obstruction may occur
- **Management:**
 - Antimicrobial agents (penicillin)
 - corticosteroid therapy
 - needle aspirations are performed to decompress the abscess
- **Nursing interventions:**
 - Assist in performing intubation, cricothyroidotomy or tracheotomy to treat airway obstruction
 - Assist in needle aspiration when indicated
 - Gentle gargling after the procedure with cool normal saline gargle may relieve discomfort
 - Provide cool liquids
 - Instruct patient to refrain from or cease smoking
 - it is also important to reinforce the need for good oral hygiene

Laryngitis

- An inflammation of the larynx, often occurs as a result of voice abuse or exposure to dust, chemicals, smoke, and other pollutants
- Most common cause are viruses, bacterial invasion may be secondary
- **Clinical manifestations:**
 - Horseness of voice (initial sign)
 - Aphonia (complete loss of voice)
 - Severe cough
 - Throat feels worse in the morning and improves when the patient is in a warmer climate
- **Management:**
 - Instruct the patient to rest the voice and avoid irritants (including smoking)
 - Inhaling cool steam or an aerosol is provided
 - Administer antibacterial therapy as ordered
 - Topical corticosteroids may be given by inhalation



- Increased OFI

Laryngeal Cancer

- **Etiology:**
 - Most tumors of the larynx are squamous cell carcinoma
 - Men > women, age 60 - 70
 - Cigarette smoking and alcohol consumption are associated with laryngeal cancer
- **Clinical manifestations:**
 - Hoarseness of voice for more than 2 weeks
 - Persistent cough and sore throat
 - Dyspnea
 - Dysphagia
 - Pain radiating to ear and burning sensation in the throat
 - Weight loss
 - Enlarged cervical lymph nodes
 - Unilateral nasal obstruction
- **Diagnostic tests:**
 - Virtual endoscopy
 - Optical imaging
 - CT Scan
 - MRI
 - Direct laryngoscopic examination
- **Management:**
 - Radiation therapy
 - Chemotherapy
 - Surgery
 - **Partial laryngectomy:** a portion of the larynx is removed, along with one vocal cord and the tumor
 - Complication: change in voice quality or hoarseness of voice
 - **Total laryngectomy:** laryngeal structures are removed, including the hyoid bone, epiglottis, cricoid cartilage and two or three rings of the trachea
 - Complication: permanent loss of voice, salivary leak, wound infection, stomal stenosis, and dysphagia
- **Nursing intervention:**
 - Arrange for clients with laryngectomies to meet with members of support groups
 - Establish a method for communication before surgery
 - Maintain airway, have suction equipment at bedside
 - Observe for signs of hemorrhage or infection
 - Teach about tracheostomy and stoma care



- Assist in period of grieving

Lower Respiratory System Disorders

Chronic Obstructive Pulmonary Disease (COPD)

Three pathological processes:

- Loss of alveoli attachment
- Inflammatory obstruction of the airway
- Excessive mucus production

Obstructive Lungs Disorder

- **Chronic bronchitis** - excessive mucus production and inflammatory obstruction of the airway
- **Emphysema** - loss of alveoli attachment

Etiology:

- Chronic cigarette smoking
- Lalaki (male)
- Exposure to irritants
- Age (old)
- Respiratory infection

Emphysema

- Destruction of alveoli, narrowing of airway, the alveoli lose their elasticity and cannot inflate and deflate as they used to. This then has an effect on the exchange of oxygen and carbon dioxide in the lungs
- **Pathophysiology:**
 - Protease (inflammatory response) -> destruction of alveoli, elastic fiber, collagen -> loss of elastic recoil, deficiency in alpha 1 anti trypsin -> air trapping -> impaired gas exchange -> hypoxia/CO2 retention -> hyperinflation of lungs -> barrel chest
- **Risk factor:**
 - Chronic cigarette smoking
 - working/living in highly polluted area
- **Signs and symptoms:**
 - Pink puffer (pathognomonic sign)
 - Barrel chest
 - Polycythemia (increased in RBC production)
 - Weight loss
 - Clubbing of fingers (chronic hypoxia)
 - Purse lip breathing (to exhale the retained or excess carbon dioxide)
 - Hypercapnia (excessive carbon dioxide)
- **Laboratory:**



- ABG (respiratory acidosis)
- CXR
- Pulse oximeter
- spirometry
- **Nursing intervention:**
 - O2, however low flow with the rate of 1-2LPM (hypoxic drive) via venturi mask/nasal cannula
 - Position: sitting/upright/orthopneic (to expand the lungs)
 - Increase fluid intake (liquify mucus secretions)
 - Diet: high CHO, CHON, vitamin C

Chronic Bronchitis

- **Criteria:**
 - Productive cough for 3 months for 2 consecutive years
- **Pathophysiology:**
 - Increase function of goblet cells -> increased mucus production -> increase air volume -> hyperinflation of the lungs -> decrease oxygen level/CO2 retention -> respiratory acidosis
- **Signs and symptoms:**
 - Productive cough
 - Cyanosis
 - Crackles
 - Wheezing
 - Hypercapnia
 - Cor pulmonale -> right ventricular hypertrophy
 - Weight gain
 - Blue bloaters (pathognomonic sign)
- **Laboratory:**
 - CXR
 - ABG
 - Pulse oximeter
- **Nursing Intervention:**
 - O2 low flow 1-2LPM via venturi mask
 - Increase fluid intake
 - CPT (check for contraindication first)
 - Increased humidification

Bronchial Asthma

- Chronic inflammatory disease of the respiratory system
 - Inflammation of the airways
 - bronchospasm



- Increased mucus secretion
 - Narrowing of airway -> impaired gas exchange -> hypoxia/CO₂ retention (Respiratory Acidosis)
- **Risk factor:**
 - Allergies
 - Smoking/Stress/Sports
 - Temperature
 - Hazardous materials
 - Microorganism
- **Signs and symptoms:**
 - Wheezing (upon expiration, prolonged exhalation)
 - Dyspnea
 - Tachypnea
 - Tachycardia
 - Cough
 - Hyperresonance
 - Restless
- **Laboratory:**
 - ABG
 - CXR
 - Spirometry
- **Intervention:**
 - Avoid exercise, emotional factor, environmental factors
- **Alert:**
 - Can't talk (obstructed)
 - Cyanosis (impaired gas exchange)
 - Inhaler won't work (fully obstructed)
 - (-) breath sounds

Treatment for COPD and Bronchial Asthma

- **Bronchodilators**
 - selective beta 2 agonist (beta 2 - stimulates bronchodilation)
 - albuterol (Proventil)
 - salbutamol (Ventolin)
 - Non-selective beta agonist (beta 1 - stimulates increased HR)
 - Isoproterenol (Isopril)
 - Terbutaline (Bricanyl)
 - Xanthine derivatives (with compounds of caffeine)
 - Aminophylline, theophylline
 - Side effect: palpitations



- **Anti-inflammatory**
 - Corticosteroids
 - Hydrocortisone (Solu-cortef), Prednisone
 - Anti-histamine
 - Benadryl
 - Side effect: drowsiness
 - Avoid driving or machinery
 - Leukotriene (inflammatory mediator) inhibitor
 - Monteleukast
- **Expectorant**
 - Gualafenesin, mucolytics
 - Antitussive given at night

Bronchiectasis

- Abnormal permanent dilation of bronchus leading to destruction of muscular and elastic tissues of alveoli
- **Predisposing factors:**
 - Recurrent lower respiratory tract infections
 - Chest trauma
 - Congenital defects
 - Related to presence of tumor
- **Signs and symptoms:**
 - productive cough
 - Dyspnea
 - cyanosis
 - anorexia and generalized body malaise
 - Hemoptysis
- **Diagnostic:**
 - ABG
 - Bronchoscopy - direct visualization of bronchus using fiberscope
 - Pre:
 - Secure informed consent and explain procedure to client
 - Maintain NPO 6-8 hours prior to procedure
 - Monitor VS and breath sounds
 - Post:
 - Feeding initiated upon return of gag reflex
 - Avoid talking, coughing and smoking may cause irritation
 - Monitor signs of laryngeal spasm - prepare tracheostomy set
- **Treatment:**



- Surgery (pneumonectomy, 1 lung removed and position on affected side)
- Segmental wedge lobectomy (promote re expansion of lungs) - unaffected lobectomy facilitate drainage

Occupational Lung Diseases

- **Asbestosis:** diffuse interstitial fibrosis of the lung caused by inhalation of asbestos dust and particles
 - Found in workers involved in manufacture, cutting and demolition of asbestos-containing materials
- **Silicosis:** chronic pulmonary fibrosis caused by inhalation of silica dust
 - Exposure to silica dust is encountered in almost any form of mining because the earth's crust is composed of silica and silicates (gold, coal, tin, copper mining); also stone cutting, quarrying, manufacture of abrasives, ceramics, pottery, and foundry work
- **Sarcoidosis:** granulomatous disease in which clumps of inflammatory epithelial cells occur in many organs, primarily in the lungs
 - Lymph node enlargement seen on chest x-ray
- **Clinical manifestations:**
 - Chronic cough; productive in silicosis
 - Dyspnea on exertion; progressive and irreversible in asbestosis
 - Susceptibility to lower respiratory tract infections
 - Bibasal crackles in asbestosis
- **Management:**
 - There is no specific management; exposure is eliminated and the patient is treated symptomatically
 - Give prophylactic isoniazid (INH) to patient with positive tuberculin test because silicosis is associated with high risk of TB
 - Persuade people who have been exposed to asbestos fiber to stop smoking to decrease risk of lung cancer
 - Keep asbestos worker under cancer surveillance; watch for changing cough, hemoptysis, weight loss, melena
 - Bronchodilators may be of some benefit if any degree of airway obstruction is present
- **Nursing interventions:**
 - Administer oxygen therapy as required
 - Administer or teach self-administration of bronchodilators as ordered
 - Encourage smoking cessation
 - Advise patient on pacing activities to prevent fatigue
 - Provide information to healthy workers on prevention of occupational lung disease



Lower Respiratory Tract Infectious Diseases

Pneumonia

- Inflammation of the lung parenchyma caused by various microorganisms, including bacteria, mycobacteria, fungi and viruses
- **Types:**
 - **Community -acquired pneumonia:** occurs either in the community setting or within the first 48 hours after hospitalization or institutionalization
 - **Hospital-acquired pneumonia:** also known as nosocomial pneumonia, is defined as the onset of pneumonia symptoms more than 48 hours after admission in patients with no evidence of infection at the time of admission
 - **Aspiration pneumonia:** refers to the entry pulmonary consequences resulting from entry of endogenous or exogenous substances into the lower airway
- **Clinical manifestations:**
 - Sudden onset, rapidly rising fever 38.3-40.5C
 - Cough productive of purulent sputum
 - Pleuritic chest pain aggravated by deep respiration/coughing
 - Dyspnea, tachypnea accompanied by respiratory grunting, nasal flaring use of accessory muscles of respiration, fatigue
 - Rapid, bounding pulse
 - Orthopnea
 - Rusty, blood-tinged sputum
 - Poor appetite, diaphoresis
- **Diagnostic tests:**
 - CXR: shows presence/extent of pulmonary disease typically consolidation
 - Gram stain and culture and sensitivity test may indicate offending organism
 - Blood culture detects bacteraemia (bloodstream invasion) occurring with bacterial pneumonia
- **Management:**
 - Administration of the appropriate antibiotic as determined by the results of gram stain
 - S. pneumonia - macrolide antibiotic (azithromycin, clarithromycin or erythromycin)
 - Pseudomonas infection - anti pneumococcal, anti pseudomonal beta-lactam
 - Treatment of viral pneumonia is primarily supportive
 - Oxygen therapy if patient has inadequate gas exchange
- **Complications:**
 - Shock and respiratory failure
 - Pleural effusion



- **Nursing interventions:**

- Encourage coughing and deep breathing after chest physiotherapy, splinting the chest if necessary
- Maintain semi-fowlers position
- Promote hydration (2-3L/day) to liquify secretions
- Teach effective coughing techniques to minimize energy expenditures; plan rest periods
- Suction if necessary
- Instruct client to cover nose and mouth when coughing
- Teach the need to continue entire course of antimicrobial therapy which is usually seven to ten days
- Teach the patient about proper administration of antibiotics and potential side effects
- Teach that findings are expected to be less within 48-72 hours of initial therapy
- Nutritionally enriched drinks or shakes maybe helpful in maintaining nutrition

Pulmonary Tuberculosis

- Tuberculosis (TB) is an infectious disease that primarily affects the lung parenchyma
- It is also transmitted to other parts of the body, including meninges, kidneys, bones and lymph nodes
- The primary infectious agent, *M. tuberculosis* is an acid fast aerobic rod that grows slowly and sensitive to heat and ultraviolet light spreads from person to person by airborne transmission
- **Clinical manifestations:**
 - Fatigue, anorexia, weight loss, low grade fever, night sweats
 - Some patients have acute febrile illness, chills and flu-like symptoms
 - Cough (insidious onset) progressing in frequency and producing mucoid or mucopurulent sputum
 - Hemoptysis, chest pain, dyspnea (indicates extensive involvement)
- **Diagnostic test:**
 - Sputum smear/sputum culture confirms a diagnosis of TB
 - CXR to determine presence and extent of disease
 - Tuberculin skin test (purified protein derivative PPD or Mantoux test)
- **Classification:**
 - Data from the history, physical examination, TB test, CXR, and microbiologic studies are used to classify TB into one of five classes
 - A classification scheme provides public health officials with a systematic way to monitor epidemiology and treatment of the disease
 - **Class 0:** no exposure; no infection
 - **Class 1:** exposure; no evidence of infection
 - **Class 2:** latent infection; no disease
 - **Class 3:** disease; clinically active



- **Class 4:** disease; not clinically active
- **Class 5:** suspected disease; diagnosis pending
- **Management:**
 - Pulmonary TB is treated primarily with antituberculosis agents for 6-12 months
 - The initial phase consists of a multiple medication regime of INH, rifampicin, pyrazinamide and ethambutol and is administered daily for 8 weeks
 - Continuation phase of treatment include INH and rifampicin and lasts for additional 4-7 months
 - Vitamin B (pyridoxine) is usually administered with INH to prevent INH-associated peripheral neuropathy

Commonly Used Agents	Adult Daily Dosage	Most common Side Effects
Isoniazid (INH)	5 mg/kg (300 mg maximum daily)	Peripheral neuritis, hepatic enzyme elevation, hepatitis, hypersensitivity
Rifampicin	10 mg/kg (600 mg maximum daily)	Hepatitis, febrile reaction, purpura (rare), nausea and vomiting
Pyrazinamide	15-30 mg/kg (2.0 g maximum daily)	Hyperuricemia, hepatotoxicity, skin rash, arthralgias, GI distress
Ethambutol	15-25 mg/kg (no maximum daily dose but based on lean body)	Optic neuritis (may lead to blindness; very rare at 15 mg/kg), skin rash

- **Nursing intervention:**
 - Instructs the patient to increase fluid intake and about correct positioning to facilitate airway drainage
 - Discuss the medications schedule and side effects of the drugs
 - Instructs the patient to take the medication either on an empty stomach or at least 1 hour before meals because food interferes with medication absorption
 - Patients taking INH should avoid food that contain tyramine and histamine because it may result in headache, flushing, hypotension, lightheadedness, palpitations, and diaphoresis
 - Monitors for side effects of anti-TB drugs
 - Encourage rest and avoidance of exertion
 - Provide nutritional plan that allows for small frequent meals
 - Instruct the patient about important hygiene measures including mouth care, covering the mouth and nose when coughing and sneezing, proper disposal of tissues and hand washing



Acute Respiratory Distress Syndrome (ARDS)

- An acute condition characterized by bilateral pulmonary infiltrates and severe hypoxemia in the absence of evidence for cardiogenic pulmonary edema
- **Causes:**
 - Viral
 - Trauma
 - Burns
 - Aspiration pneumonia
 - Overdose
 - Near drowning
- **Pathophysiology:**
 - Viral infection -> initiation of immune inflammatory response
 - -> release of a-vasoactive substance -> increased capillary permeability
 - -> pulmonary edema -> hypoxia
 - -> atelectasis -> decreased ventilation
 - -> b-proteolytic enzymes -> decreased perfusion
- **Signs and symptoms:**
 - Tachypnea (earliest sign)
 - Crackles. Dyspnea, intercostal retractor
 - Cyanosis
 - Decreased breath sounds
 - hypoxia
- **Nursing management:**
 - Identify and treat the cause of the ARDS
 - Administer oxygen as prescribed (8-10 LPM)
 - Position: High fowlers/orthopneic
 - Restrict fluid intake as prescribed (due to pulmonary edema)
 - Provide respiratory treatment as prescribed
 - Administer diuretics, anticoagulants or corticosteroids as prescribed
 - Prepare the client for intubation and mechanical ventilation using PEEP (to open up the collapsed alveoli)

Diagnostic Tests

ABG

- **Purpose:** to detect the presence of acid-base imbalance and to assess ventilation (respiratory therapist)
- **Nursing responsibilities:**



- Heparinized syringe 10 ml
- Allen's test (Normal: 5-15 seconds) - to check the adequacy of collateral circulation
- Artery "radial" - oxygenated blood
- Pressure (5-10 minutes to prevent bleeding)
- Ice - to prevent hemolysis (breakdown of red blood cells)
- Educate/elevate the site of extraction to prevent bleeding
- **ML:** 5-10 ml of blood
- **Avoid:** suctioning before ABG (it suctions also the oxygen)

Pulse Oximeter

- **Purpose:** to determine oxygen saturation in the blood
- **Normal:** 95-100%
- **Nursing key points:**
 - Avoid exposure to sunlight
 - Blanch test - press the index finger (<2 secs)
 - Color removal of nails
 - Do not place the same arm with the BP monitoring
 - Edematous hand (-)

Thoracentesis

- **Purpose:** aspiration of fluid or air from the pleural space
- **Indication:** pneumothorax & pleural effusion
- **Uses:** diagnostic (10-20 ml) & therapeutic (to alleviate the symptoms) - 1000 ml/1L (standard)
- **Preprocedure:**
 - Topical anesthesia (local)
 - Orthopneic position (sitting upward leaning forward) - 2-3 ICS midclavicular/4-5 ICS anterior axillary
 - Record initial vital signs (baseline: HR, RR, BP) -> hypovolemic shock (complication)
 - Avoid coughing/remain still (may lead to internal bleeding)
 - Consent (invasive procedure)
- **Post-procedure:**
 - Position: unaffected side (to expand the lungs, prevent leakage of fluid at thoracic cavity)
 - Apply pressure (5-10 minutes to prevent bleeding)
 - Check for expectoration of blood - lung trauma (notify physician)
 - Monitor vital signs (BP/HR/RR) hypotachytachy = hypovolemic shock
 - Monitor complications: shock, pneumothorax, pulmonary edema, air embolism



Bronchoscopy

- **Purpose:** direct inspection and observation of the larynx, trachea, and bronchi
- **Uses:** Diagnostic (collect secretion - CS) & therapeutic (to alleviate the symptoms of client)
- **Preprocedure:**
 - Consent (secure)
 - NPO (6-8 hours) to prevent aspiration and vomiting
 - Medication:
 - Atropine sulfate - reduce salivation
 - Anesthesia (local) - to reduce discomfort
 - Remove dentures (to prevent aspiration)
- **Post-procedure:**
 - Position: side lying - for drainage
 - Check for return of gag reflex (tongue depressor)
 - if (+) gag reflex, they can eat
 - WOF: sign of bronchial perforation
 - Hemoptysis
 - Cyanosis
 - SOB/dyspnea
 - Tachycardia
 - Arrhythmias
 - Hypotension

Sputum Collection

- **Purpose:** to assess for gross appearance of the sputum
- **Color:**
 - **Rusty** - pneumococcal or bacterial
 - **Green** - pseudomonas
 - **Blood-tinged sputum** - pulmonary tuberculosis
- Confirmatory test for PTB: acid-fast bacillus staining
- Culture and sensitivity
- Cytologic study - to detect abnormal lung cell
- **Key points:**
 - Best time: early morning (concentrated)
 - Rinse mouth with plain water only
 - Use sterile container
 - Assist client in expectorating sputum (chest physiotherapy, nebulization)
 - Secretion first before administering antibiotic
 - 15 ml pure sputum (no saliva)



Tracheobronchial Suctioning

- **Purpose:** to remove excess mucus secretion to maintain patent airway
- To collect specimen for diagnostic studies
- **Indication:** based on clinical findings
- **Nursing responsibilities:**
 - Sterile technique - gloves
 - Hyperventilate: 100% oxygenation before and after the procedure
 - Insert catheter: 3-5 inches in length
 - Apply suction while withdrawal of catheter in circular motion with a duration of 5-10 secs but not more 15 seconds (can lead to hypoxia if prolonged)
- **Complications of over suctioning:**
 - Hypoxia
 - Bleeding
 - Bradycardia (stimulated vagus nerve)
 - Hypotension
- **Interval:** 2-3 minutes

Chest Physiotherapy

- **Purpose:** to help clear excessive bronchial secretions for airways
- **Three method:**
 - **Percussion:** produce mechanical waves of energy that is transmitted through chest wall (with cupped hands) - upper lobe first before lower lobe to loosen secretions
 - **Vibration:** hand is paced on the client's chest while rapid and vigorous vibration
 - **Postural drainage:** use of gravity to assist in the movement of secretion
- **Contraindication:**
 - Lung trauma
 - Ribs and spinal cord injury
 - Pulmonary tuberculosis
 - Increased ICP
- **Nursing intervention:**
 - Secure consent
 - Check area of accumulation of mucus
 - Place client in each position for 10-15 minutes
 - Change position gradually (postural hypotension)
 - procedure best done in the morning or before meals
 - Provide good oral hygiene



Close Chest Tube

- **Purpose:** to restore the negative pressure and re expand the lungs
- **Three types of closed chest tube:**
 - **One-bottle system**
 - Serves as drainage bottle and water seal bottle
 - Immerse tip of the tube 3-5 cm of sterile NSS to create water seal
 - Note: observe for fluctuations of fluid along the tube, intermittent bubbling of the fluid
 - Continuous bubbling - indicate presence of air leak (apply tape)
 - **Two-bottle system**
 - Not connected to the suction apparatus
 - 1st bottle is drainage
 - 2nd bottle is water seal bottle (intermittent bubbling)
 - Connected to suction apparatus
 - 1st bottle is drainage and water seal bottle
 - 2nd bottle is suction control bottle (continuous bubbling)
 - **Three bottle system**
 - 1st bottle is drainage bottle
 - 2nd bottle is water seal bottle
 - 3rd bottle is suction-control bottle
- 2-3 feet below the level of the chest
- Absence of fluctuation in water seal
 - Obstruction or kink
 - Re-expansion (CXR)
- **Dislodgement:** apply occlusive dressing at site, 3 side taped technique (to prevent tension pneumothorax)
- **Disconnected:** submerge the tube to the water
- Avoid clamping
- Chest tube removal: physician
 - Criteria:
 - Negative fluctuation
 - Positive breath sounds
 - Positive XR (lung re-expansion)
 - Instruct: exhale deeply, hold and do valsalva maneuver
 - Position: semi-fowlers
 - Monitor RR, WOF: respiratory distress - tachypnea (1st hour after removal) - Notify MD



Hematology

Hematology - a branch of medicine focused on the study of blood-forming organs and blood-related disorders

Hematology nursing - specializing in caring for patients with blood disorders

Hematologic System

- The hematologic system consist of the blood and the and the sites where blood is produced, including the bone marrow and the reticuloendothelial system (RES)
- Blood is a specialized organ that differs from the other organs in that it exists in a fluid state. Blood is composed of plasma and various types of cells
- Plasma is the fluid portion of blood; it contains various proteins such as albumin, globulin, fibrinogen and other factors necessary for clotting as well as electrolytes, waste products, and nutrients. About 55% of blood volume is plasma
- Serum is plasma minus the clotting factors

Blood Cells

Myeloid lineage: involved in innate immunity

- Neutrophils
- Macrophages
- Monocytes
- Eosinophil and Basophil
- Platelets
- RBCs

Lymphoid lineage: involved in adaptive immunity

- B lymphocytes
- T lymphocytes
- Natural Killer cells

Cell Type	Major Function
WBC (Leukocyte) Normal: 4500-11000/mm ³	Fights infection
Granulocytes Neutrophil	Essential in preventing or limiting bacterial infection via phagocytosis
Eosinophil	Involved in allergic reactions (neutralize histamine), digest foreign proteins



Agranulocytes Monocyte T lymphocyte Plasma cell	Enters tissue as macrophage; highly phagocytic, especially against fungus, immune surveillance Responsible for cell-mediated immunity Secretes immunoglobulin (Ig/antibody) Most mature form of B lymphocytes
RBC (Erythrocyte) Hemoglobin Male: 13-18 g/dl Female: 12-16 g/dl Hematocrit Male: 42-52% Female: 35-47%	Carries hemoglobin to provide oxygen to tissues; average lifespan is 120 days
Platelet (Thrombocyte) Normal: 150000-450000/mm ³	Fragment of megakaryocyte; provides basis for coagulation to occur; maintains hemostasis; average lifespan is 10 days

Hematologic Disorders

Anemia

- Decreased availability of oxygen to tissues
- **Causes:**
 - Inadequate intake of vitamins and food rich in iron and RBC production
 - Decreased RBC production by the bone marrow (aplastic anemia)
 - Increased destruction of RBC due to severe infection (hemolytic anemia)
 - Acute, chronic blood loss
- **Diagnosis:**
 - Low mean corpuscular volume (MCV): the size of RBC (Normal: 80-100 fL) macrocytic/microcytic
 - Low Mean Corpuscular Hemoglobin Concentration (MCHC): average amount of hemoglobin in every single red blood cell (Normal: 32-36 g/dL) hypochromic/hyperchromic

Iron Deficiency Anemia (IDA)

- Results when the intake of dietary iron is inadequate for hemoglobin synthesis
- **Causes:**



- chronic blood loss due to trauma, menorrhagia, GI bleeding (hematemesis, melena, hematochezia)
- Inadequate Fe+2 in the diet
- Impaired Fe+2 absorption due to:
 - Chronic diarrhea
 - Malabsorption syndrome
 - Gastrectomy
 - celiac disease
- **Clinical manifestation:**
 - Fatigue/easy fatigability (Hallmark sign)
 - Brittle hair, spoon-shaped nails (koilonychia) due to atrophy of epidermal cells
 - Palpitations, cold sensitivity
 - Pallor
 - Smooth, sore tongue
 - Plummer Vinson's syndrome - atrophic glossitis, stomatitis, dysphagia due to atrophy of papilla of the tongue, mouth and pharyngeal cells
 - Pica - due to neuronal degeneration that affects cognitive function
 - Angular cheilosis - ulceration of the corner of the mouth
 - Cerebral hypoxia - dizziness, dyspnea
- **Diagnostic tests:**
 - RBC, hemoglobin, hematocrit, reticulocyte count, serum iron, ferritin
- **Management:**
 - Correction of chronic blood loss
 - Oral or parenteral iron therapy
 - Nursing
 - Monitor all signs of bleeding
 - CBR
 - Increase intake of iron rich foods:
 - green, leafy vegetables
 - Organ liver meat
 - Egg yolk
 - Legumes
 - california raisins
 - red meats
 - Molasses
 - Administer iron preparation as ordered
 - FeSO₄, Fe+2 Gluconate, Fe+2 fumarate
 - Taken with vitamin C to increase absorption
 - Instruct patient to avoid taking antacids and dairy products (it decreases iron absorption)



- Best absorbed on an empty stomach, in between meals to prevent GI upset
- Monitor side effects: nausea and vomiting, anorexia, abdominal pain, diarrhea/constipation, black stools

Megaloblastic: Pernicious Anemia

- Most dangerous of all chronic anemia due to deficiency of intrinsic factor leading to vitamin B12 malabsorption
- **Predisposing factors:**
 - Unknown cause
 - Subtotal gastrectomy
 - Hereditary
 - Inflammation disorders of the ileum
 - Crohn's diseases
 - Absence of intrinsic factor
 - Strict vegetarian diet
- **Clinical manifestations:**
 - Headache, dizziness, dyspnea, palpitations, cold sensitivity, general body malaise, extreme pallor
 - Sore mouth, anorexia, nausea, vomiting, loss of weight, indigestion, epigastric discomfort, recurring diarrhea or constipation
 - Red beefy tongue/glossitis - pathognomonic sign
 - Paresthesia in the extremities, difficulty maintaining their balance, lose position sense (proprioception) (+) Romberg's test
- **Diagnostic tests:**
 - CBC (decreased hemoglobin and hematocrit)
 - blood smear (large and oval shaped - macroovalocytes)
 - Schilling's test for absorption of vitamin B12 - patient receives small amount of radioactive B12 orally and 24 hour urine collection is obtained (definitive/confirmatory)
 - Positive: Vitamin B12 is absent in urine
 - Negative: Vitamin B12 is present in urine
- **Nursing management:**
 - Enforce CBR and ensure safety
 - Administer vitamin B12 (cyanocobalamin) injections at monthly intervals for lifetime as ordered (IM)
 - Diet:
 - SFF, bland soft food
 - Increase CHO, CHON, iron (0.3 mg/day) & vitamin C, folic acid 1 mg/day
 - Avoid irritating mouthwashes, use of soft bristled toothbrush is encouraged



- Avoid excessive heat and cold temperature
- Blood transfusion PRN
- Physical examinations every 6 months (hematologic studies & GI evaluation)
- Administer parenteral iron preparations as ordered
 - Administer using Z-track method to prevent discoloration and leakage to tissues
 - Do not massage the injection site, encourage ambulation instead
 - Monitor side effects:
 - Pain at injection site
 - Localized abscess
 - Lymphadenopathy
 - Fever and chills
 - Skin rashes
 - Watch out for anaphylaxis due to parenteral iron supplement

Megaloblastic: Folic Acid Deficiency

- Occurs if there is folic acid deficiency within 4 months
- **Risk factors:**
 - People who rarely eat uncooked vegetables
 - Alcoholism
 - Chronic hemolytic anemia
 - Pregnancy
 - Malabsorptive diseases of the small bowel such as sprue
- **Clinical manifestations:**
 - Fatigue, weakness
 - Pallor, dizziness, headache
 - Tachycardia
 - sore tongue, cracked lips
- **Diagnostic tests:**
 - CBC will show decreased RBC, hemoglobin, and hematocrit with increased mean corpuscular volume and mean corpuscular hemoglobin concentration
- **Management:**
 - Administer 1mg folic acid daily
 - Folic acid IM for patients with malabsorption problems
 - SFF, bland soft food if sore mouth and tongue is present
 - Diet: food rich in folic acid such as beef liver, peanut butter, red beans, oatmeal, broccoli, asparagus



Aplastic Anemia

- A rare disease caused by a decrease in or damage to marrow stem cells, damage to the microenvironment within the marrow, and replacement of the marrow with fat resulting in pancytopenia (low RBCs, WBCs and platelets)
- Hypoplasia of bone marrow
- **Predisposing factors:**
 - Congenital
 - Viral infection
 - Chemicals (benzene & its derivatives, pesticides)
 - Radiation
 - immunologic injury
 - Drug causing bone marrow depression
 - Broad spectrum antibiotics:
 - Chloramphenicol
 - Sulfonamides (Bactrim)
 - Chemo therapeutic drugs:
 - Methotrexate
 - nitrogen mustard
 - Vincristine
 - Attack of T cells against bone marrow
 - Infections and pregnancy
- **Clinical manifestations:**
 - Decreased RBC: pallor, weakness, fatigue, exertional dyspnea, palpitations, fatigue
 - Decreased WBC: Infections associated with leukopenia: fever, headache, malaise, abdominal pain, diarrhea, erythema, pain, exudates at wounds or sites of invasive procedures, lymphadenopathies, and splenomegaly
 - Decreased Platelets: Thrombocytopenia - bleeding from gums, nose, GI or GU tracts, purpura, petechiae, ecchymosis, retinal hemorrhages, oozing of blood from venipuncture sites
- **Diagnostic tests:**
 - Bone marrow aspiration shows an extremely hypoplastic or even aplastic (very few to no cells) marrow replaced with fat
 - CBC and peripheral blood smear shows decreased RBC, WBC and platelets (pancytopenia)
- **Therapeutic Management:**
 - Removal of causative agent or toxin
 - Bone Marrow Transplantation (BMT) or Peripheral Blood Stem Cell Transplant (PBSCT)
 - Immunosuppressive therapy
 - Corticosteroids



- Sandimmune (cyclosporine)
- Cytosan
- Imuran
- Supportive treatment includes platelet and RBC transfusions, antibiotics and antifungal administration
- Blood transfusion (depends on what blood cell is low)
- **Nursing Management:**
 - **Low RBC**
 - O2 administration
 - Rest
 - **Low WBC**
 - Practice hand hygiene
 - Avoid raw food
 - Administration of immunosuppressants as ordered
 - Teach patient how to minimize risk for infection
 - Reverse isolation due to leukopenia
 - Monitor for signs of infection
 - Limit visitors
 - Avoid crowded places
 - **Low Platelets**
 - Minimize risk of bleeding
 - Avoid SQ, IM. Use only soft toothbrush and electric razor
 - Avoid invasive procedures, contact sports

Sickle Cell Anemia

- Is a severe hemolytic anemia that results from inheritance of the sickle hemoglobin gene. This gene causes the hemoglobin molecule to be defective. The abnormal sickle hemoglobin (HbS) acquires a crystal-like formation when exposed to low oxygen tension
- **Autosomal recessive disorder**
 - If both parents have trait: 25%
 - If 1 parent have trait and other have the disease: 50%
 - If both have disease: 100%
- **Clinical manifestations:**
 - Severe pain in various parts of the body
 - Tachycardia, murmurs and cardiomegaly
 - Chest pain and dyspnea
 - Jaundice
 - Delayed growth
 - Fatigue



- enlarged skull and facial bones due to bone marrow expansion
- **Complications:**
 - Hypoxia, ischemia, infection, poor wound healing
 - Impotence
 - CVA
 - Renal failure
 - Heart failure
 - Pulmonary hypertension
- **Types of Cell Crisis:**
 - **Sickle crisis** - results from tissue hypoxia and necrosis due to inadequate blood flow to a specific region of tissue or organ
 - **Aplastic crisis** - results from infection with the human parvovirus
 - **Sequestration crisis** - results when other organs pool that sickled cells
 - **Vaso-occlusion crisis** - severe pain, hypoxia or ischemia
 - Eye
 - Lungs (pulmonary hypertension)
 - Kidney (kidney damage)
 - Brain (CVA)
 - Spleen (splenomegaly)
 - Hypovolemia
 - **Bone weakening**
- **Diagnosis:**
 - CBC (low RBC, hemoglobin)
 - Hemoglobin electrophoresis - to detect hemoglobin S
 - Stem cell - to produce healthy RBC
- **Treatment:**
 - bone marrow transplant
 - Hydroxyurea - anti-metabolite, avoids RBC to become sickle cell
 - Long term RBC transfusion
 - Splenectomy
 - peripheral blood stem cell transplant
- **Nursing management:**
 - **Managing pain**
 - Treat the triggering factors
 - Hypoxia: provide oxygen support
 - dehydration: infuse IV, encourage increased OFI
 - Infection control
 - Pain reliever: PRN (NSAIDs)
 - Support and elevate acutely inflamed joint
 - Relaxation techniques



- **Prevent and manage infection**
 - Monitor patient for signs and symptoms of infection
 - Initiate prompt antibiotic therapy
- **Monitor and prevent potential complications**
 - Always provide adequate hydration
 - Avoid cold temperature that may cause vasoconstriction
 - Protect leg from trauma and contamination to prevent leg ulcer
 - Aseptic technique
 - Avoid high altitude
- **Diet:**
 - High calorie, high protein, folic acid supplements

Thalassemia

- Group of hereditary anemias characterized by hypochromia (an abnormal decrease in the hemoglobin content of erythrocytes), extreme monocytosis (smaller than normal erythrocytes), destruction of blood elements (hemolysis) and variables degrees of anemia
- Associated with defective synthesis of hemoglobin; the production of one or more globulin chains within the hemoglobin molecule is reduced
- **2 classifications:**
 - **Alpha-thalassemia**
 - Occur mainly in people from Asia and Middle East
 - Milder than the beta forms and often occurs without symptoms; the erythrocytes are extremely microcytic, but the anemia if present is mild
 - **Beta-thalassemia**
 - Most prevalent in people from Mediterranean regions
 - Patients with mild forms have microcytosis and mild anemia
 - Severe beta-thalassemia (thalassemia major or Cooley's anemia) can be fatal within the first few years of life if untreated
- **Management:**
 - Bone marrow transplant
 - Blood transfusion of PRBCs
- **Thalassemia major (Cooley's anemia)**
 - Characterized by severe anemia, marked hemolysis and ineffective erythropoiesis
 - With early regular transfusion therapy, growth and development through childhood are facilitated
 - Management: PBSCT before liver damage occurs
 - WOF iron overload which results to excessive iron in multiple packed RBCs
 - Management: regular chelation therapy



Polycythemia Vera (Primary Polycythemia)

- Proliferative disorder of the myeloid stem cells
- The bone marrow is hypercellular (hyperplasia)
- Elevated levels of blood cells (erythrocytes, leukocytes, thrombocytes)
- Blood = concentration
 - -> increased blood viscosity -> thromboembolism -> coronary artery disease/CVA
 - -> organ involvement -> cerebral hypoxia/arthritis/splenomegaly/hepatomegaly
 - -> capillary over distention -> rupture -> bleeding
- **Clinical manifestations:**
 - Chest pain
 - Ruddy complexion
 - Splenomegaly/Hepatomegaly
 - headache and dizziness
 - Tinnitus, fatigue, paresthesia
 - Blurred vision
 - Increased blood viscosity: angina, claudication, dyspnea, and thrombophlebitis
 - elevated BP
 - Uric acid may be elevated resulting in gout and renal stone formation
 - Generalized pruritus
 - Erythromyalgia (burning sensation in fingers and toes)
- **Diagnostic tests:**
 - CBC
 - Bone Marrow aspiration
- **Complications:**
 - CVA
 - MI
 - bleeding due to dysfunctional large amount of platelets
- **Management:**
 - Phlebotomy - removing enough blood (initially 500 ml once or twice a week) to reduce blood viscosity and to deplete the patient's iron stores
 - Chemotherapeutic agents (hydroxyurea) can be used to suppress marrow function
 - Anagrelide (Agrylin) - inhibits platelet aggregation
 - Interferon alfa-2b (Intron-A) - for management of pruritus (WOF: flu-like symptoms and depression)
 - Antihistamine
 - Allopurinol
 - Analgesic for pain
 - Nursing management:
 - Monitor sign of thromboembolism



- Increase fluid intake to reduce blood viscosity
- Instruct the patient to avoid sedentary behaviors, crossing legs, wearing tight or restrictive clothing
- avoid aspirin and aspirin-containing medications
- Minimize alcohol intake
- Instruct patient to avoid iron supplements
- Avoid activity in high altitude (low level of oxygen -> stimulates kidney (erythropoietin) -> stimulates bone to produce blood -> increased blood viscosity)
- For pruritus:
 - Bathing in tepid or cool water
 - Use cocoa butter or oat meal based lotions and bath products
 - Dissolved baking soda in bath water

Bleeding Disorders

Hemophilia

- **Inherited bleeding disorder**
 - **Hemophilia A (Classic Hemophilia)** - caused by genetic disease that results in deficient or defective factor VIII
 - **Hemophilia B (Christmas Disease)** - genetic defect that causes deficient or defective factor IX
- Both types of hemophilia are inherited as X-linked recessive traits, so almost all affected people are males; females can be carriers
- Recognized in early childhood, usually in the toddler age group
- **Clinical manifestations:**
 - hemorrhages into various parts of the body
 - hemarthroses and hematomas
 - 75% of all bleeding occurs into joints
 - Chronic pain or ankylosis (fixation) of the joints occurs
 - spontaneous hematuria and GI bleeding
 - Intracranial or extra cranial bleeding - most dangerous
- **Management:**
 - Administration of factor VIII and factor IX concentration
 - Infusion of fresh frozen plasma
 - Plasmapheresis or concurrent immunosuppressive therapy
 - Aminocaproic acid inhibits fibrinolysis and therefore stabilize the clot
 - Desmopressin (DDVAP) - induce significant but transient rise in factor VII levels



- **Nursing management:**

- RICE: Rest, Ice/Immobilize, Compression/Elevate
- Gentle massive exercise 48 hours after acute phase A
- Avoid immobility
- Assist child in coping with the condition
- Encouraged to be self-sufficient and to maintain independence by preventing unnecessary trauma that can cause acute bleeding episodes
- Instruct patient to avoid OTC medications such as aspirin, NSAIDs, herbs, nutritional supplements and alcohol
- Nasal packing should be avoided, because bleeding frequently resumes when the packing is removed
- All injections should be avoided
- Splints and other orthopedic devices may be useful in patients with joint or muscle hemorrhages
- Warm baths promote relaxation, improve mobility
- Provide genetic testing and counselling to female carriers

Thrombocytopenia

- **Causes:**

- decreased platelet production
- Increased platelet destruction (ITP, DIC)

- **Normal:** 150,000-450,000

- **Signs and symptoms:**

- Petechiae
- Ecchymosis
- bleeding (GI, GU)

- **Management:**

- blood transfusion: platelet
- Steroids - decreased production of anti platelet antibody (IgG)
- Bleeding precaution:
 - Avoid contact sports
 - NSAIDs, aspirin

Idiopathic Thrombocytopenia Purpura

- Autoimmune

- **Management:**

- Platelet transfusion
- Steroids
- IV immunoglobulin - increases platelet count
- Bleeding precautions



von Willebrand's Factor Disease

- Inherited or acquired blood disorder
- Glycoprotein that plays a crucial role in blood clotting
- decreased level of von Willebrand's factor (needed for formation of clotting factor VIII (Hemophilia A) -> bleeding -> bruising, abnormal menstrual bleeding, bleeding in gums, nose, GIT, excessive bleeding after procedure
- **Management:**
 - Desmopressin - synthetic analogue of vasopressin (increase the level of von Willebrand's factor -> reduced bleeding)
 - Reduced fluid intake
 - S/E: decreased urination (cerebral edema -> increased ICP)
 - Factor VIII replacement: cryoprecipitate
 - Antifibrinolytic agent: Amicar - to stabilize clot formation especially before procedure
 - Bleeding precaution

Bleeding Precaution

1. **Minimize trauma**
 - a. Avoid IM unless absolutely necessary
 - b. Use the smallest gauge needle possible
 - c. Use soft toothbrushes or oral swabs for oral care
 - d. Avoid rectal temperatures, enemas, or suppositories
 - e. Use electric razors instead of blades
2. **Monitor for bleeding**
 - a. Check for signs of:
 - i. Gum bleeding
 - ii. Nosebleed
 - iii. blood in urine or stool
 - iv. Bruising or petechiae
 - b. Report persistent bleeding immediately
3. **Avoid medications that increase bleeding risk**
 - a. Avoid NSAIDS unless prescribed
 - b. Only use prescribed anticoagulants with close monitoring
 - c. Check herbal supplements that may interfere without (garlic, ginkgo)
4. **Environmental safety**
 - a. Keep the patient's environment clutter-free to avoid bumps and falls
 - b. ensure adequate lighting to prevent injury
 - c. Use bed rails if needed to prevent falls
5. **Activity restrictions**



- a. avoid contact sports or vigorous activity
- b. Use protective equipment if mobility aids are used
- c. consider limiting straining during bowel movements (use stool softeners as needed)

6. Patient education

- a. Teach patients and caregivers:
 - i. To recognize signs of internal and external bleeding
 - ii. The importance of avoiding injury
 - iii. When and how to seek medical help

Infection Precaution

1. Standard precaution

- a. Frequent hand hygiene
- b. Use of PPE
- c. Safe injection practices and sterile techniques
- d. Proper waste disposal and handling of bodily fluids

2. Neutropenic precautions (<500 cells/mm³)

- a. Protective isolation or reverse isolation
- b. Limit visitors
- c. Staff and visitors must wear surgical masks, sanitize hands and possibly gown
- d. No fresh flowers/plants or raw fruits/vegetables
- e. Use bottled or sterile water if advised
- f. daily bathing with antimicrobial soap

3. Environmental controls

- a. Patient rooms clean daily with disinfectants
- b. HEPA filtration for bone marrow transplant units
- c. Avoid shared equipment unless thoroughly disinfected
- d. No humidifiers or standing water

4. Antimicrobial prophylaxis

- a. Antibacterial (levofloxacin)
- b. Antiviral (acyclovir)
- c. Antifungal (fluconazole)
- d. Pneumocystis jirovecii (PCP) prevention (trimethoprim-sulfamethaxazole)

5. Vaccination precaution

- a. No live vaccines
- b. Ensure household contacts are up to date on vaccines
- c. Post-chemotherapy or transplant patients may need revaccination

6. Patient and family education

- a. Report fever >38C (100.4F) immediately (medical emergency in neutropenia)
- b. Avoid crowds and public spaces when counts are low



- c. Maintain good oral hygiene but avoid dental procedures unless cleared
- d. Safe food handling
- e. Encourage wearing mask in public

Perioperative/Surgery

Perioperative nursing - a specialized area of nursing focused on the care of surgical patients before, during and after their procedure

- **Preoperative phase:** begins when the decision to proceed with surgical intervention is made and ends with the transfer of the patient onto the operating room table
 - **Visit the patient** - the day or the night before the surgery; for additional queries
 - **Interview/informed consent** - one method of assessment (preoperative assessment)
 - **Preoperative assessment:** to identify and correct problems before surgery and establish a baseline data for postoperative comparison
 - **Preparation** - physical and psychological
 - **Psychological preparation**
 - Alleviate the fear of the unknown (anxiety) - most common
 - **Physical preparation**
 - Health teaching (deep breathing, performing early ambulation and coughing exercises, splinting the wound)
 - **Preoperative checklist:** tests or activities conducted before the procedure
 - What is to be removed and retained
 - Jewelries/contact lens/dentures/nail polish (removed)
 - Hearing aid (retained)
 - Methods of hair removal
 - Shaving
 - Clipping
 - Depilation (cream)
- **Intraoperative phase:** begins when the patient is transferred onto the OR table and ends with admission to the PACU (Post Anesthesia Care Unit)
 - **Safety**
 - **Aseptic-controlled environment**
 - **Traffic control in the OR**
 - Movement of personnel - kept to a minimum
 - OR zones - installing signage to clarify requirements for what must be worn in each area
 - **Cleaning in the OR**
 - **Routine cleaning**



- **Cleaning first thing the morning** - done before the patient is received in the OR
 - **Damp dusting** - performed before surgical supplies are brought to OR and are opened, use clean, low-linting cloth moistened with disinfectant
 - **End of procedure cleaning** - done between every patient during the day
 - **Terminal cleaning and disinfecting/Turnover cleaning** - done at the end of the day
- **Scheduled cleaning** - depends upon institutional protocol
- **Coordinated cleaning** - cleaning of equipment with the help of maintenance
- **Prevent sentinel events in the OR**
 - **Sentinel events** - a patient safety event that may result in death, severe or permanent harm; wrong patient, wrong procedure; wrong site
 - **Pre-procedure verification process** - multiple times whenever necessary
 - **Site marking** - laterality (extremities), paired organs, specific levels (spine surgery)
 - Do not use X mark (ambiguous)
 - **Time out/huddle** - brainstorming led by the circulating nurse; recite the information of procedure to ensure the whole team know the procedure (everyone is on the same page); it may be performed as necessary
- **Temperature and humidity level in the OR**
 - **Temperature:** 20-24C, **Humidity:** 50-55%
 - Microorganisms will proliferate if the temperature and humidity is not in the recommended range
- **Positioning**
- **I&O** (collaborative with anesthesiologist)
- **Postoperative phase:** begins with the admission of the patient to PACU and ends with the follow-up evaluation in the clinical setting or home
 - **Temperature** and other VS
 - **Oxygenation** status
 - **LOC**
 - **Drainage** and dressing
 - **Airway**



Surgical Classification

According to Magnitude

- **Major procedure**
 - High risk for morbidity and mortality
 - Extensive or prolonged
 - Large amount of blood loss
 - Presence of complications
- **Minor procedure:** procedures that are not included in the characteristics of major procedures

Note: anesthesia is not a part of the characteristics of major or minor procedure

According to Purpose

- **Diagnostic:** to establish the disease or condition (biopsy)
- **Exploratory:** to determine cause of unexplained symptoms; to determine the extent of a condition (exploratory laparotomy)
- **Preventive/Prophylactic:** stops the development of a disease (removal of pre-cancerous lesions)
- **Curative**
 - **Ablative:** removal of body part
 - **Constructive:** repair for congenital defects
 - **Reconstructive or cosmetics:** repair of a damaged or diseased organs (mammoplasty or facelift)
- **Palliative:** to reduce distressful signs and symptoms

According to the Degree of Urgency

- **Optional:** surgery is scheduled completely at the preference of the patient
- **Elective:** the approximate time for surgery is at the convenience of the patient; failure to have surgery is not catastrophic
- **Required:** the condition requires surgery within a few weeks
- **Urgent:** the surgical problem requires attention within 24 to 48 hours
- **Emergency:** the situation requires immediate surgical attention without delay

According to Invasiveness

- **Open surgery:** requires cutting of skin and tissues for a full view of the organs or structures involved
- **Minimally invasive surgery:** does not require large incision (keyhole surgery/laparoscopy surgery)

Informed Consent

Informed consent: the patient's autonomous decision about whether to undergo a surgical procedure. It is the process of informing the patient about the surgical procedure; that is risks and possible complications of surgery and anesthesia

- Consent is obtained by the surgeon; this is a legal requirement



- **Purpose:**
 - To ensure that the patient understands the nature of treatment, including potential complications
 - To indicate the patient's decision was made without pressure
 - To protect the patient against unauthorized procedure is performed on the correct body part
 - To protect the surgical team and hospital against legal action by a patient who claims that an unauthorized procedure was performed
- **Information provided to the patient**
 - Nature and magnitude of the procedure
 - Risks and benefits of the procedure
 - Alternative treatment
 - Prognosis
- **Role of Nurse**
 - Witness client's signature on the form
 - Advocate client by verifying that the client received adequate information to give consent
- **Important reminders**
 - Right of consent also includes the right of refusal
 - Right to refuse continues even after signing the consent form
 - Document client concerns or questions and notify the HCP
- **Who cannot give consent**
 - Minors
 - Unconscious
 - Mentally-ill
- **Conditions requiring Informed Consent**
 - Minor and Major procedures
 - Anesthesia administration
 - Radiographic exam with the use of contrast medium
 - Entry to body cavity

Surgical Team

1. Circulating Nurse

- a. Main responsibilities include:
 - i. Verifying consent
 - ii. Coordinating the team
 - iii. Ensuring cleanliness
 - iv. Proper temperature and humidity
 - v. Light and safe function of equipment and availability of supplies and materials



- b. The circulating nurse monitors aseptic practices to avoid breaks in technique while coordinating the movement of related personnel as well as implementing fire safety precautions
 - c. Responsible for ensuring that the second verification of the surgical procedure and site takes place is documented
- 2. Scrub Nurse**
- a. Performs surgical hand scrub
 - b. Setting up the sterile tables
 - c. Preparing sutures, ligatures and special equipments
 - d. Assisting the surgeon and the surgical assistants during the procedure by anticipating the instruments and supplies that will be required such as sponges, drains, and other equipment
 - e. Scrub nurse and circulating nurse count all needles, sponges and instruments to be sure they are accounted for and not retained as a foreign body in the patient
- 3. Surgeon**
- a. performs the surgical procedure, heads the surgical team and is specially trained and qualified
 - b. Has the ultimate responsibility for performing the surgery in an effective and safe manner
- 4. Anesthesiologist**
- a. Assess the patient before surgery, selects anesthesia, administer it, intubates patient if necessary, manages any technical problems related to the administration of the anesthetic agents and supervises the patient's condition throughout the surgical procedure
 - b. During surgery, the anesthesiologist monitors the patient's BP, PR, RR as well as ECG, O2 saturation, tidal volume, blood gas level, blood pH, alveolar gas concentration and body temperature

Surgical Environment

- The surgical suite is double doors, and access is limited to authorized personnel
- External precautions include adherence to principles of surgical asepsis; strict control of the OR environment is required including traffic pattern restriction
- To provide the best possible conditions for surgery, the OR is situated in a location that is central to all supporting services
- To help decrease microbes, the surgical area is divided into three zones:
 - **Unrestricted zone:** where street clothes are allowed; area in operating room that interfaces with other departments, includes patient reception and holding area
 - **Semi-restricted zone:** area in the OR where scrub attire is required; may include area where surgical instruments are processed
 - **Restricted zone:** scrub clothes, shoe cover caps, and masks are worn, includes OR and sterile core area



- **Other concepts in OR:**

- Time out
- Sentinel events
- Surgical conscience

- **Quintana Model of Surgical Conscience**

- Knowing what is sterile and safe
- Accepting a moral obligation to advocate safety
- Acting in accordance with the obligation to advocate safety

Surgical Attire

1. Scrub suit

- a. Two piece pant suit
- b. Worn in the semi-restricted
- c. Must fit the body properly
- d. Waistline drawstrings must be tucked in
- e. Wet or soiled garments should be changed

2. Head cover

- a. Should cover the hair completely
- b. Worn in the semi-restricted
- c. Never comb your hair when wearing a scrub suit
- d. Disposable caps are preferred
- e. Bald head also causes contamination by shedding squamous cells
- f. Net caps do not prevent contamination

3. Shoes and Shoe cover

- a. Worn in semi-restricted area
- b. Should be comfortable and puncture resistant
- c. Shoe covers are worn during procedures with expected spills/splashes of blood or body fluid
- d. Street shoes are not used
- e. Shoe covers should be disposed before leaving OR

4. Surgical mask

- a. High filtration masks decrease the risk of post wound infection
- b. Worn inside the restricted area at all times
- c. Should cover nose and mouth completely
- d. Should fit tightly
- e. Double masking - a barrier not a filter
- f. Masks are changed between patients and should not be worn outside the OR
- g. Handle the mask by ties or strings
- h. Front of the mask is contaminated



- i. Mask should never be hung on the neck or placed on top of cap
 - j. It should not be kept in the pocket after use
 - k. Should not interfere with breathing, speech or vision
- 5. Eye wear**
- a. Eye wear or face shield protects the eyes from splashing of blood and body fluids or from debris when bone drilling is performed
- 6. Laser eyewear**
- a. Protects the eyes from the intense light created by laser surgery
- 7. Gloves**
- a. Nonsterile gloves: donned for clean procedures
 - b. Sterile gloves: donned for sterile procedures

Principles for Surgical Asepsis

- All surgical supplies, instruments, needles, sutures, dressings, gloves, covers, and solutions that may come in contact with the surgical wound or exposed tissue must be sterilized before use
- The surgeon, surgical assistants, and nurses prepare themselves by scrubbing their hands and arms with anti-septic soap and water or alcohol-based product or scrubless soap is used to prepare for surgery
- During surgery, only personnel who have scrubbed, gloved, gowned touch sterilize objects
- Requires meticulous cleaning and maintenance of the OR environment
- An area of the patient's skin larger than that requiring exposure during the surgery is meticulously cleansed, and an antiseptic solution is applied

Basic Guidelines for Maintaining Surgical Asepsis

- Only sterile surfaces/articles may touch other sterile surfaces/articles
- Gowns of the surgical team are considered sterile in front from the chest to the level of the sterile field. The sleeves are also considered sterile from 2 inches above the elbow to the stockinette cuff
- Only the top surface of a draped table is considered sterile
- Sterile supplies including solutions are delivered to a sterile field or handed to a scrubbed person in such a way that the sterility of the object or fluid remains intact
- The movements of the surgical team are from sterile to sterile areas and from unsterile to unsterile areas
- Movement around the sterile field must not cause contamination of the field
- Whenever a sterile barrier is breached, the area must be considered contaminated
- Items of doubtful sterility are considered unsterile

Decontamination Process

- **Decontamination/Cleaning:** removal of foreign material from the instrument by a combination of mechanical means (scrubbing) and chemical means (enzyme and detergents)
 - **Mechanical/Physical:** the use of physical force
 - **Chemical:** uses substances to loosen up the debris from the instrument
- **Disinfection:** reduces or eliminates many pathogens but not necessarily all especially bacterial spores
 - **Levels of Disinfection**
 - **Low level:** limited/narrow spectrum (phenolic compounds)
 - **Intermediate level:** wide spectrum (70% isopropyl alcohol)
 - **High level:** 2% glutaraldehyde (high level disinfectant)
 - **Recommended immersion time:** 20 minutes (high level disinfectants); sterilization: 10 hours
 - **Physical disinfection by boiling:** uses 100C boiling water to destroy most pathogens except spores
 - **Pasteurization:**
 - Used for items such as reusable respiratory devices and anesthesia breathing circuit
 - Exposure to hot water with temperature of 60-80C for 30 mins
 - **Chemical disinfection:**
 - Items are soaked in disinfectant
 - Choice depends on compatibility and effectiveness of the instruments
 - Used for heat labile instruments that cannot be boiled or sterilized
- **Sterilization:**
 - Process which all pathogens are destroyed including spores
 - Highest level of decontamination
 - **Methods of sterilization:**
 - **Physical sterilization**
 - Exposure of instruments to radiation
 - **Chemical sterilization**
 - Ethylene oxide gas is used to sterilize items that are sensitive to heat or moisture
 - Heat and moisture labile instruments
 - **Autoclaving (Steam under pressure sterilization)**
 - Most common method
 - Uses steam with 120-121C temperature and 1 ATM pressure for 30 mins
 - **Flush sterilization** - emergency sterilization of unwrapped items
 - **Dry heat sterilization**
 - Dry heat in form of air is used



- Sterilizes anhydrous oils, petroleum products and talc powder
- **Indicators Used in Sterilization**
 - Indicators never indicate sterility; it is only a parameter that instruments have undergone sterilization
 - Event related not based on time
 - Expiration date is only an estimate
 - Shelf life depends on:
 - Amount of handling
 - The quality of packaging materials used
 - Storage conditions

General Risk Factors to Surgery

- Age - older adults have reduced physiologic reserves
- Obesity
- Poor nutrition
- Fluid and electrolyte imbalances
- Pre-existing conditions (DM, respiratory conditions)
- Alcoholism - can cause alcohol withdrawal (delirium tremors)
- Severity and urgency of surgery

Preoperative Medication

- Reduce anxiety
- Promote relaxation
- Reduce pharyngeal secretions
- Prevent laryngospasm
- Inhibit gastric secretion
- Decrease amount of anesthetic needed for induction and maintenance of anesthesia
- Anesthesia a state of narcosis, analgesia, relaxation and reflex loss

General Anesthesia

- A reversible consisting of complete loss of consciousness that provides analgesia, muscle relaxation and sedation, protective reflex are lost
- Lose the ability to maintain ventilator function and require assistance in maintaining a patent airway

Administration of General Anesthesia

- **Inhalation**



- Inhaled anesthetic agents include volatile liquid agents and gases
- Volatile liquid: anesthetic agents produce anesthesia when their vapors are inhaled
 - Halothane (Fluothane)
 - Enflurane (Ethrane)
 - Isoflurane (Forane)
 - Sevoflurane (Ultrane)
 - Desflurane (Suprane)
- Gas anesthetic agents are administered by inhalation and are always combined with oxygen
 - Nitrous oxide
- **Intravenous**
 - Barbiturates
 - Benzodiazepines
 - Non-barbiturate hypnotics
 - Dissociative agents
 - Opioid agents

Stages of General Anesthesia

- **Stage 1 (Beginning Anesthesia/Induction)**
 - Feeling of detachment
 - Drowsy/dizziness
 - Hallucinations occurs
 - Close OR doors
 - Ringing, roaring or buzzing in the ears
 - Keep quiet because exaggerated noises are heard by the patient
 - Standby to assist the client
- **Stage 2 (Excitement/Delirium)**
 - Pupils are dilated, pulse rate is rapid, and may have irregular respirations
 - Because of uncontrolled movement of the patient, restraints are necessary
- **Stage 3 (Surgical Anesthesia)**
 - Patient is unconscious and lies quietly
 - Pupils are small but reactive to light
 - Respirations are regular, the pulse and volume are normal
 - Skin is pink or slightly flushed
- **Stage 4 (Medullary Depression/Danger)**
 - Too much anesthesia has been administered
 - Shallow respirations, weak and thread pulse
 - Widely dilated pupils
 - Death may occur
 - If this stage develops, discontinue anesthesia and initiate respiratory and circulatory support



Regional Anesthesia

- anesthetic agents are injected around nerves so that the region supplied by these nerves is anesthetized
- patient receiving regional anesthesia is awake and aware of his/her surroundings unless medications are given to produce mild sedation or to relieve anxiety

Administration of Regional Anesthesia

- **Epidural Anesthesia**
 - achieved by injecting a local anesthetic agents into the epidural space that surrounds the dura mater of the spinal cord
 - advantage: absence of headache
 - Disadvantage: greater technical challenge of introducing the anesthetic agents into the epidural rather than the subarachnoid space
- **Spinal Anesthesia**
 - extensive conduction nerve block that is produced when a local anesthetic agents is introduced into the subarachnoid space at the lumbar level, usually between L4 and L5
 - it produces anesthesia of the lower extremities, perineum, and lower abdomen
- **Moderate sedation**
 - Previously referred to as conscious sedation, is form of anesthesia that involves the IV administration of sedative or analgesic medications to reduce patient's anxiety and control pain during diagnostic or therapeutic procedures
- **Monitored Anesthesia Care (MAC)**
 - Also referred to as monitored sedation is moderate sedation administered by an anesthesiologist or anesthesiologist who must be prepared and qualified to convert to general anesthesia if necessary
- **Local Anesthesia**
 - Injection of a solution containing the anesthetic agents into the tissues at the planned incision site

WHO Surgical Checklist

Sign In

- Before the actual start of surgery
- Identifying the patient, site
- Allergies, antibiotics, anticipated blood loss (if >500ml or 7ml/kg in children, need BT)
- Pulse oximeter

Time Out

- Introduction of patient and procedure by doctor
- Identify of team
- Special concerns of anesthesiologist



- Completeness and sterility of equipment by scrub nurse
- Done before skin incision

Sign Out

- Completeness of instruments
- Proper endorsement of specimen
- Must be labeled with name, age, gender, specimen/location, surgeon, pathologist
- For normal histopathology
 - On buffered formalin solution
 - After 7-10 working days
 - Labeled properly
- Frozen section
 - Clean plastic bag
 - Labeled properly
 - Labeled immediately and sent right away
- Follow up histopathology results to physician

Preoperative Phase

Preoperative Assessment

- The goal in the preoperative period is for the patient to be as healthy as possible
 - Consent
 - Health history is obtained
 - Nutritional and fluid status
 - Dentition
 - Alcohol and drug use
 - Respiratory status
 - Cardiovascular status
 - Hepatic, renal and endocrine function
 - Previous medication used

Preoperative Nursing Interventions

- **Providing patient teaching**
 - **Deep breathing, coughing and incentive**
 - Spirometry
 - Demonstrate how to take deep, slow breath and how to exhale slowly
 - Instruct patient to breathe deeply, exhale through mouth, take a short breath, and cough from deep in the lungs
 - Demonstrates how to use an incentive spirometer
- **Mobility and active body movement**

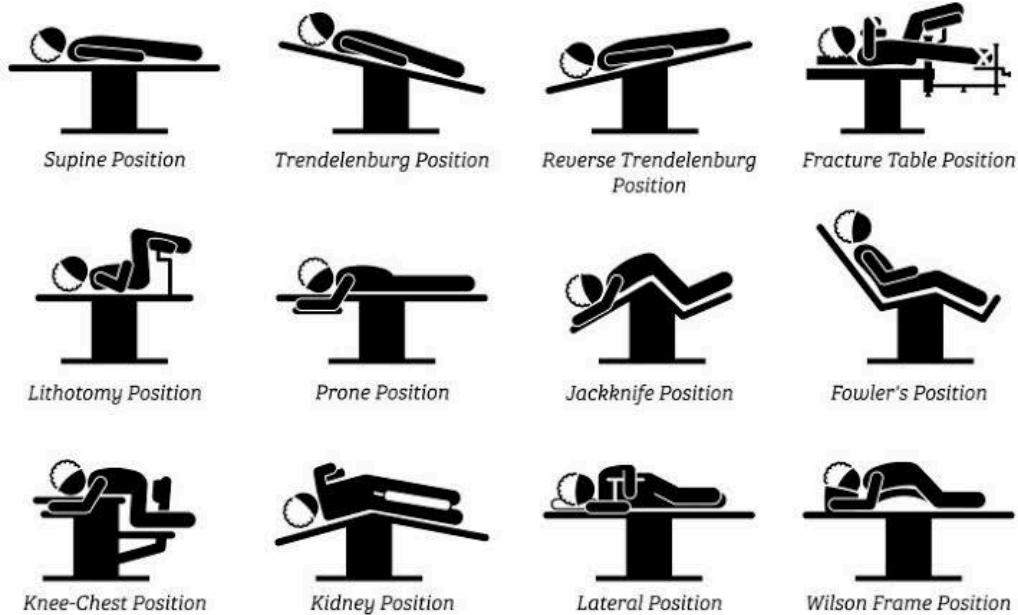


- Patient should be taught that early and frequent ambulation, exercise of the extremities and frequent change of position immediately postoperative as tolerated will help prevent complications
- **Pain management**
 - Pain is instructed to take medication as frequently as prescribed during the initial postoperative period for pain relief
- **Reducing anxiety and decreasing fear**
 - Assist patient to identify coping strategies that he/she has previously used to decrease fear
- **Managing nutrition and fluids**
 - NPO overnight or longer to prevent aspiration
 - Specific recommendations depend on the age of the patient and type of food eaten
- **Bowel preparation**
 - Enemas are not commonly prescribed preoperatively unless the patient is undergoing abdominal or pelvic surgery
 - Allow satisfactory visualization of the surgical site and to prevent trauma to the intestine or contamination of the peritoneum by fecal material
 - Cleansing enema or laxatives may be prescribed evening before the surgery and maybe repeated the morning of the surgery
- **Skin preparation**
 - Goal is to decrease bacteria without injuring the skin
 - Cleanse the skin with soap containing detergent-germicide
 - If hair must be removed, electric clippers are used

Immediate Pre-Op Nursing Interventions

- Assist the patient in changing hospital gown
- cover the head completely with a cap
- Inspect the mount, dentures, and plates are removed
- Remove all jewelries/body piercing
- If the patient objects in removing wedding ring, secure it with tape
- All valuable articles are given to family members or labelled it clearly and store it in a safe and secured place according to hospital policy
- Patient should avoid first before going to the OR
- If preanesthetic medication is administered, the patient is kept in bed with side rails raised
- The completed chart accompanies the patient to the OR with the surgical consent form attached, along with all laboratory reports and nurse's records
- Transfer the patient to the holding area/presurgical suite 30-60 minutes before anesthetic agents is administered
- Patient safety in the preoperative area is PRORITY

Surgical Positions



- **Supine (dorsal) position**
 - patient is flat on the back, both arms are positioned at the side of the table, one with the hand placed palm down; the other is carefully positioned on an armboard to facilitate IV infusion of fluids, blood, or medications
 - Used for procedures of anterior surface of the body, such as abdominal, abdominothoracic and some lower extremities procedures
 - Shoulder or anterolateral procedures: the patient is on supine position with a small sandbag/water bag/roll/pad is placed under the affected side to elevate and expose the shoulder
 - Dorsal recumbent: for vaginal or perineal procedures
 - Modified dorsal recumbent (frog leg): surgical procedures in the groin lower extremities
 - Arm extension: surgical procedures of the breast, axilla, upper extremities or hand
- **Trendelenburg's position**
 - Usually used for surgery on the lower abdomen and pelvis to obtain good exposure by displacing the intestines into the upper abdomen
- **Reverse Trendelenburg's position**
 - Used for thyroidectomy, laparoscopic gallbladder, biliary tract or stomach procedure
- **Fowler's position**
 - Used for shoulder, nasopharyngeal, facial and breast reconstruction procedure
- **Sitting position**



- Occasionally used for otorhinologic and neurosurgical procedure
- **Prone position**
 - Used for all procedures with dorsal or posterior approach
 - Modified prone procedure: used for neurosurgical and spine procedures
- **Kraske (Jack-Knife) position**
 - Hips are positioned over the center break of the operating table between the body and leg section
 - The leg section of the operating table is lowered (usually 90) and the entire operating bed is tilted head downward to elevate hips above the rest of the body
 - Done for rectal procedures
- **Knee-Chest position**
 - used for sigmoidoscopy or culdoscopy
- **Lateral position**
 - Used for renal surgery

Spaulding's Classification of Instruments

- **Non-critical:** items that come in contact with intact skin; immediate level disinfection
 - Stethoscope
 - BP cuffs
 - Tourniquet
- **Semi-critical:** items that come in contact with mucous membranes and non-intact skin; sterilization (preferred)/high level disinfection
 - Anesthesia equipment
 - GI endoscopes
 - Speculum
 - Bronchoscopies
 - Laryngoscopes
 - Thermometer
 - Respiratory therapy equipment (endotracheal tube)
- **Critical:** items that come in penetrate sterile tissues such as body cavity and vascular system; sterilization
 - Surgical instruments
 - IUD
 - Vascular catheters
 - Implants
 - Urinary catheter
 - IV needles, fluids



Classification of Instruments

- **Cutting and dissecting**
 - Used to dissect, incise, separate or excise tissues
 - **Scalpels**
 - Blade 10, 11, 12 and 15 fits handle size 3 or 7
 - Blade 20, 22, 25 fits handle size 4
 - Blade 10 is used to open the skin
 - Blade 11 makes initial skin puncture for tiny deep incisions
 - Blade 12 is commonly used for tonsillectomy
 - Blade 15 is used for shallow short controlled incisions
 - Blade 20 same with 10 but larger in size
 - **Scissors**
 - Suture scissors are used to cut sutures
 - Wire scissors are used to cut wires
 - Bandage scissors are used to cut drains and dressings and to open items such as plastic packets
 - Sharp-tipped angled scissors with short jaws used for vascular surgery
 - Mayo scissors are used for cutting heavy fascia and sutures
 - Metzenbaum scissors are more delicate than mayo scissors and are used to cut delicate tissues
 - **Curettes**
 - Tissue from bone is removed by scraping with the sharp edge of the loop or scoop on the end of the curette
- **Grasping and holding**
 - Tissues should be grasped held in position so surgeon can perform the design and the maneuver without injuring the surrounding tissues
 - **Delicate forceps** - hold fine tissues such as eye tissues
 - **Adson forcep** - used to pick up or hold soft tissues during closure
 - **Smooth (thumb) forceps** - used to prevent injury to the suture
 - **Toothed forceps** - hold on tough tissue
 - **Allis forceps** - used to hold tough tissue
 - **Babcock forceps** - end of each jaw is rounded to grasp tissue without injury
 - **Lahey forceps** - has sharp point tips to grasp tough organs or tumors
 - **Stone forceps** - grasp calculi in kidney and gallbladder
- **Clamping and occluding**
 - instruments that apply pressure by clamping or occluding
 - **Hemostatic forceps** - used for occluding blood vessels
 - **Crushing clamps** - used to crush tissues or clamp blood vessels
- **Exposing and retracting**



- Soft tissues, muscles, and other structures should be pulled aside for exposure of the surgical site
 - **Malleable retractors** - maybe bent to the desired angle and depth for retraction
 - **Hooks** - commonly used to retract skin edges during a wide-flap dissection
 - **self-retraining** - inserted to spread the edges of an incision and hold them apart
- **Sutures**
 - used for ligating, stitching, or approximating tissues
 - **Absorbable sutures**
 - **Surgical gut** - collagen derived from the submucosa or sheep's intestine
 - **Plain surgical gut** - loses strength in 5-10 days and is digested within 70 days, used to ligate small vessels and sutures subcutaneous fats
 - **Chromic surgical gut** - support the wound for about 14 days and loses tensile strength up to 21 days and is absorbed within 90 days, used for ligation of larger vessels and sutures urinary/biliary tract
 - **Synthetic absorbable polymers** - are absorbed by a slow hydrolysis process in the presence of tissue fluids
 - **Polydioxanone sutures (PDS)** - useful in tissues in which wound healing is slow, as in the fascia or where extended wound support is desirable
 - **Poliglecaprone 25 (Monocryl)** - loses all tensile strength by 21 days and absorption is between 91-119 days, used in soft tissues such as gynecologic, urologic and plastic surgery
 - **Polyglactin 910 (Vicryl)** - absorbs rapidly within 90 days, uncoated polyglactin 910 is used for ophthalmic procedures
 - **Non-absorbable sutures**
 - **Surgical silk** - loses tensile strength when wet, used frequently in the serosa of the GI tract and to close fascia in the absence of infection
 - **Surgical cotton** - gains tensile strength when wet, used in most body tissues for ligating and suturing
 - **Surgical stainless steel** - used for abdominal wall or for retention suture to reduce the danger of wound disruption/dehiscence
- **Surgical needles**
 - **Point of the needle** - honed to the configuration and sharpness desired of specific types of tissue
 - **Cutting point** is used when tissue is difficult to penetrate
 - Conventional cutting needles
 - Reverse-cutting needles
 - Side cutting needles
 - Trocar point
 - **Taper (round) point** is used when tissue such as intestines and peritoneum
 - **Blunt point** is used for suturing friable tissues



- **Body of the needle**
 - **Straight needles** are used in readily accessible tissues
 - **Curved needles** are used to approximate most tissue
 - **French eye needle** has a slit from the inside of the eye to the end of the needle through which the suture strand is drawn
 - **Eyeless needle** is a continuous unit with the suture strand, needle is swagged onto the end of the strand in the manufacturing process

Potential Intraoperative Complications

- **Nausea and vomiting**
 - administer antiemetics preoperatively or intraoperatively as ordered to counteract possible aspiration
 - If gagging occurs, turn the patient to side, the head of the table is lowered, and a basin is provided to collect the vomitus
 - Suction saliva and vomited gastric content
- **Respiratory complications**
 - May lead to brain damage if not recognized
 - Be vigilant in monitoring the patient's oxygenation status
 - Check peripheral perfusion frequently
- **Hypothermia**
 - Patient's temperature may fall during the anesthesia
 - may occur as a result of low temperature in the OR, infusions of cold fluids, inhalation of cold gases, open body wounds or cavities and decreased muscle activity
 - Environmental temperature in the OR can temporarily be set at 25-26.6C
 - Warm IV and irrigating fluids
 - Wet gowns and drapes are removed promptly and replaced with dry materials

Preventing Intraoperative Positioning Injury

- The patient should be in as a comfortable a position as possible, whether conscious or unconscious
- The operative field must be adequately exposed
- an awkward position, under pressure on a body part, or use of stirrups or traction should not obstruct the vascular supply
- Respiration should not be impeded by pressure of arms, hands, and legs or feet can cause serious injury or paralysis. Shoulder braces must be well padded to prevent irreparable nerve injury, especially when the trendelenburg position is necessary
- Precautions for patient safety must be observed, particularly with thin, elderly, or obese patient and those with a physical deformity
- The patient may need light restraint before induction in case of excitement



Post-Operative Phase

Care of Patient in the Post-Anesthesia Care Unit (PACU)

- **Phases of post-anesthesia care**
 - **Phase I:** care of surgical patients immediately after surgery and for the patient whose condition warrants close monitoring and intensive care is provided
 - **Phase II:** surgical patient's condition no longer requires close monitoring provided in phase I PACU. Patient is prepared for self-care or care in the hospital or extended care setting
 - **Phase III:** setting in which the patient is cared for immediate postoperative period and then prepared for discharge from the facility
- **Determining readiness for discharge from the PACU**
 - many hospitals use a scoring system (Aldrete score) to determine the patient's general condition and readiness for transfer from the PACU
 - Throughout the recovery period, the patient's physical signs are observed and evaluated by means of a scoring system based on the set of objective criteria
 - The patient is assessed at regular intervals and a total score is calculated and recorded
 - Aldrete score is usually 8-10 before discharge from the PACU, the patient with the score of <7 must remain in the PACU until condition improves or they transferred to ICU
 - Areas of assessment in Aldrete score includes:
 - Activity
 - Respiration
 - circulation
 - Consciousness
 - oxygen saturation
- **Priority #1:** restoration of homeostasis and prevent complications
- **Priority #2:** maintain and promote adequate airway and respiratory function
- **Priority #3:** maintain adequate cardiac function and promote tissue perfusion
- **Priority #4:** maintain adequate fluid and electrolyte balance and adequate renal function
 - Sufficient fluids to maintain extracellular fluids and blood volume
 - Prevent fluid overload with resultant
 - pulmonary congestion and edema
 - Monitor serum electrolyte
 - Accurate I&O recording
 - Instruct and support breathing exercise
 - Don't force fluid too soon
- **Priority #5:** promote comfort and rest
 - Manage pain during variety of approaches: pharmacologic, comfort measures
- **Priority #6:** promote adequate nutrition and elimination
 - normal peristalsis returns to 48 to 72 hours post op



- liquid diet
- Early ambulation to prevent abdominal distention
- **Priority #7:** promote wound healing and prevention of:
 - **Dehiscence:** total or partial disruption or separation in wound edges but underlying subcutaneous tissue has not parted
 - **Evisceration:** protrusion of viscera through an abdominal wound opening

Post-Operative Complications

Shock

- One of the most serious postoperative complications, can result to hypovolemia and decreased intravascular volume
 - Primary intervention for hypovolemic shock is volume replacement
 - Administer oxygen
 - Continuously monitor patient's condition has stabilized
 - Keep the patient warm and maintain normothermia
- **Assessment:**
 - SBP: <90 mmHg
 - Downward trend of 5 mmHg reading
- **Intervention:**
 - Administer oxygen (1-2 LPM via nasal cannula)
 - Position: modified trendelenburg

Hemorrhage

- Is copious escape of blood from blood vessel
- **Classification of hemorrhage:**
 - **Primary:** hemorrhage occurs at the time of surgery
 - **Intermediary:** hemorrhage occurs during the few hours after surgery when the rise of blood pressure to its normal level dislodges insecure clots from untied vessels
 - **Secondary:** hemorrhage may occur sometime after surgery if a suture slips because of the blood vessel was not securely tied, became infected, or was eroded by a drainage tube
- **Types of vessels:**
 - **Capillary:** hemorrhage is characterized by a slow, general ooze
 - **Venous:** darkly colored blood bubbles out quickly
 - Blood is bright red and appears in spurts with each heartbeat
- **Visibility:**
 - **Evident:** hemorrhage is on the surface and can be seen



- **Concealed:** hemorrhage is in a body cavity and cannot be seen
- **Clinical manifestations:**
 - Patient presents with hypotension, rapid, thready pulse, disorientation, restlessness, oliguria, cold and pale skin
 - Feeling of apprehension, decreased cardiac output and vascular resistance and signifies that the patient is in the early phase of shock
- **Nursing intervention:**
 - Transfusing of blood or blood products and determining the cause of hemorrhage are the initial therapeutic measures
 - Inspect surgical site and incision, if bleeding is evident, a sterile gauze pad and a pressure dressing are applied, and the site of bleeding is elevated to heart level if possible
 - Place the patient in shock position (flat on back, legs are elevated at 20 degree angle and kept straight)
 - If suspected for internal hemorrhage, patient is taken back to OR

Deep Vein Thrombosis

- DVT occurs in pelvic veins or in the deep veins of the lower extremities in postoperative patients
- DVT is common after hip surgery
- Venous thrombi located above the knee are considered as the major source of pulmonary emboli
- **Clinical manifestations:**
 - Homan's sign: calf pain upon dorsiflexion
 - Painful, swelling of the entire leg
 - Slight fever, chills and perspiration
 - Marked tenderness over the anteromedial surface of the thigh
 - Intravascular clotting without marked inflammation may develop, leading to phlebothrombosis
 - Circulation distal to the DVT may be compromised if sufficient swelling is present
- **Nursing intervention:**
 - Hydrate patient adequately postoperatively to prevent hemoconcentration
 - Encourage leg exercise and ambulate patient as soon as permitted by surgeon
 - Avoid restricting devices such as tight straps that can constrict and impair circulation
 - Avoid rubbing or massaging calves and thighs
 - Instruct to avoid standing and sitting in one place for prolonged periods and crossing legs when seated
 - Assess the distal peripheral pulses, capillary refill and sensation of lower extremities
 - Check the positive Homan's sign
 - Prevent the use of bed rocks or knee-gatch on the patients at risk because there is danger of constricting the vessels under the knee
 - Initiate anticoagulant therapy either IV, SQ, or orally as prescribed



- Prevent swelling and stagnation of venous blood by applying approximately fitting elastic stockings or wrapping the legs from the toes or the groin with elastic bandage
- Apply external pneumatic compression intraoperatively to patients of highest risk of DVT

Pulmonary Embolism

- Is caused by the obstruction of one or more pulmonary arteries by the embolus originating somewhere in the venous system or in the right side of the heart
- Postoperatively, most emboli develop in the pelvic or in the iliofemoral veins before becoming dislodged and travelling to the lungs
- **Clinical manifestations:**
 - Dyspnea is the most frequent symptom
 - Chest pain (sudden and pleuritic in origin)
 - Anxiety, fever, tachycardia, apprehension, cough, diaphoresis, hemoptysis, and syncope
 - Most frequent sign is tachypnea
- **Nursing intervention:**
 - Administer oxygen
 - Calm the patient
 - Monitor vital signs, ECG, arterial blood gases
 - Treat for shock or heart failure as directed
 - Give analgesics or sedatives as directed to control pain or apprehension
 - Prepare for anticoagulation or thrombolytic therapy or surgical intervention (management depends on the severity)

Urinary Retention

- Intervention for bladder or urinary retention: foley catheter
- Expected time patient will void: within 8 hours
- Expected amount of urine voided: after surgery atleast 200 ml
- **Bladder training start:**
 - If without urge: clamp 2 ½ hours then release for 30 minutes, if no urge: reclamp
 - Repeat steps until urge occurs
 - If with urge: remove clamp for 30 minutes then reclamp until next urge
 - Indication for removal of foley catheter: 4-5 consistent urges (successive)
 - Kegels exercise: to promote bladder control
 - Expected time of urination after FC removal: within 4 hours
- Interventions to stimulate urination:
 - Fluids: increase OFI
 - Sound: whooshing



- Faucet: running water
- Environment: comfort room as much as possible
- Bedpan or commode: warm

Wound Dehiscence/Evisceration

- **Dehiscence:** partial or total separation of wound edges
- **Evisceration:** uncontrolled exteriorization of intra-abdominal contents
- Onset: 5th to 8th post op (commonly)
- **Causes:**
 - **Surgical team:** inadequate surgical closure
 - **Patient activity:** coughing, vomiting, valsalva maneuver, bleeding down
- **Signs:**
 - **Drainage:** serosanguinous (pinkish)
 - Sensation: something let go
- **Management:**
 - Initial management
 - Dehiscence: assist patient to go back to bed
 - Evisceration: lie down where they are then sterile gauze with PNSS
 - Position: dorsal recumbent
 - Subsequent management
 - WOF: shock
 - Anticipate return to OR
 - Prevention:
 - Coughing with splinting
 - Pillow
 - Antiemetics
 - Stool softener: colace
- **Changing wound dressing**
 - Optimal times: not during meals and do it without visitors
 - Never touch dressing with ungloved hands
 - In removing tapes: moisten tape with sterile water
 - Wear gloves during
 - Clean gloves: removing old dressing
 - Sterile gloves: wound cleaning and applying new dressing
 - WOF: sanguinous, purulent drainage



Immunology

- Is the biological and medical study of an organism's immune system, focusing on its components, function, and how it protects against diseases like infections and cancer
- **Immunology nursing** involves nurses applying knowledge of the complex immune system to provide specialized care for patients, focusing on immune-related diseases and conditions like allergies, autoimmune disorders, and immunodeficiency

NURSING PRACTICE IV

Neurology

- Focused on the diagnosis and treatment of disorders affecting the nervous system

Cranial nerve - found in brain stem

Lowest part of the brain - medulla oblongata (controls RR, HR, swallowing, hiccups and vomiting)

Nervous System

- **Central Nervous System** (brain & spinal cord)
- **Peripheral Nervous System** (ganglia) - cranial and spinal nerves
 - **Voluntary** (somatic NS)
 - Efferent (motor/down) and afferent (sensory/up) nerve
 - **Involuntary** (autonomic NS)
 - **Sympathetic** - increased all except GI & GU (fight or flight)
 - **Catecholamine** (epinephrine, norepinephrine, dopamine) released by adrenal medulla
 - **Anti-cholinergic/adrenergic effects**
 - dilated pupils (mydriasis) - to be aware of surroundings
 - Increased BP & HR - vasoconstriction
 - Bronchodilation - increased oxygen
 - Increased RR
 - Dry mouth
 - Constipation & water retention
 - Fluid volume excess
 - **Parasympathetic** - decreased all except GI & GU (rest and digest)
 - **Acetylcholine** - decrease all bodily activities except GI & GU
 - **Cholinergic/vagal effects**
 - Constrict pupil (miosis)
 - Decreased BP & HR - vasodilation



- Bronchostriction - decreased RR
- Increased salivation
- Diarrhea & urinary frequency
- Fluid Volume deficit

Cranial Nerves

Cranial Nerve	Name	Sensory/Motor	Function	Abnormalities
I	Olfactory	Sensory	Smell	Anosmia (absence of smell)
II	Optic	Sensory	Vision	Blurring of vision, blindness
III	Oculomotor	Motor	Eye movement (up, down, in) eyelid up, pupil constriction	Horizontal & Vertical Diplopia, Ptosis (drooping of eyelids), Anisocoria (unequal pupil)
IV	Trochlear	Motor (smallest)	Eye movement (down, in)	Vertical Diplopia
V	Trigeminal	Both (largest)	Maxillary nerve Ophthalmic (corneal reflex) Mandibular nerve (mastication)	Trigeminal neuralgia
VI	Abducens	Motor	Eye movement (abduction)	Horizontal Diplopia
VII	Facial	Both	Facial expression and taste, Anterior $\frac{2}{3}$	Bell's palsy
VIII	Vestibulocochlear	Sensory	Hearing and balance	Hearing Loss, tinnitus, vertigo, Nystagmus
IX	Glossopharyngeal	Both	Taste, posterior $\frac{1}{3}$, gag reflex	Dysphasia (risk for aspiration)
X	Vagus	Both (longest)	Gag reflex, PNS activation	Dysphasia, abnormal PNS



			(cholenergetic) GI secretions	
XI	Accessory	Motor	Neck and shoulder movement	Accessory neuropathy
XII	Hypoglossal	Motor	Tongue movement	Dysarthria, tongue protrusion/deviation

Spinal Nerves

Spinal Nerve	Controls
Cervical (8)	Diaphragm, chest wall muscles, shoulders, arms
Thoracic (12)	Trunk stability, thoracic organs (sympathetic), heart lungs, GIT
Lumbar (5)	Lower body, bowel and bladder
Sacral (5)	
Coccyx (1)	

Common Neurologic Illness

Increased ICP

- Increased pressure within the skull due to increase in one of the intracranial components (0-15 mmHg)
- **Predisposing Factors**
 - Stroke (Hemorrhagic)
 - Tumor
 - Inflammation (Meningitis, encephalitis)
 - Trauma (surgery)
 - Cerebral edema (SIADH)
 - Hydrocephalus
- **Monro-Kellie Doctrine** - the skull is a closed vault, an increase in one component will increase the ICP
 - Skull components:
 - 80% - brain mass



- 10% - CSF
- 10% - blood
- **Pathophysiology**
 - Alteration in BM, C, B (trauma) -> brain compression/brain herniation -> MO will move to FM -> respiratory arrest
- **Signs and symptoms**
 - **Early signs**
 - Blurred vision, decreased visual acuity, diplopia
 - change or decreased LOC
 - restlessness to confusion
 - Disorientation to lethargy
 - Sluggish pupils & eye movement
 - Monoparesis/hemiparesis (weakness)
 - Throbbing/dull headache
 - **Late signs** (Change in VS)
 - Systolic hypertension (systolic high, diastolic same)
 - wide pulse pressure (N) 40-50
 - Decreased irregular RR (Cheyne's stokes - hypercapnia with periods of apnea)
 - Increased temperature
 - Projectile vomiting
 - Abnormal posturing (decorticate (cortex) - abnormal flexion/decerebrate (brain stem) - abnormal extension)
 - (+) Babinski reflex
 - Hemiplegia
 - Seizure
 - Coma

Cerebral Perfusion Pressure - 60-100mmHg

- Formula: CPP= MAP - ICP
 - MAP = SBP + (DBP) 2 / 3 (N: 70-100mmHg)

Management

- Head of Bed: elevated (semi-fowler's) 30-45 degree
- Neutral neck position for venous drainage
- Evaluate neurological status every 1-2 hours
- Airway: check RR: 12-20 cpm
 - If GCS 8 = intubation
- Avoid fever and chills (acetaminophen & blanket)
- Avoid coughing (antitussives - dextromethorphan)
- Avoid vomiting (antiemetic - ondansetron)
- Avoid valsalva maneuver (stool softener - docusate Na)
- Avoid seizure (anticonvulsant - levetiracetam)



- Drainage from ears, if CSF (+) glucose/halo sign (sterile dressing, check for s/six of meningitis)
- Safety (side rails up, no sedation or narcotics)

Medications

- Mannitol (Osmitol) - osmotic diuretic (promotes cerebral diuresis by decompressing brain tissue)
- 3% NaCl (hypertonic) - pulls water from brain cells to IVS

Cerebrovascular Accident

- Partial or complete destruction of brain's blood supply
- Death of brain cells
- 2 largest & common arteries affected
 - Middle cerebral artery (most common)
 - Anterior cerebral artery
- Common in men (2-3x high risk)
- Atherosclerotic clot (cholesterol <200) , blood clot (DM)
- **Causes**
 - Ischemic Stroke
 - Thrombosis (DVT) (+) Homan's sign
 - Embolism - dislodged clot (cerebral embolism)
 - Headache
 - Decreased LOC
 - Hemorrhagic stroke (hemorrhage)
 - Compartment syndrome - compression of nerves/artries
- **Signs and Symptoms**
 - Transient Ischemic Attack (TIA) - mini stroke - warning sign of an impending stroke attack, resolve in < 24 hours
 - Facial drooping
 - increased ICP (possible)
 - Numbness
 - Changes in speech and vision
 - Headache (1st)
 - Paresis or plegia (monoplegia - 1 extremity only)
 - Aphasia
 - Dizziness & vertigo, tinnitus
 - Stroke in evolution - progression of signs and symptoms of stroke
 - Complete stroke - cheyne-stokes respiration
 - (+) Kernig's and Brudzinski (+) hemorrhagic stroke
 - Homonymous hemianopsia - loss of half of visual field (occipital lobe) (unaffected side)
 - Unilateral neglect - unaware of paralyzed side existence (affected side)
 - Common in left body



- **Diagnostic test**
 - CT Scan (non-invasive) - to know the type of stroke
 - Cerebral arteriography - reveal and extent of malocclusion
 - Allergy test (due to dye)
 - MRI - detects brain tissue damage
 - With dye injection
 - Carotid ultrasound - shows plaque and blood flow of carotid arteries
 - Echocardiogram - to find clot source in heart
- **Assessment** (you only need one sign)
 - Balance
 - Eyes
 - Face - uneven smile
 - Arm - patient can't raise & keep arms up
 - Slurred speech
 - Time of onset, call 911 or CT scan STAT (hospital)

Neurological Tests

Glasgow Coma Scale

- LOC measurement or quick neuro check
- **3 Components**
 - Motor response (6)
 - Verbal response (5)
 - Eye opening (4)
 - Total: (15), lowest (3)
- **Levels of Consciousness**
 - Conscious (14-15)
 - Lethargy (13-11)
 - Stupor (10-8)
 - Coma (7)
 - Deep Coma (3)
- **Severity of Brain Injury**
 - Mild (15-13)
 - Moderate (12-9)
 - Severe (8-3)

NOTE: GCS 8 - intubation, decreased GCS - REPORT

AVPU

- Alert - awake
- Verbal - respond to voice



- Pain - respond to pain (trapezius squeeze)
- Unresponsive - no response

Management

- O2 if SpO2 <94%
- Check glucose -
- CT Scan within 20 minutes of arrival
- NPO
- HOB:
 - Low fowlers (15-30 degree) - Ischemic
 - High fowlers - Hemorrhagic
- Avoid valsalva maneuver - Docusate Na (Colace)
- Maintain side rails
- Prevent complications of immobility
 - Turn client every 2 hours
 - TA decubitus ulcer
 - Pneumonia
- Incentive spirometry
- Passive ROM every 4 hours (paralyzed)
- Active ROM (weakness)
- NGT - give thick fluids (to prevent aspiration due to dysphagia)
- Increased fluid and fiber (due to constipation)
- Position: side-lying (unaffected)
- Communication: picture-board (non-verbal cues)
- Egg crate mattress

Medications

- **Criteria for thrombolytics:**
 - CT Scan (-) bleeding
 - No recent surgery or trauma/anticoagulant
 - Age >18 y/o
 - BP controlled <185/100
 - Blood tests (glucose, INR (PT), platelet - 150K-450K)
 - Warfarin 2-3 secs (INR)
 - Without warfarin <1 sec (INR)
- If eligible:
 - **Thrombolytics/fibrinolytics** - to dissolve clots/ thrombus
 - Urokinase (hemorrhage)
 - Streptokinase (pruritus)
 - Tissue Plasminogen Activator (hemorrhage) *close monitoring (ICU for 24 hours)
 - give within 3-4.5 hrs of onset of s/s of CVA



- Alteplase (DOC)
 - Reteplase
 - Tenecteplase
- **Platelet aggregation inhibitors**
 - 24 hours later if tPA is used
 - Aspirin (tinnitus)
 - Clopidogrel (given to patients who are allergic to aspirin) - GI discomfort
 - Ticagrelor - GI discomfort
- **Anti-cholesterol** - after 24 hours
 - Atorvastatin (severe muscle pain)
 - Simvastatin (severe myalgia)
 - May lead to rhabdomyolysis
 - May have dark urine
- **Anti-hypertensive** (>185/110)
 - Labetadol IV
 - Nicardipine IV
 - Avoid nitroglycerine (will increase ICP)
- **Anticoagulants**
 - Heparin - short-acting (to avoid DVT) after 24-48 hours
 - Monitor PTT (30-40 secs)
 - Antidote: protamine sulfate
 - Warfarin (Coumadin) - long - acting (4-14 days post stroke)
 - PT (11-12 secs)
 - Antidote: Vitamin K, aquamephyton
 - Direct oral anticoagulant - no more bleeding monitoring
 - Dabigatran (Pradaxa)
 - Apixaban (Eliquis)

Seizure

- Sudden onset of excessive discharge of electricity in cerebral cortex
 - Seizure - 1st convulsive attack
 - Febrile seizure: normal if <5 y/o
 - Pathologic if >5 y/o
 - Epilepsy - 2nd & with history of seizure
- **Risk factors**
 - Metabolic imbalance
 - Alcohol/drug withdrawal
 - Autism (brain developmental disorder)
 - Genetics



- Infection
- Injuries of the brain
- Cerebral palsy
- Stress - physical/emotional
- **Management**
 - Calm - maintain a calm environment, provide privacy
 - Anticonvulsant - Phenytoin (Dilantin) DOC
 - Carbamazepine - trigeminal neuralgia
 - Valproic acid
 - Evaluate the type of seizure
 - First activity of the patient
 - Duration of seizure - post - ictal sleep (the longer = status epilepticus)
 - Safety
 - Turn to side (during)
 - Unrestrained
 - Remove hazards
 - Nothing in mouth
 - Padded side rails - always up
 - Oxygen after seizure
 - Don't leave patient
 - Avoid alcohol
 - + Dilantin = CNS depression
 - Aura
 - Activities - what occurred before seizure
 - Reduce stimuli (bright/flashing lights, noise)
 - Remain with patient - note the time when it began, how long it lasted
 - Reorient
- **Diagnostic Test**
 - Patient history
 - CT Scan (brain lesion)
 - EEG (electroencephalography)
 - (+) Abnormal spike-and-waves
 - Wash hair first
 - No fasting
 - No stimulants - may alter brain waves
 - No sleeping (to trigger seizure)
- **Signs and Symptoms**
 - **Generalized seizure**
 - **Tonic-Clonic seizure** - common
 - AURA - warning sign of impending seizure attack (no stimulus)



- Epigastric pain (flashes of lights, noise, taste, smell)
- Epileptic cry - fall
- Loss of consciousness (3-5 minutes)
- Tonic-clonic contractions
 - Extension of extremities - tonic (10-20 secs)
 - Contractions - clonic (30 secs - 2 mins)
- Post-ictal sleep (5-30 minutes) - lethargy, drowsy and unresponsive, exhausted
- **PRIORITY: SAFETY**
- **Petitmal/absence seizure** - common in 12 y/o
 - blank stare
 - Decreased eye blinking
 - staring into space
 - pausing amidst conversation
 - loss of consciousness (5-10 secs)
- **Status epilepticus**
 - Continuous seizure activity > 5mins (can be up to 30 minutes according to Saunders)
 - 2 or more seizure within 5 minute period
 - If untreated, coma/death
 - **PRIORITY: AIRWAY** (min. 15LPM high flow non-rebreather mask)
 - Medications: IV push slowly
 - Lorazepam (DOC)
 - Diazepam (or rectal)
 - Midazolam (IM, if no IV site)
 - If seizure > 30mins, not breathing (intubate)
 - Add IV thiamine if with alcohol withdrawal

CNS	PNS
- Brain, brain stem, & Spinal cord - MS - Alzheimer's - Parkinson's - Huntington's	- Cranial and spinal nerve - MG - ALS (muscle weakness) - GBS



CNS Disorders

Multiple Sclerosis

- Chronic asymmetrical autoimmune disease that destroys the myelin sheath (oligodendrocytes) of neurons in CNS
- Inflamed neuron -> decreased nerve impulse transmission
- Common: women (20-50 y/o)
- Cause: idiopathic
- **Pathophysiology**
 - Immune system attacks -> myelin sheath -> communication breakdown -> sensory and motor dysfunction, cognitive dysfunction
- **Signs and Symptoms**
 - **Charcot's Triad**
 - **Dysarthria** - difficulty/unclear speech (damaged in brain stem due to plaque)
 - **Nystagmus** - involuntary rapid eye movement (plaque in eye nerves)
 - Optic nerve - blurring of vision (1st)
 - Optic neuritis
 - Diplopia - double vision
 - Scotomas - blind spots
 - **Intention tremor** - plaque in motor pathway
 - Tremors
 - Ataxia
 - Muscle weakness & spasms
 - Paralysis
 - **Plaques in sensory pathway**
 - Numbness & paresthesia
 - **Plaques in ANS (autonomic NS)**
 - Urinary retention or incontinence
 - Constipation
 - Decreased sexual activity
 - **Higher order activities**
 - Cognitive: poor memory & attention, problem-solving issues
 - Psychosocial: depression, anxiety, mood swings
 - **Lhermitte's sign** - sudden electric-like shock sensation
 - From spine to limb upon bending neck forward
 - **Uhthoff's sign** - heat makes s/sx worse
- **Diagnostic Test**
 - **CSF analysis via lumbar puncture**



- Increased CSF proteins levels (oligoclonal bands)
- Means CNS inflammation
- **MRI with contrast** - confirmatory test
 - Site and extent of demyelination
- **Management**
 - **Supportive/palliative**
 - Prevent Uhthoff's sign
 - Assistive devices
 - Side rails up
 - Prevent complications of immobility
 - Assess in plasmapheresis
 - Removal of plasma (fluid part of the blood) specifically antibodies
 - **Medication: for acute exacerbation**
 - Corticosteroids (methylprednisolone IV) - decreased inflammation and prevent paralysis
 - **Symptom management**
 - Baclofen - muscle relaxant
 - Tizanidine - muscle relaxant
 - Amantadine - fatigue
 - Propranolol - tremors
 - **Chronic treatment**
 - immunosuppressants
 - Beta-interferon - increase function of regulatory T-cells
 - Avonex
 - Rebif

Alzheimer's Disease

- Atrophy of the brain due to beta amyloid plaque deposits outside neurons which disrupts synapse and cause inflammation
- Cognitive decline and neuron death
- Decreased acetylcholine (learning & memory)
- Cause: idiopathic
- Common: women (elderly)
- **Signs and symptoms**
 - Short term memory loss (antero) - initial sign
 - **Amnesia** - events
 - **Anomia** - can't name everyday objects
 - **Agnosia** - can't recognize familiar objects, persons, sounds, shapes or smell
 - **Apraxia** - cannot perform motor activities despite intact motor function



- **Aphasia**
 - **Expressive: broca's (left frontal) aphasia** - can't speak (common in stroke)
 - **Receptive: wernicke's (left temporal) aphasia** - can't understand spoken words (common in alzheimer's)
- **Diagnostic Tests**
 - CT Scan - progressive brain atrophy
 - Autopsy - confirmatory
- **Management**
 - Supportive/palliative
 - DOC: donepezil (Aricept) - take at bedtime
 - Memantine (Namenda)
 - **PRIORITY: SAFETY** (risk for injury)

Parkinson's Disease

- Degenerative disease
- Movement disorder, no weakness
- **Pathophysiology**
 - Dopamine (inhibitory) - smooth and coordinated muscle movement , acetylcholine (excitatory) - muscle contraction
 - Dopaminergic neuron degeneration (substantia nigra) -> decreased dopamine -> dopamine & acetylcholine imbalance -> resting tremor, akinesia, rigidity
- **Signs and symptoms**
 - Tremors (resting)
 - Pill rolling (1st sign)
 - Rigidity & bradykinesia
 - mask-like facies - hypomimia
 - monotonous voice - hypophonia
 - dysphagia
 - Cogwheel rigidity
 - Reduced arm swing
 - Shuffling gait
 - Akinesia - difficulty starting movement
 - Postural instability - stooped posture, impaired balance
- **Management**
 - Safety (Priority)
 - Assist in ambulation
 - non-skid shoes, no heels
 - Grab bars, walkers, canes
 - Medications (increases dopamine)



- Levodopa
- + Carbidopa
- = Sinemet
- Ropinirole (dopamine agonist) - delays levodopa use
- Amantadine - controls dyskinesias in long term use of levodopa
- Interdisciplinary
 - Physical Therapist
 - Occupational Therapist
 - Nurse
 - Speech Therapist
- Lifestyle changes
 - Exercise, high fiber, increase OFI
- Encourage gait training - use metronome or music
- Diet:
 - Decrease or no protein at all - it inhibits absorption of medication (AM)
 - Increased protein in PM to induce sleep due to tryptophan
 - Small frequent meal
 - Double swallowing
 - Chin-tuck maneuver
- Surgery: deep brain stimulation (most common surgical treatment)
- Anti-parkinsonian
 - Take before meals - to increase absorption
 - With meals if there is nausea and vomiting
 - Inform patient - urine, stool may be darkened
 - Delayed reaction (2-3 weeks)

PNS Disorders

Myasthenia Gravis

- Grave muscle weakness, chronic autoimmune asymmetrical descending muscle weakness due to destruction of acetylcholine receptors at neuromuscular junction
- Voluntary muscles
- Common: women (20-40 y/o)
- Cause: idiopathic
- **Pathophysiology**
 - Acetylcholine -> ACH receptors -> antibodies blocks the receptors -> no muscle contraction
- **Signs and Symptoms**



- Ptosis (initial sign)
- Diplopia
- Mask-like facies
- Dysarthria
- Dysphagia
- Laryngeal muscle weakness
- Respiratory muscle weakness
- Leg weakness
- Fatigue
- Paralysis
- **Diagnostic Test**
 - Tensilon test (Edrophonium Chloride)
 - Strengthens muscle (5-10 minute)
 - Short term - cholinergic
 - Antidote: atropine sulfate
 - CSF analysis - increase cholinesterase level
- **Management**
 - Supportive/palliative
 - Surgery: thymectomy (removal of thymus - T-cell maturation)
 - Medication
 - Cholinergics or anticholinesterase agents
 - Pyridostigmine
 - Neostigmine
 - Corticosteroids - to suppress immune response
 - Prednisone
 - Immunosuppressant
 - azathioprine
 - Methotrexate (RA)
- **Crisis**
 - **Myasthenic crisis**
 - **Causes**
 - Under medication
 - Stress
 - infection
 - **Signs and symptoms**
 - Ptosis & diplopia
 - Dysphagia
 - Can't breathe
 - **Management**
 - Cholinergic agent



- **Cholinergic crisis**
 - **Cause**
 - Over medication
 - **Signs and symptoms**
 - Salivation
 - Lacrimation
 - Urination
 - Defecation
 - GI distress
 - Emesis
 - **Management**
 - Atropine sulfate

Gullain-Barre Syndrome

- Acute autoimmune asymmetrical ascending paralysis due to demyelination of peripheral nerves
- Common: men (40-60 y/o)
- Cause: idiopathic
- Recovery is possible
- **Pathophysiology**
 - Immune system attacks myelin sheath -> communication breakdown -> sensory, motor, ANS dysfunction
- **Signs and symptoms**
 - Tingling sensation/clumsiness (1st sign)
 - Neuropathic or muscle pain
 - Decreased DTR
 - Leg/arm weakness
 - Dyspnea
 - Dysphagia
 - ANS changes
 - BP fluctuations
 - Arrhythmia
 - Urinary retention
 - Constipation
- **Diagnostic Test**
 - Clinical
 - Acute ascending weakness
 - Preceded by viral or bacterial infection
 - CSF analysis via lumbar puncture
 - Increased protein but normal WBC



- Pulmonary function test
 - Decreased vital capacity
- **Management**
 - O₂
 - Neuro check
 - ECG - arrhythmia
 - **Medications**
 - IV immunoglobulin - to fight GBS antibodies
 - Gabapentin - for neuropathic pain
 - Heparin - DVT prophylaxis
 - Anti-arrhythmic agents (amiodarone, beta blockers, calcium-channel blockers - verapamil & diltiazem)

Fluids and Electrolytes

- **Fluid** - 60% total body weight
 - **Intracellular Fluid** - about $\frac{2}{3}$ (70%)
 - **Extracellular Fluid** - about $\frac{1}{3}$ (30%)
 - **Intravascular** - fluid within the blood vessels (plasma)
 - **Interstitial** - fluid surrounds the cell
- **Electrolytes** - substances that dissociate into ions when mixed with water and are crucial for maintaining fluid balance, electrical charge and proper organ function
 - **ICF** - potassium and phosphate
 - **ECF** - sodium and chloride

Acid - Base Balance

- **pH:** 7.35-7.45 -> blood
- **Lungs (PaCO₂):** 35-45
- **Kidneys (HCO₃):** 22-26
- **PaO₂:** 80-100
- **ABG Analysis**
 - (1) Determine the pH (acidic/alkali)
 - (2) Look for the culprit (ROME)
 - (a) **Lungs** - respiratory
 - (b) **Kidneys** - metabolic
 - (3) look for compensation
 - (a) **Uncompensated** - pH abnormal, PaCO₂ or HCO₃ abnormal
 - (b) **Partially compensated** - all are abnormal
 - (c) **Fully compensated** - pH normal, both are abnormal



(d) **Mixed** - if all are acid/alkali

Regulatory Mechanism

- (1) **Osmosis** - movement of fluid from an area of low solute concentration to high concentration to achieve equilibrium
- (2) **Diffusion** - movement of solutes from an area of high concentration to low concentration
- (3) **Filtration** - fluid movement from arterial capillary beds (hydrostatic pressure) into the interstitial fluid, carrying water and electrolytes
- (4) **Active transport** - uses energy (ATP) to move ions against the concentration gradient

Sources of Normal Fluid Loss

- **Urine output** - 1 - 2L daily, varying with hydration status and kidney function
- **Sensible losses (visible perspiration)** - 0 - 100 ml/hr, increased with fever and activity
- **Insensible losses (evaporation)** - 600 ml/day under normal conditions, increased with fever
- **GI losses** - 100 - 200 ml/day through normal digestive processes

Homeostatic Mechanisms

(1) Renal System

- (a) Kidneys filter 170 L of plasma daily
- (b) Reabsorbs bicarbonate (HCO_3), secrete hydrogen ions
- (c) Slow compensatory response (may take several days)

(2) Pulmonary System

- (a) Lungs regulate carbon dioxide (CO_2) and carbonic acid (H_2CO_3) to maintain acid-base balance

(3) Cardiovascular System

- (a) Maintains adequate renal perfusion and supports fluid regulation
- (b) Essential for blood pressure control

(4) Endocrine System

- (a) Pituitary gland secretes ADH (promotes water retention)
- (b) Adrenal gland secretes aldosterone (causes sodium retention and potassium excretion)

(5) RAAS (Renin-Angiotensin-Aldosterone System)

- (a) Maintains blood pressure through fluid retention and vasoconstriction
- (b) Decreased BP or decreased sodium or decreased renal perfusion -> kidneys releasing Renin -> Renin acts on angiotensinogen (from liver) -> forms angiotensin I -> angiotensin I converted to angiotensin II by ACE (in lungs) -> angiotensin II effects -> vasoconstriction (increased BP) & stimulates adrenal glands (increased aldosterone) ->



aldosterone promotes sodium and water retention and potassium excretion -> increased blood volume -> increased BP

Hypovolemia

- **Fluid Volume deficit** - excessive loss of water and electrolytes in equal proportion, leading to dehydration
- **Etiology:**
 - Inadequate fluid intake
 - Excessive fluid loss: severe diarrhea, vomiting, blood loss
 - Third-space fluid shift: ascites, burns, pancreatitis
- **Clinical manifestations:**
 - Tenting skin turgor, dry mucous membranes
 - Postural hypotension, increased HR
 - Extreme thirst, dizziness, weakness
 - Renal shutdown, coma (severe cases)
- **Diagnostic findings:**
 - Urine specific gravity > 1.020
 - elevated BUN disproportionately to serum creatinine
 - Elevated hematocrit
- **Nursing management:**
 - Monitor I&O, urine specific gravity
 - Weight daily (same clothes, same time, same scale)
 - Provide hydration (oral or IV fluids)
 - Assess VS, LOC and skin color

Solution Type	Examples	Osmolarity	Indications	Mechanism
Isotonic Solutions	- 0.9% Normal Saline (NS) - Lactated Ringer's (LR)	similar to blood plasma	- Effective for extracellular fluid volume depletion due to vomiting, diarrhea, or blood loss.	- No fluid shift between Intracellular Fluid (ICF) and Extracellular Fluid (ECF). - Maintains normal fluid balance without changing cellular hydration.
Hypotonic Solutions	- 0.45% Normal Saline (½ NS) - 5% Dextrose in Water (D5W)	Lower than blood plasma	- Cellular dehydration - Rehydration of cells, lowering sodium levels.	- Water moves from ECF to ICF, hydrating cells. - Expands the ICF (cells become hydrated).
Hypertonic Solutions	- 3% Normal Saline - D5 ½ Normal Saline	Higher than blood plasma	- Severe hyponatremia, cerebral edema. - To increase extracellular volume and dilute sodium levels.	- Water moves from ICF to ECF, expanding blood volume. - Draws fluid into the bloodstream, helping restore blood volume.



Hypervolemia

- **Fluid Volume Excess** - excessive retention of water and electrolytes, leading to fluid overload
- **Etiology:**
 - Excessive fluid intake
 - Impaired fluid excretion (renal, liver disease)
 - Heart failure, corticosteroid therapy, excessive sodium intake
- **Clinical manifestations:**
 - Weight gain, dependent edema (feet, ankles, sacrum)
 - Dyspnea, crackles (especially in cardiac patients)
 - Bounding pulse, JVD
 - Change in mental status (due to hyponatremia)
- **Diagnostic findings:**
 - Decreased BUN and hematocrit
- **Nursing management:**
 - Monitor I&O, VS
 - Elevate HOB for dyspnea; encourage deep breathing
 - Administer diuretics as prescribed (furosemide, spironolactone)
 - Provide client education on sodium-restricted diet and fluid restrictions

Electrolyte Imbalances

Hyponatremia

- **Sodium deficiency** - serum sodium level <135 mEq/L (Sodium deficiency)
- **Etiology:**
 - Fluid loss: vomiting, diarrhea, diuretic therapy
 - Adrenal insufficiency (Addison's disease)
 - SIADH (syndrome of inappropriate antidiuretic hormone)
 - Excessive water gain (psychogenic polydipsia)
- **Clinical manifestations**
 - Anorexia, nausea, muscle cramps
 - Lethargy, confusion, seizures (neurologic)
- **Diagnostic findings:**
 - Serum sodium <135 mEq/L
 - Urine sodium varies: low with sodium loss, high with SIADH
- **Nursing management:**
 - Monitor I&O, weight daily
 - Administer sodium supplements



- Monitor neurological status, maintain seizure precautions

Hypernatremia

- **Sodium excess** - serum sodium level of >145 mEq/L (Sodium excess)
- **Etiology:**
 - Water loss: diarrhea, fever, diabetes insipidus
 - Inadequate water intake: common in elderly, comatose patients
 - Excessive sodium intake
- **Clinical manifestations:**
 - Thirst, tongue dry, tented skin turgor, sticky mucous membranes
 - Elevated temperature, lethargy, restlessness
- **Diagnostic findings:**
 - Serum sodium level >145 mEq/L
 - Serum osmolality >295 mOsm/kg
- **Nursing management:**
 - Monitor I&O, weight daily
 - Increased fluid intake, administer hypotonic solutions
 - Monitor serum sodium levels and neurological status
 - Protect the client from injury

Hypokalemia

- **Serum potassium level of <3.5 mEq/L (Potassium deficiency)**
- **Etiology:**
 - Inadequate potassium intake: low dietary intake of potassium-rich foods
 - Excessive loss of potassium (medications, GI losses, renal losses)
 - Metabolic alkalosis: causes potassium to shift intracellularly, leading to a reduction in extracellular potassium levels
 - Hyperaldosteronism: high aldosterone levels cause increased potassium excretion by the kidneys
- **Clinical manifestations:**
 - (1) **Neuromuscular**
 - (a) Fatigue: the most common complaint
 - (b) Muscle weakness: especially in the legs
 - (c) Leg cramps: due to muscle cell dysfunction
 - (d) Paresthesia: tingling or numbness in extremities
 - (2) **Cardiovascular**
 - (a) Cardiac arrhythmias: premature ventricular contractions (PCVs), U waves, flattened T waves, prolonged PR intervals



(b) Hypotension due to reduced cardiac output

(3) Gastrointestinal

(a) Anorexia, nausea and vomiting

(b) Decreased bowel motility, leading to ileus or abdominal distention

- Laboratory and diagnostic findings:

- Serum potassium <3.5 mEq/L (typically <3 mEq/L is considered severe)
- Electrocardiogram (ECG)
 - Flattened T wave
 - Prominent U wave
 - ST-segment depression
 - Prolonged PR interval

- Nursing management:

(1) Therapeutic intervention

(a) Parenteral potassium

- (i) Administer IV potassium slowly (NEVER IV PUSH or IM), always dilute it in at least 100 ml of fluid
- (ii) Infuse via infusion pump and monitor ECG during the infusion

(2) Client education

- (a) Dietary teaching: encourage consumption of potassium-rich foods such as bananas, oranges, cantaloupe, spinach, potatoes and tomatoes
- (b) Monitor digoxin levels: hypokalemia can increase digoxin toxicity. Clients taking digoxin should be monitored closely for signs of toxicity

(3) Monitor for complications

- (a) Cardiac arrhythmia
- (b) Increased risk for diuretics
- (c) Assess paralytic ileus: hypokalemia can lead to intestinal obstruction, causing symptoms like bloating, constipation and abdominal pain

Hyperkalemia

- Serum potassium level >5.5 mEq/L (Potassium excess)

- Etiology:

(1) Decreased renal excretion

- (a) Renal failure: the kidneys are unable to excrete potassium properly, leading to accumulation in the bloodstream
- (b) Hypoaldosteronism: aldosterone deficiency results in reduced potassium excretion

(2) Tissue injury

- (a) Severe tissue trauma: burns, massive infections, rhabdomyolysis or crush injuries can release intracellular potassium into the extracellular space



- (3) Acidosis**
 - (a) Metabolic acidosis causes potassium to shift from inside the cells to extracellular space
 - (4) Excessive potassium intake**
 - (a) Overuse of potassium supplements or potassium-containing salt substitutes
 - (b) Excessive potassium administration in IV fluids (rare, but possible)
- **Clinical manifestations**
 - (1) Cardiovascular** - cardiac arrhythmias: severe hyperkalemia can cause life-threatening arrhythmias such as ventricular fibrillation, heart block and asystole
 - (2) Neuromuscular**
 - (a) Muscle weakness, particularly in the legs
 - (b) Paresthesias or numbness (especially in the extremities)
 - (c) possible paralysis (due to muscle and nerve dysfunction)
 - (3) Gastrointestinal**
 - (a) Abdominal cramps, nausea, and vomiting
 - (b) Diarrhea: this is due to increased motility in the intestines
 - (4) Respiratory (severe hyperkalemia)**
 - (a) Shallow respirations (due to muscle weakness and impaired diaphragm function)
- **Laboratory and diagnostic findings**
 - Serum potassium >5.5 mEq/L, severity increases as potassium exceeds 6 mEq/L
 - ECG
 - Tall, tented T waves
 - Prolonged PR interval and QRS duration
 - Absence of P waves
 - ST-segment depression
- **Nursing management**
 - Therapeutic interventions
 - Cation-exchange resins (kayexalate)
 - Calcium Gluconate
 - Sodium bicarbonate
 - Insulin and glucose
 - Hemodialysis
 - Monitor for complications
 - Cardiac monitoring: closely monitor the ECG for arrhythmias
 - Muscle weakness: watch for signs of muscle weakness, which may indicate worsening hyperkalemia or paralysis
 - Lab results: regularly check potassium levels and renal function
 - Client education
 - Diet restrictions: advise patients to avoid high-potassium food such as bananas, oranges, tomatoes, dried fruits and potatoes

- Medications: warn clients on potassium-sparing diuretics (spironolactone) about the risk of hyperkalemia

Hypocalcemia

Functions of Calcium

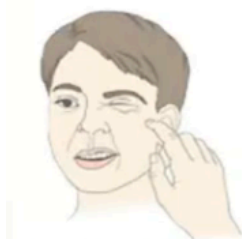
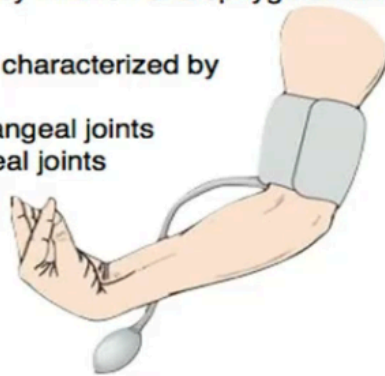
- (1) Electrical energy for the nervous system
 - (2) Electrical energy for muscular system
 - (3) Strength for skeletal system
- Serum calcium levels <8.5 mEq/L, leading to symptoms like muscle spasms, tetany and arrhythmias (Calcium deficiency)
 - **Etiology**
 - Primary or surgical hypoparathyroidism pancreatitis
 - Insufficient vitamin D intake or synthesis
 - Renal failure
 - Drug therapy
 - Insufficient dietary calcium intake

Trousseau's Sign

Induction of carpopedal spasm by inflation of a sphygmomanometer above SBP for 3 minutes

Response: Carpopedal spasm characterized by

- Adduction of the thumb
- Flexion of the metacarpophalangeal joints
- Extension of the interphalangeal joints
- Flexion of the wrist



Chvostek's sign

Contraction of the ipsilateral facial muscles elicited by tapping the facial nerve just anterior to the ear

Response: Twitching of the lip to spasm of all facial muscles

- **Clinical manifestations**
 - Tetany (tingling in the fingers and circumoral area, muscle spasms in extremities and face)
 - (+) Trousseau's and Chvostek's sign (Hallmark)
 - Carpopedal spasms
 - Hyperactive deep tendon reflexes



- Seizures due to neuromuscular irritability
- **Laboratory and diagnostic findings**
 - Serum calcium <8.5 mEq/L
 - ECG reveals prolonged QT interval
- **Nursing management**
 - (1) Prepare for therapy
 - (a) Institute seizure precautions
 - (b) Administer parenteral calcium (careful of tissue infiltration, which can cause necrosis)
 - (c) Avoid adding calcium to parenteral solutions containing bicarbonate or phosphorus
 - (d) Caution with digoxin: calcium increases its effects, risking cardiac arrest)
 - (e) Maintain a quiet, relaxed environment to promote rest
 - (2) Administer vitamin D: this helps increase calcium absorption from the GI tract
 - (3) Client education
 - (a) Encourage a calcium-rich diet (1000 to 1500 mg/day) from foods like milk, green leafy vegetables and oysters
 - (b) Avoid excessive alcohol or caffeine as they inhibit calcium absorption
 - (c) Osteoporosis prevention: ensure adequate calcium intake and exercise to prevent bone loss

Hypercalcemia

- **Serum calcium >10.5 mEq/L (Calcium excess)**
- **Etiology**
 - Excessive calcium intake or administration
 - Calcium moving from bones to serum
 - Hyperparathyroidism
 - Thiazide diuretic therapy
- **Clinical manifestations**
 - Anorexia, nausea and vomiting
 - Constipation
 - Muscular weakness and incoordination
 - Altered LOC (slurred speech, confusion, lethargy, coma)
 - Polyuria and polydipsia
 - Cardiac arrhythmias
 - Hypoactive deep tendon reflexes
- **Laboratory findings**
 - Serum calcium >10.5 mEq/L
 - ECG: shortened QT interval, bradycardia, heart blocks



- **Nursing management**

- (1) Prepare for therapy
 - (a) Administer parenteral saline solution to dilute serum calcium and inhibit reabsorption
 - (b) Administer parenteral phosphate to promote calcium deposition in bones
 - (c) Use furosemide (Lasix) to increase calcium excretion
 - (d) Treat underlying causes: use calcitonin for hyperparathyroidism, mithramycin and corticosteroids for cancer-related hypercalcemia
- (2) Maintain normal bowel elimination
 - (a) Assess dehydration and mental confusion
 - (b) Encourage fluids (3-4 quarts/day) and dietary fiber to prevent constipation
- (3) Injury prevention
 - (a) for clients with mental confusion, ensure bedside rails are up and the bed is in low position
 - (b) Reposition the client frequently and secure all invasive lines
- (4) Provide client education
 - (a) Promote early ambulation to reduce calcium loss from the bones
 - (b) Encourage daily weight bearing activities to prevent further calcium loss from bones

Hypomagnesemia

Function of Magnesium

- (1) Neuromuscular function
 - (2) Carbohydrate and protein metabolism
 - (3) Nerve function (sedative effects)
- Magnesium serum level <1.3 mEq/L (Magnesium deficiency)
- **Etiology**
- Poor nutrition: insufficient intake of magnesium-rich foods
 - Alcoholism: impaired absorption and increased renal excretion of magnesium
 - GI and renal losses: diarrhea, vomiting, or renal disorders without adequate replacement
- **Clinical manifestations**
- (1) Neuromuscular irritability: muscle spasms and tremors due to low magnesium
 - (2) (+) Trousseau's sign and Chvostek's sign: indicative of neuromuscular excitability
 - (3) Insomnia - difficulty in sleeping due to neuromuscular hyperactivity
 - (4) Mood changes: including irritability and anxiety
 - (5) Cardiac arrhythmias: irregular heartbeats due to changes in electrical activity
- **Laboratory and diagnostic findings**
- Serum magnesium <1.3 mEq/L
 - ECG: flattened T wave, prominent U wave, ST-segment depression



- **Nursing management**

(1) Prepare for therapy

- (a) Administer parenteral magnesium carefully using an infusion pump to prevent rapid administration and reduce the risk of cardiac arrest
- (b) Monitor for magnesium toxicity: look for signs such as hot, flushed skin, diaphoresis, hypotension, anxiety, lethargy, and laryngeal stridor
- (c) Monitor ECG and pulse for abnormalities

(2) Prevent injury

- (a) Institute seizure precaution to prevent injury due to neuromuscular irritability
- (b) Ensure safety measures for mental confusion
- (c) Assess swallowing ability before administering oral medications as magnesium depletion can cause dysphagia
- (d) Monitor clients on digoxin closely, as magnesium deficit increases the risk of digoxin toxicity

(3) Provide client and family education

- (a) Dietary sources of magnesium: encourage foods like nuts, whole grains, spinach, bananas, and oranges
- (b) Increase dietary magnesium to meet daily intake requirements

Hypermagnesemia

- **Elevated magnesium serum level of >2.5 mEq/L (Magnesium excess)**

- **Etiology**

- Renal failure
- Overuse of magnesium-containing antacids
- Excessive magnesium administration
- Overuse of enemas or laxatives containing magnesium
- Severe dehydration, as occurs in diabetic ketoacidosis

- **Clinical manifestations**

- Hot, flushed skin
- Hypoactive reflexes
- Hypotension, bradycardia
- Depressed respirations, lethargy, drowsiness, and absent deep tendon reflexes with increasing magnesium level
- Cardiac arrhythmias

- **Laboratory and diagnostic findings**

- Serum magnesium >2.5 mEq/L
- ECG: prolonged QT interval and atrioventricular blocks

- **Nursing management**

(1) Prepare for therapy



- (a) Discontinue all magnesium sources: stop parenteral and oral magnesium
- (b) Administer IV calcium Gluconate to counteract the neuromuscular effects of magnesium
- (c) Prepare the patient for hemodialysis to remove excess magnesium from the bloodstream
- (d) Administer diuretics and 0.45% normal saline to increase magnesium excretion
- (2) Provide client and family education
 - (a) Educate the client about the adverse effects of overusing magnesium-containing antacids or laxatives
 - (b) Read OTC drug labels carefully for magnesium contents
 - (c) Monitor dietary intake of magnesium to avoid excessive levels

Hypophosphatemia

- **Serum phosphorus level of <2.5 mEq/L (Phosphorus deficiency)**
- **Etiology**
 - Inadequate intake or administration
 - Obsessive intake or administration of carbohydrates/calories
 - Total parenteral nutrition (TPN) administration without proper phosphorus supplementation
 - Metabolic disturbances
 - Prolonged intense hyperventilation
 - Alcohol withdrawal and diabetic ketoacidosis
 - Respiratory alkalosis
 - Severe dehydration (common in diabetic ketoacidosis)
- **Clinical manifestations**
 - (1) Neurological and muscular symptoms
 - (a) Irritability, apprehension, weakness
 - (b) Numbness, confusion, seizures, coma
 - (c) Muscle weakness, muscle pain
 - (2) Systemic effects
 - (a) Hypoxia leading to increased respiratory rate and respiratory alkalosis
 - (b) chronic phosphorus loss may cause bruising and bleeding due to platelet dysfunction
 - (c) Increased susceptibility to infection
- **Laboratory and diagnostic findings**
 - Serum phosphorus level <2.5 mEq/L
 - Possible ECG changes or other abnormal findings related to cellular function due to low phosphorus levels
- **Nursing management**



- (1) Therapeutic interventions
 - (a) Administer TPN with gradual phosphorus repletion for malnourished clients to prevent rapid shifts
 - (b) Monitor for complications such as hypocalcemia during phosphorus administration
- (2) Prevent injury
 - (a) Assess VS, mental status for signs for confusion, apprehension or changes in LOC
 - (b) Institute seizure precautions to avoid injury
 - (c) Keep bedside rails up, bed brakes locked and reposition the client often
- (3) Client education
 - (a) Teach about preventing infection: due to changes in granulocytes in hypophosphatemia
 - (b) Emphasize the importance of adequate dietary phosphorus intake

Hyperphosphatemia

- **Serum phosphorus level >4.5 mEq/L (Phosphorus excess)**
- **Etiology**
 - Renal failure
 - Chemotherapy for neoplastic disease
 - High phosphorus intake
 - Profound muscle necrosis
- **Clinical Manifestation**
 - Muscle and systemic effects
 - Soft tissues calcifications, particularly in organs
 - Tetany: due to low calcium levels associated with high phosphorus
 - Anorexia, nausea, and vomiting
 - Muscle weakness, hyperreflexia, and tachycardia
 - Cardiovascular effects
 - Cardiac arrhythmias due to impact of calcium-phosphate balance on the heart
- **Laboratory and diagnostic findings**
 - Serum phosphorus level of >4.5 mEq/L
 - X-ray studies may reveal abnormal bone development or skeletal changes associated with excess phosphorus
- **Nursing management**
 - (1) Therapeutic intervention
 - (a) Administer phosphate binders to reduce absorption of phosphorus in the intestines
 - (b) Ensure adequate hydration to help with phosphate excretion



- (c) Administer calcium supplements if ordered, to balance the calcium-phosphate ratio
- (d) Prepare the client for dialysis if necessary, especially in severe renal cases
- (2) Prevent injury
 - (a) Monitor the serum phosphorus level frequently
 - (b) Institute seizure precautions and closely monitor for seizure activity
- (3) Client education
 - (a) Instruct the client to avoid foods high in phosphorus such as cream, nuts, whole-grain products, dried fruits and dried vegetables
 - (b) Avoid phosphate-containing substances like certain laxatives and enemas
 - (c) Educate about monitoring OTC medications carefully as it may contain phosphorus

Electrolyte	Normal Level	Main Role	Opposing Electrolyte	Supplementing Electrolyte
Sodium (Na^+)	135–145 mEq/L	Main extracellular cation	Potassium (K^+)	Chloride (Cl^-)
Potassium (K^+)	3.5–5.0 mEq/L	Main intracellular cation	Sodium (Na^+)	Magnesium (Mg^{2+})
Calcium (Ca^{2+})	8.5–10.5 mg/dL	Muscle, bone, clotting, nerve function	Phosphate (PO_4^{3-})	Magnesium (Mg^{2+})
Magnesium (Mg^{2+})	1.5–2.5 mEq/L	Enzyme activity, stabilizes cells	Calcium (Ca^{2+})	Potassium (K^+)
Phosphate (PO_4^{3-})	2.5–4.5 mg/dL	Energy (ATP), bone, acid-base buffer	Calcium (Ca^{2+})	Magnesium (Mg^{2+})
Bicarbonate (HCO_3^-)	22–26 mEq/L	Main blood buffer (acid-base)	Chloride (Cl^-)	Sodium (Na^+)
Chloride (Cl^-)	98–106 mEq/L	Follows sodium to maintain balance	Bicarbonate (HCO_3^-)	Sodium (Na^+)

Genito-Urinary System

Functions of Kidney

- **Acid-base balance** - regulates HCO_3
- **Water balance** - retention/excretion (hydration status)
- **Electrolyte balance** - controlling electrolytes
- **Toxin removal** - filters blood plasma -> excrete waste
- **Blood pressure regulation** - renin
- **Erythropoietin production** - increased RBC production
- **D vitamin activation**

Organs of the Urinary System

- **Kidneys:** main filtration organ
 - Cortex - glomerulus -> inside Bowman's capsule
 - Medulla - loop of Henle
 - Renal Pelvis - collects the urine -> ureters
 - **Main parts:**
 - Glomerulus
 - Bowman's capsule
 - Proximal Convolute Tubule (PCT) - reabsorption
 - Loop of Henle - concentrates urine
 - Distal Convolute Tubule (DCT) - hormone-controlled reabsorption
 - Collecting Duct - final urine processing
- **Ureters:** tubes that transport urine to bladder
- **Urinary bladder:** holds/stores urine
- **Urethra:** tube urine exits the body

Micturition reflex: bladder fills (500ml) -> stretch receptors fire -> spinal cord reflex -> detrusor muscle contracts -> internal sphincter relaxes

- **Laboratory and Diagnostic Test**
 - GFR (Glomerular Filtration Rate)
 - Key measure of kidney function
 - Normal: 90-120 ml/min
 - <60 = moderate damage; <15 = kidney failure
 - Creatinine and BUN
 - Increased creatinine = decreased kidney function
 - BUN: another indicator of filtration efficiency

Assessment

- **Health history**
 - Common risk factors
 - Smoking
 - Pregnancies
 - Exposure to toxins
 - Diabetes & hypertension

Symptom	Definition
Dysuria	Painful or burning urination



Flank pain	Pain on side/back
Hematuria	Blood in urine
Urinary discharge	Mucus, pus or abnormal fluid
Nocturia	Frequent peeing at night
Incontinence	Loss of bladder control
Hesitancy	Trouble starting urine stream or weak flow

1. Urinalysis

a. Normal values:

- Color: pale yellow to amber
- Clarity: clear (cloudy - UTI)
- Odor: aromatic (foul - UTI; frothy - ketones)
- Specific gravity: 1.010-1.030
- pH: 4.5-8
- Protein: negative (+ proteinuria)
- Glucose: negative (+ DM)
- Ketones: negative (+ stones)
- Blood: negative (+ hematuria)
- Leukocytes esterase: negative
- WBCs: 0-5 per high-power field (HPF)

2. Urine Culture and Sensitivity

- a. Purpose: identify causative organisms in UTIs and determine appropriate antibiotic therapy
- b. Procedure: midstream clean-catch urine sample

3. Blood Test

- a. Serum creatinine (N): 0.6-1.2 mg/dL
- b. Blood Urea Nitrogen (BUN) (N): 10-20 mg/dL

4. Creatinine Clearance Test

- a. Purpose: estimate glomerular filtration rate (GFR) 90-130
- b. Procedure: 24-hour urine collection and blood sample

- Invasive Studies

1. Intravenous pyelogram (IVP)

- a. Purpose: visualize urinary tract using contrast dye

2. Cystoscopy

- a. Purpose: direct visualization of urethra and bladder

3. Renal biopsy

- a. Obtain kidney tissue for histological examination



Urinary Retention and Incontinence

Feature	Urinary Retention	Urinary Incontinence
Main Problem	Inability to empty the bladder even when full	Inability to control urine flow - leakage occurs involuntarily
Urine Flow	Reduced or absent	Uncontrolled or frequent leaks
Common Causes	BPH, nerve injury, surgery, medications	Weak pelvic floor, neurological issues, stress and aging
Urgency to Void	Present but ineffective	Sudden, intense urge or unaware of leakage
Risk of Infection (UTI)	High - due to urine stasis in bladder	Moderate to high - due to skin contact, moisture, hygiene
Bladder Condition	Over distended (can be palpable or visible)	Often underfilled or leaking small amounts
More Common in	Men (especially older men with BPH)	Women, especially elderly or postpartum
Emergency status	Yes, especially in acute retention	No, unless severe or associated with infection or dehydration
Complications	Hydronephrosis, bladder damage/rupture, overflow incontinence	Skin breakdown, social isolation, emotional distress
More common in	Men (especially older men with BPH)	Women, especially elderly or postpartum
First-line Management	- Intermittent or indwelling catheterization - Treat underlying cause	- Pelvic floor exercise (kegel's) - Bladder training - Medications (anti-cholinergics)
Nursing Focus	- Monitor bladder distention - Prevent UTI - document output	- Maintain skin integrity - Promote fluid balance - emotional support

Types of Incontinence

1. **Enuresis** - involuntary urination during sleep (bed wetting in children)
2. **Stress incontinence** - leaks with pressure (laughing, coughing, sneezing); common in women
3. **Urge incontinence** - sudden, strong urge -> can't reach bathroom in time
4. **Overflow incontinence** - bladder overfills -> constant dribbling



Genito-Urinary Tract Disorders

Urinary Tract Infection

- UTI is an infection and inflammation of any part of the urinary tract
 - **Upper UTI** - pyelonephritis (kidney infection)
 - **Lower UTI** - cystitis (bladder), urethritis (urethra), ureteritis (ureter)
- Women are more prone due to shorter urethras
- Incidence increases with age, especially in women; in men, rises after age 50
- **Etiology**
 - Caused by bacteria, most commonly E.coli
- **Risk factors**
 - Sexual activity
 - Urinary catheter
 - urine retention or stasis
 - Mucosal damage
 - Poor hygiene
 - Diabetes
- **Pathophysiology**
 - An upper UTI is a bacterial infection of the renal pelvis, tubules, and interstitial tissue of one or both kidneys that occurs because of reflux of urine into the ureters
 - A lower UTI is an infection that typically ascends from the urethra to the bladder and possibly to the ureter
- **Assessment findings**
 - Signs and symptoms
 - Upper UTI
 - Flank pain, fever, chills
 - Nausea, malaise
 - Dysuria, urgency, frequency
 - Cloudy or bloody urine
 - Costovertebral angle tenderness
 - Lower UTI
 - Burning during urination
 - Frequency, urgency, nocturia
 - Painful urination
 - Inflammation at urethral opening

Cystitis

- Infection of urinary bladder
- Usually caused by ascending bacterial infection (E.coli)
- Most common route is transurethral
 - Female - shorter urethra, childbirth, anatomic proximity of urethra to rectum)
 - Male - due to epididymitis, prostatitis, renal calculi)
- **Predisposing factors:**
 - Microbial invasion - E. coli
 - High risk - women
 - Obstruction
 - Urinary retention
 - Increase estrogen levels
 - Sexual intercourse
- **Clinical manifestations:**
 - Pain - flank area
 - Hematuria
 - Nocturia
 - Dysuria
 - Pyuria
 - Fever
 - Urgency
 - Chills
 - Suprapubic pain
 - Urinary frequency
- **Diagnostic test**
 - Urine culture and sensitive: (+) E. coli
- **Management:**
 - Pharmacologic
 - Antibiotics (Co-trimazole - DOC)
 - Antispasmodics
 - Analgesics
 - Nursing intervention
 - Force fluid/hydration
 - Diet
 - Cranberry/orange juice
 - Avoid urinary tract irritants (coffee, tea, alcohol)
 - Warm sitz bath
 - Empty bladder after sexual intercourse
 - Good hygiene



- Encourage frequent voiding

Urethritis

- Inflammation of the urethra
- **Causative agents:** E. coli, staphylococcus, streptococci, pseudomonas
- Although inflammatory symptoms are similar to gonorrheal urethritis, sexual contact is not the cause
- May cause prostatitis and epididymitis
- **Signs and symptoms**
 - Burning on urination
 - Purulent urethral discharge appear 3-14 days
- **Treatment:**
 - Tetracycline or doxycycline

Nephrolithiasis/Urolithiasis

- Formation of stones at urinary tract
- **Types of stones:** acidic and alkaline

Acidic Stones	Cause/Diet
Calcium oxalate	Cabbage, beans, spinach, cranberry, nuts, tea, chocolate
Uric Acid	Anchovies, organ meat, whole grain, nuts, sardines
Cystine	Meat, milk, eggs, cheese
Alkaline Stones	
Calcium phosphate	Dairy products, meat, immobility, obesity, hyperparathyroidism
Struvite	Urea-splitting bacteria

- **Predisposing factors:**
 - Diet: increase calcium and oxalate
 - Hereditary: gout
 - Obesity
 - Sedentary lifestyle
 - Hyperparathyroidism
 - Male (3x) more common
 - Catheterization, infection, urinary stasis
 - Dehydration



- **Signs and symptoms**
 - **Nephrolithiasis**
 - Intense, deep ache in costovertebral region
 - Hematuria
 - Pyuria
 - Acute pain, nausea, vomiting, costovertebral area tenderness (renal colic)
 - Abdominal discomfort
 - Diarrhea
 - **Ureterolithiasis**
 - Acute, excruciating, colicky, wavelike pain, radiating down the thigh to the genitalia
 - Frequent desire to void, but little urine passed
 - Hematuria
 - **Urolithiasis**
 - Hematuria
 - Symptoms of irritation
 - Urinary retention
 - Possible sepsis
- **Diagnostic tests**
 - Intravenous pyelogram
 - Kidney Ureter Bladder X-ray
 - Cystoscopic exam
 - Stone analysis
 - Urinalysis
 - Ultrasound
- **Management**
 - Pharmacologic
 - Narcotic analgesics
 - Antispasmodic
 - Allopurinol (uric acid)
 - Diuretics
 - Antibiotics
 - Surgical
 - Nephrolithotomy: renal stone
 - Pyelolithotomy: renal pelvis stone
 - Uterolithotomy: ureteral stone
 - Cystolithotomy: bladder stone
 - Nursing intervention
 - I&O
 - Strain urine using gauze pad and save solid materials for analysis



- Exercise
- Warm sitz bath for comfort
- Alternate warm compress at flank area
- Diet
 - Force fluid (3L/day) to help client pass stone
 - Acidic stones: Alkaline-ash diet
 - Fruits and vegetables
 - Milk
 - Alkaline stones: Acid-ash diet
 - Cranberry
 - Prune
 - Plum
 - Meat and poultry
 - Calcium stone (low calcium diet, acid-ash diet, decrease dietary protein and sodium intake)
 - Uric acid (low purine foods, alkaline-ash diet)
 - Cystine stone (low methionine, alkaline-ash diet)
 - Phosphate stone (aluminum hydroxide gel, low in phosphorus)

Benign Prostatic Hyperplasia

- Slow enlargement of prostate gland in men > 40 years old
- Constriction of urethra and subsequent interference in urination
- Unknown cause
- **Predisposing factors:**
 - Aging process
 - Hormonal imbalance (estrogen, androgen)
- **Clinical manifestations:**
 - Frequency
 - Nocturia
 - Hesitancy
 - Residual urine
 - Decrease in force of urine stream
- **Diagnostic tests:**
 - Digital rectal exam
 - Cystoscopy
 - Renal biopsy
 - Prostatic massage
 - Relief of obstruction by insertion of indwelling catheter
- **Management:**



- Pharmacologic
 - Terazosin (Hytrin)
 - α_1 -adrenergic receptor blocker
 - Relaxes bladder sphincter
 - Finasteride (Proscar)
 - Inhibits 5-alpha reductase (blocks uptake and utilization of androgens by the prostate)
 - Reduction of glandular hyperplasia
 - Atrophy of prostate gland
- Balloon dilation
 - To relax smooth muscle of the bladder neck and prostate
- Immediate catheterization
 - If patient cannot void
- Watchful waiting
 - To monitor disease progression
- Surgical
 - Transurethral Resection of the Prostate (TURP)
 - No incision
 - Prostate resected through urethra
 - Continuous bladder irrigation (cystoclysis)
 - No incontinence
 - No impotence
 - Suprapubic prostatectomy
 - Incision over abdomen and bladder
 - With cystostomy tube and 2-way foley catheter
 - No incontinence
 - No impotence
 - Retropubic prostatectomy
 - Low abdominal incision
 - No incontinence
 - No impotence
 - Perineal prostatectomy
 - Impotence
 - Incontinence or rectal injury may be a complication
 - Post-operative nursing care
 - Increase fluid intake
 - Maintain patency of the catheter
 - If drainage is reddish, increase flow rate
 - Practice asepsis
 - Use a sterile NSS to prevent water intoxication



- Prevent thrombophlebitis
- Monitor for hemorrhage
- After removal of catheter, observe for urinary retention/dribbling
- Kegel's exercise
- Avoid anti-cholinergic
- Antihistamine
- Upon discharge avoid the following:
 - Vigorous exercise
 - Heavy lifting
 - Sexual intercourse 3 weeks after discharge
 - Driving 2 weeks after discharge
 - Straining with defecation
 - Prolonged sitting or standing
 - Crossing the legs
 - Long trips

Pyelonephritis

- Infection of kidney
- Bacteria (most common), fungal, viral
- **2 types:**
 - **Acute**
 - Bacterial contamination from urethra by instrumentation (iatrogenic) or hematogenous spread
 - E. coli/streptococcus
 - **Chronic**
 - Idiopathic, obstruction or reflex (stone, tumor or neurogenic bladder)
 - Progressive scarring of the kidney resulting in weight loss, hypertension and renal failure
- **Clinical manifestations:**
 - **Acute**
 - Fever
 - Urgency
 - Chills
 - Hematuria
 - Nocturia
 - Pyuria
 - Flank pain
 - Urinary frequency
 - Costovertebral tenderness



- Dysuria
- Malaise
- **Chronic**
 - Fatigue
 - Headache
 - Poor appetite
 - Polyuria
 - Excessive thirst
 - Weight loss
- **Diagnostic test:**
 - Urinalysis
 - Urine culture and sensitivity
 - Cystoscopy, IVP, ultrasound
 - CT scan
- **Management:**
 - Pharmacologic
 - Antibiotics
 - Antispasmodics
 - Analgesics
 - Nursing intervention:
 - Complete bed rest
 - VS, I&O, weight
 - Diet
 - Cranberry juice, orange juice
 - Force fluids (3-4 L/day)
 - Empty bladder regularly
 - Performing recommended perineal hygiene (wipe the perineum from front to back)

Acute Glomerulonephritis/Nephritic Syndrome

- Inflammatory and degenerative disorder of the glomerulus
- Damage to both kidney from filtration of trapping antibody-antigen complexes within the glomeruli resulting to decrease glomerular filtration rate
- **Types:**
 - **Acute post-streptococcal**
 - After 7-10 days after streptococcal throat infection
 - Immune reaction to the presence of an infectious organism (group A beta hemolytic streptococcus/GABHS)
 - **Chronic glomerulonephritis**



- Hypertensive nephrosclerosis
 - Heart failure
 - Chronic renal failure
- **Signs and symptoms:**
 - Pathognomonic sign: periorbital edema
 - Flank pain, costovertebral tenderness
 - Headache, visual disturbance
 - Fever, malaise, weakness, fatigue
 - Anorexia
 - Dyspnea (salt and water retention)
 - Tachycardia, hypertension
 - Oliguria
- **Assessment and Diagnostic tests:**
 - Urinalysis
 - Hematuria and proteinuria (most important indicator of glomerular injury)
 - Casts
 - Elevated BUN and creatinine
 - Positive antibody response test for streptococcus
 - Elevated erythropoietin sedimentation rate
 - Hyponatremia, hypophosphatemia
 - Hyperkalemia
- **Management:**
 - Pharmacologic
 - Diuretics
 - Antihypertensive
 - Corticosteroids
 - If residuals streptococcal infection is suspected, penicillin is the DOC
 - Nursing intervention
 - Monitor VS, I&O, daily weight and urine specific gravity
 - Dietary restriction of sodium, fluid and protein
 - Carbohydrates are given liberally to provide energy and reduce the catabolism of protein
 - Provide special skin care
 - WOF: renal failure, cardiac failure, hypertensive encephalopathy
 - Monitor urinalysis, BUN, creatinine levels
 - Promote rest and regular activity when hematuria and proteinuria resolve

Nephrotic Syndrome

- Renal pathology characterized by increased glomerular permeability and is manifested by massive proteinuria
- **Pathophysiology:**
 - Damage glomerular capillary membrane -> loss of plasma protein (albumin)
 - -> stimulates synthesis of lipoproteins -> hyperlipidemia
 - -> hypoalbuminemia -> decreased oncotic pressure -> shifting of fluids from IV to IS -> activation of RAAS -> sodium retention -> edema
 - -> proteinuria
- **Clinical manifestations:**
 - Pathognomonic sign: anasarca (generalized edema)
 - Edema (soft and pitting)
 - Periorbital edema
 - Dependent edema (sacrum, ankle, and hands)
 - Ascites
 - Irritability
 - Headache
 - Malaise
- **Assessment and Diagnostic test:**
 - Proteinuria
 - Increased WBC in urine
 - Needed biopsy of the confirmatory exam
- **Complications:**
 - Infection
 - Thromboembolism
 - Pulmonary emboli
 - Acute renal failure
 - Accelerate atherosclerosis
- **Management:**
 - Pharmacologic
 - Diuretics
 - ACE inhibitors
 - Lipid lowering agents
 - **Nursing intervention**
 - Monitor VS, I&O, daily weight and urine specific gravity
 - Dietary restriction of sodium, fluid and protein
 - Carbohydrates are given liberally to provide energy and reduce the catabolism of protein
 - Provide special skin care



- WOF: renal failure, cardiac failure, hypertensive encephalopathy
- Monitor urinalysis, BUN, creatinine levels
- Promote rest and regular activity when hematuria and proteinuria resolve

Acute Renal Failure

- Acute tubular necrosis (ATN), renal parenchymal failure, acute tubule-interstitial nephritis
- **Reversible condition characterized by:**
 - A sudden reduction or cessation of renal function
 - Retention of waste products
 - Increased BUN and creatinine
- **Causes**
 - **Pre-renal**
 - Hypoperfusion of kidney
 - Volume depletion
 - Impaired cardiac efficiency
 - Vasodilation
 - **Intra-renal**
 - Actual damage to kidney tissue
 - Prolonged renal ischemia
 - Nephrotoxic agents
 - Infectious process
 - **Post-renal**
 - Obstruction to urine flow
 - Urinary tract obstruction
 - Calculi (stones), tumors
 - Benign prostatic hyperplasia
 - Blood clots
- **Phases**
 - **Onset**
 - Benign with initial insult and ends when oliguria develops
 - Initial phase of injury 1-3 days
 - **Oliguric phase**
 - The oliguria period is accompanied by an increase in the serum concentration of substances usually excreted by the kidney
 - Urine output <400 cc/24 hours for 3 days - 2 weeks
 - Elevated BUN, creatinine
 - Edema, hypertension
 - Hyperkalemia
 - Hyponatremia



- Hyperphosphatemia
- Metabolic acidosis
- **Diuretic phase**
 - The diuresis period is marked by a gradual increase in urine output which signals that glomerular filtration has started to recover
 - Urine output (3-5 L/days for 10 days)
 - Elevated BUN and creatinine
 - Elevated BP
 - Hypokalemia
 - Last for 1 week
- **Recovery phase**
 - The recovery period signals the improvement of renal function and may take 3-12 months
 - Avoid nephrotoxic drugs
 - May lead to CRF
- **Diagnostic and laboratory findings:**
 - Hyperkalemia
 - Hyperphosphatemia
 - Hypocalcemia
 - Metabolic acidosis
 - Azotemia
 - Proteinuria
 - Urinalysis (Cast, RBC, WBC)
- **Signs and symptoms:**
 - Irritability
 - Headache
 - Anorexia
 - Tingling of extremities
 - Lethargy that can progress to stupor and coma
 - Sudden dramatic drop in urinary output
 - Restlessness, twitching, convulsions
 - Skin pallor, anemia, and increased bleeding time
 - Ammonia odor breath and perspiration
 - Generalized edema
 - Hypertension which can progress to pulmonary edema and CHF
- **Management:** correct the underlying cause
 - Pharmacologic
 - Volume expanders (dopamine) to restore renal perfusion in hypertensive client
 - Loop diuretics
 - ACE inhibitors for hypertension



- H2 blockers to prevent gastric ulcers
- Kayexalate to reduce potassium
- Sodium bicarbonate to treat acidosis
- Nursing intervention
 - Diet
 - Moderate protein restriction
 - High carbohydrates and restricted potassium
 - Total parenteral nutrition
 - Monitor I&O
 - Observe for oliguria followed by polyuria
 - Weigh patient daily and observe for edema
 - Monitor electrolyte imbalance (acidosis and hyperkalemia)
 - Assess for signs of overhydration (edema, crackles, headache, distended neck vein, hypertension)
 - Provide periods of undisturbed rest
 - Protect client from injury
 - Observed for early signs of complication
 - Provide skin care
 - Assist in peritoneal dialysis or hemodialysis

Chronic Renal Failure

- Irreversible condition of progressive damage to the nephrons and glomerulus
- Retention of waste products (uremia)
- **Most common cause:**
 - DM nephropathy-leading cause
 - Hypertension
 - Glomerulonephritis
 - SLE
- **Stages of CHF** (based on GFR; normal GFR: 125 ml/min/1.73 m²)
 - **Stage 1**
 - GFR: 90 ml/min/1.73 m²
 - Kidney damage with normal or increased GFR
 - **Stage 2**
 - GFR: 60-89 ml/min/1.73 m²
 - Mild increased in GFR
 - **Stage 3**
 - GFR: 30-59 ml/min/1.73 m²
 - Moderate increased in GFR
 - **Stage 4**



- GFR: 15-29 ml/min/1.73 m²
- Severe increased in GFR
- **Stage 5/ CKD V**
 - GFR: 15 ml/min/1.73 m²
 - Kidney failure (end stage renal disease)
- **Stages of Chronic Renal Failure**
 - **Diminished renal reserve volume**
 - Asymptomatic
 - No accumulation of waste products
 - Healthier kidney compensation
 - **Renal insufficiency**
 - Metabolic waste accumulation
 - Decrease GFR
 - Symptoms of renal failure
 - **End stage renal disease (ESRD)**
 - Excessive amounts of metabolic waste
 - Uremia
 - Life-threatening condition
- **Clinical manifestations:**
 - **GI**
 - Nausea and vomiting
 - Stomatitis
 - Uremic breath/uremic fetor
 - Diarrhea/constipation
 - **Respiratory**
 - Kussmaul's respiration
 - Deep rapid respiration
 - Decreased cough reflex
 - **F&E**
 - Hyperkalemia
 - Dilutional hyponatremia
 - Hypermagnesemia
 - Hyperphosphatemia
 - **Integumentary**
 - Pruritus
 - Dry skin
 - Uremic frost
 - Edema
 - **Cardiovascular**
 - Hypertension due to activation of RAAS



- Pericarditis due to irritation by uremic toxins
- **Hematologic**
 - Anemia
 - Thrombocytopenia
- **Musculoskeletal**
 - Muscle cramps
 - Loss of muscle strength
 - Renal osteodystrophy
 - Bone pain
 - Bone fractures
- **Reproductive**
 - Amenorrhea
 - Testicular atrophy
 - Infertility
 - Decreased libido
- **Neurologic**
 - Confusion
 - Disorientation
 - Seizures
 - Burning of soles of feet
 - Behavior changes
- **Management**
 - **Pharmacologic**
 - Calcium and phosphorus binders
 - Antihypertensive and cardiovascular agents
 - Antiseizure agents
 - Erythropoietin
 - **Nursing intervention**
 - **Nutrition therapy**
 - Regulate protein intake
 - Allow only protein with high caloric value (dairy products, eggs, meat)
 - Fluid intake to balance fluid losses
 - Fluid allowance (500-600 ml) more than the previous days 24 hour urine output
 - Some restriction of potassium
 - Adequate caloric intake and vitamin supplement
 - **Maintain F&E balance**
 - Monitor VS and I&O
 - Weight the patient daily
 - Assess presence and extent of edema



- Auscultate breath sounds
- Administer phosphate binding agent (Amphojel)
- **Prevent infection and injury**
 - Meticulous skin care
 - Encourage activity but avoid fatigue
 - Protect from exposure to infectious agent
 - Avoid aspirin
 - Soft toothbrush
- **Promote comfort**
 - Analgesics as ordered
 - Antipruritic
- Assist with the coping in lifestyle and self-concept
 - Promote hope
 - Provide opportunity to express feelings
 - Identify available community resources
 - Prepare client for dialysis or kidney transplant

Peritoneal Dialysis

- **Principles:**
 - Dialyzing solution is introduced via catheter inserted in the peritoneal cavity
 - The peritoneal membrane is used as a dialyzing membrane to remove toxic substances metabolic waste and excess fluid
 - Patient can dialyze alone in any location
 - Can be used in patients who are hemodynamically unstable
 - The peritoneal membrane that covers the abdominal organs and lines the abdominal wall serves as the semipermeable membrane
 - Once the cavity, uremic toxins such as urea and creatinine begin to be cleared from the blood through diffusion and osmosis
- **Nursing care**
 - Preparing the patient
 - Consent (patient and the family)
 - Obtain baseline VS
 - Explain the procedure
 - Empty the bladder and bowel to prevent puncture
 - Administer broad-spectrum antibiotic to prevent infection
 - Administer heparin to prevent fibrin formation
 - Warm the dialysate to dilate vessels of peritoneum
 - Note: normal color of the drainage is colorless
 - Cloudy: infection, peritonitis



- Bloody: normal at first few exchanges
- Yellowish: punctured urinary bladder
- Regulate fluid volume and drainage
- Promote comfort
- Prevent complications
 - Leaks
 - Obstruction
 - Peritonitis
- Drain exit site infection
- Monitor urine/glucose levels
- Teach client of dialysis and care of peritoneal cathete

Hemodialysis

- Client is attached (via a surgically created AV fistula or graft) to a machine that pumps blood along a semi-permeable membrane, dialyzing solution is on the other side of the membrane and osmosis, diffusion of waste, toxins, and fluid from the client occurs
- Diffusion, osmosis, and ultrafiltration are the principles in dialysis
- Hemodialysis access
 - **AV fistula**
 - Commonly in the forearm anastomosis artery to vein either side to side or end to end
 - It takes at least 14 days to mature
 - Palpate for thrills, auscultate for bruits
 - **AV graft**
 - Can be created by subcutaneously interposing a biologic, semibiologic, or synthetic graft material between an artery and vein
 - A graft is created when the patient's vessels are not suitable for fistula
 - **Vascular access devices**
 - Creation of a double-lumen large core catheter into the subclavian, internal jugular or femoral vein
- **Nursing management**
 - Protecting vascular access
 - Evaluate venous access site for bruit or thrill
 - Absence means blockage or clotting
 - Taking precautions during IV
 - The rate of administration must be as slow as possible
 - Monitoring symptoms of uremia
 - Deleterious cardiac and respiratory complications
 - Assessment must be conducted frequently



- Controlling electrolyte levels and diet
- Managing discomforts and pain
 - Antihistamines for pruritus
 - Use bath oils, superfatted soap, cream and lotion
 - Keep nails trimmed to avoid scratching and excoriation
 - Applying lotion to the skin instead of scratching also promotes comfort
- Monitoring BP
 - Antihypertensive agents must be withheld before dialysis to avoid hypertension due to combined effect of dialysis and the medications
- Preventing infection
 - Caring for the catheter site
 - Performed during showering and bathing
 - Exit site should not be submerged in bath water
 - Liquid soap is recommended
 - Make sure that the catheter remains secure to avoid tension and trauma
- **Vascular access complications:**
 - Poor blood flow
 - Clotting
 - Infection
 - Pseudoaneurysm/aneurysm
 - Ischemia of the hand
 - May contribute to congestive heart failure

Renal Transplant

- **Kidney transplantation** involves transplanting a kidney from a living donor or deceased donor to a recipient who no longer has renal function
- **Philippines' Organ Donation Act of 1991 (RA 7170)**
 - Any individual, at least 18 years of age and of sound mind, may give by way of legacy, to take effect after his death, all or part of his body for the purpose of medical or dental education, research advancement of medical or dental science, therapy or transplant
 - In all donations, the death of a person whose body organ will be removed after his death for the purpose of transplantation to a living person, shall be diagnosed separately by 2 qualified physicians neither of whom shall be:
 - A member of the team of medical practitioners who will affect the removal of the organ from the body
 - Lead of hospital or designated officer authorizing the removal of the organ
- **Where do organs come from?**
 - Living related donors
 - Living unrelated donors



- Emotionally related donors
 - husband/wife
 - Best friend
- Deceased donor
 - Acute head/neurologic trauma
 - Vehicular crash, gunshot wound
 - Blunt head injuries
 - CVA
 - Aneurysm
 - Cerebral anorexia
 - Drowning, hanging
 - Primary brain tumors
- **Pre-operative nursing care**
 - Complete physical examination is performed
 - Tissue typing
 - Blood typing
 - Antibody screening
 - Psychosocial evaluation
 - Patient teaching
 - Post-operative pulmonary hygiene
 - Pain management options
 - Dietary restrictions
 - Early ambulation
 - No blood transfusion for at least 2 weeks prior to transplantation
 - Commercially prepared erythropoietin used as substitute for management of anemia
- **Post-operative nursing care**
 - Immunosuppressive agents are administered
 - Assessing the patient for transplant rejection
 - Oliguria
 - Edema
 - Fever
 - Increased BP
 - Weight gain
 - Swelling or tenderness over the transplanted kidney
 - Increased in serum creatinine level
 - Preventing infection
 - Signs and symptoms
 - Shaking chill
 - Fever



- Tachycardia
 - Tachypnea
 - Increase or decrease WBC
- Attention to hand hygiene
- Monitor urinary function
- Addressing psychological concerns
 - Assessment of the patient's stress and coping
- Monitoring and managing potential complications
 - Breathing exercises
 - Early ambulation
 - Care of surgical incision
- **Detecting rejection**
 - Ultrasonography may be used to detect enlargement of the kidney
 - Percutaneous renal biopsy (most reliable) and x-ray techniques are used to evaluate transplant rejection
- **Renal transplant rejection**
 - **Hyperacute**
 - Within hours after surgery
 - Antibody-antigen reaction
 - No urine output
 - Blue, flaccid kidney
 - Transplanted kidney must be removed
 - Client resumes hemodialysis
 - **Acute**
 - Days to month after surgery
 - Body mounts an immune system defense against donor kidney
 - Urine output drops sharply
 - Increased BUN and creatinine
 - Fever, graft tenderness, swelling
 - Increased dosage of immunosuppressant drugs
 - **Chronic**
 - Months to years after surgery
 - Unclear cause
 - Gradual decline in kidney function
 - No specific treatment

Special Senses

Eyes

- **External layer** - protection



- Sclera - white, maintain the shape of the eyeball
- Conjunctiva - thin, transparent and protecting the front layer of the eyes
- Corneal - protects the iris and pupil, refracts light
- **Middle layer** - vascular layer, provides nutrition to the eyes
 - Iris - colored-part of the eye, regulates the pupil size
 - Pupil - central opening, controls light entry
 - Choroid - nourishes the retina, absorbs light to prevent damage to the eyes
 - Ciliary muscles - contracts and relaxes lens to focus, optical power -> accommodation
- **Inner/Sensory layer**
 - **Lens** - focuses on objects, refracts light -> retina
 - **Retina** - light-sensitive cells (photoreceptors: cones - colored, rods - b&w)
 - **Optic nerve (CN II)** - transfer visual information to the brain for processing
 - Optic disc - protects optic nerve, with blood vessels and nerves
 - **Aqueous humor** - watery clear fluid, nourishes lens and cornea, maintains IOP: 10-20 mmHg
 - **Vitreous humor** - gel - like, supports retina, maintains shape of the eye
 - **Macula lutea** - on the retina, concentrated cones - sharp detailed central vision/tunnel vision (visual acuity)
 - **Canal of schlemm** - drainage pathway/system

Assessment

- **CN II** - Optic (sensory)
- **CN III** - Oculomotor (motor)
- **CN IV** - Trochlear (motor)
- **CN VI** - Abducens (motor)

Ptosis	Drooping of eyelid
Exophthalmos	Bulging eyelid
Enophthalmos	Sunken eyes
Nystagmus	Involuntary eye movement
Strabismus	Cross eyed
Conjunctivitis	Irritation of conjunctiva
Diplopia	Double vision

PERRLA

- Pupil equal & round: 2-6mm



- Reactive to light: (CN II & CN III - brain stem)
 - Direct response: pupils constrict to light
 - Consensus response: opposite eye also constricts
- Pupillary reflex - spinal cord & brain stem
- Accommodation reflex
 - (N) Constriction, convergence of eyes

Snellen Chart - assess visual acuity

- (N) 20/20, mild: 20/40, moderate: 20/60, legally blind: 20/200
- How to read:
 - 20/60 (20 - distance from the chart, 60 - (N) vision cannot read this test at 60 feet away)

Six Cardinal Position of Gaze

- EOM - assess CN III, IV, VI

Refractive Errors

- Myopia (nearsightedness)
- Hyperopia (farsightedness)
- Astigmatism (scattered light)

Diagnostic Test

- Auto-refractor - automatically detects reaction to light
- Photoreceptor - manual, more accurate

Nursing Management

- Normal tone
- Approach: alert
- Orient:
 - Time: clock/app - verbalize time
 - Clock method (look at 6 o'clock)
 - Focal point
- Cane for the blind: white (blind), white + red tip (blind & deaf)
- Ambulating with nurse: offer your arm, let them hold elbow level, arm should be close to the body, ½ ahead of the patient

Eye Disorders

Cataract

- Opacity of the lens (cloudy appearance)
- **Signs and symptoms:**
 - Gradual, painless
 - Blurred/cloudy vision
 - Colors appear faded (Ishihara chart)



- White pupil (increase sensitivity to glare)
- **Diagnostic tests:**
 - Snellen chart
 - Ophthalmoscopic assessment
 - Mydriatics (causes dilation) - 1 hour before
 - Ishihara tet
- **Surgical management**
 - Removal of lens (one eye at a time)
 - **Phacoemulsification** - use of sound/vibration is going to break down the cataract into small pieces, then suction out replaced with intraocular lens (permanent)
 - OPD
 - Awake during the procedure
 - **Extracapsular cataract extraction (ECCE)** - dense cataracts, lens removed whole
 - **Intracapsular cataract extraction (ICCE)** - dense cataracts, lens + capsule removed

Retinal Detachment

- Medical emergency, retina should be attached within 3 days
- **Causes:**
 - Blunt - force trauma
 - Aging - vitreous humor (gel -> watery)
- **Contributing factors:**
 - Myopia
 - Family history
 - Eye surgery
 - Tumor
 - Hemorrhage
- **Signs and symptoms:**
 - Painless
 - Curtain/veil (loss of peripheral vision)
 - Flashes of light
 - Floaters/black spots (sign of bleeding)
 - Loss of visual field (peripheral & central)
 - Increased blurriness
- **Diagnostic test:**
 - Ophthalmoscopy - gold standard
 - Ocular ultrasound - if no visualization on ophthalmoscopy due to bleeding
- **Nursing intervention**
 - Provide bed rest
 - Cover eyes (eye patch) - prevent further detachment, reduce strain/movement



- Position head as prescribed
- **Surgical procedure**
 - Pneumatic retinopexy - inject a gas bubble
 - Vitrectomy - removal of vitreous humor
 - Repair tear (laser, cryotherapy)
 - Inject gas bubble/silicone oil
 - Scleral buckling
 - Holds choroid and retina together
- **Post Op Nursing Intervention**
 - Position as prescribed (sleeping - face down)
 - Walk - look down/face down

Glaucoma

- Increased IOP
- **Types**
 - **Open-angle/wide/primary** - most common
 - Gradual increase of IOP, painless
 - Tunnel vision
 - Halo around light/blurred vision
 - Irreversible blindness
 - **Close-angle/closure/primary**
 - Painful, sudden increase of IOP
 - Headache
 - Nausea and vomiting
 - Red halos around light
 - Medical emergency - faster blindness
- **Diagnostic test**
 - **Tonometry** - measures IOP
 - **Gonioscopy** - determines if wide/closed
- **Treatment**
 - Goal: Reduce IOP to prevent blindness
 - Medication:
 - Eye drops (-prosts, -lolol, -zolamide) decrease aqueous humor
 - Miotics - constriction (pilocarpine) increased outflow of AH
 - CI: mydriatic, anti-cholinergic, bronchodilators (causes dilation of pupils)
- **Surgery**
 - Laser trabeculoplasty - small opening in TM, drains AH
 - Iridotomy - small opening in iris, drains AH
- **Intervention**



- Compliance - lifetime
- Annual eye checking
- Regularly monitor IOP
- **Post-op procedure**
 - Semi-fowlers, non-operative side side lying (cataract/glaucoma)
 - Position as ordered/face down (retinal detachment)
 - Belongings: non-operative side
- **Eye protection**
 - Eye shield/patch -> sleep
 - Don't rub
 - Sunglasses
- **Prevent increase IOP**
 - (-) Valsalva
 - Mild pain: acetaminophen, REPORT if not relieve by medication

Ear Disorders

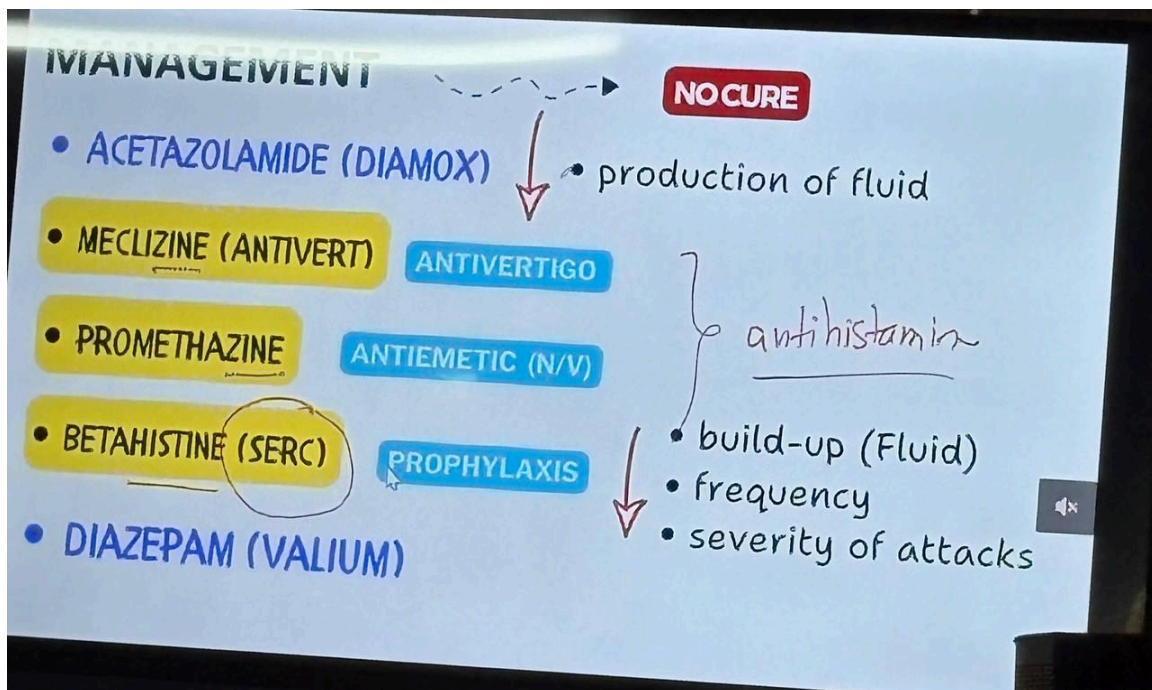
Meniere's Disease

- Inner ear
- Balance (vestibules) + hearing (cochlear)
- CN VIII - vestibulocochlear
- Excessive build up of fluid
- Tinnitus, Vertigo, Unilateral sensorineural hearing loss (Triad)
- "Drop attack"
- **Diagnostic test**
 - Romberg's test - vestibular assessment (balance)
 - "Closed eyes"
 - (+) Significant swaying, falls
 - Otoscopic exam

Conductive Hearing Loss	Sensorineural Hearing Loss
- Impacted earwax, otitis media - Temporary - Affected ear (Weber's test) - BC > AC (Rinne's)	- Damaged auditory nerves - Permanent - Unaffected ear (Weber's test) - AC > BC but no hearing/poor (Rinne's)

- **Weber's Test** - "lateralization" which side hears better
 - Tuning fork on top of the head
 - Normal - equally heard

- Conductive - louder in affected ear
- Sensorineural - louder in unaffected ear
- **Rinne's Test** - check the affected side only for bone conduction < air conduction (normal)
 - Tuning fork on mastoid bone, then ear canal
 - (+) AC > BC
 - BC > AC - Conductive Hearing Loss
 - AC > BC No hearing/poor - sensorineural hearing loss
- **Audiometry** - confirm and measures degree of hearing loss
- **Management**
 - Low sodium diet
 - Distribute evenly OFI (3L max)
 - Vertigo - assume a comfortable position



- **Surgery**
 - Labyrinthectomy - permanent hearing loss
 - Endolymphatic drainage - shunt -> drain fluid

Otitis Media

- Middle ear infection
- Caused by bacteria or virus
- Eustachian tube (connects middle ear to throat) inflamed/swell
 - Traps fluid leading to infection



- **Signs and symptoms**
 - Drainage
 - Painful
 - Fever
 - Pulling of affected ear due to pain
 - Hearing impairment
 - Issues with sleep
 - Poor appetite
- **Diagnostic test**
 - Otoscope
 - Normal eardrum/tympanic membrane: pearly gray
 - Infected: red/swollen
- **Intervention**
 - Mild: may resolve on its own
 - Moderate - severe: antibiotic: amoxicillin
 - Analgesics: paracetamol, ibuprofen, pain-relieving eardrops
- **Surgery**
 - Myringotomy
 - Recurrent
 - Hearing Loss complication due to fluid build up
 - (1) Incision - to drain fluid
 - (2) Tube - prevent further build up
- **Patient education**
 - Increased OFI
 - Elevate the head of the bed to promote drainage
 - Keep the affected ear dry
 - Warm compress -> on the ear
 - Prevent recurrence:
 - Vaccination
 - Bottle feeding -> elevate head
 - See physician immediately

Immunology

Immunology - the study of body's immune system, its functions and how it protects against disease

Immunology nursing - focuses on the care of patients with disorders of the immune system including autoimmune diseases, allergies and immunodeficiency



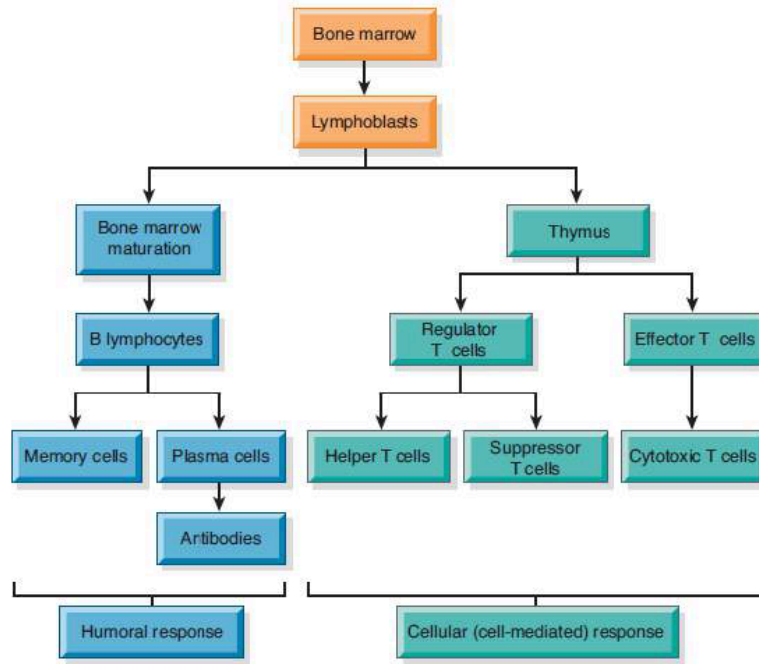
Immune System

- **Immunity:** the body's specific protective response to a foreign agent or organism
- **Immune system:** functions as the body's defense mechanism against invasion and allows a rapid response to foreign substances in a specific manner
- **Anatomy of Immune System:**
 - **Bone marrow:** the WBC involved in immunity are produced in the bone marrow. Lymphocytes are generated from stem cells (undifferentiated cells)
 - **Types of lymphocytes**
 - B cells/lymphocytes
 - T cells/lymphocytes
 - **Lymphoid tissue**
 - **Spleen:** composed of red and white pulp acts somewhat like a filter
 - Red pulp: the site where old and injured RBCs are destroyed
 - White pulp: contains concentrations of lymphocytes
 - **Lymph nodes:** remove foreign material from lymph system before it enters the bloodstream
 - Centers for immune cell proliferation
 - Contain immune cells that defend the body's mucosal surfaces against microorganisms
- **Function of the immune system**
 - Remove foreign antigens such as viruses and bacteria to maintain homeostasis
 - **General types:**
 - **Natural (innate):** or nonspecific immunity is present at birth
 - Provides a broad spectrum of defense against and resistance to infection
 - First line of host defense following antigen exposure because it protects the host without remembering prior contact with an infectious agent
 - Monocytes, macrophages, dendritic cells, natural killer cells, basophils, eosinophils and granulocytes
 - **Acquired (adaptive):** or specific immunity develops after birth
 - Usually develops due to prior exposure to an antigen through immunization or by contracting a disease, both of which generate a protective immune response
 - **Divided into two mechanism:**
 - The cell-mediated response involving T-cell activation
 - Effector mechanisms, involving B-cell maturation and production of antibodies
 - **Types of acquired immunity:**
 - **Active acquired immunity:** refers to immunologic defenses developed by the person's own body



- **Passive acquired immunity:** temporary immunity transmitted from a source outside the body that has developed immunity resulting from the transfer of antibodies from mother to an infant in utero or through breastfeeding or receiving injections of immunoglobulin
- **Three means of defense:**
 - **Phagocytic immune response:** primarily involves the WBCs (granulocytes and macrophages) which ingest foreign particles and destroy the invading agent
 - **Humoral immune response (antibody response):** begins with the B lymphocytes, which can transform themselves into plasma cells that manufacture antibodies
 - Bacterial phagocytosis and lysis
 - Anaphylaxis
 - Allergic hay fever and asthma
 - Immune complex disease
 - Bacterial and some viral infections
 - **Cellular immune response:** involves T lymphocytes which can turn into special cytotoxic (or killer) T cells that can attack the pathogens
 - Transplant rejection
 - Delayed hypersensitivity (tuberculin reaction)
 - Graft versus host disease
 - Tumor surveillance or destruction
 - Intracellular infections
 - Viral, fungal and parasitic infection

Physiology/Pathophysiology



Immune System Disorders

Disorder	Description
Autoimmunity	Normal protective immune response paradoxically turns against or attacks the body leading to tissue damage
Hypersensitivity	Body produces inappropriate or exaggerated responses to specific antigen
Gammopathies	Overproduction of immunoglobulin
Primary Immune Deficiencies	Deficiency results from improper development of immune cells or tissues usually congenital or inherited
Secondary Immune Deficiencies	Deficiency results from some interference with an already developed immune system usually acquired later in life

Immunoglobulins

- IgG



- 75% of total immunoglobulin
- Appears in serum and tissue (interstitial fluid)
- Assumes a major role in bloodborne and tissue infections
- Activates the complement system
- Enhances phagocytosis
- Crosses the placenta
- **IgA**
 - 15% of total immunoglobulin
 - Appears in body fluids (blood, saliva, tears and breast milk as well as the pulmonary, GI, prostatic and vaginal secretions)
 - Protects against respiratory, GI and GU infections
 - Prevents absorption of antigens from food
 - Passes to neonate in breastmilk for protection
- **IgM**
 - 10% of total immunoglobulin
 - Appears mostly in intravascular serum
 - Appears as the first immunoglobulin produced in response to bacterial and viral infections
 - Activates the complement system
- **IgD**
 - 0.2% of total immunoglobulin
 - Appears in small amount in serum
 - Possibly influences B lymphocytes differentiation but role is unclear
- **IgE**
 - 0.004% of total immunoglobulin
 - Appears in serum
 - Takes part in allergic and some hypersensitivity reactions
 - Combats parasitic infections

Hypersensitivity

- **Type I – Immediate (Anaphylactic) Hypersensitivity**
 - Mediated by: IgE antibodies
 - Mechanism: Upon exposure to an allergen, IgE binds to mast cells and basophils → release of histamine and other mediators.
 - Onset: Within minutes
 - Examples:
 - Anaphylaxis
 - Asthma
 - Allergic rhinitis (hay fever)
 - Urticaria (hives)



- **Type II – Cytotoxic Hypersensitivity**
 - Mediated by: IgG or IgM antibodies
 - Mechanism: Antibodies bind to antigens on cell surfaces → complement activation → cell destruction.
 - Examples:
 - Hemolytic anemia
 - Blood transfusion reactions
 - Hemolytic disease of the newborn
- **Type III – Immune Complex-Mediated Hypersensitivity**
 - Mediated by: Immune complexes (antigen-antibody)
 - Mechanism: Immune complexes deposit in tissues → inflammation and tissue damage.
 - Examples:
 - Serum sickness
 - Systemic lupus erythematosus (SLE)
 - Rheumatoid arthritis
- **Type IV – Delayed-Type (Cell-Mediated) Hypersensitivity**
 - Mediated by: T lymphocytes (not antibodies)
 - Mechanism: Sensitized T cells release cytokines → inflammation and tissue injury after 24–72 hours.
 - Examples:
 - Contact dermatitis (e.g., poison ivy)
 - Tuberculin skin test reaction
 - Graft-versus-host disease

Endocrinology

Endocrinology - a branch of physiology and medicine concerned with endocrine glands and hormones

Endocrinology nursing - a specialized field focused on the care of patients with disorders of the endocrine system, which regulates hormones and metabolisms

Major components:

- **Glands** - secrete their products directly into the chemical substances secreted by endocrine glands
- **Hormones** - chemical substances secreted by the endocrine glands
 - Regulates and integrates body's metabolic activities
 - Functions together with the nervous system
- **Target cells/receptor**



Endocrine Glands

- **Hypothalamus** - produces and secrete pro-hormones (hormones that stimulate or inhibit production/release of pituitary hormones) - controls pituitary gland
 - Hormones
 - Corticotropin-releasing hormone (CRH)
 - Thyrotropin-releasing hormone (TRH)
 - Growth hormone-releasing hormone (GHRH)
 - Gonadotropin-releasing hormone (GnRH)
- **Pituitary Gland/Hypophysis** - commonly referred as the master gland because of the influence it has on secretion of hormones other than endocrine glands
- Directly controlled: thyroid gland (TSH), adrenal cortex (adrenocorticotrophic)
 - **Anterior Pituitary Gland**
 - **Somatostatin/Growth Hormone (GH)**
 - Stimulates growth of bone and muscle size, promotes protein synthesis and fat metabolism, decreases carbohydrates metabolism
 - Increased when you are sleeping
 - Slightly increased in pedia; slightly decreased in adults
 - Used for tissue repair
 - **Adrenocorticotrophic hormone (ACTH)**
 - Stimulates synthesis and secretion of adrenal cortical hormones
 - Increased ACTH = increased adrenal cortex = increased sugar, salt, sex
 - Low SSS = addison's disease; high SSS = cushing's disease
 - **Thyroid-stimulating hormone (TSH)**
 - Stimulates synthesis and secretion of the thyroid hormones
 - Increased TSH = Increased thyroid = Increased T3,T4 - secondary hyperthyroidism
 - Decreased TSH = Decreased thyroid = Decreased T3,T4 - secondary hypothyroidism
 - **Follicle-stimulating hormone (FSH)/Sertoli cell-stimulating hormone (males)**
 - **Female:** stimulates growth of ovarian follicle/ovulation
 - **Male:** stimulates sperm production/maturation
 - **Luteinizing hormone (LH)/Leydig cell-stimulating hormone (males)**
 - **Female:** stimulates development of corpus luteum, release of oocyte, production of estrogen and progesterone
 - **Male:** stimulates secretion of testosterone, development of interstitial tissues of testes
 - **Prolactin** - prepares female breast for breastfeeding, galactorrhea (increased prolactin)
 - lactation amenorrhea method - 6 months only (decreased LH, FSH)



- **Melanocyte-stimulating hormone (MSH)** - influence skin pigmentation
 - increased MSH = dark pigment
 - Decreased MSH = light pigment
- **Posterior Pituitary Gland**
 - **Antidiuretic hormone (ADH)/Vasopressin** - increases water reabsorption by kidney
 - increased ADH = SIADH; decreased ADH = DI
 - Head injury/surgery
 - **Oxytocin** - stimulates contraction of pregnant uterus, milk ejection from breast after birth
- **Adrenal Cortex** - the outer portion of the adrenal gland; stimulated by ACTH to produce corticosteroids
 - **Mineralocorticoids (aldosterone)** - increase sodium absorption, potassium loss by kidney (SALT)
 - **Glucocorticoids (cortisol)** - affect metabolism of all nutrients; regulates blood glucose levels, affects growth, has anti-inflammatory action, and decreases effects of stress (SUGAR)
 - **Adrenal androgens** - have minimal intrinsic androgenic activity; they are converted to testosterone and dihydrotestosterone in the periphery (SEX)
- **Adrenal Medulla** - the center of the adrenal gland that reacts to autonomic NS signals to release catecholamines
 - **Epinephrine/Adrenaline**
 - serves as neurotransmitters for the SNS
 - Prepares the body for the fight or flight response converting glycogen, stored in the liver, to glucose and increasing cardiac output
 - **Norepinephrine/Noradrenaline**
 - Serves as neurotransmitters for the SNS
 - Produces effect similar to epinephrine and produce extensive vasoconstriction
- **Thyroid Gland** - butterfly-shaped organ located in the lower neck, anterior to the trachea (largest endocrine gland)
 - Thyroid hormones: triiodothyronine (T3), thyroxine (T4)
 - Increase the metabolic rate; increase protein and bone turnover
 - Regulate cellular metabolic activity
 - T3 produced predominantly from peripheral conversion of T4
 - T3 - metabolism
 - T4 - Heat
 - Increased T3, T4 = decreased TSH - Grave's disease (common type of primary hyperthyroidism)
 - **Calcitonin**
 - Lowers blood calcium and phosphate levels



- secreted in response to high blood calcium levels
- inhibits bone reabsorption
- **Parathyroid Glands** - small glands, usually four, surround the posterior thyroid tissue; they are often difficult to locate and may be removed accidentally during thyroid or other neck surgeries
 - **Parthormone (PTH, parathyroid hormone)**
 - Regulates serum calcium
 - raise blood calcium by increasing calcium resorption from kidney, intestines and bones
- **Pancreatic Islet Cells** - a slender, elongated organ lying horizontally in the posterior abdomen behind the stomach which function as an exocrine and an endocrine gland
 - **Glucagon (alpha cells)** - increases blood glucose concentration by stimulation of glycogenolysis and gluconeogenesis
 - **Glycogenolysis** - breakdown of stored glucose
 - **Gluconeogenesis** - production of new glucose from amino acids and other substances
 - **Insulin (beta cells)** - lower blood glucose by facilitating glucose transport across cell membranes of muscles, livers and adipose tissue
 - **Somatostatin (delta cells)** - delays intestinal absorption of glucose

Kidney

- Paired organs located on the either side of the vertebral column, they are between 12th thoracic and 3rd lumbar vertebrae in the posterior abdomen behind the peritoneum
 - **1,25 dihydroxy vitamin D** - stimulates calcium absorption from the intestine
 - **Renin** - activates Renin-Angiotensin-Adlosterone system
 - **Erythropoietin** - increases RBC production

Testes

- Male gonads
- Two almond-shaped organs suspended inside the scrotum; primary function is for reproduction
 - **Androgen (Testosterone)** - affect development of male sex organs and secondary sex characteristics; aid in sperm production

Ovaries

- Female gonads
- Two almond-shaped organs located at the anterior pelvis; primary function is reproduction
 - **Estrogen** - affect development of female sex organs and secondary sex characteristics



- **Progesterone** - regulates endometrium of the uterus; maintains pregnancy

Anterior Pituitary Gland Disorders

Gigantism

- **Description:**
 - Over secretion of GH results in gigantism in children; a person may be 7 or 8 feet tall
 - Noticed at puberty
 - Epiphyseal plate still open
 - Enlargement of bones of head, hands and feet
- **Causes:**
 - Tumor of somatotrophs (signs of increased ICP)
- **Diagnostic test:**
 - CT & MCRI
 - Serum levels of pituitary hormones
- **Clinical manifestations:**
 - More than 7 feet tall
 - weak and lethargic
 - severe headaches
 - Visual disturbances
 - Diplopia
 - loss of color discrimination
 - Decalcification of the skeleton
- **Management:**
 - Pharmacological
 - Bromocriptine (Parlodel) - dopamine antagonist
 - Octreotide (Sandostatin) - synthetic analogue of GH
 - Surgical - Hypophysectomy
 - Stereotactic Radiation Therapy - requires use of neurosurgery-type stereotactic frame, may be used to deliver external beam radiation therapy precisely to the pituitary tumor with minimal effect on normal tissue
- **Nursing Interventions:**
 - Record height and head circumference
 - Provide nursing care when receiving radiation therapy, perioperative care
 - prepare the client for surgical removal of pituitary tumor
 - Assist child in interacting normally with peers



Acromegaly

- **Description:** an excess of growth hormone in adults, results in bone and soft tissue deformities and enlargement of viscera without an increase in height; closed epiphyseal plate
- **Diagnostic test:**
 - CT & MRI
 - Serum levels of pituitary hormones
- **Clinical manifestations:**
 - Transverse enlargement of bone
 - Broad skull
 - Protruding jaw
 - Prognathism
 - Broadening of hands and feet
 - Thickening heel pads
 - lips become heavier
 - Enlarged tongue
 - Soft tissue enlargement (brain, heart, internal organs)
 - Coarse features
 - Organomegaly (barrel chest)
 - Dense bones (osteoarthritis)
- **Management:**
 - **Pharmacological**
 - Antihormone medications
 - Bromocriptine (Parlodel) dopamine antagonist
 - Octreotide (Somatostatin/Sandostatin) synthetic analogue of GH) DOC
 - **Surgical**
 - **Hypophysectomy** - partial/complete removal of pituitary gland
 - Craniotomy - tumor too big
 - Sublabial incision - beneath upper lip along the gumline
 - Technique: Endoscopic
 - **Postoperative:**
 - **#1 Priority:** increased in CP (altered LOC)
 - Management:
 - elevated the HOB (promote gravity drainage)
 - avoid anything that can increase ICP
 - Valsalva maneuver
 - Coughing, sneezing (sneeze with your mouth open)
 - Bending at waist
 - Heavy lifting



- Suctioning
- Mannitol (Osmitol) - osmotic diuretic
 - Does not cross the blood brain barrier
 - Give: BP normal or elevated
 - Hold: BP low <90/60
- WOF:
 - Leak CSF (clear nasal drainage - rhinorrhea)
 - Elevate HOB
 - Check for glucose - (+) CSF leak
 - Swab, then send to lab
 - Halo's sign (soak the gauze for 2-3 mins) obsolete (no longer recommended)
 - Report to MD
 - CSF infection - earliest sign: fever
 - Specific: nuchal rigidity (nape is not flexible as normally is)
 - Late symptoms: (+) Kernig's and Brudzinski's sign
 - Management:
 - Send culture to lab (CSF, blood) to determine the causative agent
 - Antibiotic as ordered
- Stereotactic Radiation therapy
- **Nursing management**
 - Prepare the client for pituitary radiation and hypophysectomy if indicated
 - Monitor post-surgical clients for signs of complications:
 - Hemorrhage
 - Transient diabetes insipidus
 - Rhinorrhea, which may indicate CSF leak
 - Adrenal insufficiency
 - Thyroid insufficiency
 - Infection (meningitis)
 - Visual disturbances, decreased visual field
 - Monitor for hyperglycemia, cardiovascular and neurologic problems
- **Nursing care:**
 - Insulin therapy
 - Medication to treat peptic ulcer
 - blood glucose monitoring
 - assessment of stools for blood



- Deep breathing is taught before the surgery
- HOB is raised for at least 2 weeks to decrease pressure on the sella turcica and to promote drainage
- Observe for post-nasal drip and check for glucose
- Patient is cautioned against engaging in activities that increases ICP
- measure I&O, daily weight

Hormonal Dwarfism

- **Description:** generalized limited growth resulting from insufficient secretion of growth hormone during childhood
- **Diagnostic tests:**
 - X-ray
 - Computed tomography and MRI
 - Blood sample
- **Clinical manifestations**
 - Overweight for height
 - Underdeveloped jaw
 - Abnormal teeth position
 - High voice
 - delayed puberty
 - Toddler-like appearance
 - Proportional to the body, (-) cannot reproduce
- **Management**
 - **Pharmacological (hormonal replacement)**
 - Somatrem (Protropin)
 - Somatropin (Humatrope)
 - **Nursing intervention**
 - provide psychologic support and acceptance for alteration of body image
 - Assist in ambulation; avoid high impact activities
- **Hyperprolactinemia**
 - **Female**
 - Prolactin-secreting tumors
 - Amenorrhea
 - Galactorrhea
 - **Male**
 - Gynemastia
 - Decreased sex drive
 - Impotence



Posterior Pituitary Gland Disorders

Syndrome of Inappropriate Anti-Diuretic Hormone (SIADH)

- **Description:** excessive ADH secretion from the pituitary gland even in the face of subnormal serum osmolality
- **Causes:**
 - Bronchogenic carcinoma
 - Severe pneumonia
 - Pneumothorax
 - Malignant tumor
 - Head injury
 - brain surgery or tumor
 - Infection
 - Some medications
- **Diagnostics tests:**
 - Decreased serum osmolality (<280 mOsm/kg) - blood
 - Elevated ADH level (NV: 0.4-7 pG/ml)
 - Plasma osmolality and serum sodium levels are decreased
 - Urinalysis detects elevated urine sodium and osmolality
 - Serum ADH level is elevated
- **Clinical manifestations:**
 - Fluid volume excess
 - weight gain (prominent) hypervolemia (high BP), edematous (crackles)
 - Urine changes
 - Less than 30-60 ml/hr (oliguria)
 - Dark amber
 - SPG: elevated >1.030 (N: 1.030)
 - Osmolality: elevated
 - Electrolytes (inversely proportional)
 - Sodium: decreased (dilutional hyponatremia)
 - Potassium: elevated (hyperkalemia)
 - Altered mental status, headache, confusion, lethargy, seizures, and coma in severe hyponatremia (cerebral edema)
 - delayed deep tendon reflex
- **Management:**
 - **Pharmacological**
 - **Diuretics** (furosemide - lasix - loop diuretics)
 - Increased urine, potassium wasting



- Give it in the morning (prevent sleep disturbance)
- Adverse effect: ototoxicity (ear toxicity - damage CN VIII)
 - Sensorineural hearing loss (permanent)
 - Earliest sign: tinnitus (ringing/buzzing/roaring in the ears)
- **Antihormone** - conivaptan, tonivaptan - inhibits ADH
- demeclocycline - drugs that render the kidneys less sensitive to ADH
- **Surgery**
 - Hypophysectomy
- **Nursing management**
 - monitor fluid intake and output, daily weight (best fluid balance indicator), urine and blood chemistries and neurologic status
 - Provide supportive measures and explanations of procedures and treatments to assist patient to deal with this disorder
 - Limit fluid intake as indicated (1-1.5L/day)
 - Regularly assess mental status

Diabetes Insipidus (DI)

- Disorder of the posterior lobe of the pituitary gland that is characterized by deficiency of ADH (vasopressin)
- **Causes:**
 - Head trauma
 - Brain tumor
 - Surgical ablation/irradiation of the pituitary gland
 - Infections of the CNS
 - Tumor
 - Failure of the renal tubules to respond to ADH, nephrogenic
- **Clinical manifestations:**
 - Excessive thirst - polydipsia
 - Dilute urine with specific gravity of 1.001 to 1.008
 - Dehydration (Fluid Volume Deficit)
 - skin tenting, dry mucosa, skin
 - Dry inside
 - Urine changes:
 - More than 30-50 ml/hr (polyuria)
 - Clear and colorless
 - USG: low
 - Osmolality: low
 - Electrolytes:
 - Sodium: elevated (hypernatremia)



- Potassium: low
- nocturia
- Weight loss
- Tachycardia
- hypotension
- Weakness
- Confusion, irritability, ALOC
- **Diagnostic test:**
 - Plasma osmolality and serum sodium levels are elevated
 - Water (fluid) deprivation test - demonstrate inability of the kidney to concentrate urine despite increased plasma osmolality and low plasma vasopressin level
 - Vasopressin test - demonstrates that the kidneys can concentrate urine after administration of ADH, this differentiates central from nephrogenic diabetes insipidus
- **Management:**
 - **Pharmacological**
 - Antihormone:
 - Desmopressin acetate (DDAVP) - intranasal synthetic vasopressin, could also administered IM
 - alternate nares to prevent nasal septum damage
 - Thiazide diuretics, mild salt depletion and prostaglandin inhibitor for nephrogenic DI
 - **Nursing Management:**
 - Replace fluids as indicated
 - Encourage the client to drink fluids in response to thirst (3-4L/day)
 - Teach patient about follow-up care and emergency measures
 - demonstrate correct medication administration
 - Advise wearing a medical identification bracelet
 - Weight patient daily
 - Same time, clothes & scale

Thyroid Gland Disorders

Goiter

- Description: thyroid tumors or enlargement sufficient visible swelling in the neck
- **Classification:**
 - **Toxic goiter** - accompanied by hyperthyroidism
 - **Non-toxic goiter** - associated with a euthyroid state
- **Types of Goiter**



- **Endemic (Iodine-Deficient) Goiter**
 - Most common type
 - Caused by iodine deficiency
 - Simple/colloid goiter
 - Usually no symptoms, only swelling; tracheal compression when excessive
 - Treatment
 - Supplementary iodine
 - Iodized salt
 - SSKI
- **Nodular Goiter**
 - Areas of hyperplasia (overgrowth)
 - Slow increase in size
 - Can cause local pressure symptoms in the thorax
 - Some are malignant or with hyperthyroid state
- **Thyroid cancer**
 - External Radiation of the neck, chest in infancy and childhood increases the risk of thyroid carcinoma

Hyperthyroidism

- **Other terms:** grave's disease/Basedow's/Parry's disease
- **Description:** results from an excessive output of the thyroid hormones caused by abnormal stimulation of the thyroid gland by circulating immunoglobulins
 - Diffuse toxic non-nodular goiter
 - Autoimmune disease (Grave's disease)
 - Everything goes up except weight, menses and sleep
- **Clinical manifestations:**
 - **Thyrotoxicosis**
 - Nervousness/Restlessness
 - Irritable and apprehensive
 - Palpitations
 - Tachycardia
 - Heat intolerance (overriding metabolism)
 - Diaphoresis
 - flush skin, warm, soft and moist
 - tremors
 - Diarrhea
 - Insomnia
 - **Exophthalmos** (Hallmark sign)
 - Bulging eyes, which produces a startled facial expression



- Von Graefe's sign: eyelid lag when looking downwards
- Dalrymple sign: upper lid retraction
- **Goiter**
 - Swelling of the thyroid gland
 - Increased appetite (voracious eater)
 - Progressive weight loss
 - Amenorrhea
 - osteoporosis
 - Myocardial hypertrophy
- **Diagnostic test:**
 - Thrill at the anterior neck
 - Bruit at the anterior neck
 - Decreased TSH
 - increased in free T4
 - Increased in radioactive iodine uptake
- **Management:**
 - Pharmacological
 - Beta Blockers - beta cells = SNS
 - Radioactive Iodine Therapy
 - Destroy the overactive thyroid cells
 - Health teachings:
 - Observe for thyroid storms
 - Propranolol may be given to control symptoms
 - CI: during pregnancy because it crosses the placenta and while breastfeeding
 - Antithyroid Medications
 - Action: block the utilization of iodine by interfering with the iodination of tyrosine and the coupling of iodotyrosines in the synthesis of thyroid hormones
 - Propylthiouracil (PTU)
 - Methimazole (Tapazole) - first line
 - Health teachings:
 - Morning, full stomach (GI irritant)
 - Agranulocytosis is the most toxic side effects (low cells count in immune cells)
 - WOF: With any sign of infection, especially with pharyngitis/sore throat, myalgia and fever or the occurrence of mouth ulcers, the patient is advised to stop the medication, notify the physician immediately and undergo hematologic studies



- Methimazole is the treatment of choice during pregnancy
- **Adjunctive Therapy**
 - Action: iodine or iodide compounds decrease the release of thyroid hormones from the thyroid gland and reduce the vascularity and size of the thyroid
 - Potassium iodide (KI)
 - Lugol's solution
 - Saturated solution of potassium iodide (SSKI)
 - Health teachings:
 - Iodide compounds are more palatable in milk or fruit juice
 - Administer through a straw to prevent staining of teeth
 - Beta-adrenergic blocking agents are important in controlling the sympathetic nervous system effects of hyperthyroidism
- **Surgical management**
 - **Subtotal thyroidectomy**
 - The surgical removal of about five-sixth of the thyroid
 - Health teaching: before surgery, PTU is administered until signs of hyperthyroidism have disappeared
 - A beta-adrenergic blocking agents (Propranolol) may be used reduce HR and other signs and symptoms of hyperthyroidism
 - Pre-op:
 - Iodine (Lugol's solution or SSKI) may be prescribed in an effort to reduce blood loss, reduced size and vascularity of the thyroid gland
 - Use a straw (to prevent staining)
 - You can mix with milk/juices (unpleasant taste)
 - Medications that may prolong clotting (Aspirin) are stopped several weeks
 - Patients receiving iodine medication must be monitored for evidence of iodine toxicity (iodism) which requires immediate withdrawal of medication
 - Symptoms of iodism
 - Swelling of the buccal mucosa
 - Excessive salivation
 - Coryza
 - Skin eruptions
 - Post-op:
 - #1 Priority - AIRWAY - Stridor (adventitious sound during inhalation - large airway obstruction)
 - Bedside: tracheostomy set, O2 and suction



- Normal: hoarseness (1-2 weeks)
- Verbal response (+) voice = good, (-) voice - laryngeal nerve damage
- Complications:
 - Hypoparathyroidism - low bone resorption = hypocalcemia
 - Earliest: numbness and tingling sensation you feel around your mouth/face
 - Late sign: Trousseau's & Chvostek's
 - Management:
 - Calcium gluconate/calcium carbonate
 - Increased calcium in diet
 - Thyroid Storm - methimazole
- **Nursing management**
 - **Improving nutritional status**
 - discourage highly seasoned foods and stimulants to reduce diarrhea
 - High calorie, high protein, high CHO, high fats, low fiber foods are encouraged
 - Weight and dietary intake are recorded
 - **Enhancing coping measures**
 - Use, calm, unhurried approach
 - the environment is kept quiet and uncluttered
 - **Improving self-esteem**
 - Eye care and protection drops or ointment
 - The patients should also be discouraged from smoking
 - **Maintaining normal body temperature**
 - maintain the environment at a cool, comfortable temperature
 - Cool baths and cool/cold fluids are encouraged
 - O2 support
 - Farthest to the station (room) - to decrease stimuli
 - Give diazepam as ordered
 - Close the eyelids at night (tape) to prevent injury and preserve moisture of the eyes; AM - sunglasses

Hypothyroidism

- Description: state of insufficient serum thyroid hormone
- Increased TSH = decreased T3, T4 - Hashimoto's thyroiditis (primary hypothyroidism)
- Everything goes down except your weight, menses and sleep (hypometabolism)
- **Cause:** autoimmune thyroiditis
- **Myxedema**



- Refers to the accumulation of mucopolysaccharides in subcutaneous and other interstitial issues
- Mucinous (mucus-filled) edema
- Extreme symptoms of severe hypothyroidism
- **Clinical manifestations:**
 - Hair loss
 - brittle nails
 - Dry nails
 - Numbness and tingling of fingers
 - Hoarseness of voice
 - Heavy menses
 - Loss of libido
 - Subnormal body temperature
 - bradycardia
 - Weight gain (increased fat)
 - thickened skin
 - Mask-like and expressionless face
 - cold intolerance
 - Slow speech
 - enlarged tongue
 - Deafness
 - Constipation
 - Low appetite
 - Hypersomnia
 - Personality and cognitive changes
 - Periorbital edema (Hallmark)
 - Myxedema coma
 - Sign of depression, diminished cognitive status (forgetful), lethargy and somnolence, depressed respiratory drive, narcosis, coma (due to SEDATION)
 - IV levothyroxine (management)
- **Management:**
 - Pharmacological
 - Atropine sulfate - increase HR
 - Dopamine - increase BP
 - Synthetic (T4) levothyroxine (Synthroid or levothroid)
 - Liothyronine (T3)
 - Give empty stomach, AM
 - Prevention of cardiac dysfunction
 - Patients may have elevated serum cholesterol, atherosclerosis, and coronary artery disease



- Angina or dysrhythmias can occur, the nurse must monitor for myocardial ischemia or infarction
- Prevention of medication interactions
 - Thyroid hormones may increase blood glucose levels
 - Bone loss and osteoporosis may also occur in thyroid therapy
- Supportive therapy
 - Arterial blood gases
 - In all patients with hypothyroidism, the effects of analgesic agents, sedatives, and anesthetic agents are prolonged
- **Nursing management**
 - **Fatigue**
 - Space activities to promote rest and exercise as tolerated
 - Assist with self-care activities when patient is fatigued
 - Provide stimulation through conversation and non-stressful activities
 - **Cold intolerance**
 - Provide extra layer of clothing or extra blanket
 - Protect from exposure to colds and drafts
 - **Constipation**
 - Encourage increased OFI within limits of fluid restriction
 - Provide foods high in fiber
 - Encourage increased mobility within patient's exercise
 - Encourage patient to use laxatives and enemas sparingly
 - O2 support
 - Diet
 - High CHON, high fiber, low calories, low proteins, low fats
 - Give written home instructions
 - Alternate warm and cold compress on the eyes (<20 minutes)
 - **NEVER SEDATE** - causes myxedema coma

Parathyroid Gland Disorders

Hyperparathyroidism

- **Description:** overproduction of parathormone by the parathyroid gland, is characterized by bone decalcification and the development of renal calculi (kidney stones) containing calcium
- **Clinical manifestations:**
 - Apathy
 - Fatigue
 - Muscle weakness



- nausea and vomiting
- Constipation
- Hypertension
- cardiac dysrhythmias
- Hypercalcemia
- renal stones
- Skeletal pain and tenderness
- Pathologic fractures
- shortening of body stature
- **Diagnostic test:**
 - Elevated serum calcium
 - Elevated concentration of parathormone
 - X-ray bone changes
 - double antibody parathyroid hormone test
 - Ultrasound
 - MRI
 - Thallium scan
 - Fine-needle biopsy
- **Complications:**
 - Hypercalcemic crisis - result in neurologic cardiovascular and renal symptoms that can be life threatening
- **Management:**
 - **Pharmacological**
 - Antacids for peptic ulcer
 - Stool softener for constipation after surgery
 - Phosphate therapy
 - Cytotoxic agents
 - Calcitonin
 - biphosphonates
 - **Surgical**
 - Parathyroidectomy
 - Criteria:
 - Younger than 50 years of age
 - unable or unlikely to participate in follow-up care
 - serum calcium level more than 1.0 mg/dl (0.25 mmol/L) above normal reference range
 - Urinary calcium level greater than 400 mg/day
 - 30% or greater decrease renal function
 - Complaints of primary hyperparathyroidism
- **Nursing Intervention**



- Hydration therapy
 - Daily fluid intake of 2000 ml or more
 - Cranberry juice
 - thiazide diuretics are avoided
- Mobility
 - Walking
 - Rocking chair
- Diet
 - avoid a diet with restricted or excess calcium
- Pre-operative care
 - Prepare the client
 - Force fluids to prevent dehydration
 - Reduce added calcium
 - Strain the urine for calculi
- Post-operative care
 - Assess for renal calculi, report hematuria or flank pain
 - protect the client
 - Assist with ADL
 - Encourage weight bearing
 - Provide relief of constipation
 - monitor nutritional status

Hypoparathyroidism

- **Description:** inadequate secretion of parathormone after interruption of blood supply or surgical removal of parathyroid tissue during thyroidectomy or radical neck dissection
- **Deficiency of parathormone results in:**
 - Increased blood phosphate (hyperphosphatemia)
 - Decreased blood calcium (hypocalcemia)
- **Clinical manifestations:**
 - Tetany
 - Numbness, tingling and cramps in the extremities
 - Stiffness in the hands and feet
 - Bronchospasm
 - Laryngospasm
 - Carpopedal spasm/Positive Trousseau's sign
 - Positive Chvostek's sign
 - Dysphagia
 - Photophobia
 - Cardiac dysrhythmia



- Seizures
- Anxiety
- Irritability
- **Diagnostic test:**
 - Serum calcium 5-6 mg/dl or lower
 - Serum phosphate levels are increased
 - X-ray calcification
- **Management:**
 - **Pharmacological**
 - V calcium Gluconate - after thyroidectomy if tetany occurs
 - parenteral parathormone - monitor allergic reaction
 - Oral tablets of calcium salts
 - Aluminum hydroxide gel
 - Aluminum carbonate - after meals to bind phosphate and promote its excretion through the GI tract
 - Vitamin D - enhanced absorption of calcium
 - **Post-operative care**
 - Calcium Gluconate at bedside
 - Be alert for possible laryngeal spasm, keep tracheostomy set available
 - Institute seizure precaution
 - Minimize environmental stimuli
 - encourage a diet high in calcium, low in phosphorus

Adrenal Gland Disorders

Pheochromocytoma

- **Description:** tumor that is usually benign and originates from the chromaffin cells of the adrenal medulla
 - It is one form of hypertension that is usually cured by surgery, however, without detection and treatment, it is usually fatal
- **Clinical manifestations:**
 - Vertigo
 - Blurring of vision
 - Diaphoresis
 - air hunger
 - Palpitations
 - tachycardia
 - Tremors



- Flushing
- Anxiety
- Nausea and vomiting
- 5 H's
 - Hypertension
 - Hyperhidrosis
 - Hypermetabolism
 - Hyperglycemia
 - Headache
- **Diagnostic test:**
 - **Vanillylmandelic acid (VMA) test**
 - A 24 hour specimen of urine is collected
 - A number of medications and foods such as coffee and tea, bananas, chocolate, vanilla, aspirin may alter the results
 - **Total plasma catecholamine (epinephrine & norepinephrine)**
 - Measures at supine and at rest for 30 minutes
 - Values of epinephrine greater than 400 pG/ml (2180 pmol/L) or norepinephrine values greater than 2000 pG/ml (11800 pmol/L) are considered diagnostic of pheochromocytoma
 - **Clonidine suppression test**
 - May be performed if the results of plasma and urine tests of catecholamine are inconclusive
 - **CT, MRI, Ultrasonography**
- **Management:**
 - **Pharmacological: antihypertensive**
 - Alpha-adrenergic blocking agents (phentolamine - Regitinel)
 - Smooth muscle relaxants (sodium nitroprusside - Nipride)
 - Phenoxybenzamine (Dibenzylamine) - a long acting alpha adrenergic blockers
 - CCB (nifedipine)
 - BB (propranolol)
 - **Surgical: adrenalectomy**
 - The patient needs to be well-hydrated before, during, after surgery to prevent hypotension
 - Corticosteroid replacement is required if bilateral adrenalectomy has been necessary
 - Hypotension and hypoglycemia may occur in the postoperative period because of the sudden withdrawal of excessive amounts of catecholamine
 - **Nursing Intervention:**
 - Advise bed rest, with HOB elevated to promote orthostatic decrease in BP



- Monitor ECG changes, arterial pressures, fluids and electrolytes balance and blood glucose levels
- Encourage patient to schedule follow-up appointments to observe return of normal BP
- Give instructions regarding long term steroid therapy, including the risk of skipping dose or stopping medication abruptly

Addison's Disease

- **Description:** adrenocortical insufficiency, occurs when adrenal cortex function is inadequate to meet the patient's need for cortical hormones
 - Autoimmune or idiopathic atrophy of the adrenal glands is responsible for the vast majority of cases
 - Decrease in SSS
- **Clinical manifestations:**
 - Bronze skin discoloration (hallmark)
 - Muscle weakness
 - Anorexia
 - High pitch
 - Feminine look
 - fatigue
 - Emaciation
 - Dark pigmentation
 - Hypotension
 - Poor stress coping
 - low blood glucose
 - Low serum sodium
 - High serum potassium
 - **Addisonian crisis** - characterized by cyanosis and the classic signs of circulatory shock (pallor, apprehension, rapid, and weak pulse, rapid respirations and low BP)
- **Diagnostic tests:**
 - Early morning serum cortisol less than 165 nmol/L and plasma ACTH more than 22.0 pmol/L
 - Decreased blood glucose
 - Decreased level of serum sodium
 - Increased level of serum potassium
 - Increased WBC
- **Management:**
 - Pharmacological (hormone replacement)
 - Steroids = cortisol



- Fludrocortisone - replace aldosterone
- Hydrocortisone (Solu-Cortef) is administered by IV, followed by 5% dextrose in normal saline - immediate treatment for crisis
- Vasopressor amines may be required if hypotension persists
- Antibiotics may be administered if infection has precipitated adrenal crisis
- During stressful procedures or significant illnesses, additional supplementary therapy with glucocorticoids is required to prevent Addisonian crisis
- **Nursing Management:**
 - Weight patient daily
 - Increased sugar, CHON, CHO
 - Stress management
 - Avoid ISIS = CRISIS
 - Select foods high in sodium during GI disturbances and in a very hot weather
 - Administer hormone therapy replacement as prescribed
 - Modify the dosage during illness and other stressful situation
 - Monitor for signs and symptoms indicative of Addison's crisis, which can include shock, hypotension and rapid respirations, weak pulse
 - Avoid unnecessary activity and stress that could precipitate another hypotensive episode
 - Instruct patient to inform other health care providers such as dentists about the use of corticosteroids
 - Wear a medical alert bracelet and to carry information at all times about the need for corticosteroids

Cushing's Disease

- **Description:** excessive rather than deficient adrenocortical activity caused by the long term use of corticosteroids medications
 - is infrequently the result of excessive corticosteroids production secondary to hyperplasia of the adrenal cortex, tumor of the pituitary gland
 - Increased sugar (cortisol), salt (aldosterone), sex
- **Clinical manifestations:**
 - Buffalo hump
 - Central obesity, thin extremities (truncal obesity)
 - Skin is thin, fragile and easily traumatized
 - Ecchymosis
 - Weakness
 - hypertension
 - hyperglycemia
 - Hyponatremia



- Hypervolemia
- hypokaemia
- Penduculous abdomen
- Purple striae (stretch marks)
- Moon faced appearance
- Acne
- weight gain
- Slow healing
- Virilization
- Hirsutism (increased testosterone)
 - Deep voice
- Breast atrophy
- Loss of libido
- **Diagnostic tests:**
 - **Dexamethasone suppression test**
 - Increase in serum sodium
 - Increase in blood glucose levels
 - Decrease in potassium
 - Reduction in the number of blood eosinophils
 - Disappearance of lymphoid tissue
 - Plasma and urinary cortisol levels increased
- **Management:**
 - **Pharmacological**
 - Adrenal enzyme inhibitors (antihormone)
 - Metyrapone (Metopirone)
 - Aminoglutethimide (Cytadren)
 - Mitotane (Lysodren) - chemotherapeutic drug
 - Use gloves (double gloves)
 - Ketoconazole (Nizoral) - DOC - can decrease SSS
 - Potassium - sparring diuretics (spironolactone - aldactone) - excrete sodium but not waste potassium
 - **Dosing (steroids)**
 - AM $\frac{2}{3}$ dose (cortisol naturally rise)
 - PM $\frac{1}{3}$ dose (cortisol naturally drop)
 - **Double/Triple dose**
 - ISIS - Infection, surgery, injury, stress = ADDISONIAN CRISIS
 - **Surgical**
 - Transsphenoidal hypophysectomy
 - Bilateral adrenalectomy - lifetime replacement of adrenal cortex hormones is necessary



- Before: IV steroids (hydrocortisone)
- Immediately after: IV steroids
- Post-op: lifelong steroids (to prevent Addisonian crisis)
 - Never skip a dose
 - Never abruptly stop (taper)
- **Radiation of pituitary gland**
- **Nursing management:**
 - Weight patient daily
 - Insulin PRN
 - Establishing a protective environment
 - Recommend foods high in protein, calcium, and vitamin D to minimize muscle wasting
 - Space rest periods throughout the day
 - Meticulous skin care
 - Low carbohydrate, low sodium diet, limit OFI

Conn's Syndrome

- Also known as primary hyperaldosteronism
- A condition where the adrenal glands produce too much of the hormone aldosterone, leading to high blood pressure and low potassium levels
- **Clinical manifestations:**
 - Hypertension
 - Hypokalemia
 - Muscle weakness
 - Fatigue
 - Headache
 - Increased thirst
 - Increased urination
- Cause: adenoma, adrenal hyperplasia
- **Diagnosis:**
 - blood test (aldosterone & renin)
 - CT Scan & MRI
- **Treatment:**
 - Pharmacological
 - Spironolactone
 - Surgery
 - Adrenalectomy
- **Prevention:**
 - Increasing physical activity



- Limiting alcohol intake
- Reducing sodium in diet
- Stop smoking

Pancreatic Islet Disorders

Diabetes Mellitus

- Group of metabolic diseases characterized by increased levels of glucose in blood resulting from defects in insulin secretion, insulin action or both (hypo endocrine condition)
- **Normal:** 70-110/120 mg/dl
- **Risk factors:**
 - Family history of diabetes
 - Obesity
 - Race (African American, Hispanic Americans, Native Americans, Asian Americans, Pacific Islanders)
 - Age > 45 y/o
 - Hypertension
 - HDL cholesterol level <35 mg/dl and triglyceride level >250 mg/dl
- **Classifications of DM**
 - Type 1 DM
 - Type 2 DM
 - Gestational diabetes
 - Diabetes mellitus associated with other conditions or syndromes
- **Diagnostic tests:**
 - Fasting plasma glucose
 - Random plasma glucose
 - Oral glucose tolerance test (OGTT)
- **Criteria of Diagnosis of DM**
 1. Symptoms of diabetes plus casual plasma glucose concentration equal to or greater than 200 mg/dl (11.1 mmol/L). Casual is defined as any time of day without regard to time since the last meal. The classic symptoms include polyuria, polydipsia and unexplained weight loss
 2. Fasting plasma glucose greater than or equal to 126 mg/dl (7.0 mmol/L). Fasting is defined as no caloric intake for 8 hours
 3. Two-hour postprandial glucose equal to or greater than 200 mg/dl (11.1 mmol/L) during an oral glucose tolerance test. The test should be performed as described by WHO, using a glucose load containing the equivalent of 75 g anhydrous glucose dissolved in water

Type 1 Diabetes Mellitus



- **Cause:** Autoimmune, congenital
- Characterized by destruction of pancreatic beta cells
- Abnormal response in which antibodies are directed against normal tissues of the body responding to these tissues as if they were foreign
- destruction of the beta cells results in decreased insulin production, unchecked glucose production by the liver, and fasting hyperglycemia
- Glucose derived from food cannot be stored in the liver but instead remains in the bloodstream and contributes to postprandial (after meals) hyperglycemia
- If the concentration of glucose in the blood exceeds the renal threshold for glucose, usually 180-200 mg/dl (9.9 to 11.1 mmol/L), the kidneys may be reabsorbed all the filtered glucose; the glucose then appears in the urine (glycosuria)
- When excess glucose is excreted in the urine, it is accompanied by excessive loss of fluids and electrolytes this is called osmotic diuresis
- Fat breakdown occur, resulting in an increase production of ketone bodies which are the byproducts of fat breakdown
- Ketone bodies are acids that disturb the acid-base balance of the body when they accumulate in excessive amounts, this results to diabetic ketoacidosis
- **Clinical manifestations:**
 - <30 years of age
 - Thin, recent weight loss
 - Hyperglycemia (>140 mg/dl)
 - Polyuria (osmotic diuresis)
 - Polydipsia (increased thirst)
 - Polyphagia (increased appetite)
 - Poor wound healing (poor circulation)
 - Blurred vision
- **Management:**
 - Test blood sugar
 - Administer insulin
 - Eat - prevent hypoglycemia
 - Activity (eat before and during)
 - Always carry simple sugar (candy/juice)
 - Hypoglycemia signs and symptoms (SNS + decreased sugar)
 - Palpitations/increased HR
 - tremors
 - Lightheadedness
 - Syncope
 - Hyperglycemia (3Ps + dehydration)

Type 2 Diabetes Mellitus



- **Description:**
 - Increased insulin resistance
 - Decreased insulin sensitivity
 - Impaired insulin secretion
- **Cause:** insulin resistance secondary to sedentary lifestyle (activity/diet)
- this is called metabolic syndrome which includes hypertension, hypercholesterolemia, and abdominal obesity
- Despite the impaired insulin secretion that is characteristic of diabetes type 2, there is enough insulin present to prevent the breakdown of fat and accompany production of ketone
- **Clinical Manifestation:**
 - >30 years old
 - Obese/overweight
 - Hyperglycemia
 - 3Ps
 - Poor wound healing
 - Blurred vision
- **Diagnosis:**
 - Self-monitoring of blood glucose - this allows for detection and prevention of hypoglycemia and hyperglycemia and plays crucial role in normalizing blood glucose
 - Glycosylated hemoglobin (HgbA1C) - blood test that reflects average blood glucose levels over a period of approximately 2-3 months
 - Testing for ketones - ketone in the urine signal that there is a deficiency of insulin and control of type 1 diabetes is deteriorating. The risk of DKA is high
- **Management:**
 - **Test blood sugar, Administer, Eat**
 - **Nutritional therapy (diet exchange list)**
 - to promote a 1-2 pounds weight loss per week
 - 500 to 1000 calories are subtracted from the daily total
 - The caloric distribution currently recommended is higher in carbohydrates than in fat and protein
 - Foods high in carbohydrates such as sucrose are not totally eliminated from the diet but should be eaten in moderation because they are typically high in fat and lack vitamins, minerals and fiber
 - Additional recommendations include limiting total intake of dietary cholesterol to less than 300 mg/day
 - Increase fiber in diet may improve blood glucose levels, decrease the need of exogenous insulin and lower total cholesterol and low-density lipoprotein levels in the blood
 - Alcohol is absorbed before other nutrients and does not require insulin for absorption. Large amounts can be converted to fats, increasing the risk for DKA



- It is important that patients read labels of “health foods” especially snack because they often contain carbohydrates and saturated fats which may be contraindicated in people with elevated blood lipid levels
- **Exercise (Activity)**
 - Exercise lower blood glucose levels by increasing the uptake of glucose by body muscles and by improving insulin utilization
 - Exercise at the same time and in the same amount each day
 - Regular aerobic daily exercise
 - Brisk walking is safe and beneficial (10K steps)
 - Eat 15 g carbohydrate snack before engaging in moderate exercise to prevent unexpected hypoglycemia
 - Use proper footwear. Avoid exercise in extreme heat or cold. Inspect feet daily after exercise. Avoid exercise during periods of poor metabolic control
- **Pharmacological**
 - **Oral hypoglycemic agents (OHA)**
 - **Sulfonylureas**
 - stimulate beta cell of the pancreas to secrete insulin; may improve binding between insulin and insulin receptors or increase the number of insulin receptors
 - Glipizide, Glicazide - increase insulin sensitivity
 - **Biguanide**
 - Inhibits production of glucose by the liver
 - Increase body tissues sensitivity to insulin
 - Decrease hepatic synthesis of cholesterol
 - the only biguanide in the market: metformin (Glucophage) - increase insulin sensitivity (DOC)
 - Never mix with alcohol (lactic acidosis & damaged kidney)
 - Never mix with contrast agent (hold 24 hours before and 48 hours after) - lactic acidosis + AKI
 - **Alpha-glucosidase inhibitor**
 - Delay absorption of complex carbohydrates in the intestine and slow entry of glucose into systemic circulation
 - **Non-sulfonylureas secretagogues (Meglitinides and phenylalanine derivatives)**
 - Stimulate pancreas to secrete insulin
 - **Thiazolidinediones (Glitazone)**
 - Sensitized body tissue to insulin; stimulate receptor sites to lower blood glucose and improve action of insulin
 - **Dipeptide-peptidase-4 (DDP-4) inhibitors**



- Increase and prolongs the action of incretin, a hormone that increases insulin release and decreases glucagon levels which result to improve glucose control
- **Meglitinides (Repaglinide)** - increase pancreatic insulin
- **Insulin (PRN)**

Cause of Morning Hyperglycemia	
Characteristic	Treatment
Insulin Warning - progressive rise in blood glucose from bedtime to morning	Increase evening (pre-dinner or bedtime) dose of intermediate acting or long acting insulin or institute a dose of insulin before the evening meal if one is not already part of treatment regimen
Dawn Phenomenon - relatively normal blood glucose until about 3 AM, when the level begins to rise	Change time of injection of evening intermediate acting insulin from dinner time to bedtime
Somogyi Effect - normal or elevated blood glucose at bedtime, a decrease at 2-3 am to hypoglycemic levels and a subsequent increase caused by the production of counter regulatory hormones	Decrease evening (pre-dinner or bedtime) dose of intermediate acting insulin or increase bed time snack

Insulin Therapy

- In type 1 diabetes, exogenous insulin must be administered for life because the body loses the ability to produce insulin

Time Course	Agent	Onset	Peak	Duration	Indication
Rapid - acting	Lispro (Humalog) Aspart (Novolog) Glulisine (Apidra)	15-30 mins	30 mins - 1 hr	2-4 hours	Used for rapid reduction of glucose Level to treat postprandial Hyperglycemia and/or to prevent nocturnal hypoglycemia
Short - acting	Regular (Humalog R, Novolin R, Iletin II Regular)	30 mins - 1 hr	1-2 hours	4-8 hours	Usually administered 20-30 minutes before meals; maybe taken alone or in combination with long - acting insulin

					Can be incorporated to an IV infusion
Intermediate - acting	NPH (Neutral Protamine Hagedorn) Humulin N, Iletin II Lente, Iletin II NPH	1-2 hours	2-4 hours	8-16 hours	Usually taken after food
Long - acting	Glargine (Lantus) Detemir (Levemir)	1-2 hours	Continuous (no peak)	24-48 hours	Use for basal dose

- Human insulin preparations have a shorter duration of action than insulin from animal sources because the presence of animal protein triggers an immune response that results in binding of animal insulin
- Short acting insulin is called regular insulin (marked R on the bottle). Regular insulin is a clear solution and is usually administered 20-30 minutes before a meal. Regular insulin is the only insulin approved for IV use
- Intermediate acting insulins are called NPH insulin or Lente insulin
- “Peakless” basal or very long acting insulin is absorbed very slowly over 24 hours and can be given once a day

Complications of Insulin Therapy

- **Local allergic reactions**
 - Redness, swelling, tenderness and induration or 2-4 cm wheal may appear at the injection site 1-2 hours after the insulin administration
- **Systemic allergic reactions**
 - When they occur, there is an immediate local skin reaction that gradually spreads into generalized urticaria (hives)
 - These are rare reactions are occasionally associated with the generalized edema or anaphylaxis
- **Insulin lipodystrophy**
 - Lipodystrophy refers to localized reaction in the form of either lipatrophy or lipohypertrophy occurring at the site of insulin injection
 - Lipatrophy is loss of subcutaneous fat; it appears as slight dimpling or more serious pitting of subcutaneous fat
- **Resistance to injected insulin**
 - Most patients have some degree of insulin resistance at one time or another. The most common being obesity, which can overcome weight loss
- **Morning hyperglycemia**



- an elevated blood glucose level on arising in the morning is caused by an insufficient level of insulin, which may be caused by several factors; the dawn phenomenon, the somogyi effect or insulin warning

Things to Take Note in Administering Insulin

- Storing insulin

- Vials not in use, including spare vials should be refrigerated
- Insulin should not be allowed to freeze and should not be kept in direct sunlight
- the insulin in use should be kept at room temperature to reduce local irritation at the injection site
- The patient should be instructed to always have spare vials of the type or types of insulin he/she uses
- Cloudy insulins should be thoroughly mixed by gently inverting the vial or rolling it between the hands before drawing the solution into a syringe or a pen
- Bottles of intermediate acting insulin should be also be inspected for flocculation which is frosted whitish coating

- Mixing insulins

- Long acting insulins must be mixed thoroughly before drawing in the syringe
- Regular insulin should be drawn first

- Withdrawing insulin

- Inject air into the bottle of insulin equivalent to the number of units of insulin to be withdrawn

- Selecting and rotating the injection site

- The four main areas for injection are abdomen, upper arm (posterior surface), thighs (anterior surface) and hips
- The speed of absorption is greatest in the abdomen and decreases progressively in the arm, thighs and hips respectively
- systemic rotation of injection within an anatomic area is recommended to prevent localized changes in fatty tissues (lipodystrophy)
- Administer each injection 0.5-1 inch away from the previous injection. Another approach to rotation
- Patient should try not to use the same site more than once in 2-3 weeks
- Insulin should not be injected into the limb that will be exercised because this will cause the drug to be absorbed faster which may result to hypoglycemia

- Preparing the skin

- They should be cautioned to allow the skin to dry after cleansing with alcohol
- The alcohol may be carried into the tissues, resulting in a localized reddened area and a burning sensation

- Inserting the needle

- For a normal or overweight person, a 90 degree angle is the best insertion angle
- Aspiration is generally not recommended with self-injection of insulin



- **Disposing of syringe and needles**
 - used sharps should be placed in a puncture-resistant container

Diabetes Mellitus Complications

Hypoglycemia

- Occurs when the blood glucose falls to less than 50-60 mg/dl (2.7 to 3.3 mmol/L) because of too much insulin or oral hypoglycemic agents, too little food, or excessive physical activity
- **Clinical manifestations:**
 - **Mild**
 - sweating
 - Tremor
 - tachycardia
 - Palpitation
 - nervousness
 - hunger
 - **Moderate**
 - Inability to concentrate
 - headache
 - Lightheadedness
 - confusion
 - memory lapses
 - Numbness of the lips and tongue
 - slurred speech
 - impaired coordination
 - Emotional changes
 - Irrational or combative behaviors
 - Double vision
 - Drowsiness
 - **Severe**
 - Patient needs the assistance of another person for treatment of hypoglycemia
 - Disoriented behavior
 - Seizures
 - Difficulty arousing from sleep
 - Loss of consciousness
- **Emergency measures:**
 - Injection of glucagon 1mg (SQ or IM)



- A concentrated source of carbohydrate followed by a snack should be given to the patient on awakening
- In hospitals and ED, for patients who are unconscious or cannot swallow, 25-50 ml of 50% dextrose in water (D50W) may be administered IV
- Assuring patency of the IV line because (D50W) is very irritating to the veins
- Taking additional food when physical activity is increased
- Routine blood glucose tests are performed
- Wear an identification bracelet or tag stating that they have diabetes
- Learn to carry some form of simple sugar with them at all times
- Refrain from eating high-calorie, high fat dessert foods

Diabetes Ketoacidosis

- caused by a severe or markedly inadequate amount of insulin. His deficit in available insulin results in disorders in the metabolism of carbohydrate, protein and fat
 - Hyperglycemia
 - dehydration and electrolyte loss
 - Acidosis
- **Clinical Manifestation:**
 - Polyuria
 - polydipsia
 - blurred vision
 - Weakness
 - headache
 - orthostatic hypotension
 - Weak and rapid pulse
 - Anorexia
 - nausea and vomiting
 - abdominal pain
 - Acetone breath
 - Kussmaul's respiration (rapid, deep breathing)
- **Diagnostic tests:**
 - Blood glucose levels may vary between 300 and 800 mg/dl
 - serum bicarbonate (0-15 mEq/L)
 - Low pH (6.8 to 7.3)
 - A low partial pressure of carbon dioxide
 - PCO₂: 10-30 mmHg
 - increased level of creatinine
 - Increased BUN
 - Increased hematocrit



- **Management:**
 - Rehydration
 - 0.9% sodium chloride (normal saline solution) 0.65 to 1U/h for 2-3 hours
 - Half strength normal saline (0.45%) - hypernatremia
 - Monitoring fluid volume status
 - VS
 - Lung assessment
 - I&O
 - Plasma expanders - severe hypotension
 - Monitor signs of overload
 - **Restoring electrolytes**
 - serum potassium levels must be monitored frequently
 - As much as 40 mEq/h may be needed for several hours
 - Frequent (every 2-4 hours initially) ECGs and laboratory measurements of potassium are necessary
 - Because a patient's serum potassium level may drop quickly as a result of rehydration and insulin treatment, potassium replacement must begin once potassium levels drop to normal
 - **Reversing acidosis**
 - Insulin is usually infused intravenously at a slow continuous rate
 - bicarbonate infusion to correct severe acidosis is avoided during the treatment of DKA because it precipitates further sudden decrease potassium levels
 - When mixing the insulin drip, it is important to flush the insulin solution through the entire IV infusion set and to discard the first 50 ml of fluid
 - Insulin molecules adhere to the inner surface of IV infusion sets, therefore the initial fluid may contain a decreased concentration of insulin

Hyperglycemic Hyperosmolar Nonketotic Syndrome (HHNS)

- Serious condition in which hyperosmolarity and hyperglycemia predominate with alterations of the sensorium (sense of awareness)
- Ketosis is usually minimal or absent
- persistent hyperglycemia causes osmotic diuresis which results in losses of water and electrolytes
- **Clinical manifestations:**
 - Hypotension
 - Profound dehydration (dry mucous membranes, poor skin turgor)
 - Tachycardia
 - Alteration in sensorium
 - Seizures
 - Hemiparesis



- **Diagnostic tests:**
 - blood glucose - 600 to 1200 mg/dl
 - Osmolality exceeds 350 mOsm/kg
- **Management:**
 - Fluid replacement
 - Correction of electrolyte imbalance
 - Insulin administration
 - fluid treatment is started with 0.9% or 0.45% NS
 - central venous or hemodynamic pressure monitoring
 - potassium is added to Iv fluids
 - Insulin plays less important role in the treatment of HHNS because it is not needed for reversal of acidosis

Characteristic	DKA	HHNS
Patient most commonly affected	Can occur in type 1 or type2 ; more common in type 1 diabetes	More common in type 2 diabetes especially elderly patients with type 2 diabetes
Precipitating event	Omission of insulin; physiologic stress (infection, surgery, CVA, MI)	Physiologic stress (infection, surgery, CVA, MI)
Onset	Rapid (<24h)	Slower (over several days)
Blood glucose levels	Usually >250 mg/dl (>3.9 mmol/L)	Usually >600 mg/dl (>33.3 mmol/L)
Arterial pH level	7.3	Normal
Serum and urine ketones	Present	Absent
Serum osmolality	300-350 mOsm/L	>350 mOsm/L
Plasma bicarbonate levels	<15 mEq/L	Normal
BUN & Creatinine	Elevated	Elevated
Mortality rate	<5%	10-40%

Gastroenterology

Gastroenterology - a branch of medicine focused on the digestive system and its diseases

Gastroenterology nursing - focuses on the care of patients with conditions affecting the gastrointestinal system including the stomach, intestines, liver, pancreas and related organs



Gastrointestinal System

Definition: 23-26 foot long pathway including the mouth, esophagus, stomach, small and large intestines, rectum and anus

- **Esophagus**
 - Located in the mediastinum, anterior to the spine and posterior to the trachea
 - approximately 25 cm in length
 - Tube connecting the mouth and stomach
- **Stomach**
 - Distensible pouch into which food bolus passes to be ingested by gastric enzymes
 - Hollow muscular organ with a capacity of approximately 1500ml
 - stores food during eating
- **Small intestine**
 - Longest segment of the GI tract where the process of absorption of nutrients take place
 - Consisting of three parts: duodenum, jejunum and ileum
- **Large intestine**
 - The portion of the GI tract into which waste materials from the small intestine passes as absorption continues and elimination begins
 - Consist of several parts: ascending colon, transverse colon, descending colon, sigmoid colon and rectum

Function of Digestive System

- **Digestion:** occurs when the digestive enzymes and secretions mix with ingested food and when proteins, fats, and sugars are broken down into their component smaller molecules
- **Absorption:** occurs when small molecules, vitamins, and minerals pass through the walls of the small and large intestine and into the bloodstream
- **Elimination:** occurs after digestion and absorption, when waste products are evacuated from the body
 - **Chewing and swallowing**
 - 1st process of digestion
 - Approximately 1.5L of saliva is secreted daily from the parotid, the submaxillary, and the sublingual glands
 - Salivary amylase: an enzyme that begins the digestion of starches
 - Swallowing begins as a voluntary act that is regulated by the swallowing center in the medulla oblongata of the CNS
 - **Gastric function**
 - Secretes highly acidic fluid in response to the presence of anticipated ingestion of food (HCl)



- Intrinsic factor: secreted by the gastric mucosa, combines with dietary vitamin B12
- Pepsin: an important enzyme for protein digestion; end product of the conversion of pepsinogen from the chief cells
- Food remain in the stomach for variable length of time, from 30 minutes to several hours, depending on the volume, osmotic pressure, and chemical composition of gastric contents
- **Small intestine function**
 - Secretions contain digestive enzymes
 - **Amylase:** aids in digesting starch
 - **Lipase:** aids in digesting fats
 - **Trypsin:** aids in digesting protein
 - **Bile:** aids in emulsifying ingested fats
 - secreted by the liver and stored in the gallbladder
 - Making them easier to digest and absorb
 - Intestinal secretions total approximately 1L/day of pancreatic juice, 0.5L/day of bile and 3L/day of secretions from gland of small intestine
 - **Types of contractions in the small intestine**
 - Segmentation contractions
 - produce mixing waves that move the intestinal contents back and forth in a churning motion
 - Intestinal peristalsis
 - Propels the contents of the small intestines toward the colon
- **Colonic function**
 - Bacteria assist in completing the breakdown of waste materials especially undigested or unabsorbed proteins and bile salts
 - The slow weak peristaltic activity along the tract allows for efficient reabsorption of water and electrolytes which is the primary purpose of the colon
 - Intermittent, strong peristaltic waves propel the contents and eventually reach the rectum, usually in about 12 hours

Physical Assessment

Inspection, Auscultation, Percussion, Palpation (IAPePa)

- **Order of palpation**
 - Right lower quadrant
 - Right upper quadrant
 - Left upper quadrant
 - Left lower quadrant

Organs in specific places



- **Right hypochondriac**
 - Right lobe of the liver
 - Gallbladder
 - Part of duodenum
 - Hepatic flexure of colon
 - Upper half of the right kidney
 - Suprarenal gland
- **Epigastric**
 - Aorta
 - Pyloric end of stomach
 - Pancreas
 - Part of liver
- **Left hypochondriac**
 - Stomach
 - spleen
 - Tail of pancreas
 - Splenic flexure of the colon
 - upper half of left kidney
 - Suprarenal gland
- **Right lumbar**
 - Ascending colon
 - Lower half of right kidney
 - Part of duodenum and jejunum
- **Umbilical**
 - Omentum
 - Mesentery
 - Lower part of duodenum
 - Part of jejunum and ileum
- **Right inguinal**
 - cecum
 - Appendix
 - lower end of ileum
 - Right ureter
 - right spermatic cord
 - Right ovary
- **Hypogastric**
 - Ileum
 - Bladder (if enlarged)
 - Uterus (if enlarged)
- **Left inguinal**



- Sigmoid colon
- Left ureter
- Left spermatic cord
- Left ovary

Diagnostic Studies

- Upper GI series

- Delineates the entire GI tract after the introduction of contrast agent (Barium swallow)
- Enables the examiner to detect or exclude anatomic or functional derangement of the upper GI organs or sphincters
- Also aids in the diagnosis of ulcers, varices, tumors, regional enteritis and malabsorption syndromes
- **Nursing interventions:**
 - Clear liquid diet with NPO from midnight the night before the study
 - Smoking, chewing gum and mints can stimulate gastric motility so the nurse advises against these practices
 - Increase fluid intake to facilitate evacuation of stool and the radiopaque liquid
 - Typically, oral medications are withheld on the morning of the study and resumed that evening but each patient's medication regimen is evaluated on an individual basis

- Lower GI series

- Visualization of the lower GI tract
- with introduction of barium enema
- The procedure usually takes about 15 minutes to 30 minutes during which time x-ray images are obtained
- The patient must be assessed for allergy to iodine or contrast agent
- **Nursing intervention:**
 - Emptying and cleansing the lower bowel prior to the procedure
 - Low residue diet 1-2 days before the test
 - clear liquid diet, NPO after midnight; and cleansing enemas until returns to clear the following morning
 - Laxative is given before and after the procedure
 - Increased fluid intake after the procedure
 - Evaluation of bowel movement for evacuation of barium

- Esophago-gastro-duodenoscopy (EGD)

- Direct visualization of esophageal, gastric and duodenal mucosa through a lighted endoscope
- After the patient is sedated, the endoscope is lubricated with a water-soluble lubricant and passed smoothly and slowly along the back of the mouth and down into the esophagus



- The procedure usually takes about 30 minutes
- The patient may experience nausea, gagging and choking
- Use of topical anesthetic agents and moderate sedation makes it important to monitor and maintain the patient's oral airway during and after the procedure
- Precautions must be taken to protect the scope, because the fiberoptic bundles can be broken if the scope is bent at an acute angle
- **Nursing interventions:**
 - The patient should be NPO for 8 hours prior to the examination
 - before the introduction of the endoscope, the patient is given a local anesthetic gargle or spray
 - Midazolam (Versed) a sedative that provides moderate sedation and relieves anxiety during the procedure
 - Atropine may be administered to reduce secretions and glucagon may be administered to relax smooth muscle
 - The patient is positioned left lateral position to facilitate clearance of pulmonary secretions and provide smooth entry of the scope
 - After gastroscopy, assessment includes LOC, VS, oxygen saturation, pain level
 - Monitor signs of perforation such as pain, bleeding, unusual difficulty of swallowing, rapidly elevated temperature
 - After the patient's gag reflex has returned, lozenges, saline gargle and oral analgesics agents may be offered to relieve minor throat discomfort
 - Patients who were sedated for the procedure must remain in bed until fully alert
- **Endoscopic retrograde cholangio-pancreatography (ERCP)**
 - Visualization of the common bile duct, the pancreatic, hepatic duct, through the ampulla of Vater in the duodenum
 - Uses the endoscope in combination with x-ray techniques to view the ductal structures of the biliary tracts
- **Colonoscopy**
 - Direct visual inspection of the large intestine
 - Therapeutically, the procedure can be used to remove all visible polyps with a special snare and cautery through the colonoscope
- **Laparoscopy**
 - Direct visualization of the organs and structures within the abdomen, permitting visualization and identification of any growths, anomalies and inflammatory processes
 - A pneumoperitoneum (injecting carbon dioxide into the peritoneal cavity to separate the intestines from the pelvic organs) is created
 - Biopsy samples can be taken from the structures and organs as necessary
 - Laparoscopy usually requires general anesthesia and sometimes requires that the stomach and bowel be decompressed



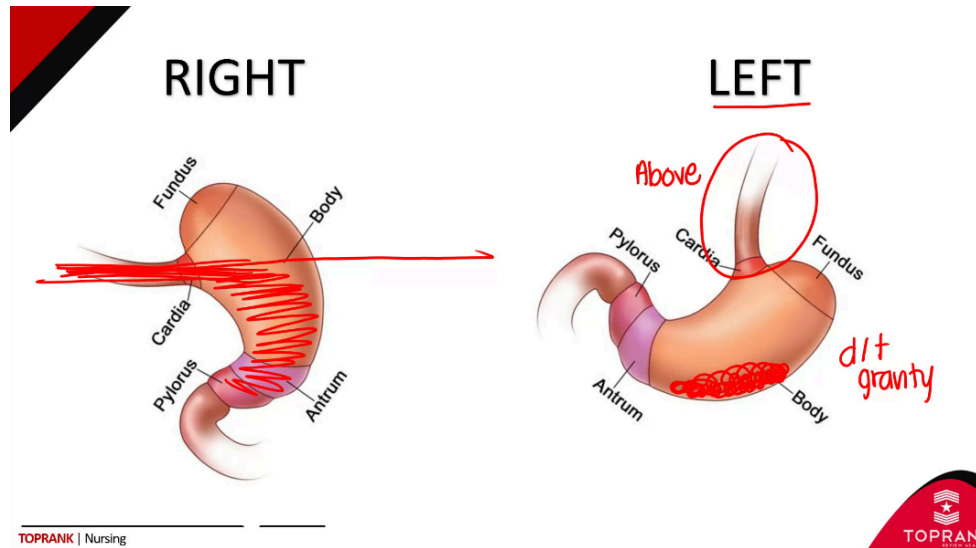
Gastrointestinal Disorders

Gastroesophageal Reflux Disease (GERD)

- Excessive backflow of gastric and duodenal contents into the esophagus due to incompetent lower esophageal sphincter
- **Causes:**
 - Weak LES (lower esophageal sphincter/cardia) - smooth muscle
 - Slow motility
 - Pyloric stenosis
 - Pyloric sphincter
 - Open: Gastric emptying (2-5 hours after meals)
 - Close: Food intake
- **Clinical manifestations:** “Indigestion burns throat, larynx and esophagus”
 - Burning sensation (pyrosis/heartburn) - commonly after meal and in the evening
 - Dyspepsia (Nausea, vomiting, stomach, bloating)
 - Dysphagia (difficulty swallowing)
 - Odinophagia (painful swallowing)
 - Globus (feeling of fullness in the throat; lump)
 - Laryngitis (hoarseness, dry cough)
- **Note:** the symptoms may mimic those of heart attack
- **Diagnostic tests:**
 - **Endoscopy**
 - Visualize the esophageal lining (check for inflammation)
 - Can also to take biopsy (barret’s esophagus)
 - **Normal:** squamous cells
 - **Abnormal:** columnar epithelial cells (intestine) - increased risk of esophageal cancer
 - **Pre-procedure:** NPO (prevent aspiration)
 - **Post-procedure:** WOF: bleeding (frequent swallowing) & perforation
 - **Esophageal pH monitoring** (probe)
 - Measures amount and frequency of acid reflux
 - **Normal esophageal pH:** >4 or slightly acidic
 - **GERD:** 24 hours
 - **Confirmation:** total of 58 minutes of exposure to acid (<4 pH)
 - **Barium Swallow:** barium sulfate (X-ray)
 - Evaluates anatomical abnormalities (weak LES)
 - **Pre-procedure:** assess for allergies, kidney function (creatinine 0.6-1.2 & BUN 10-20), NPO



- **Post-procedure:** increased OFI (2000-3000 ml for 24 hours), white stool (normal for 24-72 hours after the procedure)
- **Management:** “Food should go down”
 - **Pharmacologic**
 - **Antacids** “carbonate/hydroxide”
 - Neutralize pH
 - Examples: aluminum hydroxide
 - **H2 receptors antagonist** “tidine”
 - Decreases amount of HCl produced by stomach by blocking action of histamine receptors of parietal cells in the stomach
 - Example: ranitidine
 - **Proton pump inhibitors** “zole/prazole”
 - Decrease gastric acid secretion by slowing the ATPase pump on the surface of the parietal cells
 - Examples: omeprazole
 - **Prokinetic agents**
 - Enhancing colonic transit by increasing propulsive motor activity
 - Example: metoclopramide
 - **Nursing intervention**
 - **Diet:**
 - Fluids: in between meals (lessens abdominal distention)
 - Fiber: high (delays gastric emptying -> increases satiety (feeling of fullness) -> prevents overeating)
 - Meal: small frequent feedings (6-10 meals per day)
 - **Position:**
 - HOB: elevated at least 30-45 degrees
 - Turned to: left side lying



- **Avoid:**

- Coffee (relaxes LES)
- Citrus fruits (naturally acidic)
- Cigarette (nicotine - increases HCl; decreases saliva production)
- Carbonated drinks (causes abdominal distention)
- Chocolate (theobromine - muscle relaxant -> relaxes LES)
- Alcohol (irritates esophagus)
- Fatty foods (slowest food to digest)
- Peppermint (muscle relaxant)
- Spicy foods (worsens the signs of symptoms of GERD particularly the heartburn)
- Increases intra-abdominal pressure
 - Straining (Valsalva maneuver)
 - Tight fitting clothes
 - Heavy lifting
 - Bending below the waist
- **Drugs:** decreased motility and increased HCl
 - Aspirin (increases HCl)
 - NSAIDs (inhibits prostaglandin synthesis -> decreased mucus)
 - Anti-cholinergic/Antispasmodics (decreases motility)
 - CCB (smooth muscle relaxant)

- **Surgical management**

- **Nissen fundoplication:** most common
 - Stomach wraps fully around esophagus
- **Dor fundoplication**
 - Stomach wraps around the front of esophagus
- **Toupet fundoplication**



- Stomach wraps around the back of esophagus

Barrett's Esophagus

- A condition in which the lining of the esophageal mucosa is altered
- Associated with GERD
- Reflux causes changes in the lining of the lower esophagus
- The cells that are laid to cover the exposed area no longer squamous in origin
- Precursor to esophageal cancer
- **Clinical manifestations**
 - Burning sensation in the esophagus (pyrosis)
 - Dyspepsia
 - Dysphagia
 - Hypersalivation
 - Esophagitis
- **Diagnostic test**
 - Esophagogastroduodenoscopy (EGD)
 - Biopsy
- **Management**
 - **Photodynamic therapy** - laser thermal ablation; destroy the metaplastic cells
 - **Esophagectomy** - total resection of the esophagus with the removal of the tumor plus wide tumor-free margin of the esophagus and the lymph nodes in the area

Hiatal Hernia/Diaphragmatic Hernia

- The opening in the diaphragm through which the esophagus passes becomes enlarged and part of the upper stomach tends to move up into the lower portion of the thorax
- **Most common structure:** gastroesophageal junction - connects the esophagus to the stomach
- **Factor:**
 - Increases intra-abdominal pressure
 - Pregnancy
 - Ascites
 - obesity
 - tumor
- **Types**
 - **Sliding** - upper stomach and the gastroesophageal junction are slide displaced upward to the esophagus
 - Most common
 - May be asymptomatic
 - GERD-like symptoms



- **Paraesophageal/Rolling** - all or part of the stomach pushes through the diaphragm beside the esophagus, gastroesophageal junction stays in place
 - Rare
 - May be asymptomatic
 - (-) GERD-like symptoms
 - Fundus = herniates next to the esophagus
 - Feeling of fullness
 - Complications: strangulation -> decreased blood flow -> necrosis
- **Clinical manifestation**
 - Heartburn
 - Regurgitation
 - Dysphagia
 - Sense of fullness after eating or chest pain
- **Diagnostic tests**
 - Esophagogastroduodenoscopy (EGD) - confirmatory test
 - X-ray studies
 - Barium swallow
 - Fluoscopy
- **Management**
 - **Pharmacologic**
 - Same pharmacological management with GERD
 - **Nursing Intervention**
 - Small frequent feeding
 - Patient is advised not to recline for 1 hour after eating
 - Elevate HOB
 - Surgery is indicated in about 15% of patients
 - **Surgical: Fundoplication** (reinforce LES)
 - **Nissen fundoplication**: most common
 - Stomach wraps fully around esophagus
 - **Dor fundoplication**
 - Stomach wraps around the front of esophagus
 - **Toupet fundoplication**
 - Stomach wraps around the back of esophagus

Gastritis

- Inflammation of the stomach lining or gastric mucosa
- **Causes**
 - **Food**: highly seasoned
 - **Medications**: overuse of aspirin and other NSAIDS (aspirin/NSAID-induced gastritis)



- **Vices:** excessive alcohol intake, smoking
- **Disease:** gastric ulcer
- **Therapy:** radiation and chemo - localized inflammation
- **Substance:** ingestion of strong acid or alkali
- **Infection:** bacteria (helicobacter pylori)
- **Types:**
 - **Acute** - sudden inflammation
 - Hiccups
 - Epigastric pain
 - Anorexia
 - Dyspepsia
 - WOF: bleeding - emergency
 - **Chronic** - repeated inflammation resulting to formation of scars that destroys parietal cells
 - Halitosis (bad breath due to growth of bacteria)
 - Reflux
 - Sour taste
 - Early satiety
 - Anorexia
 - Epigastric pain
 - WOF: pernicious anemia (vitamin B12 deficiency)
- **Assessment and Diagnostic test**
 - Achlorhydria or hypochlorhydria (absence or low levels of HCl)
 - Can be determined by an upper GI series or endoscopy
 - Tissue specimen (Biopsy)
- **Management**
 - Acute: bleeding (rest the GI)
 - 1st: NPO to facilitate healing
 - 2nd: ice chips, liquid diet, solid food
 - Diet: bland - decreases HCl
 - Pharmacologic (same to PUD)
 - H2 blockers
 - Antibiotics
 - PPI
 - Antacid
 - Cytoprotective
 - Supplements: Vitamin B12 (cyanocobalamin)
 - Avoid causes/factors
 - **Surgical**



- Gastrojejunostomy - anastomosis of jejunum to stomach to detour around the pylorus

Peptic Ulcer Disease

- Excavation that forms in the mucosa wall of the stomach, in the pylorus or in the duodenum
- **Factors:**
 - Stress (chronic) -> PNS -> vagus nerves -> ACTH -> stimulates parietal, enterochromaffin & G-cells -> increases acid secretion
 - Drinks
 - Caffeinated & decaffeinated -> coffee beans -> roasted -> peptide -> increase HCl production
 - Vices
 - Smoking
 - Alcohol
 - Drugs
 - Aspirin
 - NSAIDs
 - Infection
 - H. Pyloric (raw meat) - most common cause
- **Zollinger-Ellison Syndrome** - consists of severe peptic ulcers, extreme gastric hyperacidity, and gastrin-secreting benign or malignant tumors
- **Diagnostic Test**
 - **Upper endoscopy** - direct visualization of GIT (inflammation & ulcers)
 - **Serum test** - detects antibodies to H. Pyloric
 - **Fecal Occult Blood Test (FOBT)** - detects duodenal ulcer -> hidden blood in stool
 - **Urea Breath Test** - ingest urea + specialized carbon isotope, collection bag (blows air) = labeled CO₂
 - If H. Pyloric is present: breaks down ingested agents = ammonia & labeled CO₂
- **Types:**
 - **Gastric ulcer**
 - Pain triggered by food intake (persistent gnawing pain)
 - Pain relieved by vomiting with hematemesis (vomit + blood_
 - **Duodenal ulcer**
 - Pain triggered by gastric emptying (burning pain)
 - Pain relieved by food intake with melena (old blood in stool)

	Gastric Ulcer	Duodenal Ulcer
EpigastricPain	Gnawing	Burning



Time	Food intake	Gastric emptying
Relief of Pain	Vomiting	Food intake
Bleeding	Hematemesis	Melena

- **Management:** “stomach distention is directly proportional to HCl production”
 - **Pharmacologic**
 - **H2 blockers** - reduce HCl production “tidine”
 - Time: before meals (onset: 90 minutes)
 - Ranitidine
 - **Antibiotics** - kills the H. pylori
 - Combined with PPI (reduced HCl) + bismuth (increased mucus) subsalicylate (decreased inflammation) - ex. Pepto Bismol
 - 2 antibiotics: tetracycline & metronidazole DOC
 - Amoxicillin & clarithromycin
 - **PPI** - reduce HCl production “zole, prazole”
 - Time: 30 minutes before meals
 - Omeprazole
 - **Antacid** - neutralizes pH “carbonate/hydroxide”
 - Time: 1-2 hours after meals
 - Sodium bicarbonate - S/E: metabolic alkalosis
 - Calcium carbonate - S/E: hypercalcemia, constipation
 - Aluminum hydroxide - S/E: constipation
 - Magnesium hydroxide - S/E: diarrhea
 - Note: Aluminum and Magnesium hydroxide can be taken together to decrease side effects
 - Do not drink water for 30 minutes for effectivity
 - **Gastric protestants/Cytoprotectants** - creates a viscous substance in the presence of gastric acid that forms a protective barrier, binding to the surface of the ulcer and prevents digestion of pepsin
 - **Sucralfate** - provides gastric barrier
 - Time: before meals
 - **Misoprostol** - prostaglandin analogue -> increases mucus production
 - Time: with meals
 - CI: pregnant women (can cause abortion)
 - Cytotec - dilates cervix
 - **Surgical**
 - **Vagotomy** - transecting nerves (vagus nerve) that stimulate acid secretion
 - **Gastrectomy** - removal of stomach



- **Total:** removal of esophagus to small intestine
- **Subtotal/antrectomy:** removal of lower portion of stomach
- **Anatomosis** - reconnecting/reattaching
 - **Billroth I (Gastroduodenostomy)** - removal of the lower portion of the antrum of the stomach as well as the small portion of duodenum segment; upper portion of the stomach anastomosed to duodenum (surgery of choice)
 - **Billroth II (Gastrojejunostomy)** - removal of lower portion (antrum) of the stomach with anastomosis to jejunum; a duodenal stump remains and is oversewn
- **Nursing intervention**
 - Meal: small frequent feeding
 - Diet:
 - Active phase: “painful” -> bland -> reduced HCl
 - Chew slowly and thoroughly
 - Milk restricted/limit (200-400 ml/day)
 - Avoid factors

Dumping Syndrome

- Partially the result of rapid gastric emptying, which prevents adequate mixing with the pancreatic and biliary secretions
- It is an unpleasant set of GI symptoms that sometimes occur in patients who have gastric surgery or a form of vagotomy
- **Pathophysiology:**
 - Large amount of foods -> increased concentration of food -> vasodilation -> fluids (fluid shifting)
 - -> small intestine -> increased fluid volume
 - -> decreased blood volume -> shock-like symptoms
 - Early: 5-30 minutes after meals
 - Rapid gastric emptying -> increased absorption -> elevated glucose -> pancreas -> increased insulin -> postprandial hypoglycemia
 - Late: 1-2 hours after meals
- **Clinical manifestations**
 - **Small intestine - increased fluids**
 - Distended abdomen
 - Borborygmi sound
 - Diarrhea
 - Nausea and vomiting
 - **Shock-like symptoms**



- Pallor
- Hypotension
- Tachycardia
- **Hypoglycemia**
 - Weakness
 - Diaphoresis
 - Lightheadedness
- **Management** “food should stay”
 - **Nursing intervention**
 - **Diet:**
 - Protein: high (3-4 hours after meals)
 - Fiber: high (delays gastric emptying), if with diarrhea: low
 - Carbohydrate: low (1-2 hours after meals)
 - Meals: SFF
 - Fluids: in between
 - Avoid: concentrated foods (sweet, salty, milk and coffee)
 - **Position:** supine (lie down after meals for 20-30 minutes - to delay gastric emptying)
 - **Pharmacologic**
 - Medications: anti-cholinergic/antispasmodic - reduced motility
 - Vitamin B12: given IM monthly to prevent loss of intrinsic factor (produced by parietal cells)
 - Intrinsic factor helps absorb vitamin B12

Diverticular Disease

- A sac-like herniation of the lining that of the bowel that extends through a defect in the muscle layer
- Commonly occurred in the sigmoid colon (LLQ)
- **Types:**
 - **Diverticulosis** - multiple diverticula (outpouching) are present without inflammation or symptoms
 - Cause: low fiber intake = constipation
 - Diagnostic: colonoscopy
 - **Diverticulitis** - results when food and bacteria retained in a diverticulum produce infection; diverticulosis with inflammation; alternating diarrhea (active phase) and chronic constipation
 - Cause: increased in bacterial growth
 - Diagnostics: abdominal CT scan with contrast dye/medium - to detect peritonitis
- **Clinical manifestations**



- **Inflammation:**
 - Cramping abdominal pain at LLQ
- **Infection**
 - Fever, increased WBC
- **Injury** - possibility of rupture of diverticula
 - Hematochezia (fresh blood in the stool)
- **Obstruction** - increased in gas formation
 - Bloating or flatulence
- **Management**
 - **Diverticulosis:** constipation
 - Fiber diet: high
 - Fluid intake: high
 - Medication: laxatives/lactulose
 - **Diverticulitis:** constipation & active phase (diarrhea)
 - Fiber: high, low
 - Fluid: high, high
 - Monitor for: peritonitis (rigid/board-like abdomen), paralytic ileus (no bowel sound, movement, flatus)
 - Medication: laxatives, anti-diarrheal (loperamide)

Appendicitis

- Inflammation of appendix (RLQ)
- **Causes:**
 - Fecalith/fecal obstruction
- **Ruptured appendicitis:**
 - Increased peristalsis
 - increased intra-abdominal pressure
 - Vasodilation -> increased blood flow -> increased swelling & edema
- **Diagnostics:** ultrasound & CT scan
- **Clinical manifestations:**
 - **McBurney's sign/point:** pain $\frac{1}{3}$ from the iliac region
 - **Rovsing's sign:** palpation in the LLQ will cause pain in the RLQ
 - **Dunphy's sign:** pain triggered by coughing
 - **Blumberg's sign:** rebound tenderness
 - **Psoas sign** : pain on passive extension of the right thigh
 - **Obturator sign:** eliciting pain on passive internal rotation of the hip with the knee flexed
 - **WBC:** high
 - **Bowel sound:** decreased or absent (late sign)
 - Absent - ruptured -> paralytic ileus, no pain

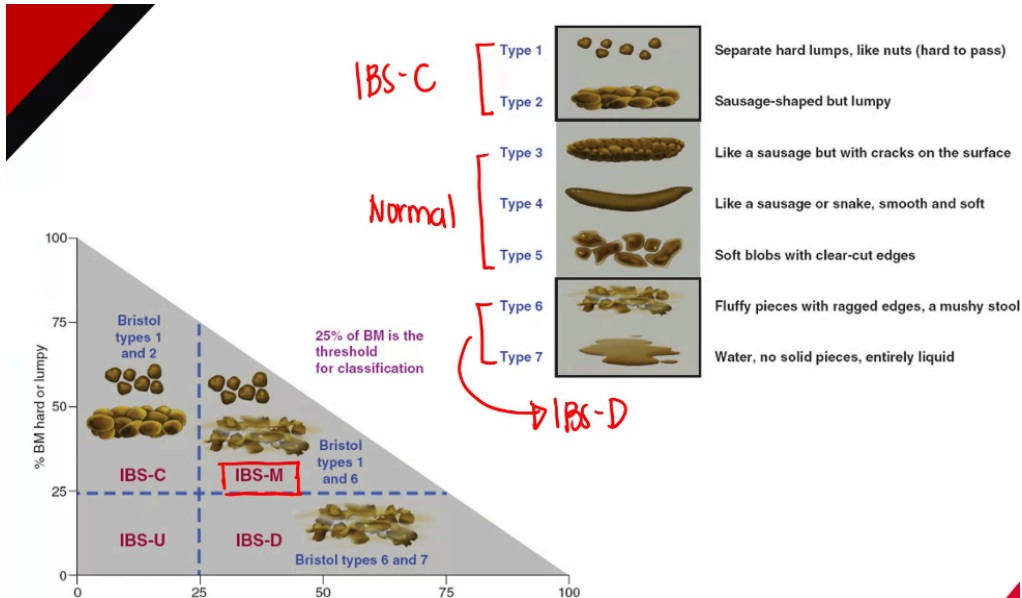


- **Fever:** present
- **Management**
 - **Suspected appendicitis**
 - (-) analgesics - it will mask the pain or rupture
 - Decreased GI activity (NPO, bed rest)
 - Avoid causes of rupture
 - (+) cold compress
 - (-) hot compress
 - Position: low/semi-fowler's - minimized movement -> decreased peristalsis
 - **Diagnosed appendicitis**
 - (+) analgesics
 - **Surgery: Appendectomy**
 - **Laparoscopic (closed)**
 - **Laparotomy (open)**
 - If there is an abscess: delay the surgery -> to avoid sepsis
 - Drain the abscess first -> continue the surgery
 - **Post-operative:**
 - Position: High fowlers - decreased tension incision -> decreased pain
 - Medication: morphine
 - Fluids: 1st 24 hours - IV
 - Foods: return of bowel movement/peristalsis (passage of flatus) & gag reflex (to prevent aspiration)
 - Ambulation: day of the surgery (1 day post op)

Irritable Bowel Syndrome (IBS)

- Functional disorder (organs are not functioning but during diagnostic tests all are normal)
- Cause: idiopathic
- **Factor:** stress -> visceral hypersensitivity -> increased pain in normal GIT activity (food intake, peristalsis, defecation)
- **Types:**
 - **IBS-C:** constipation
 - **IBS-D:** diarrhea
 - **IBS-M:** mixed
 - **IBS-U:** unidentified
- **Assessment:**
 - Abdominal pain related to/associated with:
 - Change in frequency and appearance (Bristol stool chart)
 - Defecation: before and during

- **Common complain:**
 - **Abdominal**
 - **Pain**
 - Triggered: food intake -> reristalsis (GIT activity)
 - Relieved defecation (no GIT activity)
 - **Distention**

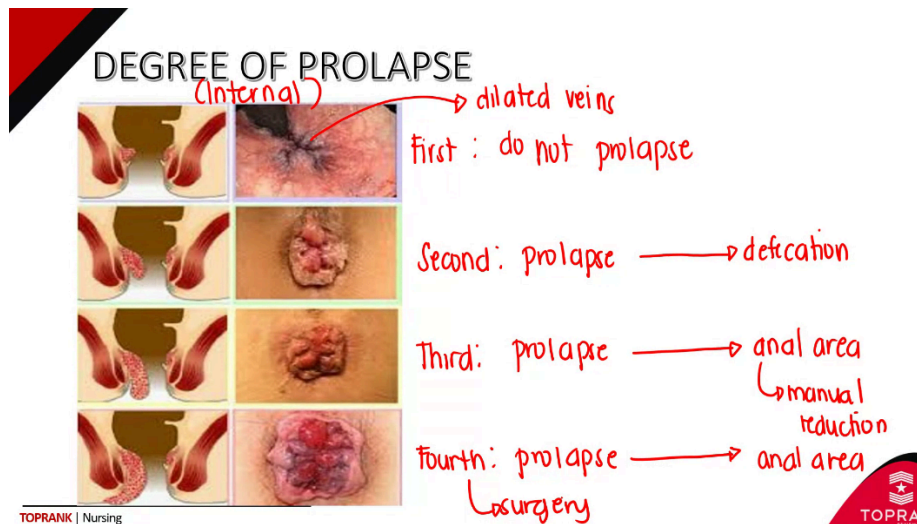


- **Management:**
 - Avoid stress
 - Diet:
 - Fiber: IBS- C - high; IBS-D: low
 - Food diary: to avoid triggers
 - Fluids: in between meals
 - **Low-FODMaP** (foods that can cause bloating/diarrhea and ferments in the colon causing gas formation/discomfort)
 - **Fermentable Oligosaccharide** (gas-forming)
 - **Disaccharides** (lactose/milk)
 - **Monosaccharides** (high fructose)
 - **Polyols** (stone fruits -> sugar alcohol -> poorly absorbed)
 - **Pharmacologic**
 - Antispasmodic: decreases motility
 - Dicyclomine
 - Peppermint oil
 - Diarrhea (IBS-D)
 - Aldosterone
 - Constipation (IBS-C)

- Lubiprostone
- Probiotics - decrease gas, diarrhea

Hemorrhoids

- Swollen, enlarged veins that form inside and outside the anus and rectum
- **Causes:**
 - Straining
 - Dilation of varicose veins (enlarged/swollen)
 - Portal hypertension
 - Increased intra-abdominal pressure (at risk pregnant women)
- **Pathophysiology:**
 - Increased venous pressure → congestion of blood → increased anorectal pressure
- **Two types:**
 - **External:** below anal sphincter, visible upon inspection
 - **Internal:** above anal sphincter



- **Manifestations**
 - Pain
 - Itching → swelling of BV → irritation of nerve endings → triggers itching sensation
 - Bleeding upon defecation (bright red)
 - External: thrombosis
 - → congestion of blood → stasis of blood → clots/thrombus
 - Internal: painless unless it prolapse
- **Management:**
 - Good personal hygiene
 - Avoid straining upon defecation (internal → causes prolapse)
 - Regular bowel movement



- Increased fiber
- increased fluids
- Ambulation/exercise
- Comfort/reduced engorgement
 - Sitz bath (after bath/defecation)
 - Cold compress (external hemorrhoids)
- **Pharmacologic**
 - Hydrophilic bulk-forming agents (stool softeners)
 - Psyllium husk
 - Analgesics ointment - topical
 - Lidocaine
 - Suppositories - prevent straining upon defecation
 - Astringents - dehydrates hemorrhoids
 - Witch hazel
- **Surgery**
 - **Sclerotherapy** - inject sclerosing agent (Phenol) to the dilated veins/tissue
 - shrink and hardening of hemorrhoids preventing prolapse
 - **Rubber band ligation** - application of rubber band in hemorrhoids -> decreases blood flow -> necrosis -> detaches itself
 - **Stapled Hemorrhoidopexy** - surgically staples prolapsed hemorrhoids
 - **Hemorrhoidectomy** - removal of hemorrhoids; indicated only if all of the conservative treatment are not working
 - **Post-operative intervention:**
 - Position: side lying/prone - decrease anal pressure
 - Ice packs: 1st 24 hours = vasoconstriction -> decreases bleeding
 - Sitz bath: temperature 31-41C, frequency: 3-4 times daily, duration: max of 20 mins to prevent burns
 - Avoid: prolonged sitting -> increased anal pressure
 - Stool softener: psyllium husk/docusate sodium (safe to pregnant women)

Hepatobiliary

Hepatobiliary nursing - focuses on the care and support of patients with disorders affecting the liver, gallbladder, bile ducts, pancreas and duodenum

Hepatobiliary System

- **Liver**



- Largest gland of the body
- Divided into four lobes
 - Right
 - Left
 - Caudate
 - Quadrate
- Contains several cell types including hepatocytes and Kupffer's cells
- Regulating blood glucose level by making glycogen which is stored in hepatocytes
- Converting ammonia produced from gluconeogenic by-products and bacteria to urea
- **Gall bladder**
 - Pear-shaped organ attached to the liver under the right lobe
 - Normally holds 30-50 ml of bile and can hold up to 70 ml when fully distended
- **Pancreas**
 - A slender, fish-shaped organ that lies horizontally in the abdomen behind the stomach and extends roughly from the duodenum to the spleen
 - **Endocrine and exocrine functions has pancreatic juice:**
 - Amylase
 - Lipase
 - Trypsin

Hepatobiliary Disorders

Hepatitis

- Inflammatory disorder of the liver parenchyma
- Occurring in Hepatitis A, B, C, D, E and toxic or drug-induced hepatitis
- Hepatocellular damage results from the body's immune response to the virus or toxin and is characterized by diffuse inflammatory infiltration with local necrosis
- **Clinical manifestations:**
 - **Pre-icteric stage**
 - Earliest symptoms are not specific
 - Flu-like symptoms
 - Malaise
 - Fatigue
 - Headache
 - Myalgias
 - Anorexia
 - Nausea and vomiting
 - Diarrhea



- **Icteric stage**
 - Few days to weeks after pre-icteric stage
 - Jaundice
 - Dark-colored urine
 - Light-colored stool
 - Steatorrhea
 - Enlarged liver
- **Post-icteric stage**
 - Convalescent stage lasting a few weeks
 - Fatigue decreases
 - Appetite returns

Viral Hepatitis	Mode of Transmission	Incubation	Outcome
Hepatitis A	Fecal-oral route	15-50 days	Usually mild with recovery
Hepatitis B	Parenterally; perinatal: sexual	28-160 days	May be severe
Hepatitis C	Blood transfusion: sexual transmission	15-160 days	Frequent occurrence of chronic hepatic cancer carrier state and chronic liver disease
Hepatitis D	Parenterally; perinatal: sexual	21-140 days	May be severe but greater likelihood of carrier state
Hepatitis E	Fecal-oral route	15-65 days	Usually mild with recovery except severe in pregnant women

- **Diagnostic tests:**
 - Liver function test results are elevated
 - Serum bilirubin level is increased
 - Urinalysis reveals increased bilirubin levels
- **Management:**
 - Administer prescribed medications which may include:
 - Immunoglobulins
 - Immunization
 - Antiviral
 - Prevent transmission of infection
 - Promote adequate rest without complications



- Encourage proper nutrition

Liver Cirrhosis

- Chronic liver disease marked by diffuse destruction and fibrotic regeneration of hepatic cells
- **Classification:**
 - **Laennec's cirrhosis**
 - Commonly caused by alcoholism and chronic nutritional deficiencies
 - **Biliary cirrhosis**
 - Caused by bile duct disorders that suppress bile flow
 - **Post-hepatic cirrhosis**
 - Caused by various types of hepatitis
- **Clinical manifestations:**
 - Enlarged, firm liver
 - Chronic dyspepsia
 - Constipation or diarrhea
 - Gradual weight loss
 - Ascites
 - Splenomegaly
 - Spider telangiectasis
 - Caput medusae
 - Dilated abdominal blood vessels
 - Portal hypertension
 - Mental deterioration
- **Laboratory and Diagnostic findings:**
 - Liver biopsy
 - Liver scan
 - Liver function test (ALT, AST)
 - Serum protein levels
 - Prothrombin time
- **Management:**
 - Administer diuretics to decrease ascites
 - Promote adequate nutrition (vitamins and nutritional supplements to promote healing of damaged liver cells)
 - Prevent threats to skin integrity
 - Minimize risk of bleeding
 - Antacid/H2 antagonist to minimize possibility of GI bleeding
 - Limit visitors and orient the client to date, time and place
 - Avoid drinking alcoholic beverages
 - Institute safety measures such as raising side rails and assisting ambulation



- **Diet:**
 - Early phase: high protein diet to promote healing of the liver
 - Late phase: low protein diet to decrease ammonia levels

Portal Hypertension

- Elevated pressure in the portal vein associated with increased resistance to blood flow through the portal venous system
- **Obstruction of portal venous flow through the liver lead to:**
 - Formation of esophageal and hemorrhoidal varicosities due to:
 - Increased venous pressure
 - Accumulation of fluid in the abdominal cavity
- **Clinical manifestations:**
 - Ascites
 - Rapid weight gain
 - Shortness of breathing
 - Fluid wave on abdominal percussion
 - Liver dullness
 - Dilated abdominal vessels radiating from the umbilicus (caput medusae)
 - Enlarged, palpable spleen
 - Fluid and electrolyte imbalance
- **Management:**
 - Bed rest
 - Administering medications which may include diuretics
 - Measure and record abdominal girth and body weight daily
 - Promote measures to prevent or reduce edema
 - Assist the health care provided with paracentesis
 - Monitor serum ammonia and electrolyte levels

Esophageal Varices

- Hemorrhagic process involving dilated, tortuous veins in the submucosa of the lower esophagus
- Caused by portal hypertension
- **Clinical manifestations:**
 - Hematemesis and melena
 - Massive hemorrhage occurs
 - Signs of hepatic encephalopathy
 - Dilated abdominal veins
 - Ascites
 - History of alcohol abuse



- **Diagnostic and laboratory tests:**
 - Endoscopy
 - AST, ALT, Bilirubin (increased)
 - Portal hypertension measurements
- **Management:**
 - Assess for ecchymosis, epistaxis, petechiae, and bleeding gums
 - Monitor level of consciousness, VS, urinary output to evaluate fluid balance
 - Monitor client during blood transfusion
 - Provide nursing care for client undergoing prescribed tamponade to control bleeding
 - **Sengstaken-Blakemore Tube**
 - **Four openings:**
 - Gastric aspirations
 - Esophageal aspiration
 - Gastric balloon inflation
 - Esophageal balloon inflation
 - The instrument at the bedside: scissors (cut the tube in case of respiratory distress)
 - The patient being treated with balloon tamponade must remain under close observation in the ICU because of the risk of serious take complications
 - Precautions must be taken to ensure that the patient not pull on or inadvertently displace the tube
 - **Vasopressin** - initial mode of therapy
 - **Sclerotherapy** - after treatment for acute hemorrhage, the patient must be observed for bleeding, perforation of the esophagus, aspiration pneumonia, and esophageal stricture
 - **Variceal band ligation**
 - A modified endoscope loaded with an elastic rubber band is passed through a band directly on the varix (or varices) to be banded
 - **Complications:**
 - Superficial ulceration
 - Dysphagia
 - Transient chest discomfort
 - Esophageal stricture

Hepatic Encephalopathy

- Neurologic syndrome that develops as a complication of liver disease
- It may be acute and self-limiting and progressing or chronic
- **Incidence is similar to cirrhosis due to:**
 - Severe liver damage



- Hepatocellular failure
- **Increased serum ammonia levels from:**
 - GI bleeding
 - High protein diet
 - Bacterial growth in the intestine
 - Uremia
- **Pathophysiology:**
 - Hepatic insufficiency -> inability to detoxify toxic by-product of metabolism (ammonia)
-> ammonia enters the brain -> excites peripheral benzodiazepine-type receptors on astrocyte cells -> stimulates GABA -> depression of CNS -> Encephalopathy
- **Clinical manifestations:**
 - Neurological dysfunction progressing from minor aberrations and motor disturbances to coma
 - **Flapping tremors/liver flap (asterixis)**
 - The patient is asked to hold the arm out with the hand held upward (dorsiflexed). Within few seconds, the hand falls forward involuntarily and then quickly returns to dorsiflexed position
 - **Fetos hepaticus**
 - A sweet, slightly fecal odor to the breath that is presumed to be of intestinal origin
 - **Constructional apraxia**
 - Deterioration of handwriting and inability to draw a simple star figure occurs with progressive hepatic encephalopathy
 - Serum ammonia level is elevated
 - Serum bilirubin level is elevated
 - Prothrombin time is prolonged
- **Management:**
 - Administer prescribed medications which may include laxatives (lactulose)
 - Ammonia is kept in the ionized state resulting in a decrease in colon pH
 - Evacuation of the bowel takes place which decreases the ammonia absorbed from the colon
 - The fecal flora are changed to organisms that do not produce ammonia from urea
 - Administer antibiotics (Neomycin)
 - Reduce levels of ammonia-forming bacteria in the colon
 - Closely monitor neurologic status for any changes
 - Evaluate serum ammonia values daily
 - Monitor signs of impending coma
 - Reduce or eliminate the client's dietary protein intake if you detect evidence of impending coma



- Monitor the client closely and administer a conservative dose of prescribed sedative or analgesic medication, because liver damage alters drug metabolism

Gallbladder Disorders

- **Risk factors:**
 - Obesity
 - Women especially who had multiple pregnancies
 - Native American or US Southwestern Hispanic ethnicity
 - Frequent changes on weight
 - Rapid weight loss
 - High dose estrogen
 - Ileal resection or disease
 - Cystic fibrosis
 - Diabetes mellitus
- **Diagnostic tests:**
 - **Abdominal X-ray**
 - **Ultrasonography** - diagnostic procedure of choice
 - **Cholescintigraphy** - radioactive agent is administered intravenously
 - **Cholecystography** - iodine containing contrast agent is administered before x-ray
 - **Endoscopic retrograde**
 - **Cholangiopancreatography (ERCP)**
 - Permits direct visualization of structures that previously could be seen only during laparotomy
 - A fiberoptic duodenoscope with side viewing apparatus is inserted into the duodenum. The ampulla of Vater is catheterized and the biliary tree is injected with contrast agent
- **Management:**
 - **Pharmacologic**
 - Ursodeoxycholic acid (UDCA, URSO, Actigal) - dissolve small radiolucent gallstone
 - Administer prescribed medication which may include analgesics (morphine sulfate) and antacids
 - **Nutritional therapy**
 - Low fat liquid
 - High in protein and carbohydrates
 - **Non-surgical approach**
 - **Intra-corporeal lithotripsy**
 - Stones in the gallbladder or common bile duct may be fragmented by means of laser pulse technology



- **Extra-corporeal shockwave lithotripsy**
 - Non- invasive procedure where repeated shockwaves are used directed at the gallstones in the gallbladder or common bile duct to fragment the stones
- **Surgical approach**
 - **Laparoscopic cholecystectomy**
 - Performed through a small incision or puncture made through the abdominal wall at the umbilicus
 - **Cholecystectomy**
 - Gallbladder is removed through an abdominal incision after the cystic duct and artery are ligated
 - A drain is placed close to the gallbladder if there is a bile leak removed after 24 hours
 - Bile duct injury serious complication
 - **Choledochostomy**
 - Making an incision in the common bile duct for removal of stones
- **Maintaining skin integrity and promoting biliary drainage**
 - If bile is not draining properly, an obstruction is probably causing the bile to be forced back into the liver or bloodstream
 - To prevent loss of bile, the physician may want the drainage tube or collection receptacle elevated above the level of the abdomen
 - Every 24 hours, the nurse measures the bile collected and records the amount, color and character of drainage
 - After several days of drainage, the tube may be clamped for 1 hour before and after each meal to deliver bile to the duodenum to aid in digestion
 - Within 7-14 days, the drainage tube is removed

Cholelithiasis

- Formation of calculi in the gallbladder
- Causes: result from the changes in bile components or bile stasis which may be associated with such factor such as infection, cirrhosis and pancreatitis
- **Clinical manifestations:**
 - Episodic, cramping pain in the RUQ of the abdomen or the epigastrium, possibly radiating to the back near the right scapular tip
 - Nausea and vomiting
 - Fat intolerance
 - Fever and leukocytosis
 - Jaundice
 - Epigastric distress



Cholecystitis

- Acute or chronic inflammation of the gallbladder
- **Causes:**
 - Obstruction of the cystic duct by impacted gallstone
 - Tissue damage due to trauma, massive burns, or surgery
 - Gram-negative septicemia
 - Overuse of opioid analgesics
- **Clinical manifestations:**
 - Biliary colic
 - Tenderness and rigidity in the RUQ elicited on palpation
 - Murphy's sign - pain on taking a deep breath when the examiner's fingers are on the approximate location of the gallbladder
 - Fever
 - Nausea and vomiting
 - Fat intolerance
 - Heartburn
 - Flatulence
 - Vitamin deficiency

Pancreatic Disorders

Acute Pancreatitis

- Self-digestion of the pancreas by its own proteolytic enzymes, principally trypsin
- Inflammation of the pancreas ranging from a relative mild, self-limiting disorder to rapidly fatal, acute hemorrhagic pancreatitis
- **Cause:**
 - Alcoholism
 - Cholecystitis
 - Surgery involving or near the pancreas
- **Clinical manifestations:**
 - Abdominal tenderness with back pain
 - GI problems such as nausea and vomiting, diarrhea, and steatorrhea
 - Fever
 - Jaundice
 - Mental confusion
 - Flank or umbilical bruising
 - Hypotension
 - Signs of hypovolemia



- Internal bleeding
 - Cullen's sign: bluish discoloration around the umbilicus
 - Turner's sign: bluish discoloration lateral of the trunk or posteriorly
- **Diagnostic tests:**
 - Elevated amylase
 - Lipase
 - Increase WBC levels
 - Hypocalcemia
- **Management:**
 - Administer prescribed medication which include opioid or non-opioid analgesics, histamine receptor antagonist, PPI
 - DOC for pain: morphine sulfate
 - The client should avoid oral intake to inhibit pancreatic stimulation and secretion of pancreatic enzymes
 - Maintain fluid and electrolyte balance
 - Promote adequate nutrition

Chronic Pancreatitis

- Progressive pancreatic inflammation resulting in permanent structural damage to pancreatic tissue
- Results from repeated episodes of acute pancreatitis
- More than half of chronic pancreatitis cases are associated with alcoholism
- Long term alcohol consumption causes hypersecretion of protein in pancreatic secretions, resulting in protein plugs and calculi within the pancreatic duct
- **Clinical manifestations:**
 - Recurring attacks of severe upper abdominal and back pain
 - Weight loss
 - Steatorrhea
 - Stools become frequent, frothy and foul-smelling because of impaired fat digestion which results in stools with high fat content
 - Anorexia
- **Assessment and Diagnostics**
 - Serum lipase and amylase elevated
 - WBC elevated
 - Endoscopic retrograde
 - Cholangiopancreatography - detects pancreatic calcification
 - Glucose tolerance test values are abnormal
- **Management:**
 - Administer prescribed medication which includes pancreatic enzymes
 - Non-opioid pain medications, antacids, H2 blockers, PPI



- Provide symptomatic treatment focusing on relieving pain, promoting comfort and treating new attacks
- Emphasize the importance of avoiding alcohol, caffeine, and foods that tend to cause abdominal discomfort
- Manage any endocrine insufficiency such as DM by initiating dietary and insulin or oral hypoglycemic therapy
- **Surgical intervention**
 - **Pancreatic jejunostomy (Roux-en-Y)**
 - Joining of the pancreatic duct to the jejunum
 - Allows drainage of the pancreatic duct to the jejunum
 - **A whipple resection (pancreaticoduodenectomy)**
 - Can be carried out to relieve the pain of chronic pancreatitis

Peritonitis

- Inflammation of the peritoneum, the serous membrane lining the abdominal cavity and covering the viscera
- **Cause:**
 - Bacterial infection
 - Injury or trauma
 - Inflammation that extends from an organ outside the peritoneal area
 - Appendicitis
 - Perforated ulcer
 - Diverticulitis
 - Bowel perforation
 - Abdominal surgical procedures
 - Peritoneal dialysis
- **Clinical manifestations:**
 - Diffuse pain, becomes constant localized and more intense on the site of maximal peritoneal irritation
 - Muscles become rigid and tender
 - Rebound tenderness
 - Paralytic ileus
 - Anorexia
 - Nausea and vomiting
 - Pyrexia
 - Increased PR
- **Diagnostic findings:**
 - Increased WBC
 - Altered levels of potassium, sodium and chloride



- Abdominal X-ray - distended bowel loops
- **Management:**
 - Fluid, colloid replacement
 - Analgesics as prescribed for pain
 - Antiemetics
 - Intestinal intubation and suction
 - Relieves abdominal distention and promotes intestinal function
 - Oxygen therapy by nasal cannula or mask
 - Antibiotic therapy
 - **Surgical intervention:**
 - Removing the infected area
 - Excision
 - Resection
 - Correcting the cause
 - Repair
 - Drainage
 - **Nursing intervention:**
 - Positioning the patient for comfort are helpful in decreasing pain
 - Patient is placed on the side with knees-flexed - decreases tension on the abdominal organs
 - Drains are frequently inserted during the surgical procedure
 - Prevent dislodging of the drain

NURSING PRACTICE V

Psychiatric Nursing

- Specialized field within nursing focused on the care of individuals with mental health disorders and substance abuse issues

Mental Health

- State of emotional, psychological, and social wellness as evidenced by satisfying interpersonal relationships, effective behavior and coping, positive self-concept and emotional stability

Factors affecting Mental Health

- (1) **Individual/Personal factor** - person's biological make up, autonomy, and independence, self-esteem, capacity for growth, vitality, ability to find meaning in life, emotional resilience or hardiness, sense of belonging, reality orientation, and coping or stress management abilities
- (2) **Interpersonal/relationship factor** - include effective communication, ability to help others, intimacy, and balance of separateness and connectedness



- (3) **Social/cultural/environmental factor** - include a sense of community, access to adequate resources, intolerance of violence, support of diversity among people, mastery of the environment and a positive yet realistic view of one's world

Mental Illness

- Includes disorders that affect mood, behavior, and thinking such as depression, schizophrenia, anxiety disorders and addictive disorders
- Causes significant distress, impaired functioning, or both

Diagnostic and Statistical Manual of Mental Disorders Fifth Edition (DSM-5)

- A taxonomy published by the **American Psychiatric Association**
- Describes all mental disorders, outlining specific diagnostic criteria for each based on clinical experience and research
- **Three purposes:**
 - To provide a standardized nomenclature and language for all mental health professionals
 - To present defining characteristics or symptoms that differentiate specific diagnoses
 - To assist in identifying the underlying causes of disorders

Therapeutic Communication

- An interpersonal interaction between the nurse and the client during which the nurse focuses on the client's specific needs to promote an effective exchange of information
- Helps the nurse understand and emphasize with the client's experience
- **Goals:**
 - Establish a therapeutic nurse-client relationship
 - Identify the most important client concern at that moment (client-centered goal)
 - Assess the client's perception of the problem as it unfolds
 - Facilitate the client's expression of emotions
 - Teach the client and family necessary self-care skills
 - Recognize the client's needs
 - Implement intervention designed to address the client's needs
 - Guide the client toward identifying a plan of action to a satisfying and socially accepted resolution
- **Communication:** exchange of information between 2 or more person
 - **Encoder:** sender
 - **Message:** content
 - **Decoder:** receiver
 - **Feedback:** response (most important)



- **Elements of Non-Verbal Communication**

- **Proxemics** - the study of distance zones between people during communication
 - **Intimate zone** (up to 18 inches) - comfortable for parents with young children, people who mutually desire personal contact, or people whispering
 - **Personal zone** (18 inches-4 ft) - comfortable between family and friends who are talking'
 - **Social zone** (9-12 feet) - acceptable for communication in social, work, and business settings
 - **Public zone** (>12 feet) - acceptable distance between a speaker and an audience, small groups and other informal functions
- **Kinetics**
 - Body language
 - Facial expressions, poise, posture, gait, movements
 - Reflects mood
- **Paralanguage**
 - Vocal cues/message delivery
- **Touch** - shows attempt to connect or relate
 - **Functional-professional touch** - used in examination or procedures
 - **Social-polite touch** - in greeting, such as handshake
 - **Friendship-warmth touch** - involves hug in greeting, an arm around thrown around the shoulder of a good friend
 - **Love-intimacy touch** - tight hugs, kisses between lovers or close relatives
 - **Sexual-arousal touch** - used by lovers
- **Silence** - encourage verbalization
- **Active listening and Observation**
 - **Active listening** - refraining from other internal mental activities and concentrating exclusively on what the client says
 - **Active observation** - watching the speaker's non-verbal actions as he or she communicates

Technique	Examples	Rationale
Clarifying - make ideas/feelings understandable	"I'm not sure I understand what you are trying to say"	To ensure accurate understanding and foster effective, patient-centered communication
Asking direct questions - closed ended	"Are you going to kill yourself?"	To obtain specific information and promote focused communication
Informing - health teaching	"Ill be your nurse for today,	To provide relevant information



	from 7:00 until 3:00 pm in the afternoon	and support the patient's understanding and decision-making
Accepting - indicate nonjudgemental acceptance of client's feelings	"Yes" "I follow what you said" nodding	An accepting response indicates the nurse has heard and followed the train of thought. It does not indicate agreement but is nonjudgmental. Facial expression, tone of voice, and so forth also must convey acceptance or the words lose their meaning.
Broad openings/Open-ended questions - allowing the client to take the initiative in introducing the topic	"Is there something you'd like to talk about?" "Where would you like to begin?"	Broad openings make explicit that the client has the lead in the interaction. For the client who is hesitant about talking, broad openings may stimulate him or her to take the initiative.
Consensual validation - searching for mutual understanding, for accord in the meaning of the words	"Tell me whether my understanding of it agrees with yours." "Are you using this word to convey that. . .?"	For verbal communication to be meaningful, it is essential that the words being used have the same meaning for both (all) participants. Sometimes, words, phrases, or slang terms have different meanings and can be easily misunderstood.
Encouraging comparison - asking that similarities and differences be noted	"Was it something like. . .?" "Have you had similar experiences?"	Comparing ideas, experiences, or relationships brings out many recurring themes. The client benefits from making these comparisons because he or she might recall past coping strategies that were effective or remember that he or she has survived a similar situation.
Encouraging description of perceptions - asking the client to verbalize what he or she perceives	"Tell me when you feel anxious." "What is happening?" "What does the voice seem to be saying?"	To understand the client, the nurse must see things from his or her perspective. Encouraging the client to describe ideas fully may relieve the tension the client



		is feeling, and he or she might be less likely to take action on ideas that are harmful or frightening.
Encouraging expression - asking the client to appraise the quality of his or her experiences	“What are your feelings in regard to. . .?” “Does this contribute to your distress?”	The nurse asks the client to consider people and events in light of his or her own values. Doing so encourages the client to make his or her own appraisal rather than to accept the opinion of others
Exploring - delving further into a subject or an idea	“Tell me more about that.” “Would you describe it more fully?” “What kind of work?”	When clients deal with topics superficially, exploring can help them examine the issue more fully. Any problem or concern can be better understood if explored in depth. If the client expresses an unwillingness to explore a subject, however, the nurse must respect his or her wishes.
Focusing - concentrating on a single point	“This point seems worth looking at more closely.” “Of all the concerns you’ve mentioned, which is most troublesome?”	The nurse encourages the client to concentrate his or her energies on a single point, which may prevent a multitude of factors or problems from overwhelming the client. It is also a useful technique when a client jumps from one topic to another.
Formulating a plan of action - asking the client to consider kinds of behavior likely to be appropriate in future situations	“What could you do to let your anger out harmlessly?” “Next time this comes up, what might you do to handle it?”	It may be helpful for the client to plan in advance what he or she might do in future similar situations. Making definite plans increases the likelihood that the client will cope more effectively in a similar situation.
General leads - giving encouragement to continue	“Go on.” “And then?”	General leads indicate that the nurse is listening and following what the client is saying without



		taking away the initiative for the interaction. They also encourage the client to continue if he or she is hesitant or uncomfortable about the topic.
Giving information - making available the facts that the client needs	"My name is. . . ." "Visiting hours are. . . ." "My purpose in being here is. . . ."	Informing the client of facts increases his or her knowledge about a topic or lets the client know what to expect. The nurse is functioning as a resource person. Giving information also builds trust with the client.
Acknowledging/Giving recognition - acknowledging, indicating awareness	"Good morning, Mr. S.. . ." "You've finished your list of things to do." "I notice that you've combed your hair."	Greeting the client by name, indicating awareness of change, or noting efforts the client has made all show that the nurse recognizes the client as a person, as an individual. Such recognition does not carry the notion of value, that is, of being "good" or "bad."
Making observations - verbalizing what the nurse perceives	what the nurse perceives "You appear tense." "Are you uncomfortable when. . . .?" "I notice that you're biting your lip."	Sometimes clients cannot verbalize or make themselves understood. Or the client may not be ready to talk.
Offering self - making oneself available	"I'll sit with you awhile." "I'll stay here with you." "I'm interested in what you think."	The nurse can offer his or her presence, interest, and desire to understand. It is important that this offer is unconditional; that is, the client does not have to respond verbally to get the nurse's attention.
Placing event in time or sequence - clarifying the relationship of events in time	"What seemed to lead up to. . . .?" "Was this before or after. . . .?" "When did this happen"	Putting events in proper sequence helps both the nurse and the client to see them in perspective. The client may gain insight into cause-and-effect behavior and consequences, or the client may be able to see that perhaps some things are not



		related. The nurse may gain information about recurrent patterns or themes in the client's behavior or relationships.
Presenting reality - offering for consideration that which is real	"I see no one else in the room." "That sound was a car backfiring." "Your mother is not here; I am a nurse."	When it is obvious that the client is misinterpreting reality, the nurse can indicate what is real. The nurse does this by calmly and quietly expressing his or her perceptions or the facts, not by way of arguing with the client or belittling his or her experience. The intent is to indicate an alternative line of thought for the client to consider, not to "convince" the client that he or she is wrong.
Reflecting - directing client actions, thoughts, and feelings back to client	Client: "Do you think I should tell the doctor. . .?" Nurse: "Do you think you should?" Client: "My brother spends all my money and then has the nerve to ask for more." Nurse: "This causes you to feel angry?"	Reflection encourages the client to recognize and accept his or her own feelings. The nurse indicates that the client's point of view has value and that the client has the right to have opinions, make decisions, and think independently.
Restating - repeating the main idea expressed	Client: "I can't sleep. I stayed awake all night." Nurse: "You have difficulty sleeping." Client: "I'm really mad, I'm really upset." Nurse: "You're really mad and upset."	The nurse repeats what the client has said in approximately or nearly the same words the client has used. This restatement lets the client know that he or she communicated the idea effectively. This encourages the client to continue. Or if the client has been misunderstood, he or she can clarify his or her thoughts.
Seeking information - seeking to make clear that which is not meaningful or that which is vague	"I'm not sure that I follow." "Have I heard you correctly?"	The nurse should seek clarification throughout interactions with clients. Doing so can help the nurse to avoid



		making assumptions that understanding has occurred when it has not. It helps the client to articulate thoughts, feelings, and ideas more clearly.
Silence - absence of verbal communication, which provides time for the client to put thoughts or feelings into words, to regain composure, or to continue talking	The nurse says nothing but continues to maintain eye contact and conveys interest.	Silence often encourages the client to verbalize, provided that it is interested and expectant. Silence gives the client time to organize thoughts, direct the topic of interaction, or focus on issues that are most important. Much nonverbal behavior takes place during silence, and the nurse needs to be aware of the client and his or her own nonverbal behavior.
Suggesting collaboration - offering to share, to strive, and to work with the client for his or her benefit	“Perhaps you and I can discuss and discover the triggers for your anxiety.” “Let’s go to your room, and I’ll help you find what you’re looking for.”	The nurse seeks to offer a relationship in which the client can identify problems in living with others, grow emotionally, and improve the ability to form satisfactory relationships. The nurse offers to do things with, rather than for, the client.
Summarizing - organizing and summing up that which has gone before	“Have I got this straight?” “You’ve said that. . .” “During the past hour, you and I have discussed. . .”	Summarization seeks to bring out the important points of the discussion and seeks to increase the awareness and understanding of both participants. It omits the irrelevant and organizes the pertinent aspects of the interaction. It allows both client and nurse to depart with the same ideas and provides a sense of closure at the completion of each discussion.
Translating into feelings - seeking to verbalize client’s feelings that he or she expresses only indirectly	Client: “I’m dead.” Nurse: “Are you suggesting that you feel lifeless?” Client: “I’m way out in the	Often what the client says, when taken literally, seems meaningless or far removed from reality. To understand, the



	ocean.” Nurse: “You seem to feel lonely or deserted.”	nurse must concentrate on what the client might be feeling to express himself or herself this way.
Verbalizing the implied - voicing what the client has hinted at or suggested	Client: “I can’t talk to you or anyone. It’s a waste of time.” Nurse: “Do you feel that no one understands?”	Putting into words what the client has implied or said indirectly tends to make the discussion less obscure. The nurse should be as direct as possible without being unfeelingly blunt or obtuse. The client may have difficulty communicating directly. The nurse should take care to express only what is fairly obvious; otherwise, the nurse may be jumping to conclusions or interpreting the client’s communication.
Voicing doubt - expressing uncertainty about the reality of the client’s perceptions	the client’s perceptions “Isn’t that unusual?” “Really?” “That’s hard to believe.”	Another means of responding to distortions of reality is to express doubt. Such expression permits the client to become aware that others do not necessarily perceive events in the same way or draw the same conclusions. This does not mean the client will alter his or her point of view, but at least the nurse will encourage the client to reconsider or reevaluate what has happened. The nurse neither agreed nor disagreed; however, he or she has not let the misperceptions and distortions pass without comment.
Supportive confrontation - acknowledge client’s feelings	“I know isn’t easy, but you can do it”	To gently challenge inconsistencies and promote self-awareness for growth



Non-Therapeutic Communication

Technique	Examples	Rationale
False Assurance - giving unwarranted or unrealistic comfort that may minimize the patient's concerns	"Don't worry, everything will be alright"	To avoid giving unrealistic hope that can diminish trust and hinder open communication
Judging - expressing approval or disapproval about a patient's thoughts, feelings or actions	"It's your own mistakes"	To avoid criticism or approval that can create defensiveness and block therapeutic communication
Advising - telling the client what to do	"I think you should. . .""Why don't you. . ."	Giving advice implies that only the nurse knows what is best for the client.
Agreeing - indicating accord with the client	"That's right." "I agree."	Approval indicates the client is "right" rather than "wrong." This gives the client the impression that he or she is "right" because of agreement with the nurse. Opinions and conclusions should be exclusively the client's. When the nurse agrees with the client, there is no opportunity for the client to change his or her mind without being "wrong."
Belittling feelings expressed - misjudging the degree of the client's discomfort	Client: "I have nothing to live for. . . I wish I was dead." Nurse: "Everybody gets down in the dumps," or "I've felt that way myself."	When the nurse tries to equate the intense and overwhelming feelings the client has expressed to "everybody" or to the nurse's own feelings, the nurse implies that the discomfort is temporary, mild, selflimiting, or not very important. The client is focused on his or her own worries and feelings; hearing the problems or feelings of others is not helpful.
Challenging - demanding proof from the client	"But how can you be president of the United States?" "If you're dead, why is your heart beating?"	Often, the nurse believes that if he or she can challenge the client to prove unrealistic ideas, the client will realize there is no



		“proof” and then will recognize reality. Actually, challenging causes the client to defend the delusions or misperceptions more strongly than before.
Defending - attempting to protect someone or something from verbal attack	“This hospital has a fine reputation.” “I’m sure your doctor has your best interests in mind.”	Defending what the client has criticized implies that he or she has no right to express impressions, opinions, or feelings. Telling the client that his or her criticism is unjust or unfounded does not change the client’s feelings but only serves to block further communication.
Disagreeing - opposing the client’s ideas	“That’s wrong.” “I definitely disagree with. . .” “I don’t believe that.”	Disagreeing implies the client is “wrong.” Consequently, the client feels defensive about his or her point of view or ideas.
Disapproving - denouncing the client’s behavior or ideas	“That’s bad.” “I’d rather you wouldn’t. . .”	Disapproval implies that the nurse has the right to pass judgment on the client’s thoughts or actions. It further implies that the client is expected to please the nurse.
Giving approval - sanctioning the client’s behavior or ideas	“That’s good.” “I’m glad that. . .”	Saying what the client thinks or feels is “good” implies that the opposite is “bad.” Approval, then, tends to limit the client’s freedom to think, speak, or act in a certain way. This can lead to the client’s acting in a particular way just to please the nurse.
Giving literal responses - responding to a figurative comment as though it were a statement of fact	Client: “They’re looking in my head with a television camera.” Nurse: “Try not to watch television” or “What channel?”	Often, the client is at a loss to describe his or her feelings, so such comments are the best he or she can do. Usually, it is helpful for the nurse to focus on the client’s feelings in response to such statements
Indicating the existence of an	source of thoughts, feelings, and	The nurse can ask, “What



external source - attributing the source of thoughts, feelings, and behavior to others or to outside influences	behavior to others or to outside influences “What makes you say that?” “What made you do that?” “Who told you that you were a prophet?”	happened?” or “What events led you to draw such a conclusion?” But to question, “What made you think that?” implies that the client was made or compelled to think in a certain way. Usually, the nurse does not intend to suggest that the source is external, but that is often what the client thinks.
Interpreting - asking to make conscious that which is unconscious; telling the client the meaning of his or her experience	“What you really mean is. . .” “Unconsciously you’re saying. . .”	The client’s thoughts and feelings are his or her own, not to be interpreted by the nurse for hidden meaning. Only the client can identify or confirm the presence of feelings.
Introducing an unrelated topic - changing the subject	Client: “I’d like to die.” Nurse: “Did you have visitors last evening?”	The nurse takes the initiative for the interaction away from the client. This usually happens because the nurse is uncomfortable, doesn’t know how to respond, or has a topic he or she would rather discuss. The client’s thoughts and feelings are his or her own, not to be interpreted by the nurse for hidden meaning. Only the client can identify or confirm the presence of feelings.
Making stereotyped comments - offering meaningless clichés or trite comments	“It’s for your own good.” “Keep your chin up.” “Just have a positive attitude and you’ll be better in no time.”	Social conversation contains many clichés and meaningless chitchat. Such comments are of no value in the nurse–client relationship. Any automatic responses lack the nurse’s consideration or thoughtfulness.
Probing - persistent questioning of the client	“Now tell me about this problem. You know I have to find out.” “Tell me your psychiatric history.”	Probing tends to make the client feel used or invaded. Clients have the right not to talk about issues or concerns if they choose. Pushing and probing by



		the nurse will not encourage the client to talk.
Reassuring - indicating there is no reason for anxiety or other feelings of discomfort	"I wouldn't worry about that." "Everything will be all right." "You're coming along just fine."	Attempts to dispel the client's anxiety by implying that there is not sufficient reason for concern completely devalue the client's feelings. Vague reassurances without accompanying facts are meaningless to the client.
Rejecting - refusing to consider or showing contempt for the client's ideas or behaviors	"Let's not discuss. . . ." "I don't want to hear about it. . . ."	When the nurse rejects any topic, he or she closes it off from exploration. In turn, the client may feel personally rejected along with his or her ideas.
Requesting an explanation - asking the client to provide reasons for thoughts, feelings, behaviors, events	"Why do you think that?" "Why do you feel that way?"	There is a difference between asking the client to describe what is occurring or has taken place and asking him to explain why. Usually, a "why" question is intimidating. In addition, the client is unlikely to know "why" and may become defensive trying to explain himself or herself.
Testing - appraising the client's degree of insight	"Do you know what kind of hospital this is?" "Do you still have the idea that. . .?"	These types of questions force the client to try to recognize his or her problems. The client's acknowledgment that he or she doesn't know these things may meet the nurse's needs but is not helpful for the client.
Using denial - refusing to admit that a problem exists	Client: "I'm nothing." Nurse: "Of course you're something —everybody's something." Client: "I'm dead." Nurse: "Don't be silly."	The nurse denies the client's feelings or the seriousness of the situation by dismissing his or her comments without attempting to discover the feelings or meaning behind them.
Ignoring - deliberately overlooking or failing to acknowledge a patient's	Patient: "I've been feeling anxious lately" Nurse: (looks away)	To avoid making the patient feel dismissed, which can damage trust and hinder communication



thoughts, feelings or concerns		
Passing the buck - shifting responsibility or avoiding the patient's concern	Patient: "I'm worried about my medication side effects, who can I talk about this?" Nurse: "Oh, you'll need to talk to the doctor about that"	It can lead the patient feel ignored, unimportant, or frustrated
Praise - causes dependency or pressure to meet expectations Criticize - expressing disapproval or judgement	"You did a great job" "You really shouldn't have skipped your medication yesterday"	To maintain genuine, unbiased communication without creating dependency and defensiveness
Why - can come across as judgemental or accusatory	"Why didn't you take your medications yesterday?"	To prevent sounding judgemental and encourage open non-defensive responses

Psychosocial Theories

Psychoanalytical Theory

- (1) Developed by **Sigmund Freud** (Father of Psychoanalysis)
- (2) Supports the notion that all human behavior is caused and can be explained
- (3) Personality components:
 - (a) **Id**: part of one's nature that reflects basic or innate desires such as pleasure-seeking, aggression and sexual impulse (starts at birth) "Devil"
 - (i) Pleasure principle
 - (ii) Unconscious
 - (iii) Sense of right and wrong (absent)
 - (iv) Internal desires
 - (b) **Superego**: a part of person's nature that reflects moral and ethical concepts, values, and parental and social expectations (starts at 3-6 year old) "Angel"
 - (i) Conscience
 - (ii) Unconscious
 - (iii) Morality
 - (c) **Ego**: represents mature and adaptive behaviour that allows a person to function successfully in the world (1-3 years old) "Self"
 - (i) Partly conscious and unconscious
 - (ii) Reality principle
 - (iii) Integrator of personality
 - (iv) Dynamic balancer



(v) Enables adaptive behavior

(4) Levels of awareness:

- (a) **Conscious:** refers to the perceptions, thoughts, and emotions that exist in the person's awareness
 - (b) **Preconscious:** thoughts and emotions are not currently in the person's awareness but he or she can recall them in some effort
 - (c) **Unconscious:** the realm of thoughts and feelings that motivates a person even though he/she is totally unaware of them
- (5) **Dream analysis** - a primary technique used in psychoanalysis involves discussing a client's dreams to discover their true meaning and significance
- (6) **Free association** - uncover the client's true thoughts and feelings by saying a word and asking the client to respond quickly with the first thing that comes to mind

Psychosexual Theory

Phase	Age	Focus
Oral (comfort)	0-1 year	Major site of tension and gratification is the mouth, lips, and tongue; includes biting and sucking activities Id is present at birth Ego develops gradually from rudimentary structure present at birth
Anal (control)	1-3 years	Anus and surrounding area are major source of interest Voluntary sphincter control (toilet training) is acquired
Phallic (Curiosity)	3-6 years	Genital is the focus of interest, stimulation, and excitement Penis is organ of interest for both sexes Masturbation is common Penny envy (wish to possess penis) is seen in girls Oedipal complex - for boys, involves desire for mother Electra complex - for girls, involves desire for father
Latency (Competition)	6-12 years old	Resolution of oedipal/electra complex Sexual drive channeled into socially accepted activities such as school work and sports Formation of superego



		Final stage of psychosexual development
Genital (Orgasm)	12-18 years old	Begins with puberty and the biologic capacity for orgasm; involves the capacity for true intimacy

Defense Mechanism

- Methods of attempting to protect the self and cope with basic drives or emotionally painful thoughts, feelings or events
- Protect ego and decrease anxiety

Defense Mechanism	Definition	Example
Compensation	Overachievement in one area to offset real or perceived deficiencies in another area	- Napoleon complex: diminutive man becoming emperor - Nurse with low self-esteem working double shifts so her supervisor will like her
Conversion	Expression of an emotional conflict through the development of a physical symptom, usually sensorimotor in nature	- Teenager forbidden to see X-rated movies is temPted to do so by friends and develops blindness, and the teenager is unconcerned about the loss of sight
Denial	Failure to acknowledge unbearable condition; failure to admit the reality of a situation or how one enables the problem to continue	- Diabetic person eating chocolate candy - Spending money freely when broke - Waiting 3 days to seek help for severe abdominal pain
Displacement	Ventilation of intense feelings toward persons less threatening than the one who aroused those feelings	- Person is mad at the boss yells at his/her spouse - Child who is harassed by a bully at school mistreats a younger sibling
Dissociation	Dealing with emotional conflict by a temporary alteration in consciousness or identity	- Amnesia that prevents recall of yesterday's auto accident - Adult remembers nothing of childhood sexual abuse



Fixation	Immobilization of a portion of a personality resulting from unsuccessful completion of tasks in a developmental stage	- Never learning to delay gratification - Lack of clear sense of identity as an adult
Idealization	Believing something is perfect	- My boyfriend is flawless despite being a thief
Identification	Modeling actions and opinions of influential others while searching for identity, or aspiring to reach a personal, social, or occupational goal (conscious copying)	- Nursing student becoming a critical care nurse because this is the specialty of an instructor she admires
Intellectualization	Separation of emotions of a painful event or situation from the facts involved; acknowledging facts but not the emotions	- Person shows no emotional expression when discussing serious car accident
Introjection	Accepting another person's attitudes, beliefs, and values as one's own (unconscious copying)	- Person dislikes guns becomes avid hunter, just like a best friend
Projection	Unconscious blaming of unacceptable inclinations or thoughts on an external object	- Man who has thought about same-sex gender sexual relationship but never had one beats a man who is gay - Person with many prejudices loudly identifies others as bigots
Rationalization	Excusing own behavior to avoid guilt, responsibility, conflict, anxiety, or loss of self-respect	- Student blames failure on teacher being mean - Man says he beats his wife because she doesn't listen to him
Reaction Formation	Acting the opposite of what one thinks or feels (plastic)	- Woman who never wanted children becomes a supermom - Person who despises the boss tells everyone what a great boss she is
Regression	Moving back to previous developmental stage to feel safe or have needs meet	- A 5 year old asks for a bottle when new baby brother is being fed



		- Man pouts like a 4 year old if he is not the center of his girlfriend's attention
Repression	Excluding emotionally painful or anxiety-provoking thoughts and feelings from conscious awareness (unconscious forgetting)	- Woman has no memory of the mugging she suffered yesterday - Woman has no memory before age 7, when she was removed from abusive parents
Resistance	Overt or covert antagonism toward remembering or processing anxiety-producing information	- Nurse is too busy with tasks to spend time talking to a dying patient - Person attends court-ordered treatment for alcoholism but refuses to participate
Splitting	Sees people as all good or all bad	- Idolizes someone when suddenly hates them
Sublimation	Substituting a socially acceptable activity for an impulse that is unacceptable	- Person who has quit smoking sucks on hard candy when the urge to smoke arises - Person goes for a 15 minute walk when tempted to eat junkfood
Substitution	Replacing the desired gratification with one that is more readily available	- Woman who would like to have her own children opens a day care center
Suppression	Conscious exclusion of unacceptable thoughts and feelings from conscious awareness (conscious forgetting)	- Student decides not to think about a parent's illness to study for a test - Woman tells a friend she cannot think about her son's death right now
Undoing	Exhibiting acceptable behavior to make up for or negate unacceptable behaviour	- Person who cheats on a spouse brings the spouse bouquet of roses - Man who is ruthless in business donates large amounts of money to charity



Transference - occurs when the client displaces onto the nurse attitudes and feelings that the client originally experienced in other relationships

Countertransference - occurs when the nurse displaces onto the client attitudes or feelings from his or her past

Psychosocial Theory

- **Erik Erikson** views psychosocial growth occurs in sequential phases and each stage is dependent on completion of the previous stage or life task

Stage	Virtue	Task
Trust vs. Mistrust (Infant) 0-1 years old	Hope	Viewing the world as safe and reliable; relationships as nurturing, stable and dependable
Autonomy vs. Shame and Doubt (Toddler) 1-3 years old	Will	Achieving a sense of control and free will
Initiative vs. Guilt (Preschool) 3-6 years old	Purpose	Beginning development of a conscience; learning to manage conflict and anxiety
Industry vs. Inferiority (School Age) 6-12 years old	Competence	Emerging confidence in own abilities; taking pleasure in accomplishments
Identity vs. Role Confusion (Adolescence) 12-18 years old	Fidelity	Formulating a sense of self and belonging
Intimacy vs. Isolation (Young Adult) 18-40 years old	Love	Forming adult, loving relationships and meaningful attachments to others
Generativity vs. Stagnation (Middle Adult) 40-65 years old	Care	Being creative and productive; establishing the next generation
Ego Integrity vs. Despair (Maturity) 65+ years old	Wisdom	Accepting responsibility for one's self and life

Interpersonal Theory

- **Harry Stack Sullivan** believed that one's personality involves more than individual characteristics, particularly how one interacts with others
- Developmental cognitive modes
 - **Prototaxic mode:** characteristic of infancy and childhood, involves brief, unconnected experiences that have no relationship to one another



- **Parataxic mode:** begins in early childhood as child begins to connect experiences in sequence
- **Syntactic mode:** which begins to appear in school-aged children and becomes more predominant in preadolescence, the person begins to perceive himself or herself and the world within the context of the environment and can analyze experiences in variety of settings
- **Therapeutic community/milieu** - the interaction among clients is seen as beneficial, and treatment emphasizes the role of this client-to-client interaction
- **Milieu therapy** - involved clients' interaction with one another, including practicing interpersonal relationship skills, giving one another feedback about behavior, and working cooperatively as a group to solve day-to-day problems

Stage	Ages	Focus
Infancy	Birth to onset of language	Primary need exists for bodily contact and tenderness Prototaxic mode dominates (no relation between experiences) Primary zone are oral and anal If needs are met, infant has sense of well-being; unmet needs lead to dread and anxiety
Childhood	Language to 5 years	Parents are viewed as source of praise and acceptance Shift to parataxic mode: experiences are connected in sequence to each other Primary zone is anal Gratification leads to positive self-esteem Moderate anxiety leads to uncertainty and insecurity; severe anxiety results in self-defeating patterns of behavior
Juvenile	5-8 years	Shift to the syntactic mode begins (thinking about self and others based on analysis of experiences in a variety of situations) Opportunities for approval and acceptance of others Learn to negotiate own needs Severe anxiety may result in a need to control or in restrictive, prejudicial attitudes
Preadolescence	8-12 years	Move to genuine intimacy with friend of the same sex Move away from family as source of satisfaction in relationships Major shift to syntactic mode occurs Capacity for attachment, love, and collaboration



		emerges or fails to develop
Adolescence	Puberty to Adulthood	Lust is added to interpersonal equation Need for special sharing relationship shifts to the opposite sex New opportunities for social experimentation lead to the consolidation of self-esteem or self-ridicule If the self-system is intact, areas of concern expand to include values, ideals, career decisions and social concerns

Therapeutic Nurse-Patient Relationships

- **Hildegard Peplau** developed four phases in this concept
 - (1) **Pre-Orientation**
 - (a) Preparing (self = nurse)
 - (b) Review chart/history
 - (c) Examine biases
 - (d) Planning first approach
 - (2) **Orientation**
 - (a) Orient rules/boundaries
 - (b) Purpose of relationship
 - (c) Establish trust/rapport
 - (d) Nurse-patient: identify goals
 - (3) **Working**
 - (a) Withdraw maladaptation
 - (b) Observe verbalization
 - (c) Resolve problems
 - (d) Keep client independence
 - (e) Longest phase
 - (4) **Termination**
 - (a) Reinforce strength
 - (b) Explore separation anxiety
 - (c) Termination of care
- **Roles of the Nurses in Therapeutic Relationship**
 - **Stranger:** offering the client the same acceptance and courtesy that the nurse would to any stranger
 - **Resource person:** providing specific answers to questions within a larger context
 - **Teacher:** helping the client to learn formally or informally
 - **Leader:** offering direction to the client or group



- **Surrogate:** serving as a substitute for another such as a parent or sibling
- **Counselor:** promoting experiences leading to health for the client such as expression of feelings

Behavioral Theories

1. **Classical conditioning by Ivan Pavlov:** behavior can be changed through conditioning with external or environmental conditions or stimuli
2. **Operant conditioning by B.F. Skinner:** people learn their behavior from their history or past experiences

Crisis

- A turning point in an individual's life that produces an overwhelming emotional response, when coping mechanism fails
- Duration: 4-6 weeks (self-limiting)
- Focus: **Gestalt therapy** (focuses on the here and the now)
- Goal: to help patient return to pre-crisis level
- **Stages of Crisis:**
 - The person is exposed to a stressor, experiences anxiety and tries to cope in a customary fashion
 - Anxiety increases when customary coping skills is ineffective
 - The person makes all possible efforts to deal with the stressor including attempts at new methods of coping
 - When coping attempts fail, the person experiences disequilibrium and significant distress
- **Three categories:**
 - **Maturational/developmental crisis:** predictable events in the normal course of life such as leaving home for the first time, getting married, having a baby, and beginning a career
 - **Situational crisis:** unanticipated or sudden events that threaten the individual's integrity such as the death of a loved one, loss of job, physical or emotional illness of a family member
 - **Adventitious/social crisis:** include natural disasters like floods, earthquakes or hurricanes, war, terrorist attacks, riots and violent crimes such as rape or murder
- **Crisis Intervention** - includes a variety of techniques based on the assessment of the individual
 - (1) **Directive intervention:** designed to assess the person's health status and promote problem-solving, such as offering the person new information, knowledge, or mean; raising the suggestions or courses of action



- (2) **Supportive intervention:** aim at dealing with the person's needs for empathetic understanding, such as encouraging the person to identify and discuss feelings, serving as sounding board for the person, affirming the person's self-worth

Mental Status Examination

- Description of all areas of the client's mental functioning as observed by the examiner at the moment

(1) Preexamination

(a) General Appearance

- (i) **Grooming and dress**
- (ii) **Hygiene**
- (iii) **Height and weight**
- (iv) **Level of eye contact**
- (v) **Hair and color texture**
- (vi) **Evidence of scars, tattoos or other distinguishing skin marks**
- (vii) **Evaluation of client's appearance** compared with chronological age, deterioration of appearance, client needs to be reminded

(b) General Mobility

(i) Posture

- 1) **Catatonia:** a group of symptoms that usually involve a lack of movement and communication
- 2) **Catatonic stupor:** the person cannot move, speak or respond to stimuli
- 3) **Catatonic rigidity:** characterized by immobile posture and cannot be easily moved
- 4) **Catatonic posturing:** spontaneous and active maintenance of a posture against gravity, mannerisms that are odd, circumstantial caricatures of normal actions
- 5) **Waxy flexibility/Cornea flexibilitas:** form of catalepsy with a high degree of suggestibility/ limbs might resist when you move them
- 6) **Catalepsy/pathologic limb rigidity:** a state marked loss of voluntary motion in which the limbs remain in whatever position they are placed
- 7) **Cataplexy:** sudden loss of muscle tone while a person is awake leads to weakness and a loss of voluntary muscle control

(ii) Gait patterns

- 1) **Any evidence of limping:** walking with difficulty
- 2) **Limitation of range of motion:** a joint or body part cannot move through its normal range of motion
- 3) **Ataxia:** the presence of abnormal, uncoordinated movements



- 4) **Shuffling:** walk by dragging one's feet along with or without lifting them fully from the ground

(c) Motor activity

- (i) **Normoactivity:** having a normal level of activity
- (ii) **Hyperactive:** abnormally or extremely active
- (iii) **Psychomotor retardation:** a slowing down or inhibition of mental and physical activity
- (iv) **Agitation:** the state of feeling irritated or restless
- (v) **Tremors:** an involuntary quivering movement
- (vi) **Tics:** sudden twitches, movements or sounds that people do repeatedly
- (vii) **Jerky or spastic movements:** quick rough motions or sudden starts and stops
- (viii) **Stereotypical movements:** rhythmic, repetitive, fixed, predictable, purposeful but purposeless movements that occur in children who are otherwise developing normally
- (ix) **Mannerisms and gestures:** habitual gesture or way of speaking or behaving
- (x) **Aggressiveness:** hostile or violent behavior
- (xi) **Echopraxia:** meaningless repetition or imitation of the movements of others
- (xii) **Bradykinesia:** slowness of movement
- (xiii) **Pacing:** walking back and forth
- (xiv) **Rocking:** moving gently to and fro or from side to side
- (xv) **Somnambulism:** sleep walking
- (xvi) **Anchoring:** dependency
- (xvii) **Anergia:** abnormal lack of energy
- (xviii) **Anhedonia:** inability to feel pleasure in normally pleasurable act
- (xix) **Regression:** return to former or less developed state
- (xx) **Compulsions:** unwanted repetitive patterns of mental acts that are intended to reduce anxiety

(d) Behavior/Nurse-Patient Interaction

- (i) **Cooperative (initially, all through out)**
- (ii) **Uninterested/apathetic**
- (iii) **Friendly**
- (iv) **Embarrassed**
- (v) **Seductive**
- (vi) **Impulsive**
- (vii) **Negativistic**
- (viii) **Indifferent**
- (ix) **Angry/hostile**
- (x) **Evasive**
- (xi) **Withdrawn**
- (xii) **Warm**



- (xiii) **Distant**
- (xiv) **Guarded/suspicious**
- (xv) **Dependent**
- (xvi) **Distracted**

(2) Stream of Talk

(a) Character

- (i) **Slowness or rapidity**
- (ii) **Intonation**
- (iii) **Volume**
- (iv) **Stuttering, hoarseness, slurring**
- (v) **Spontaneous:** occurring without apparent external cause
- (vi) **Blocking:** sudden halting of the flow of thought or speech
- (vii) **Deliberate:** being effortful in nature
- (viii) **Pressured:** uncontrollable, accelerated, excessive talking and difficult to interrupt
- (ix) **Aphasia:** a condition that robs you of the ability to communicate

(b) Organization of Talk/Form of Thought

- (i) **Relevant**
- (ii) **Irrelevant**
- (iii) **Incorrect**
- (iv) **Flight of ideas:** involves continuous, rapid speech that changes focus from moment to moment based on association, distractions, or play on words
- (v) **Loose association:** characterized by a lack of connection between ideas resulting in disorganized communication
- (vi) **Circumstantiality:** indirect speech in which the individual digresses to give unnecessary and often irrelevant details before arriving at the main point
- (vii) **Tangentiality:** a disturbance in the thought process that causes the individual to relate excessive or irrelevant details that results in never reaching the essential point or desired answer to a question
- (viii) **Neologism:** a new word or phrase of the patient's own making
- (ix) **Concrete thinking:** thinking focused on immediate experiences and specific objects or events
- (x) **Clang association:** grouping of words, usually rhyming words that are based on similar-sounding sounds, even though the words themselves don't have any logical reason to be grouped together
- (xi) **Word salad:** a jumble of extremely incoherent speech
- (xii) **Perseveration:** the repetition of a particular response regardless of the absence of cessation of a stimulus
- (xiii) **Echolalia:** repetition or echoing of words or sounds that you hear someone else say
- (xiv) **Mutism:** an inability or unwillingness to speak



- (xv) **Bradylalia:** abnormal slowness or hesitation in speech
- (xvi) **Poverty of speech:** speech problem that occurs when a person's speech is very minimal and typically must be prompted
- (xvii) **Glossolalia:** utterances that have a phonic similarity to human language but no intelligible semantic content
- (xviii) **Coprolalia:** spontaneous, unprovoked, and uncontrollable use of obscene or profane words and expressions
- (xix) **Verbigeration:** continual repetition of stereotyped phrases
- (xx) **Condensation:** fusion of multiple meanings, concepts, or emotions into one image or symbol
- (xxi) **Palilalia:** involves repetition of own words and phrases

(3) Emotional State and Reactions

- (a) **Mood** - a subjectively defined feeling state
 - (i) **Euthymic:** describe a relatively stable mood state
 - (ii) **Depression/despairing:** feeling of sadness and helplessness
 - (iii) **Euphoria:** feeling of intense excitement or happiness, stimulus comes from within
 - (iv) **Elation:** extreme feeling of happiness, external stimulus
 - (v) **Fearful**
 - (vi) **Irritable:** state of easily provoked, uncooperative, antagonistic
 - (vii) **Angry:** hostility, resentment and aggression
 - (viii) **Anxious:** feeling of tension, worried thoughts
 - (ix) **Guilty:** self-conscious emotion characterized by a painful appraisal of having done something that is wrong
 - (x) **Labile:** emotional instability
- (b) **Affect** - objectively defined, brief state of emotional tone attached to a specific idea, event or situation
 - (i) **Congruence with mood**
 - (ii) **Constricted/Blunted:** show some emotion but less than most people show
 - (iii) **Flat/Apathy:** lack of emotional response
 - (iv) **Appropriate**
 - (v) **Inappropriate:** disharmony between thought content and emotional response
 - (vi) **Ambivalent:** co-existence of 2 opposing drive towards one person, idea, object or situation
 - (vii) **Grief:** sadness suffering from a loss of love object
 - (viii) **Exaltation:** intense elation accompanied by grandeur
 - (ix) **Histrionic:** adaptation of exaggeration

(4) Thought Control/Processes

- (a) **Content of Thought**
 - (i) **Delusions:** fixed, false beliefs



- 1) **Persecutory:** believes that others intend to harm or plotting against them
 - 2) **Grandiose:** insist that they have special powers or importance
 - 3) **Ideas of reference:** believes that something in the environment has meaning intended for them
 - 4) **Thought broadcasting:** people are hearing his/her thoughts
 - 5) **Thought insertion:** putting thoughts in his/her mind
 - 6) **Thought withdrawal:** pulling out thoughts in his/her mind
 - 7) **Somatic:** believes something abnormal about bodily function or appearance
 - 8) **Nihilistic:** false idea that he/she is dead or a spirit
 - 9) **Erotomaniac:** believes that other individuals are in love with them
 - 10) **Infidelity/jealous:** believes partner is unfaithful without justification
 - 11) **Religious:** believes they have divine powers, receives messages from God or they are actually God
 - 12) **Delusion of poison:** believes that somebody is going to poison them
 - 13) **Self-accusatory:** false ideas of guilt
 - 14) **Delusions of unworthiness**
- (ii) **Suicidal Thoughts/Ideation**
- 1) **Attempt**
 - 2) **Threat**
 - 3) **Gestures**
- (iii) **Obsessions:** recurrent and persistent thoughts, impulses or images that cause distressing emotions such as anxiety or disgust
- (iv) **Magical thinking:** belief that one's idea, thoughts, actions, words can influence the course of events in the material world
- (v) **Phobia:** strong irrational fear of something
- (vi) **Poverty of content:** speech is adequate in quantity but too vague, repetitious, and lacking in content to be qualitatively adequate
- (b) **Perceptions:** the way events are interpreted through sensory stimulation
- (i) **Hallucinations:** false sensory impressions that have no external basis
- 1) **Hypnagogic:** manifested before going to sleep
 - 2) **Hypnopompic:** manifested before waking
 - 3) **Visual:** false perception of sight
 - 4) **Auditory:** false perception of sound
 - 5) **Tactile/somatic:** false perception of touch
 - 6) **Olfactory:** false perception of smell
 - 7) **Gustatory:** false perception of taste
 - 8) **Somatic/kinesthetic:** strange feeling on skin or other parts of the body
 - 9) **Micropsia (Lilliputian):** objects perceived reduced in size
 - 10) **Macropsia:** sees everything in larger size



- 11) **Trailing phenomenon:** moving objects are seen as a series of discrete discontinuous images
- (ii) **Illusions:** distortion of a real sensory stimulus
 - 1) **Visual**
 - 2) **Auditory**
 - 3) **Tactile**
 - 4) **Olfactory**
 - 5) **Gustatory**
- (iii) **Depersonalization:** feeling of estrangement of one's self
 - 1) **Derealization:** loss of reality of environment or situation
- (iv) **Preoccupation:** voluntary attention given to repetitive thoughts on particular ideas associated with affective importance
 - 1) **Ruminations:** involuntary attention given to intrusive, repetitive thoughts
 - 2) **Rituals:** compulsive behavior
 - 3) **Phobia:** intense fear
- (v) **Deja Vu:** illusion of recognition
 - 1) **Jamais Vu:** feeling of estrangement to familiar situation
- (c) **Impulse Control:** ability to control impulses
 - (i) **Aggression:** hostile or violent behavior
 - (ii) **Hostility:** unfriendly or hostile state
 - (iii) **Fear:** unpleasant emotion caused by the belief that someone or something is dangerous
 - (iv) **Guilt:** the fact of having committed a specified or implied offense or crime
 - (v) **Affection:** gentle feeling of fondness or liking
 - (vi) **Sexual feelings:** constellation of psychological sentiments that constitute desire for sexual satisfaction or release of sexual tension
- (5) **Neurovegetative Dysfunctions:** disturbance which are concomitants of psychological distress
 - (a) **Sleep**
 - (i) **Normal**
 - (ii) **Hypersomnia:** increased hours normally used for sleeping
 - (iii) **Insomnia:**
 - 1) **Early insomnia:** difficulty falling asleep and once asleep, oversleep in the morning
 - 2) **Late insomnia:** ability to go to sleep without the ability to remain asleep
 - 3) **Mixed type:** may have difficulty falling asleep and restlessness when sleeping and may have early or late insomnia
 - (iv) **Midnight Awakening (MNA)**
 - (v) **Early Morning Awakening (EMA)**
 - (vi) **Difficulty Falling Asleep (DFA)**



- (vii) **Interrupted**
- (b) **Appetite**
 - (i) **Poor/fair/good**
 - (ii) **Polyphagia**
 - (iii) **Voracious:** wanting or devouring great quantities of food
 - (iv) **Pica:** eats things not usually consumed food
 - (v) **Binge eating:** consumption of large quantities of food in a short period of time
 - (vi) **Coprophagia:** eating of feces
- (c) **Diurnal Variation:** the fluctuations that happen during the day and the variations in the day-night cycle term used to express the client's subjective description of his mood pattern during the day
- (d) **Weight**
- (e) **Libido:** sex drive
- (6) **General Sensorium and Intellectual Status:** describe impaired or unimpaired
 - (a) **Orientation:** able to describe events leading to his/her immediate situation (place, time, person, situation)
 - (b) **Memory**
 - (i) **Recent:** hours/weeks
 - (ii) **Remote:** months/years
 - (iii) **Immediate:** seconds/minutes
 - (iv) **Confabulation:** the filling of memory gaps and imaginary information in an attempt to distract from observing deficit
 - (v) **Agnosia:** unable to recognize and identify objects, persons, sounds
 - (vi) **Apraxia:** loss of ability to execute or carry out skilled movements and gestures despite having the desire and the physical ability to perform them
 - (vii) **Amnesia**
 - 1) **Hypermnnesia:** ability to recall vividly/readily
 - 2) **Anterograde:** inability to recall recent events
 - 3) **Retrograde:** inability to recall past events
 - 4) **Circumscribed:** for that particular event
 - (c) **Attention span:** the length of time for which a person is able to concentrate mentally on a particular activity
 - (d) **General information:** test client on current events
 - (e) **Abstract thinking:** ability to deal with concepts
 - (i) Similarities
 - (ii) Proverbs: ability to interpret proverbs is highly dependent on socioeconomic and cultural factors
 - (f) **Judgement/Reasoning:** ability to solve problems and make decisions, make plans for the future
- (7) **Insight:** patient's understanding of the immediate situation



Neurotransmitters

- **Dopamine** (excitatory-synapses/inhibitory-muscle cells/neuron cells)
- **Epinephrine/Norepinephrine** (excitatory)
- **Serotonin** (excitatory-synapses/inhibitory-muscle cells/neuron cells)
- **GABA** (Gamma-aminobutyric acid) - balances neurotransmitters

Cognitive Disorders

- (1) **Delirium:** a syndrome that involves a disturbance of consciousness accompanied by a change in cognition
- (2) **Dementia:** refers to a disease process marked by progressive cognitive impairment with no change in the level of consciousness
 - (a) **Aphasia:** deterioration of language function
 - (b) **Apraxia:** impaired ability to execute motor functions despite intact motor abilities
 - (c) **Agnosia:** inability to recognize or name objects despite intact sensory abilities
 - (d) **Disturbance in executive functioning:** ability to think abstractly and to plan, initiate, sequence, monitor, and stop complex behavior

Stages of Dementia

- (1) **Mild:** forgetfulness is the hallmark
- (2) **Moderate:** confusion is apparent, along with progressive memory loss
- (3) **Severe:** personality and emotional changes occur

Types of Dementia

- **Alzheimer's disease:** progressive brain disorder that has a gradual onset but causes in increasing decline in functioning, including loss of speech, loss of motor function, and profound personality and behavioral changes
 - Common signs and symptoms
 - **Apraxia:** no movement
 - **Agnosia:** cannot recognize
 - **Aphasia:** no speech
 - **Amnesia:** cannot remember
 - Medications
 - Donepezil
 - Rivastagmine
 - Amantadine
 - Galantamine
 - No open-ended question, use close-ended
 - Encourage proper diet: SFF
 - Assess for injury (Sundowning: afternoon and night)
 - Assess for severity



- **Stage 1 mild:** forgetfulness
- **Stage 2 moderate:** wandering, hallucination
- **Stage 3 severe:** bed wetting, bedridden
- **Lewy body dementia:** involves progressive cognitive impairment and extensive neuropsychiatric symptoms as well as motor symptoms
- **Vascular dementia:** similar to Alzheimer's but onset is abrupt, followed by rapid changes in functioning
- **Frontotemporal lobar degeneration (Pick's disease):** degenerative brain disease that particularly affects the frontal and temporal lobes and results in a clinical picture similar to that of Alzheimer's disease
- **Prion disease (Creutzfeldt-Jakob):** can trigger normal proteins in the brain to fold abnormally
- **HIV-associated dementia:** HIV infection can lead to dementia due to invasion of nervous tissue by HIV
- **Parkinson's disease:** slowly progressive neurologic condition characterized by tremor, rigidity, bradykinesia, and postural instability
- **Huntington's disease:** inherited, dominant gene disease that primarily involves cerebral atrophy, demyelination and enlargement of brain ventricles
- **Traumatic brain injury (TBI)** can cause dementia as a direct pathophysiologic consequence of head trauma

Indicator	Delirium	Dementia
Onset	Rapid	Gradual and insidious
Duration	Brief (hours to days)	Progressive deterioration
Level of Consciousness	Impaired, fluctuates	Not affected
Memory	Short-term memory impaired	Short-term then long-term memory impaired, eventually destroyed
Speech	May be slurred, rambling, pressured, irrelevant	Normal in early stage, progressive aphasia in later stage
Thought processes	Temporarily disorganized	Impaired thinking, eventual loss of thinking abilities
Perception	Visual or tactile hallucinations, delusions	Often absent, but can have paranoia, hallucinations, delusions
Mood	Anxious, fearful if hallucinating; weeping, irritable	Depressed and anxious in early stage, labile mood, restless pacing, angry outbursts in later stages



Psychological Disorders

Anxiety and Related Disorders

- **Anxiety** is defined as the initial response to a psychic threat (subjective)
 - **Mild anxiety (Good anxiety):** positive state of heightened awareness and sharpened senses, allowing the person to learn new behaviors and solve problems (lower extremities)
 - **Management:** DO NOTHING
 - **Moderate anxiety:** involves decreased perceptual field (focus on immediate task only); the person can learn new behavior or solve problems only with assistance (abdomen)
 - **Management:** Safety, Oral meds, Problem solving)
 - **Severe anxiety:** involves feelings of dread or terror. The person cannot be redirected to a task; he or she focuses only on scattered details and has physiological symptoms such as tachycardia, diaphoresis and chest pain (chest)
 - **Management:** Directives/Direct question, IM meds, Stay and safety
 - **Panic anxiety:** involve loss of rational thought, delusions, hallucinations and complete immobility and muteness (head)
 - **Management:** Stay (no touch), decrease stimuli, safe environment, restraint

Mild +1	Moderate +2	Severe +3	Panic +4
Widened senses Increased concentration Restlessness (stationary) Enhanced learning	Pacing back and forth Selective thinking Nausea Anorexia Vomiting Diarrhea (increased urination) Abdominal upset	Increased HR & BP Increased RR Confusion Alogia	Hallucinations Harming self or others Chest pain Syncope/Fainting

- (1) **Generalized Anxiety Disorder (GAD)** - worries excessively and feels highly anxious at least 50% of the time for 5 months or more
- (2) **Phobia** - an illogical, intense, and persistent fear of a specific object or social situation that causes extreme distress and interferes with normal functioning, fear of the known
 - (a) **Agoraphobia:** fear of open space/places
 - (b) **Acrophobia:** fear of heights
 - (c) **Astraphobia:** fear of electrical storm
 - (d) **Claustrophobia:** fear of closed spaces
 - (e) **Hydrophobia:** fear of water
 - (f) **Hematophobia:** fear of blood
 - (g) **Monophobia:** fear of being alone



- (h) **Mysophobia**: fear of dirt/germs
 - (i) **Nyctophobia**: fear of darkness
 - (j) **Pyrophobia**: fear of fire
 - (k) **Social phobia/anxiety** - the person becomes severely anxious to the point of panic or incapacitation when confronting situations involving people
 - (l) **Xenophobia**: fear of strangers
 - (m) **Zoophobia**: fear of animals
 - (n) **Ophidiophobia**: fear of snakes
 - **Defense mechanism**: displacement and avoidance
 - **Management**:
 - **Exposure therapy**
 - **Flooding**: sudden exposure
 - **Systemic desensitization**: gradual exposure
 - **Guided imagery** - concentrate on mental images to improve behaviour
 - **Medications**:
 - SSRI
 - Benzodiazepines
- (3) **Panic disorder** - composed of discrete episodes of panic attack that is 15-30 minutes of rapid, intense, escalating anxiety which the person experiences great emotional fear as well as physiological discomfort

Somatic Symptom Illness

- **Somatization** is defined as the transference of mental experiences and states into bodily symptoms
 - Can be characterized as the presence of physical symptoms that suggest a medical condition without demonstrable organic basis to account fully for them
 - **Central features**:
 - Physical complaints suggest major medical illness but have no demonstrable organic basis
 - Psychological factors and conflicts seem important in initiating, exacerbating, and maintaining the symptoms
 - Symptoms or magnified health concerns are not under the client's conscious control
- (1) **Somatoform disorder**: characterized by one or more physical symptoms that have no organic basis
- (a) **Conversion disorder**: involves unexplained, usually sudden deficits in sensory or motor function, imagined disability
 - (b) **Pain disorder**: unrelieved by analgesics and greatly affected by psychological factors in terms of onset, severity, exacerbation and maintenance



- (c) **Hypochondriasis/illness anxiety disorder**: preoccupation with the fear that one has a serious disease (disease conviction) or will get a serious disease (disease phobia), doctor hopping
- (d) **Body Dysmorphic Disorder**: imagined defect
- **With physical symptoms, no organic cause**
 - **Malingering** - faking an illness
 - **Factitious** - causing an illness to self
 - **Factitious disease by proxy** - causing an illness to others (Munchausen's syndrome)
- **Nursing management**
 - Rule out any possible organic or physiologic cause
 - Real for the patient
 - Recognize manipulation
 - Attend to physical complaints
 - Consistent caregiver must be provided
 - Encourage verbalization of feeling
- **Treatment**
 - Antidepressant - SSRIs
 - Chronic pain therapy - avoid narcotics analgesics
 - Cognitive-behavioral therapy - change the mindset of the patient, change the behavior

Obsessive-Compulsive and Related Disorders

- (1) **Obsessive-Compulsive Disorder (OCD)** - diagnosed only when these thoughts, images, and impulses consume the person or he/she is compelled to act out the behaviors to a point at which they interfere with personal, social and occupational function
 - **Obsessions**: recurrent, persistent, intrusive and unwanted thoughts, images, or impulses that marked anxiety and interfere with interpersonal, social, or occupational function
 - **Compulsions**: ritualistic or repetitive behaviors or mental acts that a person carries out continuously in an attempt to neutralize anxiety
- (2) **Excoriation Disorder/Dermatillomania** - categorized as a self-soothing behavior; that is, the behaviour is an attempt of people to soothe or comfort themselves (skin picking)
- (3) **Trichotillomania** - chronic, repetitive hair pulling, a self-soothing behavior that can cause distress and functional impairment
- (4) **Body Dysmorphic Disorder (BDD)** - preoccupation with an imagined or slight defect in physical appearance that causes significant distress for the individual and interferes with functioning in daily life
- (5) **Hoarding Disorder** - progressive, debilitating, compulsive disorder that involves excessive acquisition of animals or apparently useless things, cluttered living spaces that become uninhabitable and significant distress or impairment for the individual
- (6) **Onychophagia** - chronic nail biting



- (7) **Kleptomania** - compulsive stealing, thrill of stealing
- (8) **Oniomania** - compulsive buying

Post-Traumatic Stress Disorder (PTSD)

- A disturbing pattern of behavior demonstrated by someone who has experienced, witnessed or been confronted with a traumatic event such as natural disaster, combat or an assault (> 1 month)

Related Disorders

- **Adjustment disorder:** reaction to a stressful event that causes problems for the individuals
- **Acute stress disorder:** occurs after a traumatic event and is characterized by reexperiencing, avoidance, and hyperarousal that occur from 3 days to 4 weeks (< 1 month) following a trauma
- **Reactive attachment disorder (RAD):** exhibits minimal social and emotional response to others, lacks of positive affect and may be sad, irritable, or afraid for no apparent reason
- **Disinhibited social engagement disorder (DSED):** exhibits unselective socialization, allowing or tolerating social interaction with caregivers and strangers alike

Signs and Symptoms	Management	Psychotherapy
Severe panic attack Insomnia/appetite loss Nightmares/Flashbacks	Safety (Suicidal, self-harm, substance abuse) Self-blaming/guilt SSRI/Benzodiazepines	Talk therapy (Psychiatrist, Psychologist, Nurse)

Mood and Related Disorders

- Are also called **affective disorders**, are pervasive alterations in emotions that are manifested by depression, mania or both
 - (1) **Major Depressive Disorder** - experiences depressed mood or loss of pleasure in nearly all activities that lasts at least 2 weeks
 - (a) Everything is low & slow, it is thought to be from low levels of neurotransmitters within the brain
 - (i) Low serotonin
 - (ii) Low dopamine
 - (iii) Low norepinephrine
 - (b) **Signs of symptoms (5 or more)**
 - (i) Sleep disorders (insomnia/hypersomnia)
 - (ii) Interest deficit, anhedonia
 - (iii) Guilt (self-blaming)
 - (iv) Energy deficit
 - (v) Concentration deficit
 - (vi) Appetite disorder (gain/loss)



(vii) Psychomotor retardation (slowness of movement)

(viii) Suicidality

(c) Depressed Client

(i) Kind firmness

(ii) Silence

(iii) Offering self

(iv) Motivate - remind client of time when she/he felt better and was successful

(v) Engage in social activities

(d) **Diet:** small frequent feeding, high calorie

(e) **Catharsis:** releasing repressed emotions through art and music

(2) Bipolar Disorder - person's mood fluctuates to extremes of mania and/or depression

(a) **Mania:** a mood disorder marked by hyperactive wildly optimistic state

(b) **Depression:** the feeling of severe despondency and defection

(i) **Bipolar I:** one or more manic or mixed episodes usually accompanied by major depressive episodes

(ii) **Bipolar II:** one or more depressive episodes accompanied by at least one hypomanic episode

(c) Assessment

(i) Mood elevated

(ii) A grandiose delusion

(iii) No need for sleep or eat (fluid imbalance, imbalanced nutrition)

(iv) Inappropriate behavior

(v) Clanging, vulgar

(d) Nursing intervention (Bipolar Mania)

(i) Bakery foods (high-calorie, finger foods)

(ii) Accept client, not behavior

(iii) Keep care consistent

(iv) Establish limits

(v) Move: redirect energy (walk, dance) - avoid contact/competitive

(vi) Encourage rest (naps, PRN sedation) - haloperidol

	Bipolar 1	Bipolar 2	Manic Disorder	Major Depressive Disorder	Cyclothymia (Bipolar-like disorder) 2-3 years	Dysthymia (Minor depression 2-3 years)
Mania (psychosis)	/	X	/	X	X	X
Hypomania	/	/	/	X	/	X



Normal						
Hypo Depression (All down)	/	/	X	/	/	/
Major Depression (Suicidal)	/	/	X	/	X	X

Related Disorders

- **Persistent Depressive (Dysthymic) Disorder:** chronic, persistent mood disturbance characterized by symptoms such as insomnia, loss of appetite, feeling of hopelessness that are milder than those of depression
- **Disruptive mood dysregulation disorder:** persistent angry or irritable mood, punctuated by severe recurrent temper outbursts that are not in keeping with the provocation or situation, beginning before age 10 years
- **Cyclothymic disorder:** mild mood swings between hypomania and depression without loss of social or occupational functioning
- **Substance-induced depressive or bipolar disorder:** characterized by significant disturbance in mood that is a direct physiological consequence of ingested substances such as alcohol, other drugs or toxins
- **Seasonal affective disorders (SAD)**
 - (vii) Winter depression
 - (viii) Spring-onset
- **Postpartum or maternity blues:** mild, predictable mood disturbance occurring in the first several days after delivery of baby
- **Postpartum depression:** symptoms are consistent with those of depression with onset within 4 weeks of delivery
- **Postpartum psychosis:** severe and debilitating psychiatric illness with acute onset in the days following childbirth
- **Premenstrual dysphoric disorder:** severe form of premenstrual syndrome, and is defined as recurrent moderate psychological and physical symptoms that occur during the week before menses and resolve with menstruation
- **No suicidal self-injury:** deliberate, intentional cutting, burning, scraping, hitting or interference with wound healing

Schizophrenia and Related Disorders

- Causes distorted and bizarre thoughts, perceptions, emotions, movements, and behavior



- A long-term mental disorder involving a deteriorating breakdown in the relation between thought, emotion and behaviour
- The earlier the onset, the worse the prognosis
- **Possible causes:**
 - **Genetics** (1 parent - 15% risk, 2 parents - 35% risk, monozygotic twin - 50%)
 - **Immunovirologic** (encephalitis, meningitis)
 - **Neuroanatomic/neurochemical** (increased dopamine - synapse, decreased CSF-TBI)
 - **Substance abuse** (long-term use of substances)
- **Movement Disturbances**
 - **Posturing:** striking or weird pose to show off confuse others
 - **Apraxia:** no movement
 - **Automatism:** repeated purposeless behavior
 - **Waxy flexibility:** maintenance of awkward posture
 - **Echopraxia:** repetition of someone else's action
 - **Rigidity:** stiffness or inflexibility
- **Mood/Affect Disturbances**
 - **Apathy:** lack of interest/emotions
 - **Blunt affect:** decreased ability to express emotion
 - **Ambivalence:** two opposing emotions
 - **Flat speech:** no emotional tone when speaking
 - **Inappropriate speech:** emotional responses that are not compatible
 - **Melancholia:** deep sadness
 - **Alexithymia:** inability to recognize or describe one's own emotions
 - **Labile:** changeable mood
 - **Euphoria:** extreme happiness
- **DisorganizedDisturbed Speech**
 - **Loose association:** rapid shifts of thought with no logical connection
 - **Flight of ideas:** rapid shifts of thought with logical connection
 - **Neologisms:** making up imaginary words
 - **Clang association:** listing rhyming words other that makes no sense
 - **Word salad:** mixing words together that have no meaning except to the client
 - **Concrete thinking:** taking a statement literally
 - **Stilted language:** use of flowery words
 - **Verbigeration:** repeating phrase
 - **Perseveration:** adherence to a single topic
- **Disorganized Thought**
 - **Illusions:** false perception with stimuli
 - **Hallucination:** false sensory perception without stimuli
 - **Delusion:** the false beliefs that is inconsistent with one's knowledge and culture
 - **Jealous Delusion:** belief that partner is unfaithful without proof



- **Grandeur Delusion:** belief in being famous, powerful or important
- **Reference Delusion:** belief that random things (TV, music) are about them
- **Age (Dorian Gray) Delusion:** belief that one doesn't age while others do
- **Capgras' Syndrome:** belief a loved one is replaced by an impostor
- **Erotomaniac Delusion:** belief that famous or unattainable person is in love with them
- **Divine (Religious) Delusion:** belief in being a prophet or divinely chosen
- **Persecutory Delusion:** belief others are trying to harm, spy on, or poison them
- **Control Delusion:** belief that external forces control thoughts or actions (mind control delusion)
- Signs and symptoms are divided into 2 categories:
 - **Positive/Hard signs and symptoms** - clear symptoms visibly clear
 - **Delusions:** fixed false beliefs that have no basis in reality
 - **Hallucinations:** false sensory perceptions or perceptual experiences that do not exist in reality
 - **Auditory hallucination** - most dangerous due to commands
 - **Ideas of reference:** false impressions that external events have special meaning for the person
 - **Perseveration:** persistent adherence to a single idea or topic; verbal repetition of a sentence, word or phrase; resisting attempts to change the topic
 - **Bizarre behavior:** outlandish appearance or clothing; repetitive or stereotyped, seemingly purposeless movements; unusual social or sexual behavior
 - **Ambivalence:** holding seemingly contradictory or feelings about the same person, event or situation
 - **Associative looseness:** fragmented or poorly related thoughts and ideas
 - **Echopraxia:** imitation of the movements and gestures of another person whom the client is observing
 - **Flight of ideas:** continuous flow of verbalization in which the person jumps rapidly from one topic to another
 - **Negative/Soft signs and symptoms** - non-active symptoms
 - **Alogia:** tendency to speak little or to convey little substance of meaning (poverty of content)
 - **Anhedonia:** feeling no joy or pleasure from life or any activities or relationships
 - **Apathy:** feelings of indifference toward people, activities, and events
 - **Asociality:** social withdrawal, few or no relationships, lack of closeness
 - **Blunted affect:** restricted range of emotional feeling, tone or mood
 - **Catatonia:** psychologically induced immobility occasionally marked by periods of agitation or excitement; the client seems motionless as if in trance
 - **Flat affect:** absence of any facial expression that would indicate emotions or mood



- **Avolition:** absence of will, ambition, or drive to take action or accomplish tasks
- **Inattention:** inability to concentrate or focus on a topic or activity, regardless of importance
- **Management**
 - Hallucination must be recognized (“It must be frightening that you are hearing those voices”)
 - Assess the content (“What are the voices telling you”)
 - Reality presentation (“I don’t hear any voices”)
 - Divert the attention (“Come let’s walk to the park”)
 - Engage in reality-based activity
 - Reintegrate with the milieu (Therapeutic milieu) talk back to the voices
- **Synesthesia:** mixing of senses, hear the color, sees the sound, tastes the words
- **Delusions:** false beliefs
 - **Management:**
 - Clarify the meaning (“I’m not sure I understand what you are saying”)
 - Acknowledge the feeling (“You seem very upset today”)
 - Voice doubt (“I find that hard to believe”)
 - Engage in reality-based activities

Types of Schizophrenia	
Paranoid	Suspicious of others, delusions or grandeur, hallucinations (offer sealed foods)
Disorganized	Worst prognosis, often messy and hard to communicate (ADLs)
Catatonic	Withdrawn, isolation, assumes weird body positions
Schizoaffective	Schizophrenia + mood swings (like depression or mania) - risk for suicide
Residual	Low level or no positive symptoms, manifesting negative symptoms
Undifferentiated	Does not fit into any 5 of the categories

Related Disorders

- **Schizophreniform disorder:** acute, reactive psychosis for less than 6months
- **Catatonia:** marked psychomotor disturbance, either excessive motor activity or virtual immobility or motionlessness
- **Delusional disorder:** one or more bizarre delusions
- **Brief psychotic disorder:** sudden onset of at least one psychotic symptoms such as delusion, hallucination, or disorganized speech or behavior which lasts from 1 day to 1 month
- **Shared psychotic disorder (folie deux):** two people share a similar delusion



- **Schizotypal personality disorder:** odd, eccentric behaviors including transient psychotic symptoms

Eating and Related Disorders

- (1) **Anorexia Nervosa:** a life-threatening eating disorder characterized by the client's restriction of nutritional intakes necessary to maintain a minimally normal body weight, intense fear of gaining weight or becoming fat, significantly disturbed perception of the shape or size of the body, and steadfast inability or refusal to acknowledge the seriousness of the problem or even that one exists
 - Problem with the hypothalamus
 - Fear of obesity
 - (i) **Binge eating:** consuming large amount of food in a discrete period of usually 2 hours or less
 - (ii) **Purging:** involves compensatory behaviors designed to eliminate food by means of self-induced vomiting or misuse of laxatives, enemas and diuretics
- **Assessment:**
 - Refusal to eat/drink
 - Excessive exercise
 - Perfectionist
 - Underweight/ 15% or less than IBW
 - Signs of malnutrition (dehydration, electrolyte imbalance, bony prominence)
 - Ensure safety (suicidal) - slow suicide
- (2) **Bulimia Nervosa:** characterized by recurrent episodes of binge eating followed by inappropriate compensatory behaviors to avoid weight gain, such as purging, fasting, or excessively exercising
- **Assessment:**
 - Hoarseness of voice (frequent purging)
 - Enlarged parotid gland (saliva production)
 - Average weight
 - Russel's sign - calluses on knuckles
 - Toothache - dental caries (halitosis)
 - Metabolic acidosis and alkalosis
 - Enemas, laxatives, diuretics, and diet pills

Anorexia Nervosa	Bulimia Nervosa
- Stay with the patient (during) - Recognize manipulation - Force feeding (TPN, NGT) - Medications: TCA (Tofranil) , SSRI (Prozac) - Side effect: increased appetite, weight	- Stay with the patient (1-2 hours after meal) - Limit BRP (accompany to the restroom) - Medications: TCA (Tofranil), SSRI (Prozac) - Side effect: decreased purging



gain	
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Related Disorders

- **Binge eating disorder:** characterized by recurrent episodes of binge eating; no regular use of inappropriate compensatory behaviors such as purging or excessive exercise or abuse of laxatives; guilt, shame and disgust about eating behaviors; and marked psychological distress
- **Night eating syndrome:** characterized by morning anorexia, evening hyperphagia and nighttime awakenings to consume snacks
- **Pica:** persistent ingestion of non-food substances
- **Rumination:** or repeated regurgitation of food that is then rechewed, reswallowed or spit out

Personality Disorders

- **Personality** is defined as an ingrained, enduring pattern of behaving and relating to self, other and the environment, it includes perception, attitudes and emotions
- **3D of Personality Disorder**
 - Different from cultural norm (attitude/behaviors)
 - Distress or dysfunction (relationships)
 - Duration is long-term (unaware)
- Categories:
 - **Cluster A:**
 - Odd and eccentric behavior
 - Includes paranoid, schizoid, schizotypal personality
 - **Cluster B:**
 - Includes people appear dramatic, emotional, or erratic
 - Includes antisocial, borderline, histrionic, and narcissistic personality disorder
 - **Cluster C:**
 - Includes people who appear anxious or fearful
 - Includes avoidant, dependent, and obsessive-compulsive personality disorder

Narcissistic Personality Disorder - Need admiration - Arrogant/entitled - Really believes perfect - Controlling - Superman complex	Paranoid Personality Disorder - Suspicious/Mistrust - Intense/controlling - Lonely - Overjealous - Sensitive
Management: self-limits	Management: safety/set-limits
Histrionic Personality Disorder - Attention seeker, seductive	Dependent Personality Disorder - Wife battered syndrome



- Gratification always needed = increased self-esteem - Hyperemotional - Hysterical Management: teach social skills	- Indecisive and good follower - Fears separation (loss of function) - Enabler and co-dependent Management: develop self-reliance
Borderline Personality Disorder - Will use manipulation - Abandonment issues - Gaslighting and suicidal - Extortion (black mail) - Splitting Management: watch out for self-harm, set boundaries/limits, assign another nurse	Antisocial Personality Disorder - Good talker and charming (bad boy effect) - Manipulates others for personal gain - Aggressive (sexually - sadism, masochism) - Impulsive - Law/Rule breakers (criminal) Management: set-limits
Avoidant Personality Disorder - Avoid people - Timid/shy - Inferiority complex - Sensitive to rejection/criticism Management: develop self-esteem	Schizoid Personality Disorder - No best friend - Attach to computer and pets - I don't like people - Loner - Stands on his own Management: improve client-community relationship
Schizotypal Personality Disorder - Alone - Has special powers (superstitious belief) - Withdrawn and weird - Often tells magical stories (horror) - Overly detached Management: teach social skills	Obsessive-Compulsive Personality Disorder - Unaware - Obsession (intrusive thoughts) - Compulsion (repetitive rituals) Management: - Permit rituals at first - Allow redirection to useful tasks - Do set limits slowly - Socialize gradually (Group therapy)

Dissociative Disorders

- Occurs 2 or more identities rotate control over the client's behavior, multiple personality disorder
- Cause: PTSD (sexual/trauma)
- **Dissociative Amnesia:** sudden memory loss usually after stress or trauma



- **Fugue (Dissociative Fugue):** sudden travel or wandering with amnesia
- Pathophysiology
 - **Altered Identity (DID):** two or more personalities
 - **Depersonalization disorder:** feeling detached from self (like watching yourself from the outside)
- **Nursing management:**
 - Stay with the patient
 - Talk gently, don't force memories
 - Avoid flooding with information
 - Gather data and observe behaviors
 - Explore triggers and stressors
 - Support coping strategies

Substance Abuse Disorders

- **Substance abuse** can be defined as using a drug in a way that is inconsistent with medical or social norms and despite negative consequences
- **Substance dependence** includes problems associated with addiction such as tolerance, withdrawal and unsuccessful attempts to stop using the substance
- **Intoxication** is use of a substance that results in maladaptive behavior (overdose)
- **Withdrawal syndrome** refers to the negative psychologic and physical reactions that occurs when use of substance ceases or dramatically decreases (abrupt stop)
- **Detoxification** is the process of safely withdrawing from a substance
- **Downers:** decreased VS, LOC, anxiety, tremors, coma
 - Alcohol
 - Marijuana
 - Barbiturates
 - Opiates
 - Narcotics
- **Uppers:** increased VS, LOC, anxiety, tremors, seizures
 - Cocaine
 - Hallucinogens
 - Amphetamines

Common substances

- (1) **Alcohol:** CNS depressant that is absorbed rapidly into the blood
 - (a) Etiology: intergenerational transmission (from one generation to another generation)
 - (i) **Alcohol** -> **Blackout** (awake but unaware) -> **confabulation** (inventing stories to increase self-esteem)-> **denial** ("I am not an alcoholic")-> **dependence** ("I can't live without it") -> **enabling** (significant other tolerates abusers/co-dependency) -> **increased tolerance** (substance to achieve a previous effect)



(ii) **Wernicke-Korsakoff's syndrome** (CNS problem)

1) Cause: alcoholism - causes thiamine deficiency

Wernicke's	Korsakoff's
- Acute - Short term - Reversible - Ataxia - Confusion - Opthalmoplegia	- Chronic - Long term - Irreversible - Confabulation - Hallucination - Amnesia

- **Managements:** Vitamin B complex, nuts, green peas, pork, milk (increases thiamine)
 - **Disulfiram (Antabuse)** - Aversion therapy
 - **Avoid:** vanilla, vinegar, aftershave lotion, mouthwash, polish remover, backrub ointment, cologne, isopropyl alcohol
 - **Disulfiram reaction** (side effects)
 - Diarrhea
 - Intense headache
 - Nausea & vomiting
 - Abdominal cramps
 - **4 stages of alcohol withdrawal**
 - **Early/Initial:** fine tremors, restlessness, tachycardia, diaphoresis, hyperventilation, and nervousness
- (2) **Sedatives/hypnotics/anxiolytics:** Barbiturates
- Purpose:** to cause sedation/anesthesia
 - Commonly abused barbiturates:** barbital (phenobarbital, methohexital, thiopental)
 - Signs of abuse:** hypotension, bradycardia, bradypnea, pupil constriction
 - Route:** oral & IM/IV
 - Signs of withdrawal:** increased VS, anxiety
 - Management of overdose:** activated charcoal
- (3) **Stimulants** (Amphetamines, Cocaine): stimulates CNS
- Purpose:** to cause euphoria
 - Signs of abuse:** hypertension, tachycardia, tachypnea, insomnia, weight loss, pupil dilation
 - Route:** nasal (cocaine), amphetamine (oral) - sudden stained teeth
 - Signs of withdrawal:** decreased VS, depression
 - Commonly abused stimulants:**
 - Cocaine** (nasal excoriation)
 - Sign of abuse:** bipolar cycling
- (4) **Opioids:** desensitize the user to both physiologic and psychologic pain and induce a sense of euphoria and well-being



- (5) **Hallucinogens:** substances that distort the user's perception of reality and produce symptoms similar to psychosis (experimental drugs)
- (a) **Purpose:** to cause hallucinations/euphoria (UPPERS), causes sedation (DOWNERS)
 - (b) **Most commonly abused hallucinogens:**
 - (i) **Cannabis Sativa (Marijuana):** bloodshot eyes (increased blood flow to the eyes), weight gain
 - (c) **Route:** oral, nasal, IM/IV, topical, genital/anal
 - (d) **Other drugs:**
 - (i) LSD (Lysergic Acid Diethylamide) - synesthesia
 - (ii) PCP (Phencyclidine) - violence
 - (iii) Ecstasy - sexual aggressive
- (6) **Inhalants**
- (7) **Narcotics**
- (a) **Purpose:** to escape reality/pain
 - (b) **Commonly abused narcotics:** codeine, tramadol, oxycodone, morphine, meperidine, fentanyl
 - (c) **Signs of abuse:** hypotension, bradycardia, bradypnea, pupil constriction
 - (d) **Route:** IV (hepatitis B, HIV)
 - (e) **Signs of withdrawal:** increased VS, anxiety
 - (f) **Detoxification:** methadone - slow opioids (morphine withdrawal)
 - (g) **Antidote:** Naloxone (Narcan) (morphine intoxication)

Sexual Dysfunction

- Premature ejaculation
- Desire disorder (no urge)
- Orgasmic disorder (too less or too much)
- Genito-plevic pain disorder (painful sex)
- Erectile dysfunction
- **Management:**
 - Lubricants
 - Open talk and couples therapy
 - Viagra (WOF: low BP)
 - Exercise (sensate focus therapy)

Sexual Disorder

- Voyeurism: view "peeping tom"
- Exhibitionism: exposing



- Pedophile: pedia (sex)
- Froiteurism: friction/touching
- Management:
 - CBT
 - Open, non-judgement
 - No access to victims
 - Treat with SSRI
 - Relapse prevention
 - On psychotherapy
 - Legal involvement

Neurodevelopmental Disorders

- (1) **Autism Spectrum Disorder:** pervasive and usually severe impairment of reciprocal social interaction skills, communication deviance, and restricted stereotypical behaviors
 - **Signs and symptoms:**
 - Missing eye contact
 - Interaction lacking (gestures)
 - Likes being alone
 - Delayed speech
 - Echolalia and rituals (OC - neat)
 - Respond oddly to questions
 - (a) **Tic disorders:** sudden , rapid, recurrent, nonrhythmic, stereotyped motor movement or vocalization
 - (b) **Tourette's disorder:** multiple motor tics and one or more vocal tics which may occur many times a day for more than 1 year
 - (c) **Asperger's syndrome:** intellectual autism
- (2) **Attention Deficit Hyperactivity Disorder (ADHD):** characterized by inattentiveness, overactivity and impulsiveness
 - (a) **ADD:** Attention Deficit Disorder
 - (b) **ADHD:** Attention Deficit Hyperactivity Disorder
 - **Dopamine and norepinephrine** decreased (decreased attention span)
 - **Signs and symptoms:**
 - **Inattention**
 - Easily distracted
 - Poor concentration
 - Forgetfulness
 - **Hyperactive**
 - Fidgety, restless
 - can't sit still



- constant movement
- **Impulsive**
 - Talks excessively
 - Interrupts often
 - Acts without thinking

Routines and Consistency

- Safety
 - Structure - provide place to study, eat, play, bath
 - Schedule - time for everything
 - Set limits
 - Temper tantrums - give balloon, deep breathing exercise, haloperidols (tranquilizers)
- (3) **Intellectual Disability:** below average intellectual functioning accompanied by significant limitations in areas of adaptive functioning such as communication skills, self-care, home living, social or interpersonal skills, use of community resources, self-direction, academic skills, work, leisure, health and safety
- (a) **Profound/Imbecile:** <20 = total care
 - (b) **Moderate:** 35-49 = trainable with self-care
 - (c) **Mild:** 50-70 = teachable until grade 6
 - (d) **Borderline:** 71-84 = average knowledge
 - (e) **Average:** 90-139 = can finish college but not a genius
 - (f) **Genius/gifted:** >140 = genius

Suicide

Verbal	Non-Verbal
- I won't be a problem anymore - This is my last day on earth - I'll be gone soon	- Take this ring, it's yours (giving of valuable) - Sudden change of mood

Risk factors:

- Sex (Male)
- Unsuccessful previous attempt
- Identification with someone who committed suicide
- Chronic illness
- Depression/Dependent Personality
- Age (18-25 and >40), alcoholism
- Lethality of previous attempts/losses
- Plan - most outstanding

Nursing Interventions:

- Assess directly and monitor signs for signs/plans



- Suicidal clues, harmful object
- Safety, security and supervision
- Provide family therapy/support group and counseling

Aggressive Client

- **Decrease stimulation** - turn off television, let other clients leave the room
- **Deescalate** - assertive communication
- **Directive approach** - calm, non-threatening
- **Show of force**: visibility of 4-6 staff members

Restraint

- Danger to self & others (violent)
- **Doctor's order (Application)**: secure the order within 24 hours
- **Informed consent**: involuntary
- **Proper application**:
 - 6-8 staff members required
 - Adequate circulation must be ensured (not too tight, not too loose)
 - Anchor on a stable part of bed: bed frame
- **Doctor's order (removal)**: prerogative
- **Proper removal**:
 - Temporary - alternately, one at a time, for 10 minutes every 2 hours

Seclusion

- Inappropriate behavior, continuous thoughts of suicide
- **Informed consent**: voluntary/involuntary
- **Room**: lockable and observable from the outside
- **Purpose**: restorative, not punitive
- **Goal**: to help the client regain self-control
- **Monitoring**: one-on-one
- **Environment**: less stimulated (no visitors and phones allowed)

Abuse

- Safety first
- Assess injuries
- Find facts



- Empathy for victim
- Collect data
- Advocate for victim
- Report to authorities (Police/DSWD)
- Educate/empower

Psychopharmacology

Terms:

- **Efficacy:** refers to maximal therapeutic effect that the drug can achieve
- **Potency:** describes the amount of the drug needed to achieve maximum effects
- **Half-life:** the time it takes for half the drug to be removed from the bloodstream

Antipsychotic Drugs

- Formerly known as “**neuroleptics**” are used to treat symptoms of psychosis such as delusions and hallucinations
- Mechanism of action: block receptors for the neurotransmitter dopamine (decrease dopamine)
- Side Effects: **Extrapyramidal Symptoms (EPS)**
 - **Acute dystonia** - neck and facial spasms (temporary)
 - **Torticollis** - twisted head and neck
 - **Opisthotonus** - tightness in the entire body with the head back and arched neck
 - **Oculogyric crisis** - eyes rolled back in a locked position
 - **Pseudoparkinsonism:** drug-induced Parkinson’s disease symptoms (temporary)
 - **Akathisia:** intense need to move about/restlessness (temporary)
 - **Nursing action:**
 - Notify MD
 - Decrease dosage
 - **Management:**
 - Shift 1st generation to 2nd generation
 - Meds:
 - Akineton
 - Benadryl
 - Congentin
 - **Neuroleptic malignant syndrome (NMS):** potentially fatal idiosyncratic/adverse reaction (leads to seizures)
 - Hyperthermia (41C - malignant)
 - Hypertension (crisis)
 - Muscle spasms
 - **Nursing Action:**



- Stop the meds
- Notify MD
- **Prevention:**
 - Increased OFI
- **Medication management:**
 - Supportive
 - **Muscle relaxant:** baclofen, dantrolene
- **Tardive dyskinesia:** syndrome of permanent involuntary movement (overdose - lifetime)
 - Tongue protrusion
 - Teeth grinding
 - Lip smacking
 - **Nursing action:**
 - Notify the MD
 - **Medical management:**
 - Valbenazine (Ingrezza) - decreases symptoms
 - **Prevention:**
 - Always start with a lower dose
- **Other Side Effects:**
 - **Constipation** - increased OFI & fiber
 - **Agranulocytosis** (atypical) - report to MD
 - **Tooth decay** - sugarless hard candy or gum
 - **Dry mouth** - sugarless hard candy or gum
 - **Orthostatic** hypotension - change position gradually
 - **Galactorrhea** - wear cotton underwear
 - **Photosensitivity** - wear sunscreen SPF 25+, umbrella, hats, sunglasses
 - **Arrhythmias** - report
 - **Weight gain** - decreased sugar/calories
 - **Sedation** - avoid driving/operating heavy machineries

Psychotropic Medications	
Typical Antipsychotics (First Generation) - Chlorpromazine (Thorazine) - Fluphenazine decanoate (Prolixin decanoate) - Haloperidol (Haldol) - Thioridazine (Mellaril) “-zine” & “-dol” Usage: (+) symptoms	Atypical Antipsychotics (Second Generation) - Aripiprazole (Abilify) NEW - Clozapine (Clozaril) - Risperidone (Risperdal) - Olanzapine (Zyprexa) - Quetiapine (Seroquel) - Ziprasidone (Geodon) “-done”, “-pine”, “-zole” (new) Usage: (+) & (-) symptoms



Risk: increased EPS

Risk: (+) agranulocytosis - risk for infection

Psychotropic /Mood-Stabilizing Drugs

- Used to treat bipolar disorder by stabilizing the client's mood, preventing or minimizing the highs and lows that characterize bipolar illness, and treating acute episodes of mania
- **Lithium** (salt - 2-3 g/day)-highly toxic (mood stabilizer)
 - **Lithium carbonate** (Eskalith, Lithobid)
 - **Lithium citrate** (naps, PRN sedation)
 - **Lithium Mnemonics**
 - **Level:** 0.6-1.2 mEq/L
 - **Increased** urination (polyuria)
 - **Tremors** (early toxicity sign)
 - **Hydration** is essential (H₂O - 2-3L/day)
 - **Hypothyroidism** risk (T₃, T₄ decreased = decreased metabolism)
 - **Initial** kidney (BUN & Serum creatinine) & thyroid tests
 - **Upset** stomach (diarrhea)
 - **Mouth** dry
 - Wait for 2-4 weeks before lithium therapy takes effect
 - **Side effects:**
 - Can't see: blurred vision
 - Can't spit: dry mouth
 - Can't sweat: dry skin
 - Can pee: polyuria
 - Can poop: diarrhea
- **Valproic acid (Depakote)**
 - Vomiting
 - Very common GI upset
 - Avoid pregnancy (teratogenic)
 - Liver toxic
- **Carbamazepine**
 - CBC (agranulocytosis/aplastic)
 - Ataxia (unstable movement)
 - Rashes (Steven Johnson's syndrome)
- **Lamotrigine**
 - Life-threatening rashes
 - Added as maintenance
 - Mood stabilizer = slowest



Antidepressant Drugs

- Used in treatment of major depressive illness, anxiety disorders, the depressed phase of bipolar disorder and psychotic depression

- **4 Rules of Antidepressant**

- Increased risk of suicide
- Slow onset and slow taper off (2-4 weeks) - withdrawal symptoms if abrupt stop
- Never mix (serotonin syndrome) - leads to seizures (fatal)
- All antidepressant drugs
 - Decrease BP (hypotension)
 - Cause weight changes (weight loss-short term, weight gain-long term)

(1) Tricyclic and the related cyclic antidepressants - increases serotonin and norepinephrine

- (a) Imipramine (Tofranil)
- (b) Doxepin (Sinequan)
- (c) Amitriptyline (Elavil)

(i) **Actions:**

- 1) Two to 4th weeks onset/taper off
- 2) Can't (anticholinergic effects)
- 3) Assess for suicide

(2) Selective serotonin reuptake inhibitors (SSRIs) - increases serotonin

- (a) Citalopram (Celexa)
- (b) Fluoxetine (Prozac)
- (c) Sertraline (Zoloft)

(i) **Actions:**

- 1) Safest/Sexual dysfunction
- 2) Suicide/Serotonin syndrome
- 3) Rigidity/restlessness
- 4) II - 4 weeks onset/taper off

(3) MAO inhibitors (MAOIs) - increases serotonin, norepinephrine, dopamine

- (a) Tranylcypromine (Parnate)
- (b) Isocarboxazid (Marplan)
- (c) Phenelzine (Nardil)

(i) **Actions:**

- 1) Hypertensive crisis (cannot be treated with anti-hypertensive drugs)
- 2) Avoid alcohol, chocolates, aged cheese, avocado, soy sauce, preserved/processed foods (avoid tyramine)
 - a) Allowed cheese
 - i) Cottage cheese



- ii) Ricotta
- iii) American processed cheese
- iv) Part-skin mozzarella (white)
- 3) Avoid OTC (calcium, antacids, acetaminophen, NSAIDS)
- 4) Other antidepressant (serotonin syndrome)
- 5) Increased suicide risk

(4) Atypical Antidepressants

- (a) Mirtazapine
 - (i) Meals increase
 - (ii) Increase appetite, weight gain
 - (iii) Good for depressed/anorexic client
- (b) Buprion
 - (i) Boost up dopamine/norepinephrine
 - (ii) Smoking cessation
 - (iii) CI: seizure/eating disorder
- (c) Trazadone
 - (i) Sleeping
 - (ii) Sedation
 - (iii) WOF: priapism (prolonged painful erection)

Antianxiety Drugs (Anxiolytics)

- Used to treat anxiety and anxiety disorders, insomnia, OCD, depression, PTSD, alcohol withdrawal
- Minor tranquilizer (sedative) = increases GABA
- **Types (DOWNERS)**
 - **Benzodiazepines** - quick acting, immediate relief, for acute attack, addictive, do not abrupt stop (-pam, -lam)
 - Valium (Diazepam)
 - Xanax (Alprazolam)
 - Klonopin (Clonazepam)
 - Serax (Oxazepam)
 - Ativan (Lorazepam)
 - **Nonbenzodiazepines** - long-term, slow-acting, maintenance, no sedation, no withdrawal symptoms
 - Buspirone (Buspar) 2-4 weeks
 - Ipsapirone
- **Side Effects:**
 - **Anticholinergic side effects**
 - Can't see (blurry vision)



- Can't spit (dry mouth)
- Can't sweat (dry skin)
- Can't pee (urinary retention)
- Can't poop (constipation)
- **Antidote:** Flumazenil (Romazicon)
- **Nursing Consideration:**
 - Bed time
 - Avoid alcohol (downer)
 - Monitor your kidney (nephrotoxic) & liver (hepatotoxic)

Stimulants

- used to treat psychiatric disorders for their pronounced effects on CNS stimulation
- **ADHD Medications: UPPERS**
 - **Methylphenidate Amphetamine mixture** (Ritalin)
 - **Dextroamphetamine stimulants** (Adderall)
 - Residual ADHD grows up not antisocial
 - **Meds:** Ritalin, dexedrin, pemoline, Adderall
 - **Best time to give once a day:** morning
 - **After meals:** to prevent loss of appetite
 - **Antidote:** alprazolam

Therapies

- **Electroconvulsive Therapy**
 - Involves application of electrodes to the head of the client to deliver electrical impulse to the brain
 - It is believed that the shock stimulates brain chemistry to correct the chemical imbalance of depression
 - Induce a seizure via electrical current to treat severe or medication-resistant depression/disorder
 - **Pre:**
 - X-ray (fracture)
 - Informed consent (voluntary (patient), involuntary(family member/next of kin))
 - NPO 6-8 hours prior
 - **Meds:**
 - **Atropine** - dry mouth, oral secretions
 - **Barbiturates** - sedative/anesthesia
 - **Succinylcholine** - muscle relaxant, prevent seizure
 - **Post:**



- X-ray (fracture)
- Side-lying - lateral (drains the saliva)
- Side effect:
 - Most common side effect: temporary memory loss/confusion (reorient)
- **Nursing intervention:**
 - Safety (#1 objective)
 - Side-lying (#1 position) - after the seizure
 - Side rails up (prevent falls)
 - Support the head with a pillow after seizure
 - Stimulus (decreased) - no noise & bright light
- **Points to Remember:**
 - **Electric current:** 70-150 volts
 - **Duration of administration:** 0.5-2 secs
 - **Frequency of treatment:** 2-3 times/week
 - **Total number of treatments:** 6-12 sessions
 - **Side effect:** seizure (tonic-clonic/grand-mal): 30 secs-1 min
- **Cognitive-Behavioral Therapy**
 - Strategies designed to change the client's thinking (cognition) and actions (behavior)
- **Dialectical Behavioral Therapy**
 - Designed to help people manage intense emotions and change harmful behaviors by teaching skills in mindfulness, distress tolerance, emotion regulation, and interpersonal effectiveness
- **Cognitive Therapy**
 - Focuses on immediate thought processing - how a person perceives or interprets his/her experience and determines how he/she feels and behaves
- **Family/Group Therapy**
 - Forms of talk therapy where a trained professional guides discussions to improve relationships, communication, and problem-solving within a family or a group of individuals
- **Milieu Therapy**
 - Uses a structured, therapeutic community and everyday activities to promote healing and growth for individuals with mental health issues
- **Peer Group Therapy**
 - Involves a licensed professional facilitating a session with a group of people who share a common condition or experience, offering mutual support, shared understanding, and practical coping skills
- **Psychodynamic Theory**
 - Rooted in Freudian psychoanalysis, that explores how unconscious processes, influenced by past experiences and relationships, affect present feelings, behaviors, and thoughts
- **Rational Emotive Therapy**



- Albert Ellis identified 11 irrational beliefs that people use to make themselves unhappy
- **Logotherapy**
 - Viktor E. Frankl, a therapy designed to help individuals assume personal responsibility
- **Gestalt Therapy**
 - Frederick S. Perls, a therapy focusing on the identification of feelings in the here and now, which leads to self-acceptance
- **Systematic Desensitization**
 - Used to treat phobias and anxiety disorders by gradually exposing individuals to their fears while employing relaxation techniques to reduce anxiety responses
- **Reality Therapy**
 - William Glasser, a therapeutic focus is needed for identity through responsible behavior, individuals are challenged to examine ways in which their behavior thwarts their attempts to achieve life goals
- **Alternative medical systems**
 - Include homeopathic medicine and naturopathic medicine in Western cultures, and Traditional Chinese medicines which includes herbal and nutritional therapy, restorative physical exercises, meditation, acupuncture, remedial massage
- **Mind-body interventions**
 - Include meditation, prayer, mental healing, and creative therapies that use art, music or dance
- **Biologically based therapies**
 - Uses substances found in nature such as herbs, food and vitamins
- **Manipulative and body-based therapies**
 - Based on manipulation or movement of one or more parts of the body
- **Energy therapies**
 - Biofield therapies: intended to affect energy fields that are believed to surround and penetrate the body
 - Bioelectric-based therapies: involving the unconventional use of electromagnetic fields

Nursing Practice Beyond NP1-NP5

Professional Adjustment

- **Definition:**
 - **Profession**
 - An occupation or calling requiring advanced training and experience in some specific or specialized body of knowledge, which provides service to society in that special field
 - **Professional nurse**



- A person who has completed a basic nursing education program and is licensed in his/her country or state to practice professional nursing
- The professional nurse must have:
 - A license to practice nursing in the country
 - A bachelor of science degree in nursing
 - Be physically and mentally fit
- **License**
 - A legal document given by the government that permits a person to offer to the public his/her skills and knowledge in a particular jurisdiction, where such practice would otherwise be unlawful without a license
 - The primary purpose of registration is to protect the health of the people by establishing minimum standards which qualified practitioners must meet
 - It discourages certain persons who may be tempted to misrepresent themselves as professional nurse
 - The professional license is also called certificate of registration
- **Registration**
 - Recording of names of persons who have qualified under the law to practice their respective professions
 - Under the Philippine Nursing Act of 2002 RA 9173 section 12 and 20 state that license to practice nursing shall be issued to those who pass the licensure examination or by reciprocity
 - **Registration by reciprocity**
 - Section 20. A certificate of registration/professional license may be issued without examination to nurses registered under the laws of foreign state or country: Provided, that the requirements for registration or licensing of nurses in said country are substantially the same as those prescribed under the Act
 - **Registration by examination**
 - Section 12. All applicants for registration as a nurse and issuance of a certificate of registration and professional identification card to practice nursing shall be required to pass a written examination which shall be given by the Board in such places and dates as maybe designated by the Commission. Such examination must be in accordance with and fully compliant with RA 8981
 - **Practice through special/temporary permit**
 - It may be insured by the Board to the following persons subject to the approval of the Commission and upon payment of the prescribed fees
 - Licensed nurses from foreign countries/states whose service are either for a fee or free if they are internationally well-known



specialists or outstanding experts in any branch or specialty of nursing

- Licensed nurses from foreign countries/states on medical mission whose services shall be free in a particular hospital, center or clinic
- Licensed nurses from foreign countries/states employed by school.colleges of nursing as exchange professors in a branch or specialty of nursing. Provided, however that the special/temporary permit shall be effective only for the duration of the project, medical mission or employment contract

Continuing Professional Development in Nursing

- Consists of planned learning experiences beyond the basic education program
- Inculcation of advanced knowledge, skills, and ethical values in a post-licensure specialization or in an inter or multidisciplinary field of study, for assimilation into professional practice, self-directed research and/or lifelong learning
- **According to RA 10912 (Continuing Professional Development Act of 2016)**
 - Required CPD credit units per year: 15 units
 - Professional Identification Card (PIC) renewal: every 3 years; 45 CPD units required
 - Examples
 - Seminars
 - Conventions
 - Residency
 - Distance Learning
- **In service education**
 - Planned program provided by an employing agency to its employees
 - The major goal is not only the improvement of professional practice but also the fullest development of nurse as a person and as a contributing member of society focuses on and is designed to re-train people, to improve their performance and communication ability to get them started in the never-ending continuum of education

Legal Aspects of Nursing

- **Law**
 - Sum total of rules and regulations by which society is governed
 - It is man-made and regulates social conduct in a formal and binding way
 - Rule of conduct pronounced by controlling authority and which may be enforced
 - Classifications according to origin



- **Divine law**
 - Author is God
- **Human law**
 - Author is man
- **Public law:** a law that affects the general public
- **Private law:** a law that affects the relationship of an individual to another individual
- **Functions of Law Nursing**
 - Provides a framework for establishing what nursing actions in the care of patients are legal
 - Delineates the nurse's responsibilities from those of other health professionals
 - Helps to establish the boundaries of independent nursing actions
 - Assists in maintaining a standard of nursing practice by making persons accountable under the law
- **Torts**
 - Legally wrong, committed against a person or property independent of a contract which renders the person who commits it liable for damages in a civil action
- **Assault**
 - Imminent threat of harmful or offensive bodily contact
 - Verbal threat
- **Battery**
 - Intentional, unconsented physical touching of another person
- **False imprisonment or illegal detention**
 - Unjustifiable detention of a person without a legal warrant within boundaries fixed by the defendant by an act or violation of duty intended to result in such confinement
 - If patient is insisting on leaving the hospital, probable consequences of their action must be explained by a competent doctor or medical staff, and then he is allowed to go home against advice
- **Invasion of right of privacy and breach of confidentiality**
 - The right to be left alone, right to be free from unwanted publicity and exposure to public view
 - Privacy relates to a person or identity
 - Confidentiality relates to data/information about an individual
- **Defamation**
 - Character assassination, be it written or spoken
 - Slander (oral/spoken)
 - **Slander by deed:** defamation through actions that damage another person's reputation or dignity.
 - Libel (written)
 - There must be a third person who hears or reads the comment



Nursing Law

- **Republic Act 9173 Nursing Law or Nursing Act of October 21, 2002**
 - This repeals or changes all of RA 7164
 - Under this law, there is no more refresher course
 - A candidate can take the board exam until he passess, this is embodied in section 15 of RA 9173
 - Under this law, the upper 40% rule in no longer in effect
- **Concepts**
 - File your appeal with the Professional Regulatory Commission (PRC) within a period of 30 days after suspension or revocation order has been received (Reglementary period)
 - The Board of Nursing is under the PRC, PRC is under the Office of the President of the Philippines.
 - License revocation or suspension is an administrative case
 - Civil and criminal cases are appealed in court
- **Revocation**
 - Permanent
 - Invalidation of license
 - Under RA 9173, revocation of license is limited to four years in certain conditions:
 - Reason for revocation must have been cured
 - Reason for revocation does no longer exist
- **Suspension**
 - Temporary
 - For a fixed period
- **Illegal Dismissal**
 - Filing of cases of illegal dismissal is with Department of Labor and Employment
 - Done when there is lack of due process prior to dismissal
 - No lawyer is needed
 - If you win the case, there is back payment of salary and reinstatement
 - Reinstatement mat be in a form of paper reinstatement or payroll reinstatement

Laws affecting Nursing Practice			
EO 35	Field Health Service Information System	RA 6675	General Drug Act
EO 382	National Food Fortification Day	RA 6713	Code of Conduct and Ethical Standards for Public Employees
EO 51	Milk Code	RA 6809	Legal Age



LOI 949	Primary Health Care	RA 7160	Local Government Code
PD 1096	National Building Code of Philippines	RA 7305	Magna Carta of Public Health Workers
PD 442	Labor Code	RA 7600	Rooming-in and Breastfeeding Act
PD 603	Child and Youth Welfare	RA 10606	National Health Insurance
RA 3573	Birth Registration and Death Registration	RA 7877	Anti Sexual Harassment Act
PD 807	Civil Service Law	RA 7883	BHWs Benefits and Incentives Act
PD 825	Garbage Disposal	RA 8172	Act for Salt Ionization Nationwide Law
PD 856	Code of Sanitation	RA 8344	Emergency
RA 10028	Expanded Breastfeeding Promotion Act	RA 8423	Traditional and Alternative Medicine Act of 1997
RA 10152	Mandatory Infants and Children Health Immunization Act of 2011	RA 8746	Clean Air Act
RA 10173	Data Confidentiality Act	RA 8976	Philippine Food Fortification Act
RA 10352	Excise Tax on Alcohol and Tobacco	RA 8981	PRC Modernization Act
RA 10354	Responsible Parenthood and Reproductive Health Act of 2012	RA 9003	Ecological Solid Waste Management Act 2002
RA 10912	Continuing Professional Development Law	RA 9165	Comprehensive Dangerous Drug Act
RA 10963	Tax Reform for	RA 9173	Comprehensive



	Acceleration and Inclusion		Nursing Law
RA 11036	Mental Health Law	RA 9211	Tobacco Regulation Act of 2003
RA 11166	Philippine HIV and AIDS Policy Act	RA 9262	Violence Against Women and Children
RA 11223	Universal Health Care Law	RA 9275	Philippine Clean Water Act of 2004
RA 11332	Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern	RA 9482	Rabies Act of 2007
RA 10767	Comprehensive Philippine Plan Action to Eliminate Tuberculosis	RA9709	Universal Newborn Hearing Screening and Intervention Act of 2007
RA 4073	Act of Liberalizing the Treatment of Leprosy	RA 9710	Magna Carta of Women
RA 5901	Hours of Labor Act	RA 9711	Food and Administration Act
RA 7719	National Blood Service Act of 1994	RA 11509	Medical Scholarship and Return Service Program (Doctor Para sa Bayan Act)

Professional Negligence and Malpractice

- **Standard**
 - Desired and achievable level of performance against which actual practice is compared
 - Serves as benchmark against which to plan, to implement and assess quality of services
- **Negligence**
 - Refers to the commission or omission of an act, pursuant to a duty, that a reasonably prudent person in the same or similar circumstance would or would not do, and acting or the non-acting of which is the proximate cause of injury to another person or his property
- **Elements**
 - Existence of duty



- Failure to meet the standard of due care/breach of duty
- Foreseeability of harm/proximate cause
- Injury of the plaintiff
- **Specific examples**
 - Failure to report observations to attending physicians
 - Failure to exercise the degree of diligence which the circumstances of the particular case demands
 - Mistaken identity
 - Wrong medicine, wrong concentration, wrong route, wrong dose
 - Defects in the equipment such as stretchers, and wheelchairs may lead to falls thus injuring the patients
- **3 conditions to establish negligence without proving specific conduct**
 - Injury was of nature that it would not normally occur unless there was a negligent act
 - Injury was caused by an agency within control of the defendant
 - Plaintiff himself do not engage in any matter that would tend to bring about the injury
- **Malpractice**
 - Improper or unskillful care of a patient by a nurse; also denotes stepping beyond one's authority with serious consequences
 - Term for negligence of professional personnel (professional negligence)
 - Used properly only when it refers to a negligent act committed in the course of professional performance
- **Incompetence**
 - Lack of ability, or legal qualifications and being unfit to discharge the required duty
 - **Types of Incompetence**
 - **Gross Incompetence** – A serious lack of knowledge, skill, or judgment that endangers a patient's safety.
 - **Professional Incompetence** – Failure to meet the standards of practice expected of a nurse.
 - **Technical Incompetence** – Inability to properly perform nursing procedures or use equipment safely.
 - **Moral or Ethical Incompetence** – Failure to uphold ethical principles or professional values in nursing practice.
- **Doctrine:**
 - **Force Majeure**
 - An irresistible force, one that is unforeseen or inevitable
 - **Respondeat Superior**
 - Let the master answer for the acts of the subordinate



- Liability is shared between the employee and employer; not a shift of liability from the subordinate to the master
- **Captain of the ship:** Surgeon
- **Res Ipsa Loquitor**
 - The thing speaks for itself
 - Means that the injury could not have happened, if someone was not negligent that no further proof is required
 - **Elements:**
 - Injury will not normally occur unless negligence
 - Agency/Instrumentality within the control of the defendant/nurse
 - Patient did not participate

Consent to Medical and Surgical Procedures

- **Consent**
 - Free and rational act that presupposes knowledge of the thing to which consent is being given by a person who is legally capable to give consent
 - Must be obtained from the patient or his authorized representative who may be his parent or guardian
 - In case of emergency: consent is waived; consent does not apply
- **Nature**
 - Nurse secures consent of the patient upon admission
 - Physician obtains the consent
 - To substantiate the patient's consent, a written consent form is needed as proof against any liability
- **Elements**
 - Diagnosis and explanation of the condition
 - Fair explanation of the procedures to be done and used and the consequences
 - Description of alternative treatments or procedures
 - Description of the benefits to be expected
 - Material risks if any
 - Prognosis, if the recommended care, procedure is refused
- **Proof of consent**
 - Consent form is signed to show that the person consents and he/she understands the nature of the procedure, the risks involved and the possible consequences
- **Who must consent?**
 - **Minor**
 - Parents or someone standing in their behalf
 - If the minor is married or otherwise emancipated, parental consent is not needed
 - **Mentally ill**



- Parents or legal guardian
- **Emergency situation**
 - No consent is necessary because the inaction at such time may cause greater injury
 - Implied consent
- **Consent for sterilization**
 - Husband and wife, if the operation is primarily to accomplish sterilization
 - Patient alone, if only because of medical necessity and the sterilization is an incidental result

Medical Records

- **Purposes**
 - Saves duplication in future cases and aids in prompt treatment
 - Supplies rich material for medical and nursing research
 - Serves as a legal protection for the hospital, doctor and nurse by reflecting the disease or condition of the patient and his management
- **Remember**
 - If it was not charted, it was not observed or done
- **Nurses' notes**
 - Should be written, fully, accurately, legibly, and promptly
 - Not only includes medications and treatments but also physical and emotional symptoms exhibited by the patient
 - Aids to medical diagnosis and in understanding the patient's behavior
- **Nurses' role**
 - Safeguarding the patient's record from loss or destruction or from access by persons who are not legally authorized to read such
- **Medical records in legal proceedings**
 - It is usually the medical records librarian, by virtue of a subpoena duces tecum, who testifies that the patients' records are kept and protected from unauthorized handling and change

Advance Directives

- **Living will:** directive given by the patient as to type of treatment he wants to receive if and when he gets into respiratory arrest
- **Durable power of attorney or health care proxy:** when the patient legally assigns a person, who will decide in his behalf for his treatment



Crimes, Misdemeanors and Felonies

- **Crime:** defined as an act committed or omitted in violation of the law
- **Elements**
 - Criminal act
 - Evil/criminal intent
- **Conspiracy**
 - **Principal**
 - Direct part in the execution of the act
 - Directly force or induce others to commit it
 - Mastermind of a crime: principal by inducement
 - **Accomplice**
 - Cooperates in the execution of the offense by previous or simultaneous act
 - Had knowledge of the criminal intention of the principal
 - **Accessory**
 - Have knowledge of the commission of the crime
 - Take part subsequent to its commission by profiting themselves or assisting the offender to profit from the effects of the crime
 - Provides the exit strategy
- **Felony**
 - Committed with deceit and fault
 - Major public offense for which a convicted person is liable to be sentenced to death or to be imprisoned in a penitentiary or prison
- **According to degree to act**
 - **Consummated**
 - All the elements necessary for its execution and accomplishment are present
 - **Frustrated**
 - When the offender performs all acts or execution which will produce the felony
 - Did not produce felony by reason of causes independent of the will of the perpetrator
 - **Attempted**
 - Offender commences the commission of the same directly by overt acts
 - Does not perform all the acts or execution which shall produce the felony

Circumstances Affecting Criminal Liability

- **Justifying circumstance**
 - No criminal liability because the act itself is justified as reasonable
 - Examples
 - Self defense



- Elements
 - Unlawful aggression
 - Necessity of the means employed
 - No sufficient provocation of the attacker
 - If one of the three elements is lacking, this becomes incomplete self-defense and it becomes a mitigating circumstance
- Fulfillment of a lawful order or duty
- State of necessity
- **Exempting circumstance**
 - No criminal liability because the person who committed the crime is exempted
 - Examples
 - Insane or imbecile
 - Insane person lacks intelligence, the ability to determine what is right or what is wrong
 - Imbeciles are people who have abnormalities in development who also do not possess adequate intelligence
 - Person below 9 years old
 - Parents would pay damages
 - Force majeure/natural calamity causes injury
 - Lucid interval: person has normal state of mind during a lucid interval
- **Mitigating circumstance**
 - Decreases criminal liability because of equity and justice
 - Examples
 - Voluntary surrender
 - Confession of guilt
 - Provoked
 - Age of convict
 - Below 18 years old
 - Above 70 years old
 - Blind, deaf and mute
 - Because they cannot defend themselves in court
- **Aggravating circumstance**
 - Increase criminal liability
 - Examples
 - Abuses
 - Cruelty
 - Abuse of authority
 - Crimes in times of calamity (looting, profiteering)
 - Treachery



- Insult to authority
- Premeditation
- **Alternative circumstance**
 - Degree of education
 - Degree of intoxication
 - Relationship

Leadership and Management

Leadership - the act of influencing and motivating others toward achieving a common goal

Management - the process of planning, organizing, coordinating and directing the work of others to meet organizational goal

Leadership	Management
- Do not have delegated authority and may be employed in a variety of positions - Involves empowering others - Have common goals that may or may not reflect those of the organization - Influencing, Motivating - DFF	- Legitimate source of power due to the delegated authority - Emphasizes control, decision making and solving problems - Greater formal responsibility and accountability - Organization goal - Planning, Organizing, Directing, Controlling - Class president

Leadership Theories

Classic	
Autocratic (Authoritarian)	- Determining policies, giving orders and directions - Needs for creativity, autonomy, and -self-motivation are not met - Often used in urgent decision making when group members cannot or do not wish to participate in decision making, completing a project quickly
Democratic (Participative, Consultive)	- Group discussions and decisions making are encouraged - Assumes that individuals are internally motivated, capable of making decisions and value independence - Allows self-motivation and productivity and calls for coordination and cooperation among members



Laissez-faire (Non-directive, Permissive)	- Recognizes the group's need for autonomy and self-regulation - Assumes a "hands-off" approach - Used in groups whose members have both personal and professional maturity - "Trusts" someone who is personally and professionally mature - Judge
Bureaucratic	- Does not trust self or others to make decisions - Rely on established rules, procedures, regulations - Dura lex Sed lex "The law is hard, but the law is the law"
Situational	- Adapts leadership style to the situation - Adapts to the readiness and willingness of the individual or group members to perform the assigned task
Contemporary	
Charismatic	- Charming personality
Transactional	- Relationship with followers is based on an exchange for some resource valued by the followers (giving incentives to promote loyalty and performance)
Transformational	- Promotes creativity, risk - taking, commitment and collaboration. Involves empowering and inspiring others to contribute in the organization's vision
Shared leadership	- Co-leadership; distributes decision making among group of people (shared governance) involves principle of unity of command

Effective Leadership

- **Vision:** mental image of a possible or desirable future state
- **Power:** capacity to influence
- **Influence:** gaining cooperation through excellent communication and persuasion based on trust

Types of Power

Legitimate	Based on hierarchy in an organization, derived from formal position or office
Reward	Based on incentives (promotion, time-off from work, attractive work assignment)



Coercive	Based on fear of retribution of withholding of rewards (dismissal, demotion, low rating) - RA 7877 - Anti-Harrassment Law - Work, education, training - Moral ascendancy - Sexual
Expert	Based on high level of competence, knowledge, personality abilities or skills
Referent	Admiration and respect for the leader based on charisma and success

Levels of Management

Top Level - organizational executives; establishing goals and developing strategic plans (Chief Nurse)

Middle Level - supervise first-level managers and activities in the department they supervise (Supervisor)

- With most gathered information

First Level - manages work of non- managerial personnel and day to day activities of specific work group/groups (Head Nurse)

Principles of Management

Manager

- **Authority** - legitimate right to direct the work of others
- **Responsibility** - obligation to complete task
- **Accountability** - ability and willingness to assume responsibility for one's actions and accept consequence of one's behavior
 - **Criminal Case** - people of the Philippines (article 365) regular court
 - **Civil Case** - money, resources (damages) regular court
 - **Administrative Case** - license (suspension - 4 years sec. 23 RA 9173) PRC

Principle of Unity of Command	An employee is answerable to one boss
Principle of Requisite Authority	When a responsibility for a particular task is delegated to a subordinate, subordinate must also be given authority over resources needed for task accomplishment
Principle of Continuing Responsibility	when the performance of a task is delegated, the delegator retains accountability for the outcome
Principle of Organizational Centrality	Workers who interact with the greatest amount of work-related information (middle level)



Notable Theories of Management

Frederick Taylor - Scientific Management

Douglas McGregor - Theory X & Y

Theory X	Theory Y
Dislikes work and avoids it	Views work as natural as work and play
Avoids responsibility and seeks direction	Accepts responsibility and seeks it
Security is valued over ambition	Individuals capable of innovation
Coerced, controlled, threatened	Possesses self-direction and self-control

Max Weber - Social and economic organization

Henry Fayol - Modern Management: 14 Principles of Management

- (1) **Division of labor** - division of work
- (2) **Authority and responsibility** - principles of management
- (3) **Discipline** - applying penalty after due process/judicious application of penalty
- (4) **Remuneration** - salary
- (5) **Unity of command** - one boss
- (6) **Unity of direction** - an organization with common goal
- (7) **Subordination of individual interest** - general interest over individual interest
- (8) **Equity** - equality based on situation "less in life, more in law"
- (9) **Order** - a place for everything and everything in its place
- (10) **Scalar chain** - the greater the power, the greater the responsibility
- (11) **Degree of centralization** - person/group that has authority
- (12) **Esprit de corps** - team building
- (13) **Initiative** - take action on your own
- (14) **Stability of tenure** - regularization

General Management Roles

- **Interpersonal role**
 - Symbol: signing of papers/documents
 - Facilitator: hires, trains, fires, remunerates and judges
 - Liaison: link to community, suppliers and the organization
- **Informational role**
 - One who monitors information
 - Disseminates information from both external and internal sources
 - Spokesperson or representative of the organization
- **Decisional role**



- An entrepreneur or innovator
- A trouble shooter
- A negotiator

Fundamental Skills of Managers

- **According to Katz**
 - **Technical:** proficiently in performing an activity in the correct manner with the right technique
 - **Human:** dealing with people and how get along with them
 - **Conceptual:** ability to see individual matters as they relate to the total picture and to develop big problems and discarding irrelevant facts
- **According to Summer**
 - **Knowledge factors:** refer to ideas, concepts, or principle
 - **Attitude factors:** beliefs, feelings, and values
 - **Ability factors:** skills, art, judgement and wisdom

Management Functions

Planning	- Assessing a situation, establishing goals and objectives based on assessment of the situation, developing a plan of action that identifies priorities, delineates who is responsible, determines deadlines - Deciding what, when, where, how to do it, by whom, and what resources
Organizing	- Determining responsibilities, communicating expectations, and establishing the chain of command for authority and communication
Directing	- Process of getting the organization's work accomplished - Assigning and communicating expectations about the task to be completed, providing instructions and guidance and on-going decision-making
Coordinating	- Ensuring that plans are carried out and evaluating outcomes

Planning

- It is a continuous process of assessing, establishing goals and objectives, implementing and evaluating them and subjecting these to changes
- Deciding in advance what to do; who is to do it; how, when and where it is to be done
- **Characteristics of a good plan**
 - A well-developed plan should:



- Be precise with clearly-worded objectives
- Be guided by policies and/or procedures affecting the planned action
- Indicate priorities
- Develop actions that are flexible and realistic
- Develop a logical sequence of activities
- Include the most practical methods for achieving each objective
- Pervade the whole organization

Strategic Planning

- 3-5 years in the future
- Set overall goals and policies
- Nursing directors, chief nurses, director of nursing
- 1. **Vision:** descriptive picture of a desired FUTURE STATE
- 2. **Mission:** identifies the REASON an organization exists and what it DOES to achieve its vision
- 3. **Goals and Objectives:** ends toward which the organization is working
- 4. **Philosophy:** integrated set of beliefs and values that guides actions
- 5. **Policies and procedures**
 - **Policies:** plans reduced to instructions or statements
 - **Procedures:** process that implements a policy according to a specific task
- 6. **Standards:** professionally desirable norms
- 7. **Rules:** describe situations that allow only one choice of action; fairly inflexible

Intermediate Planning

- 6 months-2 years
- Formulation of policies, rules and regulations, methods and procedures for intermediate level planning for on-going activities is done in coordination with the top management and those with the lower level
- Nursing supervisors

Operational Planning

- Department plans, maintenance and improvement goals
- Daily and weekly plans for administration or direct patient care
- Head nurses

Budgeting: Financial Plan

- **Budget:** a financial road map that estimates future costs and a plan for utilization of resources



- **Components**
 - **Operating budget**
 - Daily revenue and expenses including
 - Salaries
 - Supplies
 - Contractual services
 - Employee benefits
 - Laundry services, drug and pharmaceuticals
 - In-service education
 - **Revenue budget**
 - Summarizes the income
 - **Expense budget**
 - Salaries, supplies, utilities, maintenance
 - **Cash budget**
 - Forecasts the amount of money received
 - In consist of the beginning cash balance, estimates of the receipts and disbursements and the estimated balance for a given period
 - Prepared by estimating the amount of money to be collected from patients and allocating it to cash disbursement required to meet obligations promptly as they come
 - **Capital expenditure budget**
 - Major expenditures
 - Consists of accumulated data for fixed assets that are expected to be acquired during the budget period
 - Replacement or expansion of the plant, major equipment and inventories
 - **Flexible budget**
 - Composed of budgets that adjust automatically over the course of the year depending on variables such as volume and labor costs
 - **Zero-based budgeting**
 - Major advantage of this type is that it forces managers to set priorities and justify
 - Disadvantage: time consuming

Organizing

- It is a process of establishing formal authorities that involves
 - Setting up the organizational structure through identification of groupings, roles and relationships
 - Determining the staff needed and distributing them in various areas
- **Elements**
 - Organization structure



- Staffing
- Scheduling
- Developing job description
- **Authority:** the right to act or make decisions without the need for approval of higher administration
 - **Terms used in organizing authority**
 - **Accountability:** taking full responsibility for the quality of work and behavior
 - **Responsibility:** obligation to perform the assigned task
- **Principles**
 - **Unity of command:** there can only be one superior
 - **Scalar principle/hierarchy/chain of command:** flow from higher to lower authority
 - **Departmentalization:** grouping of workers with similar assignments
 - **Span of control:** the number of people that reports directly to a manager
 - **Centralization:** the staff all report to one person that is higher than them
 - **Decentralization/delegation:** transferring specified decision making to lower levels of the organization
 - **Organization:** the structure and process which allows an agency to enact its philosophy and achieve its goals
 - **Types**
 - **Line organization**
 - It is the simplest and most direct type of organization
 - Each position has general authority over the lower positions in the hierarchy
 - **Advantages**
 - Maintain simplicity
 - Encourages speedy action
 - Makes a clear division of authority
 - **Disadvantages**
 - Neglects special training
 - Overworks key people
 - Depends upon retention of few key people
 - **Staff organization**
 - It is purely advisory to the line structure with no authority to put recommendations into action
 - **Advantages**
 - Frees the line executive of detailed analysis
 - Affords young specialists a mass training
 - Enables specialist to give expert advice
 - **Disadvantages**



- Reduces expert power to place recommendation into action
- Tends towards centralization of the organization
- Continues the organization even if its functions are not clear
- **Functional organization**
 - It is one where each unit is responsible for a given part of the organization's workload
 - There is a clear delineation of roles and responsibilities which are actually interrelated
 - **Advantages**
 - Relieves line executive of routine specialized decision
 - Provides frameworks for applying expert knowledge
 - **Disadvantages**
 - Makes relationship more complex
 - Makes limits of authority of each specialists a difficult coordination
- **Organizational chart**
 - It is a diagrammatic representation of the organizational structure
 - **Five major characteristics**
 - Division of work
 - Chain of command
 - Type of work to be performed
 - Grouping of work segment
 - Levels of management
 - **Forms of organizational chart**
 - **Vertical chart**
 - Also known as tall chart
 - Depicts the chief executive at the top with line of authority flowing down the hierarchy
 - **Horizontal chart**
 - Depicts structure side to side
 - Useful for showing processes or equal authority distribution
 - **Concentric chart**
 - Also known as circular chart
 - Shows outward flows of communication from center

Staffing

- It is the process of determining and providing the acceptable number and mix of nursing personnel to produce a desired level of care to meet the patient's demand
- **Purpose:** to provide each nursing unit with an appropriate and acceptable number of workers in each category to perform the nursing tasks required
- **Schedule:** a timetable showing planned workdays and shifts for nursing personnel
- **Types**
 - **Centralized**
 - Done by the chief nurse
 - He/she assigns the nursing personnel to the various units of the hospital
 - **Decentralized**
 - The shift and off duties are arranged by the nurse supervisors or senior nurse of the particular unit
 - **Cyclical**
 - Covers a designed number of weeks as one cycle, which is repeated there on
- **Advantages**
 - It is fair to all
 - Saves time
 - Enables employees to plan ahead of time
 - Scheduled leaves are more stable
 - Productivity is improved

The Hospital Nursing Service Administration Manual of the DOH has recommended the following nursing care hours for patients in the various nursing units of the hospital

Cases/Pt	NCH/Pt/Day	Prof to Non-prof Ratio
General medicine	3.5	60:40
Medical	3.4	60:40
Surgical	3.4	60:40
Obstetrics	3.0	60:40
Pediatrics	4.6	70:30
Pathologic Nursery	2.8	55:45
ER/ICU/RR	6.0	70:30
CCU	6.0	80:20



Patient Care Classification

- Method of grouping patients according to amount and complexity of nursing requirements

Levels of Care	
Level 1: Minimal	For discharge, non-emergency, 1.5 hours per day
Level 2: Moderate	Some with ADL, IV. VS 3x per shift, 3 hours per day
Level 3: Total/Intensive	Completely dependent, VS every 30 minutes, 6 hours per day
Level 4: Critical	Continuous observation, 6-9 hours or higher per day

Percentage of Patients in Various Levels of Care				
Type of Hospital	Minimal Care	Moderate Care	Intensive Care	Highly Specialized Care
Primary	70	25	5	-
Secondary	65	30	5	-
Tertiary	30	45	15	10
Special Tertiary	10	25	45	20

Total Number of Working Days per Year		
	40 hours/week	48 hours/week
Non-working days	213	265
	152	100
Working hours	1704	2120
Relievers needed	0.15	0.12

Professional to Non-Professional	
Types of Hospital	Professional to Non-Professional Ratio
Primary	55:45



Secondary	60:40
Tertiary	65:35
Specialized	70:30 or 80:20

Distribution per Shift	
AM	45%
PM	37%
Night	18%

Staffing Formula

1. Categorize according to the level of care needed. Multiply the total number of patients by the percentage of patients at each level of care
2. Find the total number of nursing care hours needed by the patients at each category level
 - a. Find the number of patients at each level by the average number of nursing care hours needed per day
 - b. Get the sum of the nursing care hours needed at the various levels
3. Find the actual number of nursing care hours needed by the given number of patients. Multiply the total nursing care hours needed per day by the total number of days in a year
4. Find the actual number of nursing care hours rendered by each nursing personnel per year
 - a. Multiply the number of hours on duty per day by the actual working days per year
5. Find the total number of nursing personnel needed
 - a. Divide the total number of nursing of nursing care needed per year by the actual number of working hours rendered by an employee per year
 - b. Find the number of relievers. Multiply the number of nursing personnel needed by 0.15 (40 hours per week) or by 0.12 (48 hours per week)
 - c. Add the number of relievers to the number of personnel need
6. Categorize the nursing personnel into professional and non-professionals. Multiply the number of personnel according to the ratio of professionals to non-professionals
7. Distribute by shifts

Directing

- Issuance of orders, assignments and instructions that enable the nursing personnel to understand what are expected



- Involves command and coordination
- **Elements**
 - Communication
 - Delegation
 - Supervision
 - Coordination
 - Staff development
 - Decision-making

Communication

- Transmission of information, opinions and intentions between and among individuals
- **Types**
 - **Verbal**
 - Oral
 - Written
 - **Non-verbal**
 - Facial expression
 - Tone of voice
 - Body language
 - Touch
- **Line of Communication**
 - **Downward**
 - From superior to subordinate
 - Examples
 - Policies
 - Rules and regulations
 - Memorandum
 - Employee
 - Handbooks
 - Performance appraisal
 - **Upward**
 - From subordinates upward
 - Examples
 - Incident report
 - Grievance report
 - **Horizontal**
 - Between peers, personnel or departments on the same level
 - Examples
 - Endorsements



- Conferences
- Nursing rounds
- **Outward**
 - From caregivers to patient and their relatives
 - Examples
 - Discharge teaching
- **Diagonal**
 - From individuals or departments that are not in the same level or the hierarchy
 - Examples
 - Nutrition department to nursing department
- **Grapevine**
 - Informal communication; often rapid and subject to much distortion
 - Example
 - Gossip
 - hearsay

Delegation

- Process by which a manager assigns specific tasks/duties to workers with commensurate authority to perform the job
- **Principles**
 - Select the right person to be delegated
 - Delegate both interesting and uninteresting tasks
 - Provide subordinates with enough time to learn
 - Delegate gradually
 - Delegate in advance
 - Consult before delegating
 - Avoid gaps and overlaps
- **What cannot be delegated**
 - Overall responsibility, authority and accountability for satisfactory completion of all activities in the unit
 - Authority to sign one's name is never delegated
 - Evaluating the staff and/or taking necessary corrective or disciplinary action
 - Responsibility for maintaining morale or the opportunity to say a few words of encouragement to the staff especially to new ones
 - Jobs that are too technical and those that involve trust and confidence
- **Responsibilities**
 - **Registered Nurse**
 - IV medications
 - Health teaching



- Assessment
- Evaluation
- Overall accountability unstable/critical patients
- **Licensed practical nurse**
 - Medications (IM, SC, ID except IV)
 - Wound cleansing
 - Blended feeding
 - Suctioning
- **Nursing assistant**
 - Routine activities
 - VS
 - Application of pulse oximeter sensor
 - Post-mortem
 - Stable/ambulatory/MGH

Nursing Care Assignments

- Can also be called modalities of nursing care, systems of nursing care, or patterns of nursing care

1. **Functional nursing**

- a. This kind of nursing modality is task-oriented in which a particular nursing function is assigned to each worker
- b. Divides work to be done and every member is responsible for his actions
- c. Best system that can be used if there are many patients and professionals

d. Advantages

- i. Work is done fast
- ii. Workers learn to work fast

e. Disadvantages

- i. Fragmentation of nursing care; therefore holistics not achieved
- ii. Patient cannot identify who their real nurse is
- iii. Nurse-patient relationship is not fully developed
- iv. Evaluation of nursing care is poor
- v. Outcomes are rarely documented
- vi. It is hard to find a specific person to answer relative question

2. **Total care nursing**

- a. One nurse is assigned to one patient for the delivery of total care

3. **Team nursing**

- a. One nurse leads a group of nursing personnel in providing nursing needs to a group of patients
- b. Decentralized system of care
- c. **Advantages**



- i. Team effort
- ii. Frees patient care coordinator to manage the unit
- iii. Nursing care conferences help problem solve and develop staff
- iv. Nursing care plan

d. Disadvantages

- i. Time needed to coordinate delegated work

4. Primary nursing

- a. RN is responsible for the total care of a small group of patients (4-6) from admission to discharge
- b. RN remains responsible for the care of those patients 24 hours per day

c. Advantages

- i. Patient and family are able to develop trusting relationship with the RN
- ii. Assures continuity of care
- iii. Improves communication with the members of the health team and eliminates the use of nursing aides in the provision of direct nursing care

d. Disadvantages

- i. High cost because there is higher RN skill
- ii. Proximity of patient assignment
- iii. Overlapping of staff function
- iv. Nurse-patient ratio must be realistic

5. Case method

- a. One is to one nurse patient ratio

b. Advantages

- i. Consistency of one individual caring for the whole shift
- ii. More opportunity to observe and monitor the patient

c. Disadvantages

- i. The nurse must not have the same patient the next day
- ii. It does not serve the purpose of decentralization

6. Modular method

- a. Also termed as district nursing
- b. Modification of team and primary nursing
- c. It is sometimes used when there are not enough RNs to practice nursing
- d. It differs from the team nursing in that the RN provides direct nursing care with the assistance of aides

Supervision

- It is providing guidelines for the accomplishment of a task or activity with initial direction and periodic inspection of the actual accomplishment coordination, and synchronization of services
- **Principles of good supervision**



- Requires adequate planning and organization which facilitate cooperation, coordination and synchronization of services
- Gives autonomy to workers depending on their competency, personality and commitment
- Stimulates the worker's ambition to grow into effectiveness
- Creates an atmosphere of cordiality and trust
- Considers the strengths and weaknesses of employees
- Strives to make the unit an effective learning situation of the task or activity
- Considers equal distribution of work considering age, physical condition and competence

Staff Development

- Providing structure and assistance for employees to learn more'

1. Orientation

- a. Done for new employees
- b. Refers to planned and guided activities of an employee in the organization, the work environment and in his job

2. In-service education

- a. Consists of on the job instruction that are given to enhance employee's recent job performance

3. Specialty courses

- a. Offered by hospitals with trained specialist, facilities, and source

4. Formal education

- a. Graduate degree: master's degree

Coordination

- Unites personnel and services toward a common objective
- Synchronization of activities among various services and department
- Prevents overlapping of functions
- Promotes good working relationships
- Work schedules are accomplished as targeted

Conflict

- A clash between two opposing parties
- It is a type of behavior involving two or more parties in opposition to each other
- **Types**
 - **Intrapersonal:** from within one person
 - **Interpersonal:** between two persons
 - **Intragroup:** within the group itself



- **Intergroup:** conflict between two groups of people in the organization

Conflict Resolution		
Avoidance	Avoid confrontation	Lose-Lose
Accommodation	One gives way to other	Win-Lose
Collaboration	Both work out ways to solve the problem	Win-Win
Compromise	Both will sacrifice	Lose-Lose
Competition	One wins	Win-Lose

Decision Making

- The process of arriving at a course of set of that is consciously chosen from alternatives
- **Decision making process**
 - Define the problem
 - Analyze the problem
 - Develop alternative solution
 - Select possible solutions'
 - Implement follow-up

Controlling

- Also called evaluating
- An on-going function of management which occurs during planning, organizing, and directing activities
- **Basic components of the control process**
 - Establish standard
 - Measure actual measurements
 - Compare performance vs. standards
 - Reinforce correct behavior
 - Implement corrective action
- **Performance appraisal**
 - A control process wherein an employee's performance is evaluated against standard
- **Characteristics of an evaluation tool**
 - Objectivity
 - Reliability
 - Validity



- Sensitivity
- **Performance appraisal tools**
 - **Checklists:** compilation of all nursing performances expected of a worker
 - **Rankings:** in simple ranking, the evaluator ranks the employees according to how he/she fared with co-workers with respect to certain aspects of performance or qualifications
 - **Rating scales:** includes a series of items representing the different tasks or activities in the nurse's job description
 - **Anecdotal recording:** describes the nurse's experience with a group or a person, or in validating technical skills and interpersonal relationships
 - **Essay:** the appraiser writes a paragraph or more about the worker's strengths, weaknesses and potentials
- **Informal Appraisal**
 - Consists of incidental observation of performance while the worker is engaged in performing nursing care
- **Formal Appraisal**
 - Collecting objective facts that can demonstrate the difference between what is expected and what was done

Quality in Healthcare

Quality Assurance (QA) Program	Quality Improvement (QI)/ Performance Improvement (PI)/ Total Quality Management (TQM)
Systematic process that measures compliance with certain standards (maintaining standards) - horizontal	Systematic approach with the intention of improving quality care (vertical)
Supporting existing good practices	Focuses on seeking areas for growth and improvement
Static	Dynamic (no stopping point)
Punishes or sanction; finds blame	Rewards innovation
Seeks to answer the question "How do I maintain?"	Seeks to answer the question "How do I improve?"
Types of evaluation: structure, process, outcome	



Quality Assurance Program

Types of Evaluation	Definition/Remark	Example
Structure	Focused on the setting wherein care is given	Equipment, staffing
Process	Focused on how care is given (manner in which the nurses use the nursing process)	Checking client's identification band before giving medication
Outcome	Focused on demonstrable changes of the client's health status as a result of nursing care	How many clients undergoing hip repair develop pneumonia?

Change

- Process of making something different from what it was
- Integral aspects of nursing (Nurses as a change agents)
- Resisting change is a natural response to some people
- Every change involves adaptation
- Who/What initiate change: system, management, patient, employee
- Phases of effective change process (Lewin): unfreezing, unmoving, refreezing
- **Strategies for affecting change**
 - **Empirical-Rational**
 - Based on the assumption that people are rational and behave according to rational self-interest
 - It follows that people should be willing to adopt a change if it is justified and if the people are shown how they can benefit from the change
 - **Normative-Reeducative**
 - Based on the assumption that people act according to their commitment to sociocultural norms
 - Manager pays attention to changes in values, attitudes, skills and relationships in addition to providing information
 - **Power-Coercive**
 - Involves compliance of less powerful people to leadership, plans, and directions of more powerful people

Kurt Lewin's Theory of Change

- **Unfreezing**
 - Development of a need for change



- Change problem is established and defined
- Factors that may cause resistance to change are considered
- **Moving**
 - Goals are established
 - Alternative possibilities are examined
 - Implement appropriate strategies, ensuring that driving forces exceed restraining forces
- **Refreezing**
 - Change is stabilized and integrated into the status quo

Types of Change

- **Coercive change**
 - Non-mutual goal setting, imbalanced power ratio, and one-sided deliberativeness characterize coercive change
- **Emulative change**
 - Transition is fostered through identification with and imitation of power figures
- **Indoctrination**
 - Uses mutual goal setting, has an imbalanced power ratio and is deliberate
- **Interactional change**
 - Mutual goal setting, fairly equal power, but not deliberativeness
- **Natural change**
 - Includes accidents and acts of God
 - Involves no goal setting deliberativeness
- **Socialization change**
 - Individual conforms to the needs of a social group
 - When there is greater deliberativeness on the power side, change becomes indoctrination
- **Technocratic change**
 - Collecting and interpreting data bring about change
 - Technocrat merely reports the findings of the analysis to bring about the change
- **Planned change**
 - Involves mutual goal setting, an equal power ratio and deliberativeness

Theories of Motivation

- **Need theory**
 - Abraham Maslow's hierarchy of needs
 - People are motivated to satisfy certain needs to complex psychological needs
- **Two-Factor theory**
 - Hygiene factors
 - Motivating factors



- Nurse managers need to use both of these factors to recruit and retain staff
- **McClelland's three basic needs theory**
 - Includes the three basic needs that people should possess in varying degrees (achievement, power and affiliation)
- **Expectancy theory**
 - **Victor Vroom's expectancy theory human motivation**
 - Indicates that felt needs individuals in work settings are increased if a person perceives positive relationship between effort and performance
- **Operant theory**
 - B.F Skinner's Operant theory
 - Suggests that an employee's work motivation is controlled by conditions in the external environment instead of internal needs and desires
- **Equity theory**
 - Perceptions about equity and inequity
 - Found that employees assess fairness by considering their input and psychological, social and financial rewards in comparison with those of others

Nursing Ethics

- Examination of all ethical and bio-ethical issues from the perspective of nursing theory and nursing
- **Teleological approach**
 - Telos or "goal or end"
 - The right thing to do is the good thing to do
 - The good resides in the promotion of happiness or the greatest net increase of pleasure over pain
- **Deontological approach or duty oriented theory**
 - A person is morally good and admirable if his actions are done from a sense of duty and reason
- **Virtue ethics approach**
 - Focused primarily on the heart of the person performing the act
 - Focuses on the traits and virtues of a good person

Universal Principles of Biomedical Ethics

- **Autonomy**
 - Voluntary decision making
 - Self-governance
 - Involves self-determination and freedom to choose and implement one's decision



- **Paternalism:** means overriding a person's choices or actions for their own good, usually with the intention of protecting them from harm or promoting their well-being.
- **Informed consent:**
 - A voluntary agreement by a patient to undergo a medical procedure, treatment, or participation in research, after being fully informed of all relevant facts.
 - It respects the ethical principle of autonomy (the patient's right to decide).
 - **Essential Elements:**
 - **Disclosure** – The patient must receive complete, accurate, and understandable information (nature of the procedure, risks, benefits, alternatives, and consequences of refusal).
 - **Comprehension** – The patient must understand the information provided.
 - **Voluntariness** – The decision must be made freely, without coercion or undue influence.
 - **Competence** – The patient must have the capacity (legal age, sound mind) to make decisions.
 - **Consent** – Expressed either verbally or in writing (usually written for major procedures)
 - **Types**
 - **Implied consent:** Assumed through actions (e.g., rolling up a sleeve for an injection).
 - **Expressed consent:** Clearly given, verbally or in writing.
 - **Written Consent:** A formal, signed document granting permission for specific medical/surgical procedures or research participation.
 - **Oral Consent:** Patient verbally agrees after being fully informed.
 - **Proxy Consent (Substituted Consent):** Given by a legal guardian, parent, or authorized representative when the patient is a minor, unconscious, or mentally incapacitated.
 - **Informed Refusal:** When a patient, after being fully informed, refuses treatment.
 - **Nursing Role**
 - Ensure the patient received accurate information from the physician.
 - Clarify any questions or misunderstandings.
 - Verify that consent is signed before procedures
 - Advocate for the patient if they feel pressured or misinformed
- **Veracity**
 - Truth telling or honesty
 - **Benevolent deception:** The act of withholding the truth or giving incomplete information to a patient with the intention of sparing them from distress, anxiety, or harm
- **Beneficence**



- Doing good
- Refers to acts of kindness and mercy that directly benefit the patient
- These acts promote the health of the patients, prevent illness or complications, alleviate suffering, and assist towards peaceful death if the inevitable comes
- **Non-maleficence**
 - Do no harm
 - Stated as an admonition in the negative form
- **Justice**
 - Just, fair, and equal treatment
- **Fidelity**
 - Keeping one's promise; loyalty
- **Advocacy**
 - The ethical responsibility of nurses to protect and promote the rights, health, and well-being of patients by acting or speaking on their behalf.
 - Rights, needs, privileges
- **Principles of Epikia**
 - Exception to the general rule
 - It is a reasonable presumption that the authority making the law will not wish to bind a person in some particular case, even though the case is covered by the letter of the law
- **Two-fold effect/Principle of double effect**
 - When a nurse is faced with a situation which may have both good and bad effects, the basis of her action may be the following:
 - Action must be morally good
 - Good effect must be willed and the bad effect merely allowed
 - Good effect must not come from an evil action but from the initial action itself directly
 - Good effect must be greater than the bad effect
- **Confidentiality, Anonymity and Privacy**
 - **Confidentiality** – The ethical and legal obligation to keep patient information secure and only share it with authorized individuals.
 - **Anonymity** – The state in which a patient's identity is not revealed or linked to the information provided.
 - **Privacy** – The patient's right to control access to their body, personal space, and personal information.
 - **Types of subpoena**
 - **Subpoena duces tecum** – A legal order requiring a person to produce documents, records, or other evidence in court.
 - **Subpoena ad testificandum** – A legal order requiring a person to appear and give oral testimony in court.



- **Expert Witness** – A person with specialized knowledge, skill, education, or experience who is permitted by the court to give an opinion about matters within their expertise, to help the judge or jury understand complex issues.

Code of Ethics and Good Governance

- **Code of ethics**
 - Systematic guides for developing behavior
 - They answer normative questions of what beliefs and values should be morally accepted
- **Code of good governance**
 - Promulgated by the PRC on July 23, 2003 states that the hallmark of all professionals is their willingness to accept a set of professional and ethical principles which they will follow in the conduct of their daily lives
 - Adopted by the PRC and the 42 professional regulatory boards to cover an environment of good governance in which all Filipino professionals shall perform their duties

BON Res. 220 S. 2004 - Code of Ethics for Registered Nurses

- **Preamble**
 - Health is a fundamental right.
 - Nurses are duty-bound to promote health, prevent illness, alleviate suffering, and assist in peaceful death when healing isn't possible.
 - Nursing care should respect cultural, social, spiritual, psychological, and environmental factors.
- **ARTICLE I – Nurses and People**
 - Respect the dignity, values, customs, and spiritual beliefs of every patient.
 - Respect patients' right to make decisions (autonomy).
 - Keep all patient information confidential.
 - Prioritize the welfare and safety of the patient at all times.
- **ARTICLE II – Nurses and Practice**
 - Value the sanctity of life and quality nursing care.
 - Maintain competence, integrity, and honesty in practice.
 - Document nursing care truthfully and accurately.
 - Act as an advocate to protect patients' rights.
- **ARTICLE III – Nurses and Co-Workers**
 - Work in harmony with other members of the health team.
 - Respect the competence and roles of co-workers.



- Avoid rivalry, unhealthy competition, and unjust criticism.
- Share knowledge and skills for professional growth.
- **ARTICLE IV – Nurses and the Profession**
 - Uphold the integrity of the nursing profession.
 - Participate in continuing education and research.
 - Contribute to the development of nursing knowledge.
 - Be active members of professional nursing organizations.
- **ARTICLE V – Nurses and Society**
 - Contribute to community health and development.
 - Support government and private health initiatives.
 - Participate in policy-making related to health and social needs.
 - Advocate for environmental preservation and sustainability.
- **ARTICLE VI – Nurses and the Environment**
 - Maintain a safe, clean, and conducive environment for health.
 - Be aware of how ecological factors affect health.
 - Promote health policies that ensure environmental protection.
- **Effectivity**
 - Approved: July 14, 2004
 - Enforced 15 days after publication in the Official Gazette or newspaper of general circulation.
 - Violations: Can result in disciplinary actions (suspension/revocation of license).

Research

Terms:

Research

- More formal, comprehensive and intensive for the discovery of new knowledge
- Months and even years

Scientific method

- informal way to discover new things
- hours or days

Nursing Research

- Focus: deals with clinical problems

Research in Nursing

- Broader study of nursing in the profession
- Issues that can affect nursing profession

Evidenced -Based Practice

- the use of current best evidence (latest findings)

Importance of Research



- Professionalism
- Accountability
- Social relevance of nursing
- Research & decision making

Ultimate Goal of Research: To improve the practice/client care/patient care/health care/nursing care/profession (if general)

Sources of knowledge:

- **Tradition** (stories from one generation to another generation)
- **Authority** (people who are experts)
- **Logical reasoning** (use of logic and reason)
- **Experience** (best teacher) **and trial & error** (try and try until succeed)
- **Scientific method/experiment** (discover something all by yourself)

Types of Research

- **According to Purpose:**
 - **Basic/Pure research**
 - To increase general knowledge
 - Formulate/refine a theory
 - General principles of human behavior
 - **Applied research**
 - Find solution to existing problem
 - Focus on intervention to achieve desired goal
 - How principles can be used to solve problems in nursing practice
- **According to Levels of Investigation (General)**
 - **Level I: Exploratory** - identify the variables (opinion) “quality”/census
 - **Level II: Descriptive** - describe whether the variables are associated or related (facts/issues/study/analysis)
 - **Descriptive** - describe the variables as it is
 - **Correlational** - describe the relationship of the variables to each other
 - **Comparative** - describe the comparison of the variables to each other (similarities/differences)
 - **Level III: Experimental** - to find out the cause and effect of the variables to each other (effect/manipulation)
- **According to Design**
 - **Non-Experimental** - without manipulation
 - **Historical** - study of the past issues (descriptive)



- **Survey** - used of interviews and questionnaires to acquire public opinion (exploratory)
- **Case study** - in depth study of persons or entities (descriptive)
- **Methodological** - study of methods, tests and procedures (descriptive)
- **Analytical** - further analysis of present event/issue (descriptive)
- **Experimental** - with manipulation
 - **True experimental** - strict control over the variables
 - **Elements of randomization** - experimental & control group
 - **Elements of manipulation** - experimental group
 - **Element of control** - control group is given placebo (without comparison)/others (with comparison)
 - to compare the results of experimental group to the control group
 - **Quasi-experimental** - just like a true manipulation
 - the control overcomes the variables
 - Lacks randomization
- **According to Time Frame**
 - **Cross-sectional design** - involves the collection of data once: the phenomena under study are captured during one period of data collection
 - **Longitudinal design** - a study in which data are collected at more than one point in time over an extended period
 - **Retrospective design** - involves collecting data on an outcome occurring in the present, and then linking it retrospectively to antecedents or determinants occurring in the past
 - **Prospective design** - information is first collected about a presumed cause or antecedent and the subsequently the effect or outcome is measured (study of present to predict the future)

Variables

- A characteristic or quality that takes on different values
 - **Independent variable** - the presumed cause of, antecedent to, or influence on the dependent variable
 - **Dependent variable** - the behavior, characteristic or outcome the research is interested in understanding, explaining or affecting, it is the presumed effect

Quantitative	Qualitative
- Positive traditions - Scientific method - Numerical data - Objective - Systematic/controlled	- Naturalistic inquiry - Human experience - Narrative description data - Subjective - Flexible/evolving



- Empiricism (uses of sense) - General/broad - Cause and effect - Deductive reasoning - Theory testing - Laboratory setting	- Analytical insight - Specific/direct - Meaning discovery - Inductive reasoning - Theory development - Field setting
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5 Major Steps of the Research Process

1. Conceptual Phase

a. Formulating and delimiting the problem

i. Identify an interesting, significant research problem and research question

ii. Research problem

1. is the situation that causes the researcher to feel apprehensive, confused and is at ease
2. Is the demarcation of a problem area within a certain context involving the who or what, the where, the when and the why of the problem situation

3. Sources of research problem

- a. Nursing literature
- b. Ideas from external sources
- c. Theory (old & existing)
- d. Experience and clinical fieldwork (richest information)
- e. Social issues
- f. Scientific research - previous researches (least being used)

4. Criteria of a Good Research Problem

- a. Significance of the study (address it to the people, society at large, profession)
- b. Availability of the subjects
- c. Limitation of subjects
- d. Time allotment and research ability
- e. Limitation of tools
- f. Interest of the researcher
- g. Novelty (newness/freshness of the idea)

iii. Problem statement

1. Articulate the problem to be addressed and indicates the need for a study through the development of an argument

iv. Research questions

1. Are the specific queries researcher wants to answer addressing the research problem



b. Reviewing the related literature

- i. Provides a foundation on which to base new evidence and usually is conducted well before any data are collected
- ii. Synthesis of previous studies
- iii. **Purpose of RRL:** To get an idea or the background of the study
- iv. **Types of Sources**
 1. primary source
 2. descriptions of studies written by the researchers who conducted them
 3. Mostly recommended
 4. secondary source - description of studies prepared by someone other than the original researcher

c. Undertaking a clinical fieldwork

- i. To get an idea or the background of the study
- ii. Complements & support RRL

d. Defining the framework and developing conceptual definition

- i. **Framework** is the overall conceptual underpinnings of the study
 1. **Conceptual framework**
 - a. Deals with the abstraction that are assembled by virtue of their relevance to common theme
 - b. It is the study that has roots in a specified conceptual mode (topic)
 2. **Theoretical framework** - it is a study based on theory
 3. **Conceptual definition** - presents the abstract or theoretical meaning of the concepts being studied
 4. **Operational definition** - specifies the operation that researchers must perform to collect and measure the required information/define the variables on how they are used on their study

e. Formulating hypothesis

- i. **Hypothesis** - is a statement of the researcher's expectations about relationships between study variable/tentative prediction of the relationship between variables
- ii. **4 Parts of Hypothesis**
 1. Independent variable (cause)
 2. Dependent variable (effect)
 3. Relationship
 4. Population to which it applies
 - a. **Population** - include all characteristics
 - b. **Set (2 or more sets)** - concentrate on characteristics that differentiate them
 - i. Set characteristic (fixed) - **independent variable**
 - ii. Vary - **dependent variable**



- c. **Set (1 set)**
 - i. Eliminate the population since 1 set
 - ii. Go back and ask what is being studied
 - iii. Fixed - independent variable
 - iv. Vary - dependent variable
 - d. **Manipulation** - anything that is given (independent variable)
 - iii. **Research hypothesis** - statement of expected relationships between variables
 - 1. **Non-directional** - predicts a relationship or difference between variables without specifying the direction of that relationship
 - 2. **Directional** - predicts the specific direction of a relationship or difference between variables
 - iv. **Null hypothesis** - state that there is no relationship between the independent and dependent variables
 - 2. **The design and planning phased** (most used of your time)
 - a. **Selecting a research design**
 - b. **Developing the protocols for the intervention** (optional, only for experimental research)
 - c. **Identifying the population to be studied**
 - i. People who provide information to the researchers or investigators are:
 - 1. Subjects
 - 2. study participants
 - 3. Respondents
 - 4. Informants (qualitative study)
 - ii. **Population** - all the individuals or objects with common defining characteristics
 - iii. **Accessible/source population** - aggregate of cases that conform to designated criteria and that are accessible as subjects for a study
 - iv. **Target population** - aggregate of cases about which the researcher would like to generalize
 - v. **Sample** - subset of population elements
 - vi. **Element** - most basic unit about which information is collected
 - vii. **Pilot study** - small-scale version or trial run designed test the methods to be used in a larger, more rigorous study called parent study
 - d. **Designing the sampling plan**
 - i. **Sampling** - the process of selecting a portion of the population to represent the entire population so that inferences about the population can be made
 - ii. **Representatives** - the key consideration in assessing a sample in quantitative study
 - 1. **Probability sampling** - involves random selection of elements
 - a. **simple random sampling** - most basic probability sampling design



- b. **Stratified random sampling** - it subdivides the population into homogenous subsets from which an appropriate number of elements are selected at random
 - c. **cluster sampling (muti-stage sampling)** - involves the successive selection of random samples from larger to smaller units by either simple random or stratified random methods
 - d. **Systematic sampling** - involves the selection of every nth case from the list, such as every tenth person on a patient list
 - 2. **Non-probability sampling** - elements are selected by non-random methods
 - a. **Convenience/accidental sampling** - entails using the most conveniently available people as study participants
 - b. **Snowball/network/chain sampling** - variant of convenience sample, every sample members are asked to refer other people to meet the eligibility criteria
 - c. **Quota sampling** - one in which the researcher identifies population strata and determines how many participants are needed from the stratum
 - d. **Purposive/judgemental sampling** - the researcher selects units to be sampled based on their knowledge and professional judgement
 - e. **Specifying the methods on how to measure research variables** - tools/instruments for data collection
 - f. **Developing a method on how to safeguard human/animal rights**
 - g. **Reviewing and finalizing the plan**
- 3. **The empirical phase (longest phase)**
 - a. **Collecting the data**
 - i. **Data saturation** - involves sampling to the point at which no new information is obtained and redundancy is achieved
 - ii. **Major Forms of Data Collection**
 - 1. **Observation** - way of gathering data by watching behavior, events, or noting physical characteristics in their natural setting
 - a. **Phenomena** (amenable to observation)
 - i. characteristics and condition of the subject
 - ii. activities and behaviors of the subjects
 - iii. skill attainment and performance (performance appraisal)
 - iv. environmental characteristics
 - v. verbal communication
 - vi. non-verbal communication



b. Types of observation

- i. **Overt** - everyone knows they are being observed
- ii. **Covert** - no one knows they are being observed and the observer is concealed

c. Types of observer

- i. **Non-participant/direct observation** - is where data are collected by observing behavior without interacting with the participants
- ii. **Participant observation** - is where data are collected by interacting with, and therefore experiencing the phenomenon being studied

d. Advantages of observation

- i. Collect data where and when an event or activity is occurring
- ii. Does not rely on people's willingness or ability to provide information
- iii. Allows you to directly see what people do rather than relying on what people say they did

e. Disadvantages of observation

- i. Susceptible to observer bias
- ii. Susceptible to Hawthorne's effect (people usually perform better when they know they are observed though indirectly)
- iii. Can be expensive and time-consuming compared to other data collection method
- iv. Does not increase your understanding of why people behave as they do

2. Self-Reports (widely used form) - disadvantage: accuracy and validity

a. Questionnaires - is a research instrument consisting of questions and other prompts for the purpose of gathering information from the respondents

i. Types of questions

- 1. **Closed-ended** - is a question format that provide respondents with a list of answer choices from which they must choose to answer the question
- 2. **Open-ended** - designed to encourage a full, meaningful answer using the subject's own knowledge or feelings

ii. Advantages



1. Cost - less costly, less time and energy to administer
 2. Anonymity - offer possibility of complete anonymity
 3. Interviewer bias - absence of an interviewer ensures that there will be no interviewer bias
- b. **Interview** - is a conversation between two people where questions obtain information from the interviewee
- i. Types of interview
 1. Structured - require adherence to a very particular set of rules. Each question that is outlined should be ready word for word by the researcher without any deviation from the protocol
 2. Semi-structured - using this type are still expected to cover every question in the protocol, they have some wiggle room to explore participant responses by asking for clarification or additional information
 3. Unstructured - need only a checklist of topics to be covered during the interview
 - ii. **Advantages**
 1. Response rates - tend to be high in face to face interview
 2. Audience - many people cannot fill up questionnaire, interview is feasible with most of the people
 3. Clarity - interviews offer some protection against ambiguous or confusing questions
 4. Depth of questioning - open-ended questions are used mostly in interviews
 5. Missing information - less likely to give missing information in the interview
 6. Order of question - in interviews, researchers have control over question ordering
3. **Biophysical measurements** (use of equipment and gadgets)
- a. **In Vivo** - measurement directly on the subject (pulse oximeter)
 - b. **In Vitro** - measurement performed outside (laboratories)
 - c. **Criteria:**
 - i. **Validity** - venture what is supposed to be measured



ii. **Reliability** - consistency or the repeatability of the tool

iii. **Sensitivity** - detect small variations or find differences

4. **Pre-existing data** (any forms of records)

b. **Preparing for data analysis**

4. **The analytic phase**

a. **Analyzing the data**

b. **Interpreting the result** - interpretation is the process of studying the results and examining their implications (inferential statistics)

i. **Parametric test** - are characterized by three attributes

1. They involve the estimation of a parameter

2. they require measurement on at least an interval scale

3. They involve several assumptions, such as the assumption that variables are normally distributed in the population

a. Interval - occurs when the researcher can specify the rank-ordering of the objects on an attribute and can assume equivalent distance between them (fahrenheit temperature scale)

b. Ratio - provides information concerning the ordering of objects on the critical attribute, the intervals between objects and the absolute magnitude of the attribute (scores in board exam)

ii. **Non-parametric tests** - they involve less restrictive assumptions about the shape of the variable's distribution than do parametric tests

1. Levels of measurement

a. Nominal - involves assigning numbers to classify characteristics into categories (gender, blood type, marital status)

b. Ordinal - involves sorting objects based on their relative ranking on an attribute (mid, moderate, severe)

iii. **Measurements**

1. Chi-square test - to test hypothesis about the proportion of cases that fall into different categories as when a contingency table has been created

2. T-test - assess whether the means of two groups are statistically different from each other

3. Analysis of variance (ANOVA) - testing differences between means when there are three or more groups

iv. **Descriptive Statistics**

1. **Measures of condense**

a. Frequency distribution - a systematic arrangement of values from lowest to highest together with a count of the number of times each value was obtained

b. Percentage - a way of expressing a number, especially a ratio as a fraction of 100



- c. Graphic presentation - transformation of data through visual methods like graphs, diagrams, maps and charts is called representation of data
- 2. **Measures of central tendency** - are indexes expressed as a single number that represent the average or typical value of a set of scores
 - a. Mode - most frequently occurring score value in distribution
 - b. Median - is the point in a distribution above which and below which 50% of cases fall
 - c. Mean - the sum of all scores, divided by the number of scores (average)
- 3. **Measures of variability** - shows how spread out the data
 - a. Range - simply the highest score minus the lowest score in a distribution
 - b. Standard deviation - indicates the average amount of deviation of values from the mean
 - c. Variance - equal to standard deviation squared
 - d. Percentile - the value of variable below which a certain percent of observations fall
- 4. **Measures of relationship**
 - a. Pearson' R - this coefficient is computed with variables measured on either an interval or ratio scale
 - b. Spearman's rho (p) - the correlation index usually used for ordinal level measures
 - c. Correlation coefficient - indicates the magnitude and direction of a relationship between to variables
- 5. **Dissemination phase**
 - a. Communicating the findings - final task of the research project is the preparation of a research report that can be shared with others
 - b. Utilizing the findings into practice

Quality of Research Criteria

- 1. **Reliability** - refers to the accuracy and consistency of information obtained in a study
- 2. **Validity** - a more complex concept that broadly concerns the soundness of the study's evidence that is whether the findings are unbiased cogent and well-grounded
- 3. **Dependability** - refers to evidence that is consistent and stable
- 4. **Conformability** - the degree to which study results are derived from characteristics of participants and the study context, not from researchers biases



5. **Credibility** - an especially important aspect of trustworthiness, is achieved to the extent that the research methods engender confidence in the truth of the data and in the researchers interpretation of the data
6. **Triangulation** - use of multiple source of reference to draw conclusions about what constitutes the truth

Research Ethics

- **Nuremberg code** - ase of ethical research principles for human experimentation
- **Declaration of Helsinki** - allowed for the inclusion of vulnerable populations in research: (1) minors, (2) indigenous people
- **Ethical principles for protecting study participants**
 - **Beneficence** - imposes a duty on researchers to minimize harm and maximize benefits
 - **Right to freedom from harm and discomfort**
 - **Right to protection from exploitation**
 - **Right to self-determination** - means that prospective participants have the right to decide voluntarily whether to participate in a study without risking any penalty or prejudicial treatment
 - **Right to full-disclosure** - researcher fully described the nature of study, the person's right to refuse participation, the researchers responsibilities and likely risks and benefits
 - **Concealment** - the collection of information without the participant's knowledge or consent
 - **Deception** - either withholding information from participants or providing false information
 - **Right to fair treatment**
 - **Justice** - connotes fairness and equity and so one aspect of the justice principle concerns the equitable distribution of benefits and burdens of research
 - **Right to Privacy** - researchers should ensure that their research is not more intrusive than it needs to be and that participant's privacy is maintained throughout the study
 - **Anonymity** - the most secure means of protecting confidentiality occurs when even the researcher cannot link the participants to their data

Pharmacology

- A branch of medicine concerned with the uses, effects, and modes of actions of drugs

Definition of Terms

- **Pharmacodynamics** - effect of the drug on the body
- **Pharmacokinetics** - effect of body on the drug



- **Absorption** - site of administration into the bloodstream
- **Distribution** - movement of drug from the blood stream to the various tissues
- **Metabolism** - alteration of substance within the body
 - **Phase 1:** reduction, oxidation, hydrolysis
 - **Phase 2:** glucuronidation, acetylation, sulfation
- **Excretion** - elimination of drugs out from the body
- **Bioavailability** - fraction of administered drug that reaches systemic circulation unchanged
- **Half-life** - time required to change the amount of drug in the body by $\frac{1}{2}$ during elimination
- **Clearance** - volume of plasma cleared of drug per unit time
- **Zero Order** - rate of elimination is constant regardless of the concentration
- **First Order** - rate of elimination is directly proportional to the drug concentration
- **Efficacy** - maximal effect of a drug can produce
- **Potency** - amount of drug needed for a given effect
- **Therapeutic Index** - measurement of drug safety

Cardiovascular Medications

Beta Blockers (-olol) - Action: inhibits beta receptors from doing their job (lower HR/BP) - Key: Beta 2 receptors work on the lungs so we may see bronchospasms (DO NOT GIVE TO ASTHMA/COPD PATIENTS) - Types: - Selective (atenolol, metoprolol) - Nonselective (propranolol, Sotalol) - Considerations: - Can be given to heart failure patients - May mask symptoms of hypoglycemia - Always check HR/BP BEFORE and AFTER giving (Hold if BP <90/HR <60) - WOF: orthostatic hypotension, wheezing, decreased HR and BP	ACE Inhibitors (-pril) - 1st choice ARB's (-sartan) - 2nd choice - Action: both inhibit the normal function of the RAAS system (lower BP by getting rid of sodium/water and increase potassium) - Key: teratogenic - With ACE - cough/angioedema (risk: airway) - increased potassium in the body - Considerations: - Potassium levels - Peak T wave, ST elevation - Potassium diet: avoid potassium rich foods - Watch BP/orthostatic hypotension
Calcium Channel Blockers - Action: blocks movement of calcium	Digoxin (for heart failure) - Can also be used as anti-arrhythmic



which relaxes smooth muscle (decreased resistance = decreased BP) - Types: - dipine (only lowers BP) - zem/amil (lowers both BP/HR) - Considerations: - do not give to heart failure patients (will worsen) - Always check HR/BP BEFORE giving (HOLD if BP <90/HR <60) - WOF orthostatic hypotension, too low HR/BP - Headache (normal)	- Action: does not affect BP, only HR - For forceful contractions - Therapeutic range: 0.5-2.0 - Considerations: - Does not waste potassium (don't change potassium diet) - Low potassium levels = increase risk of toxicity - Monitor kidney function (BUN, creatinine) decreased kidney function = increase risk of toxicity - Signs and symptoms of toxicity: vision change, dizziness, nausea and vomiting, irregular pulse, confusion
Vasodilators (can be used in heart failure) - Action: relaxes smooth muscles = decreases BP, increases O2 to major organs - Normal signs and symptoms: headache, flushing, hypotension - Considerations: - Do not take with Viagra - Hold if BP systolic <100 or drops to >30 mmHg - Monitor BP = orthostatic hypotension - Example: nitroglycerin, hydralazine, nitroprusside, isosorbide	Key Takeaway: 1. Lowers BP: all except Digoxin 2. Lowers HR: beta blockers, CCBs (zem/amil), digoxin 3. Do not give to heart failure patients: CCBs 4. Yes for heart failure patients: digoxin, BBs, vasodilators 5. Not for pregnant: ACE/ARBs 6. Watch vitals, do not stop abrupt

Diuretics Medications

Potassium Wasting (-ide) - Action: dry up the body by excreting potassium (increase urination) - Key: only give if potassium levels is in range 3.5-5.0 mEq/L - Use: heart failure, worsening crackles, new edema in the legs, rapid weight gain - Types: - Loop Diuretics (furosemide,	Potassium Sparing (spironolactone) - Action: blocks aldosterone directly - release sodium and water "saves" potassium in the body - Considerations: - WOF increased potassium levels = increased heart beats - Avoid potassium rich foods/no salt substitutes
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- bumetanide, torsemide) - Thiazide diuretics (hydrochlorothiazide, chlorothiazide) - Considerations: - WOF decreased potassium levels = decreased heart beats (flat T wave, ST depression, U wave) - Encourage potassium rich foods (bananas, green leafy vegetables, melon) - Furosemide: if given too fast (tinnitus, hypotension), if given too long (kidney damage)	- Make sure to put on a cardiac monitor
Osmotic Diuretic (Mannitol) - Action: decreased thickness of blood -> increased cerebral blood flow -> decreased BV/ICP - Use: treat increased ICP, IOP - Side effects: headache, nausea, acute renal failure (large doses) - Monitor: neurological and renal status	Key Takeways: For potassium wasting & sparring 1. Always check BP, hold if systolic <100 2. Always check BUN, creatinine, potassium levels 3. Give in the morning, not night 4. Daily weights 5. No sodium meds (cough, flu, antacids, NSAIDS) 6. Teach orthostatic hypotension 7. Risk for sunburns

Respiratory Medications

Inhaled Anticholinergics (-ropium) - Used for: COPD, bronchitis, emphysema, asthma, allergies - Action: block the parasympathetic nerve reflexes that cause the airway to constrict (bronchodilation) - Types (can be used with Albuterol) - Short-acting: Iparopium - Long-acting: Tiotropium - Side effects: dry mouth, throat irritation, hoarseness, bitter taste (give sugarless candy or increased fluids)	Leukotriene Modifiers (-lukast) - Used for: asthma, exercise induced bronchoconstriction - Action: blocks the normal function of leukotriene to dilate the smooth muscle and decreased mucus - Key: leukotriene typically causes smooth muscles to constrict and increase mucus production (we are stopping this) - Side effects: headache, sore throat, cold symptoms - NOT FOR ACUTE ATTACKS - Types: - Monteleukast - take in the evening or 2 hour before exercise
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	- Zafirlukast - take on empty stomach, check liver enzymes
Beta 2 Antagonist (-terol) - Action: enhance the effects of beta 2 in the body = bronchodilation - Types: - short-acting (for acute attacks) - Not used for daily treatment (>2x/ week = BAD) - albuterol - Long-acting (NOT FOR ACUTE ATTACKS) - Should only be taken in combo with corticosteroids (NEVER ALONE) - Salmeterol - Side effects: tachycardia, jittery, nervous, palpitations, tremors - Key: if used in combo with steroids or anticholinergic (GIVE FIRST)	Corticosteroids (-sone,-ide) - Used for asthma, rhinitis, autoimmune disorders - Action: reduce the inflammatory response by suppressing the immune system - Can be used alone or with beta 2 antagonist - Side effects: osteoporosis (affects calcium), increased glucose, decreased potassium, fluid retention, urinary retention, PUD, immunosuppression - May cause: headache, sore throat, hoarseness, candidiasis (oral thrush) - use spacer and rinse mouth - Remember: - Increased stress - may need increased dose - Never stop abruptly - TAPER - monitor signs and symptoms of infection - Avoid NSAIDS
Xanthines (Theophylline) - Used for long term asthma/COPD - Action: increased CAMP = bronchodilation - Side effects: - Minor: - GI upset - Headache - nervousness - Major: - Seizure - Dysrhythmias - Key: - Avoid caffeine - Therapeutic level: 10-20 mcg/ml	
Expectorants - Used for: nonproductive cough - Action: decrease viscosity of lung secretions this will make the cough more productive	Mucolytics - Used for: pulmonary disorders with thick mucus (CF, COPD, PNA) - Action: breaks down tenacity and viscosity of secretions



- Side effects: headache, dizziness, nausea and vomiting - Prolonged use: may mask an underlying serious condition for which you have the cough - Guaifenesin	- Side effects: bronchospasms, nausea and vomiting, headache, dizziness - WOF patients with PUD, asthma, esophageal varices - Acetyl cysteine, dornasealfa
Decongestants (oral + topical) - Used for rhinitis, sinusitis, common cold - Action: vasoconstricts the respiratory mucosa = decreased inflammation (relief of pain and congestion) - Side effects: palpitations, restlessness, weakness, hypertension, nervousness, rebound congestion - WOF if taking other OTC drugs - Pseudoephedrine, phenylephrine	Antihistamines - Used for: seasonal allergies, urticaria, angioedema, allergic conjunctivitis - Action: blocks histamine at histamine 1 receptor sites (anticholinergic & antipyretic effect) - Side effects: itching, sneezing, hives, GI upset, nausea, dry mouth, constipation, sedation, drowsiness (risk for fall) - CI: pregnant women - 1st generation: diphenhydramine - 2nd generation: loratadine/cetirizine
Nonsteroidal Anti-allergy (Cromolyn) - Used for 3rd line drugs for mastocytosis - Action: stops mast cells from secreting histamine - Side effects: burning in nose, itchy, watery eyes, bad taste in mouth - Key: Not quick = long term	Subcutaneous Anti-inflammatory (Omalizumab) - Used for: allergic asthma, chronic idiopathic urticaria - Action: blocks the role of IgE = decreased allergic response - Side effects: itching, mild rash, nausea, dizziness, ear pain, cold symptoms, pain in arms and legs - Key: Not quick, no live vaccine (immunosuppressant)

Hyperlipidemic Medications

Statins (-statin) - Used for: decrease cholesterol and reduce risk of MI or stroke - Action: blocks the enzyme HMG-CoA reductase to slow down cholesterol production - Side effects: liver toxicity (myopathy, rhabdomyolysis) - WOF LFT's and CK levels (Normal: 22-198 U/L) - metabolized in liver,	Bile Acid Sequestrants (cole-) - AKA resins - Used for: lowering cholesterol (LDL) - Action: decrease LDL by binding bile acids in the gut = pee out = pulls more cholesterol from bloodstream to make bile - Side effects: constipation, flatulence, GI disturbance, bad taste - Key: teach increase fiber/fluids - interferes with the absorption of fat soluble vitamins
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excreted by kidneys - Avoid alcohol - Take in the evening	(ADEK) - Take with food - Colesevelam - Colestpal
Cholesterol Absorption Inhibitors (Ezetimibe) - Used for lowering cholesterol (LDL) - Action: decrease LDL by inhibiting absorption of cholesterol in the intestines - Side effects: liver toxicity (myopathy, rhabdomyolysis), headache, dizziness, fatigue - WOF LFT's and CK levels (Normal: 22-198 U/L)	Niacin/Nicotinic Acid (Vitamin B3) - Used for lowering cholesterol (LDL) and triglycerides, increases HDL - Action: when given above the vitamin requirement it will reduce the release of fatty acids - Side effects: liver toxicity, gout, nausea, vomiting, diarrhea, hyperglycemia, flushing (use aspirin) - WOF LFT's and CK levels, blood glucose
Fibrates ("fib") - Used for lowering triglyceride levels, increases HDL - Action: lipoproteins lipolysis - Side effects: liver toxicity, gallstones, GI upset, nausea and vomiting - WOF LFT's and CK levels - GemFIBrozil, CloFIBrate, FenoFIBrate	Key Takeways: For all medications in this class 1. All will affect the liver and lipid profile 2. Medications will most likely not be prescribed to lower cholesterol during pregnancy 3. LFT's a. AST - 10-40 U/L b. ALT - 7-56 U/L c. ALP - 40-120 U/L d. Bilirubin - <1 mg/dl 4. Lipid Panel a. Total cholesterol: <200 b. Triglycerides: <150 c. LDL: <100 d. HDL: >60

Insulin Medications

Rapid - Acting Insulin 1. Aspart 2. Lispro 3. Glusiline Onset: 15 minutes Peak: 30-90 minutes Duration: 3-5 hours	Short - Acting Insulin (Regular Insulin) - Key: this is the ONLY insulin type that can be given via IV - Can be given with BPH at the same time with the same syringe - Can be given with long - acting at the same time in a different syringe Onset: 30-60 minutes Peak: 2-4 hours
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	Duration: 5-8 hours
Intermediate - Acting Insulin (NPH) - Key: if given with regular insulin, draw up: clear to cloudy - Regular Insulin before NPH - Given 2x a day Onset: 60-120 minutes Peak: 4-12 hours Duration: 14 hours	Long - Acting Insulin - Key: Can't be mixed with other insulin 1. Detrimir 2. Lantus 3. Glargine Onset: 60-120 minutes Peak: none Duration: 24 hours

Anticovulsants Medications

Phenytoin - Action: slows down impulses in the brain that cause seizures - Use: treat and prevent seizures - Side effects: gingival hyperplasia, diplopia, rash, low BP - Signs of toxicity: nystagmus, slurred speech, ataxia, vomiting - Therapeutic range: 10-20 mcg/ml - Key: - decreased effectiveness of oral contraceptives - Avoid milk and antacids (decrease absorption) - Take with food - regular dental check up	Lamotrigine - Action: inhibits sodium transport = decrease seizure - Use best for partial seizure, mood stabilizer - Side effects: increased risk for suicidal thoughts, double vision, headache, drowsy, dizziness, nausea and vomiting - Key: - Watch out for worsening of depression or mood changes - Do not stop abruptly - Decreased effectiveness of oral contraceptives
Barbiturates (Phenobarbital) - Action: stimulates GABA -> elevates seizure threshold -> reduce excitement in brain - Use: tonic-clonic seizure, focal, status, epilepticus - Side effects: drowsiness, ataxia, respiratory depression, low BP - Therapeutic range: 15-40 mcg/ml - Key: monitor breathing and routine neurochecks	Valproates (Valproic Acids) - Action: not fully understood, may increase levels of GABA - Use: simple and complex seizures - Side effects: nausea, vomiting, diarrhea, dizziness, drowsiness, tremor, tinnitus, increased bleeding time, low platelet count - Key: monitor WBCs, platelets, liver enzymes, LFT's
Carbamazepine - Action: inhibits sodium transport -> decreased seizures	



- Use: tonic-clonic seizure, partial, mood stabilizer
- Side effects: agranulocytosis (infection risk), hyponatremia, dizziness, drowsiness, nausea and vomiting
- Key:
 - Watch out for worsening of depression or mood change
 - Do not stop abruptly

Antipsychotics Medications

First Generation (Typical/Conventional)	Second Generation (Atypical)
- Haldol or -zine - Action: block D2 receptors on all pathways of the brain - Use: (+) symptoms of schizophrenia - Side effects: - Extrapyramidal symptoms: parkinson-like symptoms, dystonia, shuffling gait, tardive dyskinesia (give benztropine) - Neuroleptic malignant syndrome: high fever, variable BP, increased HR, muscle rigidity, confusion (give muscle relaxant) - Anticholinergic symptoms, orthostatic hypotension, seizures	- apine, -done - Action: block D2 and serotonin receptors (level out amount of dopamine) - Use: (+) & (-) symptoms of schizophrenia - Side effects: - Metabolic symptoms: hyperglycemia, increased weight, increased cholesterol - Anticholinergic symptoms, decreased libido, sedation, orthostatic hypotension - Key: - Clozapine - agranulocytosis - Risperidone - can be injected every 2 weeks via IM
Key takeaways: for all psychiatric conditions 1. Dopamine: determination for a reward, obsession, pleasure 2. Norepinephrine: increased HR/BP, on alert, recall good 3. Serotonin: sleep, emotions, good memory (+) Schizophrenia symptoms: experiencing things you shouldn't be (hallucinations, delusions) (-) Schizophrenia symptoms: lack of (characterized that should be mere)	

Mood Stabilizers Medications

Lithium	Anticonvulsants
- Action: helps reduce severity and frequency of manic episodes - Side effects: leukocytosis, increased urination/thirst, tremors, hypothyroidism,	- Valproic acid - acute mania attacks preferred over lithium due to less side effects - Carbamazepine - agranulocytosis



nausea, fatigue - Key: - Similar to salt - do not cut out salt from your diet (increased risk of toxicity) - Therapeutic range: 0.6-1.2 mEq/L - Weekly blood draws until levels are steady then 3 months interval - Monitor for salt loss (fever, sweating, diarrhea) - Teratogenic - 1-3 weeks for full effect	- Lamotrigine
Antipsychotics - olanzapine - risperidone - aripiprazole	Benzodiazepines (-pam, -lam)

Antianxiety Medications

Benzodiazepines (-pam, -lam) - action: bind to GABA receptors to enhance the action of GABA - GABA: neurotransmitter that slows (calms) the speed of brain activity - Side effects: sedation, decreased RR/BP/HR, amnesia - Key: fast acting (-pam) short term use only, can cause dependency - Antidote: flumazenil	SSRI (Selective Serotonin Reuptake Inhibitors) - Use: anxiety, depression, PTSD, OCD - Side effects: weight gain, insomnia, sexual dysfunction - Key: do not take with St. John's wort - Suffix: -tine, -line, -pram - 4 weeks for full effect
Atypical Antipsychotic (Buspirone) - Use: anxiety, PTSD, OCD, Panic - Side effects: dizziness, headache, nausea - Key: does not cause sedation or dependency - Take with food if GI upset - 1-2 weeks for full effect	Others - Tricyclic antidepressants - Beta blockers - Barbiturates (-barbital) - stays in system for 3-5 days

Antidepressants Medications

Selective Serotonin Reuptake Inhibitors (SSRI) "-tine, -line, -pram"	Serotonin - Norepinephrine Reuptake Inhibitors (SNRI)
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- Action: inhibit the Reuptake of serotonin back into the neuron -> more serotonin floating around in the brain -> improve mood - Side effects: weight gain, insomnia, sexual dysfunction - Risk: serotonin syndrome: agitation, hallucination, fever, tremors, sweaty	- Action: inhibit the reuptake of serotonin and norepinephrine into the neuron -> more floating in the brain - Side effects: loss of appetite, insomnia, sexual dysfunction
Tricyclic Antidepressants (TCA) “-tryptine, -ipramine” - Action: inhibits, serotonin and norepinephrine reuptake and alpha receptors - Use: depression, anxiety, insomnia, fibromyalgia - Side effects: orthostatic hypotension, histamine -> sedation, muscarinic receptors -> anticholinergic symptoms + urinary retention - Key: be careful in the sun (“cycle”) -> increase fiber and fluids	MonoAmine Oxidase Inhibitors (MAOI) - Action: blocks the destruction of serotonin, norepinephrine, dopamine and tyramine (increase all these in the brain) - use: depression, panic disorder - side effects: agitation, anxiety, low BP, HYPERTENSIVE CRISIS (interacts with many meds) -> DO NOT EAT TYRAMINE FOODS (chocolate, avocado, wine, aged cheese) - Example: Phenelzine, isocarboxazid
Atypical Antidepressant - Bupropion (Wellbutrin) - for depression + smoking cessation - Trazodone - for depression (high dose) + insomnia	Key Takeaways: - Take 2-4 weeks to take full effects - Risk for suicide

Antiparkinson Medications

Anticholinergic Agents (Benztropine) - Action: antagonizes acetylcholine and histamine receptors - Use: decreases muscle rigidity and tremors - Side effects: anticholinergic symptoms: dry mouth, dry eyes, blurry vision, urinary retention, constipation - Teach: chew um (dry mouth), wear sunglasses, increase fiber and fluid - Key: do not give to patient with glaucoma, no alcohol, don't stop abruptly	MAO-B Inhibitors (Rasagline aka Azilect) - Action: increase dopamine by inhibiting MAO activity - Use: improve shaking + stiffness, decrease amount of offtime (periods of slow movements) - Side effects: dizziness, drowsiness, nausea, heartburn, dry mouth - Key: avoid foods with TYRAMINE
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COMT Inhibitors (-capone) - Action: decrease wearing off of carbidopa/Levodopa by blocking COMT enzyme - Se: with carbidopa/Levodopa (slows breakdown of levodopa)	Dopamine Agonist Carbidopa/Levodopa - decrease muscle rigidity and tremors - Action: increase dopamine -> carbidopa prevents Levodopa from being converted into dopamine into the bloodstream -> increasing dopamine entering the brain - Side effects: Nausea and vomiting, drowsiness, dyskinesia, dark urine/ sweat, low BP - Key: May take 3 weeks to be effective, increased protein = increased effectiveness Ropinirole (aka Requip) - improve tremors, muscle spasms, stiffness - may also be used for restless leg syndrome - Side effects: drowsiness, confusion, orthostatic hypotension (risk for fall)
Key Takeaways: - Remember: patients have either too little dopamine or too much acetylcholine - Characteristics: - Pin rolling tremors - blank expression - Stooped posture - Shuffling gait - Muscle rigidity - Possible mental retardation	

Spasmolytics Medications

Dantrolene - Direct acting on the muscle cell - Use: decreased muscle spasticity in spinal injuries, cerebral palsy, MS, malignant hypothermia - Action: prevents calcium release in skeletal muscle = decreased muscle spasticity - Side effects: drowsiness, muscle weakness, GI upset, hepatotoxicity - Watch: liver enzymes	Baclofen - centrally acting on the CNS - Action: enhances GABA in the CNS = decreased muscle spasticity - Side effects: drowsiness, nausea, GI upset, constipation - Teach: increased fiber and fluids
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Tizanidine

- Centrally acting on alpha 2 adrenergic agonist
- Action: inhibits presynaptic motor neurons to decrease number of spasms, alleviate clonus and improve mobility
- Side effects: less muscle weakness than baclofen, dry mouth, drowsiness, dizziness

ADHD Medications

1. **Methyphenidate (Ritalin)** - short-acting
2. **Amphetamine (Adderall)**
 - a. **Short-acting:** onset 30-45 min, wear off 3-6 hours
 - b. **Long-acting (XR - extended release):** onset 45 min-1 hour, weak off 8-12 hours

Side effects:

- Weight loss
- Decreased appetite
- Insomnia
- Dysrhythmias
- Moody/irritable

Key:

- do not take at night - insomnia
- take immediately before/after a meal
- Monitor weight

Withdrawal Medications

Alcohol - As early as 6 hours, peak: 24-48 hours - Know time of last drink - Benzodiazepines: decreased withdrawal, risk of seizures and stabilize vitals - Anticonvulsants: decreased risk of seizures - Antihypertensives (Clonidine) + BB: decreased HR/BP, cravings	Opioid - Methadone (long term maintenance)
Nicotine - Bupropion - Nicotine replacement - Varenicline	Abstinence - suppress patient craving - Disulfiram - Naltrexone - Acamprosate (decreased anxiety/restlessness)



	Bad with alcohol: nausea and vomiting, palpitations
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Gastrointestinal Medications

Drug Classification	Suffix	Examples	Action
Antiemetics	-azine	promethazine	Treat and prevent nausea and vomiting
Proton Pump Inhibitors	-prazole	omeprazole	Anti-ulcer drugs that reduce gastric acid production
H ₂ - Receptor Antagonists (H ₂ Blockers)	-tidine	cimetidine	Block the action of histamine in the stomach, decreasing the production of stomach acid

Antiviral Medications

- agents used to treat the diseases caused by viruses like chicken pox and the common cold
- MOA: vir drugs inhibit viral DNA polymerase and prevent replication of viral DNA
- Key:
 - Prevent viral DNA replication
 - Do not cure (only reduce symptoms)
 - Avoid sex when lesions are present (use condom)
 - Topical (wear gloves when administering)
 - IV (watch for phlebitis)
 - May cause GI upset and nephrotoxicity
- Acyclovir
 - Used for: herpes simplex virus (cold sores, encephalitis, shingles, genital infections, varicella zoster)
 - DOC for kids (given IV, oral or topical)
 - WOF: signs of renal impairment, pregnancy
 - Increased dose = nephrotoxicity and neurotoxicity (headache, confusions, hallucinations)
- Ganciclovir
 - Used for: cytomegalovirus (infections affecting the eyes, respiratory tract and liver)
 - Given: IV
 - WOF: bone marrow suppression
 - Monitor signs of infection and CBC



Analgesics Medications

Class	Drug	Uses/Types of Pain	Side effects	Considerations
NSAIDs	Ibuprofen Naproxen Aspin Celecoxib Diclofenac Indomethacin	Pain relief Fever reduction Arthritis treatment Inflammation reduction Injury management	Stomach pain or upset Nausea and vomiting Heartburn or indigestion Diarrhea or constipation Dizziness or lightheadedness Headache	Assess for allergies and medical history Monitor for signs of bleeding Monitor for signs of stomach ulcer
Opioid Analgesics	Codeine Fentanyl Hydrocodeine Hydromorphone Meperidine Methadone	Acute pain Chronic pain Neuropathic pain	Constipation Nausea and vomiting Drowsiness or sedation Dizziness Respiratory depression	Monitor for side effects and adverse reactions Assess pain before and after administration Use caution in patients with respiratory problems
Salicylates	Aspirin Salsalte Choline salicylate Diflunisal	Pain relief Fever reduction Anti-inflammatory Blood clot prevention	Stomach upset or indigestion Nausea and vomiting Heartburn Increased risk of bleeding tinnitus	Assess patient's pain level before and after administration Monitor for GI side effects
Cox-2 Inhibitors	Celecoxib Etoricoxib	Pain relief Inflammation reduction Acute pain	Stomach upset Diarrhea or constipation Headache Elevated BP	Monitor for BP and kidney function regularly Advice patient to take with food or milk to minimize stomach upset
Antimigraine	Sumatriptan	Migraine with	Change in vision	Assessing the



Agents	Rizatriptan Zolmitriptan Naratriptan Eletriptan Almotriptan	aura Migraine without aura Chronic migraine Menstrual-related migraine Cluster headache	Slurred speech Seizure Allergic reaction	patient's medical history and any potential contraindications or drug interaction Monitoring adverse effects and providing appropriate interventions as needed
Miscellaneous Analgesics	Acetaminophen Tramadol Methadone Tapentadol Buprenorphine	Mild to moderate pain Fever Headache Migraines Menstrual cramps	Nausea and vomiting Dizziness and drowsiness Headache Constipation	Monitor for signs of overdose or toxicity Educate patient on safe use and potential side effects Use caution in patients with liver or kidney disease

Tuberculosis Medications

Key things:

- Treatment lasts approximately 6-9 months
- Sputum samples are needed every 2-4 weeks
- No longer infectious when 3 negative sputum cultures on 3 different days
- Always wear N95 mask
- Family should be tested (the baby is safe in pregnancy)

1. Rifampicin

- a. Causes: red/orange tears, sweat, urine = normal
- b. Teach wear glasses, not contacts
- c. Makes oral contraceptive ineffective
- d. Monitor jaundice - liver toxicity

2. INH (Isoniazid)

- a. Interferes with absorption of vitamin B6 (causes peripheral neuropathy)
 - i. Take B6: 25-50 mg/day
- b. Causes neuropathy - report numbness, tingling, ataxia
- c. Liver toxicity - monitor jaundice, dark urine, fatigue, elevated LFTs

3. Ethambutol



- a. “E” for eyes - report blurred vision, vision change, color change
 - b. Get routine eye exams
 - c. Generally safe for liver
- 4. Other: Pyrazinamide (hepatotoxic)**

Chemotherapy Medications

1. Leuprolide (Prostate Cancer)

- a. Action: decrease levels of testosterone
- b. Side effects: hot flushes, bone loss, low libido, gynecomastia (increased breast tissue)
- c. Key: Encourage calcium and vitamin d, assess for bone pain/urine output/bladder distention
- d. Monitor prostate specific antigen + testosterone level

2. Tamoxifen (Breast Cancer)

- a. Action: competes with estrogen for binding sites in the breast and stops the growth of estrogen dependent cancer
- b. Side effects: hot flushes, GI upset, increased calcium, increased risk for pulmonary embolism and endometrial cancer
- c. Key: Monitor: calcium levels CBC, platelets

3. Interferon Alpha-2B (Biologic Response Modifier)

- a. Use: cancer and hepatitis - interferes with cancer
- b. Action: increased immune response to tumors and viruses; decreased cancer cell production
- c. Side effects: flu-like symptoms, bone marrow suppression, cardiac toxicity, neurotoxicity, GI upset, alopecia

4. Doxorubicin (Antitumor Antibiotic)

- a. Action: binds to DNA - inhibits DNA/RNA synthesis -> death to rapidly dividing cells
- b. Side effects: GI upset, alopecia, cardiac toxicity, bone marrow suppression, anemia, neutropenia, thrombocytopenia, red discoloration of urine, sweat, tears
- c. Key:
 - i. monitor signs and symptoms of infection
 - ii. Monitor for cardiac function and CBC
 - 1. Ruby red for red discoloration, the heart and blood for bone marrow

For Variety of Cancers:

5. Vincristine (Anti-mitotic)

- a. Action: stops cell division in mitosis
- b. Side effects: peripheral neuropathy, alopecia, phlebitis, GI upset
- c. Key:
 - i. Monitor signs and symptoms of neuropathy (tingling)
 - ii. Give antiemetics



- iii. Administer best by PICC line or central line
- iv. Does not cause bone marrow suppression
- v. Too much wine you can't feel your arm and legs

6. Cyclophosphamide (Alkylating agent)

- a. Action: inhibits protein synthesis
- b. Side effects: GI upset, alopecia, bone marrow suppression, anemia, leukopenia, thrombocytopenia, hemorrhagic cystitis
- c. Key:
 - i. Increase fluid intake
 - ii. Monitor for blood in urine
 - iii. Monitor CBC, give antiemetics

Trauma Medications

1. Lidocaine

- a. Action: temporarily numbing and relieving pain
- b. IV - to treat VFib/VTach
 - i. Suppresses ventricular cells
 - ii. decreased diastolic depolarization
- c. Used for: local anesthetic (and topical), ventricular arrhythmias

2. Epinephrine

- a. Action: stimulate alpha + beta adrenergic receptors = relaxes cardiac and lung smooth muscle + dilation of vessels in skeletal muscle
- b. Used for: bronchodilation in these cases:
 - i. Anaphylaxis
 - ii. Hypersensitivity reaction
 - iii. Acute asthma attack
 - iv. Chronic simple glaucoma

3. Atropine

- a. Action: inhibits acetylcholine in lung smooth muscle + cardiac muscle blocks SA node and AV node = increased HR
- b. Used for: decreasing respiratory secretions, sinus bradycardia
- c. WOF: overdose of anticholinergic drugs

4. Narcan

- a. Action: treat overdose of narcotic meds
- b. Used for: opioid induced toxicity, respiratory depression, neonates affected by narcotics given during labor

Antibiotics Medications

Broad Spectrum (-cillin, - cef/-ceph) - Safe for pregnancy and breastfeeding - Allergy to -cillins, assess type of reaction first before giving a -ceph drug - -Cillins (Penicillin) - Watch with oral contraceptives - Take with food, shake well - Side effect: bleeding - -Cef/-ceph (Cephalosporin) - Can cause C.difficile, does not treat C. Difficile - Take with food if GI upset - Side effect: diarrhea (common) - 3rd generation cephalosporin -> ceftriaxone -> highest penetration to the BBB - Do not administer with calcium containing IV (LR) - Damage the lungs and kidneys (crystallization) - No alcohol intake - Disulfiram effect reaction - Alcoholism (aversion therapy) -> disulfiram (Antabuse) -> mere smell or taste of alcohol produces nausea and vomiting, palpitation, headache, flushing, hypotension	Highly Toxic Antibiotics (-mycins/-micin) - Use: MRSA and C.difficile - Toxic to kidney, ears and liver - Monitor peak/trough level 15-30 mins before given - Monitor signs and symptoms of toxicity - Kidney: elevated BUN, creatinine, decreased urine output - Ears: vertigo, tinnitus - Glycopeptide (Vancomycin) - Red man syndrome - not deadly (sudden onset of rash, flushing, hypotension) - Slow infusion over 1 hour - Aminoglycosides (Gentamicin, Tobramycin) - Very toxic with vancomycin - No red man syndrome
-Thromycin (Azithromycin, Erythromycin) - Big liver toxic - watch AST/ALT and when patient is taking tylenol - Increased Q interval = cardiac arrest - Take on empty stomach with full glass of water (no food) -Cyclines (Tetracycline, Doxycycline) - teratogenic - also watch or contraceptives - Tooth discoloration - Avoid the sun - When taking:	Urinary Tract Infection (sulfa/-floxacin) 1. Sulfa - stops folic acid synthesis a. Teratogenic b. Avoid sun c. WOF: increased crystals, specific gravity d. Encourage water and folic acid intake 2. Floxacin a. Avoid the sun b. Take with water - no food



- no food with water - No calcium and iron - blocks absorption - sit up for 30 mins after taking (risk for esophagitis)	c. Report muscle pain (increased risk for Achilles tendon tear) Phenazopyradine (Pyridium) - Not an antibiotics (supportive care) - Normal: red urine body fluid - WOF: jaundice (Big liver toxic)
Metronidazole (-dazole) AKA Flagyl - Use: C.difficile, STI's (Trichomoniasis) - Avoid alcohol during and 3 days after treatment - Normal: brown urine, metallic taste - Bad: new rash/skin peeling (Steven Johnson Syndrome) Antifungal (Fluconazole) - Use: candidiasis - Watch in pregnancy: teratogenic, only given when benefit outweigh risk - Side effect: GI toxicity (Watch LFTs, CBC, renal) - Contraindicated: phenytoin, digoxin, anticoagulants	Key Takeaway 1. Ineffective oral contraceptive (-cillins,-cyclins) 2. No food, with water (-thromycin, Tetracycline, -floxacin) 3. No sun (-cycline,sulfa,-floxacin) 4. Extremely toxic (mycin) 5. Teratogenic (-cycline,sulfa,antifungal) 6. Finish the prescription 7. True penicillin allergy a. Wheezing b. Fever c. Rashes

Common Vitamin Deficiency

Vitamin C (Ascorbic Acid)

- Vitamin C is essential for the production of collagen, a protein that helps maintain the health of skin, blood vessels, bones. And cartilage
- Scurvy is a condition caused by deficiency of vitamin C
 - **Symptoms:**
 - Fatigue
 - Swollen, bleeding gums
 - Easy bruising
 - Joint pain
 - Dry skin and hair
 - Anemia
 - Poor wound healing
 - Bleeding under the skin known as petechiae
 - **Causes:**



- insufficient intake of vitamin C which is found in citrus fruits, strawberries, tomatoes, bell peppers, and leafy greens
- **Treatment:**
 - Vitamin C supplements and vitamin C-rich foods
- **Prevention:**
 - ensuring a balanced diet with plenty of fruits and vegetables high in Vitamin C

Vitamin B12 (Thiamine)

- Water-soluble vitamin that plays a critical role in converting carbohydrates into energy
- **Function:**
 - **Energy production:** involved in metabolism of carbohydrates, helping the body convert them into glucose for energy
 - **Nervous system health:** essential for nerve function, supports production of neurotransmitters which allows the brain and nervous system to communicate
 - **Heart and muscle function:** important for maintaining proper heart rhythm and muscle function
- **Sources:**
 - Whole grains (brown rice, oats, wheat bread)
 - Pork
 - Legumes (beans, lentils)
 - Nuts (sunflower seeds, almonds)
 - Seeds (flaxseeds)
 - Fortified cereals and some processed foods (bread and pasta)
 - Vegetables (peas, asparagus, spinach)
- **Deficiency:**
 - **Dry Beriberi:** affects the nervous system leading to symptoms like weakness, numbness of limbs, difficulty walking
 - **Wet Beriberi:** affects cardiovascular system leading to symptoms like swelling in the legs, SOB, heart failure
 - **Wernicke-Korsakoff syndrome:** often seen in alcoholics, causes confusion, memory loss and problems with coordination
- **Symptoms:**
 - Fatigue and weakness
 - Irritability and mood changes
 - poor appetite and weight loss
 - muscle weakness and difficulty walking
 - Swelling in legs
 - Heart palpitations and SOB
- **Treatment:**



- Thiamine supplements
- **Prevention:**
 - A balance diet rich in thiamine-containing foods

Vitamin B2 (Riboflavin)

- A water-soluble vitamin that plays an important role in energy production as well as maintaining healthy skin, eyes and nervous system. It essential for the metabolism of carbohydrates, fats, and proteins
- **Functions:**
 - **Energy production:** riboflavin is a key part of the body's process of converting food into energy. It helps convert carbohydrates into glucose, which is used for energy
 - **Antioxidant properties:** it helps protect cells from oxidative stress by supporting the production of glutathione, a power antioxidant
 - **Red blood cell production:** riboflavin helps in the production of RBCs, which are crucial for transporting oxygen throughout the body
 - **Maintaining healthy skin and eyes:** it supports skin health, prevents skin conditions like cracks around the mouth and helps maintain healthy vision
- **Sources:**
 - dairy products
 - Eggs
 - Lean meats
 - Green leafy vegetables
 - Nuts and seeds
 - Fortified cereals
 - whole grains
 - Legumes
- **Deficiency:** ariboflavinosis
 - sore throat
 - Cracks or sores on the lips
 - inflamed red tongue
 - Skin rashes
 - Eye problems
 - fatigue and weakness
- **Risk factors**
 - poor diet
 - Alcoholism
 - Pregnancy and breastfeeding
 - Certain medical conditions (thyroid disease or liver disease)
- **Treatment**



- Riboflavin supplements
- **Prevention**
 - A balanced diet rich in dairy products, meats, green leafy vegetables and whole grains

Vitamin B3 (Niacin)

- is a water-soluble vitamin that is important for converting food into energy. It plays a crucial role in metabolism and in maintaining healthy skin, nerves and digestion
- **Function:**
 - **Energy production:** Niacin is involved in converting carbohydrates, fats, and proteins into usable energy for the body
 - **Support for the nervous system:** essential for nerve function and the synthesis of neurotransmitters that facilitate brain communication
 - **Cholesterol regulation:** helps lower LDL “bad” cholesterol and raise HDL “good” cholesterol levels which can help protect against heart disease
 - **Skin and digestive health:** helps maintain the health of the skin and the digestive system
- **Sources:**
 - Meat
 - Whole grains
 - Legumes
 - Nuts and seeds
 - Fortified cereals
 - Vegetables
 - Dairy products
- **Deficiency:**
 - **Pellagra**
 - **Dermatitis:** skin problems like red, scaly rash usually on areas exposed to sunlight
 - **Diarrhea:** digestive issues including frequent diarrhea
 - **Dementia:** mental symptoms such as confusion, memory loss and mood changes
 - **Causes:**
 - **Poor diet:** a diet lacking in niacin-rich foods or tryptophan
 - **Chronic alcoholism:** interferes with the absorption of niacin and its conversion from tryptophan
 - **Malabsorption conditions:** certain gastrointestinal diseases or surgeries can limit the body’s ability to absorb niacin
 - **Treatment:**
 - Niacin supplements
 - Dietary changes
 - **Niacin toxicity:**



- Flushing
- Itching or rashes
- Liver damage
- Gastrointestinal issues
- **Recommended daily intake**
 - Men: 16 mg/day
 - Women: 14 mg/day
 - Pregnant/breastfeeding women: 17-18 mg/day
- **Prevention:**
 - Eating well-balanced diet with niacin-rich foods

Vitamin B5 (Pantothenic Acid)

- Water soluble vitamin that is essential for the synthesis of coenzyme A (CoA), which plays a key role in energy production and numerous metabolic processes
- **Function:**
 - **Energy metabolism:** crucial for converting carbohydrates, fats and proteins into usable energy. It helps synthesize coenzyme A, which is involved in the citric acid cycle (Krebs cycle), central part of energy production
 - **Synthesis of hormones:** it helps in the production of adrenal hormones, including cortisol, which are important in stress response and metabolism
 - **Wound healing:** plays a role in tissue repair and skin health, it may assist in the healing of wounds
 - **Synthesis of neurotransmitters:** involved in the production of acetylcholine, a neurotransmitter for memory and muscle control
 - **Cholesterol and fat synthesis:** it helps in the synthesis of cholesterol and other fatty acids, contributing to overall metabolic health
- **Sources:**
 - meat
 - Poultry
 - Fish
 - eggs
 - Whole grains
 - Legumes
 - avocados
 - Potatoes
 - Yogurt and dairy products
 - Vegetables
- **Deficiency:** burning feet syndrome
 - Fatigue and weakness



- Irritability
- Digestive issues
- Skin problems
- Neurological problems
- Pain and tingling
- **Causes:**
 - Severe malnutrition
 - Alcoholism
- **Treatment:**
 - Pantothenic acid supplements
 - Dietary changes
- **Recommended daily intake:**
 - Adults: 5 mg/day
 - Pregnant and breastfeeding women: 6-7mg/day
- **Prevention:**
 - Consuming balanced diet with adequate pantothenic acid intake

Vitamin B6 (Pyridoxine)

- A water soluble vitamin that is essential for a wide range of physiological functions in the body
- It plays a role in metabolism, nerve function and the synthesis of neurotransmitters and hemoglobin
- Function:
 - Amino acid metabolism: crucial for the metabolism of proteins and amino acids. It helps in the conversion of amino acids, which are the building blocks of proteins, and in the synthesis of neurotransmitters like serotonin and dopamine
 - Production of neurotransmitters: it supports the synthesis of key transmitters (like serotonin, norepinephrine, and GABA) which are important for mood regulation, brain function and mental health
 - Hemoglobin production: involved in the production of hemoglobin, the protein in the red blood cells that carries oxygen throughout the body
 - Immune system support: it helps maintaining a healthy immune system, supporting the body's ability to fight infections
 - Cognitive function: plays a role in cognitive function and brain development, and has been linked to improved mood and reduced symptoms of depression
 - Hormone regulation: helps regulate hormones, including those involved in menstruation and premenstrual syndrome (PMS)
- **Sources:**
 - Poultry
 - Fish



- whole grains
- Vegetables
- Legumes
- bananas
- Nuts and seed
- Fortified cereals
- Beef liver
- Spinach
- **Deficiency:**
 - Anemia: due to impaired hemoglobin production
 - Depression and confusion
 - Peripheral neuropathy: nerve damage
 - Irritability and mood changes
 - Weakened immune function
 - Cracks or sores around the mouth (cheilosis)
 - Glossitis: inflammation of the tongue
 - Seizures: in severe cases especially in infants
- **Causes:**
 - Inadequate dietary intake
 - Chronic alcoholism
 - Certain medical conditions (kidney disease, malabsorption disorders)
 - Medications (isoniazid, certain anticonvulsants)
- **Treatment:**
 - Vitamin B6 supplements
 - Dietary changes
- **Recommended daily intake:**
 - Adults: 1.3-2.0 mg/day
 - Pregnant/Breastfeeding women: 2.0-2.2 mg/day
- **Toxicity:**
 - Nerve damage (sensory neuropathy)
 - Difficulty walking and lack of coordination
 - Nausea and gastrointestinal issues
 - Sun sensitivity and rashes
- **Prevention:**
 - balanced diet

Vitamin B7 (Biotin)

- Also known as vitamin H, a water soluble vitamin that plays an essential role in metabolism of carbohydrates, fats and proteins



- Often associated with healthy hair, skin and nails but is crucial for many bodily functions beyond beauty
- **Functions:**
 - Metabolism of macronutrients: is a coenzyme involved in the metabolism of carbohydrates, fats and proteins, helps convert food into energy by aiding enzymes that break down these macronutrients
 - Gene regulation: helps regulate genes involved in cellular processes and the production of DNA
 - Skin, nails, and hair: its role in promoting health hair, skin, and nails, helps strengthen hair and nails and may prevent hair thinning
 - Brain and nervous system health: plays a role in supporting the health of the nervous system and brain function
 - Blood sugar regulation: may assist in controlling blood sugar levels, making it beneficial for people with diabetes and those who are at risk of developing it
- **Sources:**
 - Eggs
 - Nuts and seeds
 - Legumes
 - Whole grains
 - Liver
 - Fish
 - Dairy products
 - Avocados
 - Sweet potatoes
- **Deficiency:**
 - Hair thinning or hair loss (alopecia)
 - Skin rash (seborrheic dermatitis)
 - Cracked, dry skin
 - Fatigue and general weakness
 - Depression and mood changes
 - Numbness and tingling
 - Conjunctivitis (eye inflammation)
 - Increased risk of fungal infections and skin irritation
 - Developmental delay (in infants)
- **Causes:**
 - Poor diet
 - Long-term use of certain medications (anticonvulsants)
 - Chronic alcohol consumption
 - Intestinal conditions (Crohn's or malabsorption syndromes)
 - Pregnancy



- **Treatment:**
 - Biotin supplements
 - dietary changes
- **Recommended daily intake:**
 - Infants (0-6 months): 5 mcg/day
 - Children (7 months-12 years): 6-25 mcg/day
 - Adolescents (14-18 years): 30 mcg/day
 - Adults (19 years and older): 30 mcg/day
 - Pregnancy women: 30 mcg/day
 - Breastfeeding women: 35 mcg/day
- **Prevention:**
 - Balanced diet

Vitamin 12 (Cobalamin)

- Water soluble vitamin that is essential for several functions in the body, including the production of RBCs, DNA synthesis and maintaining proper neurological function
- **Functions:**
 - RBC formation: crucial role in the production of RBCs, a deficiency can lead to megaloblastic anemia, where RBCs are larger than normal and less efficient at carrying oxygen
 - DNA synthesis: vital for division and production of cells
 - Neurological health: formation of myelin sheath that surrounds the nerves, helps in nerve transmission and maintaining cognitive function
 - Energy production: helps convert food into energy by assisting in the metabolism of fats and proteins
 - Mood regulation: synthesis of neurotransmitters like serotonin, which regulate mood
- **Sources:**
 - Meat
 - Fish
 - Poultry
 - Eggs
 - Dairy products
 - Fortified cereals
 - Fortified plant-based milk
- **Deficiency:**
 - Fatigue and weakness
 - Pale and jaundiced skin (due to anemia)
 - Numbness and tingling
 - Difficulty walking and balance problems



- Cognitive issues
- Mood changes
- Glossitis
- SOB
- Vision problems
- **Causes:**
 - Poor dietary intake
 - Malabsorption (Crohn's diseases, celiac disease, gastric bypass surgery)
 - Age (older adults)
 - Intrinsic factor deficiency (pernicious anemia)
 - Medications (PPI and metformin)
- **Treatment:**
 - Vitamin B12 supplements
 - Dietary changes
 - Fortified foods
- **Recommended daily intake:**
 - Adults: 2.4 mcg/day
 - Pregnant women: 2.6 mcg/day
 - Breastfeeding women: 2.8 mcg/day
 - Infants and children: 0.4 to 1.8 mcg/day (depending on age)
- **Prevention:**
 - Balanced diet

Geriatric Nursing

Geriatric Nursing or Gerontological Nursing is a specialized field focused on providing comprehensive care for older adults (typically 60+) to promote their health, independence and quality of life

Theories of Aging

Biological Theories

Aging - decreased cell division -> decreased growth; answers "Why"

1. **Gene theory** - biological timeline
2. **Immunologic theory** - decreased immune -> risk of disease -> cell damage -> decreased cell division
3. **Neuroendocrine/Endocrine theory** - decreased in brain function (hypothalamus -> pituitary gland) -> decreased hormone T3/T4 -> decreased metabolism
4. **Wear and tear theory** - the more you use your body, the quicker you die



5. **Free-radical theory** - unstable molecules -> cell damage
6. **Cross-linkage theory** - the manifestation of aging, tight binding of protein (muscle stiffness) -> decreased elasticity

Psychological Theories

Answers the question “how”

1. **Disengagement theory** - isolated from society -> agesism
2. **Activity theory** - increased productivity
3. **Continuity theory** - maintains values, habit, and behavior
4. **Erikson's psychosocial theory** - ego integrity (death is okay) -> satisfaction vs. despair (regrets)

Aging Changes

Physiologic Changes

General rule: decreased physiologic function, increased risk to disease

- What happen
- RN intervention/health teachings

Integumentary (decreased)

- **Epidermis** (thin, fragile skin -> soft clothes)
 - **Melanin** - acts as natural sunscreen (pale skin, white/gray hair, increased risk of skin cancer -> avoid 10am to 4pm, sunscreen >15 SPF 30 minutes before sun exposure, reapply every 2 hours, light long sleeve)
- **Dermis**
 - **Sweat/sebaceous glands** - dry skin (xerosis) -> moisturizer
 - **Blood supply** - at risk for skin ulcers -> turn patient every 2 hours, apply barrier cream
- **Hypodermis/fat** - insulator, hypothermia -> warm blankets
- **Nail growth** - thick, brittle
- **Brown spots in sun-exposed areas:** lentigo senilis
- **Bruising, bleeding under the skin in older adults:** senile purpura
- **Seborrheic dermatitis** due to xerosis (extreme dryness)

Musculoskeletal (decreased)

- **Muscles:** sarcopenia -> decreased strength, balance (risk for falls) - remove hazards (wet, rags), safety bars (CR, kitchen), adequate lighting, high seat toilet, non-skid/slip footwear, raise side rails
- **Cartilage:** dry -> increased risk of osteoarthritis
- **Bone density:** Osteoporosis

Neurological (decreased)



- **Short term memory**
- **Long term memory** (maintained)
- **Neurons**
- **Melatonin** - short sleep pattern -> drink warm milk (tryptophan = melatonin), avoid stimulants (4pm last intake), alcohol, short day naps (<20-30 minutes)
- **Conduction speed of nerve fibers**
- **Cerebral blood flow**

Endocrine (decreased)

- **Thyroid** - decreased BMR (T3,T4) -> regulates different function of the body (temperature, appetite) - SFF
- **Insulin resistance** (maintained) -> increased blood glucose
- **Sex hormones**

Special Senses (decreased)

- **Vision**
 - **Lens opacity** (increased) - transparency (senile cataract)
 - **Lens elasticity** - crystallins (proteins in the lens) -> senile cataract, presbyopia (farsightedness) -> refer eye doctor
 - **Pupil size**
 - Senile miosis - poor night vision

Mechanisms	Pupil	
Protective	Constrict	Miosis
Adaptive	Dilate	Mydriasis

- **Drusens** (increased) - macular degeneration (dry - common)
- **Hearing**
 - Progressive presbycusis - deceased hair cells in the inner ear
 - Management:
 - Face client
 - Speak slowly
 - Normal volume and low pitch (-) high frequency sounds
 - Repetition, validation
 - Inner and middle ear (Conductive)
 - Inner Ear (Sensironeural)
- **Taste**
 - Tongue - atrophy (herbs and spice)
- **Smell**
 - Olfactory bulb - atrophy (smoke detector)
- **Touch**



- Nerve conduction - decreased pain, temperature (risk for burns)

Cardiovascular (decreased)

- **Heart rate** (maintained) 60-100 bpm
- **Myocardium**: decreased contractility -> decreased cardiac output -> activity intolerance (mild to moderate weight bearing exercise)
- **Heart valves**: rigidity and thickness (with s3 - physiologic murmur)
- **Arteries**: decreased elasticity
- **Veins**: decreased elasticity -> slightly increased BP
- **Poor baroreceptor response** - responsible for BP position changes (orthostatic hypotension) -> gradually reposition

Respiratory (decreased)

- **Respiratory rate** (maintained) 12-20
- **Thoracic muscles**
 - Cough
 - Depth
- **Alveoli** - decreased oxygen, increased carbon dioxide due to small alveoli
 - Fowlers
 - Deep breathing
 - Pursed-lip breathing

Gastrointestinal (decreased)

- **Mouth**
 - **Teeth** (dysphagia) -> airway obstruction (Heimlich maneuver) - soft diet
 - **Salivary gland** - dry mouth (xerostomia) -> increased OFI, ice chips, hard sugarless candy
- **Stomach** - acid & enzymes (breakdown food) - indigestion (SFF)
- **Intestines** - decreased peristalsis (constipation), ambulation, increased OFI & fiber

Renal (decrease)

- **Kidneys** - decreased size
 - **Tubules** - concentrate urine (diluted)
 - **Nephrons** - filtration (low GFR) - accumulation of waste products in the blood, decreased drug metabolites (polypharmacy - med review) -> risk for drug toxicity (ALOC)
- **Bladder** - ability to contract the bladder (bladder tone) -> risk for urine incontinence (increased residual urine) - prone to UTI (ALOC)

Reproductive (decreased)

- **Sexual desire** (maintained)
- **Male**
 - **Testes** - smaller in size, less production of testosterone (andropause)
 - **Penis** - maintained, decreased blood supply to the penis (erectile dysfunction)
 - **Prostate** - slight increase (physiologic), BPH (pathologic)
- **Female** - low estrogen -> menopause, breast atrophy, secretions (dry) -> water-based lubricant



Psychologic Issues

Mental Health

- Isolation
- Depression
- Suicide
- Priority: SAFETY - close supervision
 - Delirium
 - Dementia

Delirium	Dementia
Temporary impairment of neurons	Death of neurons
Reversible, rapid	Irreversible, gradual
ALOC	Normal LOC
Reversible memory loss (surgery)	Irreversible memory loss (Alzheimer's disease)

Abuse

- Primary risk factor: dependent (immobility, mental conditions, physical, mental, emotional, sexual financial abuse)
- Domestic mistreatment
- Institutional mistreatment
- Self-neglect

Nursing Intervention:

- Priority: assess and treat the physical trauma
- Safety
 - NEVER leave the victim alone
 - Remove from the abuser
- Respect (therapeutic communication - "tell me more")
- Report to authorities
- Document (complete, accurate)

Fundamentals of Nursing

Definition of Nursing

- American Nursing Association (2003)
 - Nursing is the protection, promotion and optimization of health and abilities, prevention of illness and injury, alleviation of suffering through the diagnosis and treatment of



human response and advocacy in the care of individuals, families, communities and populations

- **Florence Nightingale**

- Act of utilizing “environment” of the patient to assist him in his recovery

- **Virginia Henderson**

- The unique function of the nurse is to assist the individual, sick or well, in the performance of those activities contributing to health or its recovery that he would perform unaided if he had the necessary strength, will, or knowledge and to do this in such a way as to help him gain independence as rapidly as possible

Nursing Theories

1. Florence Nightingale

- a. Considered the first nursing theorist and earned the title “**Nursing with a Lamp**”

b. Environmental Theory

- i. Pure/fresh air
- ii. Pure water
- iii. Efficient drainage
- iv. Cleanliness
- v. Light (direct sunlight)
- c. Deficiencies in this five factors produce lack of health or illness
- d. Stressed the importance of keeping the client warm, maintaining a noise free environment, attending the client’s diet

2. Virginia Henderson

a. The Nature of Nursing Model

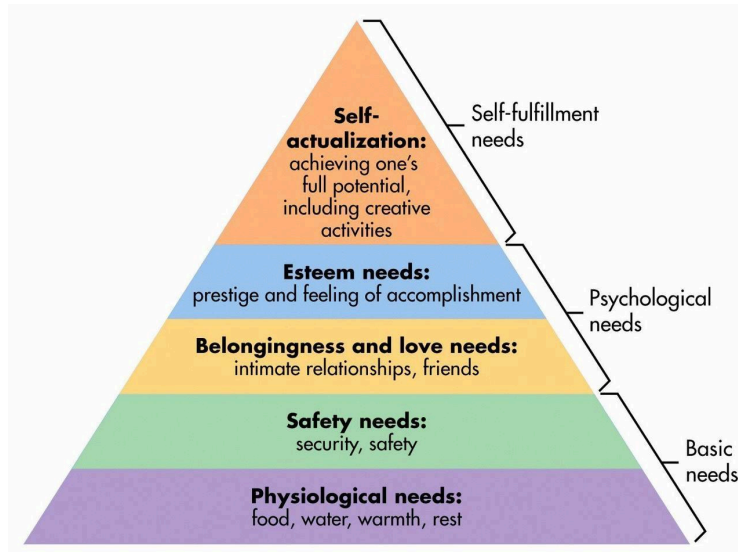
- i. Conceptualizes the nurse’s role as assisting sick or healthy individuals to gain independence in meeting the **14 fundamental needs**
 - 1. Breathing normally
 - 2. Eating and drinking adequately
 - 3. Eliminating body wastes
 - 4. Moving and maintaining a desirable position
 - 5. Sleeping and resting
 - 6. Selecting suitable clothes
 - 7. Maintaining body temperature
 - 8. Keeping the body clean and well-groomed
 - 9. Avoiding dangers and injuring others
 - 10. Communicating others
 - 11. Worshipping according to one’s faith
 - 12. Working in such a way that one feels a sense of accomplishment
 - 13. Participating in various recreation

14. Learning, discovering or satisfying the curiosity that leads to normal development and health

ii. Priority -> ABC, Maslow's Hierarchy of Needs

1. Do not call the doctor unless:

- a. post cataract retraction -> pain
- b. pots renal biopsy -> pain



3. Faye Glenn Abdellah

a. Patient-centered approaches to Nursing model

b. Identifies **21 nursing problems**

i. To maintain good hygiene and physical comfort

1. **Handwashing** -> prevent nosocomial infection

ii. To maintain good body mechanics and prevent and correct deformity

1. **Ergonomic principles**

- a. Wide base of support -> feet wide apart
- b. Bend on the knees
- c. Be as close to the object being lifted as possible
- d. Before moving contract the gluteal, abdominal, arm and leg muscles

iii. To facilitate the maintenance of elimination

1. **Fecal Occult Blood Test (FOBT)**

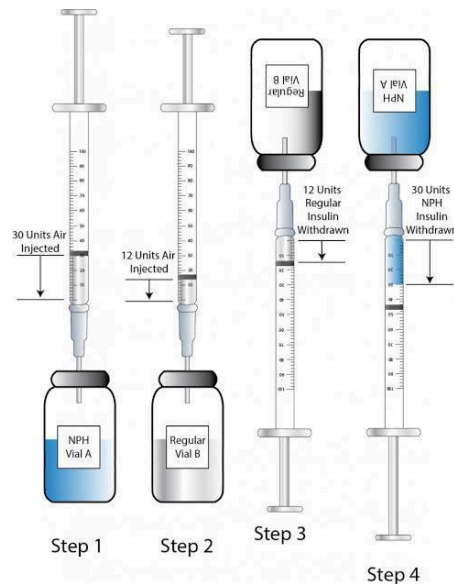
a. **False (+)**

- i. Hemoglobin free diet = 3 days
 1. No red meat, no dark colored food
- ii. Raisins
- iii. Iron
- iv. Chocolate



- v. **Anticoagulant:** prevents clot
 - 1. Heparin: IV - protamine sulfate - safe for pregnancy (PTT: 35-45 secs)
 - 2. Warfarin: oral - vitamin K/Green leafy vegetables - teratogenic (PT: 11.5-13 secs, INR: below 1)
- vi. **Thrombolytics:** dissolves clot
 - 1. Urokinase
 - 2. Streptokinase
 - 3. Alteplase
- b. **False (-)**
 - i. Vitamin
 - ii. Vitamin K
 - iii. Green leafy vegetables
- iv. To facilitate the maintenance of fluid and electrolyte balance
 - 1. **Food poisoning:** bacillus cereus
 - 2. **Sodium:** 135-145 mEq/L
 - a. **Normal:** 2000 mg/day
 - b. **Hypertension:** 1500 mg/day
 - 3. **Potassium:** 3.5-5 mEq/L
 - a. Decreased BP
 - i. **Amlodipine** - vasodilator (CCB) -> S/E: edema
 - ii. **Lozartan** -> ARBs - prevents vasoconstriction
 - b. 90% of potassium is excreted via urine
 - c. CK/RF = low urine output (oliguria) = potassium retention = hyperkalemia
- 4. **Qualification for dialysis**
 - a. **Increased BUN and creatinine**
 - i. **BUN:** 5-20 mg/dl
 - ii. **Creatinine:** 0.7-1.3 mg/dl
 - 1. Limit exercise/activity
 - b. **Fluid retention** = edema
 - i. Limit fluid
 - c. **Potassium:** 6 mEq/L
 - i. Limit fruits and vegetables
- 5. **Diabetes Mellitus:** low or absence of insulin
 - a. **Insulin (SQ):** slow release to the blood
 - i. **Insulin to burn:** to facilitate the intracellular movement of potassium
 - b. **Emergency:** IV regular insulin (short-acting)

- c. **Open vial:** room temperature, label date of opening, 28 days of potency
- d. **Unopened vial:** fridge
- e. **Fatal complication:** hypoglycemia, lipodystrophy, hypokalemia
- f. **Mixing of insulin** in a syringe, not in a vial
 - i. Short-acting (Regular = clear)
 - ii. Intermediate acting (NPH = cloudy)
 - 1. Clear first before cloudy (for assessment)
 - iii. Inject air NPH
 - iv. Inject air Regular
 - v. Withdraw Regular
 - vi. Withdraw NPH



6. Peripheral medication lock (Heplock)

- a. Sterile NSS (flushing)
- b. Administer the medication
- c. Sterile NSS
- d. Heparin
- v. To promote optimal activity: exercise, rest, sleep
- vi. To promote safety through prevention of accident, injury, or other trauma and through prevention of the spread of infection
- vii. To facilitate the maintenance of a supply of oxygen to all body cells
- viii. To facilitate of nutrition for all body cells
- ix. To recognize the physiologic responses of the body to disease condition - pathologic, physiologic, and compensatory
- x. To facilitate the maintenance of regulatory mechanisms and functions



- xi. To facilitate the maintenance of sensory function
- xii. To identify and accept positive and negative expressions, feelings, and reactions
- xiii. To identify and accept interrelatedness of emotions and organic illness
- xiv. To facilitate the maintenance of effective verbal and nonverbal communication
- xv. To promote the development of productive interpersonal relationships
- xvi. To facilitate progress toward achievement and personal spiritual goals
- xvii. To create or maintain a therapeutic environment
- xviii. To facilitate awareness of self as an individual with varying physical, emotional, and developmental needs
- xix. To accept the optimum possible goals in the light of limitations, physical and emotional
- xx. To use community resources as an aid in resolving problems that arise from illness
- xxi. To understand the role of social problems as influencing factors in the cause of illness

- c. Defines nursing as a service to individuals and families
- d. Conceptualizes nursing as an art and science that molds the attitudes, intellectual competence and technical skills of an individual nurse into the desire and ability to help people, sick or well and cope with their needs

4. Dorothy E. Johnson

a. Behavioral System Model

- b. Each person as a behavioral system is composed of **7 subsystems**:
 - i. Injective
 - ii. Eliminative
 - iii. Affiliative
 - iv. Aggressive
 - v. Dependence
 - vi. Achievement
 - vii. Sexual and role identity

5. Imogene King

a. Goal attainment theory

- b. Viewed nursing as an interaction process between patient and nurse that lead to goal attainment
- c. **Problem** (related to) -> **Etiology** (as manifested by/as evidenced by) -> **Signs and symptoms**
- d. Patient has **3 interacting system**
 - i. **Personal system** (individuals)
 - 1. Perception, self-growth and development, body image, space and time
 - ii. **Interpersonal system** (nurse-patient)
 - 1. Interaction, communication, transaction, role and stress



iii. **Social system** (health care system)

1. Organization, authority, power, status, and decision making

6. Madeleine Leininger

- a. **Transcultural Nursing Model** (cultural care diversity and universality theory)
“Ethnonursing”
- b. Emphasizes that human caring, although universal, varies among cultures in its expressions, process and patterns; it is largely culturally derived
- c. **Presents 3 intervention mode:**
 - i. Cultural care preservation and maintenance
 - ii. Culture care accommodation, negotiation or both
 - iii. Culture care restructuring and repatterning

7. Myra Estrin Levine

- a. **Four Conservation Principles**
- b. Proposed principles which are concerned with the unity and integrity of the individual
 - i. **Conservation of energy**
 1. Eat
 - ii. **Conservation of structural integrity**
 1. Avoid wounds = tetany risk
 2. Rabies - 100% preventable and fatal
 - a. Incubation period: days to years
 - b. No signs and symptoms -> until it reaches the brain -> manifest
-> fatal stage
 - iii. **Conservation of personal integrity**
 - iv. **Conservation of social integrity**

8. Betty Neuman

- a. **Health Care System Model**
- b. Asserted that nursing is unique profession in that is concerned with all the variables affecting the individual's response to stress
 - i. **Intrapersonal stressors:** within the individual
 - ii. **Interpersonal stressors:** occurs between individuals
 - iii. **Extrapersonal stressors:** outside the person, environment
- c. Nursing intervention focus on retaining or maintaining system stability

9. Dorothea Orem

- a. **Self-care and Self-care deficit theory**
- b. Defines self-care as performing activities independently by individuals throughout life to promote and maintain personal well-being
- c. **Identifies 3 types of nursing system:**
 - i. **Wholly compensatory:** for individuals who are unable to control and monitor their environment and process information



- ii. **Partly compensatory:** designed for individuals who are unable to perform some, but not all self-care activities
- iii. **Supportive-Educative:** or clients who need to learn to perform self-care measure and need assistant to do so
 - 1. **Primary:** patient is well (health promotion)
 - 2. **Secondary:** patient is at risk (screening, early defection)
 - a. Pap smear, BSE, TSE
 - b. Mammogram: 40 years old (-) deodorant (contains aluminum = cancer)
 - c. Colonoscopy: 50 years old
 - 3. **Tertiary:** prevents complications
 - a. Rehabilitation
- d. **Virchow's Triad**
 - i. **Hypercoagulability of the blood** -> DM, DHN, COC
 - ii. **Venous blood stasis** -> comato lithotomy
 - iii. **Injury to the blood vessel wall**
 - 1. **Thrombophlebitis:** warm to touch
 - 2. **Infiltration:** cold to touch
- e. **5 Cardinal signs of inflammation**
 - i. **Calor:** heat
 - ii. **Dolor:** pain
 - iii. **Tumor:** swelling
 - iv. **Rubor:** redness
 - v. **Functio laesa:** loss of function

10. Hildegard Peplau

- a. **Interpersonal model**
- b. Use of therapeutic relationship between nurse and the client
- c. **4 phases**
 - i. **Orientation:** establish rapport
 - ii. **Identification:** patient is cooperative
 - iii. **Exploitation:** takes advantage of the nurse
 - iv. **Resolution:** discharge

11. Martha Rogers

- a. **Science of unitary human beings**
- b. Views the person as an irreducible whole, the whole being is greater than the sum of its parts
- c. **According to Rogers, unitary man:**
 - i. Is an irreducible, four-dimensional energy field by pattern
 - ii. Manifests characteristics different from the sum of the parts
 - iii. Interacts continuously and creativity with the environment



- iv. Behaves as a totality
- v. As a sentient being, participates creatively in change

12. Sister Callista Roy

a. Adaptation Model

- b. Defines adaptation as the process and outcome whereby the thinking and feeling person uses conscious awareness and choice to create human and environmental integration
- c. Manipulate stressors to facilitate coping
- d. **Goal of model is to enhance life processes through adaptation in four adaptive modes:**
 - i. Physiologic mode
 - ii. Self-concept mode
 - iii. Role-function mode
 - iv. Interdependence mode

13. Lydia Hall

a. Care, Core, Cure Model

- i. **Care** - nurturance and is exclusive to nursing
- ii. **Core** - involves the therapeutic use of self and emphasize the use of reflection
- iii. **Cure** - focuses on nursing related to the physician's order

b. Elderly

- i. Stairs (red and yellow)
- ii. Blinded by blue and green
- iii. Low pitch sound -> speak slowly -> to allow lip reading

c. Infant

- i. Red, black and white: at birth
- ii. Color perception: 8 months
- iii. Hear high pitched sound with tonal variation
- iv. Taste preference: 7 months every 5-7 days

14. Ide Jean Orlando

a. The Dynamic Nurse-Patient Relationship Model

- b. Nurses provide direct assistance to meet an immediate need for help in order to avoid or to alleviate distress or helplessness
- c. **She advocated that the three elements composing the nursing situation are:**
 - i. Client behavior
 - ii. Nurse reaction
 - iii. Nurse action

15. Jean Watson

a. Human Caring Theory

- b. Practice of caring is central to nursing; it is the unifying focus for practice

c. 10 curative factors

- i. Formation of humanistic-altruism system of values



- ii. Instillation of faith and hope
- iii. Cultivation of sensitivity to one's self and others
- iv. Development of helping-trusting relationship
- v. Promoting and accepting the expression of positive and negative feelings
- vi. Systematically using the scientific problem-solving method for decision making
- vii. Promoting transpersonal teaching-learning
- viii. Provision of a supportive, protective and/or corrective mental, physical, societal and spiritual environment
- ix. Assisting with gratification of human needs
- x. Allowance for existential-phenomenological-spiritual forces

16. Rosemarie Rizzo Parse

a. Human Becoming Theory

b. Proposed 3 assumptions about human becoming:

- i. Human becoming is freely choosing personal meaning in situations in the inter subjective process of relating value priorities
- ii. Human becoming is co-creating rhythmic patterns or relating in mutual process with the universe
- iii. Human becoming is contrascending multidimensional with the emerging possibilities emphasizes how individuals choose and bear responsibility for patterns of personal health

Normal Laboratory Values

CBC WBC: 4,000-11,000 RBC: 4.5-5.5 Platelets: 150,000-450,000 Hct Female: 37-47% Hct Male: 42-52% Hgb Female: 12-16 g/dL Hgb Male: 13-18 g/dL	Coagulation PT: 10-13 seconds INR: Not on warfarin: <1 seconds On warfarin: 2-3 seconds (if <2 seconds increase dose) PTT: 25-35 seconds aPTT: 30-40 seconds (Heparin)
BMP Sodium: 135-145 mEq/L Potassium: 3.5-5.0 mEq/L Chloride: 95-105 mEq/L Magnesium: 1.5-2.5 mEq/L Phosphate: 2.5-4.5 mEq/L Calcium: 9-11 mg/dL BUN: 10-20 mg/dL Creatinine Male: 0.6-1.2 mg/dL Creatinine Female: 0.5-1.1 mg/dL	Lipid Panel Total cholesterol: <200 mg/dL Triglycerides: <150 mg/dL LDL: <100 mg/dL HDL: >60 mg/dL



Albumin: 3.4-5.4 g/dL Total protein: 6.2-8.2 g/L Glucose: 70-100 mg/dL (fasting)	
Renal Calcium: 9-11 mg/dL Magnesium: 1.5-2.5 mg/dL Phosphorus: 2.5-4.5 mg/dL	ABG pH: 7.35-7.45 PaCO ₂ : 35-45 mmHg HCO ₃ : 22-26 mmol/L PaO ₂ : 75-100 mmHg O ₂ : 95-100%
A1C Non-diabetic: 4-6% Prediabetic: 5.7-6.4% Diabetic: >6.5% Target: <6.5%	LFT's AST: 10-40 U/L ALT: 7-56 U/L ALP: 40-120 U/L Bilirubin: <1 mg/dL
Common Drugs Digoxin: 0.5-2.0 Carbamazepine: 4-10 Phenobarbital: 15-40 Lithium: 0.5-1.2 Valproic acid: 50-100 Dilantin: 10-20	Other BMI: 18.5-24.9 I&O: >30 ml/hr CD4: 500-1200 cells MAP: 70-105 mmHG

Common Signs

Signs and Symptoms	Description	Associated Diseases
Aaron's sign	Epigastric pain on palpation of McBurney's point	Appendicitis
Abbot's sign	Reversed peristalsis seen in pyloric obstruction	Pyloric stenosis
Battle's sign	Retroauricular ecchymosis (bruising behind the ear)	Basal skull fracture
Blumberg's sign	Rebound tenderness	Peritonitis
Boas' sign	Hyperesthesia below the right scapula	Acute cholecystitis
Carcinoid syndrome	Flushing, diarrhea, wheezing	Carcinoid tumor



Cecil & deBoer's sign	Pain relief when appendix ruptures	Appendiceal rupture
Charcot's triad	Fever, jaundice, RUQ pain	Cholangitis
Chovstek's sign	Facial muscle twitching on tapping facial nerve	Hypocalcemia (tetany)
Courvoisier's sign	Painless palpable gallbladder and jaundice	Periampullary carcinoma (pancreatic cancer)
Cullen's sign	Periumbilical bluish discoloration	Hemorrhagic pancreatitis, ruptured ectopic pregnancy
Dance's sign	Empty right iliac fossa	Intussusception
Danforth's sign	Shoulder pain on inspiration	Hemoperitoneum
Daughter's sign	Multiple hyperechoic liver nodules	Metastatic carcinoma
Fothergill's sign	Mass that does not cross the midline and remains palpable on rectus muscle contraction	Rectus sheath hematoma
Fox's sign	Ecchymosis over the inguinal ligament	Acute pancreatitis
Grey Turner's sign	Flank ecchymosis	Acute hemorrhagic pancreatitis
Hamman's sign	Crunching sound over the precordium in sync with heartbeats	Mediastinal emphysema (Boerhaave's syndrome, pneumomediastinum)
Homan's sign	Calf pain on dorsiflexion of foot	Deep Vein Thrombosis (DVT)
Kehr's sign	Left shoulder pain due to diaphragmatic irritation	Splenic rupture
Krukenburg tumor	Metastatic ovarian tumor from GI primary	Gastric carcinoma
Lasegue's sign	Sciatic pain on straight leg raising	Lumbar disc herniation
Lhermitte's sign	Electric shock sensation down spine on neck flexion	Spinal cord compression (Multiple sclerosis, cervical)



		myelopathy)
Lloyd's sign	Costovertebral angle tenderness	Pyelonephritis, renal stones
McBurney's sign	Tenderness at McBurney's point (1/3 from ASIS to umbilical)	Acute appendicitis
Murphy's sign	Pain and inspiratory arrest on palpation of RUQ	Acute cholecystitis
Obturator sign	RLQ pain with internal rotation of the hip	Appendicitis (pelvic appendix)
Psoas' sign	RLQ pain on passive hip extension	Retrocecal appendicitis
Rovsing's sign	RLQ pain when LLQ is palpated	Appendicitis
Sister Mary Joseph's nodule	Periumbilical metastasis	Gastric, ovarian, or pancreatic cancer
Tinel's sign	Tingling on tapping the nerve	Carpal tunnel syndrome, nerve injury
Trousseau's sign	Carpal spasm with BP cuff inflation	Hypocalcemia
Virchow's node	Left supraclavicular lymphadenopathy	Gastric cancer metastasis

Hallmark Signs and Symptoms

PTB	Low grade afternoon fever	Chicken Pox	Vesicular rash
Pneumonia	Rusty sputum	MI	Crushing stabbing pain which radiates to left shoulder, neck, arms, unrelieved by NTG
Asthma	Wheezing on expiration	LTB	Inspiratory stridor
Emphysema	Barrel chest	TEF	Coughing, choking, cyanosis, continuous drooling



Kawasaki Syndrome	Strawberry tongue	Epiglottitis	Drooling, dysphonia, dysphagia
Pernicious Anemia	Red beefy tongue	Hodgkin's Disease	Painless progressive enlargement of spleen and lymph tissues, reedstemberg cells
Down Syndrome	Protruding tongue	Infectious Mononucleosis	Sore throat, cervical lymph adenopathy, fever
Cholera	Rice watery stools	Parkinson's Disease	Pill-rolling tremors
Malaria	Step ladder like fever with chills	Firbin Hyalin	Expiratory grunt
Typhoid	Rose spots in abdomen	Cystic Fibrosis	Salty skin
Diphtheria	Pseudomembrane formation	DM	Polyuria, Polydipsia, Polyphagia
Measles	Koplik's spots	DKA	Kussmaul's breathing
SLE	Butterfly rashes	Bladder Cancer	Painless hematuria
Liver Cirrhosis	Spider-like varices	BPH	Reduced size & force of urine
Leprosy	Lining face	Pemphigus Vulgaris	(+) Nikolsky's sign
Bulimia	Chipmunk face	Retinal Detachment	Visual floaters, flashes of light, curtain vision
Appendicitis	Rebound tenderness	Glaucoma	Painful vision loss, tunnel/gun barrel/halo vision, peripheral loss
Dengue	Petechiae or (+) Heman's sign	Cataract	Painless vision loss, opacity of the lens, blurring of vision
Meningitis	(+) Kernig's sign and (+) Brudzinki's sign	Retinol Blastoma	Cat's eye reflex
Tetany	(+) Trousseau's sign and (+) Chovstek's sign	Acromegaly	Coarse facial feature



Tetanus	Risus sardonicus	Duchenne's Muscular Dystrophy	(+) Gower's sign
Pancreatitis	(+) Cullen's sign and (+) Grey Turners spot	GERD	Barrett's esophagus
Pyloric Stenosis	Olive like mass	Hepatic Encephalopathy	Flapping tremors
PDA	Machine like murmur	Hydrocephalus	(+) Bossing's sign
Addison's Disease	Bronze like skin	Increased ICP	Hypertension, Bradypnea, Bradycardia (Cushing's triad)
Cushing's Disease	Moon face & Buffalo hump	Shock	Hypotension, tachycardia, tachypnea
Grave's Disease	Exophthalmos	Meniere's Disease	Vertigo & tinnitus
Intussusception	Sausage shaped mass & (+) Dance sign	Cystitis	Burning on urination
MS	Charcot's triad	Hypocalcemia	(+) Trousseau's sign & (+) Chovstek's sign
MG	Descending muscle weakness	Ulcerative Colitis	Recurrent bloody diarrhea
GBS	Ascending muscle weakness	Lyme's Disease	Bull's eye rash
DVT	(+) Homan's sign	Basillar Fracture	Otorrhea
Angina	Crushing stubbing pain relieved by NTG	Orbital Fracture	(+) Battle's sign & Racoon eyes

Scope of Nursing Practice

- Promoting health and wellness

- A process that engages in activities and behaviors that enhance quality of life and maximize personal potential
- **Activities that enhance healthy lifestyles**
 - Improving nutrition and physical fitness



- Preventing drug and alcohol misuse
- Restricting smoking
- Preventing accidents and injury at home and workplace
- **Preventing illness**
 - **Goal of illness prevention program is to maintain optimal health by preventing disease which includes**
 - Immunization
 - Prenatal and infant care
 - Prevention of STI
- **Restoring health**
 - Focuses on the ill client and it extends from early detection of disease through helping the client during the recovery period
 - **Activities include**
 - Providing direct care to the ill client
 - Performing diagnostic and assessment procedures
 - Teaching clients about recovery activities
 - Rehabilitating clients to their optimal functional level
- **Caring for dying**
 - **Comforting and caring for people of all ages who are dying which includes**
 - Helping clients live as comfortably as possible until death
 - Helping support persons to cope with death

Standard of Nursing Practice

- **Assessment:** collect comprehensive data pertinent to the patient's health or situation
- **Diagnosis:** analyzes the assessment data to determine the diagnosis or issue
- **Outcome identification:** identifies expected outcomes for a plan individualized to the patient or the situation
- **Planning:** develops a plan that prescribe strategies and alternatives to attain expected outcomes
- **Implementation:** implements the identified plan
- **Evaluation:** evaluates progress towards attainment of outcomes
- **Quality practice:** systematically enhance the quality and effectiveness of nursing practice
- **Education:** attains knowledge and competency that reflects current nursing practice
- **Professional practice evaluation:** evaluate one's own practice in relation to professional practice standards and guidelines, relevant statutes, rules and regulation
- **Collegiality:** interacts with and contributes to the professional development of peers and colleagues
- **Collaboration:** collaborates with patients, family and others in the conduct of nursing practice
- **Ethics:** integrates ethical provisions in all areas of practice
- **Research:** integrates research findings into practice



- **Resource utilization:** considers factors related to safety, effectiveness, cost and impact on practice on the planning and delivery of nursing services
- **Leadership:** provides leadership in the professional practice setting and the profession

Roles and Function of a Nurse

- **Caregiver**
 - Function as nurturer, comforter, provider
 - **“Mothering actions”** of the nurse
 - Encompasses activities that assist the client physically and psychologically while preserving the dignity of the client
 - Nurse is primarily concerned with the client’s need
- **Communicator**
 - Communicates the identified problem of the client to other health care team
- **Teacher**
 - Nurse teaches client about their health and procedures they need to perform to restore their health
 - Encourages compliance with prescribed therapy
 - **Penicillin**
 - True penicillin allergy
 - Wheezing
 - Fever
 - Rashes
 - **Cefasporin** -> 3rd generation -> ceftriaxone -> highest penetration to BBB
 - Do not administer with calcium containing IV (LR)
 - Causes crystallization that can damage lungs and kidneys
 - Do not take with alcohol
 - Disulfiram effect reaction
 - Promotes healthy lifestyle
 - 30 minutes (2-3 months) -> 45 mins -> max. 60 mins -> stress -> cortisol -> increased BP and blood sugar
 - Any muscle movement = increased creatinine
 - **Glycemic index**
 - **High:**
 - Increased blood sugar
 - Increased glucose is irritant to the nerves
 - Rice, pasta, bread
 - Carbs = gout
 - MilkteaW
 - Watermelon



- Overripened banana
- **Middle:**
 - Grapes (10 pcs)
 - Mangoes
- **Low:**
 - Oatmeal
 - Green leafy vegetables
 - Strawberries
 - Green banana
- **Anti-colon cancer (cruciferous vegetables)**
 - Cabbage
 - Broccoli
 - Cauliflower
 - Malunggay (Moringa)
 - Lettuce
 - **NCLEX** - asparagus, kale, brussel sprouts
- Interprets information to the client
 - Trained medical interpreter
 - Someone unfamiliar with the patient
 - **2 excuses to breach of confidentiality:** (1) communicable disease, (2) endangers public safety
- **Client advocate**
 - Acts to protect the client
 - Nurse assist clients in exercising their rights and help them speak for themselves
 - Promotes what is best for the client, ensuring that the client's needs are met and protecting the client's right
 - Involves concern for and actions in behalf of the client to bring about a change
- **Counselor**
 - Nurse provides emotional, intellectual and psychological support
 - Helps client to recognize and cope with stressful psychologic or social problems; to develop an improve interpersonal relationships and to promote personal growth
 - **Family planning** -> never give advise, only provide options (IUD, condom, COC, BTL, vasectomy)
 - **PEP (post-exposure prophylaxis)** -> taken within 72 hours after exposure -> 28 days regimen (tenfovir)
 - **PrEP (pre-exposure prophylaxis)** -> taken 24-48 hours before exposure
- **Change agent**
 - Nurse assists clients to make modification in their behavior
 - CHN -> COPAR -> self-reliance
- **Leader**



- Influences others to work together to accomplish a specific goal
- **Manager**
 - Nurse plans, give direction, develop staffs, monitors operation, give rewards fairly and represents both staff members and administrations as needed
 - Makes decisions, coordinates activities of others, allocate resources, evaluate care and personnel
 - **Position of authority**
 - **Top:** chief nurse, nursing director
 - **Middle:** supervisor
 - **Low:** head nurse/team leader
 - **First line:** staff nurse
- **Case manager**
 - Works with multidisciplinary health care team to measure the effectiveness of the case management plan and monitor outcomes
- **Researcher**
 - Nurse participates in scientific investigation and uses research findings to improve client care
 - Participates in identifying significant researchable
 - **Purpose:** to enhance or improve the quality of bedside care
 - Yakult (topical use): **decubitus ulcer**
 - **Stage I:** nonblanchable erythema of the skin
 - **Stage II:** epidermis + dermis
 - **Stage III:** subcutaneous
 - **Stage IV:** involves the muscles showing the bones or shows the joint capsule
 - Increased protein intake (meat)
 - Local fruit decreases hyperactivity
 - Marang and durian -> methylphenidate (Ritalin) -> ADHD
 - Papaya (serotonin) and banana -> decreased libido
- **Collaborator**
 - Nurse works in combined effort with all those involved in care delivery

Expanded Career Roles for Nurses

- **Nurse practitioner**
 - Nurse who has advanced education and graduated from a nurse practitioner program
 - Employed in health care agencies or community based settings
 - Deals with non-emergency acute or chronic illness and provide primary ambulatory care
 - Skilled at making nursing assessments, performing PE, counseling, teaching and treating minor and self-limiting illness (viral)



- **Varicella zoster virus (chicken pox)**
 - Vesicular lesions (fluid filled)
 - Pockmarks (crater like)
 - Dew on a rose petal lesion
 - Posterior horn of the spinal cord
 - **Anterior:** motor
 - **Posterior:** sensory
 - **Steroids** - first line of treatment
 - **Herpes zoster virus (shingles)**
 - Unilateral involvement (one-sided lesion)
 - Reactivation of chicken pox
 - Direct skin to skin contact
 - Zebra-like skin (red stripes)
 - Symptom = pain
 - **Herpes simplex virus (HSV)**
 - Blister
 - STD
 - **HSV1:** lesions above the waist
 - **HSV2:** lesions below the waist
 - **Syphilis**
 - Bacterial -> Treponema pallidum
 - **Primary:** chancre sores -> painless ulcers
 - **Secondary:** condyloma latum -> wart-like
 - **Tertiary:** gummas -> sac-like
 - **Neurosyphilis** -> psychotic
 - **DOC:** penicillin
 - **Human papillomavirus (HPV)**
 - Warts
 - Cervical cancer
 - Oral cancer without history of smoking
 - **HPV4:** 4 strains
 - **HPV9:** 9 strains
 - DOH free vaccine - HPV4
 - Grade 4 female pupils
 - 9-14 year olds (boys and girls)
 - **HPV 16-18:** cervical cancer
 - **HPV 6-11:** anogenital warts
 - **Clinical nurse specialist**
 - Has an advanced degree or expertise and is considered to be an expert in a specialized area of practice



- Provides direct client care, educate others, conducts research and manages care
- **Nursing specialty certification program**
 - 10% of RNs
 - Trained by DOH (specialty centers)
 - Free - funded by PCSO, PAGCOR
 - Required to serve an Philippine Hospital for 2 years
- **Certified registered nurse anesthetist (CRNA)**
 - Completed advanced education in an accredited program in the anesthesiology
 - Carries out pre-op and post-op visits and assessment
 - Administers general anesthesia for surgery under the supervision of a physician prepared in anesthesiology and also assesses the postoperative statuses of clients
 - Only of proper training
 - Episiotomy repair -> local infiltration (lidocaine - anesthetic, anti-arrhythmic)
 - **Spinal anesthesia:** spinal headache
 - **Position during:**
 - C position
 - Fetal position
 - Shrimp position
 - **After:**
 - Supine on bed without pillow
 - Flat on the back
 - **General anesthesia**
 - **Complication:** malignant hyperthermia (hereditary disorder)
 - **Gas anesthetic**
 - Halothane
 - Desflurane
 - Isoflurane
 - Sevoflurane
 - Rapid muscle contraction -> friction between the muscles -> increased heat reduction -> malignant hyperthermia
 - **DOC:** dantrolene sodium (dantrium) - heat/light sensitive
- **Nurse-midwife**
 - RN who has completed a program in midwifery and gives prenatal and postnatal care and manages deliveries in normal pregnancies
 - May also conduct pap smears, family planning and routine breast exams
 - **Stages of Labor**
 - **First stage:** regular contraction to full cervical dilation
 - **Longest stage of labor:** Primi - 18 hours; Multi - 12 hours
 - **Latent:** 0-3 cm - longest phase



- **Active:** 4-7 cm
- **Transition:** 8-10 cm - shortest phase
- **When to go to the hospital:** when the contractions are 5 minutes apart (regular), 4 contractions in 20 minutes
- **Second stage:** full cervical dilation to delivery of fetus (1 hr)
 - **Episiotomy** - shorten the second stage of labor
 - **Epidural anesthesia** - painless delivery
 - How often will you check for fetal heart tone during the second stage: every 5 minutes (start partograph - active phase)
- **Third stage:** delivery of fetus to delivery of placenta
 - **Signs of placenta separation**
 - Gushing of blood
 - Involution
 - Rising of the fundus
 - Lengthening of the cord
- **Fourth stage:** delivery of placenta to 1-2 hours after
 - Dangerous stage of the mother -> lochia
- **Postpartum stage - Rubin**
 - **Taking - in:** delivery experience -> perineal care
 - **Taking hold:** baby care -> diaper change, breastfeeding, family planning
 - **Letting go:** accepting new role as a mother -> parental love
- **Nurse researcher**
 - Investigates nurse problems to improve nursing care and to refine and expand nursing knowledge
 - Employed in academic institutions, teaching hospitals and research center, and usually has advanced education at the doctorate level
- **Nurse administrator**
 - Manages client including the delivery of nursing services
 - Functions at various levels of management in health settings
 - **Function**
 - Budgeting
 - Staffing
 - Planning programs
- **Nurse educator**
 - Responsible for classroom and clinical teaching
 - With advanced degree, who teaches in clinical or educational settings, teaches theoretical knowledge, clinical skills and conduct research
- **Nurse entrepreneur**
 - Manages health-related businesses



Patricia Benner's Stages of Nursing Expertise

- **Stage I (Novice)**
 - 0-1 year of experience
 - Performance is limited, flexible and governed by context-free rules and regulations rather than experience
- **Stage II (Advanced beginner)**
 - 1-2 years of experience
 - Based on theories
 - Demonstrates marginally acceptable performance
 - Recognizes meaningful “aspects” of a real situation
 - Experienced enough real situations to make judgements about them
- **Stage III (Competent)**
 - 2-3 years of experience
 - Based on client need
 - Demonstrates organizational and planning abilities
 - Differentiates important factors from less important aspects of care
 - Coordinates multiple complex care demands
- **Stage IV (Proficient)**
 - 3-5 years of experience
 - Based on priorities (ABC, Maslows)
 - Perceives situations as a whole rather than in terms of parts
 - Uses maxims as guides for what to consider in a situation
 - Has holistic understanding of the client, which improves decision making
 - Focuses on long term goal
- **Stage V (Expert)**
 - > 5 years of experience
 - Intuitive instinctive
 - Performance is fluid, flexible, and highly proficient
 - No longer requires rules, guidelines, or maxims to connect an understanding of the situations to appropriate actions
 - Inclined to take a certain action because “it felt right”

Communication in Nursing

- Interchange of information between two or more people: exchange of ideas and thoughts
- In addition, thoughts are conveyed to others not only by spoken or written words but also by gestures or body actions
- **Verbal:** uses spoken or written words



- **Non-verbal:** uses gestures, facial expressions, posture/gait, body movements, physical appearance, eye contact and tone of voice
- **Components**
 - **Sender:** the person who encodes and deliver message
 - **Message:** the content of the communication, may contain verbal, non-verbal and symbolic language
 - **Receiver:** the person who receives and decodes the message
 - **Channel:** means of conveying and receiving messages through visual, auditory and tactile senses
 - **Response/feedback:** message returned by the receiver to the sender

Characteristics of Communication

- **Simplicity:** use of commonly understood words
- **Pace and intonation:** modifies the feeling and the impact of the message
- **Clarity and brevity:** message that is direct and simple
- **Timing and relevance:** require choice of time and consideration of client's interest and concern
- **Adaptability:** message needs to be altered in accordance with behavioral cues from the client
- **Credibility:** means worthiness of belief, trustworthiness and reliability
- **Humor:** used to help client adjust to difficult and painful situation

Documentation

- Written or computer based
- Served as a permanent record of client's information and progress care
- Formal, legal document that provide evidence of a client's care
- **Purposes**
 - Planning client care
 - Communication
 - For legal documents purposes
 - For research
 - For education
 - Reimbursement
 - For statistics, reporting, epidemiology
 - Auditing health agencies
 - Health care analysis
- **Types of records**
 - **Source-oriented medical record (traditional client record/SOMR)**
 - Each person or department makes notations in a separate section/s of client's chart



- Specific information is easier to locate
- **Components**
 - Admission sheet
 - Face sheet
 - Medical history and physical examination and sheet
 - Diagnostic finding sheet
 - TPR graphic sheet
 - Doctor's treatment and order sheet
 - Therapeutic sheet
 - Special flow sheet
 - Medication record
 - Nurses notes
 - Client discharge plan and referral summary
 - Initial nursing assessment
- **Problem-oriented medical record**
 - Data about the client are recorded and arranged according to the sources of the information
 - Records integrates all data about the problem, gathered by members of health team
 - **4 basic components**
 - **Database**
 - Contains all information from the patient when he first entered the agency
 - It includes nursing assessment, physician's history, social and family data, results of physician's examination
 - **Problem lists**
 - Contains all the aspects of the person's life requiring health care
 - Kept in the front of the chart
 - Problems are listed in order, which they are identified
 - Continually updated as new problems are identified and others are resolved
 - **Initial list of orders or plan of care**
 - Made with reference to the active problems and are generated by the person who lists the problem
 - **Progress notes**
 - Includes nurses narrative notes
- **Kardex**
 - Provides a concise method of organizing and recording data about the client, making information readily accessible to all members of the health care team



- May be written in a pencil to ease in recording frequent change in details of client care
- A series to flip cards usually kept in portable life

Recording

- **General guidelines**

- **Date and time**

- For legal reasons and client's safety
 - Record the time in conventional manner or according to 24 hour clock to avoid confusion about whether the time was am or pm

- **Timing**

- Adjust the frequency as per client's condition indicates
 - No recording should be done BEFORE providing nursing care
 - Documenting should be done as soon as possible after assessment/intervention

- **Legibility**

- All entries must be easy to read to prevent interpretation errors

- **Permanence**

- Records are made in the dark permanent ink

- **Use of accepted terminology**

- Use only commonly accepted abbreviation, symbols and terms that are specified by the agency

- **Correct spelling**

- Is essential for accuracy in recording
 - If unsure how to spell, look it up in a dictionary or other resource book

- **Signature**

- Each recording in nursing notes is signed by the nurse making it
 - Include name and title

- **Accuracy**

- Client's name should be written on each page of the clinical record
 - Accurate notations consists of facts/observations rather than opinions or interpretation
 - When recording MISTAKE is made, draw a line through it and write the words "mistaken entry" (avoid writing the word "error") above or next to the original entry with your initials and name
 - Do not erase, blot out or use correction fluid
 - Write every line but not between the line
 - If a blank appears in the notation, draw a through the blank space and sign the notation

- **Sequence**



- Document events in the order which they occur
- **Appropriateness**
 - Record only information that pertains to the client's health problems and care
 - Recording irrelevant information may be considered an invasion of the client privacy
- **Completeness**
 - Information needs to be complete and helpful to the client and health care professionals
 - Care that is omitted because of client's refusal of treatment must also be recorded
 - Document what and why it is omitted and who was notified
- **Conciseness**
 - Recording needs to be brief as well as completed to save time in communication
- **Legal prudence**
 - Accurate and complete documentation should be a legal protection to the client and health care team
- **Confidentiality**
 - Only the health professionals who participate in the care of the client are allowed to read the chart

Reporting

- Takes place when two or more people share information about client care, either face to face via telephone
- **Types**
 - **Change of shifts report or endorsement**
 - For continuity of care of clients by providing quick summary of health care needs and details of care to be given
 - It is not merely reciting the content of the kardex
 - **Telephone reports**
 - Provide clear, accurate and concise information
 - Date and time
 - Name of the person giving the information
 - Subject of information received
 - Name and signature of the receiver
 - Person receiving the information should repeat it back to the sender to ensure accuracy
 - **Telephone orders**
 - Only RNs may receive telephone orders
 - Another RN should listen in another telephone line to countercheck the details
 - Write the date and time the telephone order was received



- Write the complete order and read it back
- Question primary health care provider about any order that is unusual or contraindicated to client's condition
- The order should be countersigned by the physician who made the order within the prescribed period of time (within 24 hours)
- **Transfer report**
 - Done when transferring a client to other unit

Nursing Process

- **Purpose**
 - To identify client's health status
 - Actual health problem
 - Potential health problem or needs
 - To establish plans to meet identified needs
 - To deliver specific nursing care and improve the quality care
- **Characteristics of nursing care**
 - **Cyclical** (regularly repeated events) and **dynamic** (continuously changing)
 - **Client centered:** organizes the plan or care according to client's problems rather than the nursing goal
 - **Focused on problem solving:** nursing process is directed towards a client's responses to disease and illness
 - **Decision making:** involved in every phase of nursing process
 - Interpersonal and collaborative
 - Communicates with the client and family
 - Collaborates with other members of the health care team
 - **Universally applicable:** used in all types of health care setting with the clients of all age group
- Nurses must use a variety of critical thinking skills to carry out the nursing process
- **Components**
 - Assessment
 - Diagnosis
 - Planning
 - Implementation
 - Evaluation



Nursing Assessment

- A systematic and continuous collection, organization, validation and documentation of data about the client health status
- **Purpose:** establish a database
- **Activities during assessment**
 - **Data collection:** gathering information about client, considering the physical, psychological, emotional, social-cultural, and spiritual factors that may affect his/her health status
 - **Sources of data**
 - Client (primary)
 - Support people (secondary)
 - Family members, friends, and caregivers who know the client well
 - Client records
 - Medical records: past and present health and illness patterns
 - Records of therapies: social workers, nutritionist, dieticians, physical therapists
 - Laboratory record
 - Health care professionals
 - Literature
 - **Data collection methods**
 - **Observing:** gathers data by using senses
 - **2 aspects**
 - Noticing the data
 - Selecting, organizing, and interpreting data
 - **Interviewing:** is a planned communication or a conversation with a purpose
 - **2 approaches**
 - **Directive**
 - Highly structured and elicits specific information
 - Uses closed-ended questions
 - The nurse establishes the purposes and control the interview
 - Used when you need to elicit specific data
 - Used in emergency situation
 - **Non-directive (rapport building)**
 - Nurse allow the client to control the purpose, subject matter and pacing
 - Uses more open-ended questions
 - Advantage: allows the patient to explain certain information
- **Stages of interview**



- **The opening**
 - Most important
 - Establish rapport
 - Orientation
- **The body**
 - The client communicates what he/she thinks, feels, and perceives in response to the question
- **The closing**
 - Termination of interview
- **Data organization**
 - clustering /organizing of facts into group of information
 - Nurse uses a written/computerized data systematically
- **Validating data**
 - Double checking or verifying data to confirm that it is accurate and factual
- **Documenting data**
 - Accurate documentation is essential and should include all data collected about the client's health status
- **4 types of assessment**
 - **Initial assessment**
 - Perform within the specified time after admission
 - Main purpose is to create database for problem identification reference and future comparison
 - **Problem-focused assessment**
 - Integrated throughout the nursing process
 - Purpose is to determine the status of a specific health
 - **Emergency assessment**
 - Done during an acute physiologic and psychologic crisis of the client
 - Purpose is to identify life-threatening conditions and to identify new or overlooked problems
 - **Framework and principle in emergency assessment**
 - Airway
 - Breathing
 - Circulation
 - Use either Maslow's hierarchy of needs or ABC principles
 - **Time-lapsed assessment**
 - Done several months after the initial assessment
 - Purpose is to compare clients current status to baseline data previously obtained



Heart Sound

- **Stethoscope**
 - **Bell:** low pitch, cardiac murmur, S3
 - **Diaphragm:** high pitch, normal heart sounds, lung sounds
- **S1:** closure of atrioventricular valves (mitral/tricuspid)
 - RHD complication: carditis
 - **Jones criteria**
 - Joint involvement
 - Looks like the heart
 - Nodule
 - Erythematous marginatum
 - Syndamhum's
 - **GABHS infection:** (+) ASO titer
 - <200 todd unit/ml - adult
 - <100 todd unit/ml - pedia
- **S2:** closure of semilunar valve (aortic/pulmonic)
- **S3:** ventricular gallop
 - **Normal:** before 40 years old
 - **Normal:** pregnant
- **S4:** atrial gallop
 - MI
 - Cardiomyopathy
- **Location of valves on auscultation**
 - **Aortic:** 2 ICS
 - **Pulmonic:** 2 ICS
 - **Tricuspid:** 4 ICS
 - **Mitral:** 5 ICS
- **Characteristic of murmur**
 - **Aortic valve stenosis:** bruit 2 ICS at right
 - **Pulmonic valve stenosis:** bruit 2 ICS at left
 - Everything on the left except aortic

Lung Sounds

- **Wheezing:** musical sound
 - Narrowed small airway
 - Common during exhalation
 - Prominent during inhalation if severe airway obstruction (inspiratory wheezing)
 - **DOC:** salbutamol



- Suddenly stops: complete obstruction
- **Rhonchi**
 - Narrowed large airway
 - Heard on both inhalation and exhalation
- **Crackles:** bubbling sound
 - **Fine:** caused by fluid -> disease, heart failure, pulmonary edema
 - **Coarse:** caused by phlegm -> pneumonia, asthma
- **Pleural friction rub:** grating sound

Diagnosis

- 2nd phase of nursing process
- The process which results to a diagnostic statement or nursing diagnosis
- It is the clinical act of identifying problems
- **Purpose:** to identify the client's health care needs and to prepare diagnostic statement
- **Activities during diagnosing**
 - Organized cluster/group of data
 - Compare data with standard (norm)
 - Analyze data after comparing with standards
 - Identifying gaps and inconsistencies in data
 - Determine the client's health problems, risks, and strengths
 - Final output: nursing diagnosis statement
- **Nursing diagnosis** is a statement of client's potential or actual alteration of health status
 - It uses the critical thinking skills of analysis and synthesis
 - **Basic 2 part statements**
 - Problem (statement of the client's response)
 - Etiology (factors contributing to or probable causes of the responses)
 - Two parts are joined by the words "related to" (implies relationship)
 - **Basic 3 part statements**
 - Problem
 - Etiology
 - Signs and symptoms (defining characteristics by the client)
 - **1 part statement**
 - Consists of NANDA label only
 - **Collaborative problems**
 - Suggested that all collaborative problems begging with the diagnosing label "potential complications"
 - **Purpose of NANDA**
 - To define, refine and promote taxonomy of nursing diagnostic terminology of general use to professional nurses



- **Members**
 - Staff nurses
 - Clinical specialists
 - Faculty, directors of nursing
 - Deans, theorists, and researchers
- **Types**
 - **Actual diagnosis**
 - Client problem that is present at the time of the nursing assessment
 - **Risk diagnosis**
 - Clinical judgement that a problem does not exist, but the presence of risk factors indicates that a problem is likely to develop unless nurses intervene
 - **Wellness diagnosis**
 - Describes human responses to levels of wellness in an individual, family or community that have a readiness for enhancement
 - **Possible diagnosis**
 - Evidence about a health problem is incomplete or unclear
 - **Syndrome diagnosis**
 - Associated with a cluster of other diagnoses

Planning

- Deliberative, systematic phase of nursing process that involves decision making and problem solving
- **Goal setting:** to have a criteria for evaluation
- For the goal to be useful during evaluation, it should be stated in behavioral terms
- To effective, involve the patient and family
- **Types**
 - **Initial planning**
 - Done by the nurse who performs the admission assessment
 - **Ongoing planning**
 - Done by all nurses who work with the client
 - Occurs at the beginning of a shift as the nurse plans the care given that day
 - **Discharge planning**
 - Process of anticipating and planning for needs after discharge, is a crucial part of comprehensive care
 - Begins at first client contact and involves comprehensive and ongoing assessment to obtain information about client's ongoing needs
- **Planning process**
 - Setting priorities



- Establishing client goals/desired outcomes
- Selecting nursing interventions
- Writing individualized nursing interventions on care plans
- **Guidelines for writing nursing care plans**
 - Date and sign the plan
 - Use category headings
 - Use standardized/approved medical or english symbols and keywords rather than complete sentences to communicate your ideas unless the agency policy dictates otherwise
 - Be specific
 - Refer the procedure books or other sources of information rather than including all the steps on a written plan
 - Tailor the plan to the unique characteristics of the client by ensuring that the client's choices such as preferences about the times of care and methods used are included
 - Ensure that the nursing plan incorporates preventive and health maintenance aspects as well as restorative ones
 - Ensure that the plan contains interventions for ongoing assessment of the client
 - Include collaborative and coordination activities in the plan
 - Include plans for the client's discharge and home care needs
- **Characteristics of the planning process**
 - Client-centered
 - Specific
 - Measurable
 - Attainable
 - Realistic
 - Time bound

Implementation

- Doing and documenting the activities that are specific nursing actions needed to carry out the interventions
- Reassessing the client before implementing an intervention
- Determining the nurse's need for assistance
- Implementing the nursing interventions
- Supervising the delegated care
- Documenting nursing activities
- **Requirements for implementation**
 - Adequate knowledge
 - Technical skills
 - Communication skills



- Therapeutic use of self
- Right attitude

Evaluation

- Collecting data, comparing data and relating nursing activities to outcomes
- Drawing conclusions about problem status
- Continuing, modifying or terminating the NCP
- **Types**
 - **Ongoing/formative evaluation**
 - Done during or immediately after the intervention
 - Allows the nurse to decide and make on the spot modification/s in an intervention
 - **Intermittent evaluation**
 - Done at a specified time and it shows the extent of progress of the patient
 - Enables the nurse to correct deficiencies and modify the NCP
 - **Terminal/summative evaluation**
 - Done at or immediately before discharge
 - Importance: it determines whether the goals are met, partially met, or unmet
 - When goals have been partially met or when goals have not been met, two conclusions may be drawn
 - The care plan may be need to be revised, since the problem is only partially resolved
 - The care plan does not need revision, because the client merely needs more time to achieve the previously established goal/s

Concepts of Health and Illness

- **Health** is state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity (WHO)
- **Health** is the ability to maintain homeostasis or dynamic equilibrium. Homeostasis is regulated by the negative feedback mechanism (Walter Cannon)

Models of Health and Illness

- **Travis' illness-wellness continuum**
 - The illness-wellness continuum developed by Travis ranges from high-level wellness to premature death
 - The model illustrates two arrows pointing in opposite directions and joined at a neutral point



- Movement to the right of the neutral point indicates increasing levels of health and well-being for an individual
- **This is achieved in three steps**
 - Awareness
 - Education
 - Growth
- In contrast, movement to the left of the neutral point indicates progressively decreasing levels of health
- **Health belief model**
 - The model of Becker (1975) which describes the relationship between a person's belief and behavior
 - Individual perceptions and modifying factors may influence health beliefs and preventive health behavior
 - **Individual perceptions include the following**
 - Perceive susceptibility to illness
 - Perceive seriousness of an illness
 - Perceive threat of an illness
 - **Modifying factors include the following**
 - Demographic variables
 - Socio-psychologic variables
 - Structural variables
 - Cues to action
- **Smith's model of health**
 - **Clinical model:** identifies health as absence of signs or symptoms of disease or injury
 - **Role performance model:** health is identified in terms of individual's ability to perform his/her work
 - **Adaptive model:** health is a creative process; disease is a failure in adaptation; focuses on the ability of the person to cope
 - **Eudemonistic model:** health is seen as a condition of actualization or realization of person's potential
- **Leavell and Clark's Agent-Host-Environmental model (ecologic model)**
 - States that there are three interactive factors that affect health and illness
 - **Agent:** any factor or stressor that can cause or lead to illness
 - **Host:** person who may or may not be at risk of acquiring the disease
 - **Environment:** any factor external to the host that may or may not predispose the person to the development of the disease
 - **Illness:** is the state in which the person's physical, emotional, intellectual, social, developmental, or spiritual functioning is diminished or impaired compared with previous experiences



- **Disease:** an alteration in body functions resulting in reduction of capacities or a shortening of the normal life span
 - **Common causes of disease**
 - Biological agents
 - Inherited genetic defects
 - Developmental defects
 - Physical agents
 - Chemical agents
 - Tissue response to injury
 - Faulty chemical/metabolic process
 - Emotional/physical reaction to stress

Stages of Illness

- **Symptom experiences**
 - Person comes to believe something is wrong
 - **Physical:** experience of symptoms
 - **Cognitive:** the interpretation of the symptoms in terms that have some meaning to the person
 - **Emotional:** fear and anxiety
- **Assumption of the sick role**
 - Acceptance of the illness
 - Excused from normal duties and role expectations
 - Confirmation from family and friends
- **Medical care contact**
 - Seek advice of the health professionals for validation of real illness, explanation of symptoms, and reassurance or prediction of what the outcome will be
- **Dependent patient role**
 - Client becomes dependent on the health professionals for help
 - Accepts/rejects health professional's suggestions
 - Later becomes more passive and accepting
 - May regress to an earlier behavior stage
- **Recovery or rehabilitation**
 - Client is expected to relinquish the dependent role and resume former roles and responsibilities
 - **Risk factors:** any situation, habit, environmental physiologic/psychologic condition or other variable that increases the vulnerability of the individual to illness or accident
 - Genetic and physiological factors
 - Age



- Environment
- Lifestyle

Levels of Prevention

- **Primary prevention**
 - To encourage optimal health and to increase the person's resistance to illness
 - Seeks to prevent a disease or a condition at a pre-pathologic state
 - Health promotion
 - Specific protection
 - Quit smoking
 - Avoid/limit alcohol intake
 - Exercise regularly
 - Eat well-balanced diet
 - Reduce fat intake and increase fiber in the diet
- **Secondary prevention**
 - It is also known as health maintenance
 - Seeks to identify specific illnesses or conditions at an early stage with prompt intervention to prevent or limit disability
 - Early diagnosis/detection/screening
 - Prompt treatment to limit disability
 - Have annual physical examination
 - Regular pap smear test for women
 - Monthly BSE for women
- **Tertiary prevention**
 - Occurs after a disease or disability has occurred and the recovery process has begun
 - Intent to halt the disease or injury process and assist the person in obtaining an optimal health status
 - **Rehabilitation**
 - Self-monitoring of blood glucose level among diabetics
 - Physical therapy after CVA
 - Undergoing speech therapy after laryngectomy

Physiologic Responses to Stress and Illness

- **Stress**
 - Is a universal phenomenon, all person experience it
 - Is a condition in which the person responds to changes in the formal balanced state
- **Stressor**



- When a person faces stress, responses are called coping strategies, coping responses, or coping mechanism
- **Sources of stress**
 - **Internal stressor:** originate within the person
 - **External stressor:** originate outside the individual
 - **Developmental stressor:** occur at predictable times throughout an individual's life
 - **Situational stressor:** unpredictable and may occur at any time during life may be positive or negative

Response Based Model of Stress

- **Adaptation**
 - The adjustments that a person make in different situations
 - **Types**
 - **General adaptation syndrome (GAS)**
 - **Alarm reaction/stage of alarm**
 - Alerts the body's defense
 - The person becomes aware of the presence of threat or danger
 - Levels of resistance are decreased
 - Adaptive mechanism are mobilized
 - If the stress is intense enough even at the stage of alarm, death may be ensured
 - **Shock/resistance phase**
 - The stressor may be perceived consciously by the person
 - Autonomic nervous system reacts, and large amount of epinephrine and cortisol are released into the body
 - Fight or flight
 - This primary response is short, lasting from 1 minute to 24 hours
 - **Counter-shock/exhaustion phase**
 - The changes produced in the body during the shock phase are reversed
 - The person is best mobilized to react during the shock phases of the alarm reaction
 - **Stage of Resistance**
 - Is when the body's adaptation takes place
 - The body attempts to cope with the stressor and limit the stressor to the smallest area of the body that can deal with it
 - The person moves back to homeostasis
 - **Stage of exhaustion**



- The adaptation that the body made during the second stage cannot be maintained
- The ways used to cope with the stressors have been exhausted
- If adaptation has not overcome the stressors, the stress affects may spread to the entire body
- At the end of this stage, the body may either rest and return to normal, or death may be the ultimate consequence
- **Local adaptation syndrome (LAS)**
 - Man may respond to stress through a particular body part or body organ
- **Homeostasis**
 - A state of dynamic equilibrium; stability, balance; uniformity
 - It is now more commonly referred to as hemodynamics because it is characterized by constant change
 - It is regulated by negative feedback mechanism
 - **Concepts**
 - **Sympatho-adreno-medullary response (Walter Cannon)**
 - SAMR or flight response
 - Adreno-cortical response
 - Neurohypophyseal response
- **Local physiologic response to stress**
 - **Inflammation** involves mobilization of specific and nonspecific defense mechanism in response to tissue injury or infection
 - **Purpose of inflammation**
 - To localize tissue injury
 - To protect tissue from injury
 - To prepare tissue to repair
- **Cellular response**
 - **Neutrophils:** first to be launched at the site of injury
 - **Monocytes:** perform phagocytosis in chronic tissue injury
 - **Lymphocytes:** responsible for immune responses
 - **Process involved**
 - **Marginal/pavementation:** phagocytes line up at the peripheral walls of blood vessels
 - **Emigration/diapedesis:** phagocytes shift out of the blood vessels
 - **Chemotaxis:** impaired tissues release substances which exert magnet-like force to the phagocytes to bring them to the areas of injury
 - **Phagocytosis:** phagocytes ingest or engulf the antigens
- **Exudate production**
 - Inflammatory exudate is produced, consisting of fluid that escaped from the blood vessels, dead phagocytic cells, and dead tissue cells and products that they release



- Plasma protein fibrinogen, thromboplastic and platelets together form an interlacing network to wall off the area, and prevent spread of the injurious agent
- During this stage, the injurious agent is overcome, and the exudate is cleared away by lymphatic drainage
- **Healing process (reparative phase)**
 - **Regeneration:** involves replacement of damaged tissue cells by new cells which are identical in structure or function
 - **Scar formation:** involves replacement of damaged tissue cells by fibrous tissue formation
 - Granulation tissue
 - Cicatrix scar
 - **Healing may also be classified as:**
 - **First intention:** occurs in clean-cut wound. The wound edges are approximated and there is minimal or no scar tissue formation
 - **Second intention:** occurs when the wound is extensive and there is a great amount of tissue. The repair time is longer and the scarring is greater
 - **Third intention:** occurs when there is delayed surgical closure of infected wound

Vital Signs

Temperature

- Body temperature: balance between heat produced by the body and heat loss from the body
- **Types:**
 - **Core temperature:** temperature of the deep tissues of the body
 - **Surface temperature:** temperature of the skin, SQ tissues and fat, rises and falls in response to the environment
- **Processes involved in heat loss**
 - **Radiation:** transfer of heat from surface to surface of one's object to surface of another without contact
 - **Conduction:** transfer of heat from one surface to another through direct contact
 - **Convection:** dispersion of heat by air currents
 - **Evaporation:** vaporization of moisture from the respiratory tract, mucosa of the mouth and skin
- **Factors affecting body temperature**
 - Age
 - Diurnal variation (circadian rhythms)
 - Highest temperature: 4pm-6pm
 - Lowest temperature: 4am-6am



- Exercise
- Hormones
- Stress
- Environment
- **Alterations in body temperature**
 - **Pyrexia/hyperthermia/fever**
 - Body temperature is above the usual range
 - **Hyperpyrexia:** very high fever, 41C (105.8F) and above
 - **Hypothermia**
 - Core body temperature is below the lower limit of normal
 - May be caused by excessive heat loss, inadequate heat production or impaired hypothalamic thermoregulation
 - **Types of fever**
 - **Intermittent:** body temperature alternates at regular intervals between periods of fever and normal and subnormal temperature (within 24 hours)
 - **Remittent:** wide range of temperature fluctuations more than 2C for over 24 hours, all of which are above normal
 - **Relapsing:** short febrile periods of a few days are interspersed with periods of 2-3 days of normal temperature, saddle back shape
 - **Normal temperature in dengue:** low platelet (risk for bleeding)
 - **Constant fever:** body temperature fluctuates minimally but always remains above normal (40-41C: unresponsive to antipyretics)
 - **Kawasaki Disease:** unknown, autoimmune, vasculitis
 - **DOC:** aspirin + IVIG
 - **Fever spike (staircase):** temperature rises to fever level rapidly following a normal temperature then returns to normal within a few hours
- **Assessing body temperature**
 - **Oral**
 - Considered to be the most convenient and most accessible
 - Wait for 30 minutes before taking oral temperature if the client has taken cold or hot drinks/food or smoked
 - **Contraindicated to patients with:**
 - Oral lesions/surgery
 - Dyspnea
 - Cough
 - O2 therapy
 - Nausea and vomiting
 - Presence of oro-nasal pack, NGT, ET
 - Seizure prone
 - Very young children (<5 years old) -> febrile seizure



- Safety
- Airway/O2
- Take duration of seizure
- Restless, disoriented, confused (ALOC)
- Clean the thermometer before use (bulb to stem) and after use (stem to bulb)
- Place the bulb of thermometer on either side of the frenulum
- Take oral temperature for 2-3 mins
- **Normal range:** 35-37.5C
- **Rectal**
 - Considered to be very accurate
 - **Contraindicated to patients with:**
 - Anal or rectal conditions/surgery
 - Diarrhea
 - Hemorrhoids
 - Increased ICP -> at risk for brain herniation
 - **Monro-Kellie Doctrine** (3 constant in the skull)
 - **Brain volume:** meningitis/cerebral edema
 - **CSF:** continuously produced and absorbed (400-600 ml/day) -> brain herniation
 - **Intracerebral blood:** hematoma
 - Quadriplegia and myocardial infarction -> bradycardic
 - Morphine
 - Oxygen
 - Nitroglycerin
 - Aspirin
 - + laxative
 - Wear clean gloves and assists the client to assume lateral/sim's position
 - Lubricate thermometer before insertion
 - Instruct the client to take a slow deep breath during insertion
 - Never force the thermometer if resistance is felt
 - Insert 15 cm (6 in) in adults and 11 cm for children
 - Hold the thermometer in place for 2 mins
 - **Normal range:** 36-37.8C
- **Axillary**
 - Safest and non-invasive
 - Pat dry the axilla
 - Rubbing causes friction that may increase surface temperature
 - The bulb is placed in the center of axilla
 - Place the arm tightly across the chest to keep the thermometer in place and leave it for 9 mins

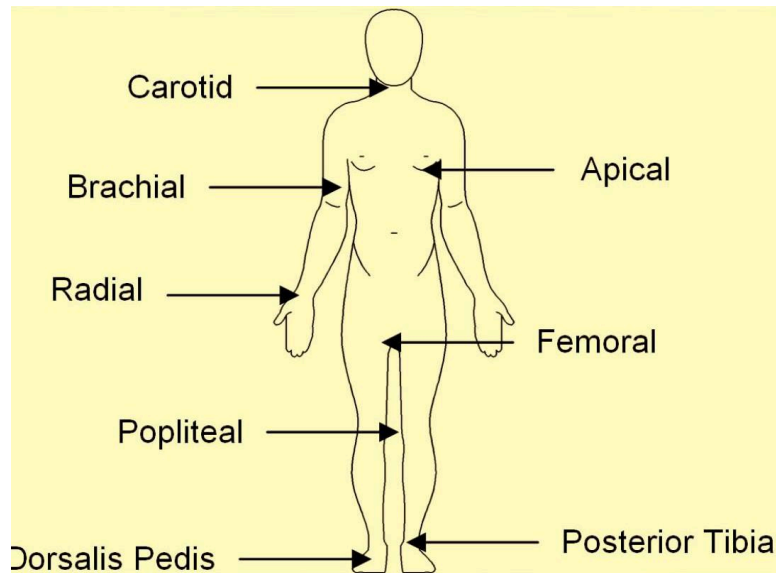


- **Contraindication:** lesion, CVA, amputation
- **Tympanic membrane**
 - Frequent site for estimating core body temperature
 - Pull pinna back and upward (adults)
 - Supine, head stabilized; pull pinna straight back and slightly downward for children <3 years old
 - Put child on adult's lap; pull pinna straight back and upward for children >3 years old
 - Point the probe slightly anteriorly, toward the eardrum
 - Insert the probe slowly using a circular motion until snug
- **Conversion of F to C**
 - $5/9 (F-32) = C$
- **Conversion of C to F**
 - $(C \times 9/5) + 32 = F$
- **Tepid sponge bath (TSB)**
 - Temperature of water: 32C
 - How to apply: done by patting
 - Rationale: to avoid friction, which increases temperature
 - Do not use alcohol when applying TSB
 - Rationale: alcohol dries skin and leads to irritation

Pulse

- Wave of blood created by contraction of the left ventricle of the heart
- Cardiac output is the volume of blood pumped into the arteries by the heart
- Normal CO is 5L of blood/min
- $CO = \text{stroke volume} \times \text{heart rate}$
- **Factors affecting the pulse**
 - Age
 - Gender (male < female)
 - Exercise
 - Fever
 - Medications
 - Hypovolemia
 - Stress
 - Position changes
 - Pathology
- **Pulse sites**
 - **Temporal:** used when radial pulse is not accessible
 - Located lateral to the eyes

- **Carotid:** used during cardiac arrest
 - Located under the mandible
 - CPR in adults
- **Radial:** readily accessible
- **Apical:** routinely used for infants and children up to 3 years old
 - Located left sternal border , midclavicular line 5th ICS
- **Brachial:** used to measure BP (3 years old) and used during cardiac arrest in infants
 - Located 1 ½ inches above the antecubital spae
- **Femoral:** used in cardiac arrest/shock and determine leg circulation
 - Check for bleeding after cardiac catheterization
- **Popliteal:** used to determine circulation on the lower leg
- **Posterior tibia and dorsal pedia:** used to determine circulation to the foot
 - Check after cardiac catheterization



- **Assessment**
 - **Pulse**
 - **Normal for adults:** 60-100 bpm
 - **Tachycardia:** excessively fast heart rate (over 100 bpm)
 - **Bradycardia:** heart rate adult is less than 60 bpm
 - **Rhythm**
 - The pattern of the beats and the intervals between beats
 - Irregular rhythm is referred to as dysrhythmia or arrhythmia
 - **Pulse volume (amplitude)**
 - A normal pulse can be felt with moderate pressure of the fingers
 - Full bounding pulse is a forceful volume that is obliterated with difficulty



- A pulse that is readily obliterated with pressure from fingers is referred to as weak or thready
- **Arterial wall elasticity**
 - A healthy, normal artery feels straight, smooth, soft and pliable
- **Pulse deficit**
 - Discrepancy between the apical and radial pulse
- **Pulse Scale**
 - 0: absence or cannot be felt
 - 1+: weak or thread
 - 2+: normal
 - 3+: full
 - 4+: strong and bounding
- **Pitting Edema**
 - 1+: barely detectable 2mm
 - 2+: 2mm - 4mm
 - 3+: 4mm - 7mm
 - 4+: above 7mm

Respirations

- **Ventilations:** movement of air in and out of the lungs
 - Inhalation (inspiration)
 - Exhalation (expiration)
- **Diffusion:** exchange of gases from higher pressure to an area of lower pressure
 - It occurs at the alveolocapillary membrane
- **Perfusion:** availability and movement of the blood for transport of gases
 - Nutrients and metabolic waste products
- **Two types of breathing**
 - **Costal (thoracic):** involves the external intercostal muscles and other accessory muscles
 - **Diaphragmatic (abdominal):** involves the contraction and relaxation of the diaphragm
- **Respiratory center**
 - **Medulla oblongata:** primary respiratory center
 - CO₂ is the primary chemical stimuli for breathing
 - **Pons:** contains pneumotaxic center that is responsible for rhythmic quality of breathing and apneustic center that is responsible for deep prolonged inspiration
 - **Carotid and aortic bodies:** contains peripheral chemoreceptors that are sensitive to O₂ and CO₂ level in the blood
- **Assessing respirations**
 - **Normal rate:** 12-20 cpm



- **Depth** is observed through the movement of the chest and describe as normal, deep or shallow
- **Rhythm** refers to the regularity of the expirations and inspirations
- **Quality of character** refers to respiratory effort and sound of breath
- **Factors affecting respiratory rate**
 - Exercise
 - Stress
 - Environment
 - Increased altitude
 - Medications
- **Terminologies**
 - **Tachypnea:** quick, shallow breaths (>20 cpm)
 - **Bradypnea:** abnormally slow breathing (<12 cpm)
 - **Apnea:** cessation of breathing
 - **Hyperventilation:** rapid, deep breathing
 - **Hypoventilation:** shallow respirations
 - **Cheyne-stokes:** gradually/progressive deep rate and depth breathing followed by temporary apnea
 - **Kussmaul's:** rapid, deep and labored breathing
 - DKA: ketones = acid (metabolic acidosis)
 - Fruity breath/acetone
 - **Dyspnea:** difficult and labored breathing
 - **Orthopnea:** ability to breathe only in upright sitting or standing position
 - **Stridor:** shrill, harsh sound heard during inspiration
 - **Stertor:** snoring or sonorous respiration
 - **Wheeze:** high pitched musical squeak or whistling sound occurring on expiration
 - **Bubbling:** gurgling sounds heard as air passes through moist secretions in the respiratory tract
 - **Biot's (cluster) respirations:** shallow breaths interrupted by apnea
 - Respiratory acidosis

Blood Pressure

- Pressure exerted by blood as it flows through the arteries
 - **Systolic:** BP as a result of ventricular contraction
 - **Diastolic:** BP when ventricles are at rest
 - **Pulse pressure:** difference between systolic and diastolic pressure
 - **Hypertension:** BP that is persistently above normal
 - **Hypotension:** BP that is below normal
 - **Orthostatic hypotension:** BP that falls when the client sits or stands



- **Determinants of BP**

- Pumping action of the heart: when the pumping action of the heart is weak, the BP decreases
- Peripheral vascular resistance: increased BP
- Blood vessel diameter: decreased blood vessel diameter (vasoconstriction) can increase BP
- Blood volume: when blood volume decreases, BP decreases
- Blood viscosity: BP increases when blood is viscous

- **Factors affecting BP**

- Age
- Exercise
- Stress
- Race
- Disease process
- Gender
- Medication
- Obesity
- Diurnal variations

- **Classification of BP**

Category	Systolic BP (mmHg)	Diastolic BP (mmHg)
Normal	<120	<80
Prehypertension	120-139	80-89
Hypertension Stage 1	140-159	90-99
Hypertension Stage 2	>160	>100

- **Korotkoff's sound**

- **Phase 1:** first faint, clear tapping or thumping sounds are heard
- **Phase 2:** heard sounds have a muffled, whooshing or swishing sound quality
- **Phase 3:** sounds become crisper and more intense, softer thumping sound
- **Phase 4:** sound become muffled and have a soft, blowing quality
- **Phase 5:** period of silence

- **Assessing BP**

- Ensure the equipment's needed
- Use appropriate size of BP cuff: 40% of the arm
- Ensure that the client has rested
- Allow 30 mins to pass if the client had engaged to exercise, had smoked or ingested caffeine before taking the BP
- Position the client in sitting or supine position



- Position the arm at the level of the heart, left arm is preferably used
- Wrap the cuff evenly around the upper arm, 1 inch above the antecubital space
 - Cover 80% of the area from the shoulder to the elbow
- Determine palpatory BP first before auscultatory BP
- Position the stethoscope appropriately
- Inflate the cuff until sphygmomanometer reads 30 mmHg above the point where the brachial pulse disappeared
- Release the valve of the cuff at the rate of 2-3 mmHg per second
- As the pressure falls, identify the manometer reading at the Korotkoff's phases
- Deflate the cuff rapidly and completely
- Wait for 1-2 mins before making further determinations
- **Errors**
 - **False low BP**
 - Bladder cuff too wide
 - Arm above the heart level
 - **False high BP**
 - Bladder cuff too narrow
 - Loose cuff
 - Arm below the heart level
 - Arm unsupported
 - Insufficient rest
- **Menopause:** decreased estrogen = decreased vasodilation = decreased BP

Pain

- Pain is an unpleasant sensory and emotional experience associated with actual and potential tissue damage
- Pain is referred as the fifth vital sign (American pain society, 2003)
- **Types**
 - **Acute pain:** lasts from seconds to 6 months
 - **Chronic pain:** constant or intermittent pain that lasts for 6 months or longer
 - Persistent, non-malignant
 - **Cancer-related pain**
- **Characteristics (PQRST)**
 - **Provoking factors:** what are you doing at the time of onset?
 - **Quality:** how does the pain feel?
 - **Radiation:** where is the pain? Is there pain anywhere else?
 - Referred pain: pain that radiates to other areas of the body
 - **Severity/intensity:** pain rating using standard scale
 - Mild to severe



- Pain score: 0-10
- **Time:** how long have you had the pain?
 - Acute/chronic
- **Assessment**
 - **Descriptive pain intensity scale**
 - No pain to worst possible pain
 - **Numeric pain scale**
 - 0-10
 - **Visual analogue scales**
 - No pain to pain as bad as it could be
 - **Face pain scale**
 - Suitable for children to describe pain

Physical Health Examination

- Conducted from head to toes (cephalocaudal technique)
- Determine the state of awareness of the client at the beginning of the physical examination
- The most important consideration during the physical examination is to prepare the client physically and psychologically
- Protect the client's privacy during the entire procedure
- Invasive procedures cause feelings of embarrassment
- Prepare the needed articles and equipment before the start of the procedure to conserve time, effort, and prevent fatigue in the client
- **Modes of examination (IPaPeA)**
 - **Inspection:** assessing the patient using the sense of sight
 - **Palpation:** examining the body using the sense of touch by using the fat pads of the fingers
 - Light (superficial) palpation should always precede by deep palpation
 - For light palpation, the nurse extends the dominant hand's fingers parallel to the skin surface and presses gently while moving the hand in a circle
 - **Percussion:** tapping the body parts to produce sound
 - **Auscultation:** listening to the body sounds with the use of stethoscope
- **Special Nursing Considerations:**
 - The sequence for examination of the abdomen (**IPaPePa**)
 - Palpate the painful quadrant of the abdomen last
 - Sequence of examining the abdomen is as follows: **right lower quadrant -> right upper quadrant -> left upper quadrant -> left lower quadrant**
 - **Not included** in assessment of **scrotum:** percussion
 - **Monthly BSE:** determine the changes from previous assessment

Alvarado score	
Feature	Score
Migration of pain	1
Anorexia	1
Nausea	1
Tenderness in right lower quadrant	2
Rebound pain	1
Elevated temperature	1
Leucocytosis	2
Shift of white blood cell count to the left	1
Total	10

Percussion Sounds and Tones			
Sound	Intensity	Quality	Example of Location
Flatness	Soft	Extremely dull	Muscle and bone
Dullness	Medium	Thud-like	Liver and heart
Resonance	Loud	Hollow	Normal lung
Hyperresonance	Very loud	Booming	Emphysematous lung
Tympany	Loud	Musical	Stomach filled with gas (air)
Normal Breath Sounds			
Type		Description	
Vesicular		Soft-intensity, low-pitched “gentle sighing” sounds created by air moving through smaller airways (bronchioles and alveoli)	
Broncho-vesicular		Moderate-intensity and moderate-pitched “blowing” sounds created by air moving through larger airway (bronchi)	



Bronchial (tubular)	High pitched loud, “harsh” sounds created by air moving through the trachea
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Positions

1. Supine

- a. the most natural “at rest” position
- b. patient is flat on his back
- c. Indication: used in surgery for abdominal, facial and extremity procedures

2. Prone

- a. patient will be lying on their stomach
- b. Patient lies face down, head is turned to one side
- c. indication: drainage of the mouth after oral or neck surgery. It also allows for full extension of the knees and legs

3. Lithotomy

- a. Common in OB/GYN whether during an exam or labor
- b. patient is lying on their back with knees elevated and hips level, often supported by stirrups
- c. Indication: gynecological procedures and childbirth

4. Sim's position

- a. prone + lateral position
- b. The patient lies on his side with his upper leg flexed and drawn in towards the chest and the upper arm flexed at the elbow
- c. Indication: administering enemas, perineal examination and comfort in pregnancy, may be used for colonoscopies

5. Lateral

- a. lying on either side
- b. Right lateral means the right side of the patient is touching the bed; left lateral means the left side of the patient is touching the bed
- c. Indications: GI issues and rectal surgery/colonoscopies

6. Fowler's position

- a. Semi fowlers - 30-40; high fowlers - 90 degrees
- b. A position which the head and trunk are raised 40-90 degrees
- c. Indications: cardiac issues, SOB related to respiratory/cardiac issues, NG placement, EGD

7. Trendelenburg

- a. “Upside down” or shock position



- b. The position involves a patient lying supine, lowering the head of the bed and raising the foot
- c. Indication: treat hypotension, during gynecological and abdominal hernia surgeries, placement of central lines

8. Reverse trendelenburg

- a. Patient is in the supine position with the head of the bed elevated and the foot of the bed down
- b. Indication: used in surgery to help promote perfusion in obese patients, helpful in treating venous air embolism and preventing pulmonary aspiration

Administration of Medications

- Ten rights of medication administration

1. Right drug
2. Right dose
3. Right client
4. Right route
5. Right time
6. Right documentation
7. Right approach
8. Right to know about the drug
9. Right to refuse
10. Right drug history

- Types of medication action

- **Therapeutic effect:** primary effect
- **Side effect:** predictable, may be harmless or potentially harmful
- **Adverse effect:** more severe side effect and may justify the discontinuation of the drug
- **Toxic effect:** overdosage or buildup of the drug in the blood because of impaired metabolism or excretion
- **Idiosyncratic effects:** unexpected and maybe individual to the client
- **Allergic reaction:** immunologic reaction

- Types of medication order

- **STAT:** an order for a single dose of medication to be given immediately
- **Single dose:** medication to be given once at a specified time
- **Standing order:** may or not have termination date may be carried out indefinitely until an order is written to cancel it or may be carried out for a specified number of days
- **PRN:** permits the nurse to give a medication when in the nurse's judgement, the client requires it



Drug Administration Route

- **Oral**
 - Considered to be the most convenient, usually less expensive, and safe because skin is not broken
 - Disadvantages include:
 - Drugs may have unpleasant taste
 - Inappropriate to client with nausea and vomiting and cannot swallow
 - Drug may stain teeth
 - Drug may irritate the gastric mucosa
 - Drugs may be aspirated by seriously ill patient
 - Never crush enteric coated or sustained release tablets
 - **Drug forms for oral administration:**
 - **Solid:** capsules, pills, tablets or powder
 - **Liquid:** syrup, suspension, emulsion, elixir
 - Syrup: sugar based liquid medication
 - Suspension: water-based liquid medication, shake the bottle before use
 - Emulsion: oil-based medication
 - Elixir: alcohol-based medication, allow 30 minutes to elapse
 - **Sublingual**
 - Drug that is placed under the tongue where it is dissolved
 - Drug is rapidly absorbed
 - If swallowed, may be inactivated by the gastric juice
 - **Buccal**
 - Medication is held in the mouth against the mucus membrane of the cheek until the drug is dissolved
 - Medication should not be chewed, swallowed or placed under the tongue
 - Swallowed drug may be inactivated by the gastric juices
- **Tropical**
 - Application of the medication to a circumscribed area of the body
 - **Dermatologic preparation:** applied to the skin
 - **Installation and immigration:** applied into the body cavities or orifices such urinary bladder, eyes, ears, nose, rectum and vagina
 - **Inhalations:** administered in the respiratory tract by a nebulizer or positive pressure breathing apparatus
- **Parenteral:** administration of medication via needle
 - **Subcutaneous:** into the subcutaneous tissue, just below the skin
 - **Intramuscular:** into the muscle
 - **Intradermal:** under the epidermis
 - **Intravenous:** into a vein



- **Intra-arterial:** into to artery
- **Intracardiac:** into the heart muscle
- **Intraosseous:** into a bone
- **Intrathecal:** into the spinal canal
- **Epidural:** into epidural space
- **Intrapleural:** into the pleural space

Dosage Calculation

- **Oral medication: solid**
 - $\text{Desired dosage/stock dosage} = \text{quantity of drugs}$
- **Oral/parenteral medication: liquid**
 - $\text{Desire dose} \times \text{dilution/stock dose} = \text{quantity of drugs}$
- **Pediatric dose**
 - **Clark's rule**
 - $\text{Wt. in lbs} \times \text{usual adult dose}/150 = \text{safe child dose}$
 - **Fried's rule**
 - $\text{Age in months} \times \text{usual adult dose}/150 = \text{child's dose}$
 - **Young's rule**
 - $\text{Age in years}/\text{Age in years} + 12 \times \text{usual adult dose} = \text{child's dose}$

IV Therapy

- IV fluid therapy is essential when clients are unable to take food and fluids orally
- An efficient and effective method of supplying fluids directly into the intravascular fluid compartment and replacing electrolyte losses
- **Types of intravascular solution**
 - **Isotonic solution:** often used used to restore vascular volume
 - 0.9% NaCl (normal saline)
 - Lactated ringer's (balanced electrolyte solution)
 - 5% dextrose in water (D5W)
 - **Hypotonic solution:** have a lesser concentration of solutes
 - 0.45% NaCl (half normal saline)
 - 0.33% NaCl (one-third normal saline)
 - **Hypertonic solution:** have greater concentration of solutes than plasma
 - 5% dextrose in normal saline (D5NS)
 - 5% dextrose in 0.45% NaCl (D5½NS)
 - 5% dextrose in lactated ringers (D5LR)



Intravenous Infusion Computation

- **Drops per minute**
 - Total infusion volume (ml) x drop factor/total time of infusion (hours) x 60 mins = gtts/min
- **Milliliters per minute**
 - Total infusion volume (ml)/total infusion time (min) = ml/min

Intravenous Infusion Complications

- **Infiltration:** needle is out of the vein and fluids accumulate in the surrounding tissues
 - **Signs:**
 - Swelling
 - Coldness, pallor
 - Pain around infusion site
 - **Management:**
 - Disconnect the IV infusion and restart at different site
 - Limit the movement of the extremity
- **Circulatory overload:** results from administration of excessive volume of IV fluids
 - **Signs:**
 - Engorged neck veins
 - Hypertension
 - Dyspnea
 - **Management:**
 - Slow the rate of the infusion
 - Notify the doctor
 - Monitor VS and rate of intravenous fluid
- **Superficial thrombophlebitis:** due to overuse of a vein, irritating solutions or drugs, clot formation or large bore catheter
 - **Signs:**
 - Local tenderness
 - Acute tenderness
 - Redness, warmth slight edema of the vein above the insertion site
 - **Management:**
 - Discontinue the infusion immediately
 - Apply warm, moist compress to the affected site
 - Avoid further use of the vein
 - Restart the infusion at a different site
- **Air embolism:** air manages to get into the circulatory system
 - **Management:**



- Position the patient on left side lying
- Notify the physician especially if the patient experiences sudden pain or difficulty of breathing
- **Infection:** invasion of pathogenic organisms in the body
 - **Signs:**
 - Fever, malaise, pain
 - Swelling at the site of infusion
 - Discharge at the IV insertion site
 - **Management:**
 - Use aseptic technique when starting an infusion
 - Change the dressing regularly
 - Always wash hands before handling the tubing
 - Administer antibiotic as ordered
- **Speed shock:** may result from administration of IV push medications rapidly
 - **Signs:**
 - Pounding headache
 - Fainting, chills
 - Rapid pulse rate, back pain
 - Apprehension, dyspnea
 - **Management:**
 - Refer to the physician and monitor VS and rate of infusion
 - Use proper tubing especially to all pediatric client

Blood Transfusion

- Introduction of whole blood or blood component into the venous circulation
- **Blood products for transfusion**
 - **Whole blood:** for extreme cases of acute hemorrhage
 - **Packed RBC:** used to increase oxygen-carrying capacity of the blood
 - **Autologous RBC:** used for blood replacement following planned elective surgery
 - **Platelets:** used in patients with bleeding disorders or platelet deficiency
 - **Fresh frozen plasma:** expands blood volume and provides clotting factor
 - **Albumin and plasma protein fraction:** blood volume expanders and provides plasma protein
 - **Clotting factors and cryoprecipitate:** used for clients with clotting factors deficiencies

Transfusion Reaction

- **Hemolytic reaction:** incompatibility between client's blood and donor's blood



- **Clinical manifestations:** chills, fever, headache, backache, dyspnea, cyanosis, chest pain, tachycardia, hypotension
- **Nursing intervention:**
 - Stop the transfusion
 - Maintain vascular and access with normal saline
 - Notify the physicians immediately
 - Monitor VS
 - Monitor I&O
 - Send the remaining blood, blood set, sample of the client's blood to the laboratory
- **Febrile reaction:** sensitivity of the client's blood to WBC, platelets or plasma proteins
 - **Clinical manifestations:** fever, chills, warm, flushed skin, headache, anxiety, muscle pain
 - **Nursing intervention:**
 - Stop transfusion immediately
 - Give antipyretic as ordered
 - Notify the physician
 - KVO with normal saline
- **Allergic reaction (mild):** sensitivity to infused plasma protein
 - **Clinical manifestation:** flushing, itching, urticaria, bronchial wheezing
 - **Nursing intervention:**
 - Stop or slow the transfusion
 - Notify the physician
 - Administer antihistamine as ordered
- **Allergic reaction (severe):** antibody-antigen reaction
 - **Clinical manifestation:** dyspnea, chest pain, circulatory collapse, cardiac arrest
 - **Nursing intervention:**
 - Stop the transfusion
 - KVO with NSS
 - Notify physician immediately
 - Monitor VS
 - Administer medications/oxygen as ordered
- **Circulatory overload:** blood administered faster than the circulation can accommodate
 - **Clinical manifestations:** cough, dyspnea, crackles, distended neck veins, tachycardia, hypertension
 - **Nursing intervention:**
 - Place the patient in upright with feet dependent
 - Stop or slow the infusion
 - Notify the physician
 - **Administer diuretics and oxygen as ordered**
- **Sepsis:** contaminated blood is administered



- **Clinical manifestations:** high fever, chills, vomiting, diarrhea, hypotension
- **Nursing intervention:**
 - Stop the transfusion
 - KVO with NSS
 - Notify the physician
 - Administer IV fluids, antibiotic
 - Obtain blood specimen for culture
 - Send the remaining blood and tubing to the laboratory

Chest Physiotherapy

- Done 1 hour before meals or at bedtime
- **Vibration:** shaking movement
- **Percussion:** deliver blows (cupped hands)
- **Postural drainage:** gravity -> elevate the affected side

Suctioning

- 10 - 15 secs: adult
- No more than 10 secs: pedia
- Hyperventilate = BVM (ambubag)
 - **3 infiltrates:** pedia
 - **3 hyperinfiltrate:** adult
- Suction withdrawal using circular motion with intermittent suction
- Pinna: down back (child)
- Pinna: up back (adult)

Catheter

1. **Straight:** relieve urinary retention
 2. **Indwelling (foley):** long-term use (72 hours)
 3. **Coude:** small lumen (BPH)
 4. **Condom:** urinary incontinence
 5. **3-way: cystoclysis** (continuous bladder irrigation)
 - a. 24 hours (pink), if not increase rate
- **Catheter associated UTI**
 - Maintain a closed sterile drainage system (no disconnection, no flushing)
 - Encourage a liberal/generous amount of fluid intake
 - Decreased urine = palpate the hypogastrium/bladder



- Residual urine = catheterize the patient immediately after voiding
 - Measure both the voided and catheterized amounts